



dsPIC33CK512MPT608 Family

Digital Signal Controllers with Secure Subsystem

Description

The dsPIC33CK512MPT608 family devices are Secure Digital Signal Controllers (DSCs) that are intended for automotive, industrial or commercial systems. The devices support implementing code authentication (secure boot), Message Authentication Code (MAC) generation, trusted firmware updates, mutual node authentication and multiple key management protocols, including Transport Layer Security (TLS) and other roots of trust-based operations. This family offers features that enable implementing motor control, digital power, general purpose and robust applications with security.

Operating Conditions

- 3V to 3.6V, -40°C to +125°C:
 - DC to 100 MIPS

Core: dsPIC33CK CPU

- 256-512 Kbytes of Program Flash with ECC and 64 Kbytes of Data RAM
- Fast Six-Cycle Divide
- LiveUpdate
- Code-Efficient (C and Assembly) Architecture
- 40-Bit Wide Accumulators
- Single-Cycle (MAC/MPY) with Dual Data Fetch
- Single-Cycle, Mixed-Sign MUL Plus Hardware Divide
- 32-Bit Multiply Support
- Five Sets of Interrupt Context Selected Registers for Fast Interrupt Response
- Zero Overhead Looping
- RAM Memory Built-In Self-Test (MBIST)

Advanced Flash Features

- Dual Partition Flash with LiveUpdate Capability for Firmware Updates
- Secure Use Case Includes:
 - Capability to factory program the Flash with secure bootloader and enable the Flash OTP, while allowing end user to program application code using the bootloader
 - Immutable secure boot implementation to support code authentication and secure firmware

Security Features

- Secure Subsystem:
 - Secure Subsystem with Advanced Crypto Engine (ACE) for execution of various cryptography commands
 - Fast Crypto Engine for SHA-256, HMAC and AES-CMAC algorithms
 - Physical Protection Security Measures, including voltage tamperers, temperature tamperers and active shield circuitry
 - Sign/verify support:
 - ECDSA – P224, P256, P384 and 256-bit Brainpool elliptic curves
 - ECDSA – SECP256K1 (bitcoin/blockchain) curve
 - RSA 2048-bit signature generation and verification
 - RSA 3072-bit signature verification only
 - ECDH/ECDHE/ECBD key agreement support:
 - Elliptic Curve Diffie-Hellman (ECDH) support for P224, P256, P384 and 256-bit Brainpool
 - Elliptic Curve Burmester-Desmedt (ECBD) support for P224 curve
 - Internal symmetric and asymmetric key generation and derivation:
 - P224, P256, P384 and 256-bit Brainpool
 - 2048-bit RSA keys
 - AES 16-byte keys
 - AES and RSA encryption/decryption support:
 - AES ECB/GCM encryption/decryption supported directly
 - RSA 1024-bit and 2048-bit keys encryption/decryption support
 - NIST SP800-90 A/B/C Random Number Generator (RNG)
 - 16 MHz SPI interface to communicate security commands between the core and the Secure Subsystem
 - Secure Subsystem's authorization sessions can be used to prevent various kinds of attacks or denial of service
 - Secure Subsystem's Advanced Crypto Engine algorithms have achieved JIL HIGH rating and are certified by FIPS as per Cryptographic Algorithm Validation Program (CAVP)
 - Secure Subsystem with FIPS 140-2 Level 2 with Physical Security Level 3 certification as per Cryptographic Module Validation Program (CMVP) [in progress]
- Advanced Flash Security with CodeGuard™ Flash Security and Flash OTP:
 - Secure Boot Segment (BS) with immutability:
 - CodeGuard Flash Security and Flash OTP by Write Inhibit facilitates realizing secure boot with immutability and external tools (Debugger/ICSP™) will not have access to read/write/erase the Boot Segment:
 - Optionally, as per user configuration, it is possible to allow an external tool to read the Flash content
 - User code in General Segment (GS) will not be capable of reading/writing/erasing code in Boot Segment (with high-security setting) via Run-Time Flash Programming/Reading:
 - Ensures integrity of the boot code
 - Code in Boot Segment cannot modify itself with high security and write protection enabled
- Secure firmware upgrade with CodeGuard™
 - CodeGuard provides protection for on-chip code Flash memory
 - Provides an environment to support secure methods for transmitting and receiving data in the customer end product
- Additional Security Features Include:
 - Does not allow execution from RAM
 - Restricts ICSP programming/erasing operations for all of Flash memory when Flash OTP by Write Inhibit is activated, and cannot be deactivated

Safety Features

- DMT (Deadman Timer)
- ECC (Error Correcting Code) for Flash Memory
- WDT (Watchdog Timer)
- CodeGuard™ Security
- CRC (Cyclic Redundancy Check)
- ICSP™ Write Inhibit
- RAM Memory Built-In Self Test (MBIST)
- Two-Speed Start-up
- Fail-Safe Clock Monitoring (FSCM)
- Backup FRC (BFRC)
- Capless Internal Voltage Regulator
- Virtual Pins for Redundancy and Monitoring

Clock Management

- Fast RC (FRC)
- Internal Oscillator
- Programmable PLLs and Oscillator Clock Sources
- Reference Clock Output
- Fail-Safe Clock Monitor (FSCM)
- Fast Wake-up and Start-up
- 8 MHz Backup FRC (BFRC) with a Divider (244 decimal) to provide a Nominal 32.768 kHz Output with a 50% Duty Cycle

Power Management

- Low-Power Management Modes (Sleep, Idle, Doze)
- Integrated Power-on Reset and Brown-out Reset

High-Resolution PWM with Fine Edge Placement

- Up to Six PWM Channels
- 250 ps PWM Resolution
- Applications Include:
 - DC/DC Converters
 - AC/DC power supplies
 - Uninterruptable Power Supply (UPS)
 - Motor Control: BLDC, PMSM, SR, ACIM

Timers/Output Compare/Input Capture

- One General Purpose 16-Bit Timer
- Peripheral Trigger Generator (PTG) Module
- Nine MCCP/SCCP Modules which Include Timer, Capture/Compare and PWM:
 - One MCCP
 - Eight SCCPs
 - 16 or 32-bit time base

- 16 or 32-bit capture
- Four-deep capture buffer
- Fully asynchronous operation available in Sleep modes

Advanced Analog Features

- Five ADC Modules:
 - 12-bit, 3.5 Msps ADC
 - Up to 24 conversion channels
 - 250 ns conversion latency
- Six DAC/Analog Comparator Modules:
 - 12-bit DACs with hardware slope compensation
 - 15 ns analog comparators
- Shared DAC/Analog Output:
 - DAC/analog comparator outputs
- Three Op Amp Modules – 20 MHz GBW:
 - 40 V/s Slew Rate
 - ± 1 mV offset

Communication Interfaces

- Three UART Modules:
 - Support for DMX, LIN/J2602 protocols
- Two 4-Wire SPI/I²S Modules⁽¹⁾
- Two CAN Flexible Data-Rate (FD) Modules
- Three I²C Modules:
 - Support for SMBus
- PPS to Allow Function Remap
- Two SENT Modules

Note:

1. This family of devices has three SPIs. However, one SPI is used for communication between the core and Secure Subsystem. Only two are available for general purpose.

Direct Memory Access (DMA)

- Eight DMA Channels

Additional Peripheral Features

- Three Quadrature Encoder Interfaces (QEI):
 - Four inputs: Phase A, Phase B, Home, Index
- Four Configurable Logic Cells (CLCs) with Internal Connections to Select Peripherals and PPS
- Two Current Bias Generators (CBGs)

Debugger Development Support

- In-Circuit and In-Application Programming
- Three Complex, Five Simple Breakpoints
- IEEE 1149.2 Compatible (JTAG) Boundary Scan

- Trace Buffer and Run-Time Watch

Functional Safety Readiness – ISO 26262/IEC 61508/IEC 60730

To learn about the Functional Safety Readiness of this device family and various Functional Safety standards an application can target using this device family, visit www.microchip.com/dsPIC33-Functional-Safety.

Qualification Support

- AEC-Q100 REV-H (Grade 1: -40°C to +125°C) Compliant

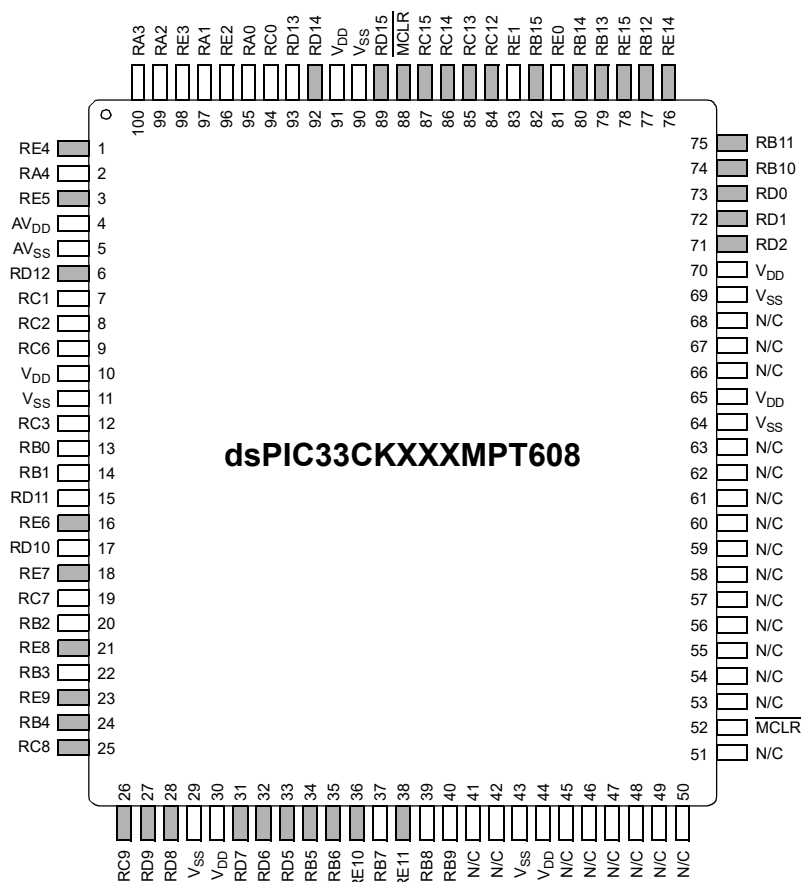
dsPIC33CK512MPT608 Product Families

The device names, pin counts, memory sizes and peripheral availability of each device are listed in [Table 1](#). The following pages show their pinout diagrams.

Table 1. dsPIC33CK512MPT608 Secure Digital Signal Controllers⁽¹⁾

Product	Pins	Flash(PEAM)	Data RAM	ADC Modules	ADC Channels	16-Bit Timers	SCCP/MCCP	CAN FD	SENT	UART	SPI/I ² S	I ² C	QEI	CLC	CRC	PWM (High Resolution)	DAC/Comparator	REFO
dsPIC33CK512MPT608	100	512K	64K	5	24	1	8/1	2	2	3	3	3	3	4	1	6	6	1
dsPIC33CK256MPT608	100	256K	64K	5	24	1	8/1	2	2	3	3	3	3	4	1	6	6	1
Note: 1. Not all pins have I/O functionality. Please refer to the Pin Diagram section for more information on the number of pins with I/O functionality.																		

Pin Diagram

100-Pin TQFP⁽¹⁾

Note:

1. Shaded pins are up to 5 V tolerant.

Table 2. 100-Pin TQFP

Pin #	Function	Pin #	Function
1	RE4	51	N/C
2	OA3OUT/AN4/ANB1/ANB2/CMP3B/ IBIAS3/RA4	52	MCLR
3	RE5	53	N/C
4	AV _{DD}	54	N/C
5	AV _{SS}	55	N/C
6	RP76/RD12	56	N/C

Legend: RPn and RPin represent remappable pins for Peripheral Pin Select (PPS) functions.

Notes:

1. A pull-up resistor is connected to this pin during programming.
2. This pin is toggled during programming.

.....continued

Pin #	Function	Pin #	Function
7	OA3IN-/AN13/CMP1B/ISRC0/ RP49 /RC1	57	N/C
8	OA3IN+/AN14/CMP2B/ISRC1/ RP50 /RC2	58	N/C
9	AN17/ANN1/CMP4B/IBIAS1/ RP54 /RC6	59	N/C
10	V _{DD}	60	N/C
11	V _{SS}	61	N/C
12	AN15/ANN2/CMP2A/IBIAS2/ RP51 /RC3	62	N/C
13	OSCI/CLKI/AN5/ RP32 /RB0	63	N/C
14	OSCO/CLKO/AN6/ RP33 /RB1 ⁽²⁾	64	V _{SS}
15	AN19/ANB0/CMP2C/ RP75 /RD11	65	V _{DD}
16	RE6	66	N/C
17	AN18/ANC2/CMP3C/ISRC3/ RP74 /RD10	67	N/C
18	RE7	68	N/C
19	DACOUT2/AN16/CMP4C/ISRC2/ RP55 /RC7	69	V _{SS}
20	OA2OUT/AN1/AN7/ANA0/ANA2/ANA3/CMP1D/ CMP2D/CMP3D/CMP4D/CMP5D/CMP6D/ RP34 /SCL3/INT0/RB2	70	V _{DD}
21	RE8	71	RP66 /RD2
22	PGD2/OA2IN-/AN8/CMP4A/ RP35 /RB3	72	RP65 /PWM4H/RD1
23	RE9	73	RP64 /PWM4L/RD0
24	PGC2/OA2IN+/ RP36 /RB4	74	TMS/ RP42 /PWM3H/RB10 ⁽¹⁾
25	RP56 /ASDA1/SCK2/RC8	75	TCK/ RP43 /PWM3L/RB11
26	RP57 /ASCL1/SDI2/RC9	76	RE14
27	RP73 /PCI20/RD9	77	TDI/ RP44 /PWM2H/RB12
28	RP72 /SDO2/PCI19/RD8	78	RE15
29	V _{SS}	79	RP45 /PWM2L/RB13
30	V _{DD}	80	RP46 /PWM1H/RB14
31	RP71 /RD7	81	AN20/ANC0/CMP5C/RE0
32	RP70 /RD6	82	RP47 /PWM1L/RB15
33	RP69 /RD5	83	AN21/ANC1/CMP6B/RE1
34	PGD3/ RP37 /SDA2/RB5	84	RP60 /PWM8H/RC12
35	PGC3/ RP38 /SCL2/RB6	85	RP61 /PWM8L/RC13
36	RE10	86	RP62 /PWM6H/RC14

Legend: **RPn** and **RPIn** represent remappable pins for Peripheral Pin Select (PPS) functions.

Notes:

1. A pull-up resistor is connected to this pin during programming.
2. This pin is toggled during programming.

.....continued

Pin #	Function	Pin #	Function
37	TDO/AN2/AN26/CMP3A/ RP39 /SDA3/RB7	87	RP63 /PWM6L/RC15
38	RE11	88	MCLR
39	PGD1/AN10/CMP6A/ RP40 /SCL1/RB8	89	RP79 /PCI22/RD15
40	PGC1/AN11/CMP5A/ RP41 /SDA1/RB9	90	V _{SS}
41	N/C	91	V _{DD}
42	N/C	92	RP78 /PCI21/RD14
43	V _{SS}	93	ANN4/CMP5B/ RP77 /RD13
44	V _{DD}	94	AN12/ANN0/ RP48 /RC0
45	N/C	95	OA1OUT/AN0/CMP1A/IBIAS0/RA0
46	N/C	96	AN22/ANB3/CMP6C/RE2
47	N/C	97	OA1IN-/ANA1/RA1
48	N/C	98	AN23/ANN3/RE3
49	N/C	99	OA1IN+/AN9/RA2
50	N/C	100	DACOUT1/AN27/AN3/CMP1C/RA3

Legend: **RPn** and **RPIn** represent remappable pins for Peripheral Pin Select (PPS) functions.

Notes:

1. A pull-up resistor is connected to this pin during programming.
2. This pin is toggled during programming.

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Referenced Sources

This device data sheet is based on the following individual chapters of the “dsPIC33/PIC24 Family Reference Manual”. These documents should be considered as the general reference for the operation of a particular module or device feature.

Note: To access the documents listed below, browse to the documentation section of the dsPIC33CK512MPT608 product page of the Microchip website (www.microchip.com) or select a family reference manual section from the following list.

In addition to parameters, features and other documentation, the resulting page provides links to the related family reference manual sections.

- “Introduction” (www.microchip.com/DS70573)
- “Enhanced CPU” (www.microchip.com/DS70005158)
- “dsPIC33/PIC24 Program Memory” (www.microchip.com/DS70000613)
- “Data Memory” (www.microchip.com/DS70595)
- “Dual Partition Flash Program Memory” (www.microchip.com/DS70005156)
- “Flash Programming” (www.microchip.com/DS70000609)
- “Reset” (www.microchip.com/DS70602)
- “Interrupts” (www.microchip.com/DS70000600)
- “I/O Ports with Edge Detect” (www.microchip.com/DS70005322)
- “Oscillator Module with High-Speed PLL” (www.microchip.com/DS70005255)
- “Direct Memory Access Controller (DMA)” (www.microchip.com/DS30009742)
- “CAN Flexible Data-Rate (FD) Protocol Module” (www.microchip.com/DS70005340)
- “High-Resolution PWM with Fine Edge Placement” (www.microchip.com/DS70005320)
- “12-Bit High-Speed, Multiple SARs A/D Converter (ADC)” (www.microchip.com/DS70005213)
- “High-Speed Analog Comparator Module with Slope Compensation DAC” (www.microchip.com/DS70005280)
- “Quadrature Encoder Interface (QEI)” (www.microchip.com/DS70000601)
- “Multiprotocol Universal Asynchronous Receiver Transmitter (UART) Module” (www.microchip.com/DS70005288)
- “Serial Peripheral Interface (SPI) with Audio Codec Support” (www.microchip.com/DS70005136)
- “Inter-Integrated Circuit (I²C)” (www.microchip.com/DS70000195)
- “Secure Subsystem” (This family reference manual is under NDA. Please contact local Microchip sales team for more information)
- “Single-Edge Nibble Transmission (SENT) Module” (www.microchip.com/DS70005145)
- “Timer1 Module” (www.microchip.com/DS70005279)
- “Capture/Compare/PWM/Timer (MCCP and SCCP)” (www.microchip.com/DS30003035)
- “Configurable Logic Cell (CLC)” (www.microchip.com/DS70005298)
- “Peripheral Trigger Generator (PTG)” (www.microchip.com/DS70000669)
- “32-Bit Programmable Cyclic Redundancy Check (CRC)” (www.microchip.com/DS30009729)
- “Current Bias Generator (CBG)” (www.microchip.com/DS70005253)
- “Deadman Timer” (www.microchip.com/DS70005155)
- “Watchdog Timer and Power-Saving Modes” (www.microchip.com/DS70615)
- “CodeGuard™ Intermediate Security” (www.microchip.com/DS70005182)
- “Dual Watchdog Timer” (www.microchip.com/DS70005250)
- “Programming and Diagnostics” (www.microchip.com/DS70608)

Terminology Cross Reference

Table 3 provides updated terminology for depreciated naming conventions. Register and bit names remain unchanged, however, descriptions and usage guidance may have been updated.

Table 3. Terminology Cross References

Use Case	Depreciated Term	New Term
CPU	Master	Initiator
DMA	Master	Initiator
I ² C	Master	Host
	Slave	Client
SPI	Master	Host
	Slave	Client
UART, LIN Mode	Master	Commander
	Slave	Responder
PWM	Master	Host
	Slave	Client

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1. Device Overview

Notes:

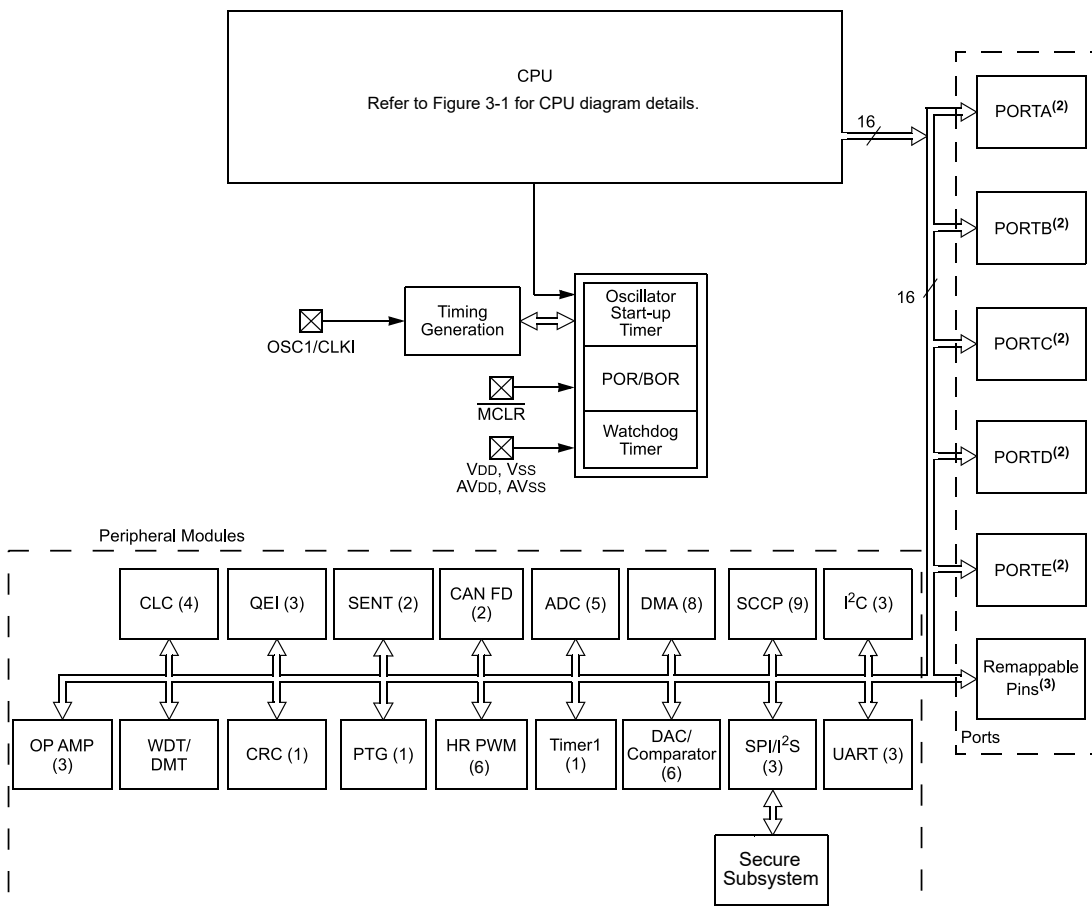
1. This data sheet summarizes the features of the dsPIC33CK512MPT608 family of devices. It is not intended to be a comprehensive resource. To complement the information in this data sheet, refer to the related section of the “dsPIC33/PIC24 Family Reference Manual”, which is available from the Microchip website (www.microchip.com)
2. Some registers and associated bits described in this section may not be available on all devices. Refer to [4. Memory Organization](#) in this data sheet for device-specific register and bit information.

This document contains device-specific information for the dsPIC33CK512MPT608 Digital Signal Controller (DSC) devices.

dsPIC33CK512MPT608 devices contain extensive Digital Signal Processor (DSP) functionality with a high-performance MCU architecture.

[Figure 1-1](#) shows a general block diagram of the core and peripheral modules of the dsPIC33CK512MPT608 family.

Figure 1-1. dsPIC33CK512MPT608 Family Block Diagram⁽¹⁾



Notes:

1. The numbers in the parentheses are the number of instantiations of the module indicated.
2. Not all I/O pins or features are implemented on all device pinout configurations. See [Table 1-1](#) for specific implementations by pin count.
3. Some peripheral I/Os are only accessible through Peripheral Pin Select (PPS).

Table 1-1. Pinout I/O Descriptions

Pin Name ⁽¹⁾	Pin Type	Buffer Type	PPS	Description
AN0-AN23, AN26, AN27	I	Analog	No	Analog input channels
ANA0-ANA3	I	Analog	No	Analog alternate inputs
ANB0-ANB3	I	Analog	No	Analog alternate “B” inputs
ANC0-ANC2	I	Analog	No	Analog alternate “C” inputs
ANN0-ANN4	I	Analog	No	Analog negative inputs
ADTRG31	I	ST	Yes	ADC Trigger Input 31
CAN1RX	I	ST	Yes	CAN1 receive input
CAN1	O	—	Yes	CAN1 transmit output
CAN2RX	I	ST	Yes	CAN2 receive input
CAN2	O	—	Yes	CAN2 transmit output
CLKI	I	ST/CMOS	No	External Clock (EC) source input. Always associated with OSCI pin function.
CLKO	O	—	No	Oscillator crystal output. Connects to crystal or resonator in Crystal Oscillator mode. Optionally functions as CLKO in RC and EC modes. Always associated with OSCO pin function.
OSCI	I	ST/CMOS	No	Oscillator crystal input. ST buffer when configured in RC mode; CMOS otherwise.
OSCO	I/O	—	No	Oscillator crystal output. Connects to crystal or resonator in Crystal Oscillator mode. Optionally functions as CLKO in RC and EC modes.
REFOI	I	ST	Yes	Reference clock input
REFCLKO	O	—	Yes	Reference clock output
INT0	I	ST	No	External Interrupt 0
INT1	I	ST	Yes	External Interrupt 1
INT2	I	ST	Yes	External Interrupt 2
INT3	I	ST	Yes	External Interrupt 3

Legend: CMOS = CMOS compatible input or output; Analog = Analog input; P = Power; ST = Schmitt Trigger input with CMOS levels; O = Output; I = Input; PPS = Peripheral Pin Select; TTL = TTL input buffer

Notes:

1. Not all pins are available in all package variants. See the [Pin Diagram](#) section for pin availability.
2. These pins are remappable as well as dedicated.
3. This device has two $\overline{\text{MCLR}}$ pins.

.....continued

Pin Name ⁽¹⁾	Pin Type	Buffer Type	PPS	Description
IOCA[4:0]	I	ST	No	Interrupt-on-Change input for PORTA
IOCB[15:0]	I	ST	No	Interrupt-on-Change input for PORTB
IOCC[15:12], IOCC[9:6], IOCC[3:0]	I	ST	No	Interrupt-on-Change input for PORTC
IOCD[15:5], IOCD[2:0]	I	ST	No	Interrupt-on-Change input for PORTD
IOCE[15:14], IOCE[12:0]	I	ST	No	Interrupt-on-Change input for PORTE
QEIA1	I	ST	Yes	QEI Input A1
QEIB1	I	ST	Yes	QEI Input B1
QEINDX1	I	ST	Yes	QEI Index 1 input
QEIHOM1	I	ST	Yes	QEI Home 1 input
QEICMP	O	—	Yes	QEI comparator output
QEIA2	I	ST	Yes	QEI Input A2
QEIB2	I	ST	Yes	QEI Input B2
QEINDX2	I	ST	Yes	QEI Index 2 input
QEIHOM2	I	ST	Yes	QEI Home 2 input
QEICMP	O	—	Yes	QEI comparator output
QEIA3	I	ST	Yes	QEI Input A3
QEIB3	I	ST	Yes	QEI Input B3
QEINDX3	I	ST	Yes	QEI Index 3 input
QEIHOM3	I	ST	Yes	QEI Home 3 input
QEICMP	O	—	Yes	QEI comparator output
RA0-RA4	I/O	ST	No	PORTA is a bidirectional I/O port
RB0-RB15	I/O	ST	No	PORTB is a bidirectional I/O port
RC0-RC3, RC6-RC9, RC12-RC15	I/O	ST	No	PORTC is a bidirectional I/O port
RD0-RD2, RD5-RD15	I/O	ST	No	PORTD is a bidirectional I/O port
RE0-RE11, RE14-RE15	I/O	ST	No	PORTE is a bidirectional I/O port
T1CK	I	ST	Yes	Timer1 external clock input

Legend: CMOS = CMOS compatible input or output; Analog = Analog input; P = Power; ST = Schmitt Trigger input with CMOS levels; O = Output; I = Input;

PPS = Peripheral Pin Select; TTL = TTL input buffer

Notes:

1. Not all pins are available in all package variants. See the [Pin Diagram](#) section for pin availability.
2. These pins are remappable as well as dedicated.
3. This device has two $\overline{\text{MCLR}}$ pins.

.....continued

Pin Name ⁽¹⁾	Pin Type	Buffer Type	PPS	Description
U1CTS	I	ST	Yes	UART1 Clear-to-Send
U1RTS	O	—	Yes	UART1 Request-to-Send
U1RX	I	ST	Yes	UART1 receive
U1TX	O	—	Yes	UART1 transmit
U1DSR	I	ST	Yes	UART1 Data-Set-Ready
U1DTR	O	—	Yes	UART1 Data-Terminal-Ready
U2CTS	I	ST	Yes	UART2 Clear-to-Send
U2RTS	O	—	Yes	UART2 Request-to-Send
U2RX	I	ST	Yes	UART2 receive
U2TX	O	—	Yes	UART2 transmit
U2DSR	I	ST	Yes	UART2 Data-Set-Ready
U2DTR	O	—	Yes	UART2 Data-Terminal-Ready
U3CTS	I	ST	Yes	UART3 Clear-to-Send
U3RTS	O	—	Yes	UART3 Request-to-Send
U3RX	I	ST	Yes	UART3 receive
U3TX	O	—	Yes	UART3 transmit
U3DSR	I	ST	Yes	UART3 Data-Set-Ready
U3DTR	O	—	Yes	UART3 Data-Terminal-Ready
SENT1	I	ST	Yes	SENT1 input
SENT2	I	ST	Yes	SENT2 input
SENT1OUT	O	—	Yes	SENT1 output
SENT2OUT	O	—	Yes	SENT2 output
PTGTRG24	O	—	Yes	PTG Trigger Output 24
PTGTRG25	O	—	Yes	PTG Trigger Output 25
TCKI1-TCKI9	I	ST	Yes	SCCP/MCCP Timer Inputs 1 through 9
ICM1-ICM9	I	ST	Yes	SCCP/MCCP Capture Inputs 1 through 9
OCFA-OCFD	I	—	Yes	SCCP/MCCP Fault Inputs A through D
OCM1-OCM9	O	—	Yes	SCCP/MCCP Compare Outputs 1 through 9

Legend: CMOS = CMOS compatible input or output; Analog = Analog input; P = Power; ST = Schmitt Trigger input with CMOS levels; O = Output; I = Input; PPS = Peripheral Pin Select; TTL = TTL input buffer

Notes:

1. Not all pins are available in all package variants. See the [Pin Diagram](#) section for pin availability.
2. These pins are remappable as well as dedicated.
3. This device has two MCLR pins.

.....continued

Pin Name ⁽¹⁾	Pin Type	Buffer Type	PPS	Description
SCK1	I/O	ST	Yes	Synchronous serial clock I/O for SPI1
SDI1	I	ST	Yes	SPI1 data in
SDO1	O	—	Yes	SPI1 data out
$\overline{SS1}$	I/O	ST	Yes	SPI1 Client synchronization or frame pulse I/O
SCK2	I/O	ST	Yes	Synchronous serial clock I/O for SPI2
SDI2	I	ST	Yes	SPI2 data in
SDO2	O	—	Yes	SPI2 data out
$\overline{SS2}$	I/O	ST	Yes	SPI2 Client synchronization or frame pulse I/O
SCK3	I/O	ST	Yes	Synchronous serial clock I/O for SPI3
SDI3	I	ST	Yes	SPI3 data in
SDO3	O	—	Yes	SPI3 data out
$\overline{SS3}$	I/O	ST	Yes	SPI3 Client synchronization or frame pulse I/O
SCL1	I/O	ST	No	Synchronous serial clock I/O for I2C1
SDA1	I/O	ST	No	Synchronous serial data I/O for I2C1
ASCL1	I/O	ST	No	Alternate synchronous serial clock I/O for I2C1
ASDA1	I/O	ST	No	Alternate synchronous serial data I/O for I2C1
SCL2	I/O	ST	No	Synchronous serial clock I/O for I2C2
SDA2	I/O	ST	No	Synchronous serial data I/O for I2C2
SCL3	I/O	ST	No	Synchronous serial clock I/O for I2C3
SDA3	I/O	ST	No	Synchronous serial data I/O for I2C3
TMS	I	ST	No	JTAG Test mode select pin
TCK	I	ST	No	JTAG test clock input pin
TDI	I	ST	No	JTAG test data input pin
TDO	O	—	No	JTAG test data output pin

Legend: CMOS = CMOS compatible input or output; Analog = Analog input; P = Power; ST = Schmitt Trigger input with CMOS levels; O = Output; I = Input;

PPS = Peripheral Pin Select; TTL = TTL input buffer

Notes:

1. Not all pins are available in all package variants. See the [Pin Diagram](#) section for pin availability.
2. These pins are remappable as well as dedicated.
3. This device has two MCLR pins.

.....continued

Pin Name ⁽¹⁾	Pin Type	Buffer Type	PPS	Description
PCI8-PCI18	I	ST	Yes	PWM Inputs 8 through 18
PWMEA-PWMEF	O	—	Yes	PWM Event Outputs A through F
PCI19 - PCI22	I	ST	Yes	PWM Inputs 19 through 22
PWM1L-PWM4L ⁽²⁾	O	—	No	PWM Low Outputs 1 through 4
PWM1H-PWM4H ⁽²⁾	O	—	No	PWM High Outputs 1 through 4
CLCINA-CLCIND	I	ST	Yes	CLC Inputs A through D
CLC1OUT-CLC4OUT	O	—	Yes	CLC Outputs 1 through 4
CMP1	O	—	Yes	Comparator 1 output
CMP1A-CMP3A	I	Analog	No	Comparator Channels 1A through 3A inputs
CMP1B-CMP3B	I	Analog	No	Comparator Channels 1B through 3B inputs
CMP1C-CMP3C	I	Analog	No	Comparator Channels 1C through 3C inputs
CMP1D-CMP3D	I	Analog	No	Comparator Channels 1D through 3D inputs
DACOUT1	O	—	No	DAC1 output voltage
DACOUT2	O	—	No	DAC2 output voltage
IBIAS3, IBIAS2, IBIAS1, IBIAS0/ISRC3, ISRC2, ISRC1, ISRC0	O	Analog	No	Constant-Current Outputs 0 through 3
OA1IN+	I	—	No	Op Amp 1+ input
OA1IN-	I	—	No	Op Amp 1- input
OA1OUT	O	—	No	Op Amp 1 output
OA2IN+	I	—	No	Op Amp 2+ input
OA2IN-	I	—	No	Op Amp 2- input
OA2OUT	O	—	No	Op Amp 2 output
OA3IN+	I	—	No	Op Amp 3+ input
OA3IN-	I	—	No	Op Amp 3- input
OA3OUT	O	—	No	Op Amp 3 output

Legend: CMOS = CMOS compatible input or output; Analog = Analog input; P = Power; ST = Schmitt Trigger input with CMOS levels; O = Output; I = Input;
PPS = Peripheral Pin Select; TTL = TTL input buffer

Notes:

1. Not all pins are available in all package variants. See the [Pin Diagram](#) section for pin availability.
2. These pins are remappable as well as dedicated.
3. This device has two MCLR pins.

.....continued

Pin Name ⁽¹⁾	Pin Type	Buffer Type	PPS	Description
PGD1	I/O	ST	No	Data I/O pin for Programming/Debugging Communication Channel 1
PGC1	I	ST	No	Clock input pin for Programming/Debugging Communication Channel 1
PGD2	I/O	ST	No	Data I/O pin for Programming/Debugging Communication Channel 2
PGC2	I	ST	No	Clock input pin for Programming/Debugging Communication Channel 2
PGD3	I/O	ST	No	Data I/O pin for Programming/Debugging Communication Channel 3
PGC3	I	ST	No	Clock input pin for Programming/Debugging Communication Channel 3
MCLR ⁽³⁾	I/P	ST	No	Master Clear (Reset) input. This pin is an active-low Reset to the device.
AV _{DD}	P	P	No	Positive supply for analog modules. This pin must be connected at all times.
AV _{SS}	P	P	No	Ground reference for analog modules. This pin must be connected at all times.
V _{DD}	P	—	No	Positive supply for peripheral logic and I/O pins
V _{SS}	P	—	No	Ground reference for logic and I/O pins

Legend: CMOS = CMOS compatible input or output; Analog = Analog input; P = Power; ST = Schmitt Trigger input with CMOS levels; O = Output; I = Input; PPS = Peripheral Pin Select; TTL = TTL input buffer

Notes:

1. Not all pins are available in all package variants. See the [Pin Diagram](#) section for pin availability.
2. These pins are remappable as well as dedicated.
3. This device has two $\overline{\text{MCLR}}$ pins.

2. Guidelines for Getting Started with Digital Signal Controllers

2.1 Basic Connection Requirements

Getting started with the family devices of the dsPIC33CK512MPT608 requires attention to a minimal set of device pin connections before proceeding with development. The following is a list of pin names which must always be connected:

- All V_{DD} and V_{SS} pins
(see [2.2. Decoupling Capacitors](#))
- All AV_{DD} and AV_{SS} pins
regardless if ADC module is not used (see [2.2. Decoupling Capacitors](#))
- $\overline{MCLR}$ pin⁽¹⁾ (see [2.3. Master Clear \(MCLR\) Pin](#))
- PGC_x/PGD_x pins
used for In-Circuit Serial Programming™ (ICSP™) and debugging purposes (see [2.4. ICSP Pins](#))
- $OSCI$ and $OSCO$ pins
when an external oscillator source is used (see [2.5. External Oscillator Pins](#))

Note:

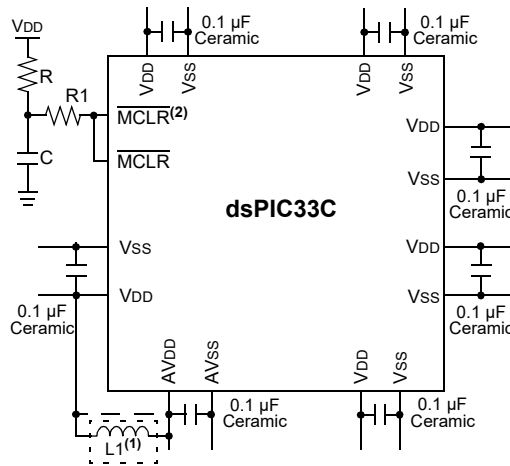
1. This device has two $\overline{MCLR}$ pins.

2.2 Decoupling Capacitors

The use of decoupling capacitors on every pair of power supply pins, such as V_{DD} , V_{SS} , AV_{DD} and AV_{SS} is required. Consider the following criteria when using decoupling capacitors:

- **Value and type of capacitor:** Recommendation of 0.1 μF (100 nF), 10-20V. This capacitor should be a low-ESR and have resonance frequency in the range of 20 MHz and higher. It is recommended to use ceramic capacitors.
- **Placement on the Printed Circuit Board:** The decoupling capacitors should be placed as close to the pins as possible. It is recommended to place the capacitors on the same side of the board as the device. If space is constricted, the capacitor can be placed on another layer on the PCB using a via; however, ensure that the trace length from the pin to the capacitor is within one-quarter inch (6 mm) in length.
- **Handling high-frequency noise:** If the board is experiencing high-frequency noise above tens of MHz, add a second ceramic-type capacitor in parallel to the above described decoupling capacitor. The value of the second capacitor can be in the range of 0.01 μF to 0.001 μF . Place this second capacitor next to the primary decoupling capacitor. In high-speed circuit designs, consider implementing a decade pair of capacitances as close to the power and ground pins as possible. For example, 0.1 μF in parallel with 0.001 μF .
- **Maximizing performance:** On the board layout from the power supply circuit, run the power and return traces to the decoupling capacitors first, and then to the device pins. This ensures that the decoupling capacitors are first in the power chain. Equally important is to keep the trace length between the capacitor and the power pins to a minimum, thereby reducing PCB track inductance.

Figure 2-1. Recommended Minimum Connection



Note 1: As an option, instead of a hard-wired connection, an inductor (L1) can be substituted between VDD and AVDD to improve ADC noise rejection. The inductor impedance should be less than 1Ω and the inductor capacity greater than 10 mA.

2: This device has two $\overline{\text{MCLR}}$ pins.

Where:

$$f = \frac{F_{CNV}}{2} \quad (\text{i.e., ADC conversion rate}/2)$$

$$f = \frac{1}{(2\pi\sqrt{LC})}$$

$$L = \left(\frac{1}{(2\pi f\sqrt{C})} \right)^2$$

2.3 Master Clear ($\overline{\text{MCLR}}$) Pin

The $\overline{\text{MCLR}}$ pin provides two specific device functions:

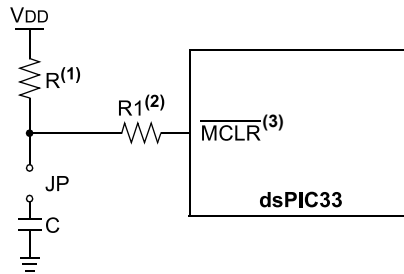
- Device Reset
- Device Programming and Debugging.

During device programming and debugging, the resistance and capacitance that can be added to the pin must be considered. Device programmers and debuggers drive the $\overline{\text{MCLR}}$ pin. Consequently, specific voltage levels (V_{IH} and V_{IL}) and fast signal transitions must not be adversely affected. Therefore, specific values of R and C will need to be adjusted based on the application and PCB requirements.

For example, as shown in [Figure 2-2](#), it is recommended that the capacitor, C, be isolated from the $\overline{\text{MCLR}}$ pin during programming and debugging operations.

Place the components, as shown in [Figure 2-2](#), within one-quarter inch (6 mm) from the $\overline{\text{MCLR}}$ pin.

Figure 2-2. Example of MCLR Pin Connections



Notes:

1. $R \leq 10 \text{ k}\Omega$ is recommended. A suggested starting value is $10 \text{ k}\Omega$. Ensure that the $\overline{\text{MCLR}}$ pin V_{IH} and V_{IL} specifications are met.
2. $R1 \leq 470\Omega$ will limit any current flowing into $\overline{\text{MCLR}}$ from the external capacitor, C, in the event of $\overline{\text{MCLR}}$ pin breakdown due to Electrostatic Discharge (ESD) or Electrical Overstress (EOS). Ensure that the $\overline{\text{MCLR}}$ pin V_{IH} and V_{IL} specifications are met.
3. This device has two $\overline{\text{MCLR}}$ pins.

2.4 ICSP Pins

The PGCx and PGDx pins are used for ICSP and debugging purposes. It is recommended to keep the trace length between the ICSP connector and the ICSP pins on the device as short as possible. If the ICSP connector is expected to experience an ESD event, a series resistor is recommended, with the value in the range of a few tens of Ohms, not to exceed 100 Ohms.

Pull-up resistors, series diodes and capacitors on the PGCx and PGDx pins are not recommended as they will interfere with the programmer/debugger communications to the device. If such discrete components are an application requirement, they should be removed from the circuit during programming and debugging. Alternatively, refer to the AC/DC characteristics and timing requirements information in the respective device Flash programming specification for information on capacitive loading limits and pin Voltage Input High (V_{IH}) and Voltage Input Low (V_{IL}) requirements.

Ensure that the “Communication Channel Select” (i.e., PGCx/PGDx pins) programmed into the device matches the physical connections for the ICSP to PICKit™ 3, MPLAB® ICD 3 or MPLAB REAL ICE™ emulator.

For more information on MPLAB ICD 2, MPLAB ICD 3 and REAL ICE emulator connection requirements, refer to the following documents that are available on the Microchip website.

- “Using MPLAB® ICD 3 In-Circuit Debugger” (poster) (DS51765)
- “Development Tools Design Advisory” (DS51764)
- “MPLAB® REAL ICE™ In-Circuit Emulator User’s Guide for MPLAB X IDE” (DS50002085)
- “Using MPLAB® REAL ICE™ In-Circuit Emulator” (poster) (DS51749)

2.5 External Oscillator Pins

When the Primary Oscillator (POSC) circuit is used to connect a crystal oscillator, special care and consideration is needed to ensure proper operation. The POSC circuit should be tested across the environmental conditions that the end product is intended to be used. The load capacitors specified in the crystal oscillator data sheet can be used as a starting point, however, the parasitic capacitance from the PCB traces can affect the circuit, and the values may need to be altered to ensure proper start-up and operation. Excessive trace length and other physical interaction can lead to poor signal quality. Poorly tuned oscillator circuits can have reduced amplitude, incorrect frequency (runt pulses), distorted waveforms and long start-up times that may result in unpredictable application behavior, such as instruction misexecution, illegal opcode fetch, etc. Ensure that the crystal oscillator circuit is at full amplitude and correct frequency before the system begins to execute code. In planning the application’s routing and I/O assignments, ensure that adjacent port pins, and other signals in close proximity to the oscillator, do not have

high frequencies, short rise and fall times, and other similar noise. For further information on the Primary Oscillator, see [10.4. Primary Oscillator \(POSC\)](#).

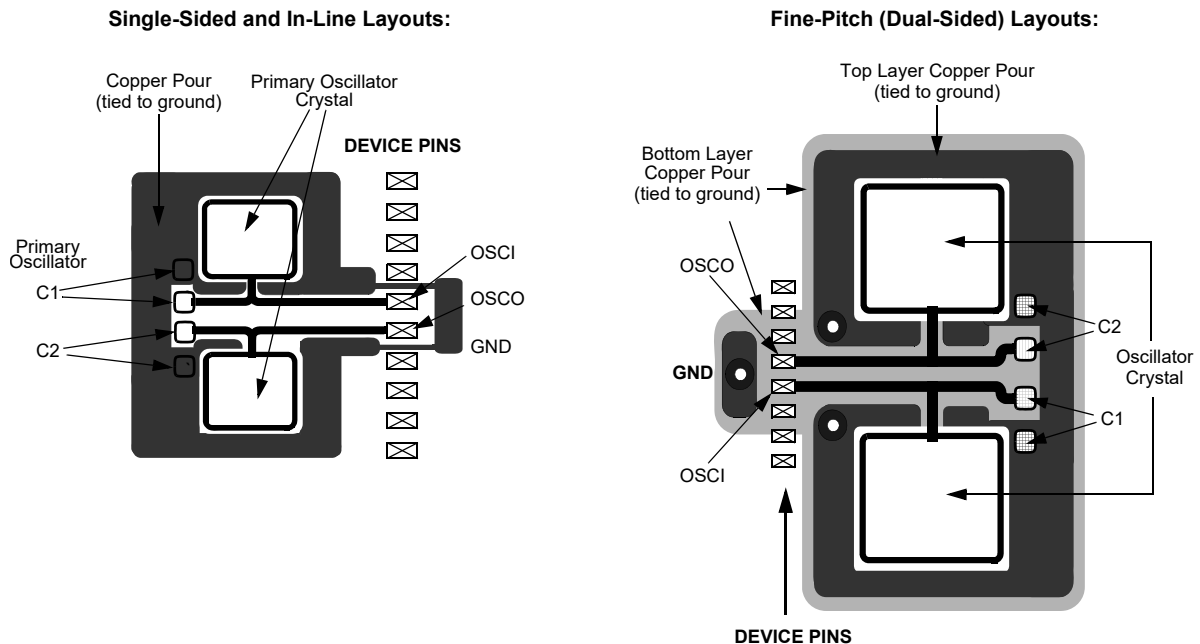
2.6 External Oscillator Layout Guidance

Use best practices during PCB layout to ensure robust start-up and operation. The oscillator circuit should be placed on the same side of the board as the device. Also, place the oscillator circuit close to the respective oscillator pins, not exceeding one-half inch (12 mm) distance between them. The load capacitors should be placed next to the oscillator itself, on the same side of the board. Use a grounded copper pour around the oscillator circuit to isolate them from surrounding circuits. The grounded copper pour should be routed directly to the MCU ground. Do not run any signal traces or power traces inside the ground pour. If using a two-sided board, avoid any traces on the other side of the board where the crystal is placed. A suggested layouts are shown in [Figure 2-3](#). With fine-pitch packages, it is not always possible to completely surround the pins and components. A suitable solution is to tie the broken guard sections to a mirrored ground layer. In all cases, the guard trace(s) must be returned to ground.

For additional information and design guidance on oscillator circuits, please refer to these Microchip Application Notes, available at the Microchip website (www.microchip.com):

- AN943, “*Practical PICmicro® Oscillator Analysis and Design*”
- AN949, “*Making Your Oscillator Work*”
- AN1798, “*Crystal Selection for Low-Power Secondary Oscillator*”

Figure 2-3. Suggested Placement of the Oscillator Circuit



2.7 Oscillator Value Conditions on Device Start-up

If the PLL of the target device is enabled and configured for the device start-up oscillator, the maximum oscillator source frequency must be limited to a certain frequency (see [10. Oscillator with High-Frequency PLL](#)) to comply with device PLL Start-up conditions. This means that if the external oscillator frequency is outside this range, the application must start up in the FRC mode first. The default PLL settings after a POR with an oscillator frequency outside this range will violate the device operating speed.

Once the device powers up, the application firmware can initialize the PLL SFRs, CLKDIV and PLLFBD, to a suitable value, and then perform a clock switch to the Oscillator + PLL clock source. Note that clock switching must be enabled in the device Configuration Word.

2.8 Unused I/Os

Unused I/O pins should be configured as outputs and driven to a Logic Low state.

Alternatively, connect a 1k to 10k resistor between V_{SS} and unused pins, and drive the output to logic low.

2.9 Bulk Capacitors

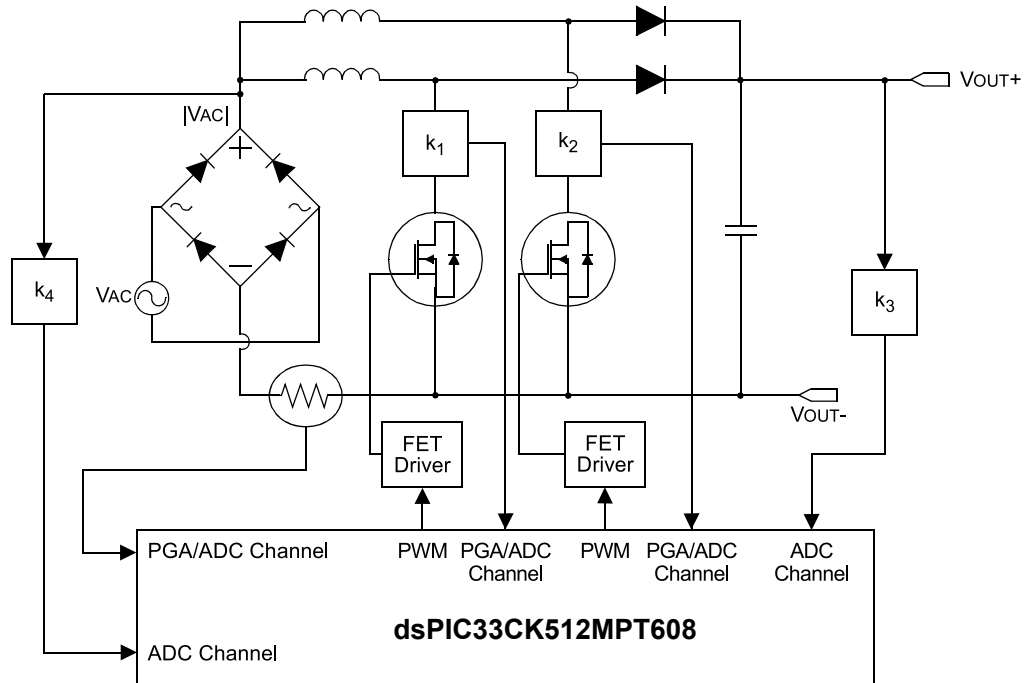
On boards with power traces running longer than six inches in length, it is suggested to use a bulk capacitor for integrated circuits, including DSCs, to supply a local power source. The value of the bulk capacitor should be determined based on the trace resistance that connects the power supply source to the device and the maximum current drawn by the device in the application. In other words, select the bulk capacitor so that it meets the acceptable voltage sag at the device. Typical values range from 4.7 μF to 47 μF .

2.10 Targeted Applications

- Power Factor Correction (PFC):
 - Interleaved PFC
 - Critical Conduction PFC
 - Bridgeless PFC
- DC/DC Converters:
 - Buck, Boost, Forward, Flyback, Push-Pull
 - Half/Full-Bridge
 - Phase-Shift Full-Bridge
 - Resonant Converters
- DC/AC:
 - Half/Full-Bridge Inverter
 - Resonant Inverter
- Motor Control:
 - BLDC
 - PMSM
 - SR
 - ACIM
- Security Related Applications:
 - Full and Partial Secure Boot
 - Secure Firmware Update
 - CAN Message Authentication
 - WPC 1.3 Qi High Power Transmitter Authentication
 - High-Bandwidth Digital Content Protection (HDCP) Cryptographic Support
 - Network Authentication and Session Establishment using TLS
 - Electric Vehicle (EV) Battery Authentication

Examples of typical application connections are shown in [Figure 2-4](#) through [Figure 2-6](#).

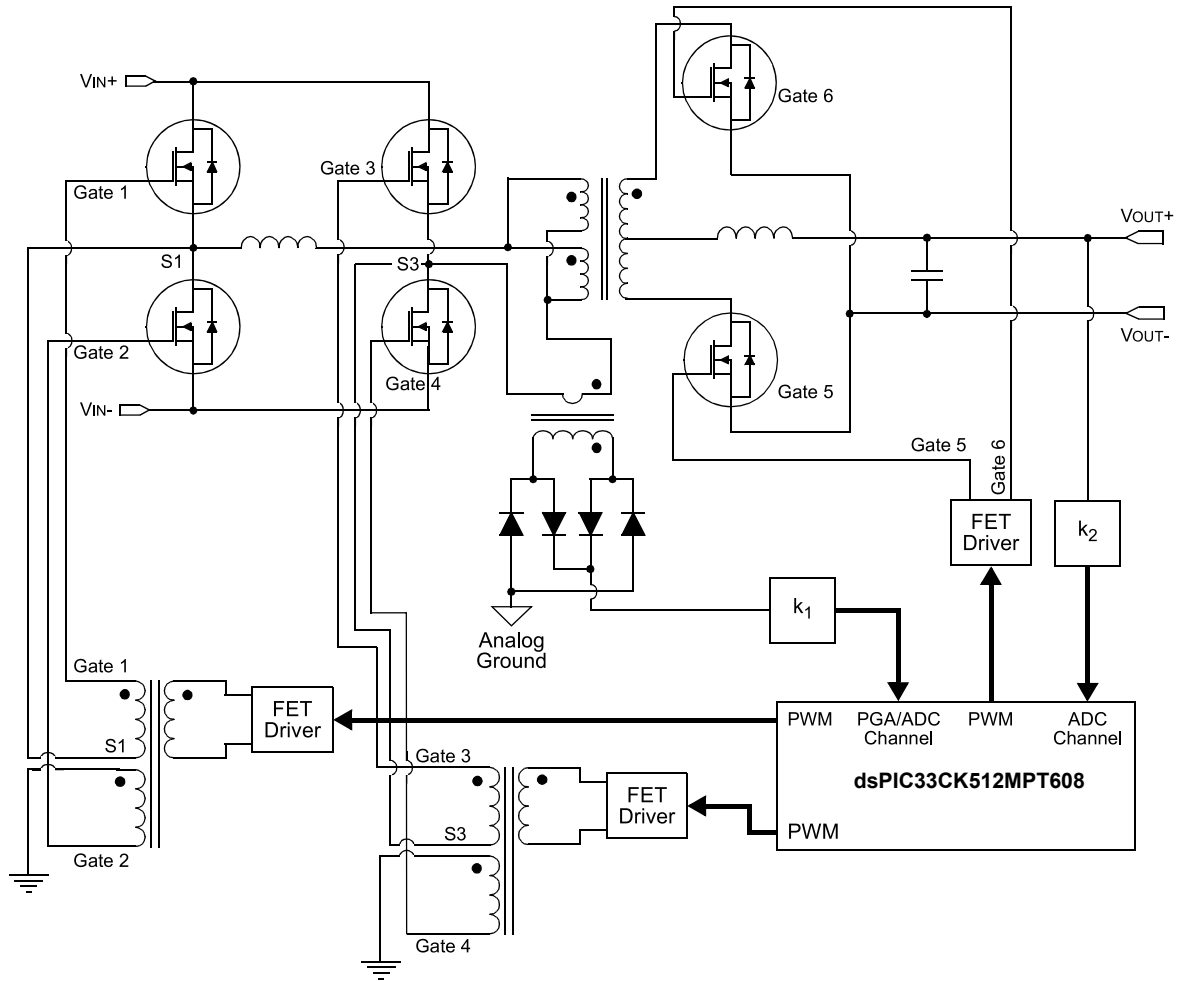
Figure 2-4. Interleaved PFC



dsPIC33CK512MPT608 Family

Guidelines for Getting Started with Digital ...

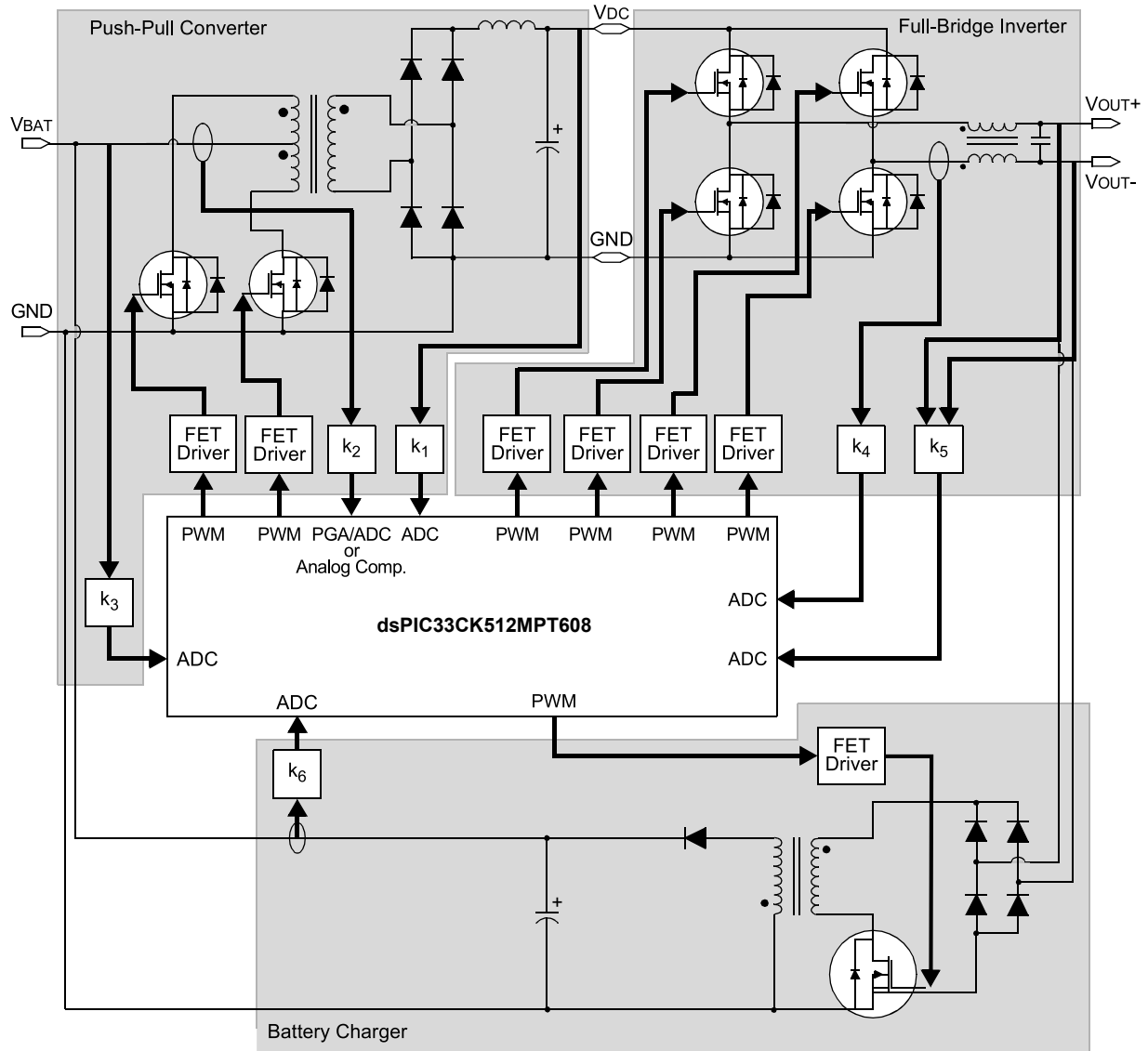
Figure 2-5. Phase-Shifted Full-Bridge Converter



dsPIC33CK512MPT608 Family

Guidelines for Getting Started with Digital ...

Figure 2-6. Off-Line UPS



3. CPU

Note: This data sheet summarizes the features of the dsPIC33CK512MPT608 family of devices. It is not intended to be a comprehensive reference source. To complement the information in this data sheet, refer to “**Enhanced CPU**” (www.microchip.com/DS70005158) in the “dsPIC33/PIC24 Family Reference Manual”.

The dsPIC33CK512MPT608 family CPU has a (data) modified Harvard architecture with an enhanced instruction set, including significant support for Digital Signal Processing (DSP). The CPU has a 24-bit instruction word with a variable length opcode field. The Program Counter (PC) is 23 bits wide and addresses up to 4M x 24 bits of user program memory space.

An instruction prefetch mechanism helps maintain throughput and provides predictable execution. Most instructions execute in a single-cycle effective execution rate, with the exception of instructions that change the program flow, the double-word move (`MOV.D`) instruction, PSV accesses and the table instructions. Overhead-free program loop constructs are supported using the `DO` and `REPEAT` instructions, both of which are interruptible at any point.

3.1 Registers

The dsPIC33CK512MPT608 devices have sixteen, 16-bit Working registers in the programmer's model. Each of the Working registers can act as a Data, Address or Address Offset register. The 16th Working register (W15) operates as a Software Stack Pointer (SSP) for interrupts and calls.

In addition, the dsPIC33CK512MPT608 devices include four Alternate Working register sets, which consist of W0 through W14. The Alternate Working registers can be made persistent to help reduce the saving and restoring of register content during Interrupt Service Routines (ISRs). The Alternate Working registers can be assigned to a specific Interrupt Priority Level (IPL1 through IPL6) by configuring the `CTXTx[2:0]` bits in the `FALTREG` Configuration register. The Alternate Working registers can also be accessed manually by using the `CTXTSWP` instruction. The `CCTXI[2:0]` and `MCTXI[2:0]` bits in the `CTXTSTAT` register can be used to identify the current, and most recent, manually selected Working register sets.

3.2 Instruction Set

The instruction set for dsPIC33CK512MPT608 devices has two classes of instructions: the MCU class of instructions and the DSP class of instructions. These two instruction classes are seamlessly integrated into the architecture and execute from a single execution unit. The instruction set includes many addressing modes and was designed for optimum C compiler efficiency.

Refer to the [31. Instruction Set Summary](#) for more information.

3.3 Data Space Addressing

The base Data Space can be addressed as up to 4K words or 8 Kbytes, and is split into two blocks, referred to as X and Y data memory. Each memory block has its own independent Address Generation Unit (AGU). The MCU class of instructions operates solely through the X memory AGU, which accesses the entire memory map as one linear Data Space. Certain DSP instructions operate through the X and Y AGUs to support dual operand reads, which splits the data address space into two parts. The X and Y Data Space boundary is device-specific.

The upper 32 Kbytes of the Data Space memory map can optionally be mapped into Program Space (PS) at any 16K program word boundary. The program-to-Data Space mapping feature, known as Program Space Visibility (PSV), lets any instruction access Program Space as if it were Data Space. Refer to “**Data Memory**” (www.microchip.com/DS70595) in the “dsPIC33/PIC24 Family Reference Manual” for more details on PSV and table accesses.

On dsPIC33CK512MPT608 family devices, overhead-free circular buffers (Modulo Addressing) are supported in both X and Y address spaces. The Modulo Addressing removes the software boundary checking overhead for DSP algorithms. The X AGU Circular Addressing can be used with any of the MCU class of instructions. The X AGU also supports Bit-Reversed Addressing to greatly simplify input or output data re-ordering for radix-2 FFT algorithms.

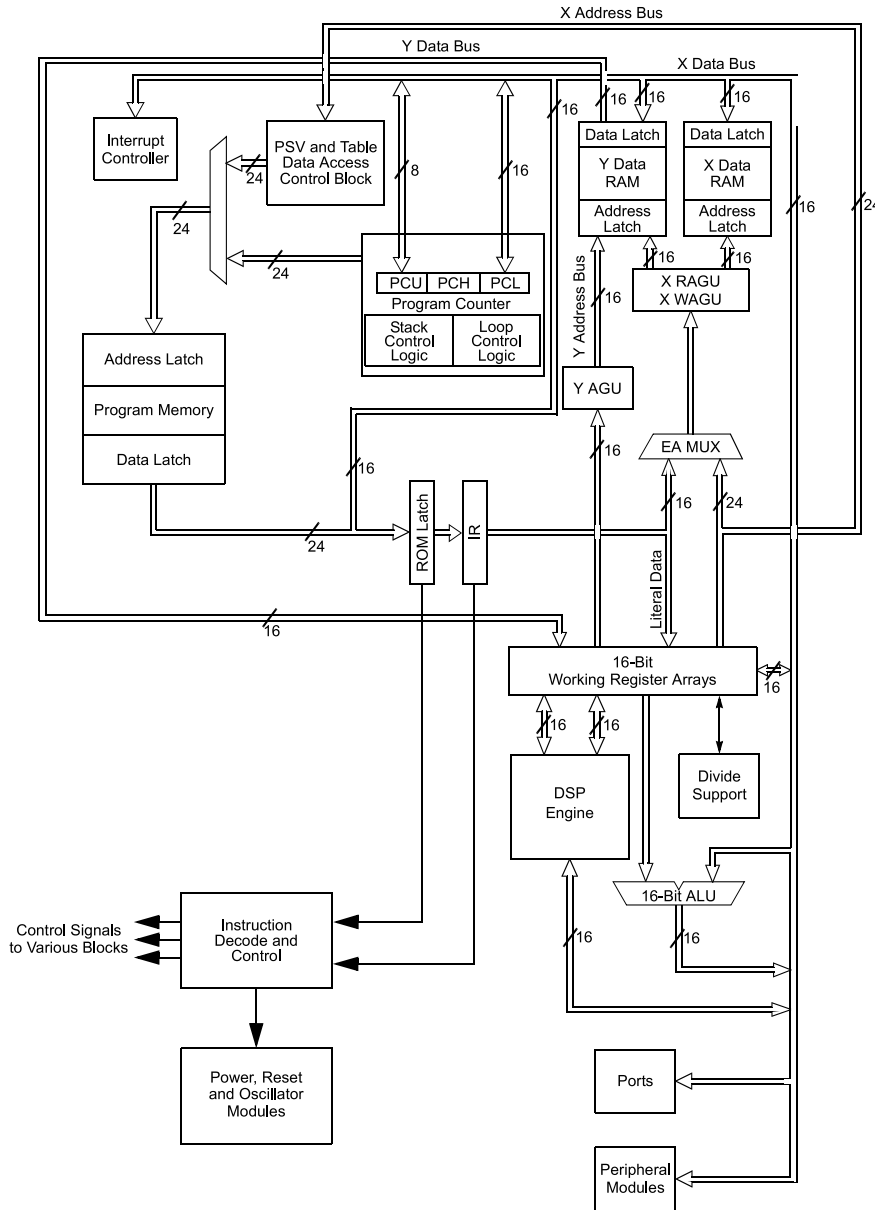
3.4 Addressing Modes

The CPU supports these addressing modes:

- Inherent (no operand)
- Relative
- Literal
- Memory Direct
- Register Direct
- Register Indirect

Each instruction is associated with a predefined addressing mode group, depending upon its functional requirements. As many as six addressing modes are supported for each instruction.

Figure 3-1. dsPIC33CK512MPT608 Family CPU Block Diagram



3.4.1 Programmer's Model

The programmer's model for the dsPIC33CK512MPT608 family is shown in [Figure 3-2](#). All registers in the programmer's model are memory-mapped and can be manipulated directly by instructions. [Table 3-1](#) lists a description of each register.

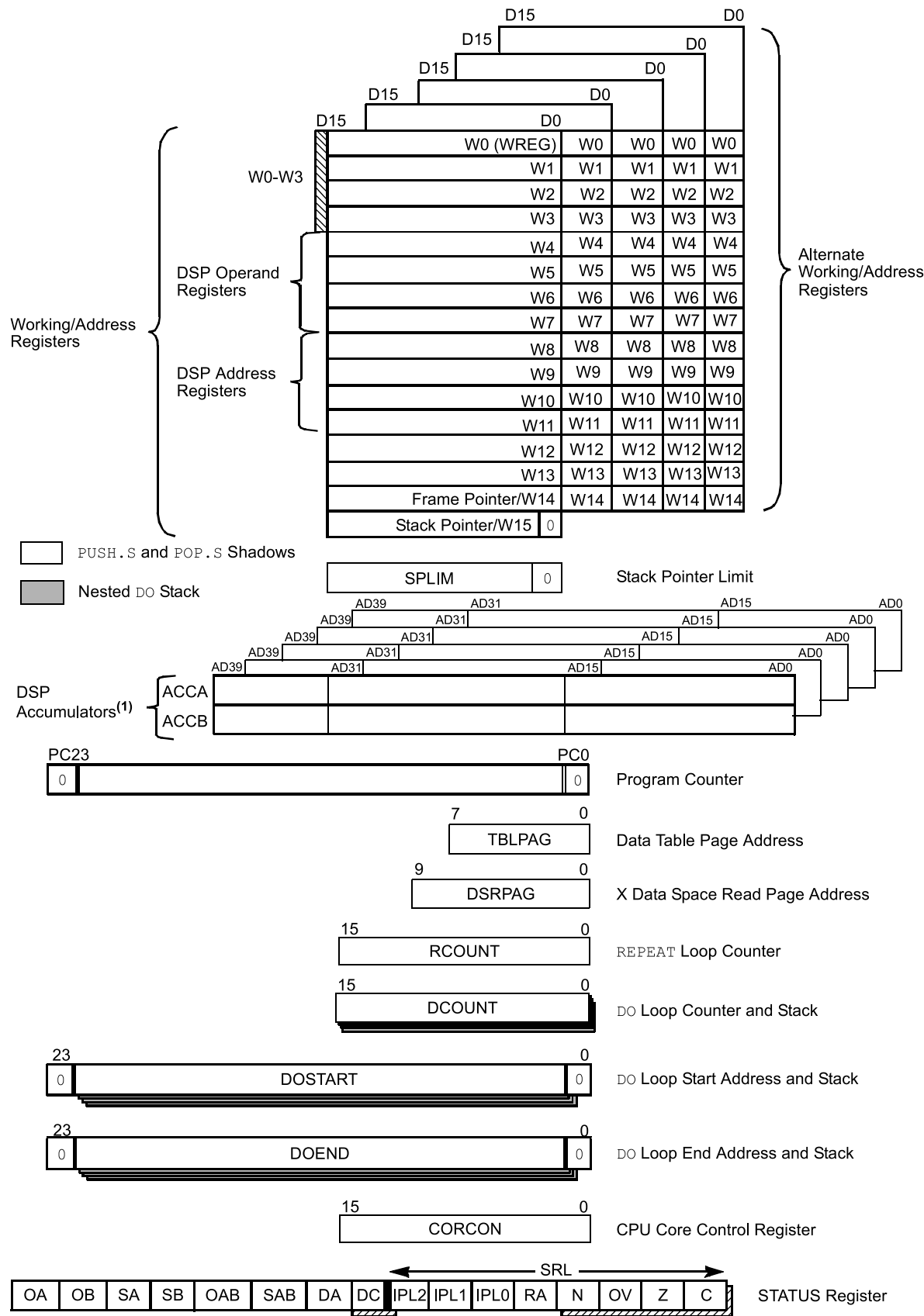
In addition to the registers contained in the programmer's model, the dsPIC33CK512MPT608 devices contain control registers for Modulo Addressing, Bit-Reversed Addressing and interrupts. These registers are described in subsequent sections of this document.

All registers associated with the programmer's model are memory-mapped, as shown in [Figure 3-2](#).

Table 3-1. Programmer's Model Register Descriptions

Register(s) Name	Description
W0 through W15 ⁽¹⁾	Working Register Array
W0 through W14 ⁽¹⁾	Alternate Working Register Array 1
W0 through W14 ⁽¹⁾	Alternate Working Register Array 2
W0 through W14 ⁽¹⁾	Alternate Working Register Array 3
W0 through W14 ⁽¹⁾	Alternate Working Register Array 4
ACCA, ACCB	40-Bit DSP Accumulators (Additional 4 Alternate Accumulators)
PC	23-Bit Program Counter
SR	ALU and DSP Engine STATUS Register
SPLIM	Stack Pointer Limit Value Register
TBLPAG	Table Memory Page Address Register
DSRPAG	Extended Data Space (EDS) Read Page Register
RCOUNT	REPEAT Loop Counter Register
DCOUNT	DO Loop Counter Register
DOSTARTH, DOSTARTL ⁽²⁾	DO Loop Start Address Register (High and Low)
DOENDH, DOENDL	DO Loop End Address Register (High and Low)
CORCON	Contains DSP Engine, DO Loop Control and Trap Status bits
Notes:	
1. Memory-mapped W0 through W14 represent the value of the register in the currently active CPU context.	
2. The DOSTARTH and DOSTARTL registers are read-only.	

Figure 3-2. Programmer's Model



3.4.2 CPU Resources

Many useful resources are provided on the main product page of the Microchip website for the devices listed in this data sheet. This product page contains the latest updates and additional information.

3.5 CPU Control/Status Registers

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x00	WREG0	15:8					WREG0[15:8]			
		7:0					WREG0[7:0]			
0x02	WREG1	15:8					WREG1[15:8]			
		7:0					WREG1[7:0]			
0x04	WREG2	15:8					WREG2[15:8]			
		7:0					WREG2[7:0]			
0x06	WREG3	15:8					WREG3[15:8]			
		7:0					WREG3[7:0]			
0x08	WREG4	15:8					WREG4[15:8]			
		7:0					WREG4[7:0]			
0x0A	WREG5	15:8					WREG5[15:8]			
		7:0					WREG5[7:0]			
0x0C	WREG6	15:8					WREG6[15:8]			
		7:0					WREG6[7:0]			
0x0E	WREG7	15:8					WREG7[15:8]			
		7:0					WREG7[7:0]			
0x10	WREG8	15:8					WREG8[15:8]			
		7:0					WREG8[7:0]			
0x12	WREG9	15:8					WREG9[15:8]			
		7:0					WREG9[7:0]			
0x14	WREG10	15:8					WREG10[15:8]			
		7:0					WREG10[7:0]			
0x16	WREG11	15:8					WREG11[15:8]			
		7:0					WREG11[7:0]			
0x18	WREG12	15:8					WREG12[15:8]			
		7:0					WREG12[7:0]			
0x1A	WREG13	15:8					WREG13[15:8]			
		7:0					WREG13[7:0]			
0x1C	WREG14	15:8					WREG14[15:8]			
		7:0					WREG14[7:0]			
0x1E	WREG15	15:8					WREG15[15:8]			
		7:0					WREG15[7:0]			
0x20	SPLIM	15:8					SPLIM[15:8]			
		7:0					SPLIM[7:0]			
0x22	ACCAL	15:8					ACCAL[15:8]			
		7:0					ACCAL[7:0]			
0x24	ACCAH	15:8					ACCAH[15:8]			
		7:0					ACCAH[7:0]			
0x26	ACCAU	15:8					ACCA39[7:0]			
		7:0					ACCAU[7:0]			
0x28	ACCB L	15:8					ACCB L[15:8]			
		7:0					ACCB L[7:0]			
0x2A	ACCBH	15:8					ACCBH[15:8]			
		7:0					ACCBH[7:0]			
0x2C	ACCBU	15:8					ACCB39[7:0]			
		7:0					ACCBU[7:0]			
0x2E	PCL	15:8					PCL[15:8]			
		7:0					PCL[7:0]			
0x30	PCH	15:8								
		7:0	PCH[7:0]							
0x32	DSRPAG	15:8							DSRPAG[9:8]	
		7:0	DSRPAG[7:0]							
0x33	DSWPAG	15:8							DSWPAG[8]	
		7:0	DSWPAG[7:0]							
0x35	Reserved									

dsPIC33CK512MPT608 Family

CPU

.....continued

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x36	RCOUNT	15:8	RCOUNT[15:8]							
		7:0	RCOUNT[7:0]							
0x38	DCOUNT	15:8	DCOUNT[15:8]							
		7:0	DCOUNT[7:0]							
0x3A	DOSTARTL	15:8	DOSTARTL[15:8]							
		7:0	DOSTARTL[7:0]							
0x3C	DOSTARTH	15:8								
		7:0		DOSTARTH[6:0]						
0x3E	DOENDL	15:8	DOENDL[15:8]							
		7:0	DOENDL[7:0]							
0x40	DOENDH	15:8								
		7:0		DOENDH[6:0]						
0x42	SR	15:8	OA	OB	SA	SB	OAB	SAB	DA	DC
		7:0	IPL[2:0]			RA	N	OV	Z	C
0x44	CORCON	15:8	VAR		US[1:0]		EDT	DL[2:0]		
		7:0	SATA	SATB	SATDW	ACCSAT	IPL3	SFA	RND	IF
0x46	MODCON	15:8	XMODEN	YMODEN			BWM[3:0]			
		7:0	YWM[3:0]				XWM[3:0]			
0x48	XMODSRT	15:8	XS[15:8]							
		7:0	XS[7:0]							
0x4A	XMODEND	15:8	XE[15:8]							
		7:0	XE[7:0]							
0x4C	YMODSRT	15:8	YS[15:8]							
		7:0	YS[7:0]							
0x4E	YMODEND	15:8	YE[15:8]							
		7:0	YE[7:0]							
0x50	XBREV	15:8	BREN		XB[14:8]					
		7:0	XB[7:0]							
0x52	DISICNT	15:8			DISICNT[13:8]					
		7:0	DISICNT[7:0]							
0x54	TBLPAG	15:8								
		7:0	TBLPAG[7:0]							
0x56	YPAG	15:8								
		7:0	YPAG[7:0]							
0x58	MSTRPR	15:8								
		7:0			DMAPR	CANPR	CAN2PR			NVMPR
0x5A	CTXTSTAT	15:8						CCTXI[2:0]		
		7:0						MCTXI[2:0]		

3.5.1 Working Register x

Name: WREGx

Offset: 0x00, 0x02, 0x04, 0x06, 0x08, 0x0A, 0x0C, 0x0E, 0x10, 0x12, 0x14, 0x16, 0x18, 0x1A, 0x1C, 0x1E

Bit	15	14	13	12	11	10	9	8
	WREGx[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	WREGx[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – WREGx[15:0] Data bits

3.5.2 Stack Pointer Limit Value Register

Name: SPLIM
Offset: 0x20

Bit	15	14	13	12	11	10	9	8
	SPLIM[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	SPLIM[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – SPLIM[15:0] Stack Limit Address bits

3.5.3 Accumulator A Low Register

Name: ACCAL
Offset: 0x22

Bit	15	14	13	12	11	10	9	8
	ACCAL[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	ACCAL[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – ACCAL[15:0] Accumulator A Low Register bits

3.5.4 Accumulator A High Register

Name: ACCAH
Offset: 0x24

Bit	15	14	13	12	11	10	9	8
	ACCAH[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	ACCAH[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – ACCAH[15:0] Accumulator A High Register bits

3.5.5 Accumulator A Upper Register

Name: ACCAU
Offset: 0x26

Bit	15	14	13	12	11	10	9	8
	ACCA39[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	ACCAU[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – ACCA39[7:0] Accumulator A bits

Bits 7:0 – ACCAU[7:0] Accumulator A bits

3.5.6 Accumulator B Low Register

Name: ACCBL
Offset: 0x28
Property: R/W

Bit	15	14	13	12	11	10	9	8
	ACCBL[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	ACCBL[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – ACCBL[15:0] Accumulator B Low Register bits

3.5.7 Accumulator B High Register

Name: ACCBH
Offset: 0x2A
Property: R/W

Bit	15	14	13	12	11	10	9	8
	ACCBH[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	ACCBH[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – ACCBH[15:0] Accumulator B High Register bits

3.5.8 Accumulator B Upper Address Register

Name: ACCBU

Offset: 0x2C

Bit	15	14	13	12	11	10	9	8
	ACCB39[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	ACCBU[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – ACCB39[7:0] Accumulator B bits

Bits 7:0 – ACCBU[7:0] Accumulator B bits

3.5.9 Program Counter Low Register

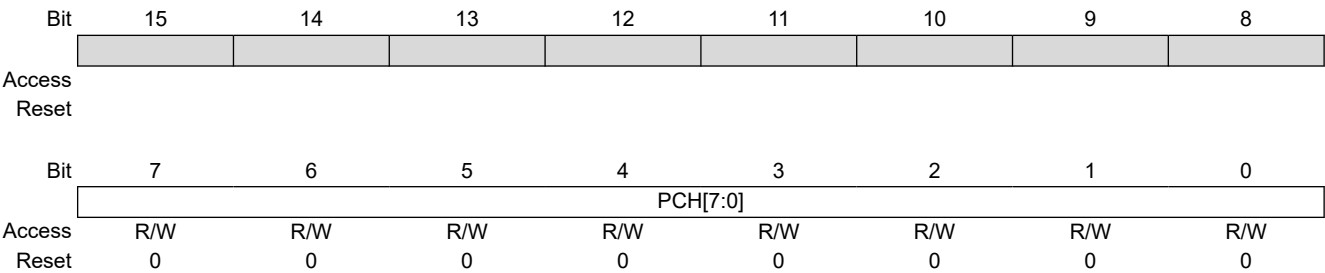
Name: PCL
Offset: 0x2E

Bit	15	14	13	12	11	10	9	8
	PCL[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PCL[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – PCL[15:0] Program Counter Low Value bits

3.5.10 Program Counter High Register

Name: PCH
Offset: 0x30



Bits 7:0 – PCH[7:0] Program Counter High Value bits

3.5.11 Data Space Read Page Register

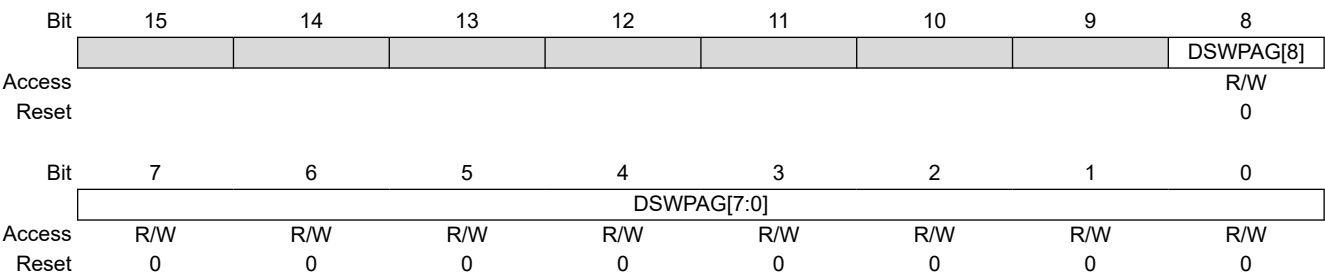
Name: DSRPAG
Offset: 0x32

Bit	15	14	13	12	11	10	9	8
							DSRPAG[9:8]	
Access							R/W	R/W
Reset							0	0
Bit	7	6	5	4	3	2	1	0
	DSRPAG[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 9:0 – DSRPAG[9:0] Data Space Read Page Value bits

3.5.12 Data Space Write Page Register

Name: DSWPAG
Offset: 0x33



Bits 8:0 – DSWPAG[8:0] Data Space Write Page Value bits

3.5.13 REPEAT Loop Counter Register

Name: RCOUNT
Offset: 0x36

Bit	15	14	13	12	11	10	9	8
	RCOUNT[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	RCOUNT[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – RCOUNT[15:0] Current Loop Counter Value for REPEAT Instruction bits

3.5.14 DO Loop Iteration Count Register

Name: DCOUNT
Offset: 0x38

Bit	15	14	13	12	11	10	9	8
	DCOUNT[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	DCOUNT[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – DCOUNT[15:0] DO Loop Iteration Count Register bits

3.5.15 DO Loop Start Address Register Low

Name: DOSTARTL
Offset: 0x3A

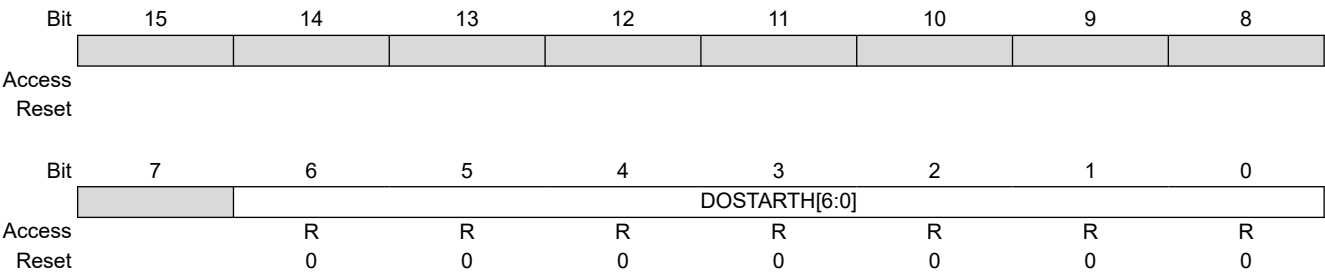
Bit	15	14	13	12	11	10	9	8
	DOSTARTL[15:8]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	DOSTARTL[7:0]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – DOSTARTL[15:0] Current DO Loop Start Address bits

Note: DOSTARTL[0] always reads as '0'; DOSTARTL is a read-only register.

3.5.16 DO Loop Start Address Register High

Name: DOSTARTH
Offset: 0x3C



Bits 6:0 – DOSTARTH[6:0] Current DO Loop Start Address bits
Note: DOSTARTH[0] always reads as ‘0’; DOSTARTH is a read-only register.

3.5.17 DO Loop End Address Register Low

Name: DOENDL
Offset: 0x3E

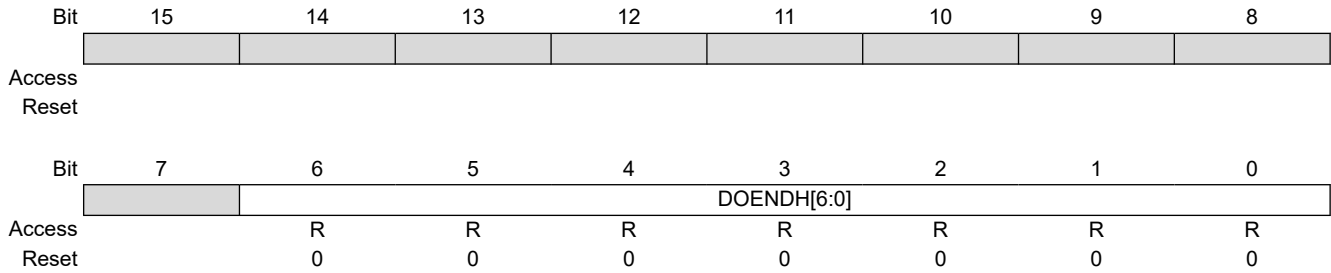
Bit	15	14	13	12	11	10	9	8
	DOENDL[15:8]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	DOENDL[7:0]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – DOENDL[15:0] Current DO Loop End Address bits

Note: DOENDL[0] always reads as '0'.

3.5.18 DO Loop End Address Register High

Name: DOENDH
Offset: 0x40



Bits 6:0 – DOENDH[6:0] Current DO Loop End Address bits

Note: DOENDH[0] always reads as '0'.

3.5.19 CPU STATUS Register

Name: SR
Offset: 0x42

Notes:

1. The IPL[2:0] bits are concatenated with the IPL[3] bit (CORCON[3]) to form the CPU Interrupt Priority Level. The value in parentheses indicates the IPL, if IPL[3] = 1. User interrupts are disabled when IPL[3] = 1.
2. The IPL[2:0] Status bits are read-only when the NSTDIS bit (INTCON1[15]) = 1.
3. A data write to the SR register can modify the SA and SB bits by either a data write to SA and SB or by clearing the SAB bit. To avoid a possible SA or SB Bit Write Race condition, the SA and SB bits should not be modified using bit operations.

Legend: C = Clearable bit

Bit	15	14	13	12	11	10	9	8
	OA	OB	SA	SB	OAB	SAB	DA	DC
Access	R/W	R/W	R/W	R/W	R/C	R/C	R	R/W
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	IPL[2:0]			RA	N	OV	Z	C
Access	R/W	R/W	R/W	R	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – OA Accumulator A Overflow Status bit

Value	Description
1	Accumulator A has overflowed
0	Accumulator A has not overflowed

Bit 14 – OB Accumulator B Overflow Status bit

Value	Description
1	Accumulator B has overflowed
0	Accumulator B has not overflowed

Bit 13 – SA Accumulator A Saturation ‘Sticky’ Status bit⁽³⁾

Value	Description
1	Accumulator A is saturated or has been saturated at some time
0	Accumulator A is not saturated

Bit 12 – SB Accumulator B Saturation ‘Sticky’ Status bit⁽³⁾

Value	Description
1	Accumulator B is saturated or has been saturated at some time
0	Accumulator B is not saturated

Bit 11 – OAB OA || OB Combined Accumulator Overflow Status bit

Value	Description
1	Accumulator A or B has overflowed
0	Neither Accumulator A or B has overflowed

Bit 10 – SAB SA || SB Combined Accumulator ‘Sticky’ Status bit

Value	Description
1	Accumulator A or B is saturated or has been saturated at some time
0	Neither Accumulator A or B is saturated

Bit 9 – DA DO Loop Active bit

Value	Description
1	DO loop is in progress
0	DO loop is not in progress

Bit 8 – DC MCU ALU Half Carry/Borrow bit

Value	Description
1	A carry-out from the 4th low-order bit (for byte-sized data) or 8th low-order bit (for word-sized data) of the result occurred
0	No carry-out from the 4th low-order bit (for byte-sized data) or 8th low-order bit (for word-sized data) of the result occurred

Bits 7:5 – IPL[2:0] CPU Interrupt Priority Level Status bits^(1,2)

Value	Description
111	CPU Interrupt Priority Level is 7 (15); user interrupts are disabled
110	CPU Interrupt Priority Level is 6 (14)
101	CPU Interrupt Priority Level is 5 (13)
100	CPU Interrupt Priority Level is 4 (12)
011	CPU Interrupt Priority Level is 3 (11)
010	CPU Interrupt Priority Level is 2 (10)
001	CPU Interrupt Priority Level is 1 (9)
000	CPU Interrupt Priority Level is 0 (8)

Bit 4 – RA REPEAT Loop Active bit

Value	Description
1	REPEAT loop is in progress
0	REPEAT loop is not in progress

Bit 3 – N MCU ALU Negative bit

Value	Description
1	Result was negative
0	Result was non-negative (zero or positive)

Bit 2 – OV MCU ALU Overflow bit

This bit is used for signed arithmetic (two's complement). It indicates an overflow of the magnitude that causes the sign bit to change state.

Value	Description
1	Overflow occurred for signed arithmetic (in this arithmetic operation)
0	No overflow occurred

Bit 1 – Z MCU ALU Zero bit

Value	Description
1	An operation that affects the Z bit has set it at some time in the past
0	The most recent operation that affects the Z bit has cleared it (i.e., a non-zero result)

Bit 0 – C MCU ALU Carry/Borrow bit

Value	Description
1	A carry-out from the Most Significant bit of the result occurred
0	No carry-out from the Most Significant bit of the result occurred

3.5.20 Core Control Register

Name: CORCON
Offset: 0x44

Notes:

1. This bit is always read as '0'.
2. The IPL3 bit is concatenated with the IPL[2:0] bits (SR[7:5]) to form the CPU Interrupt Priority Level.

Legend: C = Clearable bit

Bit	15	14	13	12	11	10	9	8
	VAR		US[1:0]		EDT	DL[2:0]		
Access	R/W		R/W	R/W	R/W	R	R	R
Reset	0		0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	SATA	SATB	SATDW	ACCSAT	IPL3	SFA	RND	IF
Access	R/W	R/W	R/W	R/W	R/C	R	R/W	R/W
Reset	0	0	1	0	0	0	0	0

Bit 15 – VAR Variable Exception Processing Latency Control bit

Value	Description
1	Variable exception processing is enabled
0	Fixed exception processing is enabled

Bits 13:12 – US[1:0] DSP Multiply Unsigned/Signed Control bits

Value	Description
11	Reserved
10	DSP engine multiplies are mixed sign
01	DSP engine multiplies are unsigned
00	DSP engine multiplies are signed

Bit 11 – EDT Early DO Loop Termination Control bit⁽¹⁾

Value	Description
1	Terminates executing DO loop at the end of the current loop iteration
0	No effect

Bits 10:8 – DL[2:0] DO Loop Nesting Level Status bits

Value	Description
111	7 DO loops are active
. . .	
001	1 DO loop is active
000	0 DO loops are active

Bit 7 – SATA ACCA Saturation Enable bit

Value	Description
1	Accumulator A saturation is enabled
0	Accumulator A saturation is disabled

Bit 6 – SATB ACCB Saturation Enable bit

Value	Description
1	Accumulator B saturation is enabled
0	Accumulator B saturation is disabled

Bit 5 – SATDW Data Space Write from DSP Engine Saturation Enable bit

Value	Description
1	Data Space write saturation is enabled
0	Data Space write saturation is disabled

Bit 4 – ACCSAT Accumulator Saturation Mode Select bit

Value	Description
1	9.31 saturation (super saturation)
0	1.31 saturation (normal saturation)

Bit 3 – IPL3 CPU Interrupt Priority Level Status bit 3⁽²⁾

Value	Description
1	CPU Interrupt Priority Level is greater than 7
0	CPU Interrupt Priority Level is 7 or less

Bit 2 – SFA Stack Frame Active Status bit

Value	Description
1	Stack frame is active; W14 and W15 address 0x0000 to 0xFFFF, regardless of DSRPAG
0	Stack frame is not active; W14 and W15 address the base Data Space

Bit 1 – RND Rounding Mode Select bit

Value	Description
1	Biased (conventional) rounding is enabled
0	Unbiased (convergent) rounding is enabled

Bit 0 – IF Integer or Fractional Multiplier Mode Select bit

Value	Description
1	Integer mode is enabled for DSP multiply
0	Fractional mode is enabled for DSP multiply

3.5.21 Modulo and Bit-Reversed Addressing Control Register

Name: MODCON
Offset: 0x46

Bit	15	14	13	12	11	10	9	8
	XMODEN	YMODEN			BWM[3:0]			
Access	R/W	R/W			R/W	R/W	R/W	R/W
Reset	0	0			0	0	0	0
Bit	7	6	5	4	3	2	1	0
	YWM[3:0]				XWM[3:0]			
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – XMODEN X RAGU and X WAGU Modulus Addressing Enable bit

Value	Description
1	X AGU Modulus Addressing enabled
0	X AGU Modulus Addressing disabled

Bit 14 – YMODEN Y AGU Modulus Addressing Enable bit

Value	Description
1	Y AGU Modulus Addressing enabled
0	Y AGU Modulus Addressing disabled

Bits 11:8 – BWM[3:0] X WAGU Register Select for Bit-Reversed Addressing bits

Value	Description
0000	W0 selected for Bit-Reversed Addressing
1110	W14 selected for Bit-Reversed Addressing
1111	W15 Bit-Reversed Addressing disabled

Bits 7:4 – YWM[3:0] Y AGU W Register Select for Modulo Addressing bits

Value	Description
0000	W0 selected for Modulo Addressing
1110	W14 selected for Modulo Addressing
1111	W15 Modulo Addressing disabled

Bits 3:0 – XWM[3:0] X RAGU and X WAGU W Register Select for Modulo Addressing bits

Value	Description
0000	W0 selected for Modulo Addressing
1110	W14 selected for Modulo Addressing
1111	W15 Modulo Addressing disabled

3.5.22 X AGU Modulo Addressing Start Register

Name: XMODSRT

Offset: 0x48

Bit	15	14	13	12	11	10	9	8
	XS[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	XS[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – XS[15:0] X RAGU and X WAGU Modulo Addressing Start Address bits

3.5.23 X AGU Modulo Addressing End Register

Name: XMODEND
Offset: 0x4A

Bit	15	14	13	12	11	10	9	8
	XE[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	XE[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – XE[15:0] X RAGU and X WAGU Modulo Addressing End Address bits

3.5.24 Y AGU Modulo Addressing Start Register

Name: YMODSRT
Offset: 0x4C

Bit	15	14	13	12	11	10	9	8
	YS[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	YS[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – YS[15:0] Y AGU Modulo Addressing Start Address bits

3.5.25 Y AGU Modulo Addressing End Register

Name: YMODEND
Offset: 0x4E

Bit	15	14	13	12	11	10	9	8
	YE[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	YE[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – YE[15:0] X AGU Modulo Addressing End Address bits

3.5.26 X AGU Bit Reversal Addressing Control Register

Name: XBREV

Offset: 0x50

Bit	15	14	13	12	11	10	9	8
	BREN		XB[14:8]					
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	x	x	x	x	x	x	x
Bit	7	6	5	4	3	2	1	0
	XB[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	x	x	x	x	x	x	x	x

Bit 15 – BREN Bit-Reversed Addressing (X AGU) Enable bit

Value	Description
1	Bit-Reversed Addressing enabled
0	Bit-Reversed Addressing disabled

Bits 14:0 – XB[14:0] X AGU Bit-Reversed Modifier bits

3.5.27 Disable Interrupt Count Register

Name: DISICNT

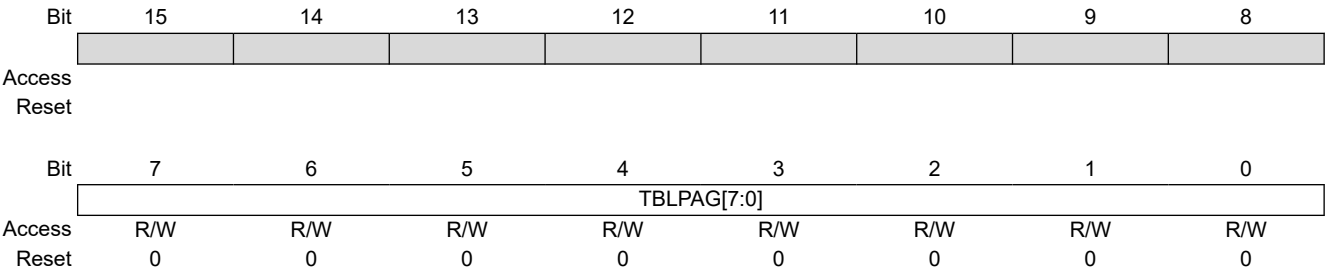
Offset: 0x52

Bit	15	14	13	12	11	10	9	8
			DISICNT[13:8]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	DISICNT[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 13:0 – DISICNT[13:0] Current Counter Value for `DISI` Instruction bits

3.5.28 Table Memory Page Address Register

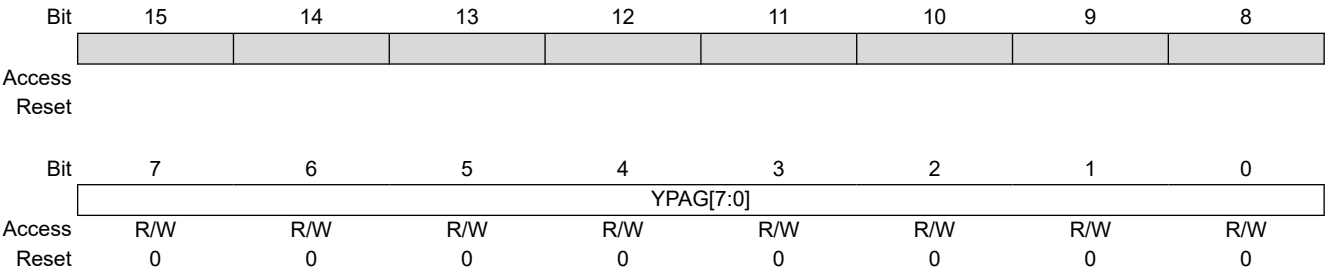
Name: TBLPAG
Offset: 0x54



Bits 7:0 – TBLPAG[7:0] Table Memory Page Value bits

3.5.29 Y Page Register

Name: YPAG
Offset: 0x56



Bits 7:0 – YPAG[7:0] Y Page bits

Note: When implemented, YPAG is a R/W SFR register which resets to 0x0001. When not implemented, YPAG is a read-only SFR register which will always return the fixed Y RAM page value, 0x0001.

3.5.30 EDS Bus Initiator Priority Control Register

Name: MSTRPR
Offset: 0x58

Bit	15	14	13	12	11	10	9	8
Access								
Reset								

Bit	7	6	5	4	3	2	1	0
			DMAPR	CANPR	CAN2PR			NVMPR
Access			R/W	R/W	R/W			R/W
Reset			0	0	0			0

Bit 5 – DMAPR Modify DMA Controller Bus Initiator Priority Relative to CPU bit

Value	Description
1	Raises DMA Controller bus Initiator priority to above that of the CPU
0	No change to DMA Controller bus Initiator priority

Bit 4 – CANPR Modify CAN1 Bus Initiator Priority Relative to CPU bit

Value	Description
1	Raises CAN1 bus Initiator priority to above that of the CPU
0	No change to CAN1 bus Initiator priority

Bit 3 – CAN2PR Modify CAN2 Bus Initiator Priority Relative to CPU bit

Value	Description
1	Raises CAN2 bus Initiator priority to above that of the CPU
0	No change to CAN2 bus Initiator priority

Bit 0 – NVMPR Modify NVM Controller Bus Initiator Priority Relative to CPU bit

Value	Description
1	Raises NVM Controller bus Initiator priority to above that of the CPU
0	No change to NVM Controller bus Initiator priority

3.5.31 CPU W Register Context Status Register

Name: CTXTSTAT
Offset: 0x5A

Bit	15	14	13	12	11	10	9	8
							CCTXI[2:0]	
Access						R	R	R
Reset						0	0	0

Bit	7	6	5	4	3	2	1	0
							MCTXI[2:0]	
Access						R	R	R
Reset						0	0	0

Bits 10:8 – CCTXI[2:0] Current (W Register) Context Identifier bits

Value	Description
111	Reserved
. . .	
100	Alternate Working Register Set 4 is currently in use
011	Alternate Working Register Set 3 is currently in use
010	Alternate Working Register Set 2 is currently in use
001	Alternate Working Register Set 1 is currently in use
000	Default register set is currently in use

Bits 2:0 – MCTXI[2:0] Manual (W Register) Context Identifier bits

Value	Description
111	Reserved
. . .	
100	Alternate Working Register Set 4 was most recently manually selected
011	Alternate Working Register Set 3 was most recently manually selected
010	Alternate Working Register Set 2 was most recently manually selected
001	Alternate Working Register Set 1 was most recently manually selected
000	Default register set was most recently manually selected

3.6 Arithmetic Logic Unit (ALU)

The dsPIC33CK512MPT608 family ALU is 16 bits wide and is capable of addition, subtraction, bit shifts and logic operations. Unless otherwise mentioned, arithmetic operations are two's complement in nature. Depending on the operation, the ALU can affect the values of the Carry (C), Zero (Z), Negative (N), Overflow (OV) and Digit Carry (DC) Status bits in the SR register. The C and DC Status bits operate as Borrow and Digit Borrow bits, respectively, for subtraction operations.

The ALU can perform 8-bit or 16-bit operations, depending on the mode of the instruction that is used. Data for the ALU operation can come from the W register array or data memory, depending on the addressing mode of the instruction. Likewise, output data from the ALU can be written to the W register array or a data memory location.

Refer to the “16-Bit MCU and DSC Programmer's Reference Manual” (www.microchip.com/DS70000157) for information on the SR bits affected by each instruction.

The core CPU incorporates hardware support for both multiplication and division. This includes a dedicated hardware multiplier and support hardware for 16-bit divisor division.

3.6.1 Multiplier

Using the high-speed, 17-bit x 17-bit multiplier, the ALU supports unsigned, signed or mixed-sign operation in several MCU Multiplication modes:

- 16-bit x 16-bit signed
- 16-bit x 16-bit unsigned
- 16-bit signed x 5-bit (literal) unsigned
- 16-bit signed x 16-bit unsigned
- 16-bit unsigned x 5-bit (literal) unsigned
- 16-bit unsigned x 16-bit signed
- 8-bit unsigned x 8-bit unsigned

3.6.2 Divider

The divide block supports 32-bit/16-bit and 16-bit/16-bit signed and unsigned integer divide operations with the following data sizes:

- 32-bit signed/16-bit signed divide
- 32-bit unsigned/16-bit unsigned divide
- 16-bit signed/16-bit signed divide
- 16-bit unsigned/16-bit unsigned divide

The 16-bit signed and unsigned `DIV` instructions can specify any W register for both the 16-bit divisor (W_n) and any W register (aligned) pair ($W(m+1):W_m$) for the 32-bit dividend. The divide algorithm takes one cycle per bit of divisor, so both 32-bit/16-bit and 16-bit/16-bit instructions take the same number of cycles to execute. There are additional instructions: `DIV2` and `DIVF2`. Divide instructions will complete in six cycles.

3.7 DSP Engine

The DSP engine consists of a high-speed 17-bit x 17-bit multiplier, a 40-bit barrel shifter and a 40-bit adder/subtractor (with two target accumulators, round and saturation logic).

The DSP engine can also perform inherent accumulator-to-accumulator operations that require no additional data. These instructions are: `ADD`, `SUB`, `NEG`, `MIN` and `MAX`.

The DSP engine has options selected through bits in the CPU Core Control register (`CORCON`), as listed below:

- Fractional or integer DSP multiply (`IF`)
- Signed, unsigned or mixed-sign DSP multiply (`USx`)
- Conventional or convergent rounding (`RND`)
- Automatic saturation on/off for `ACCA` (`SATA`)
- Automatic saturation on/off for `ACCB` (`SATB`)
- Automatic saturation on/off for writes to data memory (`SATDW`)
- Accumulator Saturation mode selection (`ACCSAT`)

Table 3-2. DSP Instructions Summary

Instruction	Algebraic Operation	ACC Write-Back
CLR	$A = 0$	Yes
ED	$A = (x - y)^2$	No
EDAC	$A = A + (x - y)^2$	No
MAC	$A = A + (x \cdot y)$	Yes
MAC	$A = A + x^2$	No
MOVSAC	No change in A	Yes
MPY	$A = x \cdot y$	No
MPY	$A = x^2$	No
MPY.N	$A = -x \cdot y$	No
MSC	$A = A - x \cdot y$	Yes

4. Memory Organization

Note: This data sheet summarizes the features of the dsPIC33CK512MPT608 family of devices. It is not intended to be a comprehensive reference source. To complement the information in this data sheet, refer to “**dsPIC33/PIC24 Program Memory**” (www.microchip.com/DS70000613) in the “*dsPIC33/PIC24 Family Reference Manual*”.

The dsPIC33CK512MPT608 family architecture features separate program and data memory spaces, and buses. This architecture also allows the direct access of program memory from the Data Space (DS) during code execution.

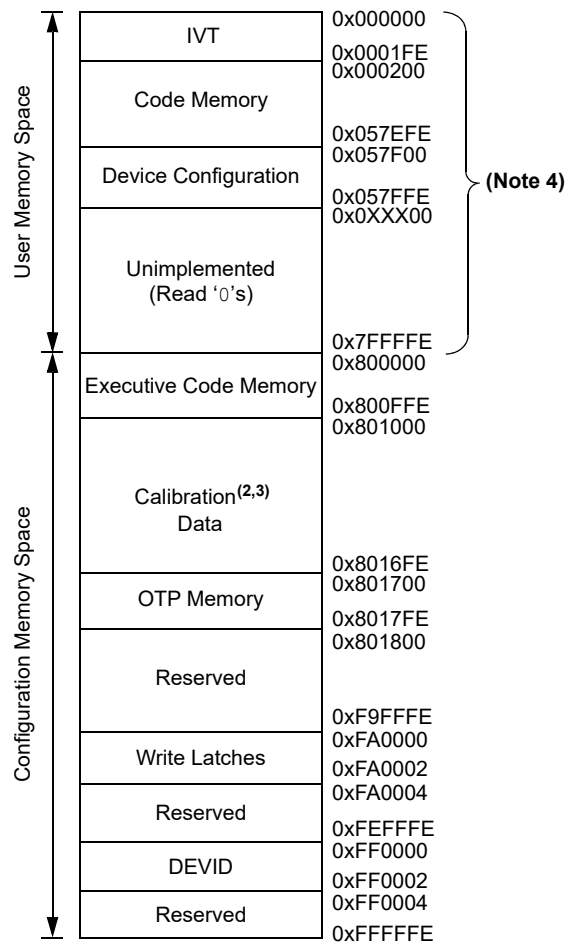
4.1 Program Address Space

The program address memory space of the dsPIC33CK512MPT608 family devices is 4M instructions. The space is addressable by a 24-bit value derived either from the 23-bit PC during program execution, or from table operation or Data Space remapping, as described in 4.4.5. [Interfacing Program and Data Memory Spaces](#).

User application access to the program memory space is restricted to the lower half of the address range (0x000000 to 0x7FFFFFFF). The exception is the use of **TBLRD** operations, which use **TBLPAG[7]** to permit access to calibration data and Device ID sections of the configuration memory space.

The program memory maps for the dsPIC33CK512MPT608 devices are shown in [Figure 4-1](#) through [Figure 4-3](#).

Figure 4-1. Program Memory Map for dsPIC33CK512MPT608 Device⁽¹⁾



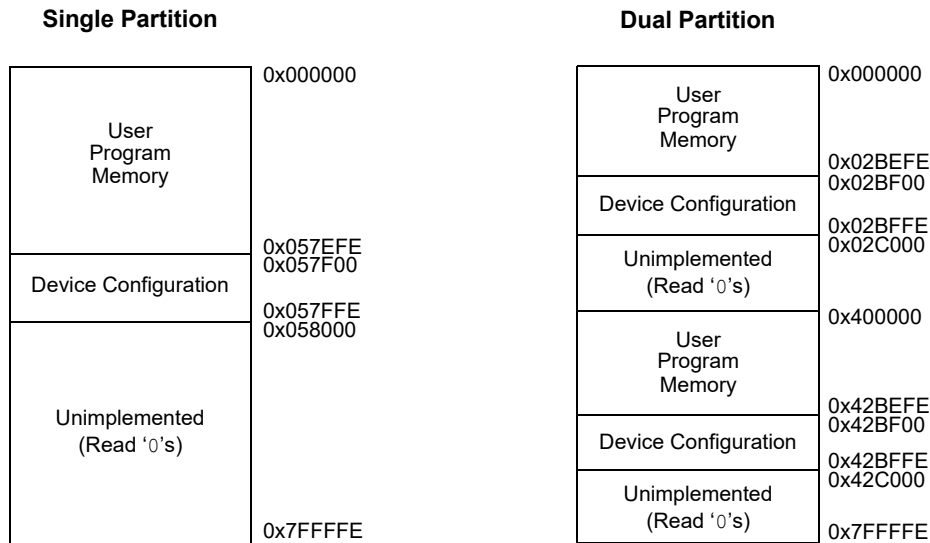
dsPIC33CK512MPT608 Family

Memory Organization

Notes:

1. Memory areas are not shown to scale.
2. Calibration data area must be maintained during programming.
3. Calibration data area includes UDID, ICSP™ Write Inhibit and FBOOT registers' locations.
4. See [Figure 4-2](#) and [Figure 4-3](#) for details

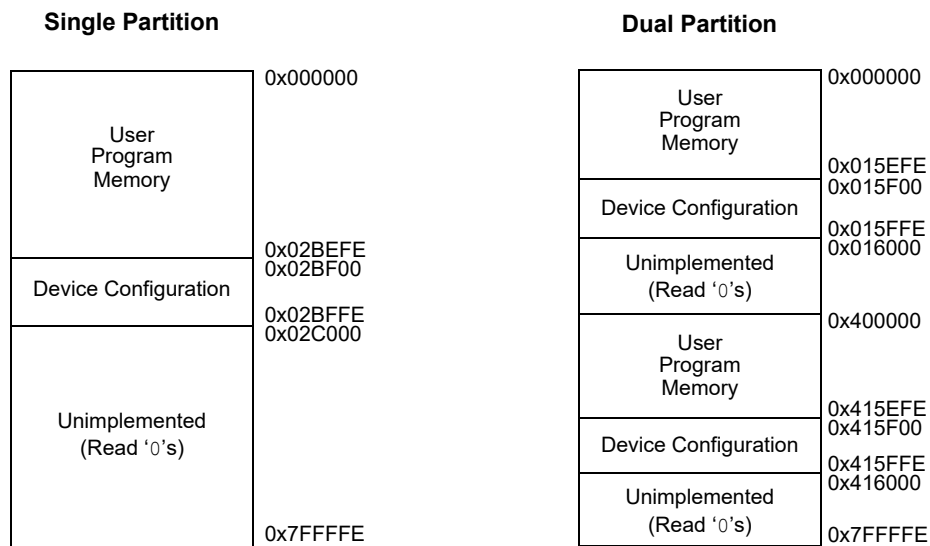
Figure 4-2. Program Memory Map for dsPIC33CK512MPT608 Device⁽¹⁾



Note:

1. Memory areas are not shown to scale.

Figure 4-3. Program Memory Map for dsPIC33CK256MPT608 Device⁽¹⁾



Note:

1. Memory areas are not shown to scale.

4.1.1 Program Memory Organization

The program memory space is organized in word-addressable blocks. Although it is treated as 24 bits wide, it is more appropriate to think of each address of the program memory as a lower and upper word, with the upper byte of the upper word being unimplemented. The lower word always has an even address, while the upper word has an odd address ([4.1.2. Interrupt and Trap Vectors](#)).

dsPIC33CK512MPT608 Family

Memory Organization

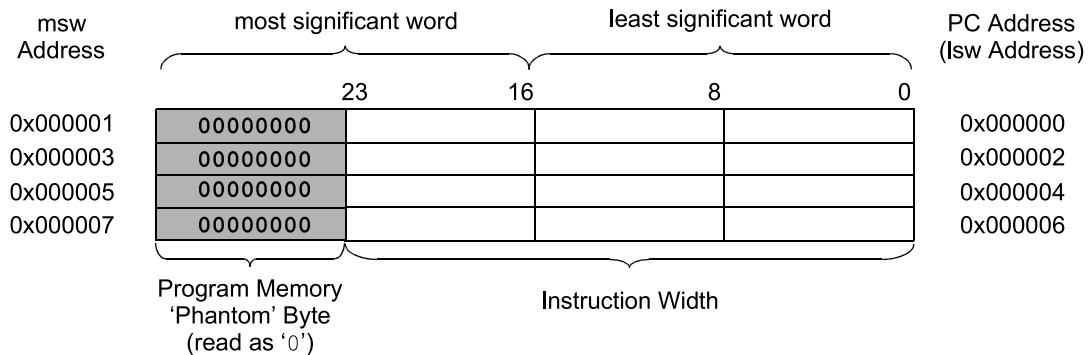
Program memory addresses are always word-aligned on the lower word, and addresses are incremented or decremented, by two, during code execution. This arrangement provides compatibility with data memory space addressing and makes data in the program memory space accessible.

4.1.2 Interrupt and Trap Vectors

All dsPIC33CK512MPT608 family devices reserve the addresses between 0x000000 and 0x000200 for hard-coded program execution vectors. A hardware Reset vector is provided to redirect code execution from the default value of the PC on device Reset to the actual start of code. A GOTO instruction is programmed by the user application at address, 0x000000, of Flash memory, with the actual address for the start of code at address, 0x000002, of Flash memory.

A more detailed discussion of the Interrupt Vector Tables (IVTs) is provided in [8. Interrupt Controller](#).

Figure 4-4. Program Memory Organization



4.1.3 Unique Device Identifier (UDID)

All dsPIC33CK512MPT608 family devices are individually encoded during final manufacturing with a Unique Device Identifier or UDID. The UDID cannot be erased by a bulk erase command or any other user-accessible means. This feature allows for manufacturing traceability of Microchip Technology devices in applications where this is a requirement. It may also be used by the application manufacturer for any number of things that may require unique identification, such as:

- Tracking the device
- Unique identifying number
- Unique security key

The UDID comprises five 24-bit program words. When taken together, these fields form a unique 120-bit identifier.

The UDID is stored in five read-only locations, located between 0x801200 and 0x801208 in the device configuration space. [Table 4-1](#) lists the addresses of the identifier words and shows their contents.

Table 4-1. UDID Addresses

UDID	Address	Description
UDID1	0x801200	UDID Word 1
UDID2	0x801202	UDID Word 2
UDID3	0x801204	UDID Word 3
UDID4	0x801206	UDID Word 4
UDID5	0x801208	UDID Word 5

4.2 Data Address Space

The dsPIC33CK512MPT608 family CPU has a separate 16-bit wide data memory space. The Data Space is accessed using separate Address Generation Units (AGUs) for read and write operations. The data memory map is shown in [Figure 4-5](#).

All Effective Addresses (EAs) in the data memory space are 16 bits wide and point to bytes within the Data Space. This arrangement gives a base Data Space address range of 64 Kbytes or 32K words.

The lower half of the data memory space (i.e., when $EA[15] = 0$) is used for implemented memory addresses, while the upper half ($EA[15] = 1$) is reserved for the Program Space Visibility (PSV).

The dsPIC33CK512MPT608 family devices implement up to 128 Kbytes of data memory. If an EA points to a location outside of this area, an all-zero word or byte is returned.

4.2.1 Data Memory Organization and Alignment

To maintain backward compatibility with PIC[®] MCU devices and improve Data Space memory usage efficiency, the dsPIC33CK512MPT608 family instruction set supports both word and byte operations. As a consequence of byte accessibility, all Effective Address calculations are internally scaled to step through word-aligned memory. For example, the core recognizes that Post-Modified Register Indirect Addressing mode $[Ws++]$ results in a value of $Ws + 1$ for byte operations and $Ws + 2$ for word operations.

A data byte read, reads the complete word that contains the byte, using the LSB of any EA to determine which byte to select. The selected byte is placed onto the LSB of the data path. That is, data memory and registers are organized as two parallel, byte-wide entities with shared (word) address decode, but separate write lines. Data byte writes only write to the corresponding side of the array or register that matches the byte address.

All word accesses must be aligned to an even address. Misaligned word data fetches are not supported, so care must be taken when mixing byte and word operations, or translating from 8-bit MCU code. If a misaligned read or write is attempted, an address error trap is generated. If the error occurred on a read, the instruction underway is completed. If the error occurred on a write, the instruction is executed but the write does not occur. In either case, a trap is then executed, allowing the system and/or user application to examine the machine state prior to execution of the address Fault.

All byte loads into any W register are loaded into the LSB; the MSB is not modified.

A Sign-Extend (SE) instruction is provided to allow user applications to translate 8-bit signed data to 16-bit signed values. Alternatively, for 16-bit unsigned data, user applications can clear the MSB of any W register by executing a Zero-Extend (ZE) instruction on the appropriate address.

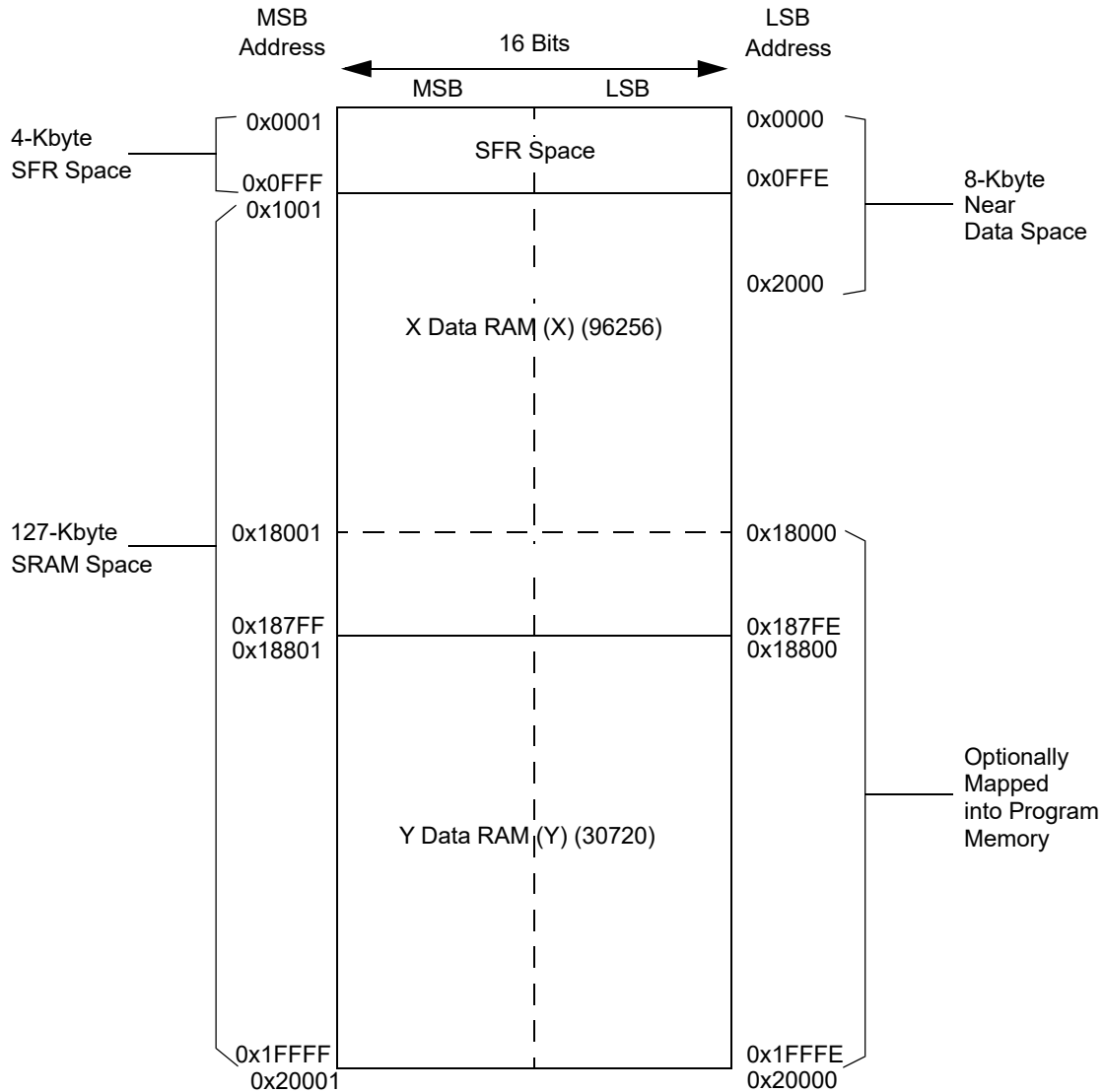
4.2.2 Near Data Space

The 8-Kbyte area, between 0x0000 and 0x1FFF, is referred to as the Near Data Space. Locations in this space are directly addressable through a 13-bit absolute address field within all memory direct instructions. Additionally, the whole Data Space is addressable using MOV instructions, which support Memory Direct Addressing mode with a 16-bit address field, or by using Indirect Addressing mode using a Working register as an Address Pointer.

dsPIC33CK512MPT608 Family

Memory Organization

Figure 4-5. Data Memory Map for dsPIC33CKXXXMPT608 Devices



Note: Memory areas are not shown to scale.

4.2.3 X and Y Data Spaces

The dsPIC33CK512MPT608 family core has two Data Spaces, X and Y. These Data Spaces can be considered either separate (for some DSP instructions) or as one unified linear address range (for MCU instructions). The Data Spaces are accessed using two Address Generation Units (AGUs) and separate data paths. This feature allows certain instructions to concurrently fetch two words from RAM, thereby enabling efficient execution of DSP algorithms, such as Finite Impulse Response (FIR) filtering and Fast Fourier Transform (FFT).

The X Data Space is used by all instructions and supports all addressing modes. X Data Space has separate read and write data buses. The X read data bus is the read data path for all instructions that view Data Space as combined X and Y address space. It is also the X data prefetch path for the dual operand DSP instructions (MAC class).

The Y Data Space is used in concert with the X Data Space by the MAC class of instructions (CLR, ED, EDAC, MAC, MOVSAC, MPY, MPY.N and MSC) to provide two concurrent data read paths.

Both the X and Y Data Spaces support Modulo Addressing mode for all instructions, subject to addressing mode restrictions. Bit-Reversed Addressing mode is only supported for writes to X Data Space.

All data memory writes, including in DSP instructions, view Data Space as combined X and Y address space. The boundary between the X and Y Data Spaces is device-dependent and is not user-programmable.

4.3 BIST Overview

The dsPIC33CK512MPT608 family features a data memory Built-In Self-Test (BIST) that has the option to be run at start-up or run time. The memory test checks that all memory locations are functional and provides a pass/fail status of the RAM that can be used by software to take action if needed. If a failure is reported, the specific location(s) are not identified.

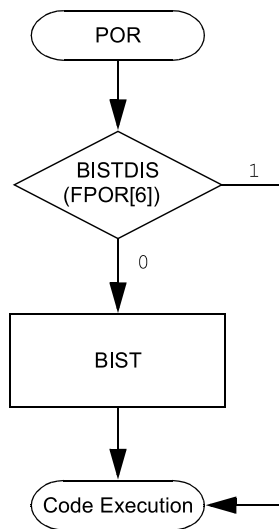
The MBISTCON register (4.3.3. MBISTCON) contains control and status bits for BIST operation. The MBISTDONE bit (MBISTCON[7]) indicates if a BIST was run since the last Reset and the MBISTSTAT bit (MBISTCON[4]) provides the pass/fail result.

4.3.1 BIST at Start-up

The BIST can be configured to automatically run on a POR-type Reset, as shown in Figure 4-6. By default, when BISTDIS (FPOR[6]) = 1, the BIST is disabled and will not be part of device start-up. If the BISTDIS bit is cleared during device programming, the BIST will run after all Configuration registers have been loaded and before code execution begins.

BIST will always run on FRC+PLL with PLL settings resulting in a 125 MHz clock rate.

Figure 4-6. BIST Flowchart



4.3.2 BIST at Run Time

The BIST can also be run at any time during code execution. Note that a BIST will corrupt all of the RAM contents, including the Stack Pointer, and requires a subsequent Reset. The system should be prepared for a Reset before a BIST is performed. The BIST is invoked by setting the MBISTEN bit (MBISTCON[0]). The MBISTCON register is protected against accidental writes and requires an unlock sequence prior to writing. Only one bit can be set per unlock sequence. The procedure for a run-time BIST is as follows:

1. Execute the unlock sequence by consecutively writing 0x55 and 0xAA to the NVMKEY register.
2. Write 0x0001 to the MBISTCON SFR.
3. Execute a Software Reset command.
4. Verify a Software Reset has occurred by reading SWR (RCON[6]) (optional).
5. Verify that the MBISTDONE bit is set.
6. Take action depending on test result indicated by MBISTSTAT.

4.3.2.1 Fault Simulation

A mechanism is available to simulate a BIST failure to allow testing of Fault handling software. When the FLTINJ bit is set during a run-time BIST, the MBISTSTAT bit will be set regardless of the test result. The procedure for a BIST Fault simulation is as follows:

1. Execute the unlock sequence by consecutively writing 0x55 and 0xAA to the NVMKEY register.
2. Set the MBISTEN bit (MBISTCON[0]).
3. Execute 2nd unlock sequence by consecutively writing 0x55 and 0xAA to the NVMKEY register.
4. Set the FLTINJ bit (MBISTCON[8]).
5. Execute a Software Reset command.
6. Verify a Software Reset has occurred by reading SWR (RCON[6]) (optional).
7. Verify the MBISTDONE, MBSITSTAT and FLTINJ bits are all set.

4.3.3 MBIST Control Register

Name: MBISTCON
Offset: 0EFC

Notes:

1. Resets only on a true POR Reset.
2. This bit will self-clear when the MBIST test is complete.

Legend: HS = Hardware Settable bit; HC = Hardware Clearable bit

Bit	15	14	13	12	11	10	9	8
								FLTINJ
Access								R/W
Reset								0

Bit	7	6	5	4	3	2	1	0
	MBISTDONE			MBISTSTAT				MBISTEN
Access	R/W/HS			R				R/W/HC
Reset	0			0				0

Bit 8 – FLTINJ MBIST Fault Inject Control bit⁽¹⁾

Value	Description
1	The MBIST test will complete and sets MBISTSTAT = 1, simulating an SRAM test failure
0	The MBIST test will execute normally

Bit 7 – MBISTDONE MBIST Done Status bit

Value	Description
1	An MBIST operation has been executed
0	No MBIST operation has occurred on the last Reset sequence

Bit 4 – MBISTSTAT MBIST Status bit

Value	Description
1	The last MBIST failed
0	The last MBIST passed; all memory may not have been tested

Bit 0 – MBISTEN MBIST Enable bit⁽²⁾

Value	Description
1	MBIST test is armed; an MBIST test will execute at the next device Reset
0	MBIST test is disarmed

4.4 Memory Resources

Many useful resources are provided on the main product page of the Microchip website for the devices listed in this data sheet. This product page contains the latest updates and additional information.

4.4.1 Paged Memory Scheme

The dsPIC33CK512MPT608 architecture extends the available Data Space through a paging scheme, which allows the available Data Space to be accessed using MOV instructions in a linear fashion for pre- and post-modified Effective Addresses (EAs). The upper half of the base Data Space address is used in conjunction with the Data Space Read Page (DSRPAG) register to form the Program Space Visibility (PSV) address.

The Data Space Read Page (DSRPAG) register is located in the SFR space. When DSRPAG[9] = 1 and the base address bit, EA[15] = 1, the DSRPAG[8:0] bits are concatenated onto EA[14:0] to form the 24-bit PSV read address.

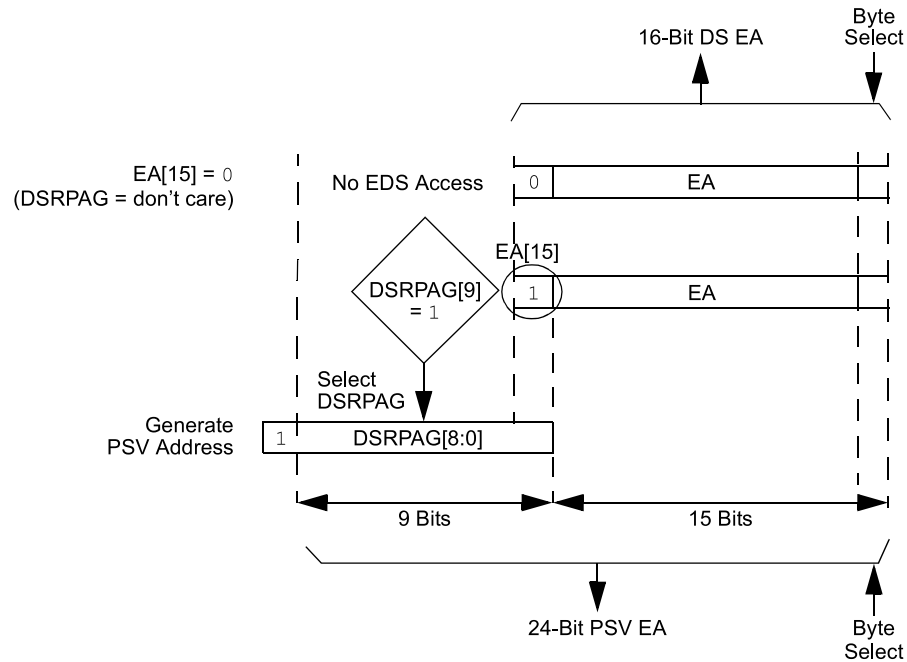
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The paged memory scheme provides access to multiple 32-Kbyte windows in the PSV memory. The Data Space Read Page (DSRPAG) register, in combination with the upper half of the Data Space address, can provide up to 8 Mbytes of PSV address space. The paged data memory space is shown in [Figure 4-8](#).

The Program Space (PS) can be accessed with a DSRPAG of 0x200 or greater. Only reads from PS are supported using the DSRPAG.

Figure 4-7. Program Space Visibility (PSV) Read Address Generation

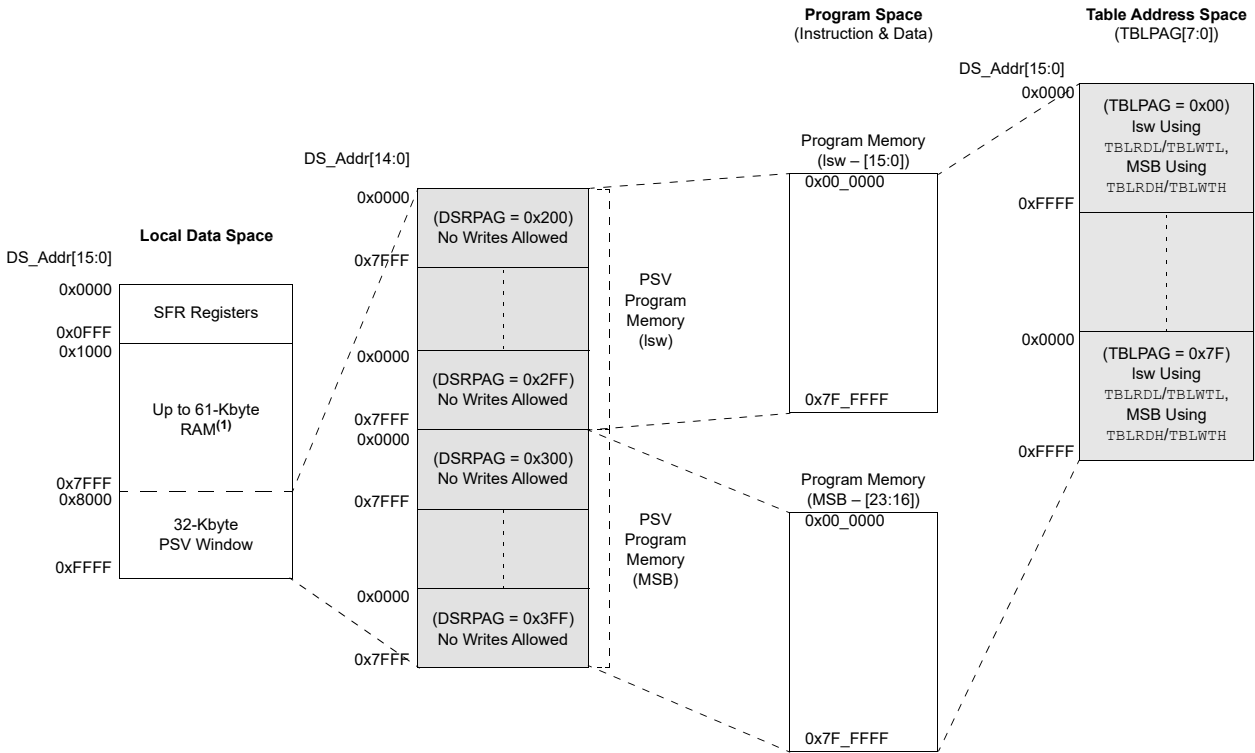


Note: DS read access when DSRPAG = 0x000 will force an address error trap.

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Figure 4-8. Paged Data Memory Space



When a PSV page overflow or underflow occurs, EA[15] is cleared as a result of the register indirect EA calculation. An overflow or underflow of the EA in the PSV pages can occur at the page boundaries when:

- The initial address, prior to modification, addresses the PSV page
- The EA calculation uses Pre- or Post-Modified Register Indirect Addressing; however, this does not include Register Offset Addressing

In general, when an overflow is detected, the DSRPA G register is incremented and the EA[15] bit is set to keep the base address within the PSV window. When an underflow is detected, the DSRPA G register is decremented and the EA[15] bit is set to keep the base address within the PSV window. This creates a linear PSV address space, but only when using Register Indirect Addressing modes.

Exceptions to the operation described above arise when entering and exiting the boundaries of Page 0 and PSV spaces. [Table 4-2](#) lists the effects of overflow and underflow scenarios at different boundaries.

In the following cases, when overflow or underflow occurs, the EA[15] bit is set and the DSRPA G is not modified; therefore, the EA will wrap to the beginning of the current page:

- Register Indirect with Register Offset Addressing
- Modulo Addressing
- Bit-Reversed Addressing

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Table 4-2. Overflow and Underflow Scenarios at Page 0 and PSV Space Boundaries^(2,3,4)

O/U, R/W	Operation	Before			After		
		DSRPAG	DS EA[15]	Page Description	DSRPAG	DS EA[15]	Page Description
O, Read	[++Wn] or	DSRPAG = 0x2FF	1	PSV: Last lsw page	DSRPAG = 0x300	1	PSV: First MSB page
O, Read	[Wn++]	DSRPAG = 0x3FF	1	PSV: Last MSB page	DSRPAG = 0x3FF	0	See Note 1
U, Read	[--Wn] or	DSRPAG = 0x001	1	PSV page	DSRPAG = 0x001	0	See Note 1
U, Read	[Wn--]	DSRPAG = 0x200	1	PSV: First lsw page	DSRPAG = 0x200	0	See Note 1
U, Read		DSRPAG = 0x300	1	PSV: First MSB page	DSRPAG = 0x2FF	1	PSV: Last lsw page

Legend: O = Overflow, U = Underflow, R = Read, W = Write

Notes:

1. The Register Indirect Addressing now addresses a location in the base Data Space (0x0000-0x8000).
2. An EDS access, with DSRPAG = 0x000, will generate an address error trap.
3. Only reads from PS are supported using DSRPAG.
4. Pseudolinear Addressing is not supported for large offsets.

4.4.1.1 Extended X Data Space

The lower portion of the base address space range, between 0x0000 and 0x7FFF, is always accessible, regardless of the contents of the Data Space Read Page register. It is indirectly addressable through the register indirect instructions. It can be regarded as being located in the default EDS Page 0 (i.e., EDS address range of 0x000000 to 0x007FFF with the base address bit, EA[15] = 0, for this address range). However, Page 0 cannot be accessed through the upper 32 Kbytes, 0x8000 to 0xFFFF, of base Data Space in combination with DSRPAG = 0x00. Consequently, DSRPAG is initialized to 0x001 at Reset.

Notes:

1. DSRPAG should not be used to access Page 0. An EDS access with DSRPAG set to 0x000 will generate an address error trap.
2. Clearing the DSRPAG in software has no effect.

The remaining PSV pages are only accessible using the DSRPAG register in combination with the upper 32 Kbytes, 0x8000 to 0xFFFF, of the base address, where the base address bit, EA[15] = 1.

4.4.1.2 Software Stack

The W15 register serves as a dedicated Software Stack Pointer (SSP), and is automatically modified by exception processing, subroutine calls and returns; however, W15 can be referenced by any instruction in the same manner as all other W registers. This simplifies reading, writing and manipulating the Stack Pointer (for example, creating stack frames).

Note: To protect against misaligned stack accesses, W15[0] is fixed to '0' by the hardware.

W15 is initialized to 0x1000 during all Resets. This address ensures that the SSP points to valid RAM in all dsPIC33CK512MPT608 devices and permits stack availability for non-maskable trap exceptions. These can occur before the SSP is initialized by the user software. You can reprogram the SSP during initialization to any location within Data Space.

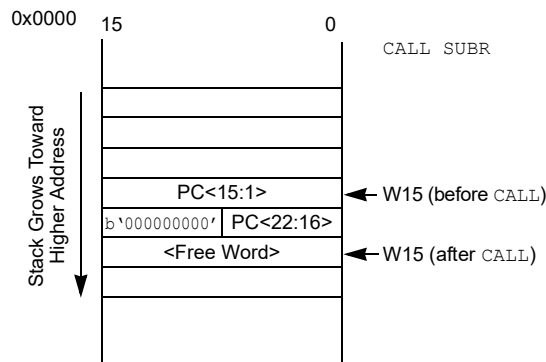
The Software Stack Pointer always points to the first available free word and fills the software stack, working from lower toward higher addresses. Figure 4-9 illustrates how it pre-decrements for a stack pop (read) and post-increments for a stack push (writes).

When the PC is pushed onto the stack, PC[15:0] are pushed onto the first available stack word, then PC[22:16] are pushed into the second available stack location. For a PC push during any `CALL` instruction, the MSB of the PC is zero-extended before the push, as shown in Figure 4-9. During exception processing, the MSB of the PC is concatenated with the lower eight bits of the CPU STATUS Register, SR. This allows the contents of SRL to be preserved automatically during interrupt processing.

Notes:

1. To maintain system Stack Pointer (W15) coherency, W15 is never subject to (EDS) paging, and is therefore, restricted to an address range of 0x0000 to 0xFFFF. The same applies to the W14 when used as a Stack Frame Pointer (SFA = 1).
2. As the stack can be placed in, and can access X and Y spaces, care must be taken regarding its use, particularly with regard to local automatic variables in a C development environment.

Figure 4-9. CALL Stack Frame



4.4.2 Instruction Addressing Modes

The addressing modes shown in Table 4-3 form the basis of the addressing modes optimized to support the specific features of individual instructions. The addressing modes provided in the `MAC` class of instructions differ from those in the other instruction types.

Table 4-3. Fundamental Addressing Modes Supported

Addressing Mode	Description
File Register Direct	The address of the file register is specified explicitly.
Register Direct	The contents of a register are accessed directly.
Register Indirect	The contents of Wn form the Effective Address (EA).
Register Indirect Post-Modified	The contents of Wn form the EA. Wn is post-modified (incremented or decremented) by a constant value.
Register Indirect Pre-Modified	Wn is pre-modified (incremented or decremented) by a signed constant value to form the EA.
Register Indirect with Register Offset (Register Indexed)	The sum of Wn and Wb forms the EA.
Register Indirect with Literal Offset	The sum of Wn and a literal forms the EA.

4.4.2.1 File Registration Instructions

Most file register instructions use a 13-bit address field (f) to directly address data present in the first 8192 bytes of data memory (Near Data Space). Most file register instructions employ a Working register, W0, which is denoted as WREG in these instructions. The destination is typically either the same file register or WREG (with the exception

of the `MUL` instruction), which writes the result to a register or register pair. The `MOV` instruction allows additional flexibility and can access the entire Data Space.

4.4.2.2 MCU Instructions

The three-operand MCU instructions are of the form:

Operand 3 = Operand 1 [function] Operand 2

where `Operand 1` is always a Working register (that is, the addressing mode can only be Register Direct), which is referred to as `Wb`. `Operand 2` can be a W register fetched from data memory or a 5-bit literal. The result location can either be a W register or a data memory location. The following addressing modes are supported by MCU instructions:

- Register Direct
- Register Indirect
- Register Indirect Post-Modified
- Register Indirect Pre-Modified
- 5-Bit or 10-Bit Literal

Note: Not all instructions support all the addressing modes given above. Individual instructions can support different subsets of these addressing modes.

4.4.2.3 Move and Accumulator Instructions

Move instructions, and the DSP accumulator class of instructions, provide a greater degree of addressing flexibility than other instructions. In addition to the addressing modes supported by most MCU instructions, move and accumulator instructions also support Register Indirect with Register Offset Addressing mode, also referred to as Register Indexed mode.

Note: For the `MOV` instructions, the addressing mode specified in the instruction can differ for the source and destination EA. However, the 4-bit `Wb` (Register Offset) field is shared by both source and destination (but typically only used by one).

In summary, the following addressing modes are supported by move and accumulator instructions:

- Register Direct
- Register Indirect
- Register Indirect Post-Modified
- Register Indirect Pre-Modified
- Register Indirect with Register Offset (Indexed)
- Register Indirect with Literal Offset
- 8-Bit Literal
- 16-Bit Literal

Note: Not all instructions support all the addressing modes given above. Individual instructions may support different subsets of these addressing modes.

4.4.2.4 MAC Instructions

The dual source operand DSP instructions (`CLR`, `ED`, `EDAC`, `MAC`, `MPY`, `MPY.N`, `MOVSAC` and `MSC`), also referred to as `MAC` instructions, use a simplified set of addressing modes to allow the user application to effectively manipulate the Data Pointers through register indirect tables.

The two-source operand prefetch registers must be members of the set {W8, W9, W10, W11}. For data reads, W8 and W9 are always directed to the X RAGU, and W10 and W11 are always directed to the Y AGU. The Effective Addresses generated (before and after modification) must therefore, be valid addresses within X Data Space for W8 and W9, and Y Data Space for W10 and W11.

Note: Register Indirect with Register Offset Addressing mode is available only for W9 (in X space) and W11 (in Y space).

In summary, the following addressing modes are supported by the MAC class of instructions:

- Register Indirect
- Register Indirect Post-Modified by 2
- Register Indirect Post-Modified by 4
- Register Indirect Post-Modified by 6
- Register Indirect with Register Offset (Indexed)

4.4.2.5 Other Instructions

Besides the addressing modes outlined previously, some instructions use literal constants of various sizes. For example, **BRA** (branch) instructions use 16-bit signed literals to specify the branch destination directly, whereas the **DISI** instruction uses a 14-bit unsigned literal field. In some instructions, such as **ULNK**, the source of an operand or result is implied by the opcode itself. Certain operations, such as a **NOP**, do not have any operands.

4.4.3 Modulo Addressing

Modulo Addressing mode is a method of providing an automated means to support circular data buffers using hardware. The objective is to remove the need for software to perform data address boundary checks when executing tightly looped code, as is typical in many DSP algorithms.

Modulo Addressing can operate in either Data or Program Space (since the Data Pointer mechanism is essentially the same for both). One circular buffer can be supported in each of the X (which also provides the pointers into Program Space) and Y Data Spaces. Modulo Addressing can operate on any W Register Pointer. However, it is not advisable to use W14 or W15 for Modulo Addressing since these two registers are used as the Stack Frame Pointer and Stack Pointer, respectively.

In general, any particular circular buffer can be configured to operate in only one direction, as there are certain restrictions on the buffer start address (for incrementing buffers) or end address (for decrementing buffers), based upon the direction of the buffer.

The only exception to the usage restrictions is for buffers that have a power-of-two length. As these buffers satisfy the start and end address criteria, they can operate in a Bidirectional mode (that is, address boundary checks are performed on both the lower and upper address boundaries).

4.4.3.1 Start and End Address

The Modulo Addressing scheme requires that a starting and ending address be specified and loaded into the 16-bit Modulo Buffer Address registers: XMODSRT, XMODEND, YMODSRT and YMODEND.

Note: Y space Modulo Addressing EA calculations assume word-sized data (LSb of every EA is always clear).

The length of a circular buffer is not directly specified. It is determined by the difference between the corresponding start and end addresses. The maximum possible length of the circular buffer is 32K words (64 Kbytes).

4.4.3.2 W Address Register Selection

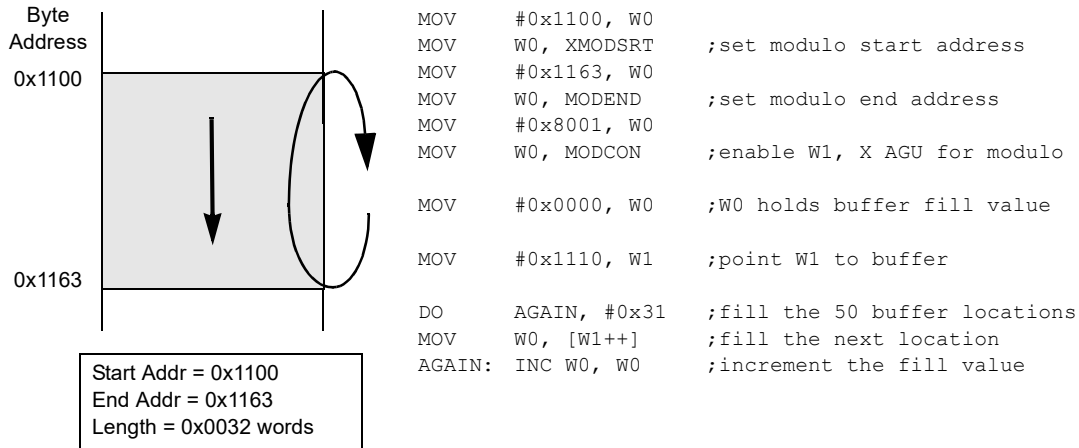
The Modulo and Bit-Reversed Addressing Control register, MODCON[15:0], contains enable flags, as well as a W register field to specify the W Address registers. The XWM and YWM fields select the registers that operate with Modulo Addressing:

- If XWM = 1111, X RAGU and X WAGU Modulo Addressing is disabled
- If YWM = 1111, Y AGU Modulo Addressing is disabled

The X Address Space Pointer W (XWM) register, to which Modulo Addressing is to be applied, is stored in MODCON[3:0]. Modulo Addressing is enabled for X Data Space when XWM is set to any value other than '1111' and the XMODEN bit is set (MODCON[15]).

The Y Address Space Pointer W (YWM) register, to which Modulo Addressing is to be applied, is stored in MODCON[7:4]. Modulo Addressing is enabled for Y Data Space when YWM is set to any value other than '1111' and the YMODEN bit (MODCON[14]) is set.

Figure 4-10. Modulo Addressing Operation Example



4.4.4 Bit-Reversed Addressing

Bit-Reversed Addressing mode is intended to simplify data reordering for radix-2 FFT algorithms. It is supported by the X AGU for data writes only.

The modifier, which can be a constant value or register contents, is regarded as having its bit order reversed. The address source and destination are kept in normal order. Thus, the only operand requiring reversal is the modifier.

4.4.4.1 Bit-Reversed Addressing Implementation

Bit-Reversed Addressing mode is enabled in any of these situations:

- BWMx bits (W register selection) in the MODCON register are any value other than '1111' (the stack cannot be accessed using Bit-Reversed Addressing)
- The BREN bit is set in the XBREV register
- The addressing mode used is Register Indirect with Pre-Increment or Post-Increment

If the length of a bit-reversed buffer is $M = 2^N$ bytes, the last 'N' bits of the data buffer start address must be zeros.

The XB[14:0] bits are the Bit-Reversed Addressing modifier, or 'pivot point', which is typically a constant. In the case of an FFT computation, its value is equal to half of the FFT data buffer size.

Note: All bit-reversed EA calculations assume word-sized data (LSb of every EA is always clear). The XB value is scaled accordingly to generate compatible (byte) addresses.

When enabled, Bit-Reversed Addressing is executed only for Register Indirect with Pre-Increment or Post-Increment Addressing and word-sized data writes. It does not function for any other addressing mode or for byte-sized data and normal addresses are generated instead. When Bit-Reversed Addressing is active, the W Address Pointer is always added to the address modifier (XB) and the offset associated with the Register Indirect Addressing mode is ignored. In addition, as word-sized data are a requirement, the LSb of the EA is ignored (and always clear).

Note: Modulo Addressing and Bit-Reversed Addressing can be enabled simultaneously using the same W register, but Bit-Reversed Addressing operation will always take precedence for data writes when enabled.

If Bit-Reversed Addressing has already been enabled by setting the BREN (XBREV[15]) bit, a write to the XBREV register should not be immediately followed by an indirect read operation using the W register that has been designated as the Bit-Reversed Pointer.

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Figure 4-11. Bit-Reversed Addressing Example

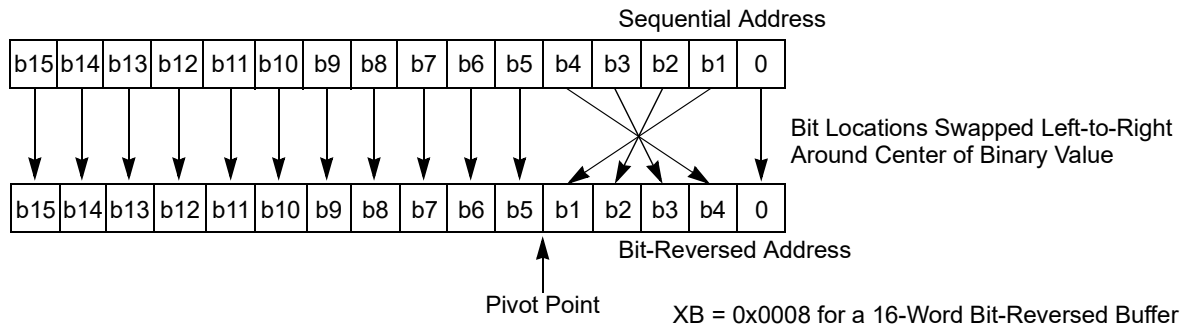


Table 4-4. Bit-Reversed Addressing Sequence (16-Entry)

Normal Address					Bit-Reversed Address				
A3	A2	A1	A0	Decimal	A3	A2	A1	A0	Decimal
0	0	0	0	0	0	0	0	0	0
0	0	0	1	1	1	0	0	0	8
0	0	1	0	2	0	1	0	0	4
0	0	1	1	3	1	1	0	0	12
0	1	0	0	4	0	0	1	0	2
0	1	0	1	5	1	0	1	0	10
0	1	1	0	6	0	1	1	0	6
0	1	1	1	7	1	1	1	0	14
1	0	0	0	8	0	0	0	1	1
1	0	0	1	9	1	0	0	1	9
1	0	1	0	10	0	1	0	1	5
1	0	1	1	11	1	1	0	1	13
1	1	0	0	12	0	0	1	1	3
1	1	0	1	13	1	0	1	1	11
1	1	1	0	14	0	1	1	1	7
1	1	1	1	15	1	1	1	1	15

4.4.5 Interfacing Program and Data Memory Spaces

The dsPIC33CK512MPT608 family architecture uses a 24-bit wide Program Space (PS) and a 16-bit wide Data Space (DS). The architecture is also a modified Harvard scheme, meaning that data can also be present in the Program Space. To use these data successfully, they must be accessed in a way that preserves the alignment of information in both spaces.

Aside from normal execution, the architecture of the dsPIC33CK512MPT608 family devices provides two methods by which Program Space can be accessed during operation:

- Using table instructions to access individual bytes or words anywhere in the Program Space
- Remapping a portion of the Program Space into the Data Space (Program Space Visibility)

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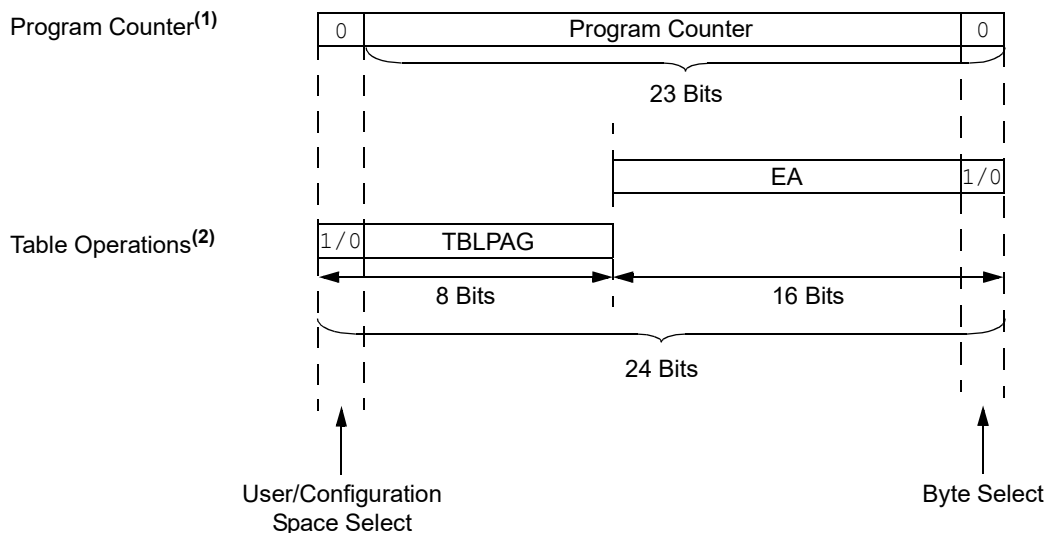
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Table instructions allow an application to read or write to small areas of the program memory. This capability makes the method ideal for accessing data tables that need to be updated periodically. It also allows access to all bytes of the program word. The remapping method allows an application to access a large block of data on a read-only basis, which is ideal for look-ups from a large table of static data. The application can only access the least significant word of the program word.

Table 4-5. Program Space Address Construction

Access Type	Access Space	Program Space Address				
		[23]	[22:16]	[15]	[14:1]	[0]
Instruction Access (Code Execution)	User	0	PC[22:1]			0
		0xxx xxxx xxxx xxxx xxxx xxx0				
TBLRD/TBLWT (Byte/Word Read/Write)	User	TBLPAG[7:0]		Data EA[15:0]		
		0xxx xxxx xxxx xxxx xxxx xxxx				
	Configuration	TBLPAG[7:0]		Data EA[15:0]		
		1xxx xxxx xxxx xxxx xxxx xxxx				

Figure 4-12. Data Access from Program Space Address Generation



- Note 1:** The Least Significant bit (LSb) of Program Space addresses is always fixed as '0' to maintain word alignment of data in the Program and Data Spaces.
- 2:** Table operations are not required to be word-aligned. Table Read operations are permitted in the configuration memory space.

4.4.5.1 Data Access from Program Memory Using Table Instructions

The **TBLRDL** and **TBLWTL** instructions offer a direct method of reading or writing the lower word of any address within the Program Space without going through Data Space. The **TBLRDH** and **TBLWTH** instructions are the only method to read or write the upper eight bits of a Program Space word as data.

The PC is incremented by two for each successive 24-bit program word. This allows program memory addresses to directly map to Data Space addresses. Program memory can thus be regarded as two 16-bit wide word address spaces, residing side by side, each with the same address range. **TBLRDL** and **TBLWTL** access the space that contains the least significant data word. **TBLRDH** and **TBLWTH** access the space that contains the upper data byte.

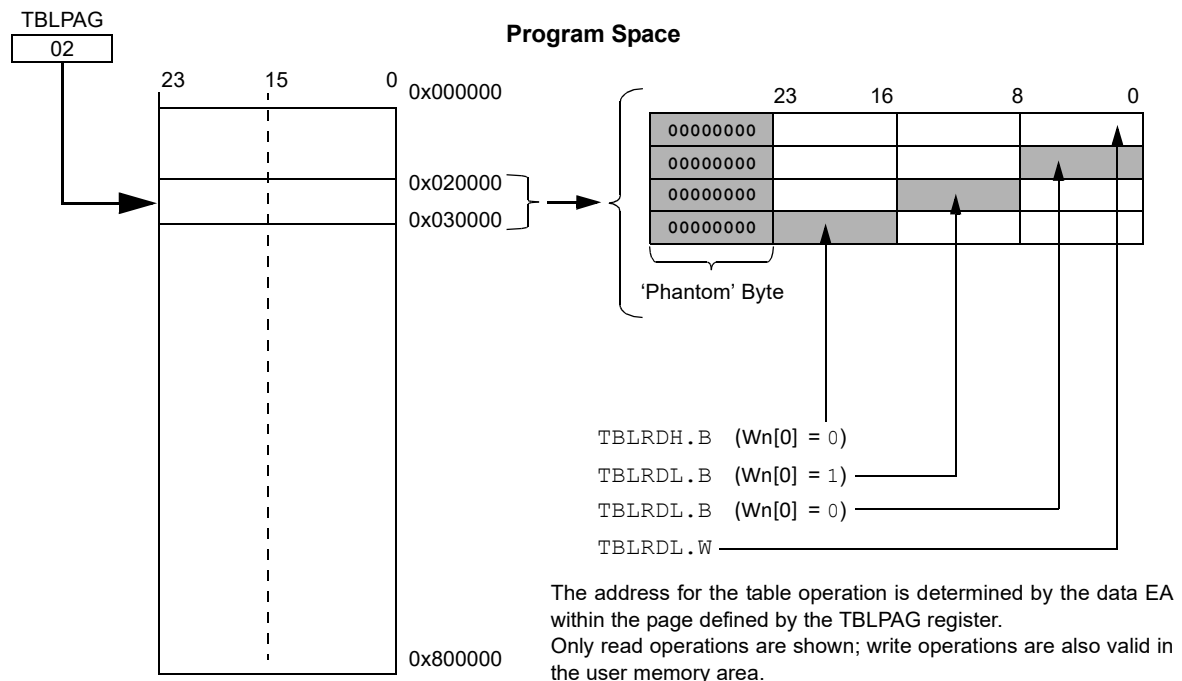
Two table instructions are provided to move byte or word-sized (16-bit) data to and from Program Space. Both function as either byte or word operations.

- **TBLRDL (Table Read Low):**
 - In Word mode, this instruction maps the lower word of the Program Space location (P[15:0]) to a data address (D[15:0]).
 - In Byte mode, either the upper or lower byte of the lower program word is mapped to the lower byte of a data address. The upper byte is selected when Byte Select is '1'; the lower byte is selected when it is '0'.
- **TBLRDH (Table Read High):**
 - In Word mode, this instruction maps the entire upper word of a program address (P[23:16]) to a data address. The 'phantom' byte (D[15:8]) is always '0'.
 - In Byte mode, this instruction maps the upper or lower byte of the program word to D[7:0] of the data address in the TBLRDL instruction. The data are always '0' when the upper 'phantom' byte is selected (Byte Select = 1).

In a similar fashion, two table instructions, TBLWTH and TBLWTL, are used to write individual bytes or words to a Program Space address. The details of their operation are explained in [5. Flash Program Memory](#).

For all table operations, the area of program memory space to be accessed is determined by the Table Page register (TBLPAG). TBLPAG covers the entire program memory space of the device, including user application and configuration spaces. When TBLPAG[7] = 0, the table page is located in the user memory space. When TBLPAG[7] = 1, the page is located in configuration space.

Figure 4-13. Accessing Program Memory with Table Instructions



5. Flash Program Memory

Note: This data sheet summarizes the features of the dsPIC33CK512MPT608 family of devices. It is not intended to be a comprehensive reference source. To complement the information in this data sheet, refer to “**Dual Partition Flash Program Memory**” (www.microchip.com/DS70005156) in the “dsPIC33/PIC24 Family Reference Manual”.

Some registers and associated bits described in this section may not be available on all devices.

The dsPIC33CK512MPT608 family devices contain internal Flash program memory for storing and executing application code. The memory is readable, writable and erasable during normal operation over the entire V_{DD} range.

Flash memory can be programmed in three ways:

- In-Circuit Serial Programming™ (ICSP™) programming capability
- Enhanced In-Circuit Serial Programming (Enhanced ICSP)
- Run-Time Self-Programming (RTSP)

ICSP allows for a dsPIC33CK512MPT608 family device to be serially programmed while in the end application circuit. This is done with a Programming Clock and Programming Data (PGCx/PGDx) line, and three other lines for power (V_{DD}), ground (V_{SS}) and Master Clear ($\overline{MCLR}$). This allows customers to manufacture boards with unprogrammed devices and then program the device just before shipping the product. This also allows the most recent firmware or a custom firmware to be programmed.

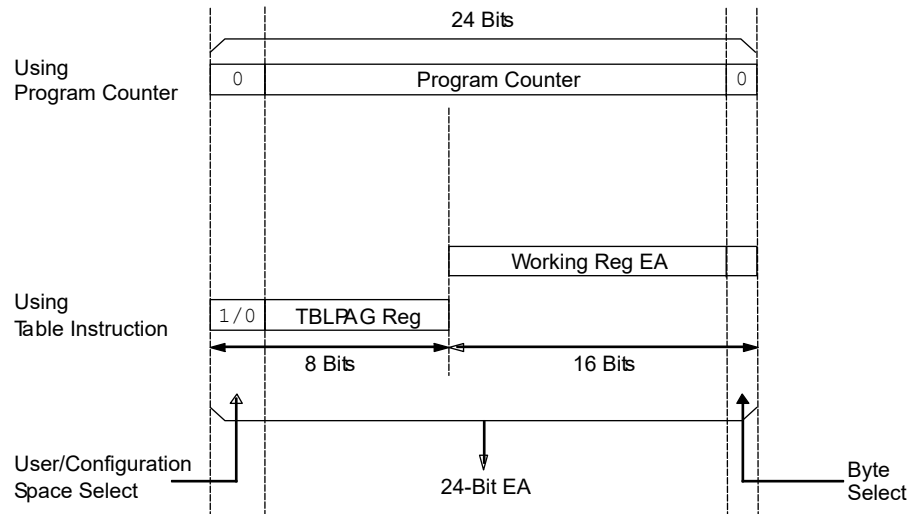
Enhanced In-Circuit Serial Programming uses an on-board bootloader, known as the Program Executive, to manage the programming process. Using an SPI data frame format, the Program Executive can erase, program and verify program memory. For more information on Enhanced ICSP, see the device programming specification.

RTSP allows the Flash user application code to update itself during run time. The feature is capable of writing a single program memory word (two instructions) or an entire row as needed.

5.1 Table Instructions and Flash Programming

Regardless of the method used, all programming of Flash memory is done with the Table Read and Table Write instructions. These allow direct read and write access to the program memory space from the data memory while the device is in normal operating mode. The 24-bit target address in the program memory is formed using bits[7:0] of the TBLPAG register and the Effective Address (EA) from a W register, specified in the table instruction, as shown in [Figure 5-1](#). The TBLRDL and TBLWTL instructions are used to read or write to bits[15:0] of program memory. TBLRDL and TBLWTL can access program memory in both Word and Byte modes. The TBLRDH and TBLWTH instructions are used to read or write to bits[23:16] of program memory. TBLRDH and TBLWTH can also access program memory in Word or Byte mode.

Figure 5-1. Addressing for Table Registers



5.2 RTSP Operation

RTSP allows the user application to program one double instruction word or one row at a time. The double instruction word write blocks and single row write blocks are edge-aligned, from the beginning of program memory, on boundaries of one double instruction word and 64 double instruction words, respectively.

The basic sequence for RTSP programming is to first load two 24-bit instructions into the NVM write latches found in configuration memory space. Then, the WR bit in the NVMCON register is set to initiate the write process. The processor stalls (waits) until the programming operation is finished. The WR bit is automatically cleared when the operation is finished.

Double instruction word writes are performed by manually loading both write latches, using TBLWTL and TBLWTH instructions, and then initiating the NVM write while the NVMOPx bits are set to '0x1'. The program space destination address is defined by the NVMADR/U registers.

Row programming is performed by first loading 128 instructions into data RAM and then loading the address of the first instruction in that row into the NVMSRCADRL/H registers. Once the write has been initiated, the device will automatically load two instructions into the write latches and write them to the program space destination address defined by the NVMADR/U registers.

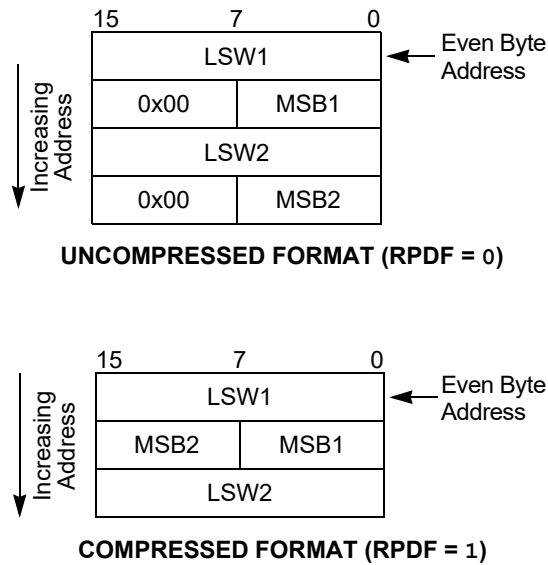
The operation will increment the NVMSRCADRL/H and the NVMADR/U registers until all double instruction words have been programmed.

The RPDF bit (NVMCON[9]) selects the format of the stored data in RAM to be either compressed or uncompressed. See [Figure 5-2](#) for data formatting.

Compressed data help to reduce the amount of required RAM by using the upper byte of the second word for the MSB of the second instruction.

All erase and program operations may optionally use the NVM interrupt to signal the successful completion of the operation.

Figure 5-2. Uncompressed/Compressed Format



Example 5-1. Flash Write/Read

```

//////////Flash write //////////
//Sample code for writing 0x123456 to address locations 0x10000 / 10002
NVMCON = 0x4001;
TBLPAG = 0xFA;           // write latch upper address
NVMADR = 0x0000;         // set target write address of general
segment
NVMADRU = 0x0001;
__builtin_tblwtl(0, 0x3456); // load write latches
__builtin_tblwth (0,0x12);
__builtin_tblwtl(2, 0x3456); // load write latches
__builtin_tblwth (2,0x12);
asm volatile ("disi #5");
__builtin_write_NVM();
while( _WR == 1 ) ;

//////////Flash Read//////////
//Sample code to read the Flash content of address 0x10000
// readDataL/ readDataH variables need to defined
TBLPAG = 0x0001;
readDataL = __builtin_tblrld(0x0000);
readDataH = __builtin_tblrhd(0x0000);

```

5.3 Error Correcting Code (ECC)

In order to improve program memory performance and durability, the devices include Error Correcting Code functionality (ECC) as an integral part of the Flash memory controller. ECC can determine the presence of single-bit errors in program data, including which bit is in error, and correct the data automatically without user intervention. ECC cannot be disabled.

When data are written to program memory, ECC generates a 7-bit Hamming code parity value for every two (24-bit) instruction words. The data are stored in blocks of 48 data bits and seven parity bits; parity data are not memory-mapped and are inaccessible. When the data are read back, the ECC calculates the parity on them and compares it to the previously stored parity value. If a parity mismatch occurs, there are two possible outcomes:

- Single-bit error has occurred and has been automatically corrected on read-back.
- Double-bit error has occurred and the read data are not changed.

Single-bit error occurrence can be identified by the state of the ECCSBEIF (IFS0[13]) bit. An interrupt can be generated when the corresponding interrupt enable bit is set, ECCSBEIE (IEC0[13]). The ECCSTATL register contains the parity information for single-bit errors. The SECOUT[7:0] bits field contains the expected calculated SEC parity and the SECIN[7:0] bits contain the actual value from a Flash read operation. The SECSYNDx bits (ECCSTATH[7:0]) indicate the bit position of the single-bit error within the 48-bit pair of instruction words. When no error is present, SECINx equals SECOUTx and SECSYNDx is zero. The ECCSTATL and ECCSTATH registers will only update and be valid when an error has occurred, or when included Fault injection is enabled and an ECCADDR match occurs.

Double-bit errors result in a generic hard trap. The ECCDBE bit (INTCON4[1]) will be set to identify the source of the hard trap. If no Interrupt Service Routine is implemented for the hard trap, a device Reset will also occur. The ECCSTATH register contains double-bit error status information. The DEDOUT bit is the expected calculated DED parity and DEDIN is the actual value from a Flash read operation. When no error is present, DEDIN equals DEDOUT.

5.4 ECC Fault Injection

To test Fault handling, an EEC error can be generated. Both single and double-bit errors can be generated in both the read and write data paths. Read path Fault injection first reads the Flash data and then modifies them prior to entering the ECC logic. Write path Fault injection modifies the actual data prior to them being written into the target Flash and will cause an EEC error on a subsequent Flash read. The following procedure is used to inject a Fault:

1. Load the Flash target address into the ECCADDR register.
2. Select 1st Fault bit determined by FLT1PTRx (ECCCONH[7:0]). The target bit is inverted to create the Fault.
3. If a double Fault is desired, select the 2nd Fault bit determined by FLT2PTRx (ECCCONH[15:8]); otherwise, set to all '1's.
4. Write 0x55 to NVMKEY.
5. Write 0xAA to NVMKEY.
6. Set the FLTINJ bit (ECCCONL[0]) in a single operation to enable the ECC Fault injection logic.
7. Perform a read or write to the Flash target address.

5.5 Flash OTP by ICSP™ Write Inhibit

ICSP Write Inhibit is an access restriction feature, that when activated, restricts all of Flash memory. Once activated, ICSP Write Inhibit permanently prevents ICSP Flash programming and erase operations, and cannot be deactivated. This feature is intended to prevent alteration of Flash memory contents, with behavior similar to One-Time-Programmable (OTP) devices.

RTSP, including erase and programming operations, is not restricted when ICSP Write Inhibit is activated; however, code to perform these actions must be programmed into the device before ICSP Write Inhibit is activated. This allows for a bootloader-type application to alter Flash contents with ICSP Write Inhibit activated.

Entry into ICSP and Enhanced ICSP modes is not affected by ICSP Write Inhibit. In these modes, it will continue to be possible to read configuration memory space and any user memory space regions which are not code-protected. With ICSP writes inhibited, an attempt to set WR (NVMCON[15]) = 1 will maintain WR = 0, and instead, set WRERR

(NVMCON[13]) = 1. All Enhanced ICSP erase and programming commands will have no effect with self-checked programming commands returning a FAIL response opcode (PASS if the destination already exactly matched the requested programming data).

Once ICSP Write Inhibit is activated, it is not possible for a device executing in Debug mode to erase/write Flash, nor can a debug tool switch the device to Production mode. ICSP Write Inhibit should therefore only be activated on devices programmed for production.

5.6 Dual Partition Flash Configuration

For dsPIC33CK512MPT608 devices operating in Dual Partition Flash Program Memory modes, the Inactive Partition can be erased and programmed without stalling the processor. The same programming algorithms are used for programming and erasing the Flash in the Inactive Partition, as described in [5.2. RTSP Operation](#). On top of the page erase option, the entire Flash memory of the Inactive Partition can be erased by configuring the NVMOP[3:0] bits in the NVMCON register.

Note: The application software to be loaded into the Inactive Partition will have the address of the Active Partition. The bootloader firmware will need to offset the address by 0x400000 in order to write to the Inactive Partition.

5.6.1 Flash Partition Swapping

The Boot Sequence Number is used for determining the Active Partition at start-up and is encoded within the FBTSEQ Configuration register bits. Unlike most Configuration registers, which only utilize the lower 16 bits of the program memory, FBTSEQ is a 24-bit Configuration Word. The Boot Sequence Number (BSEQ) is a 12-bit value and is stored in FBTSEQ twice. The true value is stored in bits, FBTSEQ[11:0], and its complement is stored in bits, FBTSEQ[23:12]. At device Reset, the sequence numbers are read and the partition with the lowest sequence number becomes the Active Partition. If one of the Boot Sequence Numbers is invalid, the device will select the partition with the valid Boot Sequence Number, or default to Partition 1 if both sequence numbers are invalid. See [30. Special Features](#) for more information.

The `BOOTSWP` instruction provides an alternative means of swapping the Active and Inactive Partitions (soft swap) without the need for a device Reset. The `BOOTSWP` must always be followed by a `GOTO` instruction. The `BOOTSWP` instruction swaps the Active and Inactive Partitions, and the PC vectors to the location specified by the `GOTO` instruction in the newly Active Partition.

It is important to note that interrupts should temporarily be disabled while performing the soft swap sequence and that after the partition swap, all peripherals and interrupts which were enabled remain enabled. Additionally, the RAM and stack will maintain state after the switch. As a result, it is recommended that applications using soft swaps jump to a routine that will reinitialize the device in order to ensure the firmware runs as expected. The Configuration registers will have no effect during a soft swap.

For robustness of operation, in order to execute the `BOOTSWP` instruction, it is necessary to execute the NVM unlocking sequence as follows:

1. Write 0x55 to NVMKEY.
2. Write 0xAA to NVMKEY.
3. Execute the `BOOTSWP` instruction.

If the unlocking sequence is not performed, the `BOOTSWP` instruction will be executed as a forced `NOP` and a `GOTO` instruction, following the `BOOTSWP` instruction, will be executed, causing the PC to jump to that location in the current operating partition.

The `SFTSWP` and `P2ACTIV` bits in the NVMCON register are used to determine a successful swap of the Active and Inactive Partitions, as well as which partition is active. After the `BOOTSWP` and `GOTO` instructions, the `SFTSWP` bit should be polled to verify the partition swap has occurred and then cleared for the next partition swap event.

5.6.2 Dual Partition Modes

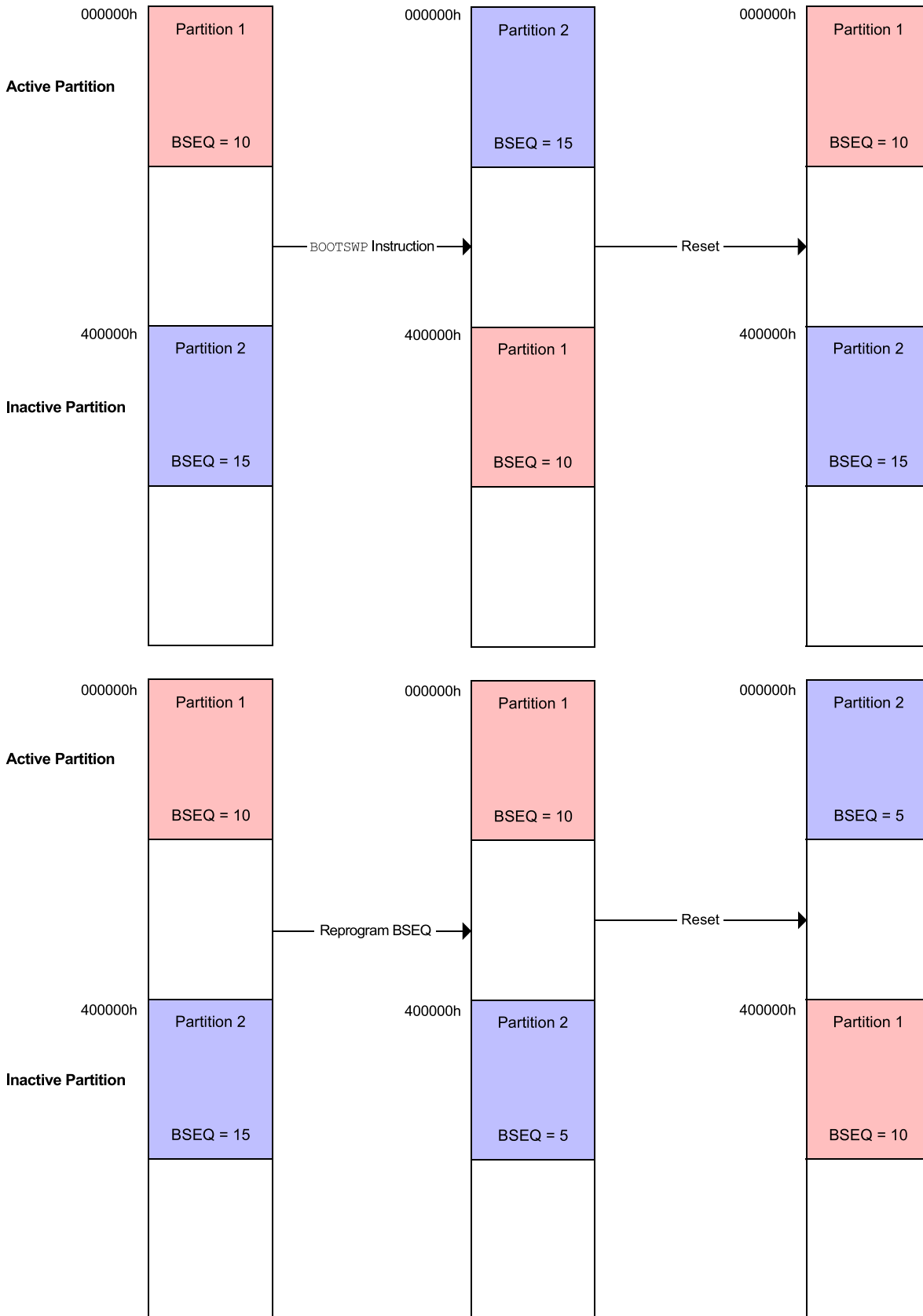
While operating in Dual Partition mode, the dsPIC33CK512MPT608 family devices have the option for both partitions to have their own defined security segments, as shown in [30.10. Code Protection and CodeGuard™ Security](#). Alternatively, the device can operate in Protected Dual Partition mode, where Partition 1 becomes permanently erase/write-protected. Protected Dual Partition mode allows for a “Factory Default” mode, which provides a fail-safe backup image to be stored in Partition 1.

dsPIC33CK512MPT608 family devices can also operate in Privileged Dual Partition mode, where additional security protections are implemented to allow for protection of intellectual property when multiple parties have software within the device. In Privileged Dual Partition mode, both partitions place additional restrictions on the FBSLIM register. These prevent changes to the size of the Boot Segment and General Segment, ensuring that neither segment will be altered.

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Flash Program Memory

Figure 5-3. Relationship Between Partitions 1/2 and Active/Inactive Partitions



5.7 NVM/ECC Control Registers

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x00 ... 0xEF	Reserved									
0xF0	ECCCONL	15:8 7:0								FLTINJ
0xF2	ECCCONH	15:8 7:0	FLT2PTR[7:0] FLT1PTR[7:0]							
0xF4	ECCADDRL	15:8 7:0	ECCADDR[15:8] ECCADDR[7:0]							
0xF6	ECCADDRH	15:8 7:0	ECCADDR[23:16]							
0xF8	ECCSTATL	15:8 7:0	SECOUT[7:0] SECIN[7:0]							
0xFA	ECCSTATH	15:8 7:0							DEDOUT	DEDIN
0xFC ... 0x08CF	Reserved									
0x08D0	NVMCON	15:8 7:0	WR	WREN	WRERR	NVMSIDL	SFTSWP	P2ACTIV	RPDF	URERR
0x08D2	NVMADR	15:8 7:0	NVMADR[15:8] NVMADR[7:0]							
0x08D4	NVMADRU	15:8 7:0	NVMADRU[23:16]							
0x08D6	NVMKEY	15:8 7:0	NVMKEY[7:0]							
0x08D8	NVMSRCADR	15:8 7:0	NVMSRCADR[15:8] NVMSRCADR[7:0]							
0x08DA	NVMSRCADRH	15:8 7:0	NVMSRCADR[23:16]							

5.7.1 Nonvolatile Memory (NVM) Control Register

Name: NVMCON
Offset: 0x8D0

Notes:

1. These bits can only be reset on a POR.
2. If this bit is set, there will be minimal power savings (I_{IDLE}), and upon exiting Idle mode, there is a delay (T_{VREG}) before Flash memory becomes operational.
3. All other combinations of NVMOP[3:0] are unimplemented.
4. Execution of the `PWRSVAV` instruction is ignored while any of the NVM operations are in progress.
5. Two adjacent words on a 4-word boundary are programmed during execution of this operation.

Legend: C = Clearable bit; SO = Settable Only bit

Bit	15	14	13	12	11	10	9	8
	WR	WREN	WRERR	NVMSIDL	SFTSWP	P2ACTIV	RPDF	URERR
Access	R/SO	R/W	R/C	R/W	R/C	R	R/W	R/C
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
					NVMOP[3:0]			
Access					R/W	R/W	R/W	R/W
Reset					0	0	0	0

Bit 15 – WR Write Control bit⁽¹⁾

Value	Description
1	Initiates a Flash memory program or erase operation; the operation is self-timed and the bit is cleared by hardware once the operation is complete
0	Program or erase operation is complete and inactive

Bit 14 – WREN Write Enable bit⁽¹⁾

Value	Description
1	Enables Flash program/erase operations
0	Inhibits Flash program/erase operations

Bit 13 – WRERR Write Sequence Error Flag bit⁽¹⁾

Value	Description
1	An improper program or erase sequence attempt, or termination has occurred (bit is set automatically on any set attempt of the WR bit)
0	The program or erase operation completed normally

Bit 12 – NVMSIDL NVM Stop in Idle Control bit⁽²⁾

Value	Description
1	Flash voltage regulator goes into Standby mode during Idle mode
0	Flash voltage regulator is active during Idle mode

Bit 11 – SFTSWP Partition Soft Swap Status bit

Value	Description
1	Partitions have been successfully swapped using the <code>BOOTSWP</code> instruction (soft swap)
0	Awaiting successful partition swap using the <code>BOOTSWP</code> instruction or a device Reset will determine the Active Partition based on the FBTSEQ register

Bit 10 – P2ACTIV Partition 2 Active Status bit

Value	Description
1	Partition 2 Flash is mapped into the active region

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Value	Description
0	Partition 1 Flash is mapped into the active region

Bit 9 – RPDF Row Programming Data Format bit

Value	Description
1	Row data to be stored in RAM are in compressed format
0	Row data to be stored in RAM are in uncompressed format

Bit 8 – URERR Row Programming Data Underrun Error bit

Value	Description
1	Indicates row programming operation has been terminated
0	No data underrun error is detected

Bits 3:0 – NVMOP[3:0] NVM Operation Select bits^(1,3,4)

Value	Description
1111	Reserved
1110	User memory bulk erase operation
1101	Reserved
1100	Reserved
1011	Reserved
1010	Reserved
1001	Reserved
1000	Boot mode (FBOOT) double-word program operation
0111	Reserved
0101	Reserved
0100	Inactive Partition memory erase operation
0011	Memory page erase operation
0010	Memory row program operation
0001	Memory double-word operation ⁽⁵⁾
0000	Reserved

5.7.2 Nonvolatile Memory Lower Address Register

Name: NVMADR

Offset: 0x8D2

Legend: x = Bit is unknown

Bit	15	14	13	12	11	10	9	8
	NVMADR[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	x	x	x	x	x	x	x	x
Bit	7	6	5	4	3	2	1	0
	NVMADR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	x	x	x	x	x	x	x	x

Bits 15:0 – NVMADR[15:0] Nonvolatile Memory Lower Write Address bits

Selects the lower 16 bits of the location to program or erase in Program Flash Memory. This register may be read or written to by the user application.

5.7.3 Nonvolatile Memory Upper Address Register

Name: NVMADRU
Offset: 0x8D4

Legend: x = Bit is unknown

Bit	15	14	13	12	11	10	9	8
Access								
Reset								

Bit	7	6	5	4	3	2	1	0
	NVMADRU[23:16]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	x	x	x	x	x	x	x	x

Bits 7:0 – NVMADRU[23:16] Nonvolatile Memory Upper Write Address bits
Selects the upper eight bits of the location to program or erase in Program Flash Memory. This register may be read or written to by the user application.

5.7.4 Nonvolatile Memory Key Register

Name: NVMKEY
Offset: 0x8D6

Bit	15	14	13	12	11	10	9	8
Access								
Reset								
Bit	7	6	5	4	3	2	1	0
	NVMKEY[7:0]							
Access	W	W	W	W	W	W	W	W
Reset	0	0	0	0	0	0	0	0

Bits 7:0 – NVMKEY[7:0] NVM Key Register bits (write-only)

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5.7.5 NVM Source Data Address Register Low

Name: NVMSRCADRL
Offset: 0x8D8

Bit	15	14	13	12	11	10	9	8
	NVMSRCADR[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	NVMSRCADR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – NVMSRCADR[15:0] NVM Source Data Address bits

The RAM address of the data to be programmed into Flash when the NVMOP[3:0] bits are set to row programming.

5.7.6 NVM Source Data Address Register High

Name: NVMSRCADRH
Offset: 0x8DA

Bit	15	14	13	12	11	10	9	8
Access								
Reset								
Bit	7	6	5	4	3	2	1	0
	NVMSRCADR[23:16]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 7:0 – NVMSRCADR[23:16] NVM Source Data Address bits
The RAM address of the data to be programmed into Flash when the NVMOP[3:0] bits are set to row programming.

5.7.7 ECC Fault Injection Configuration Register Low

Name: ECCCONL
Offset: 0x0F0

Bit	15	14	13	12	11	10	9	8
Access								
Reset								

Bit	7	6	5	4	3	2	1	0
								FLTINJ
Access								R/W
Reset								0

Bit 0 – FLTINJ Fault Injection Sequence Enable bit

Value	Description
1	Enabled
0	Disabled

5.7.8 ECC Fault Injection Configuration Register High

Name: ECCCONH
Offset: 0x0F2

Bit	15	14	13	12	11	10	9	8
	FLT2PTR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	FLT1PTR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – FLT2PTR[7:0] ECC Fault Injection Bit Pointer 2 bits

Value	Description
11111111-00111000	No Fault injection occurs
00110111	Fault injection (bit inversion) occurs on bit 55 of ECC bit order
. . .	
00000001	Fault injection (bit inversion) occurs on bit 1 of ECC bit order
00000000	Fault injection (bit inversion) occurs on bit 0 of ECC bit order

Bits 7:0 – FLT1PTR[7:0] ECC Fault Injection Bit Pointer 1 bits

Value	Description
11111111-00111000	No Fault injection occurs
00110111	Fault injection occurs on bit 55 of ECC bit order
. . .	
00000001	Fault injection occurs on bit 1 of ECC bit order
00000000	Fault injection occurs on bit 0 of ECC bit order

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5.7.9 ECC Fault Inject Address Compare Register Low

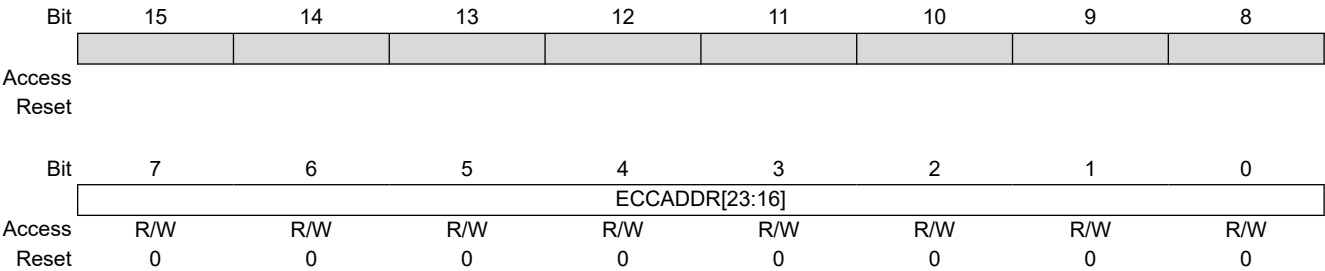
Name: ECCADDRL
Offset: 0x0F4

Bit	15	14	13	12	11	10	9	8
	ECCADDR[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	ECCADDR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – ECCADDR[15:0] ECC Fault Injection NVM Address Match Compare bits

5.7.10 ECC Fault Inject Address Compare Register High

Name: ECCADDRH
Offset: 0x00F6



Bits 7:0 – ECCADDR[23:16] ECC Fault Injection NVM Address Match Compare bits

5.7.11 ECC System Status Display Register Low

Name: ECCSTATL
Offset: 0x0F8

Bit	15	14	13	12	11	10	9	8
	SECOUT[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	SECIN[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – SECOUT[7:0] Calculated Single Error Correction Parity Value bits

Bits 7:0 – SECIN[7:0] Read Single Error Correction Parity Value bits
 SECIN[7:0] bits are the actual parity value of a Flash read operation.

5.7.12 ECC System Status Display Register High

Name: ECCSTATH
Offset: 0x00FA

Bit	15	14	13	12	11	10	9	8
							DEDOUT	DEDIN
Access							R/W	R/W
Reset							0	0
Bit	7	6	5	4	3	2	1	0
							SECSYND[7:0]	
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

- Bit 9 – DEDOUT** Calculated Dual Bit Error Detection Parity bit
- Bit 8 – DEDIN** Read Dual Bit Error Detection Parity bit
DEDIN is the actual parity value of a Flash read operation.
- Bits 7:0 – SECSYND[7:0]** Calculated ECC Syndrome Value bits
Indicates the bit location that contains the error.

6. Secure Subsystem

The information available in this section is only a summary of the Secure Subsystem. For a detailed reference manual, which is under NDA, contact your local Microchip sales office or obtain additional support at www.microchip.com/en-us/support/design-help/client-support-services

6.1 Overview

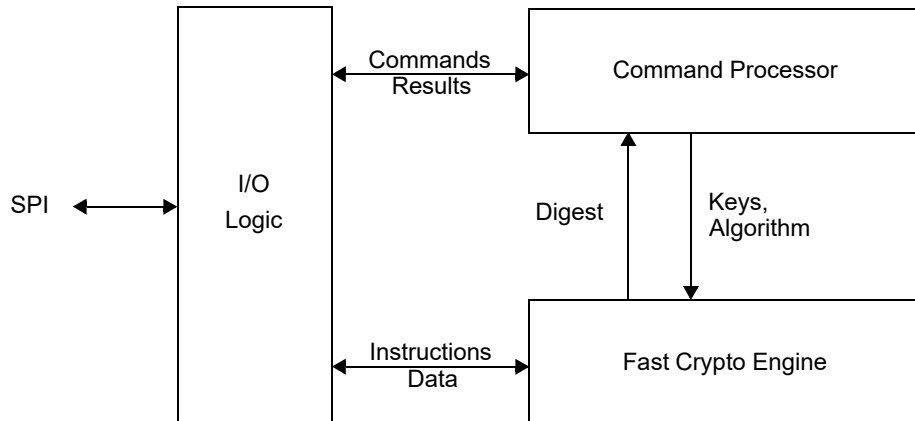
The Secure Subsystem, combined with various device features, provides a hardened root of trust with symmetric and asymmetric computation ability to facilitate a number of security-related capabilities within an automotive, industrial and consumer system.

- Secure Boot Support:
 - Code image and signature validation
 - Secure encryption key storage and image encryption
 - Authenticated update of the code validation public key
- X.509 Certificate Storage, Parsing, Validation and Revocation, Supporting both ECC and RSA
- Fully Internal Random Key Generation for RSA, ECC and AES
- Monotonic Counters Protected Against Tearing
- Elliptic Curves Support:
 - P224 – ECDSA sign, verify, ECDH and ECB
 - P256 – ECDSA sign, verify and ECDH
 - SECP256K1 (bitcoin/blockchain) – ECDSA support
 - 256-bit Brainpool – ECDSA and ECDH
 - P384 – ECDSA sign, verify and ECDH
- RSA Support:
 - 1024-bit and 2048-bit RSA OAEP/MGF encrypt/decrypt
 - 2048-bit RSA signature generation and verification
 - 3072-bit RSA verification
- ECDH Key Management Capability with Integrated KDF, either PRF or HKDF
- NIST SP800-90 A/B/C High-Quality Cryptographic Random Number Generation (RNG)
- AES-CMAC Calculation and Validation
- AES-ECB and GCM Encrypt/Decrypt for General Purpose Use
- SHA-256 and SHA-HMAC Digest Calculation
- Input/Output Encryption and Authentication using AES-GCM, AES-CMAC and/or SHA-HMAC
- Flexible Self-Test Support to meet FIPS 140 Requirements

The Secure Subsystem contains two processing blocks:

- A main command processor that implements an Advanced Crypto Engine along with the management and session establishment functionality. The ACE can implement all symmetric and asymmetric crypto functions.
- A Fast Crypto Engine capable of implementing AES and SHA calculations in parallel with the operation of the main command processor.

Figure 6-1. Secure Subsystem Block Diagram



6.2 Features

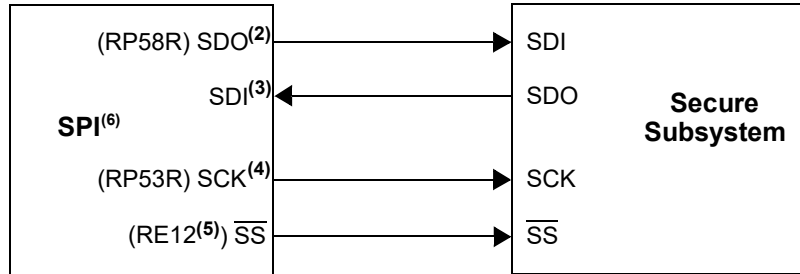
- Secure Subsystem includes protection against both active (invasive) and passive (noninvasive) attacks on the certificates, private and symmetric keys stored within the device. Specific hardware and firmware elements are included in this peripheral to prevent environmental (voltage, temperature and frequency) attacks, emissions attacks, Fault attacks, physical attacks, cloning and many other attack methodologies. All internal memory for private/symmetric keys or other secret data are encrypted.
- Advanced Crypto Engine (ACE) for Execution of All Cryptography Commands
- Physical Protection Security Measures, including Voltage Tamperers, Temperature Tamperers and Active Shield Circuitry
- Sign/Verify Support:
 - ECDSA – P224, P256, P384 and 256-bit Brainpool elliptic curves
 - ECDSA – SECP256K1 (bitcoin/blockchain) curve
 - RSA 2048-bit signature generation and verification
 - RSA 3072-bit signature verification only
- ECDH/ECDHE/ECBD Key Agreement Support:
 - Elliptic-Curve Diffie-Hellman (ECDH) support for P224, P256, P384 and 256-bit Brainpool
 - Elliptic-Curve Burmester-Desmedt (ECBD) support for P224 curve
- Internal Symmetric and Asymmetric Key Generation and Derivation:
 - P224, P256, P384 and 256-Bit Brainpool
 - 2048-bit RSA keys
 - AES 16-byte keys
- AES and RSA Encryption/Decryption Support:
 - AES ECB/GCM encryption/decryption supported directly
 - RSA 1024-bit and 2048-bit keys encryption/decryption support
- NIST SP800-90 A/B/C Random Number Generator (RNG)
- 16 MHz SPI Interface to communicate Security Commands between the core and the Secure Subsystem
- The Secure Subsystem Advanced Crypto Engine algorithms have achieved **JIL HIGH** Rating and are certified by FIPS as per *Cryptographic Algorithm Validation Program (CAVP)*
- Secure Subsystem with FIPS 140-2 Level 2 with Physical Security Level 3 certification as per *Cryptographic Module Validation Program (CMVP)* [in progress].

6.3 SPI Clock Configuration

The SPI clock configuration used for communication between the core and Secure Subsystem is as follows:

1. For communication in Mode 0, CKP = 0, CKE = 1.
2. For communication in Mode 3, CKP = 1, CKE = 0.

Figure 6-2. SPI Communication with Secure Subsystem⁽¹⁾



Notes:

1. All the connections are internal to the device. The corresponding PPS registers must be configured to establish communication with the Secure Subsystem.
2. Refer to [9.12.57. RPOR13](#) for more information on configuring the PPS register.
3. The corresponding SDIxR register bits of SPI should be set to 52.
4. Refer to [9.12.54. RPOR10](#) for more information on configuring the PPS register.
5. RE12 I/O pin is not available for general purpose and its associated I/O registers are available only for communication with the Secure Subsystem.
6. Ensure that the SPI signals are not left floating to avoid unintended current consumption.

6.4 Applications

- Full and Partial Secure Boot
- Secure Firmware Update
- CAN Message Authentication
- WPC 1.3 Qi High-Power Transmitter Authentication
- High Bandwidth Digital Content Protection (HDCP) Cryptographic Support
- Network Authentication and Session Establishment using TLS
- Electric Vehicle (EV) Battery Authentication

7. Resets

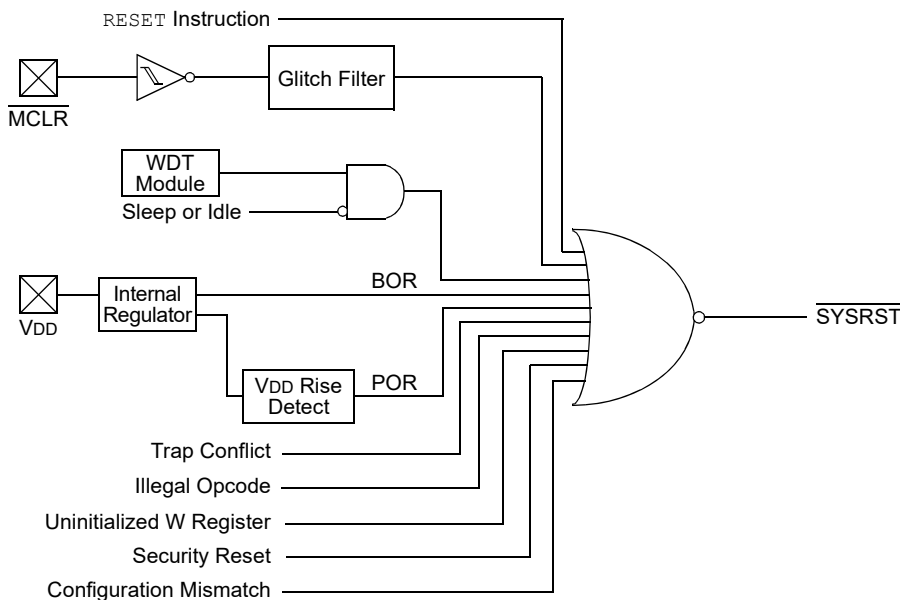
Note: This data sheet summarizes the features of the dsPIC33CK512MPT608 family of devices. It is not intended to be a comprehensive reference source. To complement the information in this data sheet, refer to “Reset” (www.microchip.com/DS70602) in the “dsPIC33/PIC24 Family Reference Manual”.

The Reset module combines all Reset sources and controls the device Reset Signal, $\overline{\text{SYSRST}}$. The following is a list of device Reset sources:

- POR: Power-on Reset
- BOR: Brown-out Reset
- MCLR: Master Clear Pin Reset
- SWR: RESET Instruction
- WDTO: Watchdog Timer Time-out Reset
- CM: Configuration Mismatch Reset
- TRAPR: Trap Conflict Reset
- IOPUWR: Illegal Condition Device Reset
 - Illegal Opcode Reset
 - Uninitialized W Register Reset
 - Security Reset

A simplified block diagram of the Reset module is shown in Figure 7-1.

Figure 7-1. Reset System Block Diagram



Any active source of Reset will make the $\overline{\text{SYSRST}}$ signal active. On system Reset, some of the registers associated with the CPU and peripherals are forced to a known Reset state, and some are unaffected.

Note: Refer to the specific peripheral section or 4. Memory Organization of this data sheet for register Reset states.

All types of device Reset set a corresponding status bit in the RCON register to indicate the type of Reset.

A POR clears all the bits, except for the BOR and POR bits ($\text{RCON}[1:0]$) that are set. The user application can set or clear any bit, at any time, during code execution. The RCON bits only serve as status bits. Setting a particular Reset status bit in software does not cause a device Reset to occur.

The RCON register also has other bits associated with the Watchdog Timer and device Power-Saving states. The function of these bits is discussed in other sections of this manual.

Note: The status bits in the RCON register should be cleared after they are read so that the next RCON register value after a device Reset is meaningful.

For all Resets, the default clock source is determined by the FNOSC[2:0] bits in the FOSCSEL Configuration register. The value of the FNOSCx bits is loaded into the NOSC[2:0] (OSCCON[10:8]) bits on Reset, which in turn, initializes the system clock.

Note: SWR, WDTO, CM, TRAPR and IOPUWR Reset sources have no effect on the Secure Subsystem.

7.1 Reset Resources

Many useful resources are provided on the main product page of the Microchip website for the devices listed in this data sheet. This product page contains the latest updates and additional information.

7.1.1 Key Resources

- “Reset” (www.microchip.com/DS70602) in the “dsPIC33/PIC24 Family Reference Manual”
- Code Samples
- Application Notes
- Software Libraries
- Webinars
- All Related “dsPIC33/PIC24 Family Reference Manual” sections
- Development Tools

7.1.2 Reset Control Register

Name: RCON⁽¹⁾

Offset: 0xF80

Notes:

1. All of the Reset status bits can be set or cleared in software. Setting one of these bits in software does not cause a device Reset.
2. If the FWDTEN Configuration bit is '1' (unprogrammed), the WDT is always enabled, regardless of the SWDTEN bit setting.

Bit	15	14	13	12	11	10	9	8
	TRAPR	IOPUWR					CM	VREGS
Access	R/W	R/W					R/W	R/W
Reset	0	0					0	0

Bit	7	6	5	4	3	2	1	0
	EXTR	SWR	SWDTEN	WDTO	SLEEP	IDLE	BOR	POR
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	1	1

Bit 15 – TRAPR Trap Reset Flag bit

Value	Description
1	A Trap Conflict Reset has occurred
0	A Trap Conflict Reset has not occurred

Bit 14 – IOPUWR Illegal Opcode or Uninitialized W Register Access Reset Flag bit

Value	Description
1	An Illegal Opcode, an Illegal Address mode or Uninitialized W Register used as an Address Pointer caused a Reset
0	An Illegal Opcode or Uninitialized W Register Reset has not occurred

Bit 9 – CM Configuration Mismatch Flag bit

Value	Description
1	A Configuration Mismatch Reset has occurred
0	A Configuration Mismatch Reset has not occurred

Bit 8 – VREGS Voltage Regulator Standby During Sleep bit

Value	Description
1	Voltage regulator is active during Sleep
0	Voltage regulator goes into Standby mode during Sleep

Bit 7 – EXTR External Reset (MCLR) Pin bit

Value	Description
1	A Master Clear (pin) Reset has occurred
0	A Master Clear (pin) Reset has not occurred

Bit 6 – SWR Software RESET (Instruction) Flag bit

Value	Description
1	A RESET instruction has been executed
0	A RESET instruction has not been executed

Bit 5 – SWDTEN Software Enable/Disable of WDT bit⁽²⁾

Value	Description
1	WDT is enabled
0	WDT is disabled

Bit 4 – WDTO Watchdog Timer Time-out Flag bit

Value	Description
1	WDT time-out has occurred
0	WDT time-out has not occurred

Bit 3 – SLEEP Wake-up from Sleep Flag bit

Value	Description
1	Device has been in Sleep mode
0	Device has not been in Sleep mode

Bit 2 – IDLE Wake-up from Idle Flag bit

Value	Description
1	Device has been in Idle mode
0	Device has not been in Idle mode

Bit 1 – BOR Brown-out Reset Flag bit

Value	Description
1	A Brown-out Reset has occurred
0	A Brown-out Reset has not occurred

Bit 0 – POR Power-on Reset Flag bit

Value	Description
1	A Power-on Reset has occurred
0	A Power-on Reset has not occurred

8. Interrupt Controller

Note: This data sheet summarizes the features of the dsPIC33CK512MPT608 family of devices. It is not intended to be a comprehensive reference source. To complement the information in this data sheet, refer to “**Interrupts**” (www.microchip.com/DS70000600) in the “*dsPIC33/PIC24 Family Reference Manual*”.

The dsPIC33CK512MPT608 family interrupt controller reduces the numerous peripheral interrupt request signals to a single interrupt request signal to the dsPIC33CK512MPT608 family CPU.

The interrupt controller has the following features:

- Six Processor Exceptions and Software Traps
- Seven User-Selectable Priority Levels
- Interrupt Vector Table (IVT) with a Unique Vector for each Interrupt or Exception Source
- Fixed Priority within a Specified User Priority Level
- Fixed Interrupt Entry and Return Latencies
- Alternate Interrupt Vector Table (AIVT) for Debug Support

8.1 Interrupt Vector Table

The dsPIC33CK512MPT608 family Interrupt Vector Table (IVT), shown in [Figure 8-1](#), resides in program memory, starting at location, 000004h. The IVT contains six non-maskable trap vectors and up to 246 sources of interrupts. In general, each interrupt source has its own vector. Each interrupt vector contains a 24-bit wide address. The value programmed into each interrupt vector location is the starting address of the associated Interrupt Service Routine (ISR).

Figure 8-1. dsPIC33CK512MPT608 Family Interrupt Vector Table (IVT)⁽¹⁾

Decreasing Natural Order Priority ↓ IVT ↓ ↓	Reset – GOTO Instruction	0x000000	See Note 2 for Interrupt Vector Details
	Reset – GOTO Address	0x000002	
	Oscillator Fail Trap Vector	0x000004	
	Address Error Trap Vector	0x000006	
	Generic Hard Trap Vector	0x000008	
	Stack Error Trap Vector	0x00000A	
	Math Error Trap Vector	0x00000C	
	Reserved	0x00000E	
	Generic Soft Trap Vector	0x000010	
	Reserved	0x000012	
	Interrupt Vector 0	0x000014	
	Interrupt Vector 1	0x000016	
	:	:	
	:	:	
	:	:	
	Interrupt Vector 52	0x00007C	
	Interrupt Vector 53	0x00007E	
	Interrupt Vector 54	0x000080	
	:	:	
	:	:	
	:	:	
	Interrupt Vector 116	0x0000FC	
	Interrupt Vector 117	0x0000FE	
	Interrupt Vector 118	0x000100	
	Interrupt Vector 119	0x000102	
	Interrupt Vector 120	0x000104	
	:	:	
	:	:	
	:	:	
	Interrupt Vector 244	0x0001FC	
Interrupt Vector 245	0x0001FE		
START OF CODE	0x000200		

Notes:

1. In Dual Partition modes, each partition has a dedicated Interrupt Vector Table.
2. See Trap Vector Details.

Interrupt vectors are prioritized in terms of their natural priority. This priority is linked to their position in the vector table. Lower addresses generally have a higher natural priority. For example, the interrupt associated with Vector 0 takes priority over interrupts at any other vector address.

8.2 Alternate Interrupt Vector Table

The Alternate Interrupt Vector Table (AIVT), shown in [Figure 8-2](#), is available only when the Boot Segment (BS) is defined and the AIVT has been enabled. To enable the Alternate Interrupt Vector Table, the Configuration bit, AIVTDIS in the FSEC register, must be programmed and the AIVTEN bit must be set (INTCON2[8] = 1). When the AIVT is enabled, all interrupt and exception processes use the alternate vectors instead of the default vectors. The

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AIVT begins at the start of the last page of the Boot Segment, defined by BSLIM[12:0]. The second half of the page is no longer usable space. The Boot Segment must be at least two pages to enable the AIVT.

Note: Although the Boot Segment must be enabled in order to enable the AIVT, application code does not need to be present inside of the Boot Segment. The AIVT (and IVT) will inherit the Boot Segment code protection.

Figure 8-2. dsPIC33CK512MPT608 Alternate Interrupt Vector Table⁽²⁾

Decreasing Natural Order Priority AIVT	Reserved	$BSLIM<12:0>^{(1)} + 0x000000$
	Reserved	$BSLIM<12:0>^{(1)} + 0x000002$
	Oscillator Fail Trap Vector	$BSLIM<12:0>^{(1)} + 0x000004$
	Address Error Trap Vector	$BSLIM<12:0>^{(1)} + 0x000006$
	Generic Hard Trap Vector	$BSLIM<12:0>^{(1)} + 0x000008$
	Stack Error Trap Vector	$BSLIM<12:0>^{(1)} + 0x00000A$
	Math Error Trap Vector	$BSLIM<12:0>^{(1)} + 0x00000C$
	Reserved	$BSLIM<12:0>^{(1)} + 0x00000E$
	Generic Soft Trap Vector	$BSLIM<12:0>^{(1)} + 0x000010$
	Reserved	$BSLIM<12:0>^{(1)} + 0x000012$
	Interrupt Vector 0	$BSLIM<12:0>^{(1)} + 0x000014$
	Interrupt Vector 1	$BSLIM<12:0>^{(1)} + 0x000016$
	:	:
	:	:
	:	:
	Interrupt Vector 52	$BSLIM<12:0>^{(1)} + 0x00007C$
	Interrupt Vector 53	$BSLIM<12:0>^{(1)} + 0x00007E$
	Interrupt Vector 54	$BSLIM<12:0>^{(1)} + 0x000080$
	:	:
	:	:
	:	:
	Interrupt Vector 116	$BSLIM<12:0>^{(1)} + 0x0000FC$
	Interrupt Vector 117	$BSLIM<12:0>^{(1)} + 0x0000FE$
	Interrupt Vector 118	$BSLIM<12:0>^{(1)} + 0x000100$
	Interrupt Vector 119	$BSLIM<12:0>^{(1)} + 0x000102$
	Interrupt Vector 120	$BSLIM<12:0>^{(1)} + 0x000104$
	:	:
	:	:
	:	:
	Interrupt Vector 244	$BSLIM<12:0>^{(1)} + 0x0001FC$
	Interrupt Vector 245	$BSLIM<12:0>^{(1)} + 0x0001FE$

See **Note 3** for Interrupt Vector Details

Notes:

1. The address depends on the size of the Boot Segment defined by BSLIM[12:0]: $[(BSLIM[12:0] - 1) \times 0x800] + \text{Offset}$.
2. In Dual Partition modes, each partition has a dedicated Alternate Interrupt Vector Table (if enabled).
3. See [Table 8-1](#).

The AIVT supports debugging by providing a means to switch between an application and a support environment without requiring the interrupt vectors to be reprogrammed. This feature also enables switching between applications for evaluation of different software algorithms at run time.

8.3 Reset Sequence

A device Reset is not a true exception because the interrupt controller is not involved in the Reset process. The dsPIC33CK512MPT608 family devices clear their registers in response to a Reset, which forces the PC to zero. The device then begins program execution at location, 0x000000. A GOTO instruction at the Reset address can redirect program execution to the appropriate start-up routine.

Note: Any unimplemented or unused vector locations in the IVT should be programmed with the address of a default interrupt handler routine that contains a RESET instruction.

Table 8-1. Trap Vector Details

Trap Description	MPLAB [®] XC16 Trap ISR Name	Vector #	IVT Address	Trap Bit Location			Priority
				Interrupt Flag	Type	Enable	
Oscillator Failure	_OscillatorFail	0	0x000004	INTCON1[1]	—	—	15
Address Error	_AddressError	1	0x000006	INTCON1[3]	—	—	14
ECC Double-Bit Error	_HardTrapError	2	0x000008	INTCON4[1]	—	—	13
Software Generated Trap	_HardTrapError	2	0x000008	INTCON4[0]	—	INTCON2[13]	13
Stack Error	_StackError	3	0x00000A	INTCON1[2]	—	—	12
Overflow Accumulator A	_MathError	4	0x00000C	INTCON1[4]	INTCON1[14]	INTCON1[10]	11
Overflow Accumulator B	_MathError	4	0x00000C	INTCON1[4]	INTCON1[13]	INTCON1[9]	11
Catastrophic Overflow Accumulator A	_MathError	4	0x00000C	INTCON1[4]	INTCON1[12]	INTCON1[8]	11
Catastrophic Overflow Accumulator B	_MathError	4	0x00000C	INTCON1[4]	INTCON1[11]	INTCON1[8]	11
Shift Accumulator Error	_MathError	4	0x00000C	INTCON1[4]	INTCON1[7]	INTCON1[8]	11
Divide-by-Zero Error	_MathError	4	0x00000C	INTCON1[4]	INTCON1[6]	INTCON1[8]	11
Reserved	_Reserved	5	0x00000E	—	—	—	—
Deadman Timer	_SoftTrapError	6	0x000010	INTCON3[15]	—	—	9
CAN Address Error	_SoftTrapError	6	0x000010	INTCON3[9]	—	—	—
NVM Address Error	_SoftTrapError	6	0x000010	INTCON3[8]	—	—	9
CAN2 Address Error	_SoftTrapError	6	0x000010	INTCON3[6]	—	—	9
DMA Address Error	_SoftTrapError	6	0x000010	INTCON3[5]	—	—	9
D0 Stack Overflow	_SoftTrapError	6	0x000010	INTCON3[4]	—	—	9
APLL Loss of Lock	_SoftTrapError	6	0x000010	INTCON3[0]	—	—	9
Reserved	Reserved	7	0x000012	—	—	—	—

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Table 8-2. Interrupt Vector Details

Interrupt Source	MPLAB [®] XC16 ISR Name	Vector #	IRQ #	IVT Address	Interrupt Bit Location		
					Flag	Enable	Priority
External Interrupt 0	_INT0Interrupt	8	0	0x000014	IFS0[0]	IEC0[0]	IPC0[2:0]
Timer1	_T1Interrupt	9	1	0x000016	IFS0[1]	IEC0[1]	IPC0[6:4]
Change Notice Interrupt A	_CNAInterrupt	10	2	0x000018	IFS0[2]	IEC0[2]	IPC0[10:8]
Change Notice Interrupt B	_CNBInterrupt	11	3	0x00001A	IFS0[3]	IEC0[3]	IPC0[14:12]
DMA Channel 0	_DMA0Interrupt	12	4	0x00001C	IFS0[4]	IEC0[4]	IPC1[2:0]
Change Notice Interrupt F	_CNFInterrupt	13	5	0x00001E	IFS0[5]	IEC0[5]	IPC1[4:6]
Input Capture/ Output Compare 1	_CCP1Interrupt	14	6	0x000020	IFS0[6]	IEC0[6]	IPC1[10:8]
CCP1 Timer	_CCT1Interrupt	15	7	0x000022	IFS0[7]	IEC0[7]	IPC1[14:12]
DMA Channel 1	_DMA1Interrupt	16	8	0x000024	IFS0[8]	IEC0[8]	IPC2[2:0]
SPI1 Receiver	_SPI1RXInterrupt	17	9	0x000026	IFS0[9]	IEC0[9]	IPC2[6:4]
SPI1 Transmitter	_SPI1TXInterrupt	18	10	0x000028	IFS0[10]	IEC0[10]	IPC2[10:8]
UART1 Receiver	_U1RXInterrupt	19	11	0x00002A	IFS0[11]	IEC0[11]	IPC2[14:12]
UART1 Transmitter	_U1TXInterrupt	20	12	0x00002C	IFS0[12]	IEC0[12]	IPC3[2:0]
ECC Single-Bit Error	_ECCSBEInterrupt	21	13	0x00002E	IFS0[13]	IEC0[13]	IPC3[6:4]
NVM Write Complete	_NVMInterrupt	22	14	0x000030	IFS0[14]	IEC0[14]	IPC3[10:8]
External Interrupt 1	_INT1Interrupt	23	15	0x000032	IFS0[15]	IEC0[15]	IPC3[14:12]
I2C1 Client Event	_SI2C1Interrupt	24	16	0x000034	IFS1[0]	IEC1[0]	IPC4[2:0]
I2C1 Host Event	_MI2C1Interrupt	25	17	0x000036	IFS1[1]	IEC1[1]	IPC4[6:4]
DMA Channel 2	_DMA2Interrupt	26	18	0x000038	IFS1[2]	IEC1[2]	IPC4[10:8]
Change Notice Interrupt C ⁽¹⁾	_CNCInterrupt	27	19	0x00003A	IFS1[3]	IEC1[3]	IPC4[14:12]
External Interrupt 2	_INT2Interrupt	28	20	0x00003C	IFS1[4]	IEC1[4]	IPC5[2:0]
DMA Channel 3	_DMA3Interrupt	29	21	0x00003E	IFS1[5]	IEC1[5]	IPC5[6:4]
DMA Channel 4	_DMA4Interrupt	30	22	0x000040	IFS1[6]	IEC1[6]	IPC5[10:8]
Input Capture/ Output Compare 2	_CCP2Interrupt	31	23	0x000042	IFS1[7]	IEC1[7]	IPC5[14:12]
CCP2 Timer	_CCT2Interrupt	32	24	0x000044	IFS1[8]	IEC1[8]	IPC6[2:0]
CAN1 Combined Error	_CAN1Interrupt	33	25	0x000046	IFS1[9]	IEC1[9]	IPC6[6:4]
External Interrupt 3	_INT3Interrupt	34	26	0x000048	IFS1[10]	IEC1[10]	IPC6[10:8]

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.....continued							
Interrupt Source	MPLAB [®] XC16 ISR Name	Vector #	IRQ #	IVT Address	Interrupt Bit Location		
					Flag	Enable	Priority
U2RX – UART2 Receiver	_U2RXInterrupt	35	27	0x00004A	IFS1[11]	IEC1[11]	IPC6[14:12]
U2TX – UART2 Transmitter	_U2TXInterrupt	36	28	0x00004C	IFS1[12]	IEC1[12]	IPC7[2:0]
SPI2 Receiver	_SPI2RXInterrupt	37	29	0x00004E	IFS1[13]	IEC1[13]	IPC7[6:4]
SPI2 Transmitter	_SPI2TXInterrupt	38	30	0x000050	IFS1[14]	IEC1[14]	IPC7[10:8]
CAN1 RX Data Ready ⁽²⁾	_C1RXInterrupt	39	31	0x000052	IFS1[15]	IEC1[15]	IPC7[14:12]
CAN2 RX Data Ready ⁽²⁾	_C2RXInterrupt	40	32	0x000054	IFS2[0]	IEC2[0]	IPC8[2:0]
CAN2 Combined Error	_CAN2Interrupt	41	33	0x000056	IFS2[1]	IEC2[1]	IPC8[6:4]
DMA Channel 5	_DMA5Interrupt	42	34	0x000058	IFS2[2]	IEC2[2]	IPC8[10:8]
Input Capture/ Output Compare 3	_CCP3Interrupt	43	35	0x00005A	IFS2[3]	IEC2[3]	IPC8[14:12]
CCP3 Timer	_CCT3Interrupt	44	36	0x00005C	IFS2[4]	IEC2[4]	IPC9[2:0]
I2C2 Client Event	_SI2C2Interrupt	45	37	0x00005E	IFS2[5]	IEC2[5]	IPC9[6:4]
I2C2 Host Event	_MI2C2Interrupt	46	38	0x000060	IFS2[6]	IEC2[6]	IPC9[10:8]
Reserved	Reserved	47	39	0x000062	—	—	—
Input Capture/ Output Compare 4	_CCP4Interrupt	48	40	0x000064	IFS2[8]	IEC2[8]	IPC10[2:0]
CCP4 Timer	_CCT4Interrupt	49	41	0x000066	IFS2[9]	IEC2[9]	IPC10[6:4]
Reserved	Reserved	50	42	0x000068	—	—	—
Input Capture/ Output Compare 5	_CCP5Interrupt	51	43	0x00006A	IFS2[11]	IEC2[11]	IPC10[14:12]
CCP5 Timer	_CCT5Interrupt	52	44	0x00006C	IFS2[12]	IEC2[12]	IPC11[2:0]
Deadman Timer	_DMTInterrupt	53	45	0x00006E	IFS2[13]	IEC2[13]	IPC11[6:4]
Input Capture/ Output Compare 6	_CCP6Interrupt	54	46	0x000070	IFS2[14]	IEC2[14]	IPC11[10:8]
CCP6 Timer	_CCT6Interrupt	55	47	0x000072	IFS2[15]	IEC2[15]	IPC11[14:12]
QE1 1 Position Counter Compare	_QE1Interrupt	56	48	0x000074	IFS3[0]	IEC3[0]	IPC12[2:0]
UART1 Error	_U1EInterrupt	57	49	0x000076	IFS3[1]	IEC3[1]	IPC12[6:4]
UART2 Error	_U2EInterrupt	58	50	0x000078	IFS3[2]	IEC3[2]	IPC12[10:8]
CRC Generator	_CRCInterrupt	59	51	0x00007A	IFS3[3]	IEC3[3]	IPC12[14:12]
CAN1 TX Data Request ⁽²⁾	_C1TXInterrupt	60	52	0x00007C	IFS3[4]	IEC3[4]	IPC13[2:0]
CAN2 TX Data Request ⁽²⁾	_C2TXInterrupt	61	53	0x00007E	IFS3[5]	IEC3[5]	IPC13[6:4]

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Interrupt Controller

.....continued							
Interrupt Source	MPLAB® XC16 ISR Name	Vector #	IRQ #	IVT Address	Interrupt Bit Location		
					Flag	Enable	Priority
QE12 Position Counter Compare	_QE12Interrupt	62	54	0x000080	IFS3[6]	IEC3[6]	IPC13[10:8]
Reserved	Reserved	63	55	0x000082	—	—	—
UART3 Error	_U3EInterrupt	64	56	0x000084	IFS3[8]	IEC3[8]	IPC14[2:0]
UART3 Receiver	_U3RXInterrupt	65	57	0x000086	IFS3[9]	IEC3[9]	IPC14[6:4]
UART3 Transmitter	_U3TXInterrupt	66	58	0x000088	IFS3[10]	IEC3[10]	IPC14[10:8]
SPI3 Receiver	_SPI3RXInterrupt	67	59	0x00008A	IFS3[11]	IEC3[11]	IPC14[14:12]
SPI3 Transmitter	_SPI3TXInterrupt	68	60	0x00008C	IFS3[12]	IEC3[12]	IPC15[2:0]
In-Circuit Debugger	_ICDInterrupt	69	61	0x00008E	IFS3[13]	IEC3[13]	IPC15[6:4]
PTG Step	_PTGSTePInterrupt	71	63	0x000092	IFS3[15]	IEC3[15]	IPC15[14:12]
I2C1 Bus Collision	_I2C1BCInterrupt	72	64	0x000094	IFS4[0]	IEC4[0]	IPC16[2:0]
I2C2 Bus Collision	_I2C2BCInterrupt	73	65	0x000096	IFS4[1]	IEC4[1]	IPC16[6:4]
QE13 Position Counter Compare	_QE13Interrupt	74	66	0x000098	IFS4[2]	IEC4[2]	IPC16[10:8]
PWM Generator 1	_PWM1Interrupt	75	67	0x00009A	IFS4[3]	IEC4[3]	IPC16[14:12]
PWM Generator 2	_PWM2Interrupt	76	68	0x00009C	IFS4[4]	IEC4[4]	IPC17[2:0]
PWM Generator 3	_PWM3Interrupt	77	69	0x00009E	IFS4[5]	IEC4[5]	IPC17[6:4]
PWM Generator 4	_PWM4Interrupt	78	70	0x0000A0	IFS4[6]	IEC4[6]	IPC17[10:8]
Reserved	Reserved	79	71	0x0000A2	—	—	—
PWM Generator 6	_PWM6Interrupt	80	72	0x0000A4	IFS4[8]	IEC4[8]	IPC18[2:0]
Reserved	Reserved	81	73	0x0000A6	—	—	—
PWM Generator 8	_PWM8Interrupt	82	74	0x0000A8	IFS4[10]	IEC4[10]	IPC18[10:8]
Change Notice D ⁽¹⁾	_CNDInterrupt	83	75	0x0000AA	IFS4[11]	IEC4[11]	IPC18[14:12]
Change Notice E ⁽¹⁾	CNEInterrupt	84	76	0x0000AC	IFS4[12]	IEC4[12]	IPC19[2:0]
Comparator 1	_CMP1Interrupt	85	77	0x0000AE	IFS4[13]	IEC4[13]	IPC19[6:4]
Comparator 2	_CMP2Interrupt	86	78	0x0000B0	IFS4[14]	IEC4[14]	IPC19[10:8]
Comparator 3	_CMP3Interrupt	87	79	0x0000B2	IFS4[15]	IEC4[15]	IPC19[14:2]
Comparator 4	_CMP4Interrupt	88	80	0x0000B4	IFS5[0]	IEC5[0]	IPC20[2:0]
PTG Watchdog Timer Time-out	_PTGWDTInterrupt	89	81	0x0000B6	IFS5[1]	IEC5[1]	IPC20[6:4]
PTG Trigger 0	_PTG0Interrupt	90	82	0x0000B8	IFS5[2]	IEC5[2]	IPC20[10:8]
PTG Trigger 1	_PTG1Interrupt	91	83	0x0000BA	IFS5[3]	IEC5[3]	IPC20[14:12]
PTG Trigger 2	_PTG2Interrupt	92	84	0x0000BC	IFS5[4]	IEC5[4]	IPC21[2:0]
PTG Trigger 3	_PTG3Interrupt	93	85	0x0000BE	IFS5[5]	IEC5[6]	IPC21[6:4]
SENT1 TX/RX	_SENT1Interrupt	94	86	0x0000C0	IFS5[6]	IEC5[6]	IPC21[10:8]

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Interrupt Controller

.....continued							
Interrupt Source	MPLAB® XC16 ISR Name	Vector #	IRQ #	IVT Address	Interrupt Bit Location		
					Flag	Enable	Priority
SENT1 Error	_SENT1Interrupt	95	87	0x0000C2	IFS5[7]	IEC5[7]	IPC21[14:12]
SENT2 TX/RX	_SENT2Interrupt	96	88	0x0000C4	IFS[8]	IEC5[8]	IPC22[2:0]
SENT2 Error	_SENT2EInterrupt	97	89	0x0000C6	IFS[9]	IEC5[9]	IPC22[6:4]
ADC Global Interrupt	_ADCInterrupt	98	90	0x0000C8	IFS5[10]	IEC5[10]	IPC22[10:8]
ADC AN0 Interrupt	_ADCAN0Interrupt	99	91	0x0000CA	IFS5[11]	IEC5[11]	IPC22[14:12]
ADC AN1 Interrupt	_ADCAN1Interrupt	100	92	0x0000CC	IFS5[12]	IEC5[12]	IPC23[2:0]
ADC AN2 Interrupt	_ADCAN2Interrupt	101	93	0x0000CE	IFS5[13]	IEC5[13]	IPC23[6:4]
ADC AN3 Interrupt	_ADCAN3Interrupt	102	94	0x0000D0	IFS5[14]	IEC5[14]	IPC23[10:8]
ADC AN4 Interrupt	_ADCAN4Interrupt	103	95	0x0000D2	IFS5[15]	IEC5[15]	IPC23[14:12]
ADC AN5 Interrupt	_ADCAN5Interrupt	104	96	0x0000D4	IFS6[0]	IEC6[0]	IPC24[2:0]
ADC AN6 Interrupt	_ADCAN6Interrupt	105	97	0x0000D6	IFS6[1]	IEC6[1]	IPC24[6:4]
ADC AN7 Interrupt ⁽³⁾	_ADCAN7Interrupt	106	98	0x0000D8	IFS6[2]	IEC6[2]	IPC24[10:8]
ADC AN8 Interrupt	_ADCAN8Interrupt	107	99	0x0000DA	IFS6[3]	IEC6[3]	IPC24[14:12]
ADC AN9 Interrupt	_ADCAN9Interrupt	108	100	0x0000DC	IFS6[4]	IEC6[4]	IPC25[2:0]
ADC AN10 Interrupt	_ADCAN10Interrupt	109	101	0x0000DE	IFS6[5]	IEC6[5]	IPC25[6:4]
ADC AN11 Interrupt	_ADCAN11Interrupt	110	102	0x0000E0	IFS6[6]	IEC6[6]	IPC25[10:8]
ADC AN12 Interrupt ⁽³⁾	_ADCAN12Interrupt	111	103	0x0000E2	IFS6[7]	IEC6[7]	IPC25[14:12]
ADC AN13 Interrupt ⁽³⁾	_ADCAN13Interrupt	112	104	0x0000E4	IFS6[8]	IEC6[8]	IPC26[2:0]
ADC AN14 Interrupt ⁽³⁾	_ADCAN14Interrupt	113	105	0x0000E6	IFS6[9]	IEC6[9]	IPC26[6:4]
ADC AN15 Interrupt ⁽³⁾	_ADCAN15Interrupt	114	106	0x0000E8	IFS6[10]	IEC6[10]	IPC26[10:8]
ADC AN16 Interrupt	_ADCAN16Interrupt	115	107	0x0000EA	IFS6[11]	IEC6[11]	IPC26[14:12]
ADC AN17 Interrupt	_ADCAN17Interrupt	116	108	0x0000EC	IFS6[12]	IEC6[12]	IPC27[2:0]
ADC AN18 Interrupt ⁽³⁾	_ADCAN18Interrupt	117	109	0x0000EE	IFS6[13]	IEC6[13]	IPC27[6:4]
ADC AN19 Interrupt ⁽³⁾	_ADCAN19Interrupt	118	110	0x0000F0	IFS6[14]	IEC6[14]	IPC27[10:8]
ADC AN20 Interrupt ⁽³⁾	_ADCAN20Interrupt	119	111	0x0000F2	IFS6[15]	IEC6[15]	IPC27[14:12]
ADC AN21 Interrupt ⁽³⁾	_ADCAN21Interrupt	120	112	0x0000F4	IFS7[0]	IEC7[0]	IPC28[2:0]
ADC AN22 Interrupt ⁽³⁾	_ADCAN22Interrupt	121	113	0x0000F6	IFS7[1]	IEC7[1]	IPC28[6:4]
ADC AN23 Interrupt ⁽³⁾	_ADCAN23Interrupt	122	114	0x0000F8	IFS7[2]	IEC7[2]	IPC28[10:8]

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.....continued							
Interrupt Source	MPLAB [®] XC16 ISR Name	Vector #	IRQ #	IVT Address	Interrupt Bit Location		
					Flag	Enable	Priority
ADC Fault	_ADFLTInterrupt	123	115	0x0000FA	IFS7[3]	IEC7[3]	IPC28[14:12]
ADC Digital Comparator 0	_ADCMPOInterrupt	124	116	0x0000FC	IFS7[4]	IEC7[4]	IPC29[2:0]
ADC Digital Comparator 1	_ADCMP1Interrupt	125	117	0x0000FE	IFS7[5]	IEC7[5]	IPC29[6:4]
ADC Digital Comparator 2	_ADCMP2Interrupt	126	118	0x000100	IFS7[6]	IEC7[6]	IPC29[10:8]
ADC Digital Comparator 3	_ADCMP3Interrupt	127	119	0x000102	IFS7[7]	IEC7[7]	IPC29[14:12]
ADC Oversample Filter 0	_ADFLTR0Interrupt	128	120	0x000104	IFS7[8]	IEC7[8]	IPC30[2:0]
ADC Oversample Filter 1	_ADFLTR1Interrupt	129	121	0x000106	IFS7[9]	IEC7[9]	IPC30[6:4]
ADC Oversample Filter 2	_ADFLTR2Interrupt	130	122	0x000108	IFS7[10]	IEC7[10]	IPC30[10:8]
ADC Oversample Filter 3	_ADFLTR3Interrupt	131	123	0x00010A	IFS7[11]	IEC7[11]	IPC30[14:12]
CLC1 Positive Edge	_CLC1PInterrupt	132	124	0x00010C	IFS7[12]	IEC7[12]	IPC31[2:0]
CLC2 Positive Edge	_CLC2PInterrupt	133	125	0x00010E	IFS7[13]	IEC7[13]	IPC31[6:4]
SPI1 Error	_SPI1Interrupt	134	126	0x000110	IFS7[14]	IEC7[14]	IPC31[10:8]
SPI2 Error	_SPI2Interrupt	135	127	0x000112	IFS7[15]	IEC7[15]	IPC31[14:12]
SPI3 Error	_SPI3Interrupt	136	128	0x000114	IFS8[0]	IEC8[0]	IPC32[2:0]
Reserved	Reserved	137-149	129-141	0x000116-0x00012E	—	—	—
I2C3 Client Event	_SI2C3Interrupt	150	142	0x000130	IFS8[14]	IEC8[14]	IPC35[10:8]
I2C3 Host Event	_MI2C3Interrupt	151	143	0x000132	IFS8[15]	IEC8[15]	IPC35[14:12]
I2C3 Bus Collision	_I2C3BInterrupt	152	144	0x000134	IFS9[0]	IEC9[0]	IPC36[2:0]
ADC AN27 Interrupt	_ADCAN27Interrupt	153	145	0x000136	IFS9[1]	IEC9[1]	IPC36[6:4]
ADC AN28 Interrupt	_ADCAN28Interrupt	154	146	0x000138	IFS9[2]	IEC9[2]	IPC36[10:8]
ADC AN29 Interrupt	_ADCAN29Interrupt	155	147	0x00013A	IFS9[3]	IEC9[3]	IPC36[14:12]
ADC AN30 Interrupt	_ADCAN30Interrupt	156	148	0x00013C	IFS9[4]	IEC9[4]	IPC37[2:0]
Input Capture/ Output Compare 7	_CCP7Interrupt	157	149	0x00013E	IFS9[5]	IEC9[5]	IPC37[6:4]
CCP7 Timer	_CCT7Interrupt	158	150	0x000140	IFS9[6]	IEC9[6]	IPC37[10:8]
ADC AN26 Interrupt	_ADCAN26Interrupt	159	151	0x000142	IFS9[7]	IEC9[7]	IPC37[14:12]
Input Capture/ Output Compare 8	_CCP8Interrupt	160	152	0x000144	IFS9[8]	IEC9[8]	IPC38[2:0]
CCP8 Timer	_CCT8Interrupt	161	153	0x000146	IFS9[9]	IEC9[9]	IPC38[6:4]
DMA Channel 6	_DMA6Interrupt	162	154	0x000148	IFS9[10]	IEC9[10]	IPC38[10:8]
DMA Channel 7	_DMA7Interrupt	163	155	0x00014A	IFS9[11]	IEC9[11]	IPC38[14:12]

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Interrupt Controller

.....continued

Interrupt Source	MPLAB [®] XC16 ISR Name	Vector #	IRQ #	IVT Address	Interrupt Bit Location		
					Flag	Enable	Priority
Reserved	Reserved	164-176	156-168	0x00014C-0x000164	—	—	—
PEVTA – PWM Event A	_PEVTAInterrupt	177	169	0x000166	IFS10[9]	IEC10[9]	IPC42[6:4]
PEVTB – PWM Event B	_PEVTBInterrupt	178	170	0x000168	IFS10[10]	IEC10[10]	IPC42[10:8]
PEVTC – PWM Event C	_PEVTCInterrupt	179	171	0x00016A	IFS10[11]	IEC10[11]	IPC42[14:12]
PEVTD – PWM Event D	_PEVTDInterrupt	180	172	0x00016C	IFS10[12]	IEC10[12]	IPC43[2:0]
PEVTE – PWM Event E	_PEVTEInterrupt	181	173	0x00016E	IFS10[13]	IEC10[13]	IPC43[6:4]
PEVTF – PWM Event F	_PEVTFInterrupt	182	174	0x000170	IFS10[14]	IEC10[14]	IPC43[10:8]
CLC3 Positive Edge	_CLC3PInterrupt	183	175	0x000172	IFS10[15]	IEC10[15]	IPC43[14:12]
CLC4 Positive Edge	_CLC4PInterrupt	184	176	0x000174	IFS11[0]	IEC11[0]	IPC44[2:0]
CLC1 Negative Edge	_CLC1NInterrupt	185	177	0x000176	IFS11[1]	IEC11[1]	IPC44[6:4]
CLC2 Negative Edge	_CLC2NInterrupt	186	178	0x000178	IFS11[2]	IEC11[2]	IPC44[10:8]
CLC3 Negative Edge	_CLC3NInterrupt	187	179	0x00017A	IFS11[3]	IEC11[3]	IPC44[14:12]
CLC4 Negative Edge	_CLC4NInterrupt	188	180	0x00017C	IFS11[4]	IEC11[4]	IPC45[2:0]
Input Capture/ Output Compare 9	_CCP9Interrupt	189	181	0x00017E	IFS11[5]	IEC11[5]	IPC45[6:4]
CCP9 Timer	_CCT9Interrupt	190	182	0x000180	IFS11[6]	IEC11[6]	IPC45[10:8]
Reserved	Reserved	191-194	183-186	0x000182-0x000188	—	—	—
Comparator 5	_CMP5Interrupt	195	187	0x00018A	IFS11[11]	IEC11[11]	IPC46[14:12]
Comparator 6	_CMP6Interrupt	196	188	0x00018C	IFS11[12]	IEC11[12]	IPC47[2:0]
UART1 Event	_U1EVTInterrupt	197	189	0x00018E	IFS11[13]	IEC11[13]	IPC47[6:4]
UART2 Event	_U2EVTInterrupt	198	190	0x000190	IFS11[14]	IEC11[14]	IPC47[10:8]
UART3 Event	_U3EVTInterrupt	199	191	0x000192	IFS11[15]	IEC11[15]	IPC47[14:12]
AN24 Done	_ADCAN24Interrupt	200	192	0x000194	IFS12[0]	IEC12[0]	IPC48[2:0]
AN25 Done	_ADCAN25Interrupt	201	193	0x000196	IFS12[1]	IEC12[1]	IPC48[6:4]
Reserved	Reserved	202-255	194-247	0x000198-0x0001FE	—	—	—

Note:

1. Availability dependent on supported I/O ports. Refer to [9.1. Parallel I/O \(PIO\) Ports](#) for availability on device variants.
2. Availability dependent on supported peripherals, refer to [Table 1](#).
3. Availability dependent on number of supported ADC channels. Refer to [Table 1](#) for ADC channel availability on device variants.

8.4 Interrupt Resources

Many useful resources are provided on the main product page of the Microchip website for the devices listed in this data sheet. This product page contains the latest updates and additional information.

8.5 Interrupt Control and Status Registers

The dsPIC33CK512MPT608 family devices implement the following registers for the interrupt controller:

- INTCON1
- INTCON2
- INTCON3
- INTCON4
- INTTREG

8.5.1 INTCON1 through INTCON4

Global interrupt control functions are controlled from INTCON1, INTCON2, INTCON3 and INTCON4.

INTCON1 contains the Interrupt Nesting Disable bit (NSTDIS), as well as the control and status flags for the processor trap sources.

The INTCON2 register controls external interrupt request signal behavior, contains the Global Interrupt Enable bit (GIE) and the Alternate Interrupt Vector Table Enable bit (AIVTEN).

INTCON3 contains the status flags for the Auxiliary PLL and DO stack overflow status trap sources.

The INTCON4 register contains the Software Generated Hard Trap Status bit (SGHT).

8.5.1.1 IFSx

The IFSx registers maintain all of the interrupt request flags. Each source of interrupt has a status bit, which is set by the respective peripherals or external signal and is cleared via software.

8.5.1.2 IECx

The IECx registers maintain all of the interrupt enable bits. These control bits are used to individually enable interrupts from the peripherals or external signals.

8.5.1.3 IPCx

The IPCx registers are used to set the Interrupt Priority Level (IPL) for each source of interrupt. Each user interrupt source can be assigned to one of seven priority levels.

8.5.1.4 INTTREG

The INTTREG register contains the associated interrupt vector number and the new CPU Interrupt Priority Level, which are latched into the Vector Number (VECNUM[7:0]) and Interrupt Level bits (ILR[3:0]) fields in the INTTREG register. The new Interrupt Priority Level is the priority of the pending interrupt.

The interrupt sources are assigned to the IFSx, IECx and IPCx registers in the same sequence as they are listed in [Table 8-2](#). For example, INT0 (External Interrupt 0) is shown as having Vector Number 8 and a natural order priority of 0. Thus, the INT0IF bit is found in IFS0[0], the INT0IE bit in IEC0[0] and the INT0IP[2:0] bits in the first position of IPC0 (IPC0[2:0]).

8.6 Status/Control Registers

Although these registers are not specifically part of the interrupt control hardware, two of the CPU Control registers contain bits that control interrupt functionality. For more information on these registers, refer to “**Enhanced CPU**” (www.microchip.com/DS70005158) in the “dsPIC33/PIC24 Family Reference Manual”.

- The CPU STATUS Register, SR, contains the IPL[2:0] bits (SR[7:5]). These bits indicate the current CPU Interrupt Priority Level. The user software can change the current CPU Interrupt Priority Level by writing to the IPLx bits.

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- The CORCON register contains the IPL3 bit, which together with IPL[2:0], also indicates the current CPU priority level. IPL3 is a read-only bit so that trap events cannot be masked by the user software.

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8.7 Status/Control Registers

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x00 ... 0x41	Reserved									
0x42	SR	15:8								
		7:0	IPL[2:0]							
0x44	CORCON	15:8	VAR							
		7:0					IPL3			
0x46 ... 0x07FF	Reserved									
0x0800	IFS0	15:8	INT1IF	NVMIF	ECCSBEIF	U1TXIF	U1RXIF	SPI1TXIF	SPI1RXIF	DMA1IF
		7:0	CCT1IF	CCP1IF		DMA0IF	CNBIF	CNAIF	T1IF	INT0IF
0x0802	IFS1	15:8	C1RXIF	SPI2TXIF	SPI2RXIF	U2TXIF	U2RXIF	INT3IF	C1IF	CCT2IF
		7:0	CCP2IF	DMA4IF	DMA3IF	INT2IF	CNCIF	DMA2IF	MI2C1IF	SI2C1IF
0x0804	IFS2	15:8	CCT6IF	CCP6IF	DMTIF	CCT5IF	CCP5IF		CCT4IF	CCP4IF
		7:0		MI2C2	SI2C2	CCT3IF	CCP3IF	DMA5IF	C2IF	C2RXIF
0x0806	IFS3	15:8	PTGSTIEIF	JTAGIF	ICDIF	SPI3TXIF	SPI3RXIF	U3TXIF	U3RXIF	U3EIF
		7:0		QE12IF	C2TXIF	C1TXIF	CRCIF	U2EIF	U1EIF	QE11IF
0x0808	IFS4	15:8	CMP3IF	CMP2IF	CMP1IF	CNEIF	CNDIF	PWM8IF		PWM6IF
		7:0		PWM4IF	PWM3IF	PWM2IF	PWM1IF	QE13IF	I2C2BCIF	I2C1BCIF
0x080A	IFS5	15:8	ADCAN4IF	ADCAN3IF	ADCAN2IF	ADCAN1IF	ADCAN0IF	ADCIF	SENT2EIF	SENT2IF
		7:0	SENT1EIF	SENT1IF	PTG3IF	PTG2IF	PTG1IF	PTG0IF	PTGWDIF	CMP4IF
0x080C	IFS6	15:8	ADCAN20IF	ADCAN19IF	ADCAN18IF	ADCAN17IF	ADCAN16IF	ADCAN15IF	ADCAN14IF	ADCAN13IF
		7:0	ADCAN12IF	ADCAN11IF	ADCAN10IF	ADCAN9IF	ADCAN8IF	ADCAN7IF	ADCAN6IF	ADCAN5IF
0x080E	IFS7	15:8	SPI2GIF	SPI1GIF	CLC2PIF	CLC1PIF	ADFLTR4IF	ADFLTR3IF	ADFLTR2IF	ADFLTR1IF
		7:0	ADCMP3IF	ADCMP2IF	ADCMP1IF	ADCMP0IF	ADFLTIF	ADCAN23IF	ADCAN22IF	ADCAN21IF
0x0810	IFS8	15:8	MI2C3IF	SI2C3IF						
		7:0								SPI3IF
0x0812	IFS9	15:8					DMA7IF	DMA6IF	CCT8IF	CCP8IF
		7:0	ADCAN26IF	CCT7IF	CCP7IF				ADCAN27IF	I2C3BCIF
0x0814	IFS10	15:8	CLC3PIF	PEVTFIF	PEVTEIF	PEVTDIF	PEVTCIF	PEVTBIF	PEVTAIF	
		7:0					ADC3EIF	ADC2EIF	ADC1EIF	ADC0EIF
0x0816	IFS11	15:8	U3ENTIF	U2ENTIF	U1ENTIF	CMP6IF	CMP5IF			
		7:0		CCT9IF	CCP9IF	CLC4NIF	CLC3NIF	CLC2NIF	CLC1NIF	CLC4PIF
0x0818	IFS12	15:8								
		7:0							ADCAN25IF	ADCAN24IF
0x081A ... 0x081F	Reserved									
0x0820	IEC0	15:8	INT1IE	NVMIE	ECCSBEIE	U1TXIE	U1RXIE	SPI1TXIE	SPI1RXIE	DMA1IE
		7:0	CCT1IE	IOC1IE		DMA0IE	CNBIE	CNAIE	T1IE	INT0IE
0x0822	IEC1	15:8	C1RXIE	SPI2TXIE	SPI2RXIE	U2TXIE	U2RXIE	INT3IE	C1IE	CCT2IE
		7:0	CCP2IE	DMA4IE	DMA3IE	INT2IE	CNCIE	DMA2IE	MI2C1IE	SI2C1IE
0x0824	IEC2	15:8	CCT6IE	CCP6IE	DMTIE	CCT5IE	CCP5IE		CCT4IE	CCP4IE
		7:0		MI2C2IE	SI2C2IE	CCT3IE	CCP3IE	DMA5IE	C2IE	C2RXIE
0x0826	IEC3	15:8	PTGSTIE	JTAGIE	ICDIE	SPI3TXIE	SPI3RXIE	U3TXIE	U3RXIE	U3EIE
		7:0		QE12IE	C2TXIE	C1TXIE	CRCIE	U2EIE	U1EIE	QE11IE
0x0828	IEC4	15:8	CMP3IE	CMP2IE	CMP1IE	CNEIE	CNDIE	PWM8IE		PWM6IE
		7:0		PWM4IE	PWM3IE	PWM2IE	PWM1IE	QE13IE	I2C2BCIE	I2C1BCIE
0x082A	IEC5	15:8	ADCAN4IE	ADCAN3IE	ADCAN2IE	ADCAN1IE	ADCAN0IE	ADCIE	SENT2EIE	SENT2IE
		7:0	SENT1EIE	SENT1IE	PTG3IE	PTG2IE	PTG1IE	PTG0IE	PTGWDIE	CMP4IE
0x082C	IEC6	15:8	ADCAN20IE	ADCAN19IE	ADCAN18IE	ADCAN17IE	ADCAN16IE	ADCAN15IE	ADCAN14IE	ADCAN13IE
		7:0	ADCAN12IE	ADCAN11IE	ADCAN10IE	ADCAN9IE	ADCAN8IE	ADCAN7IE	ADCAN6IE	ADCAN5IE
0x082E	IEC7	15:8	SPI2GIE	SPI1GIE	CLC2PIE	CLC1PIE	ADCFLTR4IE	ADCFLTR3IE	ADCFLTR2IE	ADCFLTR1IE
		7:0	ADCS1CMP3IE	ADCS1CMP2IE	ADCS1CMP1IE	ADCS1CMP0IE	ADCFLTIE	ADCAN23IE	ADCAN22IE	ADCAN21IE

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.....continued

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x0830	IEC8	15:8	MI2C3IE	SI2C3IE						
		7:0								SPI3IE
0x0832	IEC9	15:8					DMA7IE	DMA6IE	CCT8IE	CCP8IE
		7:0	ADCAN26IE	CCT7IE	CCP7IE					I2C3BCIE
0x0834	IEC10	15:8	CLC3PIE	PEVTFIE	PEVTEIE	PEVTDIE	PEVTCIE	PEVTBIE	PEVTAIE	
		7:0					ADC3EIE	ADC2EIE	ADC1EIE	ADC0EIE
0x0836	IEC11	15:8	U3EVTIE	U2EVTIE	U1EVTIE	CMP6IE	CMP5IE			
		7:0		CCT9IE	CCP9IE	CLC4NIE	CLC3NIE	CLC2NIE	CLC1NIE	CLC4PIE
0x0838	IEC12	15:8								
		7:0							ADCAN25IE	ADCAN24IE
0x083A ... 0x083F	Reserved									
0x0840	IPC0	15:8			CNBIP[2:0]				CNAIP[2:0]	
		7:0			T1IP[2:0]				INT0IP[2:0]	
0x0842	IPC1	15:8			CCT1IP[2:0]				CCP1IP[2:0]	
		7:0			CNFIP[2:0]				DMA0IP[2:0]	
0x0844	IPC2	15:8			U1RXIP[2:0]				SPI1TXIP[2:0]	
		7:0			SPI1RXIP[2:0]				DMA1IP[2:0]	
0x0846	IPC3	15:8			INT1IP[2:0]				NVMIP[2:0]	
		7:0			ECCSBEIP[2:0]				U1TXIP[2:0]	
0x0848	IPC4	15:8			CNCIP[2:0]				DMA2IP[2:0]	
		7:0			MI2C1IP[2:0]				SI2C1IP[2:0]	
0x084A	IPC5	15:8			CCP2IP[2:0]				DMA4IP[2:0]	
		7:0			DMA3IP[2:0]				INT2IP[2:0]	
0x084C	IPC6	15:8			U2RXIP[2:0]				INT3IP[2:0]	
		7:0			C1IP[2:0]				CCT2IP[2:0]	
0x084E	IPC7	15:8			C1RXIP[2:0]				SPI2TXIP[2:0]	
		7:0			SPI2RXIP[2:0]				U2TXIP[2:0]	
0x0850	IPC8	15:8			CCP3IP[2:0]				DMA5IP[2:0]	
		7:0			C2IP[2:0]				C2RXIP[2:0]	
0x0852	IPC9	15:8							MI2C2IP[2:0]	
		7:0			SI2C2IP[2:0]				CCT3IP[2:0]	
0x0854	IPC10	15:8			CCP5IP[2:0]					
		7:0			CCT4IP[2:0]				CCP4IP[2:0]	
0x0856	IPC11	15:8			CCT6IP[2:0]				CCP6IP[2:0]	
		7:0			DMTIP[2:0]				CCT5IP[2:0]	
0x0858	IPC12	15:8			CRCIP[2:0]				U2EIP[2:0]	
		7:0			U1EIP[2:0]				QE1IP[2:0]	
0x085A	IPC13	15:8							QE2IP[2:0]	
		7:0			C2TXIP[2:0]				C1TXIP[2:0]	
0x085C	IPC14	15:8			SPI3RXIP[2:0]				U3TXIP[2:0]	
		7:0			U3RXIP[2:0]				U3EIP[2:0]	
0x085E	IPC15	15:8			PTGSTEIP[2:0]				JTAGIP[2:0]	
		7:0			ICDIP[2:0]				SPI3TXIP[2:0]	
0x0860	IPC16	15:8			PWM1IP[2:0]				QE3IP[2:0]	
		7:0			I2C2BCIP[2:0]				I2C1BCIP[2:0]	
0x0862	IPC17	15:8							PWM4IP[2:0]	
		7:0			PWM3IP[2:0]				PWM2IP[2:0]	
0x0864	IPC18	15:8			CNDIP[2:0]				PWM8IP[2:0]	
		7:0							PWM6IP[2:0]	
0x0866	IPC19	15:8			CMP3IP[2:0]				CMP2IP[2:0]	
		7:0			CMP1IP[2:0]				CNEIP[2:0]	
0x0868	IPC20	15:8			PTG1IP[2:0]				PTG0IP[2:0]	
		7:0			PTGWDIP[2:0]				CMP4IP[2:0]	
0x086A	IPC21	15:8			SENT1EIP[2:0]				SENT1IP[2:0]	
		7:0			PTG3IP[2:0]				PTG2IP[2:0]	
0x086C	IPC22	15:8			ADCAN0IP[2:0]				ADCIP[2:0]	
		7:0			SENT2EIP[2:0]				SENT2IP[2:0]	

dsPIC33CK512MPT608 Family

Interrupt Controller

.....continued

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x086E	IPC23	15:8			ADCAN4IP[2:0]				ADCAN3IP[2:0]	
		7:0			ADCAN2IP[2:0]				ADCAN1IP[2:0]	
0x0870	IPC24	15:8			ADCAN8IP[2:0]				ADCAN7IP[2:0]	
		7:0			ADCAN6IP[2:0]				ADCAN5IP[2:0]	
0x0872	IPC25	15:8			ADCAN12IP[2:0]				ADCAN11IP[2:0]	
		7:0			ADCAN10IP[2:0]				ADCAN9IP[2:0]	
0x0874	IPC26	15:8			ADCAN16IP[2:0]				ADCAN15IP[2:0]	
		7:0			ADCAN14IP[2:0]				ADCAN13IP[2:0]	
0x0876	IPC27	15:8			ADCAN20IP[2:0]				ADCAN19IP[2:0]	
		7:0			ADCAN18IP[2:0]				ADCAN17IP[2:0]	
0x0878	IPC28	15:8			ADFLTIP[2:0]				ADCAN23IP[2:0]	
		7:0			ADCAN22IP[2:0]				ADCAN21IP[2:0]	
0x087A	IPC29	15:8			ADCMP3IP[2:0]				ADCMP2IP[2:0]	
		7:0			ADCMP1IP[2:0]				ADCMP0IP[2:0]	
0x087C	IPC30	15:8			ADFLTR3IP[2:0]				ADFLTR2IP[2:0]	
		7:0			ADFLTR1IP[2:0]				ADFLTR0IP[2:0]	
0x087E	IPC31	15:8			SPI2EIP[2:0]				SPI1EIP[2:0]	
		7:0			CLC2PEIP[2:0]				CLC1PEIP[2:0]	
0x0880	IPC32	15:8								
		7:0							SPI3IP[2:0]	
0x0882 ... 0x0885	Reserved									
0x0886	IPC35	15:8			MI2C3IP[2:0]				SI2C3IP[2:0]	
		7:0								
0x0888	IPC36	15:8			ADCAN29IP[2:0]				ADCAN28IP[2:0]	
		7:0			ADCAN27IP[2:0]				I2C3BCIP[2:0]	
0x088A	IPC37	15:8			ADCAN26IP[2:0]				CCT7IP[2:0]	
		7:0			CCP7IP[2:0]				ADCAN30IP[2:0]	
0x088C	IPC38	15:8			DMA7IP[2:0]				DMA6IP[2:0]	
		7:0			CCT8IP[2:0]				CCP8IP[2:0]	
0x088E ... 0x0893	Reserved									
0x0894	IPC42	15:8			PEVTCIP[2:0]				PEVTBIP[2:0]	
		7:0			PEVTAIP[2:0]					
0x0896	IPC43	15:8			CLC3PEIP[2:0]				PEVTFIP[2:0]	
		7:0			PEVTEIP[2:0]				PEVTDIP[2:0]	
0x0898	IPC44	15:8			CLC3NEIP[2:0]				CLC2NEIP[2:0]	
		7:0			CLC1NEIP[2:0]				CLC4PEIP[2:0]	
0x089A	IPC45	15:8							CCT9IP[2:0]	
		7:0			CCP9IP[2:0]				CLC4NEIP[2:0]	
0x089C	IPC46	15:8			CMP5IP[2:0]					
		7:0								
0x089E	IPC47	15:8			U3EVTIP[2:0]				U2EVTIP[2:0]	
		7:0			U1EVTIP[2:0]				CMP6IP[2:0]	
0x08A0	IPC48	15:8								
		7:0			ADCAN25IP[2:0]				ADCAN24IP[2:0]	
0x08A2 ... 0x08BF	Reserved									
0x08C0	INTCON1	15:8	NSTDIS	OVAERR	OVBERR	COVAERR	COVBERR	OVATE	OVATE	COVTE
		7:0	SFTACERR	DIV0ERR		MATHERR	ADDRERR	STKERR	OSCFail	
0x08C2	INTCON2	15:8	GIE	DISI	SWTRAP					AIVTEN
		7:0					INT3EP	INT2EP	INT1EP	INT0EP
0x08C4	INTCON3	15:8	DMT						CAN	NAE
		7:0		CAN2	DAE	DOOVR				APLL
0x08C6	INTCON4	15:8								
		7:0							ECCDBE	SGHT

dsPIC33CK512MPT608 Family

Interrupt Controller

.....continued

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x08C8	INTTREG	15:8			VHOLD		ILR[3:0]			
		7:0	VECNUM[7:0]							

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.1 Interrupt Request Flags Register 0

Name: IFS0
Offset: 0x800

Bit	15	14	13	12	11	10	9	8
	INT1IF	NVMIF	ECCSBEIF	U1TXIF	U1RXIF	SPI1TXIF	SPI1RXIF	DMA1IF
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	CCT1IF	CCP1IF		DMA0IF	CNBIF	CNAIF	T1IF	INT0IF
Access	R/W	R/W		R/W	R/W	R/W	R/W	R/W
Reset	0	0		0	0	0	0	0

Bit 15 – INT1IF External Interrupt 1 bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 14 – NVMIF Nonvolatile Memory Write Complete Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 13 – ECCSBEIF ECC Single-Bit Error Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 12 – U1TXIF UART1 Transmitter Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 11 – U1RXIF UART1 Receiver Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 10 – SPI1TXIF SPI1 Transmit Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 9 – SPI1RXIF SPI1 Receive Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 8 – DMA1IF Direct Memory Access 1 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

dsPIC33CK512MPT608 Family

Interrupt Controller

Bit 7 – CCT1IF Capture/Compare/Timer1 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 6 – CCP1IF Input Capture/Output Compare 1 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 4 – DMA0IF Direct Memory Access 0 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 3 – CNBIF Change Notice Interrupt B bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 2 – CNAIF Change Notice Interrupt A bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 1 – T1IF Timer1 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 0 – INT0IF External Interrupt 0 bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.2 Interrupt Request Flags Register 1

Name: IFS1
Offset: 0x802

Bit	15	14	13	12	11	10	9	8
	C1RXIF	SPI2TXIF	SPI2RXIF	U2TXIF	U2RXIF	INT3IF	C1IF	CCT2IF
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	CCP2IF	DMA4IF	DMA3IF	INT2IF	CNCIF	DMA2IF	MI2C1IF	SI2C1IF
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – C1RXIF CAN1 RX Data Ready Interrupt bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 14 – SPI2TXIF SPI2 Transfer Done Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 13 – SPI2RXIF SPI2 Receiver Done Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 12 – U2TXIF UART2 Transmitter Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 11 – U2RXIF UART2 Receiver Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 10 – INT3IF External Interrupt 3 bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 9 – C1IF CAN1 Combined Error Interrupt Bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 8 – CCT2IF Capture/Compare/Timer2 Interrupt bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

dsPIC33CK512MPT608 Family

Interrupt Controller

Bit 7 – CCP2IF Input Capture/Output Compare 2 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 6 – DMA4IF Direct Memory Access 4 Interrupt bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 5 – DMA3IF Direct Memory Access 3 Interrupt bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 4 – INT2IF External Interrupt 2 bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 3 – CNCIF Change Notice Interrupt C bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 2 – DMA2IF Direct Memory Access 2 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 1 – MI2C1IF I2C1 Host Event Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 0 – SI2C1IF I2C1 Client Event Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

8.7.3 Interrupt Request Flags Register 2

Name: IFS2
Offset: 0x804

Bit	15	14	13	12	11	10	9	8
	CCT6IF	CCP6IF	DMTIF	CCT5IF	CCP5IF		CCT4IF	CCP4IF
Access	R/W	R/W	R/W	R/W	R/W		R/W	R/W
Reset	0	0	0	0	0		0	0

Bit	7	6	5	4	3	2	1	0
		MI2C2	SI2C2	CCT3IF	CCP3IF	DMA5IF	C2IF	C2RXIF
Access		R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset		0	0	0	0	0	0	0

Bit 15 – CCT6IF Capture/Compare/Timer6 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 14 – CCP6IF Input Capture/Output Compare 6 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 13 – DMTIF Deadman Timer Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 12 – CCT5IF Capture/Compare/Timer5 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 11 – CCP5IF Input Capture/Output Compare 5 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 9 – CCT4IF Capture/Compare/Timer4 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 8 – CCP4IF Input Capture/Output Compare 4 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 6 – MI2C2 I2C2 Host Event Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

dsPIC33CK512MPT608 Family

Interrupt Controller

Bit 5 – SI2C2 I2C2 Client Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 4 – CCT3IF Capture/Compare/Timer3 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 3 – CCP3IF Input Capture/Output Compare 3 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 2 – DMA5IF Direct Memory Access 5 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 1 – C2IF CAN2 Combined Error Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 0 – C2RXIF CAN2 RX Data Ready Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.4 Interrupt Request Flags Register 3

Name: IFS3
Offset: 0x806

Bit	15	14	13	12	11	10	9	8
	PTGSTEPF	JTAGIF	ICDIF	SPI3TXIF	SPI3RXIF	U3TXIF	U3RXIF	U3EIF
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
		QE12IF	C2TXIF	C1TXIF	CRCIF	U2EIF	U1EIF	QE11IF
Access		R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset		0	0	0	0	0	0	0

Bit 15 – PTGSTEPF PTG Step Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 14 – JTAGIF JTAG Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 13 – ICDIF In-Circuit Debugger Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 12 – SPI3TXIF SPI3 Transmit Done Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 11 – SPI3RXIF SPI3 Receive Done Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 10 – U3TXIF UART3 Transmitter Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 9 – U3RXIF UART3 Receiver Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 8 – U3EIF UART3 Error Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

dsPIC33CK512MPT608 Family

Interrupt Controller

Bit 6 – QE12IF QE12 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 5 – C2TXIF CAN2 TX Data Request Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 4 – C1TXIF CAN1 TX Data Request Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 3 – CRCIF Cyclic Redundancy Check Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 2 – U2EIF UART2 Error Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 1 – U1EIF UART1 Error Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 0 – QE11IF QE11 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.5 Interrupt Request Flags Register 4

Name: IFS4
Offset: 0x808

Bit	15	14	13	12	11	10	9	8
	CMP3IF	CMP2IF	CMP1IF	CNEIF	CNDIF	PWM8IF		PWM6IF
Access	R/W	R/W	R/W	R/W	R/W	R/W		R/W
Reset	0	0	0	0	0	0		0

Bit	7	6	5	4	3	2	1	0
		PWM4IF	PWM3IF	PWM2IF	PWM1IF	QE13IF	I2C2BCIF	I2C1BCIF
Access		R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset		0	0	0	0	0	0	0

Bit 15 – CMP3IF Comparator 3 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 14 – CMP2IF Comparator 2 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 13 – CMP1IF Comparator 1 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 12 – CNEIF Change Notice E Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 11 – CNDIF Change Notice D Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 10 – PWM8IF PWM Generator 8 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 8 – PWM6IF PWM Generator 6 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 6 – PWM4IF PWM Generator 4 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

dsPIC33CK512MPT608 Family

Interrupt Controller

Bit 5 – PWM3IF PWM Generator 3 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 4 – PWM2IF PWM Generator 2 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 3 – PWM1IF PWM Generator 1 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 2 – QE13IF QE13 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 1 – I2C2BCIF I2C2 Bus Collision Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 0 – I2C1BCIF I2C1 Bus Collision Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.6 Interrupt Request Flags Register 5

Name: IFS5
Offset: 0x80A

Bit	15	14	13	12	11	10	9	8
	ADCAN4IF	ADCAN3IF	ADCAN2IF	ADCAN1IF	ADCAN0IF	ADCIF	SENT2EIF	SENT2IF
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	SENT1EIF	SENT1IF	PTG3IF	PTG2IF	PTG1IF	PTG0IF	PTGWDTIF	CMP4IF
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – ADCAN4IF ADC AN4 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 14 – ADCAN3IF ADC AN3 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 13 – ADCAN2IF ADC AN2 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 12 – ADCAN1IF ADC AN1 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 11 – ADCAN0IF ADC AN0 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 10 – ADCIF ADC Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 9 – SENT2EIF SENT2 Error Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 8 – SENT2IF SENT2 TX/RX Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 7 – SENT1EIF SENT1 Error Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 6 – SENT1IF SENT1 TX/RX Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 5 – PTG3IF PTG3 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 4 – PTG2IF PTG2 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 3 – PTG1IF PTG Trigger 1 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 2 – PTG0IF PTG Trigger 0 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 1 – PTGWDTIF PTG Watchdog Timer Time-out Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 0 – CMP4IF Comparator 4 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

8.7.7 Interrupt Request Flags Register 6

Name: IFS6
Offset: 0x80C

Bit	15	14	13	12	11	10	9	8
	ADCAN20IF	ADCAN19IF	ADCAN18IF	ADCAN17IF	ADCAN16IF	ADCAN15IF	ADCAN14IF	ADCAN13IF
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	ADCAN12IF	ADCAN11IF	ADCAN10IF	ADCAN9IF	ADCAN8IF	ADCAN7IF	ADCAN6IF	ADCAN5IF
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – ADCAN20IF ADC AN20 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 14 – ADCAN19IF ADC AN19 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 13 – ADCAN18IF ADC AN18 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 12 – ADCAN17IF ADC AN17 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 11 – ADCAN16IF ADC AN16 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 10 – ADCAN15IF ADC AN15 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 9 – ADCAN14IF ADC AN14 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 8 – ADCAN13IF ADC AN13 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

dsPIC33CK512MPT608 Family

Interrupt Controller

Bit 7 – ADCAN12IF ADC AN12 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 6 – ADCAN11IF ADC AN11 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 5 – ADCAN10IF ADC AN10 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 4 – ADCAN9IF ADC AN9 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 3 – ADCAN8IF ADC AN8 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 2 – ADCAN7IF ADC AN7 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 1 – ADCAN6IF ADC AN6 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 0 – ADCAN5IF ADC AN5 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.8 Interrupt Request Flags Register 7

Name: IFS7
Offset: 0x80E

Bit	15	14	13	12	11	10	9	8
	SPI2GIF	SPI1GIF	CLC2PIF	CLC1PIF	ADFLTR4IF	ADFLTR3IF	ADFLTR2IF	ADFLTR1IF
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	ADCMP3IF	ADCMP2IF	ADCMP1IF	ADCMP0IF	ADFLTIF	ADCAN23IF	ADCAN22IF	ADCAN21IF
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – SPI2GIF SPI2 Error Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 14 – SPI1GIF SPI1 Error Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 13 – CLC2PIF CLC2 Positive Edge Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 12 – CLC1PIF CLC1 Positive Edge Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 11 – ADFLTR4IF ADC Oversample Filter 4 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 10 – ADFLTR3IF ADC Oversample Filter 3 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 9 – ADFLTR2IF ADC Oversample Filter 2 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 8 – ADFLTR1IF ADC Oversample Filter 1 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 7 – ADCMP3IF ADC Digital Comparator 3 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 6 – ADCMP2IF ADC Digital Comparator 2 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 5 – ADCMP1IF ADC Digital Comparator 1 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 4 – ADCMP0IF ADC Digital Comparator 0 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 3 – ADFLTIF ADC Fault Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 2 – ADCAN23IF ADC AN23 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 1 – ADCAN22IF ADC AN22 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 0 – ADCAN21IF ADC AN21 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

8.7.9 Interrupt Request Flags Register 8

Name: IFS8
Offset: 0x810

Bit	15	14	13	12	11	10	9	8
	MI2C3IF	SI2C3IF						
Access	R/W	R/W						
Reset	0	0						

Bit	7	6	5	4	3	2	1	0
								SPI3IF
Access								R/W
Reset								0

Bit 15 – MI2C3IF I2C3 Host Event Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 14 – SI2C3IF I2C3 Client Event Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 0 – SPI3IF SPI3 Error Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.10 Interrupt Request Flags Register 9

Name: IFS9
Offset: 0x812

Bit	15	14	13	12	11	10	9	8
					DMA7IF	DMA6IF	CCT8IF	CCP8IF
Access					R/W	R/W	R/W	R/W
Reset					0	0	0	0

Bit	7	6	5	4	3	2	1	0
	ADCAN26IF	CCT7IF	CCP7IF				ADCAN27IF	I2C3BCIF
Access	R/W	R/W	R/W				R/W	R/W
Reset	0	0	0				0	0

Bit 11 – DMA7IF Direct Memory Access 7 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 10 – DMA6IF Direct Memory Access 6 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 9 – CCT8IF Capture/Compare/Timer8 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 8 – CCP8IF Input Capture/Output Compare 8 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 7 – ADCAN26IF ADC AN26 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 6 – CCT7IF Capture/Compare/Timer7 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 5 – CCP7IF Input Capture/Output Compare 7 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 1 – ADCAN27IF ADC AN27 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

dsPIC33CK512MPT608 Family

Interrupt Controller

Bit 0 – I2C3BCIF I2C3 Bus Collision Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.11 Interrupt Request Flags Register 10

Name: IFS10

Offset: 0x814

Bit	15	14	13	12	11	10	9	8
	CLC3PIF	PEVTFIF	PEVTEIF	PEVTDIF	PEVTCIF	PEVTBIF	PEVTAIF	
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Reset	0	0	0	0	0	0	0	

Bit	7	6	5	4	3	2	1	0
					ADC3EIF	ADC2EIF	ADC1EIF	ADC0EIF
Access					R/W	R/W	R/W	R/W
Reset					0	0	0	0

Bit 15 – CLC3PIF CLC3 Positive Edge Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 14 – PEVTFIF PWM Event F Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 13 – PEVTEIF PWM Event E Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 12 – PEVTDIF PWM Event D Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 11 – PEVTCIF PWM Event C Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 10 – PEVTBIF PWM Event B Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 9 – PEVTAIF PWM Event A Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 3 – ADC3EIF ADC Enable 3 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 2 – ADC2EIF ADC Enable 2 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 1 – ADC1EIF ADC Enable 1 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 0 – ADC0EIF ADC Enable 0 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

8.7.12 Interrupt Request Flags Register 11

Name: IFS11
Offset: 0x816

Bit	15	14	13	12	11	10	9	8
	U3ENTIF	U2ENTIF	U1ENTIF	CMP6IF	CMP5IF			
Access	R/W	R/W	R/W	R/W	R/W			
Reset	0	0	0	0	0			

Bit	7	6	5	4	3	2	1	0
		CCT9IF	CCP9IF	CLC4NIF	CLC3NIF	CLC2NIF	CLC1NIF	CLC4PIF
Access		R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset		0	0	0	0	0	0	0

Bit 15 – U3ENTIF UART3 Event Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 14 – U2ENTIF UART2 Event Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 13 – U1ENTIF UART1 Event Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 12 – CMP6IF Comparator 6 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 11 – CMP5IF Comparator 5 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 6 – CCT9IF Capture/Compare/Timer9 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 5 – CCP9IF Input Capture/Output Compare 9 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 4 – CLC4NIF CLC4 Negative Edge Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

dsPIC33CK512MPT608 Family

Interrupt Controller

Bit 3 – CLC3NIF CLC3 Negative Edge Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 2 – CLC2NIF CLC2 Negative Edge Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 1 – CLC1NIF CLC1 Negative Edge Interrupt bit

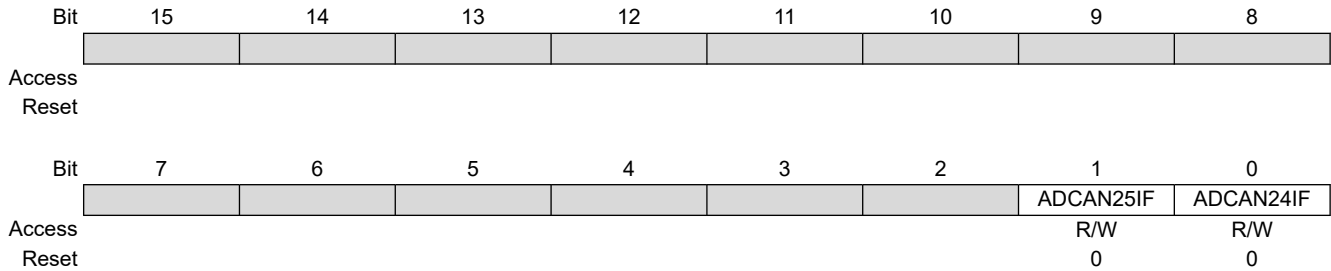
Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 0 – CLC4PIF CLC4 Positive Edge Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

8.7.13 Interrupt Request Flags Register 12

Name: IFS12
Offset: 0x818



Bit 1 – ADCAN25IF ADC AN25 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 0 – ADCAN24IF ADC AN24 Interrupt bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.14 Interrupt Enable Register 0

Name: IEC0
Offset: 0x820

Bit	15	14	13	12	11	10	9	8
	INT1IE	NVMIE	ECCSBEIE	U1TXIE	U1RXIE	SPI1TXIE	SPI1RXIE	DMA1IE
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	CCT1IE	IOC1IE		DMA0IE	CNBIE	CNAIE	T1IE	INT0IE
Access	R/W	R/W		R/W	R/W	R/W	R/W	R/W
Reset	0	0		0	0	0	0	0

Bit 15 – INT1IE External Interrupt 1 Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 14 – NVMIE NVM Program/Erase Complete Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 13 – ECCSBEIE ECC Single-Bit Error Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 12 – U1TXIE UART1 Transmitter Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 11 – U1RXIE UART1 Receiver Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 10 – SPI1TXIE SPI1 Transmit Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 9 – SPI1RXIE SPI1 Receive Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 8 – DMA1IE Direct Memory Access 1 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

dsPIC33CK512MPT608 Family

Interrupt Controller

Bit 7 – CCT1IE Capture/Compare/Timer1 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 6 – IOC1IE Interrupt-on-Change 1 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 4 – DMA0IE Direct Memory Access 0 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 3 – CNBIE Change Notice B Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 2 – CNAIE Change Notice A Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 1 – T1IE Timer1 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 0 – INT0IE External Interrupt 0 Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.15 Interrupt Enable Register 1

Name: IEC1
Offset: 0x822

Bit	15	14	13	12	11	10	9	8
	C1RXIE	SPI2TXIE	SPI2RXIE	U2TXIE	U2RXIE	INT3IE	C1IE	CCT2IE
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	CCP2IE	DMA4IE	DMA3IE	INT2IE	CNCIE	DMA2IE	MI2C1IE	SI2C1IE
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – C1RXIE CAN1 RX Data Ready Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 14 – SPI2TXIE SPI2 Transmitter Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 13 – SPI2RXIE SPI2 Receiver Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 12 – U2TXIE UART2 Transmitter Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 11 – U2RXIE UART2 Receiver Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 10 – INT3IE External Interrupt 3 Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 9 – C1IE CAN1 Combined Error Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 8 – CCT2IE Capture/Compare/Timer2 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

dsPIC33CK512MPT608 Family

Interrupt Controller

Bit 7 – CCP2IE Interrupt-on-Change 2 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 6 – DMA4IE Input Capture/Output Compare 2 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 5 – DMA3IE Direct Memory Access 3 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 4 – INT2IE External Interrupt 2 Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 3 – CNCIE Change Notice C Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 2 – DMA2IE Direct Memory Access 2 Interrupt bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 1 – MI2C1IE I2C1 Host Events Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 0 – SI2C1IE I2C1 Client Events Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

8.7.16 Interrupt Enable Register 2

Name: IEC2
Offset: 0x824

Bit	15	14	13	12	11	10	9	8
	CCT6IE	CCP6IE	DMTIE	CCT5IE	CCP5IE		CCT4IE	CCP4IE
Access	R/W	R/W	R/W	R/W	R/W		R/W	R/W
Reset	0	0	0	0	0		0	0

Bit	7	6	5	4	3	2	1	0
		MI2C2IE	SI2C2IE	CCT3IE	CCP3IE	DMA5IE	C2IE	C2RXIE
Access		R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset		0	0	0	0	0	0	0

Bit 15 – CCT6IE Capture/Compare/Timer6 Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 14 – CCP6IE Input Capture/Output Compare 6 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 13 – DMTIE Deadman Timer Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 12 – CCT5IE Capture/Compare/Timer5 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 11 – CCP5IE Input Capture/Output Compare 5 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 9 – CCT4IE Capture/Compare/Timer4 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 8 – CCP4IE Input Capture/Output Compare 4 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 6 – MI2C2IE I2C2 Host Event Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

dsPIC33CK512MPT608 Family

Interrupt Controller

Bit 5 – SI2C2IE I2C2 Client Event Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 4 – CCT3IE Capture/Compare/Timer3 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 3 – CCP3IE Input Capture/Output Compare 3 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 2 – DMA5IE Direct Memory Access 5 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 1 – C2IE CAN2 Combined Error Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 0 – C2RXIE CAN2 RX Data Ready Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.17 Interrupt Enable Register 3

Name: IEC3
Offset: 0x826

Bit	15	14	13	12	11	10	9	8
	PTGSTEPIE	JTAGIE	ICDIE	SPI3TXIE	SPI3RXIE	U3TXIE	U3RXIE	U3EIE
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
		QE12IE	C2TXIE	C1TXIE	CRCIE	U2EIE	U1EIE	QE11IE
Access		R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset		0	0	0	0	0	0	0

Bit 15 – PTGSTEPIE PTG Step Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 14 – JTAGIE JTAG Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 13 – ICDIE ICD Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 12 – SPI3TXIE SPI3 Transmitter Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 11 – SPI3RXIE SPI3 Receiver Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 10 – U3TXIE UART3 Transmitter Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 9 – U3RXIE UART3 Receiver Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 8 – U3EIE UART3 Error Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

dsPIC33CK512MPT608 Family

Interrupt Controller

Bit 6 – QE12IE QE12 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 5 – C2TXIE CAN2 TX Data Request Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 4 – C1TXIE CAN1 TX Data Request Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 3 – CRCIE CRC Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 2 – U2EIE UART2 Error Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 1 – U1EIE UART1 Error Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 0 – QE11IE QE11 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.18 Interrupt Enable Register 4

Name: IEC4
Offset: 0x828

Bit	15	14	13	12	11	10	9	8
	CMP3IE	CMP2IE	CMP1IE	CNEIE	CNDIE	PWM8IE		PWM6IE
Access	R/W	R/W	R/W	R/W	R/W	R/W		R/W
Reset	0	0	0	0	0	0		0

Bit	7	6	5	4	3	2	1	0
		PWM4IE	PWM3IE	PWM2IE	PWM1IE	QE13IE	I2C2BCIE	I2C1BCIE
Access		R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset		0	0	0	0	0	0	0

Bit 15 – CMP3IE Comparator 3 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 14 – CMP2IE Comparator 2 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 13 – CMP1IE Comparator 1 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 12 – CNEIE Change Notice E Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 11 – CNDIE Change Notice D Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 10 – PWM8IE Pulse-Width Modulation 8 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 8 – PWM6IE Pulse-Width Modulation 6 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 6 – PWM4IE Pulse-Width Modulation 4 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

dsPIC33CK512MPT608 Family

Interrupt Controller

Bit 5 – PWM3IE Pulse-Width Modulation 3 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 4 – PWM2IE Pulse-Width Modulation 2 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 3 – PWM1IE Pulse-Width Modulation 1 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 2 – QEI3IE QEI3 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 1 – I2C2BCIE I2C2 Bus Collision Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 0 – I2C1BCIE I2C1 Bus Collision Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.19 Interrupt Enable Register 5

Name: IEC5
Offset: 0x82A

Bit	15	14	13	12	11	10	9	8
	ADCAN4IE	ADCAN3IE	ADCAN2IE	ADCAN1IE	ADCAN0IE	ADCIE	SENT2EIE	SENT2IE
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	SENT1EIE	SENT1IE	PTG3IE	PTG2IE	PTG1IE	PTG0IE	PTGWDTIE	CMP4IE
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – ADCAN4IE ADC AN4 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 14 – ADCAN3IE ADC AN3 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 13 – ADCAN2IE ADC AN2 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 12 – ADCAN1IE ADC AN1 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 11 – ADCAN0IE ADC AN0 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 10 – ADCIE ADC Global Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 9 – SENT2EIE SENT2 Error Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 8 – SENT2IE SENT2 TX/RX Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

dsPIC33CK512MPT608 Family

Interrupt Controller

Bit 7 – SENT1IE SENT1 Error Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 6 – SENT1IE SENT1 TX/RX Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 5 – PTG3IE PTG3 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 4 – PTG2IE PTG2 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 3 – PTG1IE PTG1 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 2 – PTG0IE PTG0 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 1 – PTGWDTIE PTG Watchdog Timer Time-out Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 0 – CMP4IE Comparator 4 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.20 Interrupt Enable Register 6

Name: IEC6
Offset: 0x82C

Bit	15	14	13	12	11	10	9	8
	ADCAN20IE	ADCAN19IE	ADCAN18IE	ADCAN17IE	ADCAN16IE	ADCAN15IE	ADCAN14IE	ADCAN13IE
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	ADCAN12IE	ADCAN11IE	ADCAN10IE	ADCAN9IE	ADCAN8IE	ADCAN7IE	ADCAN6IE	ADCAN5IE
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – ADCAN20IE ADC AN20 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 14 – ADCAN19IE ADC AN19 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 13 – ADCAN18IE ADC AN18 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 12 – ADCAN17IE ADC AN17 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 11 – ADCAN16IE ADC AN16 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 10 – ADCAN15IE ADC AN15 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 9 – ADCAN14IE ADC AN14 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 8 – ADCAN13IE ADC AN13 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 7 – ADCAN12IE ADC AN12 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 6 – ADCAN11IE ADC AN11 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 5 – ADCAN10IE ADC AN10 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 4 – ADCAN9IE ADC AN9 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 3 – ADCAN8IE ADC AN8 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 2 – ADCAN7IE ADC AN7 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 1 – ADCAN6IE ADC AN6 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 0 – ADCAN5IE ADC AN5 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.21 Interrupt Enable Register 7

Name: IEC7
Offset: 0x82E

Bit	15	14	13	12	11	10	9	8
	SPI2GIE	SPI1GIE	CLC2PIE	CLC1PIE	ADCFLTR4IE	ADCFLTR3IE	ADCFLTR2IE	ADCFLTR1IE
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	ADCS1CMP3IE	ADCS1CMP2IE	ADCS1CMP1IE	ADCS1CMP0IE	ADCFLTIE	ADCAN23IE	ADCAN22IE	ADCAN21IE
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – SPI2GIE SPI2 Error Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 14 – SPI1GIE SPI1 Error Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 13 – CLC2PIE CLC2 Positive Edge Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 12 – CLC1PIE CLC1 Positive Edge Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 11 – ADCFLTR4IE ADC Oversample Filter 4 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 10 – ADCFLTR3IE ADC Oversample Filter 3 Interrupt bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 9 – ADCFLTR2IE ADC Oversample Filter 2 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 8 – ADCFLTR1IE ADC Oversample Filter 1 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

dsPIC33CK512MPT608 Family

Interrupt Controller

Bit 7 – ADCS1CMP3IE ADC Digital Comparator 3 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 6 – ADCS1CMP2IE ADC Digital Comparator 2 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 5 – ADCS1CMP1IE ADC Digital Comparator 1 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 4 – ADCS1CMP0IE ADC Digital Comparator 0 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 3 – ADCFLTIE ADC Fault Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 2 – ADCAN23IE ADC AN23 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 1 – ADCAN22IE ADC AN22 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 0 – ADCAN21IE ADC AN21 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.22 Interrupt Enable Register 8

Name: IEC8
Offset: 0x830

Bit	15	14	13	12	11	10	9	8
	MI2C3IE	SI2C3IE						
Access	R/W	R/W						
Reset	0	0						

Bit	7	6	5	4	3	2	1	0
								SPI3IE
Access								R/W
Reset								0

Bit 15 – MI2C3IE I2C3 Host Event Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 14 – SI2C3IE I2C3 Client Event Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 0 – SPI3IE SPI3 Error Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.23 Interrupt Enable Register 9

Name: IEC9
Offset: 0x832

Bit	15	14	13	12	11	10	9	8
					DMA7IE	DMA6IE	CCT8IE	CCP8IE
Access					R/W	R/W	R/W	R/W
Reset					0	0	0	0

Bit	7	6	5	4	3	2	1	0
	ADCAN26IE	CCT7IE	CCP7IE					I2C3BCIE
Access	R/W	R/W	R/W					R/W
Reset	0	0	0					0

Bit 11 – DMA7IE Direct Memory Access 7 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 10 – DMA6IE Direct Memory Access 6 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 9 – CCT8IE Capture/Compare/Timer8 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 8 – CCP8IE Interrupt-on-Change 8 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 7 – ADCAN26IE ADC AN26 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 6 – CCT7IE Capture/Compare/Timer7 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 5 – CCP7IE Input Capture/Output Compare 7 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 0 – I2C3BCIE I2C3 Bus Collision Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.24 Interrupt Enable Register 10

Name: IEC10

Offset: 0x834

Bit	15	14	13	12	11	10	9	8
	CLC3PIE	PEVTFIE	PEVTEIE	PEVTDIE	PEVTCIE	PEVTBIE	PEVTAIE	
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Reset	0	0	0	0	0	0	0	

Bit	7	6	5	4	3	2	1	0
					ADC3EIE	ADC2EIE	ADC1EIE	ADC0EIE
Access					R/W	R/W	R/W	R/W
Reset					0	0	0	0

Bit 15 – CLC3PIE CLC3 Positive Edge Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 14 – PEVTFIE PWM Event F Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 13 – PEVTEIE PWM Event E Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 12 – PEVTDIE PWM Event D Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 11 – PEVTCIE PWM Event C Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 10 – PEVTBIE PWM Event B Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 9 – PEVTAIE PWM Event A Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 3 – ADC3EIE ADC Enable 3 Interrupt Enable bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 2 – ADC2EIE ADC Enable 2 Interrupt Enable bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 1 – ADC1EIE ADC Enable 1 Interrupt Enable bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

Bit 0 – ADC0EIE ADC Enable 0 Interrupt Enable bit

Value	Description
1	Interrupt has occurred
0	Interrupt has not occurred

8.7.25 Interrupt Enable Register 11

Name: IEC11
Offset: 0x836

Bit	15	14	13	12	11	10	9	8
	U3EVTIE	U2EVTIE	U1EVTIE	CMP6IE	CMP5IE			
Access	R/W	R/W	R/W	R/W	R/W			
Reset	0	0	0	0	0			

Bit	7	6	5	4	3	2	1	0
		CCT9IE	CCP9IE	CLC4NIE	CLC3NIE	CLC2NIE	CLC1NIE	CLC4PIE
Access		R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset		0	0	0	0	0	0	0

Bit 15 – U3EVTIE UART3 Event Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 14 – U2EVTIE UART2 Event Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 13 – U1EVTIE UART1 Event Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 12 – CMP6IE Comparator 6 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 11 – CMP5IE Comparator 5 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 6 – CCT9IE Capture/Compare/Timer9 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 5 – CCP9IE Input Capture/Output Compare 9 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 4 – CLC4NIE CLC4 Negative Edge Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

dsPIC33CK512MPT608 Family

Interrupt Controller

Bit 3 – CLC3NIE CLC3 Negative Edge Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 2 – CLC2NIE CLC2 Negative Edge Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 1 – CLC1NIE CLC1 Negative Edge Interrupt Enable bit

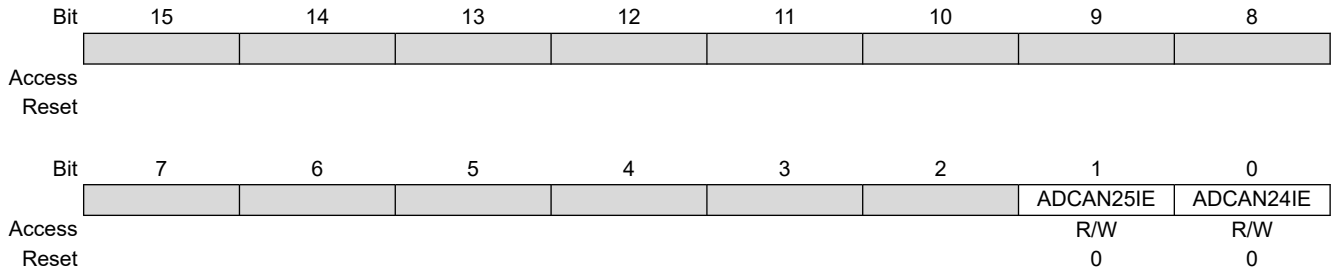
Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 0 – CLC4PIE CLC4 Positive Edge Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

8.7.26 Interrupt Enable Register 12

Name: IEC12
Offset: 0x838



Bit 1 – ADCAN25IE ADC AN25 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

Bit 0 – ADCAN24IE ADC AN24 Interrupt Enable bit

Value	Description
1	Interrupt enabled
0	Interrupt not enabled

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.27 Interrupt Priority Register 0

Name: IPC0
Offset: 0x840

Bit	15	14	13	12	11	10	9	8
		CNBIP[2:0]				CNAIP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0
Bit	7	6	5	4	3	2	1	0
		T1IP[2:0]				INT0IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – CNBIP[2:0] Change Notice B Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – CNAIP[2:0] Change Notice A Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – T1IP[2:0] Timer1 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – INT0IP[2:0] External Interrupt 0 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

dsPIC33CK512MPT608 Family

Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

8.7.28 Interrupt Priority Register 1

Name: IPC1
Offset: 0x842

Bit	15	14	13	12	11	10	9	8
		CCT1IP[2:0]				CCP1IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bit	7	6	5	4	3	2	1	0
		CNFIP[2:0]				DMA0IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – CCT1IP[2:0] Capture/Compare/Timer1 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – CCP1IP[2:0] Input Capture/Output Compare 1 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – CNFIP[2:0] Change Notice F Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – DMA0IP[2:0] Direct Memory Access 0 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

dsPIC33CK512MPT608 Family

Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.29 Interrupt Priority Register 2

Name: IPC2
Offset: 0x844

Bit	15	14	13	12	11	10	9	8
	U1RXIP[2:0]					SPI1TXIP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bit	7	6	5	4	3	2	1	0
	SPI1RXIP[2:0]					DMA1IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – U1RXIP[2:0] UART1 Receiver Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – SPI1TXIP[2:0] SPI1 Transfer Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – SPI1RXIP[2:0] SPI1 Receiver Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – DMA1IP[2:0] Direct Memory Access 1 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

dsPIC33CK512MPT608 Family

Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.30 Interrupt Priority Register 3

Name: IPC3
Offset: 0x846

Bit	15	14	13	12	11	10	9	8
		INT1IP[2:0]				NVMIP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0
Bit	7	6	5	4	3	2	1	0
		ECCSBEIP[2:0]				U1TXIP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – INT1IP[2:0] External Interrupt 1 Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – NVMIP[2:0] NVM Program/Erase Complete Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – ECCSBEIP[2:0] Error Correcting Code Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – U1TXIP[2:0] UART1 Transmitter Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

dsPIC33CK512MPT608 Family

Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.31 Interrupt Priority Register 4

Name: IPC4
Offset: 0x848

Bit	15	14	13	12	11	10	9	8
		CNCIP[2:0]				DMA2IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bit	7	6	5	4	3	2	1	0
		MI2C1IP[2:0]				SI2C1IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – CNCIP[2:0] Change Notification C Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – DMA2IP[2:0] Direct Memory Access 2 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – MI2C1IP[2:0] I2C1 Host Events Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – SI2C1IP[2:0] I2C1 Client Events Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

dsPIC33CK512MPT608 Family

Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.32 Interrupt Priority Register 5

Name: IPC5
Offset: 0x84A

Bit	15	14	13	12	11	10	9	8
		CCP2IP[2:0]				DMA4IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bit	7	6	5	4	3	2	1	0
		DMA3IP[2:0]				INT2IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – CCP2IP[2:0] Input Capture/Output Compare 2 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – DMA4IP[2:0] Direct Memory Access 4 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – DMA3IP[2:0] Direct Memory Access 3 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – INT2IP[2:0] External Interrupt 2 Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

dsPIC33CK512MPT608 Family

Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

8.7.33 Interrupt Priority Register 6

Name: IPC6
Offset: 0x84C

Bit	15	14	13	12	11	10	9	8
		U2RXIP[2:0]				INT3IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bit	7	6	5	4	3	2	1	0
		C1IP[2:0]				CCT2IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – U2RXIP[2:0] UART2 Receiver Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – INT3IP[2:0] External Interrupt 3 Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – C1IP[2:0] CAN1 Combined Error Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – CCT2IP[2:0] Capture/Compare/Timer2 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

dsPIC33CK512MPT608 Family

Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

8.7.34 Interrupt Priority Register 7

Name: IPC7
Offset: 0x84E

Bit	15	14	13	12	11	10	9	8
		C1RXIP[2:0]				SPI2TXIP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bit	7	6	5	4	3	2	1	0
		SPI2RXIP[2:0]				U2TXIP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – C1RXIP[2:0] CAN1 RX Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – SPI2TXIP[2:0] SPI2 Transmitter Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – SPI2RXIP[2:0] SPI2 Receiver Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – U2TXIP[2:0] UART2 Transmitter Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

dsPIC33CK512MPT608 Family

Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.35 Interrupt Priority Register 8

Name: IPC8
Offset: 0x850

Bit	15	14	13	12	11	10	9	8
	CCP3IP[2:0]					DMA5IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bit	7	6	5	4	3	2	1	0
		C2IP[2:0]				C2RXIP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – CCP3IP[2:0] Input Capture/Output Compare 3 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – DMA5IP[2:0] Direct Memory Access 5 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – C2IP[2:0] CAN2 Combined Error Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – C2RXIP[2:0] CAN2 RX Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

dsPIC33CK512MPT608 Family

Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

8.7.36 Interrupt Priority Register 9

Name: IPC9
Offset: 0x852

Bit	15	14	13	12	11	10	9	8
							MI2C2IP[2:0]	
Access						R/W	R/W	R/W
Reset						1	0	0

Bit	7	6	5	4	3	2	1	0
		SI2C2IP[2:0]					CCT3IP[2:0]	
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 10:8 – MI2C2IP[2:0] Host I2C2 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – SI2C2IP[2:0] Client I2C2 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – CCT3IP[2:0] Capture/Compare/Timer3 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

8.7.37 Interrupt Priority Register 10

Name: IPC10

Offset: 0x854

Bit	15	14	13	12	11	10	9	8
		CCP5IP[2:0]						
Access		R/W	R/W	R/W				
Reset		1	0	0				

Bit	7	6	5	4	3	2	1	0
		CCT4IP[2:0]					CCP4IP[2:0]	
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – CCP5IP[2:0] Input Capture/Output Compare 5 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – CCT4IP[2:0] Capture/Compare/Timer4 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – CCP4IP[2:0] Input Capture/Output Compare 4 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

8.7.38 Interrupt Priority Register 11

Name: IPC11
Offset: 0x856

Bit	15	14	13	12	11	10	9	8
		CCT6IP[2:0]				CCP6IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bit	7	6	5	4	3	2	1	0
		DMTIP[2:0]				CCT5IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – CCT6IP[2:0] Capture/Compare/Timer6 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – CCP6IP[2:0] Input Capture/Output Compare 6 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – DMTIP[2:0] Deadman Timer Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – CCT5IP[2:0] Capture/Compare/Timer5 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

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Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

8.7.39 Interrupt Priority Register 12

Name: IPC12
Offset: 0x858

Bit	15	14	13	12	11	10	9	8
		CRCIP[2:0]				U2EIP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bit	7	6	5	4	3	2	1	0
		U1EIP[2:0]				QE11IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – CRCIP[2:0] Cyclic Redundancy Check Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – U2EIP[2:0] UART2 Error Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – U1EIP[2:0] UART1 Error Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – QE11IP[2:0] QE11 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

dsPIC33CK512MPT608 Family

Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

8.7.40 Interrupt Priority Register 13

Name: IPC13
Offset: 0x85A

Bit	15	14	13	12	11	10	9	8
							QEI2IP[2:0]	
Access						R/W	R/W	R/W
Reset						1	0	0

Bit	7	6	5	4	3	2	1	0
		C2TXIP[2:0]				C1TXIP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 10:8 – QEI2IP[2:0] QEI2 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – C2TXIP[2:0] CAN2 TX Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – C1TXIP[2:0] CAN1 TX Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.41 Interrupt Priority Register 14

Name: IPC14
Offset: 0x85C

Bit	15	14	13	12	11	10	9	8
		SPI3RXIP[2:0]				U3TXIP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bit	7	6	5	4	3	2	1	0
		U3RXIP[2:0]				U3EIP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – SPI3RXIP[2:0] SPI3 Receiver Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – U3TXIP[2:0] UART3 Transmitter Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – U3RXIP[2:0] UART3 Receiver Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – U3EIP[2:0] UART3 External Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

dsPIC33CK512MPT608 Family

Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.42 Interrupt Priority Register 15

Name: IPC15
Offset: 0x85E

Bit	15	14	13	12	11	10	9	8
		PTGSTEP[2:0]				JTAGIP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bit	7	6	5	4	3	2	1	0
		ICDIP[2:0]				SPI3TXIP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – PTGSTEP[2:0] PTG Step Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – JTAGIP[2:0] JTAG Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – ICDIP[2:0] ICD Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – SPI3TXIP[2:0] SPI3 Transmitter Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

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Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

8.7.43 Interrupt Priority Register 16

Name: IPC16
Offset: 0x860

Bit	15	14	13	12	11	10	9	8
		PWM1IP[2:0]				QE13IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bit	7	6	5	4	3	2	1	0
		I2C2BCIP[2:0]				I2C1BCIP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – PWM1IP[2:0] Pulse-Width Modulation 1 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – QE13IP[2:0] QE13 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – I2C2BCIP[2:0] I2C2 Bus Collision Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – I2C1BCIP[2:0] I2C1 Bus Collision Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

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Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

8.7.44 Interrupt Priority Register 17

Name: IPC17
Offset: 0x862

Bit	15	14	13	12	11	10	9	8
						PWM4IP[2:0]		
Access						R/W	R/W	R/W
Reset						1	0	0

Bit	7	6	5	4	3	2	1	0
		PWM3IP[2:0]				PWM2IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 10:8 – PWM4IP[2:0] Pulse-Width Modulation 4 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – PWM3IP[2:0] Pulse-Width Modulation 3 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – PWM2IP[2:0] Pulse-Width Modulation 2 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.45 Interrupt Priority Register 18

Name: IPC18

Offset: 0x864

Bit	15	14	13	12	11	10	9	8
		CNDIP[2:0]				PWM8IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0
Bit	7	6	5	4	3	2	1	0
						PWM6IP[2:0]		
Access						R/W	R/W	R/W
Reset						1	0	0

Bits 14:12 – CNDIP[2:0] Change Notice D Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – PWM8IP[2:0] Pulse-Width Modulation 8 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – PWM6IP[2:0] Pulse-Width Modulation 6 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.46 Interrupt Priority Register 19

Name: IPC19

Offset: 0x866

Bit	15	14	13	12	11	10	9	8
		CMP3IP[2:0]				CMP2IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		0	0	0		0	0	0

Bit	7	6	5	4	3	2	1	0
		CMP1IP[2:0]				CNEIP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		0	0	0		0	0	0

Bits 14:12 – CMP3IP[2:0] Comparator 3 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – CMP2IP[2:0] Comparator 2 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – CMP1IP[2:0] Comparator 1 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – CNEIP[2:0] Change Notice E Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

dsPIC33CK512MPT608 Family

Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

8.7.47 Interrupt Priority Register 20

Name: IPC20
Offset: 0x868

Bit	15	14	13	12	11	10	9	8
	PTG1IP[2:0]					PTG0IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bit	7	6	5	4	3	2	1	0
	PTGWDTIP[2:0]					CMP4IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		0	0	0

Bits 14:12 – PTG1IP[2:0] Peripheral Trigger Generator 1 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – PTG0IP[2:0] Peripheral Trigger Generator 0 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – PTGWDTIP[2:0] Watchdog Timer Time-out Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – CMP4IP[2:0] Comparator 4 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

dsPIC33CK512MPT608 Family

Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

8.7.48 Interrupt Priority Register 21

Name: IPC21
Offset: 0x86A

Bit	15	14	13	12	11	10	9	8
		SENT1EIP[2:0]				SENT1IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bit	7	6	5	4	3	2	1	0
		PTG3IP[2:0]				PTG2IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – SENT1EIP[2:0] SENT1 External Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – SENT1IP[2:0] SENT1 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – PTG3IP[2:0] Peripheral Trigger Generator 3 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – PTG2IP[2:0] Peripheral Trigger Generator 2 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

dsPIC33CK512MPT608 Family

Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

8.7.49 Interrupt Priority Register 22

Name: IPC22
Offset: 0x86C

Bit	15	14	13	12	11	10	9	8
		ADCAN0IP[2:0]				ADCIP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		0	0	0

Bit	7	6	5	4	3	2	1	0
		SENT2EIP[2:0]				SENT2IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – ADCAN0IP[2:0] ADC AN0 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – ADCIP[2:0] ADC Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – SENT2EIP[2:0] SENT2 Error Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – SENT2IP[2:0] SENT2 TX/RX Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

dsPIC33CK512MPT608 Family

Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

8.7.50 Interrupt Priority Register 23

Name: IPC23
Offset: 0x86E

Bit	15	14	13	12	11	10	9	8
		ADCAN4IP[2:0]				ADCAN3IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bit	7	6	5	4	3	2	1	0
		ADCAN2IP[2:0]				ADCAN1IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – ADCAN4IP[2:0] ADC AN4 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – ADCAN3IP[2:0] ADC AN3 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – ADCAN2IP[2:0] ADC AN2 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – ADCAN1IP[2:0] ADC AN1 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

dsPIC33CK512MPT608 Family

Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

8.7.51 Interrupt Priority Register 24

Name: IPC24
Offset: 0x870

Bit	15	14	13	12	11	10	9	8
		ADCAN8IP[2:0]				ADCAN7IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bit	7	6	5	4	3	2	1	0
		ADCAN6IP[2:0]				ADCAN5IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – ADCAN8IP[2:0] ADC AN8 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – ADCAN7IP[2:0] ADC AN7 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – ADCAN6IP[2:0] ADC AN6 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – ADCAN5IP[2:0] ADC AN5 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

dsPIC33CK512MPT608 Family

Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

8.7.52 Interrupt Priority Register 25

Name: IPC25

Offset: 0x872

Bit	15	14	13	12	11	10	9	8
		ADCAN12IP[2:0]				ADCAN11IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bit	7	6	5	4	3	2	1	0
		ADCAN10IP[2:0]				ADCAN9IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – ADCAN12IP[2:0] ADC AN12 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – ADCAN11IP[2:0] ADC AN11 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – ADCAN10IP[2:0] ADC AN10 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – ADCAN9IP[2:0] ADC AN9 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

dsPIC33CK512MPT608 Family

Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

8.7.53 Interrupt Priority Register 26

Name: IPC26
Offset: 0x874

Bit	15	14	13	12	11	10	9	8
		ADCAN16IP[2:0]				ADCAN15IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bit	7	6	5	4	3	2	1	0
		ADCAN14IP[2:0]				ADCAN13IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – ADCAN16IP[2:0] ADC AN16 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – ADCAN15IP[2:0] ADC AN15 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – ADCAN14IP[2:0] ADC AN14 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – ADCAN13IP[2:0] ADC AN13 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

dsPIC33CK512MPT608 Family

Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

8.7.54 Interrupt Priority Register 27

Name: IPC27
Offset: 0x876

Bit	15	14	13	12	11	10	9	8
		ADCAN20IP[2:0]				ADCAN19IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bit	7	6	5	4	3	2	1	0
		ADCAN18IP[2:0]				ADCAN17IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – ADCAN20IP[2:0] ADC AN20 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – ADCAN19IP[2:0] ADC AN19 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – ADCAN18IP[2:0] ADC AN18 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – ADCAN17IP[2:0] ADC AN17 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

dsPIC33CK512MPT608 Family

Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.55 Interrupt Priority Register 28

Name: IPC28
Offset: 0x878

Bit	15	14	13	12	11	10	9	8
		ADFLTIP[2:0]				ADCAN23IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bit	7	6	5	4	3	2	1	0
		ADCAN22IP[2:0]				ADCAN21IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – ADFLTIP[2:0] ADC Fault Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – ADCAN23IP[2:0] ADC AN23 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – ADCAN22IP[2:0] ADC AN22 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – ADCAN21IP[2:0] ADC AN21 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

dsPIC33CK512MPT608 Family

Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.56 Interrupt Priority Register 29

Name: IPC29
Offset: 0x87A

Bit	15	14	13	12	11	10	9	8
	ADCMP3IP[2:0]					ADCMP2IP[2:0]		
Access	R/W			R/W		R/W	R/W	R/W
Reset	1			0		1	0	0
Bit	7	6	5	4	3	2	1	0
	ADCMP1IP[2:0]					ADCMP0IP[2:0]		
Access	R/W			R/W		R/W	R/W	R/W
Reset	1			0		1	0	0

Bits 14:12 – ADCMP3IP[2:0] ADC Digital Comparator 3 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – ADCMP2IP[2:0] ADC Digital Comparator 2 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – ADCMP1IP[2:0] ADC Digital Comparator 1 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – ADCMP0IP[2:0] ADC Digital Comparator 0 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

dsPIC33CK512MPT608 Family

Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.57 Interrupt Priority Register 30

Name: IPC30
Offset: 0x87C

Bit	15	14	13	12	11	10	9	8
	ADFLTR3IP[2:0]					ADFLTR2IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bit	7	6	5	4	3	2	1	0
	ADFLTR1IP[2:0]					ADFLTR0IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – ADFLTR3IP[2:0] ADC Oversample Filter 3 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – ADFLTR2IP[2:0] ADC Oversample Filter 2 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – ADFLTR1IP[2:0] ADC Oversample Filter 1 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – ADFLTR0IP[2:0] ADC Oversample Filter 0 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

dsPIC33CK512MPT608 Family

Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

8.7.58 Interrupt Priority Register 31

Name: IPC31
Offset: 0x87E

Bit	15	14	13	12	11	10	9	8
	SPI2EIP[2:0]					SPI1EIP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bit	7	6	5	4	3	2	1	0
	CLC2PEIP[2:0]					CLC1PEIP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – SPI2EIP[2:0] SPI2 Error Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – SPI1EIP[2:0] SPI1 Error Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – CLC2PEIP[2:0] CLC2 Positive Edge Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – CLC1PEIP[2:0] CLC1 Positive Edge Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

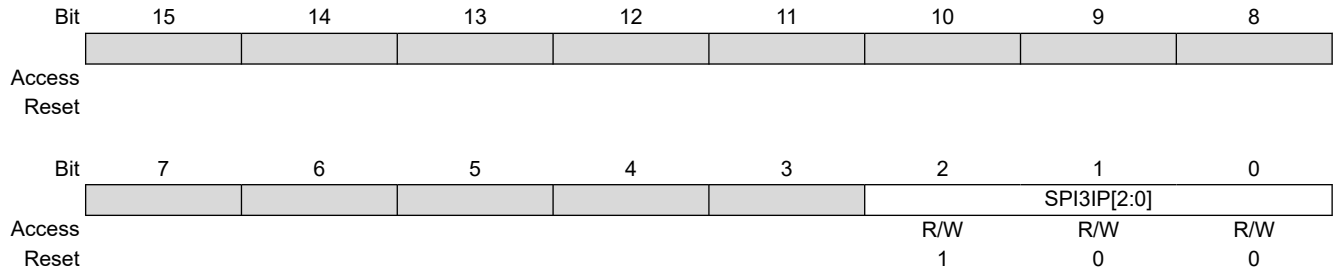
dsPIC33CK512MPT608 Family

Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

8.7.59 Interrupt Priority Register 32

Name: IPC32
Offset: 0x880



Bits 2:0 – SPI3IP[2:0] SPI3 Error Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.60 Interrupt Priority Register 35

Name: IPC35

Offset: 0x886

Bit	15	14	13	12	11	10	9	8
		MI2C3IP[2:0]				SI2C3IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		0	0	0
Bit	7	6	5	4	3	2	1	0
Access								
Reset								

Bits 14:12 – MI2C3IP[2:0] I2C3 Host Event Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – SI2C3IP[2:0] I2C3 Client Event Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.61 Interrupt Priority Register 36

Name: IPC36

Offset: 0x888

Bit	15	14	13	12	11	10	9	8
		ADCAN29IP[2:0]				ADCAN28IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bit	7	6	5	4	3	2	1	0
		ADCAN27IP[2:0]				I2C3BCIP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – ADCAN29IP[2:0] ADC AN29 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – ADCAN28IP[2:0] ADC AN28 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – ADCAN27IP[2:0] ADC AN27 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – I2C3BCIP[2:0] I2C3 Bus Collision Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

dsPIC33CK512MPT608 Family

Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.62 Interrupt Priority Register 37

Name: IPC37
Offset: 0x88A

Bit	15	14	13	12	11	10	9	8
		ADCAN26IP[2:0]				CCT7IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bit	7	6	5	4	3	2	1	0
		CCP7IP[2:0]				ADCAN30IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – ADCAN26IP[2:0] ADC AN26 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – CCT7IP[2:0] Capture/Compare/Timer7 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – CCP7IP[2:0] Input Capture/Output Compare 7 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – ADCAN30IP[2:0] ADC AN30 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

dsPIC33CK512MPT608 Family

Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.63 Interrupt Priority Register 38

Name: IPC38
Offset: 0x88C

Bit	15	14	13	12	11	10	9	8
	DMA7IP[2:0]					DMA6IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bit	7	6	5	4	3	2	1	0
	CCT8IP[2:0]					CCP8IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – DMA7IP[2:0] Direct Memory Access 7 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – DMA6IP[2:0] Direct Memory Access 6 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – CCT8IP[2:0] Capture/Compare/Timer8 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – CCP8IP[2:0] Input Capture/Output Compare 8 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

dsPIC33CK512MPT608 Family

Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

8.7.64 Interrupt Priority Register 42

Name: IPC42
Offset: 0x894

Bit	15	14	13	12	11	10	9	8
		PEVTCIP[2:0]				PEVTBIP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bit	7	6	5	4	3	2	1	0
		PEVTAIP[2:0]						
Access		R/W	R/W	R/W				
Reset		1	0	0				

Bits 14:12 – PEVTCIP[2:0] PWM Event C Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – PEVTBIP[2:0] PWM Event B Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – PEVTAIP[2:0] PWM Event A Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

8.7.65 Interrupt Priority Register 43

Name: IPC43
Offset: 0x896

Bit	15	14	13	12	11	10	9	8
		CLC3PEIP[2:0]				PEVTFIP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bit	7	6	5	4	3	2	1	0
		PEVTEIP[2:0]				PEVTDIP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – CLC3PEIP[2:0] CLC3 Positive Edge Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – PEVTFIP[2:0] PWM Event F Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – PEVTEIP[2:0] PWM Event E Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – PEVTDIP[2:0] PWM Event D Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

dsPIC33CK512MPT608 Family

Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

dsPIC33CK512MPT608 Family

Interrupt Controller

8.7.66 Interrupt Priority Register 44

Name: IPC44
Offset: 0x898

Bit	15	14	13	12	11	10	9	8
		CLC3NEIP[2:0]				CLC2NEIP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0
Bit	7	6	5	4	3	2	1	0
		CLC1NEIP[2:0]				CLC4PEIP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – CLC3NEIP[2:0] CLC3 Negative Edge Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – CLC2NEIP[2:0] CLC2 Negative Edge Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – CLC1NEIP[2:0] CLC1 Negative Edge Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – CLC4PEIP[2:0] CLC4 Positive Edge Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

dsPIC33CK512MPT608 Family

Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

8.7.67 Interrupt Priority Register 45

Name: IPC45
Offset: 0x89A

Bit	15	14	13	12	11	10	9	8
							CCT9IP[2:0]	
Access						R/W	R/W	R/W
Reset						1	0	0

Bit	7	6	5	4	3	2	1	0
		CCP9IP[2:0]				CLC4NEIP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 10:8 – CCT9IP[2:0] Capture/Compare/Timer9 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – CCP9IP[2:0] Input Capture/Output Compare 9 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – CLC4NEIP[2:0] CLC4 Negative Edge Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

8.7.68 Interrupt Priority Register 46

Name: IPC46
Offset: 0x89C

Bit	15	14	13	12	11	10	9	8
		CMP5IP[2:0]						
Access		R/W	R/W	R/W				
Reset		0	0	0				
Bit	7	6	5	4	3	2	1	0
Access								
Reset								

Bits 14:12 – CMP5IP[2:0] Comparator 5 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

8.7.69 Interrupt Priority Register 47

Name: IPC47
Offset: 0x89E

Bit	15	14	13	12	11	10	9	8
	U3EVTIP[2:0]					U2EVTIP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bit	7	6	5	4	3	2	1	0
	U1EVTIP[2:0]					CMP6IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 14:12 – U3EVTIP[2:0] UART3 Event Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 10:8 – U2EVTIP[2:0] UART2 Event Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 6:4 – U1EVTIP[2:0] UART1 Event Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – CMP6IP[2:0] Comparator 6 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3

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Interrupt Controller

Value	Description
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

8.7.70 Interrupt Priority Register 48

Name: IPC48
Offset: 0x8A0

Bit	15	14	13	12	11	10	9	8
Access								
Reset								

Bit	7	6	5	4	3	2	1	0
		ADCAN25IP[2:0]				ADCAN24IP[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		1	0	0

Bits 6:4 – ADCAN25IP[2:0] ADC AN25 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

Bits 2:0 – ADCAN24IP[2:0] ADC AN24 Interrupt Priority bits

Value	Description
7	Interrupt Priority Level 7 (highest)
6	Interrupt Priority Level 6
5	Interrupt Priority Level 5
4	Interrupt Priority Level 4 (default)
3	Interrupt Priority Level 3
2	Interrupt Priority Level 2
1	Interrupt Priority Level 1
0	Interrupt Priority Level 0 (lowest)

8.7.71 CPU STATUS Register

Name: SR
Offset: 0x42

Notes:

1. The IPL[2:0] bits are concatenated with the IPL[3] bit (CORCON[3]) to form the CPU Interrupt Priority Level. The value in parentheses indicates the IPL, if IPL[3] = 1. User interrupts are disabled when IPL[3] = 1.
2. The IPL[2:0] Status bits are read-only when the NSTDIS bit (INTCON1[15]) = 1.

Legend: C = Clearable bit

Bit	15	14	13	12	11	10	9	8
Access								
Reset								

Bit	7	6	5	4	3	2	1	0
		IPL[2:0]						
Access	R/W	R/W	R/W					
Reset	0	0	0					

Bits 7:5 – IPL[2:0] CPU Interrupt Priority Level Status bits^(1,2)

Value	Description
111	CPU Interrupt Priority Level is 7 (15); user interrupts are disabled
110	CPU Interrupt Priority Level is 6 (14)
101	CPU Interrupt Priority Level is 5 (13)
100	CPU Interrupt Priority Level is 4 (12)
011	CPU Interrupt Priority Level is 3 (11)
010	CPU Interrupt Priority Level is 2 (10)
001	CPU Interrupt Priority Level is 1 (9)
000	CPU Interrupt Priority Level is 0 (8)

8.7.72 Core Control Register

Name: CORCON
Offset: 0x44

Note:

- The IPL3 bit is concatenated with the IPL[2:0] bits (SR[7:5]) to form the CPU Interrupt Priority Level.

Legend: C = Clearable bit

Bit	15	14	13	12	11	10	9	8
	VAR							
Access	R/W							
Reset	0							

Bit	7	6	5	4	3	2	1	0
					IPL3			
Access					R/C			
Reset					0			

Bit 15 – VAR Variable Exception Processing Latency Control bit

Value	Description
1	Variable exception processing is enabled
0	Fixed exception processing is enabled

Bit 3 – IPL3 CPU Interrupt Priority Level Status bit 3⁽¹⁾

Value	Description
1	CPU Interrupt Priority Level is greater than 7
0	CPU Interrupt Priority Level is 7 or less

8.7.73 Interrupt Control Register 1

Name: INTCON1

Offset: 0x8C0

Bit	15	14	13	12	11	10	9	8
	NSTDIS	OVAERR	OVBERR	COVAERR	COVBERR	OVATE	OVBTBTE	COVTE
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	SFTACERR	DIV0ERR		MATHERR	ADDRERR	STKERR	OSCFAIL	
Access	R/W	R/W		R/W	R/W	R/W	R/W	
Reset	0	0		0	0	0	0	

Bit 15 – NSTDIS Interrupt Nesting Disable bit

Value	Description
1	Interrupt nesting is disabled
0	Interrupt nesting is enabled

Bit 14 – OVAERR Accumulator A Overflow Trap Flag bit

Value	Description
1	Trap was caused by overflow of Accumulator A
0	Trap was not caused by overflow of Accumulator A

Bit 13 – OVBERR Accumulator B Overflow Trap Flag bit

Value	Description
1	Trap was caused by overflow of Accumulator B
0	Trap was not caused by overflow of Accumulator B

Bit 12 – COVAERR Accumulator A Catastrophic Overflow Trap Flag bit

Value	Description
1	Trap was caused by catastrophic overflow of Accumulator A
0	Trap was not caused by catastrophic overflow of Accumulator A

Bit 11 – COVBERR Accumulator B Catastrophic Overflow Trap Flag bit

Value	Description
1	Trap was caused by catastrophic overflow of Accumulator B
0	Trap was not caused by catastrophic overflow of Accumulator B

Bit 10 – OVATE Accumulator A Overflow Trap Enable bit

Value	Description
1	Trap overflow of Accumulator A
0	Trap is disabled

Bit 9 – OVBTBTE Accumulator B Overflow Trap Enable bit

Value	Description
1	Trap overflow of Accumulator B
0	Trap is disabled

Bit 8 – COVTE Catastrophic Overflow Trap Enable bit

Value	Description
1	Trap on catastrophic overflow of Accumulator A or B is enabled
0	Trap is disabled

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Bit 7 – SFTACERR Shift Accumulator Error Status bit

Value	Description
1	Math error trap was caused by an invalid accumulator shift
0	Math error trap was not caused by an invalid accumulator shift

Bit 6 – DIV0ERR Divide-by-Zero Error Status bit

Value	Description
1	Math error trap was caused by a divide-by-zero
0	Math error trap was not caused by a divide-by-zero

Bit 4 – MATHERR Math Error Status bit

Value	Description
1	Math error trap has occurred
0	Math error trap has not occurred

Bit 3 – ADDRERR Address Error Trap Status bit

Value	Description
1	Address error trap has occurred
0	Address error trap has not occurred

Bit 2 – STKERR Stack Error Trap Status bit

Value	Description
1	Stack error trap has occurred
0	Stack error trap has not occurred

Bit 1 – OSCFAIL Oscillator Failure Trap Status bit

Value	Description
1	Oscillator failure trap has occurred
0	Oscillator failure trap has not occurred

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8.7.74 Interrupt Control Register 2

Name: INTCON2
Offset: 0x8C2

Bit	15	14	13	12	11	10	9	8
	GIE	DISI	SWTRAP					AIVTEN
Access	R/W	R/W	R/W					R/W
Reset	1	0	0					0

Bit	7	6	5	4	3	2	1	0
					INT3EP	INT2EP	INT1EP	INT0EP
Access					R/W	R/W	R/W	R/W
Reset					0	0	0	0

Bit 15 – GIE Global Interrupt Enable bit

Value	Description
1	Interrupts and associated IE bits are enabled
0	Interrupts are disabled, but traps are still enabled

Bit 14 – DISI DISI Instruction Status bit

Value	Description
1	DISI instruction is active
0	DISI instruction is not active

Bit 13 – SWTRAP Software Trap Status bit

Value	Description
1	Software trap is enabled
0	Software trap is disabled

Bit 8 – AIVTEN Alternate Interrupt Vector Table Enable bit

Value	Description
1	Alternate Interrupt Vector Table enabled (AIVTDIS = 0 also required)
0	Alternate Interrupt Vector Table disabled

Bit 3 – INT3EP External Interrupt 3 Edge Detect Polarity Select bit

Value	Description
1	Interrupt on negative edge
0	Interrupt on positive edge

Bit 2 – INT2EP External Interrupt 2 Edge Detect Polarity Select bit

Value	Description
1	Interrupt on negative edge
0	Interrupt on positive edge

Bit 1 – INT1EP External Interrupt 1 Edge Detect Polarity Select bit

Value	Description
1	Interrupt on negative edge
0	Interrupt on positive edge

Bit 0 – INT0EP External Interrupt 0 Edge Detect Polarity Select bit

Value	Description
1	Interrupt on negative edge
0	Interrupt on positive edge

8.7.75 Interrupt Control Register 3

Name: INTCON3
Offset: 0x8C4

Bit	15	14	13	12	11	10	9	8
	DMT						CAN	NAE
Access	R/W						R/W	R/W
Reset	0						0	0

Bit	7	6	5	4	3	2	1	0
		CAN2	DAE	DOOVR				APLL
Access		R/W	R/W	R/W				R/W
Reset		0	0	0				0

Bit 15 – DMT Deadman Timer Expiration bit

Value	Description
1	DMT Soft Trap has occurred
0	DMT Soft Trap has not occurred

Bit 9 – CAN CAN Address Error Soft Trap Status bit

Value	Description
1	CAN address error soft trap has occurred
0	CAN address error soft trap has not occurred

Bit 8 – NAE NVM Address Error Soft Trap Status bit

Value	Description
1	NVM address error soft trap has occurred
0	NVM address error soft trap has not occurred

Bit 6 – CAN2 CAN2 Address Error Soft Trap Status bit

Value	Description
1	CAN2 address error soft trap has occurred
0	CAN2 address error soft trap has not occurred

Bit 5 – DAE DMA Address Error (Soft) Trap Status bit

Value	Description
1	DMA address error trap has occurred
0	DMA address error trap has not occurred

Bit 4 – DOOVR DO Stack Overflow Soft Trap Status bit

Value	Description
1	DO stack overflow soft trap has occurred
0	DO stack overflow soft trap has not occurred

Bit 0 – APLL Auxiliary PLL Loss of Lock Soft Trap Status bit

Value	Description
1	APLL lock soft trap has occurred
0	APLL lock soft trap has not occurred

8.7.76 Interrupt Control Register 4

Name: INTCON4
Offset: 0x8C6

Bit	15	14	13	12	11	10	9	8
Access								
Reset								

Bit	7	6	5	4	3	2	1	0
							ECCDBE	SGHT
Access							R/W	R/W
Reset							0	0

Bit 1 – ECCDBE ECC Double-Bit Error Trap bit

Value	Description
1	ECC double-bit error trap has occurred
0	ECC double-bit error trap has not occurred

Bit 0 – SGHT Software Generated Hard Trap Status bit

Value	Description
1	Software generated hard trap has occurred
0	Software generated hard trap has not occurred

8.7.77 Interrupt Control and Status Register

Name: INTTREG
Offset: 0x8C8

Bit	15	14	13	12	11	10	9	8
			VHOLD				ILR[3:0]	
Access			R		R	R	R	R
Reset			0		0	0	0	0

Bit	7	6	5	4	3	2	1	0
								VECNUM[7:0]
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit 13 – VHOLD Vector Number Capture Enable bit

Value	Description
1	VECNUM[7:0] bits read current value of vector number encoding tree (i.e., highest priority pending interrupt)
0	Vector number latched into VECNUM[7:0] at Interrupt Acknowledge and retained until next IACK

Bits 11:8 – ILR[3:0] New CPU Interrupt Priority Level bits

Value	Description
1111	CPU Interrupt Priority Level is 15
. . .	
0001	CPU Interrupt Priority Level is 1
0000	CPU Interrupt Priority Level is 0

Bits 7:0 – VECNUM[7:0] Vector Number of Pending Interrupt bits

Value	Description
11111111	255, Reserved; do not use
. . .	
00001001	9, IC1 – Input Capture 1
00001000	8, INT0 – External Interrupt 0
00000111	7, Reserved; do not use
00000110	6, Generic soft error trap
00000101	5, Reserved; do not use
00000100	4, Math error trap
00000011	3, Stack error trap
00000010	2, Generic hard trap
00000001	1, Address error trap
00000000	0, Oscillator fail trap

9. I/O Ports

Note: This data sheet summarizes the features of the dsPIC33CK512MPT608 family of devices. It is not intended to be a comprehensive source. To complement the information in this data sheet, refer to “I/O Ports with Edge Detect” (www.microchip.com/DS70005322) in the “dsPIC33/PIC24 Family Reference Manual”.

Many of the device pins are shared among the peripherals and the Parallel I/O ports. All I/O input ports feature Schmitt Trigger inputs for improved noise immunity.

Some of the key features of the I/O ports are:

- Individual Output Pin Open-Drain Enable/Disable
- Individual Input Pin Weak Pull-up and Pull-Down
- Monitor Selective Inputs and Generate Interrupt when Change in Pin State is Detected
- Operation during Sleep and Idle modes

9.1 Parallel I/O (PIO) Ports

All port pins have 12 registers directly associated with their operation as digital I/Os. The Data Direction register (TRISx) determines whether the pin is an input or an output. If the data direction bit is a ‘1’, then the pin is an input.

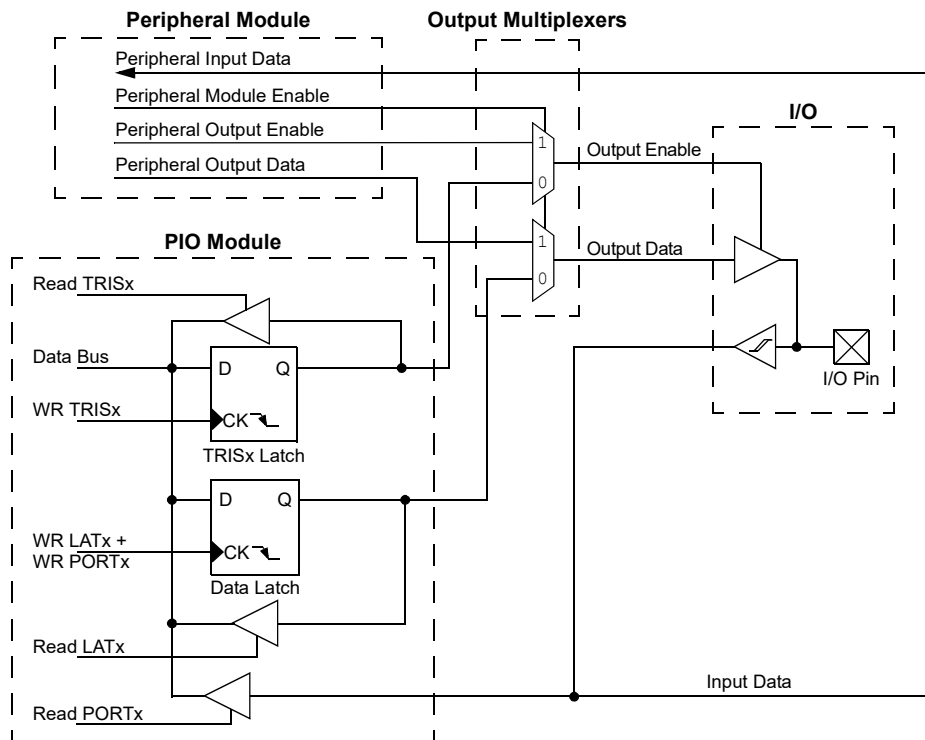
All port pins are defined as inputs after a Reset. Reads from the latch (LATx), read the latch. Writes to the latch, write the latch. Reads from the port (PORTx), read the port pins, while writes to the port pins, write the latch. Any bit and its associated data and control registers that are not valid for a particular device are disabled. This means the corresponding LATx and TRISx registers, and the port pin are read as zeros.

When a pin is shared with another peripheral or function that is defined as an input only, it is nevertheless regarded as a dedicated port because there is no other competing source of outputs. [Table 9-1](#) shows the pin availability. [Figure 9-1](#) shows the 5V input tolerant pins across this device.

Table 9-1. Pin and ANSELx Availability

Device	Rx15	Rx14	Rx13	Rx12	Rx11	Rx10	Rx9	Rx8	Rx7	Rx6	Rx5	Rx4	Rx3	Rx2	Rx1	Rx0
PORTA																
dsPIC33CKXXXMPT608-I/PT	—	—	—	—	—	—	—	—	—	—	—	X	X	X	X	X
ANSELA	—	—	—	—	—	—	—	—	—	—	—	X	X	X	X	X
PORTB																
dsPIC33CKXXXMPT608-I/PT	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
ANSELB	—	—	—	—	—	—	X	X	X	—	—	—	X	X	X	X
PORTC																
dsPIC33CKXXXMPT608-I/PT	X	X	X	X	—	—	X	X	X	X	—	—	X	X	X	X
ANSELC	—	—	—	—	—	—	—	—	X	X	—	—	X	X	X	X
PORTD																
dsPIC33CKXXXMPT608-I/PT	X	X	X	X	X	X	X	X	X	X	X	—	—	X	X	X
ANSELD	—	—	X	—	X	X	—	—	—	—	—	—	—	—	—	—
PORTE																
dsPIC33CKXXXMPT608-I/PT	X	X	—	X	X	X	X	X	X	X	X	X	X	X	X	X
ANSELE	—	—	—	—	—	—	—	—	—	—	—	—	X	X	X	X

Figure 9-1. Block Diagram of a Typical Shared Port Structure



9.1.1 Open-Drain Configuration

In addition to the PORTx, LATx and TRISx registers for data control, port pins can also be individually configured for either digital or open-drain output. This is controlled by the Open-Drain Enable for PORTx register, ODCx, associated with each port. Setting any of the bits configures the corresponding pin to act as an open-drain output.

The open-drain feature allows the generation of outputs, other than V_{DD} , by using external pull-up resistors. The maximum open-drain voltage allowed on any pin is the same as the maximum V_{IH} specification for that particular pin.

9.2 Configuring Analog and Digital Port Pins

The ANSELx registers control the operation of the analog port pins. The port pins that are to function as analog inputs or outputs must have their corresponding ANSELx and TRISx bits set. In order to use port pins for I/O functionality with digital modules, such as timers, UARTs, etc., the corresponding ANSELx bit must be cleared.

The ANSELx registers have a default value of 0xFFFF; therefore, all pins that share analog functions are analog (not digital) by default.

Pins with analog functions affected by the ANSELx registers are listed with a buffer type of analog in the Pinout I/O Descriptions (see [1. Device Overview](#)).

If the TRISx bit is cleared (output) while the ANSELx bit is set, the digital output level (V_{OH} or V_{OL}) is converted by an analog peripheral, such as the ADC module or comparator module.

When the PORTx register is read, all pins configured as analog input channels are read as cleared (a low level).

Pins configured as digital inputs do not convert an analog input. Analog levels on any pin, defined as a digital input (including the ANx pins), can cause the input buffer to consume current that exceeds the device specifications.

9.2.1 I/O Port Write/Read Timing

One instruction cycle is required between a port direction change or port write operation and a read operation of the same port. Typically, this instruction would be a NOP.

9.2.2 Port Controls/Status Registers

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x00 ... 0x0DFF	Reserved									
0x0E00	ANSELA	15:8	ANSELA[15:8]							
		7:0	ANSELA[7:0]							
0x0E02	TRISA	15:8	TRISA[15:8]							
		7:0	TRISA[7:0]							
0x0E04	PORTA	15:8	PORTA[15:8]							
		7:0	PORTA[7:0]							
0x0E06	LATA	15:8	LATA[15:8]							
		7:0	LATA[7:0]							
0x0E08	ODCA	15:8	ODCA[15:8]							
		7:0	ODCA[7:0]							
0x0E0A	CNPUA	15:8	CNPUA[15:8]							
		7:0	CNPUA[7:0]							
0x0E0C	CNPDA	15:8	CNPDA[15:8]							
		7:0	CNPDA[7:0]							
0x0E0E	CNCONA	15:8	ON				CNSTYLE			
		7:0								
0x0E10	CNEN0A	15:8	CNEN0A[15:8]							
		7:0	CNEN0A[7:0]							
0x0E12	CNSTATA	15:8	CNSTATA[15:8]							
		7:0	CNSTATA[7:0]							
0x0E14	CNEN1A	15:8	CNEN1A[15:8]							
		7:0	CNEN1A[7:0]							
0x0E16	CNFA	15:8	CNFA[15:8]							
		7:0	CNFA[7:0]							
0x0E18 ... 0x0E1B	Reserved									
0x0E1C	ANSELB	15:8	ANSELB[15:8]							
		7:0	ANSELB[7:0]							
0x0E1E	TRISB	15:8	TRISB[15:8]							
		7:0	TRISB[7:0]							
0x0E20	PORTB	15:8	PORTB[15:8]							
		7:0	PORTB[7:0]							
0x0E22	LATB	15:8	LATB[15:8]							
		7:0	LATB[7:0]							
0x0E24	ODCB	15:8	ODCB[15:8]							
		7:0	ODCB[7:0]							
0x0E26	CNPUB	15:8	CNPUB[15:8]							
		7:0	CNPUB[7:0]							
0x0E28	CNPDB	15:8	CNPDB[15:8]							
		7:0	CNPDB[7:0]							
0x0E2A	CNCONB	15:8	ON				CNSTYLE			
		7:0								
0x0E2C	CNEN0B	15:8	CNEN0B[15:8]							
		7:0	CNEN0B[7:0]							
0x0E2E	CNSTATB	15:8	CNSTATB[15:8]							
		7:0	CNSTATB[7:0]							
0x0E30	CNEN1B	15:8	CNEN1B[15:8]							
		7:0	CNEN1B[7:0]							
0x0E32	CNFB	15:8	CNFB[15:8]							
		7:0	CNFB[7:0]							
0x0E34	TRISC	15:8	TRISC[15:8]							
		7:0	TRISC[7:0]							

.....continued

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x0E36 ... 0x0E37	Reserved									
0x0E38	ANSELC	15:8	ANSELC[15:8]							
		7:0	ANSELC[7:0]							
0x0E3A ... 0x0E3B	Reserved									
0x0E3C	PORTC	15:8	PORTC[15:8]							
		7:0	PORTC[7:0]							
0x0E3E	LATC	15:8	LATC[15:8]							
		7:0	LATC[7:0]							
0x0E40	ODCC	15:8	ODCC[15:8]							
		7:0	ODCC[7:0]							
0x0E42	CNPUC	15:8	CNPUC[15:8]							
		7:0	CNPUC[7:0]							
0x0E44	CNPDC	15:8	CNPDC[15:8]							
		7:0	CNPDC[7:0]							
0x0E46	CNCONC	15:8	ON				CNSTYLE			
		7:0								
0x0E48	CNEN0C	15:8	CNEN0C[15:8]							
		7:0	CNEN0C[7:0]							
0x0E4A	CNSTATC	15:8	CNSTATC[15:8]							
		7:0	CNSTATC[7:0]							
0x0E4C	CNEN1C	15:8	CNEN1C[15:8]							
		7:0	CNEN1C[7:0]							
0x0E4E	CNFC	15:8	CNFC[15:8]							
		7:0	CNFC[7:0]							
0x0E50 ... 0x0E53	Reserved									
0x0E54	ANSELD	15:8	ANSELD[15:8]							
		7:0	ANSELD[7:0]							
0x0E56	TRISD	15:8	TRISD[15:8]							
		7:0	TRISD[7:0]							
0x0E58	PORTD	15:8	PORTD[15:8]							
		7:0	PORTD[7:0]							
0x0E5A	LATD	15:8	LATD[15:8]							
		7:0	LATD[7:0]							
0x0E5C	ODCD	15:8	ODCD[15:8]							
		7:0	ODCD[7:0]							
0x0E5E	CNPUD	15:8	CNPUD[15:8]							
		7:0	CNPUD[7:0]							
0x0E60	CNPDD	15:8	CNPDD[15:8]							
		7:0	CNPDD[7:0]							
0x0E62	CNCOND	15:8	ON				CNSTYLE			
		7:0								
0x0E64	CNEN0D	15:8	CNEN0D[15:8]							
		7:0	CNEN0D[7:0]							
0x0E66	CNSTATD	15:8	CNSTATD[15:8]							
		7:0	CNSTATD[7:0]							
0x0E68	CNEN1D	15:8	CNEN1D[15:8]							
		7:0	CNEN1D[7:0]							
0x0E6A	CNFD	15:8	CNFD[15:8]							
		7:0	CNFD[7:0]							
0x0E6C ... 0x0E6F	Reserved									

.....continued

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x0E70	ANSELE	15:8					ANSELE[15:8]			
		7:0					ANSELE[7:0]			
0x0E72	TRISE	15:8					TRISE[15:8]			
		7:0					TRISE[7:0]			
0x0E74	PORTE	15:8					PORTE[15:8]			
		7:0					PORTE[7:0]			
0x0E76	LATE	15:8					LATE[15:8]			
		7:0					LATE[7:0]			
0x0E78	ODCE	15:8					ODCE[15:8]			
		7:0					ODCE[7:0]			
0x0E7A	CNPUE	15:8					CNPUE[15:8]			
		7:0					CNPUE[7:0]			
0x0E7C	CNPDE	15:8					CNPDE[15:8]			
		7:0					CNPDE[7:0]			
0x0E7E	CNCONE	15:8	ON				CNSTYLE			
		7:0								
0x0E80	CNEN0E	15:8					CNEN0E[15:8]			
		7:0					CNEN0E[7:0]			
0x0E82	CNSTATE	15:8					CNSTATE[15:8]			
		7:0					CNSTATE[7:0]			
0x0E84	CNEN1E	15:8					CNEN1E[15:8]			
		7:0					CNEN1E[7:0]			
0x0E86	CNFE	15:8					CNFE[15:8]			
		7:0					CNFE[7:0]			

9.2.2.1 Analog Select for PORTx Register**Name:** ANSELx**Offset:** 0xE00, 0xE1C, 0xE38, 0xE54, 0xE70

Bit	15	14	13	12	11	10	9	8
	ANSELx[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	1	1	1	1	1	1	1
Bit	7	6	5	4	3	2	1	0
	ANSELx[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	1	1	1	1	1	1	1

Bits 15:0 – ANSELx[15:0] Analog Select for PORTx bits

Value	Description
1	Analog input is enabled and digital input is disabled on the PORTx[n] pin
0	Analog input is disabled and digital input is enabled on the PORTx[n] pin

9.2.2.2 Output Enable for PORTx Register**Name:** TRISx**Offset:** 0xE02, 0xE1E, 0xE34, 0xE56, 0xE72

Bit	15	14	13	12	11	10	9	8
	TRISx[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	1	1	1	1	1	1	1
Bit	7	6	5	4	3	2	1	0
	TRISx[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	1	1	1	1	1	1	1

Bits 15:0 – TRISx[15:0] Output Enable for PORTx bits

Value	Description
1	LATx[n] is not driven on the PORTx[n] pin
0	LATx[n] is driven on the PORTx[n] pin

9.2.2.3 Input Data for PORTx Register**Name:** PORTx**Offset:** 0xE04, 0xE20, 0xE3C, 0xE58, 0xE74

Bit	15	14	13	12	11	10	9	8
	PORTx[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	1	1	1	1	1	1	1
Bit	7	6	5	4	3	2	1	0
	PORTx[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	1	1	1	1	1	1	1

Bits 15:0 – PORTx[15:0] PORTx Data Input Value bits

9.2.2.4 Output Data for PORTx Register**Name:** LATx**Offset:** 0xE06, 0xE22, 0xE3E, 0xE5A, 0xE76**Legend:** x = Bit is unknown

Bit	15	14	13	12	11	10	9	8
	LATx[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	x	x	x	x	x	x	x	x
Bit	7	6	5	4	3	2	1	0
	LATx[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	x	x	x	x	x	x	x	x

Bits 15:0 – LATx[15:0] PORTx Data Output Value bits

9.2.2.5 Open-Drain Enable for PORTx Register**Name:** ODCx**Offset:** 0xE08, 0xE24, 0xE40, 0xE5C, 0xE78

Bit	15	14	13	12	11	10	9	8
	ODCx[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	ODCx[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – ODCx[15:0] PORTx Open-Drain Enable bits

Value	Description
1	Open-drain is enabled on the PORTx pin
0	Open-drain is disabled on the PORTx pin

9.2.2.6 Change Notification Pull-up Enable for PORTx Register**Name:** CNPUx**Offset:** 0xE0A, 0xE26, 0xE42, 0xE5E, 0xE7A

Bit	15	14	13	12	11	10	9	8
	CNPUx[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	CNPUx[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – CNPUx[15:0] Change Notification Pull-up Enable for PORTx bits

Value	Description
1	The pull-up for PORTx[n] is enabled – takes precedence over the pull-down selection
0	The pull-up for PORTx[n] is disabled

9.2.2.7 Change Notification Pull-Down Enable for PORTx Register**Name:** CNPDx**Offset:** 0xE0C, 0xE28, 0xE44, 0xE60, 0xE7C

Bit	15	14	13	12	11	10	9	8
	CNPDx[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	CNPDx[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – CNPDx[15:0] Change Notification Pull-Down Enable for PORTx bits

Value	Description
1	The pull-down for PORTx[n] is enabled (if the pull-up for PORTx[n] is not enabled)
0	The pull-down for PORTx[n] is disabled

9.2.2.8 Change Notification Control for PORTx Register**Name:** CNCONx**Offset:** 0xE0E, 0xE2A, 0xE46, 0xE62, 0xE7E

Bit	15	14	13	12	11	10	9	8
	ON				CNSTYLE			
Access	R/W				R/W			
Reset	0				0			

Bit	7	6	5	4	3	2	1	0
Access								
Reset								

Bit 15 – ON Change Notification (CN) Control for PORTx On bit

Value	Description
1	CN is enabled
0	CN is disabled

Bit 11 – CNSTYLE Change Notification Style Selection bit

Value	Description
1	Edge style (detects edge transitions, CNF _x [15:0] bits are used for a Change Notification event)
0	Mismatch style (detects change from last port read, CNSTAT _x [15:0] bits are used for a Change Notification event)

9.2.2.9 Interrupt Change Notification Enable for PORTx Register**Name:** CNEN0x**Offset:** 0xE10, 0xE2C, 0xE48, 0xE64, 0xE80

Bit	15	14	13	12	11	10	9	8
	CNEN0x[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	CNEN0x[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – CNEN0x[15:0] Interrupt Change Notification Enable for PORTx bits

Value	Description
1	Interrupt-on-change (from the last read value) is enabled for PORTx[n]
0	Interrupt-on-change is disabled for PORTx[n]

9.2.2.10 Interrupt Change Notification Status for PORTx Register**Name:** CNSTATx**Offset:** 0xE12, 0xE2E, 0xE4A, 0xE66, 0xE82

Bit	15	14	13	12	11	10	9	8
	CNSTATx[15:8]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	CNSTATx[7:0]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – CNSTATx[15:0] Interrupt Change Notification Status for PORTx bits

When CNSTYLE (CNCONx[11]) = 0:

Value	Description
1	Change occurred on PORTx[n] since last read of PORTx[n]
0	Change did not occur on PORTx[n] since last read of PORTx[n]

9.2.2.11 Interrupt Change Notification Edge Select for PORTx Register**Name:** CNEN1x**Offset:** 0xE14, 0xE30, 0xE4C, 0xE68, 0xE84

Bit	15	14	13	12	11	10	9	8
	CNEN1x[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	CNEN1x[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – CNEN1x[15:0] Interrupt Change Notification Edge Select for PORTx bits

9.2.2.12 Interrupt Change Notification Flag for PORTx Register**Name:** CNFx**Offset:** 0xE16, 0xE32, 0xE4E, 0xE6A, 0xE86

Bit	15	14	13	12	11	10	9	8
	CNFx[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	CNFx[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – CNFx[15:0] Interrupt Change Notification Flag for PORTx bits

When CNSTYLE (CNCONx[11]) = 1:

Value	Description
1	An enabled edge event occurred on the PORTx[n] pin
0	An enabled edge event did not occur on the PORTx[n] pin

9.3 Input Change Notification (ICN)

The Input Change Notification function of the I/O ports allows the dsPIC33CK512MPT608 family devices to generate interrupt requests to the processor in response to a Change-of-State (COS) on selected input pins. This feature can detect input Change-of-States, even in Sleep mode, when the clocks are disabled. Every I/O port pin can be selected (enabled) for generating an interrupt request on a Change-of-State. Five control registers are associated with the Change Notification (CN) functionality of each I/O port. To enable the Change Notification feature for the port, the ON bit (CNCONx[15]) must be set.

The CNEN0x and CNEN1x registers contain the CN interrupt enable control bits for each of the input pins. The setting of these bits enables a CN interrupt for the corresponding pins. Also, these bits, in combination with the CNSTYLE bit (CNCONx[11]), define a type of transition when the interrupt is generated. Possible CN event options are listed in [Table 9-2](#).

Table 9-2. Change Notification Event Options

CNSTYLE Bit (CNCONx[11])	CNEN1x Bit	CNEN0x Bit	Change Notification Event Description
0	Does not matter	0	Disabled
0	Does not matter	1	Detects a mismatch between the last read state and the current state of the pin
1	0	0	Disabled
1	0	1	Detects a positive transition only (from '0' to '1')
1	1	0	Detects a negative transition only (from '1' to '0')
1	1	1	Detects both positive and negative transitions

The CNSTATx register indicates whether a change occurred on the corresponding pin since the last read of the PORTx bit. In addition to the CNSTATx register, the CNF_x register is implemented for each port. This register contains flags for Change Notification events. These flags are set if the valid transition edge, selected in the CNEN0_x and CNEN1_x registers, is detected. CNF_x stores the occurrence of the event. CNF_x bits must be cleared in software to get the next Change Notification interrupt. The CN interrupt is generated only for the I/Os configured as inputs (corresponding TRIS_x bits must be set).

Note: Pull-ups and pull-downs on Input Change Notification pins should always be disabled when the port pin is configured as a digital output.

9.4 Peripheral Pin Select (PPS)

A major challenge in general purpose devices is providing the largest possible set of peripheral features, while minimizing the conflict of features on I/O pins. The challenge is even greater on low pin count devices. In an application where more than one peripheral needs to be assigned to a single pin, inconvenient work arounds in application code, or a complete redesign, may be the only option.

Peripheral Pin Select configuration provides an alternative to these choices by enabling peripheral set selection and placement on a wide range of I/O pins. By increasing the pinout options available on a particular device, users can better tailor the device to their entire application, rather than trimming the application to fit the device.

The Peripheral Pin Select configuration feature operates over a fixed subset of digital I/O pins. Users may independently map the input and/or output of most digital peripherals to any one of these I/O pins. Hardware safeguards are included that prevent accidental or spurious changes to the peripheral mapping once it has been established.

9.4.1 Available Pins

The number of available pins is dependent on the particular device and its pin count. Pins that support the Peripheral Pin Select feature include the label, "RP_n", in their full pin designation, where "n" is the remappable pin number. "RP" is used to designate pins that support both remappable input and output functions.

9.4.2 Available Peripherals

The peripherals managed by the Peripheral Pin Select are all digital only peripherals. These include general serial communications (UART and SPI), general purpose timer clock inputs, timer-related peripherals (input capture and output compare) and interrupt-on-change inputs.

In comparison, some digital only peripheral modules are never included in the Peripheral Pin Select feature. This is because the peripheral's function requires special I/O circuitry on a specific port and cannot be easily connected to multiple pins. One example includes I²C modules. A similar requirement excludes all modules with analog inputs, such as the A/D Converter (ADC)

A key difference between remappable and non-remappable peripherals is that remappable peripherals are not associated with a default I/O pin. The peripheral must always be assigned to a specific I/O pin before it can be used. In contrast, non-remappable peripherals are always available on a default pin, assuming that the peripheral is active and not conflicting with another peripheral.

When a remappable peripheral is active on a given I/O pin, it takes priority over all other digital I/Os and digital communication peripherals associated with the pin. Priority is given regardless of the type of peripheral that is mapped. Remappable peripherals never take priority over any analog functions associated with the pin.

9.4.3 Controlling Configuration Changes

Because peripheral mapping can be changed during run time, some restrictions on peripheral remapping are needed to prevent accidental configuration changes. The dsPIC33CK512MPT608 devices have implemented the control register lock sequence.

9.4.3.1 Control Register Lock

Under normal operation, writes to the RPINR_x and RPOR_x registers are not allowed. Attempted writes will appear to execute normally, but the contents of the registers will remain unchanged. To change these registers, they must be unlocked in hardware. The register lock is controlled by the IOLOCK bit (RPCON[11]). Setting IOLOCK prevents writes to the control registers; clearing IOLOCK allows writes.

To set or clear IOLOCK, the NVMKEY unlock sequence must be executed:

1. Write 0x55 to NVMKEY.
2. Write 0xAA to NVMKEY.
3. Clear (or set) IOLOCK as a single operation.

IOLOCK remains in one state until changed. This allows all of the Peripheral Pin Selects to be configured with a single unlock sequence, followed by an update to all of the control registers. Then, IOLOCK can be set with a second lock sequence.

Note: MPLAB® XC16 provides a built-in C language function for unlocking and modifying the RPCON register:

```
__builtin_write_RPCON(value);
```

For more information, see the MPLAB® XC16 Help files.

9.5 Considerations for Peripheral Pin Selection

The ability to control Peripheral Pin Selection introduces several considerations into application design that most users would never think of otherwise. This is particularly true for several common peripherals, which are only available as remappable peripherals.

The main consideration is that the Peripheral Pin Selects are not available on default pins in the device's default (Reset) state. More specifically, because all RPINRx registers reset to '1's and RPORx registers reset to '0's, this means all PPS inputs are tied to V_{SS} , while all PPS outputs are disconnected. This means that before any other application code is executed, the user must initialize the device with the proper peripheral configuration. Because the IOLOCK bit resets in the Unlocked state, it is not necessary to execute the unlock sequence after the device has come out of Reset. For application safety, however, it is always better to set IOLOCK and lock the configuration after writing to the control registers.

The NVMKEY unlock sequence must be executed as an Assembly language routine. If the bulk of the application is written in C, or another high-level language, the unlock sequence should be performed by writing in-line assembly or by using the `__builtin_write_RPCON(value)` function provided by the compiler.

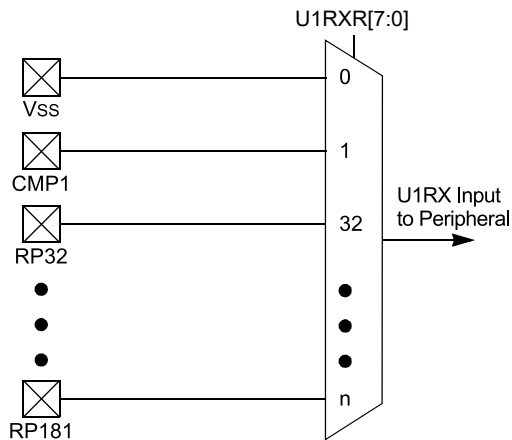
Choosing the configuration requires a review of all Peripheral Pin Selects and their pin assignments, particularly those that will not be used in the application. In all cases, unused pin-selectable peripherals should be disabled completely. Unused peripherals should have their inputs assigned to an unused RPn pin function. I/O pins with unused RPn functions should be configured with the null peripheral output.

9.6 Input Mapping

The inputs of the Peripheral Pin Select options are mapped on the basis of the peripheral. That is, a control register associated with a peripheral dictates the pin it will be mapped to. The RPINRx registers are used to configure peripheral input mapping. Each register contains sets of 8-bit fields, with each set associated with one of the remappable peripherals. Programming a given peripheral's bit field with an appropriate 8-bit index value maps the RPn pin with the corresponding value, or internal signal, to that peripheral. See [Table 9-3](#) for a list of available inputs.

For example, [Figure 9-2](#) illustrates remappable pin selection for the U1RX input.

Figure 9-2. Remappable Input for U1RX



Example 9-1 provides a configuration for bidirectional communication with flow control using UART1. The following input and output functions are used:

- Input Functions: U1RX, $\overline{U1CTS}$
- Output Functions: U1TX, $\overline{U1RTS}$

Table 9-3. Remappable Pin Inputs

RPINRx[15:8] or RPINRx[7:0]	Function	Available on Ports
0	V _{SS}	Internal
1	Comparator 1	Internal
2	Comparator 2	Internal
3	Comparator 3	Internal
4-5	RP4-RP5	Reserved
6	PTG Trigger 26	Internal
7	PTG Trigger 27	Internal
8-10	RP8-RP10	Reserved
11	PWM Event Out C	Internal
12	PWM Event Out D	Internal
13	PWM Event Out E	Internal
14-31	RP14-RP31	Reserved
32	RP32	Port Pin RB0
33	RP33	Port Pin RB1
34	RP34	Port Pin RB2
35	RP35	Port Pin RB3
36	RP36	Port Pin RB4
37	RP37	Port Pin RB5
38	RP38	Port Pin RB6
39	RP39	Port Pin RB7

.....continued

RPINRx[15:8] or RPINRx[7:0]	Function	Available on Ports
40	RP40	Port Pin RB8
41	RP41	Port Pin RB9
42	RP42	Port Pin RB10
43	RP43	Port Pin RB11
44	RP44	Port Pin RB12
45	RP45	Port Pin RB13
46	RP46	Port Pin RB14
47	RP47	Port Pin RB15
48	RP48	Port Pin RC0
49	RP49	Port Pin RC1
50	RP50	Port Pin RC2
51	RP51	Port Pin RC3
52-53	RP52-RP53	Reserved
54	RP54	Port Pin RC6
55	RP55	Port Pin RC7
56	RP56	Port Pin RC8
57	RP57	Port Pin RC9
58-59	RP58-RP59	Reserved
60	RP60	Port Pin RC12
61	RP61	Port Pin RC13
62	RP62	Port Pin RC14
63	RP63	Port Pin RC15
64	RP64	Port Pin RD0
65	RP65	Port Pin RD1
66	RP66	Port Pin RD2
67-68	RP67-RP68	Reserved
69	RP69	Port Pin RD5
70	RP70	Port Pin RD6
71	RP71	Port Pin RD7
72	RP72	Port Pin RD8
73	RP73	Port Pin RD9
74	RP74	Port Pin RD10
75	RP75	Port Pin RD11
76	RP76	Port Pin RD12
77	RP77	Port Pin RD13

.....continued

RPINRx[15:8] or RPINRx[7:0]	Function	Available on Ports
78	RP78	Port Pin RD14
79	RP79	Port Pin RD15
80-157	RP80-RP175	Reserved
158	DAC6 pwm_req_on	Internal
159	DAC6 pwm_req_off	Internal
160	DAC5 pwm_req_on	Internal
161	DAC5 pwm_req_off	Internal
162	DAC4 pwm_req_on	Internal
163	DAC4 pwm_req_off	Internal
164	DAC3 pwm_req_on	Internal
165	DAC3 pwm_req_off	Internal
166	DAC2 pwm_req_on	Internal
167	DAC2 pwm_req_off	Internal
168	DAC1 pwm_req_on	Internal
169	DAC1 pwm_req_off	Internal
176	RP176	Virtual RPV0
177	RP177	Virtual RPV1
178	RP178	Virtual RPV2
179	RP179	Virtual RPV3
180	RP180	Virtual RPV4
181	RP181	Virtual RPV5

Example 9-1. Configuring UART1 Input and Output Functions

```
//
// *****
// Unlock Registers
// *****
__builtin_write_RPCON(0x0000);
// *****
// Configure Input Functions (See Table 9-3)
// Assign U1Rx To Pin RP35
// *****
U1RXR = 35;
// Assign U1CTS To Pin RP36
// *****
U1CTSR = 36;
// *****
// Configure Output Functions (See Table 9-6)
// *****
// Assign U1Tx To Pin RP37
// *****
RP37 = 1;
// *****
// Assign U1RTS To Pin RP38
// *****
RP38 = 2;
// *****
// Lock Registers
// *****
__builtin_write_RPCON(0x0800);
```

9.7 Virtual Connections

The dsPIC33CK512MPT608 devices support six virtual RPN pins (RP176-RP181), which are identical in functionality to all other RPN pins, with the exception of pinouts. These six pins are internal to the devices and are not connected to a physical device pin.

These pins provide a simple way for inter-peripheral connection without utilizing a physical pin. For example, the output of the analog comparator can be connected to RP176 and the PWM Fault input can be configured for RP176 as well. This configuration allows the analog comparator to trigger PWM Faults without the use of an actual physical pin on the device.

Table 9-4. Selectable Input Sources (Maps Input to Function)

Input Name ⁽¹⁾	Function Name	Register	Register Bits
External Interrupt 1	INT1	RPINR0	INT1R[7:0]
External Interrupt 2	INT2	RPINR1	INT2R[7:0]
External Interrupt 3	INT3	RPINR1	INT3R[7:0]
Timer1 External Clock	T1CK	RPINR2	T1CK[7:0]
SCCP Timer1	TCKI1	RPINR3	TCKI1R[7:0]
SCCP Capture 1	ICM1	RPINR3	ICM1R[7:0]
SCCP Timer2	TCKI2	RPINR4	TCKI2R[7:0]
SCCP Capture 2	ICM2	RPINR4	ICM2R[7:0]
SCCP Timer3	TCKI3	RPINR5	TCKI3R[7:0]
SCCP Capture 3	ICM3	RPINR5	ICM3R[7:0]

.....continued

Input Name ⁽¹⁾	Function Name	Register	Register Bits
SCCP Timer4	TCKI4	RPINR6	TCKI4R[7:0]
SCCP Capture 4	ICM4	RPINR6	ICM4R[7:0]
SCCP Timer5	TCKI5	RPINR7	TCKI5R[7:0]
SCCP Capture 5	ICM5	RPINR7	ICM5R[7:0]
SCCP Timer6	TCKI6	RPINR8	TCKI6R[7:0]
SCCP Capture 6	ICM6	RPINR8	ICM6R[7:0]
SCCP Timer7	TCKI7	RPINR9	TCKI7R[7:0]
SCCP Capture 7	ICM7	RPINR9	ICM7R[7:0]
SCCP Timer8	TCKI8	RPINR10	TCKI8R[7:0]
SCCP Capture 8	ICM8	RPINR10	ICM8R[7:0]
xCCP Fault A	OCFA	RPINR11	OCFAR[7:0]
xCCP Fault B	OCFB	RPINR11	OCFBR[7:0]
PWM PCI 8	PCI8	RPINR12	PCI8R[7:0]
PWM PCI 9	PCI9	RPINR12	PCI9R[7:0]
PWM PCI 10	PCI10	RPINR13	PCI10R[7:0]
PWM PCI 11	PCI11	RPINR13	PCI11R[7:0]
QE1 Input A	QEIA1	RPINR14	QEIA1R[7:0]
QE1 Input B	QEIB1	RPINR14	QEIB1R[7:0]
QE1 Index 1 Input	QEINDX1	RPINR15	QEINDX1R[7:0]
QE1 Home 1 Input	QEIHOM1	RPINR15	QEIHOM1R[7:0]
QE2 Input A	QEIA2	RPINR16	QEIA2R[7:0]
QE2 Input B	QEIB2	RPINR16	QEIB2R[7:0]
QE2 Index 1 Input	QEINDX2	RPINR17	QEINDX2R[7:0]
QE2 Home 1 Input	QEIHOM2	RPINR17	QEIHOM2R[7:0]
UART1 Receive	U1RX	RPINR18	U1RXR[7:0]
UART1 Data-Set-Ready	$\overline{U1DSR}$	RPINR18	U1DSRR[7:0]
UART2 Receive	U2RX	RPINR19	U2RXR[7:0]
UART2 Data-Set-Ready	$\overline{U2DSR}$	RPINR19	U2DSRR[7:0]
SPI1 Data Input	SDI1	RPINR20	SDI1R[7:0]
SPI1 Clock Input	SCK1IN	RPINR20	SCK1R[7:0]
SPI1 Client Select	$\overline{SS1}$	RPINR21	SS1R[7:0]
Reference Clock Input	REFOI	RPINR21	REFOIR[7:0]
SPI2 Data Input	SDI2	RPINR22	SDI2R[7:0]
SPI2 Clock Input	SCK2IN	RPINR22	SCK2R[7:0]
SPI2 Client Select	$\overline{SS2}$	RPINR23	SS2R[7:0]

.....continued

Input Name ⁽¹⁾	Function Name	Register	Register Bits
QEI3 Input A	QEIA3	RPINR24	QEIA3R[7:0]
QEI3 Input B	QEIB3	RPINR24	QEIB3R[7:0]
QEI3 Index 1 Input	QEINDX3	RPINR25	QEINDX3R[7:0]
QEI3 Home 1 Input	QEIHOM3	RPINR25	QEIHOM3R[7:0]
CAN1 Input	CAN1RX	RPINR26	CAN1RXR[7:0]
CAN2 Input	CAN2RX	RPINR26	CAN2RXR[7:0]
UART3 Receive	U3RX	RPINR27	U3RXR[7:0]
UART3 Data-Set-Ready	$\overline{U3DSR}$	RPINR27	U3DSRR[7:0]
SPI3 Data Input	SDI3	RPINR29	SDI3R[7:0]
SPI3 Clock Input	SCK3IN	RPINR29	SCK3R[7:0]
SPI3 Client Select	$\overline{SS3}$	RPINR30	SS3R[7:0]
MCCP Timer9	TCKI9	RPINR32	TCKI9R[7:0]
MCCP Capture 9	ICM9	RPINR33	ICM9R[7:0]
xCCP Fault C	OCFC	RPINR37	OCFCR[7:0]
PWM Input 17	PCI17	RPINR37	PCI17R[7:0]
PWM Input 18	PCI18	RPINR38	PCI18R[7:0]
PWM Input 12	PCI12	RPINR42	PCI12R[7:0]
PWM Input 13	PCI13	RPINR42	PCI13R[7:0]
PWM Input 14	PCI14	RPINR43	PCI14R[7:0]
PWM Input 15	PCI15	RPINR43	PCI15R[7:0]
PWM Input 16	PCI16	RPINR44	PCI16R[7:0]
SENT1 Input	SENT1	RPINR44	SENT1R[7:0]
SENT2 Input	SENT2	RPINR45	SENT2R[7:0]
CLC Input A	CLCINA	RPINR45	CLCINAR[7:0]
CLC Input B	CLCINB	RPINR46	CLCINBR[7:0]
CLC Input C	CLCINC	RPINR46	CLCINCR[7:0]
CLC Input D	CLCIND	RPINR47	CLCINDR[7:0]
ADC Trigger Input (ADTRIG31)	ADCTRG	RPINR47	ADCTRGR[7:0]
xCCP Fault D	OCFD	RPINR48	OCFDR[7:0]
UART1 Clear-to-Send	$\overline{U1CTS}$	RPINR48	U1CTSR[7:0]
UART2 Clear-to-Send	$\overline{U2CTS}$	RPINR49	U2CTSR[7:0]
UART3 Clear-to-Send	$\overline{U3CTS}$	RPINR49	U3CTSR[7:0]

Note:

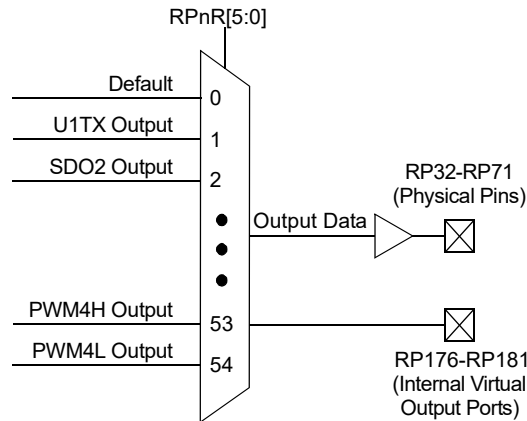
1. Unless otherwise noted, all inputs use the Schmitt Trigger input buffers.

9.8 Output Mapping

In contrast to inputs, the outputs of the Peripheral Pin Select options are mapped on the basis of the pin. In this case, a control register associated with a particular pin dictates the peripheral output to be mapped. The RPORx registers are used to control output mapping. Each register contains sets of 6-bit fields, with each set associated with one RPn pin (see 9.12.2. [RPINR0](#) through 9.12.80. [RPOR36](#)). The value of the bit field corresponds to one of the peripherals and that peripheral's output is mapped to the pin (see 9.9. [Mapping Limitations](#) and [Figure 9-3](#)).

A null output is associated with the output register Reset value of '0'. This is done to ensure that remappable outputs remain disconnected from all output pins by default.

Figure 9-3. Multiplexing Remappable Outputs for RPn



Note: There are six virtual output ports which are not connected to any I/O ports (RP176-RP181). These virtual ports can be accessed by RPOR20, RPOR21 and RPOR22.

9.9 Mapping Limitations

The control schema of the peripheral select pins is not limited to a small range of fixed peripheral configurations. There are no mutual or hardware-enforced lockouts between any of the peripheral mapping SFRs. Literally, any combination of peripheral mappings, across any or all of the RPn pins, is possible. This includes both many-to-one and one-to-many mappings of peripheral inputs, and outputs to pins. While such mappings may be technically possible from a configuration point of view, they may not be supportable from an electrical point of view (see [Table 9-5](#)).

Table 9-5. Remappable Output Pin Registers⁽¹⁾

Register	RP Pin	I/O Port
RPOR0[5:0]	RP32	Port Pin RB0
RPOR0[13:8]	RP33	Port Pin RB1
RPOR1[5:0]	RP34	Port Pin RB2
RPOR1[13:8]	RP35	Port Pin RB3
RPOR2[5:0]	RP36	Port Pin RB4
RPOR2[13:8]	RP37	Port Pin RB5
RPOR3[5:0]	RP38	Port Pin RB6
RPOR3[13:8]	RP39	Port Pin RB7

Note:

- Not all RPn pins are available on all packages. Make sure the selected device variant has the feature available on the device.

.....continued

Register	RP Pin	I/O Port
RPOR4[5:0]	RP40	Port Pin RB8
RPOR4[13:8]	RP41	Port Pin RB9
RPOR5[5:0]	RP42	Port Pin RB10
RPOR5[13:8]	RP43	Port Pin RB11
RPOR6[5:0]	RP44	Port Pin RB12
RPOR6[13:8]	RP45	Port Pin RB13
RPOR7[5:0]	RP46	Port Pin RB14
RPOR7[13:8]	RP47	Port Pin RB15
RPOR8[5:0]	RP48	Port Pin RC0
RPOR8[13:8]	RP49	Port Pin RC1
RPOR9[5:0]	RP50	Port Pin RC2
RPOR9[13:8]	RP51	Port Pin RC3
Reserved	RP52-RP53	Reserved
RPOR11[5:0]	RP54	Port Pin RC6
RPOR11[13:8]	RP55	Port Pin RC7
RPOR12[5:0]	RP56	Port Pin RC8
RPOR12[13:8]	RP57	Port Pin RC9
Reserved	RP58-RP59	Reserved
RPOR14[5:0]	RP60	Port Pin RC12
RPOR14[13:8]	RP61	Port Pin RC13
RPOR15[5:0]	RP62	Port Pin RC14
RPOR15[13:8]	RP63	Port Pin RC15
RPOR16[5:0]	RP64	Port Pin RD0
RPOR16[13:8]	RP65	Port Pin RD1
RPOR17[5:0]	RP66	Port Pin RD2
Reserved	RP67-RP68	Reserved
RPOR18[13:8]	RP69	Port Pin RD5
RPOR19[5:0]	RP70	Port Pin RD6
RPOR19[13:8]	RP71	Port Pin RD7
RPOR20[5:0]	RP72	Port Pin RD8
RPOR20[13:8]	RP73	Port Pin RD9
RPOR21[5:0]	RP74	Port Pin RD10
RPOR21[13:8]	RP75	Port Pin RD11

Note:

1. Not all RPN pins are available on all packages. Make sure the selected device variant has the feature available on the device.

.....continued

Register	RP Pin	I/O Port
RPOR22[5:0]	RP76	Port Pin RD12
RPOR22[13:8]	RP77	Port Pin RD13
RPOR23[5:0]	RP78	Port Pin RD14
RPOR23[13:8]	RP79	Port Pin RD15
Reserved	RP80-RP175	Reserved
RPOR34[5:0]	RP176	Virtual Pin RPV0
RPOR34[13:8]	RP177	Virtual Pin RPV1
RPOR35[5:0]	RP178	Virtual Pin RPV2
RPOR35[13:8]	RP179	Virtual Pin RPV3
RPOR36[5:0]	RP180	Virtual Pin RPV4
RPOR36[13:8]	RP181	Virtual Pin RPV5

Note:

1. Not all RPn pins are available on all packages. Make sure the selected device variant has the feature available on the device.

Table 9-6. Output Selection for Remappable Pins (RPn)⁽¹⁾

Function	RPnR[5:0]	Output Name
Default Port	0	RPn tied to Default Pin
U1TX	1	RPn tied to UART1 Transmit
U1RTS	2	RPn tied to UART1 Request-to-Send
U2TX	3	RPn tied to UART2 Transmit
U2RTS	4	RPn tied to UART2 Request-to-Send
SDO1	5	RPn tied to SPI1 Data Output
SCK1	6	RPn tied to SPI1 Clock Output
SS1	7	RPn tied to SPI1 Client Select
SDO2	8	RPn tied to SPI2 Data Output
SCK2	9	RPn tied to SPI2 Clock Output
SS2	10	RPn tied to SPI2 Client Select
SDO3	11	RPn tied to SPI3 Data Output
SCK3	12	RPn tied to SPI3 Clock output
SS3	13	RPn tied to SPI3 Client Select
REFCLKO	14	RPn tied to Reference Clock Output
OCM1	15	RPn tied to SCCP1 Output
OCM2	16	RPn tied to SCCP2 Output

Note:

1. Not all RPn pins are available on all packages. Make sure the selected device variant has the feature available on the device.

.....continued

Function	RPnR[5:0]	Output Name
OCM3	17	RPn tied to SCCP3 Output
OCM4	18	RPn tied to SCCP4 Output
OCM5	19	RPn tied to SCCP5 Output
OCM6	20	RPn tied to SCCP6 Output
CAN1	21	RPn tied to CAN1 Output
CAN2	22	RPn tied to CAN2 Output
CMP1	23	RPn tied to Comparator 1 Output
CMP2	24	RPn tied to Comparator 2 Output
CMP3	25	RPn tied to Comparator 3 Output
CMP4	26	RPn tied to Comparator 4 Output
U3TX	27	RPn tied to UART3 Transmit
U3RTS	28	RPn tied to UART3 Request-to-Send
CMP5	32	RPn tied to Comparator 5 Output
CMP6	33	RPn tied to Comparator 6 Output
PWM4H	34	RPn tied to PWM4H Output
PWM4L	35	RPn tied to PWM4L Output
PWMEA	36	RPn tied to PWM Event A Output
PWMEB	37	RPn tied to PWM Event B Output
QEICMP1	38	RPn tied to QE11 Comparator Output
QEICMP2	39	RPn tied to QE12 Comparator Output
CLC1OUT	40	RPn tied to CLC1 Output
CLC2OUT	41	RPn tied to CLC2 Output
OCM7	42	RPn tied to SCCP7 Output
OCM8	43	RPn tied to SCCP8 Output
PWMEC	44	RPn tied to PWM Event C Output
PWMED	45	RPn tied to PWM Event D Output
PTGTRG24	46	PTG Trigger Output 24
PTGTRG25	47	PTG Trigger Output 25
SENT1OUT	48	RPn tied to SENT1 Output
SENT2OUT	49	RPn tied to SENT2 Output
MCCP9A	50	RPn tied to MCCP9 Output A
MCCP9B	51	RPn tied to MCCP9 Output B
MCCP9C	52	RPn tied to MCCP9 Output C

Note:

1. Not all RPn pins are available on all packages. Make sure the selected device variant has the feature available on the device.

.....continued		
Function	RPnR[5:0]	Output Name
MCCP9D	53	RPn tied to MCCP9 Output D
MCCP9E	54	RPn tied to MCCP9 Output E
MCCP9F	55	RPn tied to MCCP9 Output F
QEICMP3	58	RPn tied to QEI3 Comparator Output
CLC3OUT	59	RPn tied to CLC3 Output
CLC4OUT	60	RPn tied to CLC4 Output
U1DTR	61	RPn tied to UART1 DTR
U2DTR	62	RPn tied to UART2 DTR
U3DTR	63	RPn tied to UART3 DTR
Note: 1. Not all RPn pins are available on all packages. Make sure the selected device variant has the feature available on the device.		

9.10 I/O Helpful Tips

1. In some cases, certain pins, as defined in [33.1. DC Characteristics](#) under “Injection Current”, have internal protection diodes to V_{DD} and V_{SS} . The term, “Injection Current”, is also referred to as “Clamp Current”. On designated pins, with sufficient external current-limiting precautions by the user, I/O pin input voltages are allowed to be greater or lesser than the data sheet absolute maximum ratings, with respect to the V_{SS} and V_{DD} supplies. Note that when the user application forward biases either of the high or low-side internal input clamp diodes, that the resulting current being injected into the device that is clamped internally by the V_{DD} and V_{SS} power rails, may affect the ADC accuracy by four to six counts.
2. I/O pins that are shared with any analog input pin (i.e., ANx) are always analog pins, by default, after any Reset. Consequently, configuring a pin as an analog input pin automatically disables the digital input pin buffer and any attempt to read the digital input level by reading PORTx or LATx will always return a ‘0’, regardless of the digital logic level on the pin. To use a pin as a digital I/O pin on a shared ANx pin, the user application needs to configure the Analog Select for PORTx registers in the I/O ports module (i.e., ANSELx) by setting the appropriate bit that corresponds to that I/O port pin to a ‘0’.

Note: Although it is not possible to use a digital input pin when its analog function is enabled, it is possible to use the digital I/O output function, TRISx = 0x0, while the analog function is also enabled. However, this is not recommended, particularly if the analog input is connected to an external analog voltage source, which would create signal contention between the analog signal and the output pin driver.

3. Most I/O pins have multiple functions. Referring to the device pin diagrams in this data sheet, the priorities of the functions allocated to any pins are indicated by reading the pin name, from left-to-right. The left most function name takes precedence over any function to its right in the naming convention. For example: AN16/T2CK/T7CK/RC1; this indicates that AN16 is the highest priority in this example and will supersede all other functions to its right in the list. Those other functions to its right, even if enabled, would not work as long as any other function to its left was enabled. This rule applies to all of the functions listed for a given pin.
4. Each pin has an internal weak pull-up resistor and pull-down resistor that can be configured using the CNPUx and CNPDx registers, respectively. These resistors eliminate the need for external resistors in certain applications. The internal pull-up is up to $\sim(V_{DD} - 0.8)$, not V_{DD} . This value is still above the minimum V_{IH} of CMOS and TTL devices.
5. When driving LEDs directly, the I/O pin can source or sink more current than what is specified in the V_{OH}/I_{OH} and V_{OL}/I_{OL} DC characteristics specification. The respective I_{OH} and I_{OL} current rating only applies to maintaining the corresponding output at or above the V_{OH} , and at or below the V_{OL} levels. However, for LEDs, unlike digital inputs of an externally connected device, they are not governed by the same minimum V_{IH}/V_{IL} levels. An I/O pin output can safely sink or source any current less than that listed in the Absolute Maximum Ratings in [33. Electrical Characteristics](#) of this data sheet. For example:

$V_{OH} = 2.4V @ I_{OH} = -8 \text{ mA}$ and $V_{DD} = 3.3V$

The maximum output current sourced by any 8 mA I/O pin = 12 mA.

LED source current < 12 mA is technically permitted.

6. The Peripheral Pin Select (PPS) pin mapping rules are as follows:
 - a. Only one “output” function can be active on a given pin at any time, regardless if it is a dedicated or remappable function (one pin, one output).
 - b. It is possible to assign a “remappable output” function to multiple pins and externally short or tie them together for increased current drive.
 - c. If any “dedicated output” function is enabled on a pin, it will take precedence over any remappable “output” function.
 - d. If any “dedicated digital” (input or output) function is enabled on a pin, any number of “input” remappable functions can be mapped to the same pin.
 - e. If any “dedicated analog” function(s) are enabled on a given pin, “digital input(s)” of any kind will all be disabled, although a single “digital output”, at the user’s cautionary discretion, can be enabled and active as long as there is no signal contention with an external analog input signal. For example, it is possible for the ADC to convert the digital output logic level, or to toggle a digital output on a comparator or ADC input, provided there is no external analog input, such as for a Built-In Self-Test.
 - f. Any number of “input” remappable functions can be mapped to the same pin(s) at the same time, including to any pin with a single output from either a dedicated or remappable “output”.
 - g. The TRISx registers control only the digital I/O output buffer. Any other dedicated or remappable active “output” will automatically override the TRISx setting. The TRISx register does not control the digital logic “input” buffer. Remappable digital “inputs” do not automatically override TRISx settings, which means that the TRISx bit must be set to input for pins with only remappable input function(s) assigned.
 - h. All analog pins are enabled by default after any Reset and the corresponding digital input buffer on the pin has been disabled. Only the Analog Select for PORTx (ANSELx) registers control the digital input buffer, not the TRISx register. The user must disable the analog function on a pin using the Analog Select for PORTx registers in order to use any “digital input(s)” on a corresponding pin, no exceptions.

9.11 I/O Ports Resources

Many useful resources are provided on the main product page of the Microchip website for the devices listed in this data sheet. This product page contains the latest updates and additional information.

9.11.1 Key Resources

- “I/O Ports with Edge Detect”
(www.microchip.com/DS70005322) in the “dsPIC33/PIC24 Family Reference Manual”
- Code Samples
- Application Notes
- Software Libraries
- Webinars
- All Related “dsPIC33/PIC24 Family Reference Manual” sections
- Development Tools

9.12 Peripheral Pin Select Control Registers

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x00 ... 0x0CFF	Reserved									
0x0D00	RPCON(1)	15:8					IOLOCK			
		7:0								
0x0D02 ... 0x0D03	Reserved									
0x0D04	RPINR0	15:8	INT1R[7:0]							
		7:0								
0x0D06	RPINR1	15:8	INT3R[7:0]							
		7:0	INT2R[7:0]							
0x0D08	RPINR2	15:8	T1CKR[7:0]							
		7:0								
0x0D0A	RPINR3	15:8	ICM1R[7:0]							
		7:0	TCK1R[7:0]							
0x0D0C	RPINR4	15:8	ICM2R[7:0]							
		7:0	TCKI2R[7:0]							
0x0D0E	RPINR5	15:8	ICM3R[7:0]							
		7:0	TCKI3R[7:0]							
0x0D10	RPINR6	15:8	ICM4R[7:0]							
		7:0	TCKI4R[7:0]							
0x0D12	RPINR7	15:8	ICM5R[7:0]							
		7:0	TCKI5R[7:0]							
0x0D14	RPINR8	15:8	ICM6R[7:0]							
		7:0	TCKI6R[7:0]							
0x0D16	RPINR9	15:8	ICM7R[7:0]							
		7:0	TCKI7R[7:0]							
0x0D18	RPINR10	15:8	ICM8R[7:0]							
		7:0	TCKI8R[7:0]							
0x0D1A	RPINR11	15:8	OCFBR[7:0]							
		7:0	OCFAR[7:0]							
0x0D1C	RPINR12	15:8	PCI9R[7:0]							
		7:0	PCI8R[7:0]							
0x0D1E	RPINR13	15:8	PCI11R[7:0]							
		7:0	PCI10R[7:0]							
0x0D20	RPINR14	15:8	QEIB1R[7:0]							
		7:0	QEIA1R[7:0]							
0x0D22	RPINR15	15:8	QEIHOM1R[7:0]							
		7:0	QEINDX1R[7:0]							
0x0D24	RPINR16	15:8	QEIB2R[7:0]							
		7:0	QEIA2R[7:0]							
0x0D26	RPINR17	15:8	QEIHOM2R[7:0]							
		7:0	QEINDX2R[7:0]							
0x0D28	RPINR18	15:8	U1DSRR[7:0]							
		7:0	U1RXR[7:0]							
0x0D2A	RPINR19	15:8	U2DSRR[7:0]							
		7:0	U2RXR[7:0]							
0x0D2C	RPINR20	15:8	SCK1R[7:0]							
		7:0	SDI1R[7:0]							
0x0D2E	RPINR21	15:8	REFOIR[7:0]							
		7:0	SS1R[7:0]							
0x0D30	RPINR22	15:8	SCK2R[7:0]							
		7:0	SDI2R[7:0]							
0x0D32	RPINR23	15:8								
		7:0	SS2R[7:0]							

.....continued										
Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x0D34	RPINR24	15:8	QEIB3R[7:0]							
		7:0	QEIA3R[7:0]							
0x0D36	RPINR25	15:8	QEIHOM3R[7:0]							
		7:0	QEINDX3R[7:0]							
0x0D38	RPINR26	15:8	CAN2RXR[7:0]							
		7:0	CAN1RXR[7:0]							
0x0D3A	RPINR27	15:8	U3DSRR[7:0]							
		7:0	U3RXR[7:0]							
0x0D3C ... 0x0D3D	Reserved									
0x0D3E	RPINR29	15:8	SCK3R[7:0]							
		7:0	SDI3R[7:0]							
0x0D40	RPINR30	15:8								
		7:0	SS3R[7:0]							
0x0D42 ... 0x0D43	Reserved									
0x0D44	RPINR32	15:8	TCKI9R[7:0]							
		7:0								
0x0D46	RPINR33	15:8								
		7:0	ICM9R[7:0]							
0x0D48 ... 0x0D4D	Reserved									
0x0D4E	RPINR37	15:8	PCI17R[7:0]							
		7:0	OCFCR[7:0]							
0x0D50	RPINR38	15:8								
		7:0	PCI18R[7:0]							
0x0D52 ... 0x0D57	Reserved									
0x0D58	RPINR42	15:8	PCI13R[7:0]							
		7:0	PCI12R[7:0]							
0x0D5A	RPINR43	15:8	PCI15R[7:0]							
		7:0	PCI14R[7:0]							
0x0D5C	RPINR44	15:8	SENT1R[7:0]							
		7:0	PCI16R[7:0]							
0x0D5E	RPINR45	15:8	CLCINAR[7:0]							
		7:0	SENT2R[7:0]							
0x0D60	RPINR46	15:8	CLCINCR[7:0]							
		7:0	CLCINBR[7:0]							
0x0D62	RPINR47	15:8	ADCTRGR[7:0]							
		7:0	CLCINDR[7:0]							
0x0D64	RPINR48	15:8	U1CTSR[7:0]							
		7:0	OCFDR[7:0]							
0x0D66	RPINR49	15:8	U3CTSR[7:0]							
		7:0	U2CTSR[7:0]							
0x0D68 ... 0x0D7F	Reserved									
0x0D80	RPOR0	15:8			RP33R[5:0]					
		7:0			RP32R[5:0]					
0x0D82	RPOR1	15:8			RP35R[5:0]					
		7:0			RP34R[5:0]					
0x0D84	RPOR2	15:8			RP37R[5:0]					
		7:0			RP36R[5:0]					
0x0D86	RPOR3	15:8			RP39R[5:0]					
		7:0			RP38R[5:0]					

.....continued

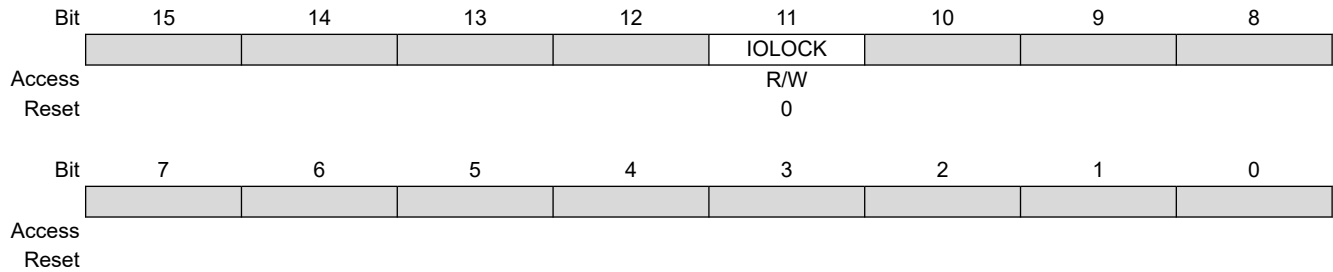
Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x0D88	RPOR4	15:8						RP41R[5:0]		
		7:0						RP40R[5:0]		
0x0D8A	RPOR5	15:8						RP43R[5:0]		
		7:0						RP42R[5:0]		
0x0D8C	RPOR6	15:8						RP45R[5:0]		
		7:0						RP44R[5:0]		
0x0D8E	RPOR7	15:8						RP47R[5:0]		
		7:0						RP46R[5:0]		
0x0D90	RPOR8	15:8						RP49R[5:0]		
		7:0						RP48R[5:0]		
0x0D92	RPOR9	15:8						RP51R[5:0]		
		7:0						RP50R[5:0]		
0x0D94	RPOR10	15:8						RP53R[5:0]		
		7:0						Reserved[5:0]		
0x0D96	RPOR11	15:8						RP55R[5:0]		
		7:0						RP54R[5:0]		
0x0D98	RPOR12	15:8						RP57R[5:0]		
		7:0						RP56R[5:0]		
0x0D9A	RPOR13	15:8						RP59R[5:0]		
		7:0						RP58R[5:0]		
0x0D9C	RPOR14	15:8						RP61R[5:0]		
		7:0						RP60R[5:0]		
0x0D9E	RPOR15	15:8						RP63R[5:0]		
		7:0						RP62R[5:0]		
0x0DA0	RPOR16	15:8						RP65R[5:0]		
		7:0						RP64R[5:0]		
0x0DA2	RPOR17	15:8						Reserved[5:0]		
		7:0						RP66R[5:0]		
0x0DA4	RPOR18	15:8						RP69R[5:0]		
		7:0						Reserved[5:0]		
0x0DA6	RPOR19	15:8						RP71R[5:0]		
		7:0						RP70R[5:0]		
0x0DA8	RPOR20	15:8						RP73R[5:0]		
		7:0						RP72R[5:0]		
0x0DAA	RPOR21	15:8						RP75R[5:0]		
		7:0						RP74R[5:0]		
0x0DAC	RPOR22	15:8						RP77R[5:0]		
		7:0						RP76R[5:0]		
0x0DAE	RPOR23	15:8						RP79R[5:0]		
		7:0						RP78R[5:0]		
0x0DB0	RPOR24	15:8						RP81R[5:0]		
		7:0						RP80R[5:0]		
0x0DB2	RPOR25	15:8						RP83R[5:0]		
		7:0						RP82R[5:0]		
0x0DB4	RPOR26	15:8						RP85R[5:0]		
		7:0						RP84R[5:0]		
0x0DB6	RPOR27	15:8						RP87R[5:0]		
		7:0						RP86R[5:0]		
0x0DB8	RPOR28	15:8						RP89R[5:0]		
		7:0						RP88R[5:0]		
0x0DBA	RPOR29	15:8						RP91R[5:0]		
		7:0						RP90R[5:0]		
0x0DBC	RPOR30	15:8						RP93R[5:0]		
		7:0						RP92R[5:0]		
0x0DBE	RPOR31	15:8						RP95R[5:0]		
		7:0						RP94R[5:0]		
0x0DC0	RPOR32	15:8						RP97R[5:0]		
		7:0						RP96R[5:0]		

.....continued

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x0DC2	RPOR33	15:8			RP99R[5:0]					
		7:0			RP98R[5:0]					
0x0DC4	RPOR34	15:8			RP177R[5:0]					
		7:0			RP176R[5:0]					
0x0DC6	RPOR35	15:8			RP179R[5:0]					
		7:0			RP178R[5:0]					
0x0DC8	RPOR36	15:8			RP181R[5:0]					
		7:0			RP180R[5:0]					

9.12.1 Peripheral Remapping Configuration Register**Name:** RPCON⁽¹⁾**Offset:** 0xD00**Note:**

- Writing to this register needs an unlock sequence.

**Bit 11 – IOLOCK** Peripheral Remapping Register Lock bit

Value	Description
1	All Peripheral Remapping registers are locked and cannot be written
0	All Peripheral Remapping registers are unlocked and can be written

9.12.2 Peripheral Pin Select Input Register 0**Name:** RPINR0**Offset:** 0xD04

Bit	15	14	13	12	11	10	9	8
	INT1R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
Access								
Reset								

Bits 15:8 – INT1R[7:0] Assign External Interrupt 1 (INT1) to the Corresponding RPn Pin bits

9.12.3 Peripheral Pin Select Input Register 1**Name:** RPINR1**Offset:** 0xD06

Bit	15	14	13	12	11	10	9	8
	INT3R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	INT2R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – INT3R[7:0] Assign External Interrupt 3 (INT3) to the Corresponding RPn Pin bits**Bits 7:0 – INT2R[7:0]** Assign External Interrupt 2 (INT2) to the Corresponding RPn Pin bits

9.12.4 Peripheral Pin Select Input Register 2

Name: RPINR2
Offset: 0xD08

Bit	15	14	13	12	11	10	9	8
	T1CKR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
Access								
Reset								

Bits 15:8 – T1CKR[7:0] Assign Timer1 External Clock (T1CK) to the Corresponding RPn Pin bits

9.12.5 Peripheral Pin Select Input Register 3**Name:** RPINR3**Offset:** 0xD0A

Bit	15	14	13	12	11	10	9	8
	ICM1R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	TCKI1R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – ICM1R[7:0] Assign SCCP Capture 1 (ICM1) Input to the Corresponding RPn Pin bits**Bits 7:0 – TCKI1R[7:0]** Assign SCCP Timer1 (TCKI1) Input to the Corresponding RPn Pin bits

9.12.6 Peripheral Pin Select Input Register 4**Name:** RPINR4**Offset:** 0xD0C

Bit	15	14	13	12	11	10	9	8
	ICM2R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	TCKI2R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – ICM2R[7:0] Assign SCCP Capture 2 (ICM2) Input to the Corresponding RPn Pin bits**Bits 7:0 – TCKI2R[7:0]** Assign SCCP Timer2 (TCKI2) Input to the Corresponding RPn Pin bits

9.12.7 Peripheral Pin Select Input Register 5**Name:** RPINR5**Offset:** 0xD0E

Bit	15	14	13	12	11	10	9	8
	ICM3R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	TCKI3R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – ICM3R[7:0] Assign SCCP Capture 3 (ICM3) Input to the Corresponding RPn Pin bits**Bits 7:0 – TCKI3R[7:0]** Assign SCCP Timer3 (TCKI3) Input to the Corresponding RPn Pin bits

9.12.8 Peripheral Pin Select Input Register 6**Name:** RPINR6**Offset:** 0xD10

Bit	15	14	13	12	11	10	9	8
	ICM4R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	TCKI4R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – ICM4R[7:0] Assign SCCP Capture 4 (ICM4) Input to the Corresponding RPn Pin bits**Bits 7:0 – TCKI4R[7:0]** Assign SCCP Timer4 (TCKI4) Input to the Corresponding RPn Pin bits

9.12.9 Peripheral Pin Select Input Register 7**Name:** RPINR7**Offset:** 0xD12

Bit	15	14	13	12	11	10	9	8
	ICM5R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	TCKI5R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – ICM5R[7:0] Assign SCCP Capture 5 (ICM5) Input to the Corresponding RPn Pin bits**Bits 7:0 – TCKI5R[7:0]** Assign SCCP Timer5 (TCKI5) Input to the Corresponding RPn Pin bits

9.12.10 Peripheral Pin Select Input Register 8**Name:** RPINR8**Offset:** 0xD14

Bit	15	14	13	12	11	10	9	8
	ICM6R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	TCKI6R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – ICM6R[7:0] Assign SCCP Capture 6 (ICM6) Input to the Corresponding RPn Pin bits**Bits 7:0 – TCKI6R[7:0]** Assign SCCP Timer6 (TCKI6) Input to the Corresponding RPn Pin bits

9.12.11 Peripheral Pin Select Input Register 9**Name:** RPINR9**Offset:** 0xD16

Bit	15	14	13	12	11	10	9	8
	ICM7R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	TCKI7R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – ICM7R[7:0] Assign SCCP Capture 7 (ICM7) Input to the Corresponding RPn Pin bits**Bits 7:0 – TCKI7R[7:0]** Assign SCCP Timer7 (TCKI7) Input to the Corresponding RPn Pin bits

9.12.12 Peripheral Pin Select Input Register 10**Name:** RPINR10**Offset:** 0xD18

Bit	15	14	13	12	11	10	9	8
	ICM8R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	TCKI8R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – ICM8R[7:0] Assign SCCP Capture 8 (ICM8) Input to the Corresponding RPn Pin bits**Bits 7:0 – TCKI8R[7:0]** Assign SCCP Timer8 (TCKI8) Input to the Corresponding RPn Pin bits

9.12.13 Peripheral Pin Select Input Register 11**Name:** RPINR11**Offset:** 0xD1A

Bit	15	14	13	12	11	10	9	8
	OCFBR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	OCFAR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – OCFBR[7:0] Assign SCCP Fault B (OCFB) Input to the Corresponding RPn Pin bits**Bits 7:0 – OCFAR[7:0]** Assign SCCP Fault A (OCFA) Input to the Corresponding RPn Pin bits

9.12.14 Peripheral Pin Select Input Register 12**Name:** RPINR12**Offset:** 0xD1C

Bit	15	14	13	12	11	10	9	8
	PCI9R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PCI8R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – PCI9R[7:0] Assign PWM Input 9 (PCI9) to the Corresponding RPn Pin bits**Bits 7:0 – PCI8R[7:0]** Assign PWM Input 8 (PCI8) to the Corresponding RPn Pin bits

9.12.15 Peripheral Pin Select Input Register 13**Name:** RPINR13**Offset:** 0xD1E

Bit	15	14	13	12	11	10	9	8
	PCI11R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PCI10R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – PCI11R[7:0] Assign PWM Input 11 (PCI11) to the Corresponding RPn Pin bits**Bits 7:0 – PCI10R[7:0]** Assign PWM Input 10 (PCI10) to the Corresponding RPn Pin bits

9.12.16 Peripheral Pin Select Input Register 14**Name:** RPINR14**Offset:** 0xD20

Bit	15	14	13	12	11	10	9	8
	QEIB1R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	QEIA1R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – QEIB1R[7:0] Assign QEI Input B (QEIB1) to the Corresponding RPn Pin bits**Bits 7:0 – QEIA1R[7:0]** Assign QEI Input A (QEIA1) to the Corresponding RPn Pin bits

9.12.17 Peripheral Pin Select Input Register 15**Name:** RPINR15**Offset:** 0xD22

Bit	15	14	13	12	11	10	9	8
	QEIHOM1R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	QEINDX1R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – QEIHOM1R[7:0] Assign QEI Home 1 Input (QEIHOM1) to the Corresponding RPn Pin bits**Bits 7:0 – QEINDX1R[7:0]** Assign QEI Index 1 Input (QEINDX1) to the Corresponding RPn Pin bits

9.12.18 Peripheral Pin Select Input Register 16**Name:** RPINR16**Offset:** 0xD24

Bit	15	14	13	12	11	10	9	8
	QEIB2R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	QEIA2R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – QEIB2R[7:0] Assign QEI2 Input B (QEIB2) to the Corresponding RPn Pin bits**Bits 7:0 – QEIA2R[7:0]** Assign QEI2 Input A (QEIA2) to the Corresponding RPn Pin bits

9.12.19 Peripheral Pin Select Input Register 17**Name:** RPINR17**Offset:** 0xD26

Bit	15	14	13	12	11	10	9	8
	QEIHOM2R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	QEINDX2R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – QEIHOM2R[7:0] Assign QEI Home 2 Input (QEIHOM2) to the Corresponding RPn Pin bits**Bits 7:0 – QEINDX2R[7:0]** Assign QEI Index 2 Input (QEINDX2) to the Corresponding RPn Pin bits

9.12.20 Peripheral Pin Select Input Register 18**Name:** RPINR18**Offset:** 0xD28

Bit	15	14	13	12	11	10	9	8
	U1DSRR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	U1RXR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – U1DSRR[7:0] Assign UART1 Data-Set-Ready ($\overline{U1DSR}$) to the Corresponding RPn Pin bits**Bits 7:0 – U1RXR[7:0]** Assign UART1 Receive (U1RX) to the Corresponding RPn Pin bits

9.12.21 Peripheral Pin Select Input Register 19**Name:** RPINR19**Offset:** 0xD2A

Bit	15	14	13	12	11	10	9	8
	U2DSRR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	U2RXR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – U2DSRR[7:0] Assign UART2 Data-Set-Ready ($\overline{U2DSR}$) to the Corresponding RPn Pin bits**Bits 7:0 – U2RXR[7:0]** Assign UART2 Receive (U2RX) to the Corresponding RPn Pin bits

9.12.22 Peripheral Pin Select Input Register 20**Name:** RPINR20**Offset:** 0xD2C**Note:**

- These register bits are assigned with value of 52 to allow communication with the Secure Subsystem.

Bit	15	14	13	12	11	10	9	8
	SCK1R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	SDI1R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – SCK1R[7:0] Assign SPI1 Clock Input (SCK1IN) to the Corresponding RPn Pin bits**Bits 7:0 – SDI1R[7:0]** Assign SPI1 Data Input (SDI1) to the Corresponding RPn Pin bits⁽¹⁾

9.12.23 Peripheral Pin Select Input Register 21**Name:** RPINR21**Offset:** 0xD2E

Bit	15	14	13	12	11	10	9	8
	REFOIR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	SS1R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – REFOIR[7:0] Assign Reference Clock Input (REFOI) to the Corresponding RPn Pin bits**Bits 7:0 – SS1R[7:0]** Assign SPI1 Client Select ($\overline{SS1}$) to the Corresponding RPn Pin bits

9.12.24 Peripheral Pin Select Input Register 22**Name:** RPINR22**Offset:** 0xD30**Note:**

- These register bits are assigned with value of 52 to allow communication with the Secure Subsystem.

Bit	15	14	13	12	11	10	9	8
	SCK2R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	SDI2R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – SCK2R[7:0] Assign SPI2 Clock Input (SCK2IN) to the Corresponding RPn Pin bits**Bits 7:0 – SDI2R[7:0]** Assign SPI2 Data Input (SDI2) to the Corresponding RPn Pin bits⁽¹⁾

9.12.25 Peripheral Pin Select Input Register 23

Name: RPINR23
Offset: 0xD32

Bit	15	14	13	12	11	10	9	8
Access								
Reset								
Bit	7	6	5	4	3	2	1	0
	SS2R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 7:0 – SS2R[7:0] Assign SPI2 Client Select ($\overline{SS2}$) to the Corresponding RPn Pin bits

9.12.26 Peripheral Pin Select Input Register 24**Name:** RPINR24**Offset:** 0xD34

Bit	15	14	13	12	11	10	9	8
	QEIB3R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	QEIA3R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – QEIB3R[7:0] Assign QEI Input B3 (QEIB3) to the Corresponding RPn Pin bits**Bits 7:0 – QEIA3R[7:0]** Assign QEI Input A3 (QEIA3) to the Corresponding RPn Pin bits

9.12.27 Peripheral Pin Select Input Register 25**Name:** RPINR25**Offset:** 0xD36

Bit	15	14	13	12	11	10	9	8
	QEIHOM3R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset								
Bit	7	6	5	4	3	2	1	0
	QEINDX3R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset								

Bits 15:8 – QEIHOM3R[7:0] Assign QEI Home 3 Input (QEIHOM3) to the Corresponding RPn Pin bits**Bits 7:0 – QEINDX3R[7:0]** Assign QEI Index 3 Input (QEINDX3) to the Corresponding RPn Pin bits

9.12.28 Peripheral Pin Select Input Register 26**Name:** RPINR26**Offset:** 0xD38

Bit	15	14	13	12	11	10	9	8
	CAN2RXR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	CAN1RXR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – CAN2RXR[7:0] Assign CAN2 Input (CAN2RX) to the Corresponding RPn Pin bits**Bits 7:0 – CAN1RXR[7:0]** Assign CAN1 Input (CAN1RX) to the Corresponding RPn Pin bits

9.12.29 Peripheral Pin Select Input Register 27**Name:** RPINR27**Offset:** 0xD3A

Bit	15	14	13	12	11	10	9	8
	U3DSRR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	U3RXR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – U3DSRR[7:0] Assign UART3 Data-Set-Ready ($\overline{\text{U3DSR}}$) to the Corresponding RPn Pin bits**Bits 7:0 – U3RXR[7:0]** Assign UART3 Receive (U3RX) to the Corresponding RPn Pin bits

9.12.30 Peripheral Pin Select Input Register 29**Name:** RPINR29**Offset:** 0xD3E**Note:**

- These register bits are assigned with value of 52 to allow communication with the Secure Subsystem.

Bit	15	14	13	12	11	10	9	8
	SCK3R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	SDI3R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – SCK3R[7:0] Assign SPI3 Clock Input (SCK3IN) to the Corresponding RPn Pin bits**Bits 7:0 – SDI3R[7:0]** Assign SPI3 Data Input (SDI3) to the Corresponding RPn Pin bits⁽¹⁾

9.12.31 Peripheral Pin Select Input Register 30**Name:** RPINR30**Offset:** 0xD40

Bit	15	14	13	12	11	10	9	8
Access								
Reset								
Bit	7	6	5	4	3	2	1	0
	SS3R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 7:0 – SS3R[7:0] Assign SPI3 Client Select ($\overline{\text{SS3}}$) to the Corresponding RPn Pin bits

9.12.32 Peripheral Pin Select Input Register 32**Name:** RPINR32**Offset:** 0xD44

Bit	15	14	13	12	11	10	9	8
	TCKI9R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
Access								
Reset								

Bits 15:8 – TCKI9R[7:0] Assign MCCP Timer9 Input (TCKI9) to the Corresponding RPn Pin bits

9.12.33 Peripheral Pin Select Input Register 33**Name:** RPINR33**Offset:** 0xD46

Bit	15	14	13	12	11	10	9	8
Access								
Reset								
Bit	7	6	5	4	3	2	1	0
	ICM9R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 7:0 – ICM9R[7:0] Assign MCCP Capture 9 Input (ICM9) to the Corresponding RPn Pin bits

9.12.34 Peripheral Pin Select Input Register 37**Name:** RPINR37**Offset:** 0xD4E

Bit	15	14	13	12	11	10	9	8
	PCI17R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	OCFCR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – PCI17R[7:0] Assign PWM Input 17 (PCI17) to the Corresponding RPn Pin bits**Bits 7:0 – OCFCR[7:0]** Assign xCCP Fault C (OCFC) to the Corresponding RPn Pin bits

9.12.35 Peripheral Pin Select Input Register 38**Name:** RPINR38**Offset:** 0xD50

Bit	15	14	13	12	11	10	9	8
Access								
Reset								
Bit	7	6	5	4	3	2	1	0
	PCI18R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 7:0 – PCI18R[7:0] Assign PWM Input 18 (PCI18) to the Corresponding RPn Pin bits

9.12.36 Peripheral Pin Select Input Register 42**Name:** RPINR42**Offset:** 0xD58

Bit	15	14	13	12	11	10	9	8
	PCI13R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PCI12R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – PCI13R[7:0] Assign PWM Input 13 (PCI13) to the Corresponding RPn Pin bits**Bits 7:0 – PCI12R[7:0]** Assign PWM Input 12 (PCI12) to the Corresponding RPn Pin bits

9.12.37 Peripheral Pin Select Input Register 43**Name:** RPINR43**Offset:** 0xD5A

Bit	15	14	13	12	11	10	9	8
	PCI15R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PCI14R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – PCI15R[7:0] Assign PWM Input 15 (PCI15) to the Corresponding RPn Pin bits**Bits 7:0 – PCI14R[7:0]** Assign PWM Input 14 (PCI14) to the Corresponding RPn Pin bits

9.12.38 Peripheral Pin Select Input Register 44**Name:** RPINR44**Offset:** 0xD5C

Bit	15	14	13	12	11	10	9	8
	SENT1R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PCI16R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – SENT1R[7:0] Assign SENT1 Input (SENT1) to the Corresponding RPn Pin bits**Bits 7:0 – PCI16R[7:0]** Assign PWM Input 16 (PCI16) to the Corresponding RPn Pin bits

9.12.39 Peripheral Pin Select Input Register 45**Name:** RPINR45**Offset:** 0xD5E

Bit	15	14	13	12	11	10	9	8
	CLCINAR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	SENT2R[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – CLCINAR[7:0] Assign CLC Input A (CLCINA) to the Corresponding RPn Pin bits**Bits 7:0 – SENT2R[7:0]** Assign SENT2 Input (SENT2) to the Corresponding RPn Pin bits

9.12.40 Peripheral Pin Select Input Register 46**Name:** RPINR46**Offset:** 0xD60

Bit	15	14	13	12	11	10	9	8
	CLCINCR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	CLCINBR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – CLCINCR[7:0] Assign CLC Input C (CLCINC) to the Corresponding RPn Pin bits**Bits 7:0 – CLCINBR[7:0]** Assign CLC Input B (CLCINB) to the Corresponding RPn Pin bits

9.12.41 Peripheral Pin Select Input Register 47**Name:** RPINR47**Offset:** 0xD62

Bit	15	14	13	12	11	10	9	8
	ADCTRGR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	CLCINDR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – ADCTRGR[7:0] Assign ADC Trigger Input (ADCTRG) to the Corresponding RPn Pin bits**Bits 7:0 – CLCINDR[7:0]** Assign CLC Input D (CLCIND) to the Corresponding RPn Pin bits

9.12.42 Peripheral Pin Select Input Register 48**Name:** RPINR48**Offset:** 0xD64

Bit	15	14	13	12	11	10	9	8
	U1CTSR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	OCFDR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – U1CTSR[7:0] Assign UART1 Clear-to-Send ($\overline{U1CTS}$) to the Corresponding RPn Pin bits**Bits 7:0 – OCFDR[7:0]** Assign xCCP Fault D (OCFD) to the Corresponding RPn Pin bits

9.12.43 Peripheral Pin Select Input Register 49**Name:** RPINR49**Offset:** 0xD66

Bit	15	14	13	12	11	10	9	8
	U3CTSR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	U2CTSR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – U3CTSR[7:0] Assign UART3 Clear-to-Send ($\overline{U3CTS}$) to the Corresponding RPn Pin bits**Bits 7:0 – U2CTSR[7:0]** Assign UART2 Clear-to-Send ($\overline{U2CTS}$) to the Corresponding RPn Pin bits

9.12.44 Peripheral Pin Select Output Register 0**Name:** RPOR0**Offset:** 0xD80

Bit	15	14	13	12	11	10	9	8
			RP33R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP32R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP33R[5:0] Peripheral Output Function is Assigned to RP33 Output Pin bits**Bits 5:0 – RP32R[5:0]** Peripheral Output Function is Assigned to RP32 Output Pin bits

9.12.45 Peripheral Pin Select Output Register 1**Name:** RPOR1**Offset:** 0xD82

Bit	15	14	13	12	11	10	9	8
			RP35R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP34R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP35R[5:0] Peripheral Output Function is Assigned to RP35 Output Pin bits**Bits 5:0 – RP34R[5:0]** Peripheral Output Function is Assigned to RP34 Output Pin bits

9.12.46 Peripheral Pin Select Output Register 2**Name:** RPOR2**Offset:** 0xD84

Bit	15	14	13	12	11	10	9	8
			RP37R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP36R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP37R[5:0] Peripheral Output Function is Assigned to RP37 Output Pin bits**Bits 5:0 – RP36R[5:0]** Peripheral Output Function is Assigned to RP36 Output Pin bits

9.12.47 Peripheral Pin Select Output Register 3**Name:** RPOR3**Offset:** 0xD86

Bit	15	14	13	12	11	10	9	8
			RP39R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP38R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP39R[5:0] Peripheral Output Function is Assigned to RP39 Output Pin bits**Bits 5:0 – RP38R[5:0]** Peripheral Output Function is Assigned to RP38 Output Pin bits

9.12.48 Peripheral Pin Select Output Register 4**Name:** RPOR4**Offset:** 0xD88

Bit	15	14	13	12	11	10	9	8
			RP41R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP40R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP41R[5:0] Peripheral Output Function is Assigned to RP41 Output Pin bits**Bits 5:0 – RP40R[5:0]** Peripheral Output Function is Assigned to RP40 Output Pin bits

9.12.49 Peripheral Pin Select Output Register 5**Name:** RPOR5**Offset:** 0xD8A

Bit	15	14	13	12	11	10	9	8
			RP43R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP42R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP43R[5:0] Peripheral Output Function is Assigned to RP43 Output Pin bits**Bits 5:0 – RP42R[5:0]** Peripheral Output Function is Assigned to RP42 Output Pin bits

9.12.50 Peripheral Pin Select Output Register 6**Name:** RPOR6**Offset:** 0xD8C

Bit	15	14	13	12	11	10	9	8
			RP45R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP44R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP45R[5:0] Peripheral Output Function is Assigned to RP45 Output Pin bits**Bits 5:0 – RP44R[5:0]** Peripheral Output Function is Assigned to RP44 Output Pin bits

9.12.51 Peripheral Pin Select Output Register 7**Name:** RPOR7**Offset:** 0xD8E

Bit	15	14	13	12	11	10	9	8
			RP47R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP46R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP47R[5:0] Peripheral Output Function is Assigned to RP47 Output Pin bits**Bits 5:0 – RP46R[5:0]** Peripheral Output Function is Assigned to RP46 Output Pin bits

9.12.52 Peripheral Pin Select Output Register 8**Name:** RPOR8**Offset:** 0xD90

Bit	15	14	13	12	11	10	9	8
			RP49R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP48R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP49R[5:0] Peripheral Output Function is Assigned to RP49 Output Pin bits**Bits 5:0 – RP48R[5:0]** Peripheral Output Function is Assigned to RP48 Output Pin bits

9.12.53 Peripheral Pin Select Output Register 9**Name:** RPOR9**Offset:** 0xD92

Bit	15	14	13	12	11	10	9	8
			RP51R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP50R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP51R[5:0] Peripheral Output Function is Assigned to RP51 Output Pin bits**Bits 5:0 – RP50R[5:0]** Peripheral Output Function is Assigned to RP50 Output Pin bits

9.12.54 Peripheral Pin Select Output Register 10**Name:** RPOR10**Offset:** 0xD94**Note:**

1. These register bits are used only for communication with Secure Subsystem and are not available for general purpose.
2. These register bits are remapped to SCKx of corresponding SPI which is used for communication with Secure Subsystem. Refer to [Table 9-6](#) for remap value of corresponding SCKx.

Bit	15	14	13	12	11	10	9	8
			RP53R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			Reserved[5:0]					
Access			r	r	r	r	r	r
Reset			0	0	0	0	0	0

Bits 13:8 – RP53R[5:0] Peripheral Output Function is Assigned to RP53 Output Pin bits^(1,2)**Bits 5:0 – Reserved[5:0]**

9.12.55 Peripheral Pin Select Output Register 11**Name:** RPOR11**Offset:** 0xD96

Bit	15	14	13	12	11	10	9	8
			RP55R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP54R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP55R[5:0] Peripheral Output Function is Assigned to RP55 Output Pin bits**Bits 5:0 – RP54R[5:0]** Peripheral Output Function is Assigned to RP54 Output Pin bits

9.12.56 Peripheral Pin Select Output Register 12**Name:** RPOR12**Offset:** 0xD98

Bit	15	14	13	12	11	10	9	8
			RP57R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP56R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP57R[5:0] Peripheral Output Function is Assigned to RP57 Output Pin bits**Bits 5:0 – RP56R[5:0]** Peripheral Output Function is Assigned to RP56 Output Pin bits

9.12.57 Peripheral Pin Select Output Register 13**Name:** RPOR13**Offset:** 0xD9A**Notes:**

1. These register bits are used only for communication with Secure Subsystem and are not available for general purpose.
2. These register bits are remapped to SDOx of corresponding SPI which is used for communication with Secure Subsystem. Refer to [Table 9-6](#) for remap value of corresponding SDOx.

Bit	15	14	13	12	11	10	9	8
	RP59R[5:0]							
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	RP58R[5:0]							
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP59R[5:0] Peripheral Output Function is Assigned to RP59 Output Pin bits**Bits 5:0 – RP58R[5:0]** Peripheral Output Function is Assigned to RP58 Output Pin bits^(1,2)

9.12.58 Peripheral Pin Select Output Register 14**Name:** RPOR14**Offset:** 0xD9C

Bit	15	14	13	12	11	10	9	8
			RP61R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP60R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP61R[5:0] Peripheral Output Function is Assigned to RP61 Output Pin bits**Bits 5:0 – RP60R[5:0]** Peripheral Output Function is Assigned to RP60 Output Pin bits

9.12.59 Peripheral Pin Select Output Register 15**Name:** RPOR15**Offset:** 0xD9E

Bit	15	14	13	12	11	10	9	8
			RP63R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP62R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP63R[5:0] Peripheral Output Function is Assigned to RP63 Output Pin bits**Bits 5:0 – RP62R[5:0]** Peripheral Output Function is Assigned to RP62 Output Pin bits

9.12.60 Peripheral Pin Select Output Register 16**Name:** RPOR16**Offset:** 0xDA0

Bit	15	14	13	12	11	10	9	8
			RP65R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP64R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP65R[5:0] Peripheral Output Function is Assigned to RP65 Output Pin bits**Bits 5:0 – RP64R[5:0]** Peripheral Output Function is Assigned to RP64 Output Pin bits

9.12.61 Peripheral Pin Select Output Register 17**Name:** RPOR17**Offset:** 0xDA2**Legend:** r = Reserved bit

Bit	15	14	13	12	11	10	9	8
			Reserved[5:0]					
Access			r	r	r	r	r	r
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP66R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – Reserved[5:0]**Bits 5:0 – RP66R[5:0]** Peripheral Output Function is Assigned to RP66 Output Pin bits

9.12.62 Peripheral Pin Select Output Register 18**Name:** RPOR18**Offset:** 0xDA4**Legend:** r = Reserved bit

Bit	15	14	13	12	11	10	9	8
			RP69R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			Reserved[5:0]					
Access			r	r	r	r	r	r
Reset			0	0	0	0	0	0

Bits 13:8 – RP69R[5:0] Peripheral Output Function is Assigned to RP69 Output Pin bits**Bits 5:0 – Reserved[5:0]**

9.12.63 Peripheral Pin Select Output Register 19**Name:** RPOR19**Offset:** 0xDA6

Bit	15	14	13	12	11	10	9	8
			RP71R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP70R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP71R[5:0] Peripheral Output Function is Assigned to RP71 Output Pin bits**Bits 5:0 – RP70R[5:0]** Peripheral Output Function is Assigned to RP70 Output Pin bits

9.12.64 Peripheral Pin Select Output Register 20**Name:** RPOR20**Offset:** 0xDA8**Note:**

- These are virtual output ports.

Bit	15	14	13	12	11	10	9	8
			RP73R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP72R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP73R[5:0] Peripheral Output Function is Assigned to RP73 Output Pin bits⁽¹⁾**Bits 5:0 – RP72R[5:0]** Peripheral Output Function is Assigned to RP72 Output Pin bits⁽¹⁾

9.12.65 Peripheral Pin Select Output Register 21

Name: RPOR21
Offset: 0xDAA

Note:
1. These are virtual output ports.

Bit	15	14	13	12	11	10	9	8
			RP75R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP74R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP75R[5:0] Peripheral Output Function is Assigned to RP75 Output Pin bits⁽¹⁾

Bits 5:0 – RP74R[5:0] Peripheral Output Function is Assigned to RP74 Output Pin bits⁽¹⁾

9.12.66 Peripheral Pin Select Output Register 22**Name:** RPOR22**Offset:** 0xDAC**Note:**

- These are virtual output ports.

Bit	15	14	13	12	11	10	9	8
			RP77R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP76R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP77R[5:0] Peripheral Output Function is Assigned to RP77 Output Pin bits⁽¹⁾**Bits 5:0 – RP76R[5:0]** Peripheral Output Function is Assigned to RP76 Output Pin bits⁽¹⁾

9.12.67 Peripheral Pin Select Output Register 23**Name:** RPOR23**Offset:** 0xDAE

Bit	15	14	13	12	11	10	9	8
			RP79R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP78R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP79R[5:0] Peripheral Output Function is Assigned to RP79 Output Pin bits**Bits 5:0 – RP78R[5:0]** Peripheral Output Function is Assigned to RP78 Output Pin bits

9.12.68 Peripheral Pin Select Output Register 24

Name: RPOR24
Offset: 0xDB0

Note:
1. These are virtual output ports.

Bit	15	14	13	12	11	10	9	8
			RP81R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP80R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP81R[5:0] Peripheral Output Function is Assigned to RP81 Output Pin bits⁽¹⁾

Bits 5:0 – RP80R[5:0] Peripheral Output Function is Assigned to RP80 Output Pin bits⁽¹⁾

9.12.69 Peripheral Pin Select Output Register 25

Name: RPOR25
Offset: 0xDB2

Note:
1. These are virtual output ports.

Bit	15	14	13	12	11	10	9	8
			RP83R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP82R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP83R[5:0] Peripheral Output Function is Assigned to RP83 Output Pin bits⁽¹⁾

Bits 5:0 – RP82R[5:0] Peripheral Output Function is Assigned to RP82 Output Pin bits⁽¹⁾

9.12.70 Peripheral Pin Select Output Register 26

Name: RPOR26
Offset: 0xDB4

Note:
1. These are virtual output ports.

Bit	15	14	13	12	11	10	9	8
			RP85R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP84R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP85R[5:0] Peripheral Output Function is Assigned to RP85 Output Pin bits⁽¹⁾

Bits 5:0 – RP84R[5:0] Peripheral Output Function is Assigned to RP84 Output Pin bits⁽¹⁾

9.12.71 Peripheral Pin Select Output Register 27**Name:** RPOR27**Offset:** 0xDB6**Note:**

- These are virtual output ports.

Bit	15	14	13	12	11	10	9	8
			RP87R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP86R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP87R[5:0] Peripheral Output Function is Assigned to RP87 Output Pin bits⁽¹⁾**Bits 5:0 – RP86R[5:0]** Peripheral Output Function is Assigned to RP86 Output Pin bits⁽¹⁾

9.12.72 Peripheral Pin Select Output Register 28

Name: RPOR28
Offset: 0xDB8

Note:
1. These are virtual output ports.

Bit	15	14	13	12	11	10	9	8
			RP89R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP88R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP89R[5:0] Peripheral Output Function is Assigned to RP89 Output Pin bits⁽¹⁾

Bits 5:0 – RP88R[5:0] Peripheral Output Function is Assigned to RP88 Output Pin bits⁽¹⁾

9.12.73 Peripheral Pin Select Output Register 29

Name: RPOR29
Offset: 0xDBA

Note:
1. These are virtual output ports.

Bit	15	14	13	12	11	10	9	8
			RP91R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP90R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP91R[5:0] Peripheral Output Function is Assigned to RP91 Output Pin bits⁽¹⁾

Bits 5:0 – RP90R[5:0] Peripheral Output Function is Assigned to RP90 Output Pin bits⁽¹⁾

9.12.74 Peripheral Pin Select Output Register 30**Name:** RPOR30**Offset:** 0xDBC**Note:**

- These are virtual output ports.

Bit	15	14	13	12	11	10	9	8
			RP93R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP92R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP93R[5:0] Peripheral Output Function is Assigned to RP93 Output Pin bits⁽¹⁾**Bits 5:0 – RP92R[5:0]** Peripheral Output Function is Assigned to RP92 Output Pin bits⁽¹⁾

9.12.75 Peripheral Pin Select Output Register 31**Name:** RPOR31**Offset:** 0xDBE**Note:**

- These are virtual output ports.

Bit	15	14	13	12	11	10	9	8
			RP95R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP94R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP95R[5:0] Peripheral Output Function is Assigned to RP95 Output Pin bits⁽¹⁾**Bits 5:0 – RP94R[5:0]** Peripheral Output Function is Assigned to RP94 Output Pin bits⁽¹⁾

9.12.76 Peripheral Pin Select Output Register 32**Name:** RPOR32**Offset:** 0xDC0**Note:**

- These are virtual output ports.

Bit	15	14	13	12	11	10	9	8
			RP97R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP96R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP97R[5:0] Peripheral Output Function is Assigned to RP97 Output Pin bits⁽¹⁾**Bits 5:0 – RP96R[5:0]** Peripheral Output Function is Assigned to RP96 Output Pin bits⁽¹⁾

9.12.77 Peripheral Pin Select Output Register 33

Name: RPOR33
Offset: 0xDC2

Note:
1. These are virtual output ports.

Bit	15	14	13	12	11	10	9	8
			RP99R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP98R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP99R[5:0] Peripheral Output Function is Assigned to RP99 Output Pin bits⁽¹⁾

Bits 5:0 – RP98R[5:0] Peripheral Output Function is Assigned to RP98 Output Pin bits⁽¹⁾

9.12.78 Peripheral Pin Select Output Register 34**Name:** RPOR34**Offset:** 0xDC4**Note:**

- These are virtual output ports.

Bit	15	14	13	12	11	10	9	8
			RP177R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP176R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP177R[5:0] Peripheral Output Function is Assigned to RP177 Output Pin bits⁽¹⁾**Bits 5:0 – RP176R[5:0]** Peripheral Output Function is Assigned to RP176 Output Pin bits⁽¹⁾

9.12.79 Peripheral Pin Select Output Register 35**Name:** RPOR35**Offset:** 0xDC6**Note:**

- These are virtual output ports.

Bit	15	14	13	12	11	10	9	8
			RP179R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP178R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP179R[5:0] Peripheral Output Function is Assigned to RP179 Output Pin bits⁽¹⁾**Bits 5:0 – RP178R[5:0]** Peripheral Output Function is Assigned to RP178 Output Pin bits⁽¹⁾

9.12.80 Peripheral Pin Select Output Register 36**Name:** RPOR36**Offset:** 0xDC8**Note:**

- These are virtual output ports.

Bit	15	14	13	12	11	10	9	8
			RP181R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
			RP180R[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 13:8 – RP181R[5:0] Peripheral Output Function is Assigned to RP181 Output Pin bits⁽¹⁾**Bits 5:0 – RP180R[5:0]** Peripheral Output Function is Assigned to RP180 Output Pin bits⁽¹⁾

10. Oscillator with High-Frequency PLL

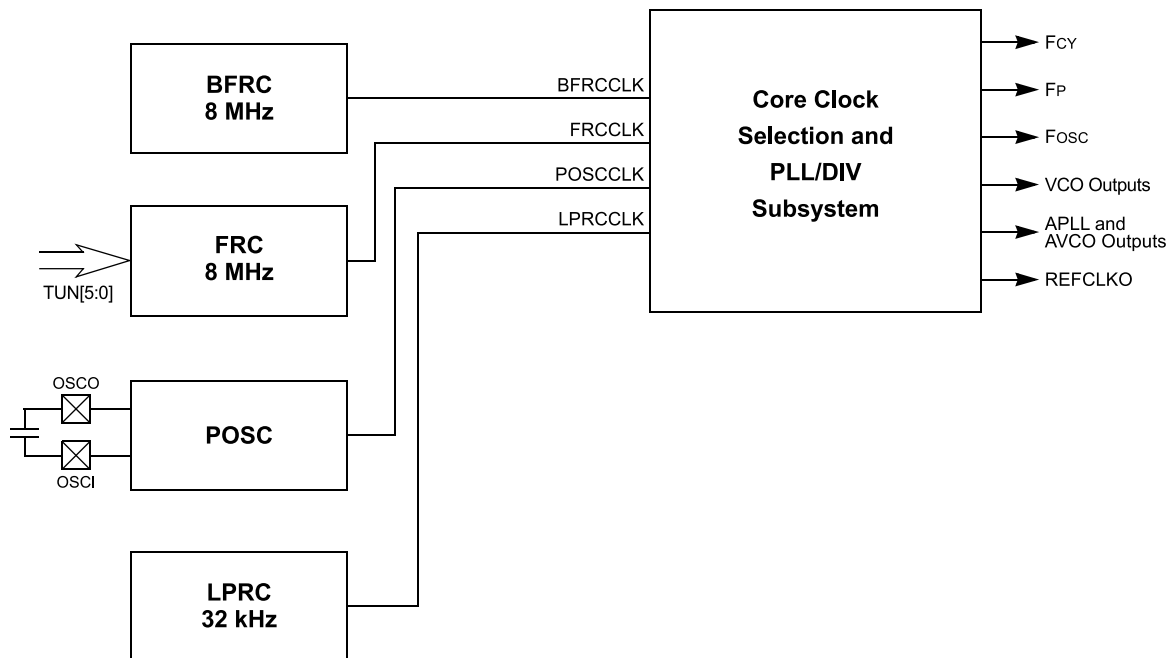
Note: This data sheet summarizes the features of the dsPIC33CK512MPT608 family of devices. It is not intended to be a comprehensive reference source. To complement the information in this data sheet, refer to “**Oscillator Module with High-Speed PLL**” (www.microchip.com/DS70005255) in the “dsPIC33/PIC24 Family Reference Manual”.

The dsPIC33CK512MPT608 family oscillator with high-frequency PLL includes these characteristics:

- On-Chip Phase-Locked Loop (PLL) to Boost Internal Operating Frequency on Select Internal and External Oscillator Sources
- Auxiliary PLL (APLL) Clock Generator to Boost Operating Frequency for Peripherals
- Doze mode for System Power Savings
- Scalable Reference Clock Output (REFCLKO)
- On-the-Fly Clock Switching between Various Clock Sources
- Fail-Safe Clock Monitoring (FSCM) that Detects Clock Failure and Permits Safe Application Recovery or Shutdown

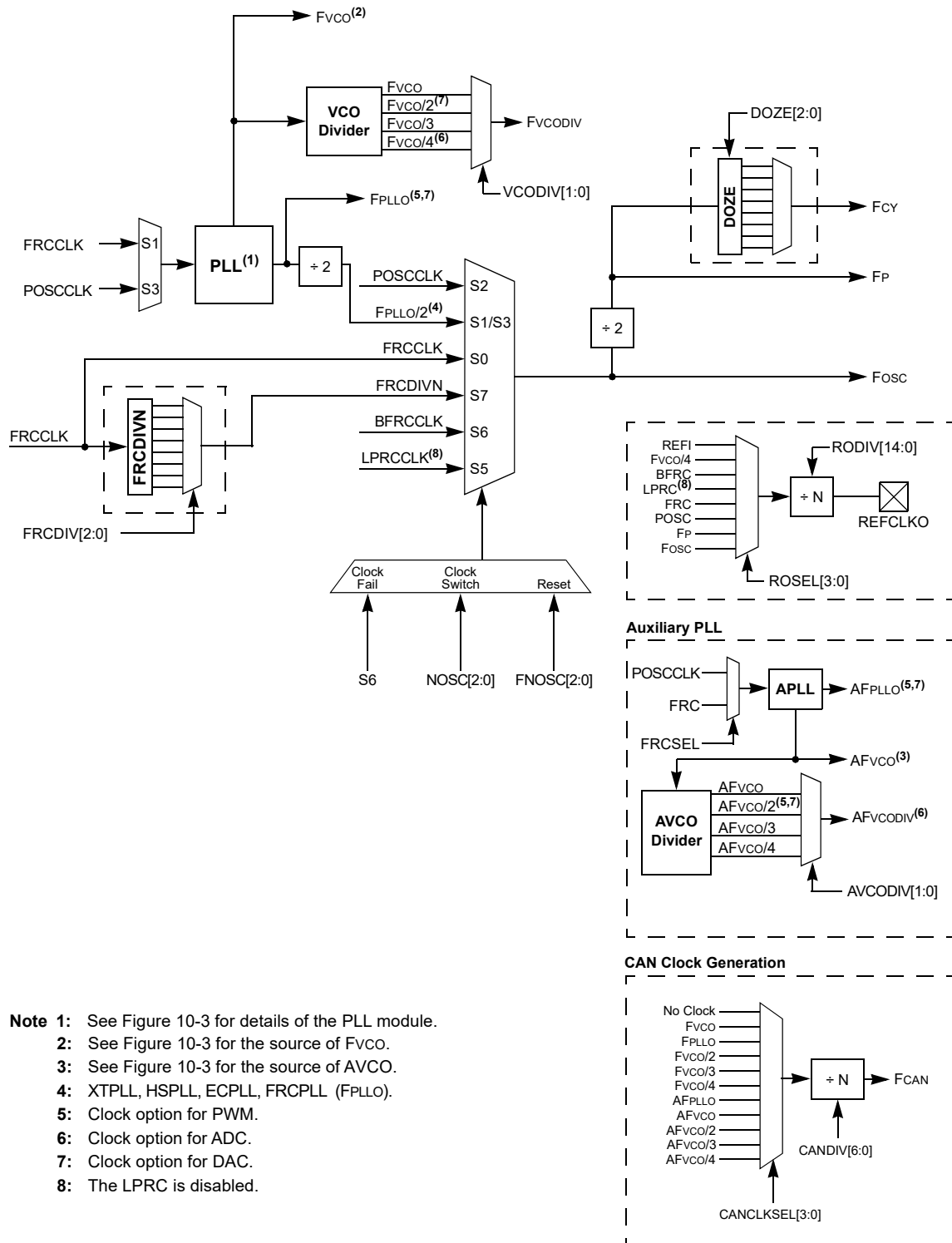
A block diagram of the dsPIC33CK512MPT608 oscillator system is shown in [Figure 10-1](#).

Figure 10-1. dsPIC33CK512MPT608 Core Clock Sources Block Diagram



Oscillator with High-Frequency PLL

Figure 10-2. dsPIC33CK512MPT608 Oscillator Subsystem



Note 1: See Figure 10-3 for details of the PLL module.

2: See Figure 10-3 for the source of FVCO.

3: See Figure 10-3 for the source of AVCO.

4: XTPLL, HSPLL, ECPLL, FRCPLL (F_{PLLO}).

5: Clock option for PWM.

6: Clock option for ADC.

7: Clock option for DAC.

8: The LPRC is disabled.

10.1 Primary PLL

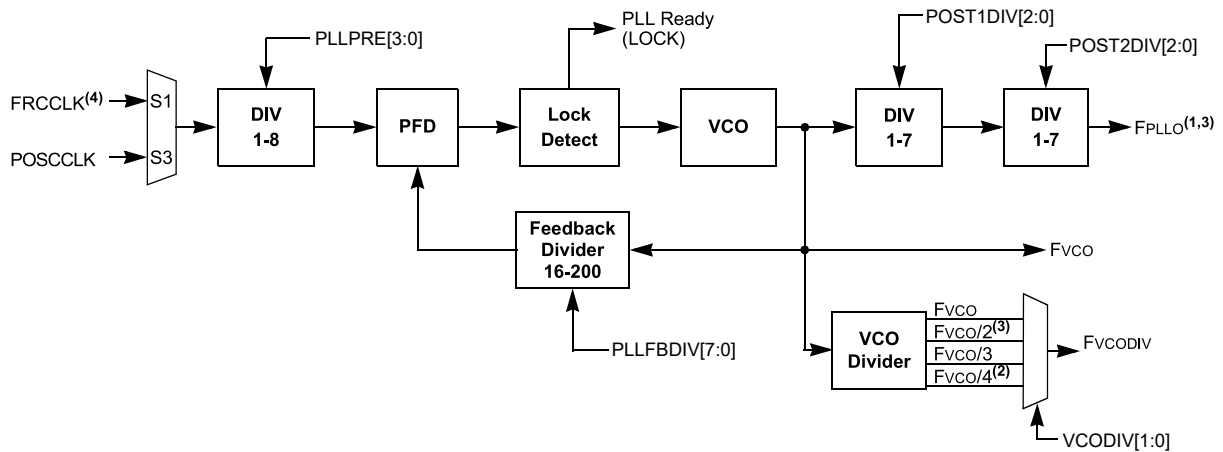
The Primary Oscillator and internal FRC Oscillator sources can optionally use an on-chip PLL to obtain higher operating speeds. Figure 10-3 illustrates a block diagram of the Primary PLL module.

For PLL operation, the following requirements must be met at all times without exception:

- The PLL Input Frequency (F_{PLLI}) must be in the range of 8 MHz to 64 MHz
- The PFD Input Frequency (F_{PFD}) must be in the range of 8 MHz to ($F_{VCO}/16$) MHz

The VCO Output Frequency (F_{VCO}) must be in the range of 400 MHz to 1600 MHz

Figure 10-3. Primary Core PLL And VCO Detail



- Note 1:** Clock option for PWM.
Note 2: Clock option for ADC.
Note 3: Clock option for DAC.
Note 4: PLL source is always FRC unless FNOSC is the Primary Oscillator with PLL.

Equation 10-1 provides the relationship between the PLL Input Frequency (F_{PLLI}) and VCO Output Frequency (F_{VCO}).

Equation 10-1. Primary Core F_{VCO} Calculation

$$F_{VCO} = F_{PLLI} \times \left(\frac{M}{N1} \right) = F_{PLLI} \times \left(\frac{PLLFBDIV[7:0]}{PLLPRE[3:0]} \right)$$

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Oscillator with High-Frequency PLL

Equation 10-2 provides the relationship between the PLL Input Frequency (F_{PLLI}) and PLL Output Frequency (F_{PLLO}).

Equation 10-2. Primary Core F_{PLLO} Calculation

$$F_{PLLO} = F_{PLLI} \times \left(\frac{M}{N1 \times N2 \times N3} \right) = F_{PLLI} \times \left(\frac{PLLFBDIV[7:0]}{PLLPRE[3:0] \times POST1DIV[2:0] \times POST2DIV[2:0]} \right)$$

Where:

$M = PLLFBDIV[7:0]$

$N1 = PLLPRE[3:0]$

$N2 = POST1DIV[2:0]$

$N3 = POST2DIV[2:0]$

Note: The PLL Phase Detector Input Divider Select (PLLPREx) bits and the PLL Feedback Divider (PLLFBDIVx) bits should not be changed when operating in PLL mode. Therefore, the user must start on either a non-PLL source or clock switch to a non-PLL source (e.g., internal FRC Oscillator) to make any necessary changes and then clock switch to the desired PLL source.

Using Two-Speed Start-up (IESO (FOSCSEL[7])) with a PLL source will start the device on the FRC while preparing the PLL. Once the PLL is ready, the device will switch automatically to the new source. This mode should not be used if changes are needed to the PLLPREx and PLLFBDIVx bits because the PLL may be running before user code execution begins.

It is not permitted to directly clock switch from one PLL clock source to a different PLL clock source. The user would need to transition between PLL clock sources with a clock switch to a non-PLL clock source.

Example 10-1. Code for Using Primary PLL with 8 MHz Internal FRC

```
//code example for 50 MIPS system clock using 8MHz FRC
// Select Internal FRC at POR
_FOSCSEL(FNOSC_FRC & IESO_OFF);
// Enable Clock Switching
_FOSC(FCKSM_CSECMD);
int main()
{
    // Configure PLL prescaler, both PLL postscalers, and PLL feedback
    divider
    CLKDIVbits.PLLPRE = 1;           // N1=1
    PLLFBDbits.PLLFBDIV = 125;       // M = 125
    PLLDIVbits.POST1DIV = 5;         // N2=5
    PLLDIVbits.POST2DIV = 1;         // N3=1
    // Initiate Clock Switch to FRC with PLL (NOSC=0b001)
    __builtin_write_OSCCONH(0x01);
    __builtin_write_OSCCONL(OSCCON | 0x01);
    // Wait for Clock switch to occur
    while (OSCCONbits.OSWEN != 0);
}
```

Note: $F_{PLLO} = F_{PLLI} * M / (N1 * N2 * N3)$; $F_{PLLI} = 8$; $M = 125$; $N1 = 1$; $N2 = 5$; $N3 = 1$;
so $F_{PLLO} = 8 * 125 / (1 * 5 * 1) = 200$ MHz or 50 MIPS.

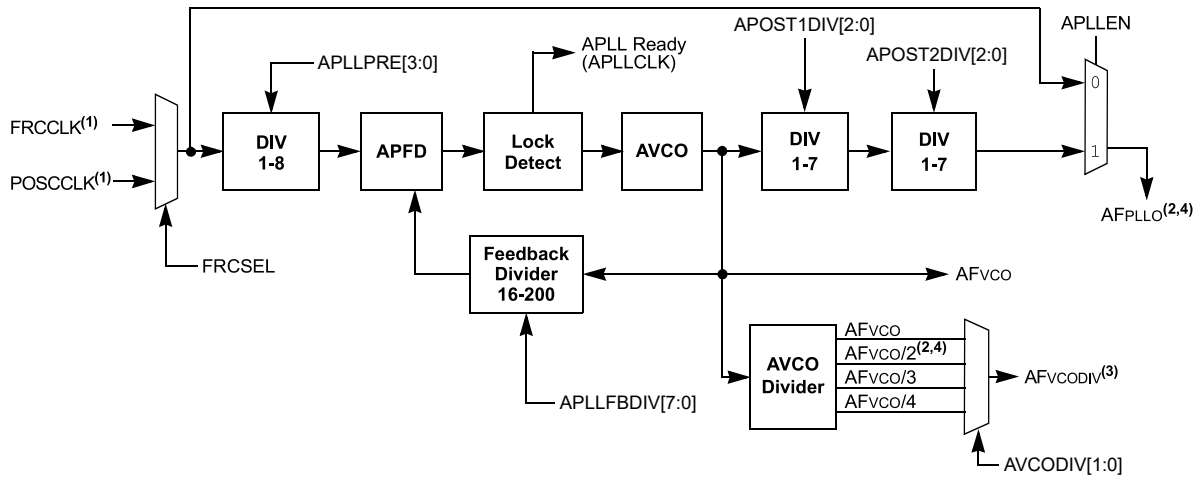
10.2 Auxiliary PLL

The dsPIC33CK512MPT608 device family implements an Auxiliary PLL (APLL) module for each core present. The APLL is used to generate various peripheral clock sources independent of the system clock. Figure 10-4 shows a block diagram of the Auxiliary core APLL module.

For APLL operation, the following requirements must be met at all times without exception:

- The APLL Input Frequency (AF_{PLL_i}) must be in the range of 8 MHz to 64 MHz
- The APFD Input Frequency (AF_{PFD}) must be in the range of 8 MHz to ($AF_{VCO}/16$) MHz
- The AVCO Output Frequency (AF_{VCO}) must be in the range of 400 MHz to 1600 MHz

Figure 10-4. Auxiliary Core APLL and VCO Detail



Equation 10-3 provides the relationship between the APLL Input Frequency (AF_{PLL_i}) and the AVCO Output Frequency (AF_{VCO}).

Equation 10-3. Auxiliary Core AF_{VCO} Calculation

$$F_{VCO} = F_{PLL_i} \times \left(\frac{M}{N1} \right) = F_{PLL_i} \times \left(\frac{PLLFBDIV [7:0]}{PLLPRE [3:0]} \right)$$

Equation 10-4 provides the relationship between the APLL Input Frequency (AF_{PLL_i}) and APLL Output Frequency (AF_{PLLO}).

Equation 10-4. Auxiliary Core AF_{PLLO} Calculation

$$AF_{PLLO} = AF_{PLL_i} \times \left(\frac{M}{N1 \times N2 \times N3} \right) = AF_{PLL_i} \times \left(\frac{APLLFBDIV [7:0]}{APLLPRE [3:0] \times APOST1DIV [2:0] \times APOST2DIV [2:0]} \right)$$

Where:

$M = APLLFBDIV[7:0]$

$N1 = APLLPRE[3:0]$

$N2 = APOST1DIV[2:0]$

$N3 = APOST2DIV[2:0]$

Example 10-2. Code for Using Auxiliary PLL with the Internal FRC Oscillator

```
//code example for AFVCO = 1 GHz and AFPLLO = 500 MHz using 8 MHz
//internal FRC
// Configure the source clock for the APLL
ACLKCON1bits.FRCSEL = 1;          // Select internal FRC as the clock
// source
// Configure the APLL prescaler, APLL feedback divider, and both APLL
// postscalers.
ACLKCON1bits.APLLPRE = 1;          // N1 = 1
APLLFBD1bits.APLLFBDIV = 125;      // M = 125
APLLDIV1bits.APOST1DIV = 2;        // N2 = 2
APLLDIV1bits.APOST2DIV = 1;        // N3 = 1
// Enable APLL
ACLKCON1bits.APLEN = 1;
```

Note: Even with the APLEN bit set, another peripheral must generate a clock request before the APLL will start.

10.3 CPU Clocking

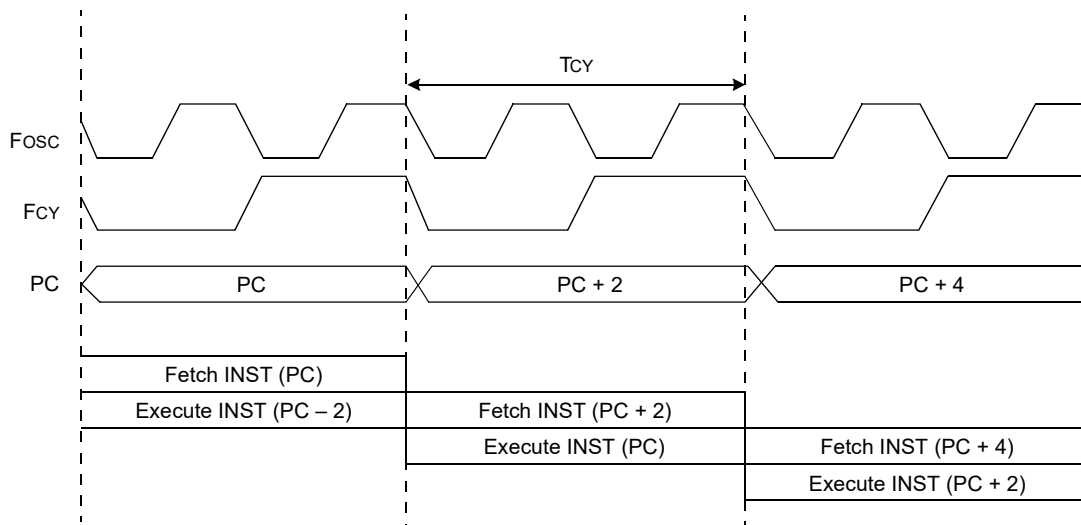
The dsPIC33CK512MPT608 devices can be configured to use any of the following clock configurations:

- Primary Oscillator (POSC) on the OSCI and OSCO pins
- Internal Fast RC (FRC) Oscillator with optional clock divider
- Internal Low-Power RC Oscillator
- Primary Oscillator with PLL (ECPLL, HSPLL, XTPLL)
- Internal Fast RC Oscillator with PLL (FRCPLL)
- Backup Internal Fast RC Oscillator (BFRC)

The system clock source is divided by two to produce the internal instruction cycle clock. In this document, the instruction cycle clock is denoted by F_{CY} . The timing diagram in [Figure 10-5](#) illustrates the relationship between the system clock (F_{OSC}), the instruction cycle clock (F_{CY}) and the Program Counter (PC).

The internal instruction cycle clock (F_{CY}) can be output on the OSCO I/O pin if the Primary Oscillator mode (POSCMD[1:0]) is not configured as HS/XT. For more information, see [10. Oscillator with High-Frequency PLL](#).

Figure 10-5. Clock and Instruction Cycle Timing



10.4 Primary Oscillator (POSC)

The dsPIC33CK512MPT608 family devices contain one instance of the Primary Oscillator (POSC). The Primary Oscillator is available on the OSCI and OSCO pins of the dsPIC33CK devices. This connection enables an external crystal (or ceramic resonator) to provide the clock to the device. The Primary Oscillator provides three modes of operation:

- **Medium Speed Oscillator (XT Mode):**
The XT mode is a Medium Gain, Medium Frequency mode used to work with crystal frequencies of 3.5 MHz to 10 MHz.
- **High-Speed Oscillator (HS Mode):**
The HS mode is a High-Gain, High-Frequency mode used to work with crystal frequencies of 10 MHz to 32 MHz.
- **External Clock Source Operation (EC Mode):**
If the on-chip oscillator is not used, the EC mode allows the internal oscillator to be bypassed. The device clocks are generated from an external source (0 MHz to up to 64 MHz) and input on the OSCI pin.

Example 10-3 illustrates code for using the PLL (50 MIPS) with the Primary Oscillator.

Example 10-3. Code for Using PLL (50 MIPS) with Primary Oscillator (POSC)

```
//code example for 50 MIPS system clock using POSC with 10 MHz external
crystal
// Select FRC on POR
#pragma config FNOSC = FRC // Oscillator Source Selection (Internal
Fast RC (FRC))
#pragma config IESO = OFF
/// Enable Clock Switching and Configure POSC in XT mode
#pragma config POSCMD = XT
#pragma config FCKSM = CSECMD
int main()
{
    // Configure PLL prescaler, both PLL postscalers, and PLL feedback
    divider
    CLKDIVbits.PLLPRE = 1; // N1=1
    PLLFBDbits.PLLFBDIV = 100; // M = 100
    PLLDIVbits.POST1DIV = 5; // N2=5
    PLLDIVbits.POST2DIV = 1; // N3=1
    // Initiate Clock Switch to Primary Oscillator with PLL (NOSC=0b011)
    __builtin_write_OSCCONH(0x03);
    __builtin_write_OSCCONL(OSCCON | 0x01);
    // Wait for Clock switch to occur
    while (OSCCONbits.OSWEN != 0);
    // Wait for PLL to lock
    while (OSCCONbits.LOCK != 1);
}
```

10.4.1 Primary Oscillator Pin Functionality

The Primary Oscillator pins (OSCI and OSCO) can be used for other functions when the Primary Oscillator is not being used. The POSCMD[1:0] Configuration bits in the Oscillator Configuration register (FOSC[1:0]) determine the oscillator pin function. The OSCIOFNC bit (FOSC[2]) determines the OSCO/CLKO pin function. By default, the CLKO function is active and the pin will output a clock frequency of F_{CY} . A clock signal is present on the OSCO/CLKO pin when device is unprogrammed or during the programming sequence. Care should be taken when the OSCO/CLKO pin is used to drive other circuitry.

10.5 Internal Fast RC (FRC) Oscillator

The dsPIC33CK512MPT608 family devices contain one instance of the internal Fast RC (FRC) Oscillator. The FRC Oscillator provides a nominal 8 MHz clock without requiring an external crystal or ceramic resonator, which results in system cost savings for applications that do not require a precise clock reference.

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Oscillator with High-Frequency PLL

The application software can tune the frequency of the oscillator using the FRC Oscillator Tuning bits (TUN[5:0]) in the FRC Oscillator Tuning register (OSCTUN[5:0]).

10.6 Low-Power RC Oscillator

The dsPIC33CK512MPT608 family devices contain one instance of the Low-Power RC (LPRC) Oscillator, which provides a nominal clock frequency of 32.768 kHz. The dsPIC33CK512MPT608 family devices implement the LPRC function with the BFRC and post-divider to yield a 50% duty cycle output.

The LPRC is the clock source for the Watchdog Timer (WDT) and Fail-Safe Clock Monitor (FSCM) circuits in the clock subsystem. The LPRC Oscillator is shut off in Sleep mode.

The LPRC Oscillator remains enabled under these conditions:

- The FSCM is enabled
- The WDT is enabled
- The LPRC Oscillator is selected as the system clock

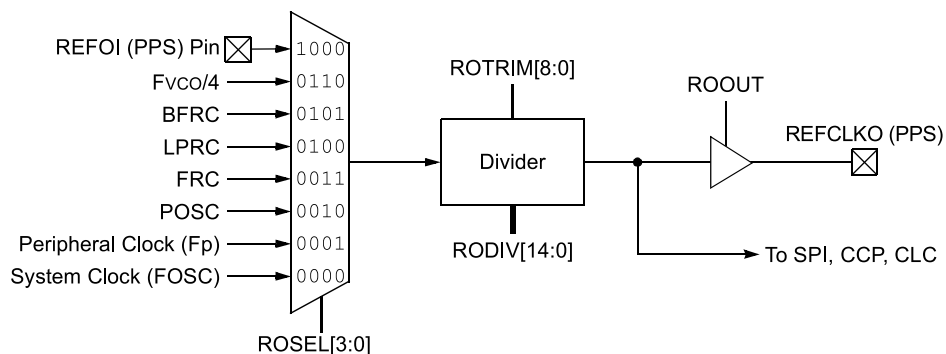
10.7 Backup Internal Fast RC (BFRC) Oscillator

The oscillator block provides a stable reference clock source for the Fail-Safe Clock Monitor (FSCM). When FSCM is enabled in the FCKSM[1:0] Configuration bits (FOSC[7:6]), it constantly monitors the main clock source against a reference signal from the 8 MHz Backup Internal Fast RC (BFRC) Oscillator. In case of a clock failure, the Fail-Safe Clock Monitor switches the clock to the BFRC Oscillator, allowing for continued low-speed operation or a safe application shutdown.

10.8 Reference Clock Output

In addition to the CLKO output ($F_{OSC}/2$), the dsPIC33CK512MPT608 family devices can be configured to provide a reference clock output signal to a port pin. This feature is available in all oscillator configurations and allows the user to select a greater range of clock submultiples to drive external devices in the application. CLKO is enabled by Configuration bit, OSCIOFNC, and is independent of the REFCLKO reference clock. REFCLKO is mappable to any I/O pin that has mapped output capability. The reference clock output module block diagram is shown in [Figure 10-6](#).

Figure 10-6. Reference Clock Generator



This reference clock output is controlled by the REFOCONL and REFOCONH registers. Setting the ROEN bit (REFOCONL[15]) makes the clock signal available on the REFCLKO pin. The RODIV[14:0] bits (REFOCONH[14:0]) and ROTRIM[8:0] bits (REFOTRIM[15:7]) enable the selection of different clock divider options. The formula for determining the final frequency output is shown in [Equation 10-5](#). The ROSWEN bit (REFOCONL[9]) indicates that the clock divider has been successfully switched. In order to switch the REFCLKO divider, the user should ensure that this bit reads as '0'. Write the updated values to the RODIV[14:0] or ROTRIM[8:0] bits, set the ROSWEN bit and then wait until it is cleared before assuming that the REFCLKO clock is valid.

Equation 10-5. Calculating Frequency Output

$$F_{REFOUT} = \frac{F_{REFIN}}{2 \cdot (RODIV[14:0] + ROTRIM[8:0]/512)}$$

Where: F_{REFOUT} = Output Frequency
 F_{REFIN} = Input Frequency
 When $RODIV[14:0] = 0$, the output clock is the same as the input clock.

The ROSEL[3:0] bits (REFOCONL[3:0]) determine which clock source is used for the reference clock output. The ROSLP bit (REFOCONL[11]) determines if the reference source is available on REFCLKO when the device is in Sleep mode.

To use the reference clock output in Sleep mode, both the ROSLP bit must be set and the clock selected by the ROSEL[3:0] bits must be enabled for operation during Sleep mode, if possible. Clearing the ROSEL[3:0] bits allows the reference output frequency to change, as the system clock changes during any clock switches. The ROOUT bit enables/disables the reference clock output on the REFCLKO pin.

The ROACTIV bit (REFOCONL[8]) indicates that the module is active; it can be cleared by disabling the module (setting ROEN to '0'). The user must not change the reference clock source, or adjust the divider when the ROACTIV bit indicates that the module is active. To avoid glitches, the user should not disable the module until the ROACTIV bit is '1'.

10.9 Oscillator Configuration

The oscillator system has both Configuration registers and SFRs to configure, control and monitor the system. The FOSCSEL and FOSC Configuration registers (FOSCSEL and FOSC, respectively) are used for initial setup.

Table 10-1 lists the configuration settings that select the device's oscillator source and operating mode at a Power-on Reset (POR).

Table 10-1. Configuration Bit Values for Clock Selection

Oscillator Source	Oscillator Mode	FNOSC[2:0] Value	POSCMD[1:0] Value	Notes
S0	Fast RC Oscillator (FRC)	000	xx	Note 1
S1	Fast RC Oscillator with PLL (FRCPLL)	001	xx	Note 1
S2	Primary Oscillator (EC)	010	00	Note 1
S2	Primary Oscillator (XT)	010	01	
S2	Primary Oscillator (HS)	010	10	
S3	Primary Oscillator with PLL (ECPLL)	011	00	Note 1
S3	Primary Oscillator with PLL (XTPLL)	011	01	
S3	Primary Oscillator with PLL (HSPLL)	011	10	
S4	Reserved	100	xx	
S5	Low-Power RC Oscillator (LPRC)	101	xx	Note 1

Notes:

1. The OSCO pin function is determined by the OSCIOFNC Configuration bit.
2. This is the default oscillator mode for an unprogrammed (erased) device.

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.....continued

Oscillator Source	Oscillator Mode	FNOSC[2:0] Value	POSCMD[1:0] Value	Notes
S6	Backup FRC (BFRC)	110	xx	Note 1
S7	Fast RC Oscillator with ÷ N Divider (FRCDIVN)	111	xx	Note 1, Note 2

Notes:

1. The OSCO pin function is determined by the OSCIOFNC Configuration bit.
2. This is the default oscillator mode for an unprogrammed (erased) device.

10.10 OSCCON Unlock Sequence

The OSCCON register is protected against unintended writes through a lock mechanism. The upper and lower bytes of OSCCON have their own unlock sequence, and both must be used when writing to both bytes of the register.

Before OSCCON can be written to, the following unlock sequence must be used:

1. Execute the unlock sequence for the OSCCON high byte. In two back-to-back instructions:
 - Write 0x78 to OSCCON[15:8]
 - Write 0x9A to OSCCON[15:8]
2. In the instruction immediately following the unlock sequence, the OSCCON[15:8] bits can be modified.
3. Execute the unlock sequence for the OSCCON low byte. In two back-to-back instructions:
 - Write 0x46 to OSCCON[7:0]
 - Write 0x57 to OSCCON[7:0]
4. In the instruction immediately following the unlock sequence, the OSCCON[7:0] bits can be modified.

Note: MPLAB® XC16 provides a built-in C language function, including the unlocking sequence to modify high and low bytes in the OSCCON register:

```
__builtin_write_OSCCONH(value)
__builtin_write_OSCCONL(value)
```


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10.11 Oscillator Control Registers

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x00 ... 0x0F83	Reserved									
0x0F84	OSCCON(1)	15:8			COSC[2:0]			NOSC[2:0]		
		7:0	CLKLOCK		LOCK		CF			OSWEN
0x0F86	CLKDIV	15:8	ROI		DOZE[2:0]		DOZEN		FRCDIV[2:0]	
		7:0			Reserved[1:0]			PLLPRE[3:0]		
0x0F88 ... 0x0F89	Reserved									
0x0F8A	PLLFBD	15:8						Reserved[3:0]		
		7:0				PLLFBDIV[7:0]				
0x0F8A	PLLDIV	15:8							VCODIV[1:0]	
		7:0			POST1DIV[2:0]			POST2DIV[2:0]		
0x0F8C	OSCTUN	15:8								
		7:0					TUN[5:0]			
0x0F8E	ACLKCON1	15:8	APLLEN	APLLCK						FRCSEL
		7:0			Reserved[1:0]			APLLPRE[3:0]		
0x0F90	APLLFBD1	15:8						Reserved[3:0]		
		7:0				APLLFBDIV[7:0]				
0x0F92	APLLDIV1	15:8							AVCODIV[1:0]	
		7:0			APOST1DIV[2:0]			APOST2DIV[2:0]		
0x0F94 ... 0x0F99	Reserved									
0x0F9A	CANCLKCON	15:8	CANCLKEN					CANCLKSEL[3:0]		
		7:0					CANCLKDIV[6:0]			
0x0F9C ... 0x0FB7	Reserved									
0x0FB8	REFOCONL	15:8	ROEN		ROSIDL	ROOUT	ROSLP		ROSWEN	ROACTIV
		7:0						ROSEL[3:0]		
0x0FBA	REFOCONH	15:8					RODIV[14:8]			
		7:0					RODIV[7:0]			
0x0FBC ... 0x0FBD	Reserved									
0x0FBE	REFOTRIMH	15:8					ROTRIM[8:0]			
		7:0	ROTRIM[8:0]							

10.11.1 Oscillator Control Register

Name: OSCCON⁽¹⁾
Offset: 0xF84

Notes:

- Writes to this register require an unlock sequence.
- Direct clock switches between any Primary Oscillator mode with PLL and FRCPLL mode are not permitted. This applies to clock switches in either direction. In these instances, the application must switch to FRC mode as a transitional clock source between the two PLL modes.
- This bit should only be cleared in software. Setting the bit in software (= 1) will have the same effect as an actual oscillator failure and will trigger an oscillator failure trap.

Legend: y = Value set from Configuration bits on POR

Bit	15	14	13	12	11	10	9	8
		COSC[2:0]				NOSC[2:0]		
Access		R	R	R		R/W	R/W	R/W
Reset		0	0	0		y	y	y

Bit	7	6	5	4	3	2	1	0
	CLKLOCK		LOCK		CF			OSWEN
Access	R/W		R		R/W			R/W
Reset	0		0		0			0

Bits 14:12 – COSC[2:0] Current Oscillator Selection bits (read-only)

Value	Description
111	Fast RC Oscillator (FRC) with Divide-by-n (FRCDIVN)
110	Backup FRC (BFRC)
101	Low-Power RC Oscillator (LPRC)
100	Reserved – default to FRC
011	Primary Oscillator (XT, HS, EC) with PLL (XTPLL, HSPLL, ECPLL)
010	Primary Oscillator (XT, HS, EC)
001	Fast RC Oscillator (FRC) with PLL (FRCPLL)
000	Fast RC Oscillator (FRC)

Bits 10:8 – NOSC[2:0] New Oscillator Selection bits⁽²⁾

Value	Description
111	Fast RC Oscillator (FRC) with Divide-by-n (FRCDIVN)
110	Backup FRC (BFRC)
101	Low-Power RC Oscillator (LPRC)
100	Reserved – default to FRC
011	Primary Oscillator (XT, HS, EC) with PLL (XTPLL, HSPLL, ECPLL)
010	Primary Oscillator (XT, HS, EC)
001	Fast RC Oscillator (FRC) with PLL (FRCPLL)
000	Fast RC Oscillator (FRC)

Bit 7 – CLKLOCK Clock Lock Enable bit

Value	Description
1	If (FCKSM0 = 1), then clock and PLL configurations are locked; if (FCKSM0 = 0), then clock and PLL configurations may be modified
0	Clock and PLL selections are not locked, configurations may be modified

Bit 5 – LOCK PLL Lock Status bit (read-only)

Value	Description
1	Indicates that PLL is in lock or PLL start-up timer is satisfied

Value	Description
0	Indicates that PLL is out of lock, start-up timer is in progress or PLL is disabled

Bit 3 – CF Clock Fail Detect bit⁽³⁾

Value	Description
1	FSCM has detected a clock failure
0	FSCM has not detected a clock failure

Bit 0 – OSWEN Oscillator Switch Enable bit

Value	Description
1	Requests oscillator switch to the selection specified by the NOSC[2:0] bits
0	Oscillator switch is complete

10.11.2 Clock Divider Register

Name: CLKDIV
Offset: 0xF86

Notes:

1. The DOZE[2:0] bits can only be written to when the DOZEN bit is clear. If DOZEN = 1, any writes to DOZE[2:0] are ignored.
2. This bit is cleared when the ROI bit is set and an interrupt occurs.
3. The DOZEN bit cannot be set if DOZE[2:0] = 000. If DOZE[2:0] = 000, any attempt by user software to set the DOZEN bit is ignored.
4. PLLPRE[3:0] may be updated while the PLL is operating, but the VCO may overshoot.

Legend: r = Reserved bit

Bit	15	14	13	12	11	10	9	8
	ROI	DOZE[2:0]			DOZEN	FRCDIV[2:0]		
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	1	1	0	0	0	0

Bit	7	6	5	4	3	2	1	0
			Reserved[1:0]		PLLPRE[3:0]			
Access			r	r	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	1

Bit 15 – ROI Recover on Interrupt bit

Value	Description
1	Interrupts will clear the DOZEN bit and the processor clock, and the peripheral clock ratio is set to 1:1
0	Interrupts have no effect on the DOZEN bit

Bits 14:12 – DOZE[2:0] Processor Clock Reduction Select bits⁽¹⁾

Value	Description
111	F _P divided by 128
110	F _P divided by 64
101	F _P divided by 32
100	F _P divided by 16
011	F _P divided by 8 (default)
010	F _P divided by 4
001	F _P divided by 2
000	F _P divided by 1

Bit 11 – DOZEN Doze Mode Enable bit^(2,3)

Value	Description
1	DOZE[2:0] field specifies the ratio between the peripheral clocks and the processor clocks
0	Processor clock and peripheral clock ratio is forced to 1:1

Bits 10:8 – FRCDIV[2:0] Internal Fast RC Oscillator Postscaler bits

Value	Description
111	FRC divided by 256
110	FRC divided by 64
101	FRC divided by 32
100	FRC divided by 16
011	FRC divided by 8
010	FRC divided by 4
001	FRC divided by 2
000	FRC divided by 1 (default)

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Bits 5:4 – Reserved[1:0] Read as '0'

Bits 3:0 – PLLPRE[3:0] PLL Phase Detector Input Divider Select bits⁽⁴⁾
(also denoted as 'N1', PLL prescaler)

Value	Description
1111	Reserved
. . .	
1001	Reserved
1000	Input divided by 8
0111	Input divided by 7
0110	Input divided by 6
0101	Input divided by 5
0100	Input divided by 4
0011	Input divided by 3
0010	Input divided by 2
0001	Input divided by 1 (power-on default selection)
0000	Reserved

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10.11.3 PLL Feedback Divider Register

Name: PLLFBD

Offset: 0xF8A

Note:

- The allowed range is 16-200 (decimal). The rest of the values are reserved and should be avoided. The power on the default feedback divider is 150 (decimal) with an 8 MHz FRC input clock. The VCO frequency is 1.2 GHz.

Legend: r = Reserved bit

Bit	15	14	13	12	11	10	9	8
					Reserved[3:0]			
Access					r	r	r	r
Reset					0	0	0	0

Bit	7	6	5	4	3	2	1	0
	PLLFBDIV[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	0	0	1	0	1	1	0

Bits 11:8 – Reserved[3:0] Maintain as '0'

Bits 7:0 – PLLFBDIV[7:0] PLL Feedback Divider bits (also denoted as 'M', PLL multiplier)

Value	Description
11111111	Reserved
. . .	
11001000	200 Maximum ⁽¹⁾
. . .	
10010110	150 (default)
00000001	Reserved
00000000	Reserved

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10.11.4 FRC Oscillator Tuning Register

Name: OSCTUN

Offset: 0xF8C

Bit	15	14	13	12	11	10	9	8
Access								
Reset								

Bit	7	6	5	4	3	2	1	0
			TUN[5:0]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 5:0 – TUN[5:0] FRC Oscillator Tuning bits

Value	Description
011111	Maximum frequency deviation of 1.45% (MHz)
011110	Center frequency + 1.40% (MHz)
. . .	
000001	Center frequency + 0.047% (MHz)
000000	Center frequency (8.00 MHz nominal)
111111	Center frequency – 0.047% (MHz)
. . .	
100001	Center frequency – 1.45% (MHz)
100000	Minimum frequency deviation of -1.50% (MHz)

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10.11.5 PLL Output Divider Register

Name: PLLDIV

Offset: 0xF8A

Notes:

1. The POST1DIVx and POST2DIVx divider values must not be changed while the PLL is operating.
2. The default values for POST1DIVx and POST2DIVx are 4 and 1, respectively, yielding a 150 MHz system source clock.

Bit	15	14	13	12	11	10	9	8
							VCODIV[1:0]	
Access							R/W	R/W
Reset							0	0

Bit	7	6	5	4	3	2	1	0
		POST1DIV[2:0]				POST2DIV[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		0	0	0		0	0	0

Bits 9:8 – VCODIV[1:0] PLL VCO Output Divider Select bits

Value	Description
11	F_{VCO}
10	$F_{VCO}/2$
01	$F_{VCO}/3$
00	$F_{VCO}/4$

Bits 6:4 – POST1DIV[2:0] PLL Output Divider #1 Ratio bits^(1,2)

POST1DIV[2:0] can have a valid value, from 1 to 7 (POST1DIVx value should be greater than or equal to the POST2DIVx value). The POST1DIVx divider is designed to operate at higher clock rates than the POST2DIVx divider.

Bits 2:0 – POST2DIV[2:0] PLL Output Divider #2 Ratio bits^(1,2)

POST2DIV[2:0] can have a valid value, from 1 to 7 (POST1DIVx value should be greater than or equal to the POST2DIVx value). The POST1DIVx divider is designed to operate at higher clock rates than the POST2DIVx divider.

10.11.6 Auxiliary Clock Control Register

Name: ACLKCON1
Offset: 0xF8E

Note:

- Even with the APLEN bit set, another peripheral must generate a clock request before the APLL will start.

Legend: r = Reserved bit

Bit	15	14	13	12	11	10	9	8
	APLEN	APLLCK						FRCSEL
Access	R/W	R/W						R/W
Reset	0	0						0

Bit	7	6	5	4	3	2	1	0
			Reserved[1:0]		APLLPRE[3:0]			
Access			r	r	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bit 15 – APLEN Auxiliary PLL Enable/Bypass Select bit⁽¹⁾

Value	Description
1	AF _{PLLO} is connected to APLL post-divider output (bypass is disabled)
0	AF _{PLLO} is connected to APLL input clock (bypass is enabled)

Bit 14 – APLLCK APLL Phase-Locked State Status bit

Value	Description
1	Auxiliary PLL is in lock
0	Auxiliary PLL is not in lock

Bit 8 – FRCSEL FRC Clock Source Select bit

Bits 5:4 – Reserved[1:0] Read as '0'

Bits 3:0 – APLLPRE[3:0] Auxiliary PLL Phase Detector Input Divider bits

Value	Description
1111	Reserved
. . .	
1001	Reserved
1000	Input divided by 8
0111	Input divided by 7
0110	Input divided by 6
0101	Input divided by 5
0100	Input divided by 4
0011	Input divided by 3
0010	Input divided by 2
0001	Input divided by 1 (power-on default selection)
0000	Reserved

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10.11.7 APLL Feedback Divider Register

Name: APLLFBD1

Offset: 0xF90

Note:

- The allowed range is 16-200 (decimal). The rest of the values are reserved and should be avoided. The power-on default feedback divider is 150 (decimal) with an 8 MHz FRC input clock; the VCO frequency is 1.2 GHz.

Legend: r = Reserved bit

Bit	15	14	13	12	11	10	9	8
					Reserved[3:0]			
Access					r	r	r	r
Reset					0	0	0	0

Bit	7	6	5	4	3	2	1	0
	APLLFBDIV[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	0	0	1	0	1	1	0

Bits 11:8 – Reserved[3:0] Maintain as '0'

Bits 7:0 – APLLFB DIV[7:0] APLL Feedback Divider bits

Value	Description
11111111	Reserved
. . .	
11001000	200 maximum ⁽¹⁾
. . .	
10010110	150 (default)
. . .	
00010000	16 minimum ⁽¹⁾
. . .	
00000010	Reserved
00000001	Reserved
00000000	Reserved

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10.11.8 APLL Output Divider Register

Name: APLLDIV1

Offset: 0xF92

Notes:

1. The APOST1DIVx and APOST2DIVx values must not be changed while the PLL is operating.
2. The default values for APOST1DIVx and APOST2DIVx are 4 and 1, respectively, yielding a 150 MHz system source clock.

Bit	15	14	13	12	11	10	9	8
							AVCODIV[1:0]	
Access							R/W	R/W
Reset							0	0

Bit	7	6	5	4	3	2	1	0
		APOST1DIV[2:0]				APOST2DIV[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		1	0	0		0	0	1

Bits 9:8 – AVCODIV[1:0] APLL VCO Output Divider Select bits

Value	Description
11	AF_{VCO}
10	$AF_{VCO}/2$
01	$AF_{VCO}/3$
00	$AF_{VCO}/4$

Bits 6:4 – APOST1DIV[2:0] APLL Output Divider #1 Ratio bits^(1,2)

APOST1DIV[2:0] can have a valid value, from 1 to 7 (the APOST1DIVx value should be greater than or equal to the APOST2DIVx value). The APOST1DIVx divider is designed to operate at higher clock rates than the APOST2DIVx divider.

Bits 2:0 – APOST2DIV[2:0] APLL Output Divider #2 Ratio bits^(1,2)

APOST2DIV[2:0] can have a valid value, from 1 to 7 (the APOST2DIVx value should be less than or equal to the APOST1DIVx value). The APOST1DIVx divider is designed to operate at higher clock rates than the APOST2DIVx divider.

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10.11.9 CAN Clock Control Register

Name: CANCLKCON

Offset: 0xF9A

Notes:

1. The user must ensure the input clock source is 640 MHz or less. Operation with input reference frequency above 640 MHz will result in unpredictable behavior.
2. The CANCLKDIVx divider value must not be changed during CAN module operation.
3. The user must ensure the maximum clock output frequency of the divider is 80 MHz or less.

Bit	15	14	13	12	11	10	9	8
	CANCLKEN					CANCLKSEL[3:0]		
Access	R/W				R/W	R/W	R/W	R/W
Reset	0				0	0	0	0

Bit	7	6	5	4	3	2	1	0
		CANCLKDIV[6:0]						
Access		R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset		0	0	0	0	0	0	0

Bit 15 – CANCLKEN Enables the CAN Clock Generator bit

Value	Description
1	CAN clock generation circuitry is enabled
0	CAN clock generation circuitry is disabled

Bits 11:8 – CANCLKSEL[3:0] CAN Clock Source Select bits⁽¹⁾

Value	Description
1011-1111	Reserved (no clock selected)
1010	$AF_{VCO}/4$
1001	$AF_{VCO}/3$
1000	$AF_{VCO}/2$
0111	AF_{VCO}
0110	AF_{PLLO}
0101	$F_{VCO}/4$
0100	$F_{VCO}/3$
0011	$F_{VCO}/2$
0010	F_{PLLO}
0001	F_{VCO}
0000	0 (no clock selected)

Bits 6:0 – CANCLKDIV[6:0] CAN Clock Divider Select bits^(2,3)

Value	Description
1111111	Divide-by-128
. . .	
0000010	Divide-by-3
0000001	Divide-by-2
0000000	Divide-by-1

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10.11.10 Reference Clock Control Low Register

Name: REFOCONL
Offset: 0xFB8

Legend: HC = Hardware Clearable bit; HSC = Hardware Settable/Clearable bit

Bit	15	14	13	12	11	10	9	8
	ROEN		ROSIDL	ROOUT	ROSLP		ROSWEN	ROACTIV
Access	R/W		R/W	R/W	R/W		R/W/HC	R/HSC
Reset	0		0	0	0		0	0

Bit	7	6	5	4	3	2	1	0
					ROSEL[3:0]			
Access					R/W	R/W	R/W	R/W
Reset					0	0	0	0

Bit 15 – ROEN Reference Clock Enable bit

Value	Description
1	Reference Oscillator is enabled on the REFCLKO pin
0	Reference Oscillator is disabled

Bit 13 – ROSIDL Reference Clock Stop in Idle bit

Value	Description
1	Reference Oscillator is disabled in Idle mode
0	Reference Oscillator continues to run in Idle mode

Bit 12 – ROOUT Reference Clock Output Enable bit

Value	Description
1	Reference clock external output is enabled and available on the REFCLKO pin
0	Reference clock external output is disabled

Bit 11 – ROSLP Reference Clock Stop in Sleep bit

Value	Description
1	Reference Oscillator continues to run in Sleep modes
0	Reference Oscillator is disabled in Sleep modes

Bit 9 – ROSWEN Reference Clock Output Enable bit

Value	Description
1	Clock divider change (requested by changes to RODIVx) is requested or is in progress (set in software, cleared by hardware upon completion)
0	Clock divider change has completed or is not pending

Bit 8 – ROACTIV Reference Clock Status bit

Value	Description
1	Reference clock is active; do not change clock source
0	Reference clock is stopped; clock source and configuration may be safely changed

Bits 3:0 – ROSEL[3:0] Reference Clock Source Select bits

Value	Description
1111	Reserved
. . .	Reserved
1000	Reserved
0111	REFI pin
0110	Fvco/4
0101	BFRC

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Value	Description
0100	LPRC
0011	FRC
0010	Primary Oscillator
0001	Peripheral clock (F_P)
0000	System clock (F_{OSC})

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10.11.11 Reference Clock Control High Register

Name: REFOCONH
Offset: 0xFBA

Bit	15	14	13	12	11	10	9	8
	RODIV[14:8]							
Access		R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset		0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	RODIV[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 14:0 – RODIV[14:0] Reference Clock Integer Divider Select bits

Divider for the selected input clock source is two times the selected value.

Value	Description
111 1111 1111 1111	Base clock value divided by 65,534 ($2 * 7FFFh$)
111 1111 1111 1110	Base clock value divided by 65,532 ($2 * 7FFEh$)
111 1111 1111 1101	Base clock value divided by 65,530 ($2 * 7FFDh$)
. . .	
000 0000 0000 0010	Base clock value divided by 4 ($2 * 2$)
000 0000 0000 0001	Base clock value divided by 2 ($2 * 1$)
000 0000 0000 0000	Base clock value

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Oscillator with High-Frequency PLL

10.11.12 Reference Clock Trim Register

Name: REFOTRIMH
Offset: 0xFBFE

Bit	15	14	13	12	11	10	9	8
	ROTRIM[8:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	ROTRIM[8:0]							
Access	R/W							
Reset	0							

Bits 15:7 – ROTRIM[8:0] REFO Trim bits

These bits provide a fractional additive to the RODIV[14:0] value for the 1/2 period of the REFO clock.

Value	Description
11111111	511/512 (0.998046875 divisor added to the RODIV[14:0] value)
11111110	510/512 (0.99609375 divisor added to the RODIV[14:0] value)
. . .	
10000000	256/512 (0.5000 divisor added to the RODIV[14:0] value)
. . .	
00000010	2/512 (0.00390625 divisor added to the RODIV[14:0] value)
00000001	1/512 (0.001953125 divisor added to the RODIV[14:0] value)
00000000	0/512 (0.0 divisor added to the RODIV[14:0] value)

11. Direct Memory Access (DMA) Controller

Note: This data sheet summarizes the features of the dsPIC33CK512MPT608 family of devices. It is not intended to be a comprehensive reference source. For more information, refer to “**Direct Memory Access Controller (DMA)**” (www.microchip.com/DS30009742) in the “dsPIC33/PIC24 Family Reference Manual”.

The Direct Memory Access (DMA) Controller is designed to service high data throughput peripherals operating on the SFR bus, allowing them to access data memory directly and alleviating the need for CPU-intensive management. By allowing these data-intensive peripherals to share their own data path, the main data bus is also deloaded, resulting in additional power savings.

The DMA Controller functions both as a peripheral and a direct extension of the CPU. It is located on the microcontroller data bus, between the CPU and DMA-enabled peripherals, with direct access to SRAM. This partitions the SFR bus into two buses, allowing the DMA Controller access to the DMA-capable peripherals located on the new DMA SFR bus. The controller serves as an Initiator device on the DMA SFR bus, controlling data flow from DMA-capable peripherals.

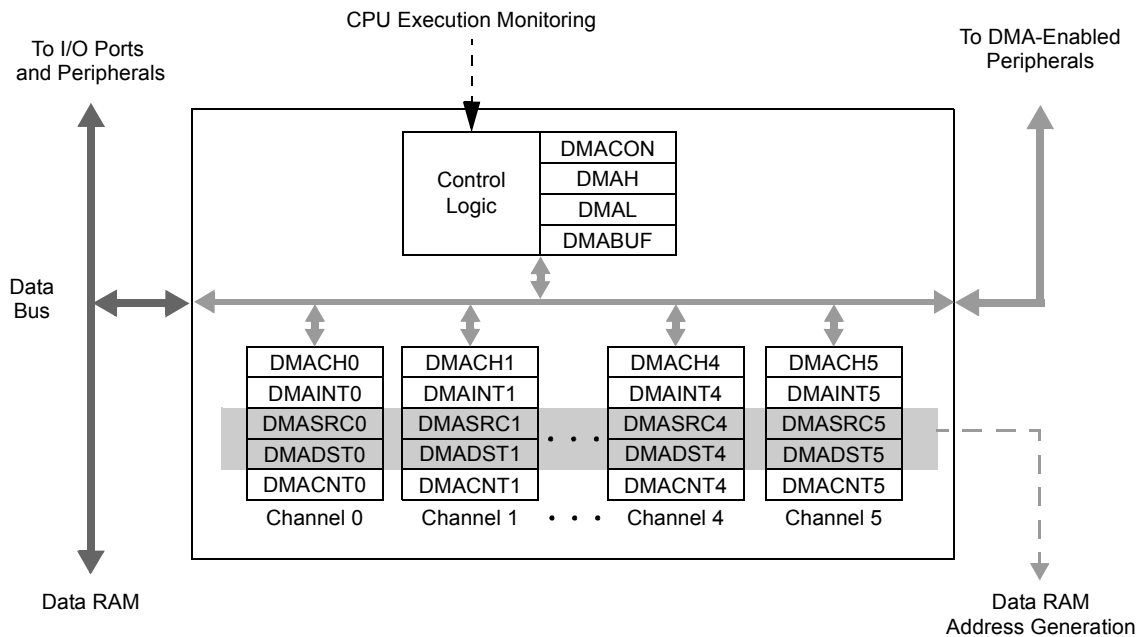
The controller also monitors CPU instruction processing directly, allowing it to be aware of when the CPU requires access to peripherals on the DMA bus and automatically relinquishing control to the CPU as needed. This increases the effective bandwidth for handling data without DMA operations, causing a processor Stall. This makes the controller essentially transparent to the user.

The DMA Controller has these features:

- A Total of Eight Independently Programmable Channels
- Concurrent Operation with the CPU (no DMA caused Wait states)
- DMA Bus Arbitration
- Five Programmable Address modes
- Four Programmable Transfer modes
- Four Flexible Internal Data Transfer modes
- Byte or Word Support for Data Transfer
- 16-Bit Source and Destination Address Register for each Channel, Dynamically Updated and Reloadable
- 16-Bit Transaction Count Register, Dynamically Updated and Reloadable
- Upper and Lower Address Limit Registers
- Counter Half-Full Level Interrupt
- Software Triggered Transfer
- Null Write mode for Symmetric Buffer Operations

A simplified block diagram of the DMA Controller is shown in [Figure 11-1](#).

Figure 11-1. DMA Functional Block Diagram



11.1 Summary of DMA Operations

The DMA Controller is capable of moving data between addresses according to a number of different parameters. Each of these parameters can be independently configured for any transaction. In addition, any or all of the DMA channels can independently perform a different transaction at the same time. Transactions are classified by these parameters:

- Source and destination (SFRs and data RAM)
- Data size (byte or word)
- Trigger source
- Transfer mode (One-Shot, Repeated or Continuous)
- Addressing modes (Fixed Address or Address Blocks with or without Address Increment/Decrement)

In addition, the DMA Controller provides channel priority arbitration for all channels.

11.1.1 Source and Destination

Using the DMA Controller, data may be moved between any two addresses in the Data Space. The SFR space (0000h to 0FFFh), or the data RAM space (1000h to 4FFFh), can serve as either the source or the destination. Data can be moved between these areas in either direction or between addresses in either area. The four different combinations are shown in [11.1.5. Addressing Modes](#).

If it is necessary to protect areas of data RAM, the DMA Controller allows the user to set upper and lower address boundaries for operations in the Data Space above the SFR space. The boundaries are set by the DMAH and DMAL Limit registers. If a DMA channel attempts an operation outside of the address boundaries, the transaction is terminated and an interrupt is generated.

11.1.2 Data Size

The DMA Controller can handle both 8-bit and 16-bit transactions. Size is user-selectable using the SIZE bit (DMACHn[1]). By default, each channel is configured for word-size transactions. When byte-size transactions are chosen, the LSB of the source and/or destination address determines if the data represent the upper or lower byte of the data RAM location.

11.1.3 Trigger Source

The DMA Controller can use 82 of the device's interrupt sources to initiate a transaction. The DMA trigger sources occur in reverse order from their natural interrupt priority and are shown in [11.1.5. Addressing Modes](#).

Since the source and destination addresses for any transaction can be programmed independently of the trigger source, the DMA Controller can use any trigger to perform an operation on any peripheral. This also allows DMA channels to be cascaded to perform more complex transfer operations.

11.1.4 Transfer Mode

The DMA Controller supports four types of data transfers, based on the volume of data to be moved for each trigger.

- One-Shot: A single transaction occurs for each trigger.
- Continuous: A series of back-to-back transactions occur for each trigger; the number of transactions is determined by the DMACNTn transaction counter.
- Repeated One-Shot: A single transaction is performed repeatedly, once per trigger, until the DMA channel is disabled.
- Repeated Continuous: A series of transactions are performed repeatedly, one cycle per trigger, until the DMA channel is disabled.

All transfer modes allow the option to have the source and destination addresses, and counter value, automatically reloaded after the completion of a transaction.

11.1.5 Addressing Modes

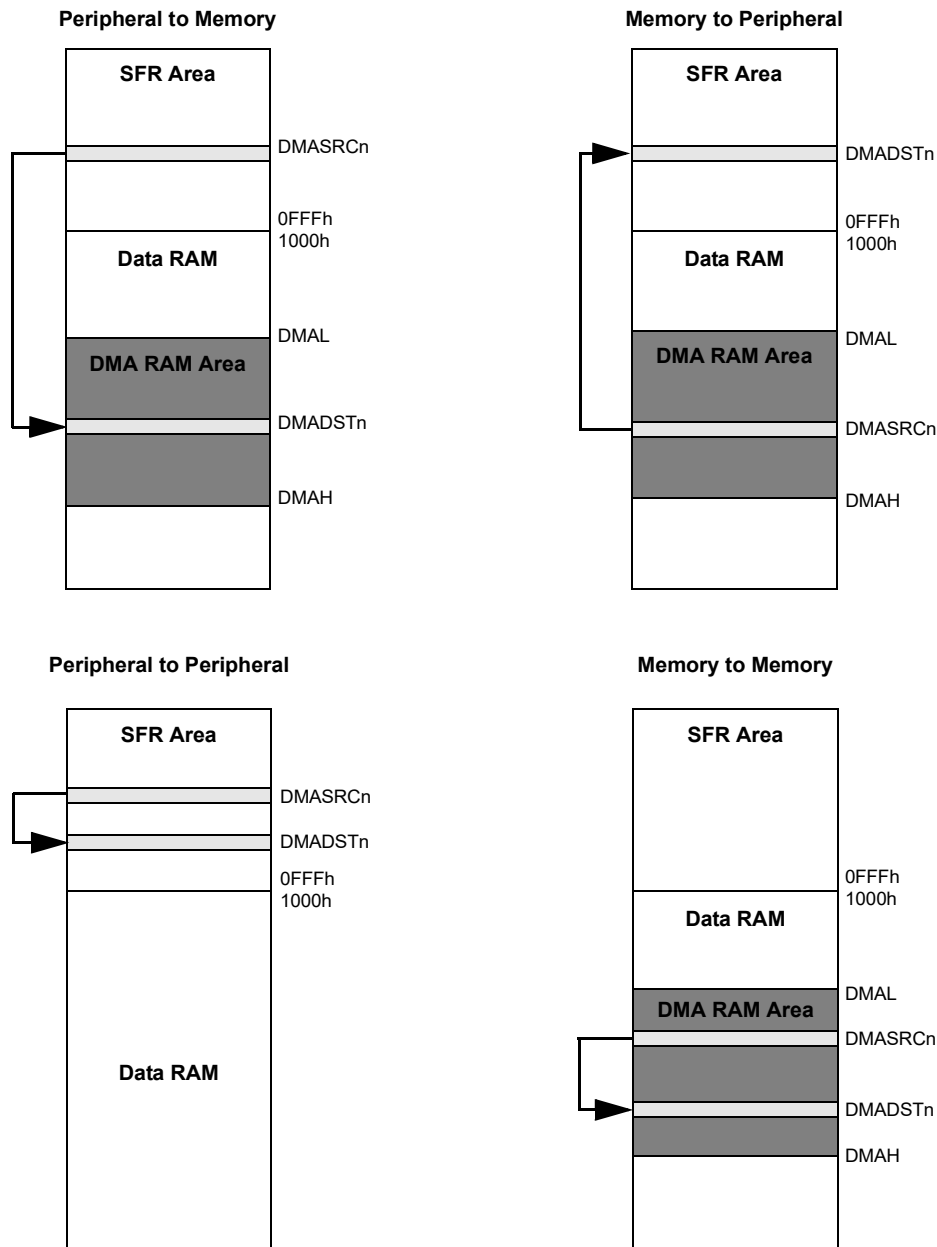
The DMA Controller also supports transfers between single addresses or address ranges. The four basic options are:

- Fixed-to-Fixed: Between two constant addresses
- Fixed-to-Block: From a constant source address to a range of destination addresses
- Block-to-Fixed: From a range of source addresses to a single, constant destination address
- Block-to-Block: From a range of source addresses to a range of destination addresses

The option to select auto-increment or auto-decrement of source and/or destination addresses is available for Block Addressing modes.

In addition to the four basic modes, the DMA Controller also supports Peripheral Indirect Addressing (PIA) mode, where the source or destination address is generated jointly by the DMA Controller and a PIA-capable peripheral. When enabled, the DMA channel provides a base source and/or destination address, while the peripheral provides a fixed range offset address.

Figure 11-2. Types of DMA Data Transfers



Note: Relative sizes of memory areas are not shown to scale.

11.1.6 Channel Priority

Each DMA channel functions independently of the others, but also competes with the others for access to the data and DMA buses. When access collisions occur, the DMA Controller arbitrates between the channels using a user-selectable priority scheme. Two schemes are available:

- Round Robin: When two or more channels collide, the lower numbered channel receives priority on the first collision. On subsequent collisions, the higher numbered channels each receive priority based on their channel number.
- Fixed: When two or more channels collide, the lowest numbered channel always receives priority, regardless of past history; however, any channel being actively processed is not available for an immediate retrigger. If a higher priority channel is continually requesting service, it will be scheduled for service after the next lower priority channel with a pending request.

11.2 Typical Setup

To set up a DMA channel for a basic data transfer:

1. Enable the DMA Controller (DMAEN = 1) and select an appropriate channel priority scheme by setting or clearing PRSSEL.
2. Program DMAH and DMAL with appropriate upper and lower address boundaries for data RAM operations.
3. Select the DMA channel to be used and disable its operation (CHEN = 0).
4. Program the appropriate source and destination addresses for the transaction into the channel's DMASRCn and DMADSTn registers. For PIA mode addressing, use the base address value.
5. Program the DMACNTn register for the number of triggers per transfer (One-Shot or Continuous modes) or the number of words (bytes) to be transferred (Repeated modes).
6. Set or clear the SIZE bit to select the data size.
7. Program the TRMODE[1:0] bits to select the Data Transfer mode.
8. Program the SAMODE[1:0] and DAMODE[1:0] bits to select the addressing mode.
9. Enable the DMA channel by setting CHEN.
10. Enable the trigger source interrupt.

11.2.1 Peripheral Module Disable

The channels of the DMA Controller can be individually powered down using the Peripheral Module Disable (PMD) registers.

11.2.2 DMA Registers

The DMA Controller uses a number of registers to control its operation. The number of registers depends on the number of channels implemented for a particular device.

There are always four module-level registers (one control and three buffer/address):

- DMACON: DMA Engine Control Register ([11.2.3.1. DMACON](#))
- DMAH and DMAL: DMA High and Low Address Limit Registers
- DMABUF: DMA Transfer Data Buffer

Each of the DMA channels implements five registers (two control and three buffer/address):

- DMACHn: DMA Channel n Control Register ([11.2.3.2. DMACHn](#))
- DMAINTn: DMA Channel n Interrupt Register ([11.2.3.3. DMAINTn](#))
- DMASRCn: DMA Data Source Address Pointer for Channel n Register
- DMADSTn: DMA Data Destination Source for Channel n Register
- DMACNTn: DMA Transaction Counter for Channel n Register

For devices, there are a total of 34 registers.

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Direct Memory Access (DMA) Controller

11.2.3 DMA Control Registers

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x00 ... 0x0AA7	Reserved									
0x0AA8	DMACON	15:8	DMAEN		DMASIDL					
		7:0								PRSSEL
0x0AAA ... 0x0AAF	Reserved									
0x0AB0	DMACHn	15:8				Reserved		NULLW	RELOAD	CHREQ
		7:0	SAMODE[1:0]		DAMODE[1:0]		TRMODE[1:0]		SIZE	CHEN
0x0AB2	DMAINTn	15:8	DBUFWF				CHSEL[6:0]			
		7:0	HIGHIF	LOWIF	DONEIF	HALFIF	OVRUNIF			HALFEN

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Direct Memory Access (DMA) Controller

11.2.3.1 DMA Engine Control Register

Name: DMACON
Offset: 0xAA8

Bit	15	14	13	12	11	10	9	8
	DMAEN		DMASIDL					
Access	R/W		R/W					
Reset	0		0					

Bit	7	6	5	4	3	2	1	0
								PRSEL
Access								R/W
Reset								0

Bit 15 – DMAEN DMA Module Enable bit

Value	Description
1	Enables module
0	Disables module and terminates all active DMA operation(s)

Bit 13 – DMASIDL DMA Stop in Idle bit

Value	Description
1	DMA continues to run in Idle mode
0	DMA is disabled in Idle mode

Bit 0 – PRSEL Channel Priority Scheme Selection bit

Value	Description
1	Round robin scheme
0	Fixed priority scheme

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Direct Memory Access (DMA) Controller

11.2.3.2 DMA Channel n Control Register

Name: DMACHn
Offset: 0xAB0

Notes:

1. Only the original DMACNTn is required to be stored to recover the original DMASRCn and DMADSTn values.
2. DMACNTn will always be reloaded in Repeated mode transfers, regardless of the state of the RELOAD bit.
3. The number of transfers executed while CHREQ is set depends on the configuration of TRMODE[1:0].

Legend: r = Reserved bit

Bit	15	14	13	12	11	10	9	8
				Reserved		NULLW	RELOAD	CHREQ
Access				r		R/W	R/W	R/W
Reset				0		0	0	0

Bit	7	6	5	4	3	2	1	0
	SAMODE[1:0]		DAMODE[1:0]		TRMODE[1:0]		SIZE	CHEN
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 12 – Reserved Maintain as ‘0’

Bit 10 – NULLW Null Write Mode bit

Value	Description
1	A dummy write is initiated to DMASRCn for every write to DMADSTn
0	No dummy write is initiated

Bit 9 – RELOAD Address and Count Reload bit⁽¹⁾

Value	Description
1	DMASRCn, DMADSTn and DMACNTn registers are reloaded to their previous values upon the start of the next operation
0	DMASRCn, DMADSTn and DMACNTn are not reloaded on the start of the next operation ⁽²⁾

Bit 8 – CHREQ DMA Channel Software Request bit⁽³⁾

Value	Description
1	A DMA request is initiated by software; automatically cleared upon completion of a DMA transfer
0	No DMA request is pending

Bits 7:6 – SAMODE[1:0] Source Address Mode Selection bits

Value	Description
11	DMASRCn is used in Peripheral Indirect Addressing and remains unchanged
10	DMASRCn is decremented based on the SIZE bit after a transfer completion
01	DMASRCn is incremented based on the SIZE bit after a transfer completion
00	DMASRCn remains unchanged after a transfer completion

Bits 5:4 – DAMODE[1:0] Destination Address Mode Selection bits

Value	Description
11	DMADSTn is used in Peripheral Indirect Addressing and remains unchanged
10	DMADSTn is decremented based on the SIZE bit after a transfer completion
01	DMADSTn is incremented based on the SIZE bit after a transfer completion
00	DMADSTn remains unchanged after a transfer completion

Bits 3:2 – TRMODE[1:0] Transfer Mode Selection bits

Value	Description
11	Repeated Continuous

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Direct Memory Access (DMA) Controller

Value	Description
10	Continuous
01	Repeated One-Shot
00	One-Shot

Bit 1 – SIZE Data Size Selection bit

Value	Description
1	Byte (8-bit)
0	Word (16-bit)

Bit 0 – CHEN DMA Channel Enable bit

Value	Description
1	The corresponding channel is enabled
0	The corresponding channel is disabled

dsPIC33CK512MPT608 Family

Direct Memory Access (DMA) Controller

11.2.3.3 DMA Channel n Interrupt Register

Name: DMAINTn
Offset: 0xAB2

Notes:

- Setting these flags in software does not generate an interrupt.
- Testing for address limit violations (DMASRCn or DMADSTn is either greater than DMAH or less than DMAL) is NOT done before the actual access.

Bit	15	14	13	12	11	10	9	8
	DBUFWF	CHSEL[6:0]						
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset								

Bit	7	6	5	4	3	2	1	0
	HIGHIF	LOWIF	DONEIF	HALFIF	OVRUNIF			HALFEN
Access	R/W	R/W	R/W	R/W	R/W			R/W
Reset								0

Bit 15 – DBUFWF DMA Buffered Data Write Flag bit⁽¹⁾

Value	Description
1	The content of the DMA buffer has not been written to the location specified in DMADSTn or DMASRCn in Null Write mode
0	The content of the DMA buffer has been written to the location specified in DMADSTn or DMASRCn in Null Write mode

Bits 14:8 – CHSEL[6:0] DMA Channel Trigger Selection bits

Bit 7 – HIGHIF DMA High Address Limit Interrupt Flag bit^(1,2)

Value	Description
1	The DMA channel has attempted to access an address higher than DMAH or the upper limit of the data RAM space
0	The DMA channel has not invoked the high address limit interrupt

Bit 6 – LOWIF DMA Low Address Limit Interrupt Flag bit^(1,2)

Value	Description
1	The DMA channel has attempted to access the DMA SFR address lower than DMAL, but above the SFR range (07FFh)
0	The DMA channel has not invoked the low address limit interrupt

Bit 5 – DONEIF DMA Complete Operation Interrupt Flag bit⁽¹⁾

Value	Description
	If CHEN = 1:
1	The previous DMA session has ended with completion
0	The current DMA session has not yet completed
	If CHEN = 0:
1	The previous DMA session has ended with completion
0	The previous DMA session has ended without completion

Bit 4 – HALFIF DMA 50% Watermark Level Interrupt Flag bit⁽¹⁾

Value	Description
1	DMACNTn has reached the halfway point to 0000h
0	DMACNTn has not reached the halfway point

Bit 3 – OVRUNIF DMA Channel Overrun Flag bit⁽¹⁾

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Direct Memory Access (DMA) Controller

Value	Description
1	The DMA channel is triggered while it is still completing the operation based on the previous trigger
0	The Overrun condition has not occurred

Bit 0 – HALFEN Halfway Completion Watermark bit

Value	Description
1	Interrupts are invoked when DMACNTn has reached its halfway point and at completion
0	An interrupt is invoked only at the completion of the transfer

11.2.4 DMA Trigger Sources

Table 11-1. DMA Channel Trigger Sources

CHSEL[6:0]	Trigger (Interrupt)	CHSEL[6:0]	Trigger (Interrupt)	CHSEL[6:0]	Trigger (Interrupt)
00h	INT0 – External Interrupt 0	25h	SENT1 TX/RX	49h	ADC Done AN27
01h	SCCP1 IC/OC Interrupt	26h	SENT2 TX/RX	4Ah	(Reserved, do not use)
02h	SPI1 Receiver	27h	ADC1 Group Convert Done	4Bh	(Reserved, do not use)
03h	SPI1 Transmitter	28h	ADC Done AN0	4Ch	(Reserved, do not use)
04h	UART1 Receiver	29h	ADC Done AN1	4Dh	(Reserved, do not use)
05h	UART1 Transmitter	2Ah	ADC Done AN2	4Eh	(Reserved, do not use)
06h	ECC Single-Bit Error	2Bh	ADC Done AN3	4Fh	(Reserved, do not use)
07h	NVM Write Complete	2Ch	ADC Done AN4	50h	(Reserved, do not use)
08h	INT1 – External Interrupt 1	2Dh	ADC Done AN5	51h	(Reserved, do not use)
09h	SI2C1 – I2C1 Client Event	2Eh	ADC Done AN6	52h	(Reserved, do not use)
0Ah	MI2C1 – I2C1 Host Event	2Fh	ADC Done AN7	53h	(Reserved, do not use)
0Bh	INT2 – External Interrupt 2	30h	ADC Done AN8	54h	(Reserved, do not use)
0Ch	SCCP2 IC/OC Interrupt	31h	ADC Done AN9	55h	(Reserved, do not use)
0Dh	INT3 – External Interrupt 3	32h	ADC Done AN10	56h	(Reserved, do not use)
0Eh	UART2 Receiver	33h	ADC Done AN11	57h	PWM Event D
0Fh	UART2 Transmitter	34h	ADC Done AN12	58h	PWM Event E
10h	SPI2 Receiver	35h	ADC Done AN13	59h	PWM Event F
11h	SPI2 Transmitter	36h	ADC Done AN14	(Reserved, do not use)	(Reserved, do not use)

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Direct Memory Access (DMA) Controller

.....continued					
CHSEL[6:0]	Trigger (Interrupt)	CHSEL[6:0]	Trigger (Interrupt)	CHSEL[6:0]	Trigger (Interrupt)
12h	SCCP3 IC/OC Interrupt	37h	ADC Done AN15	(Reserved, do not use)	(Reserved, do not use)
13h	SI2C2 – I2C2 Client Event	38h	ADC Done AN16	5Ch	SCCP7 Interrupt
14h	MI2C2 – I2C2 Host Event	39h	ADC Done AN17	5Dh	SCCP8 Interrupt
15h	SCCP4 IC/OC Interrupt	3Ah	ADC Done AN18	5Eh	(Reserved, do not use)
16h	SCCP5 IC/OC Interrupt	3Bh	ADC Done AN19	5Fh	(Reserved, do not use)
17h	SCCP6 IC/OC Interrupt	3Ch	ADC Done AN20	60h	CLC3 Positive Edge Interrupt
18h	CRC Generator Interrupt	3Dh	ADC Done AN21	61h	CLC4 Positive Edge Interrupt
19h	PWM Event A	3Eh	ADC Done AN22	62h	SPI3 Receiver
1Bh	PWM Event B	3Fh	ADC Done AN23	63h	SPI3 Transmitter
1Ch	PWM Generator 1	40h	AD1FLTR1 – Oversample Filter 1	64h	SI2C3 – I2C3 Client Event
1Dh	PWM Generator 2	41h	AD1FLTR2 – Oversample Filter 2	65h	MI2C3 – I2C3 Host Event
1Eh	PWM Generator 3	42h	AD1FLTR3 – Oversample Filter 3	66h	SPI3 – Fault Interrupt
1Fh	PWM Generator 4	43h	AD1FLTR4 – Oversample Filter 4	67h	MCCP9
20h	(Reserved, do not use)	44h	CLC1 Positive Edge Interrupt	68h	UART3 Receiver
21h	PWM Generator 6	45h	CLC2 Positive Edge Interrupt	69h	UART3 Transmitter
22h	(Reserved, do not use)	46h	SPI1 – Fault Interrupt	6Ah	ADC Done AN24
23h	PWM Generator 8	47h	SPI2 – Fault Interrupt	6Bh	ADC Done AN25
24h	PWM Event C	48h	ADC Done AN26	6Ch-7Fh	(Reserved, do not use)

12. Controller Area Network Flexible Data-Rate (CAN FD) Modules

Note: This data sheet summarizes the features of the dsPIC33CK512MPT608 family of devices. It is not intended to be a comprehensive reference source. To complement the information in this data sheet, refer to “**CAN Flexible Data-Rate (FD) Protocol Module**” (www.microchip.com/DS70005340) in the “*dsPIC33/PIC24 Family Reference Manual*”.

12.1 Features

The CAN FD modules have the following features:

General

- Nominal (Arbitration) Bit Rate up to 1 Mbps
- Data Bit Rate up to 8 Mbps
- CAN FD Controller modes:
 - Mixed CAN 2.0B and CAN FD mode
 - CAN 2.0B mode
- Conforms to ISO11898-1:2015

Message FIFOs

- Seven FIFOs, Configurable as Transmit or Receive FIFOs
- One Transmit Queue (TXQ)
- Transmit Event FIFO (TEF) with 32-Bit Timestamp

Message Transmission

- Message Transmission Prioritization:
 - Based on priority bit field and/or
 - Message with lowest ID gets transmitted first using the TXQ
- Programmable Automatic Retransmission Attempts:
 - Unlimited, three attempts or disabled

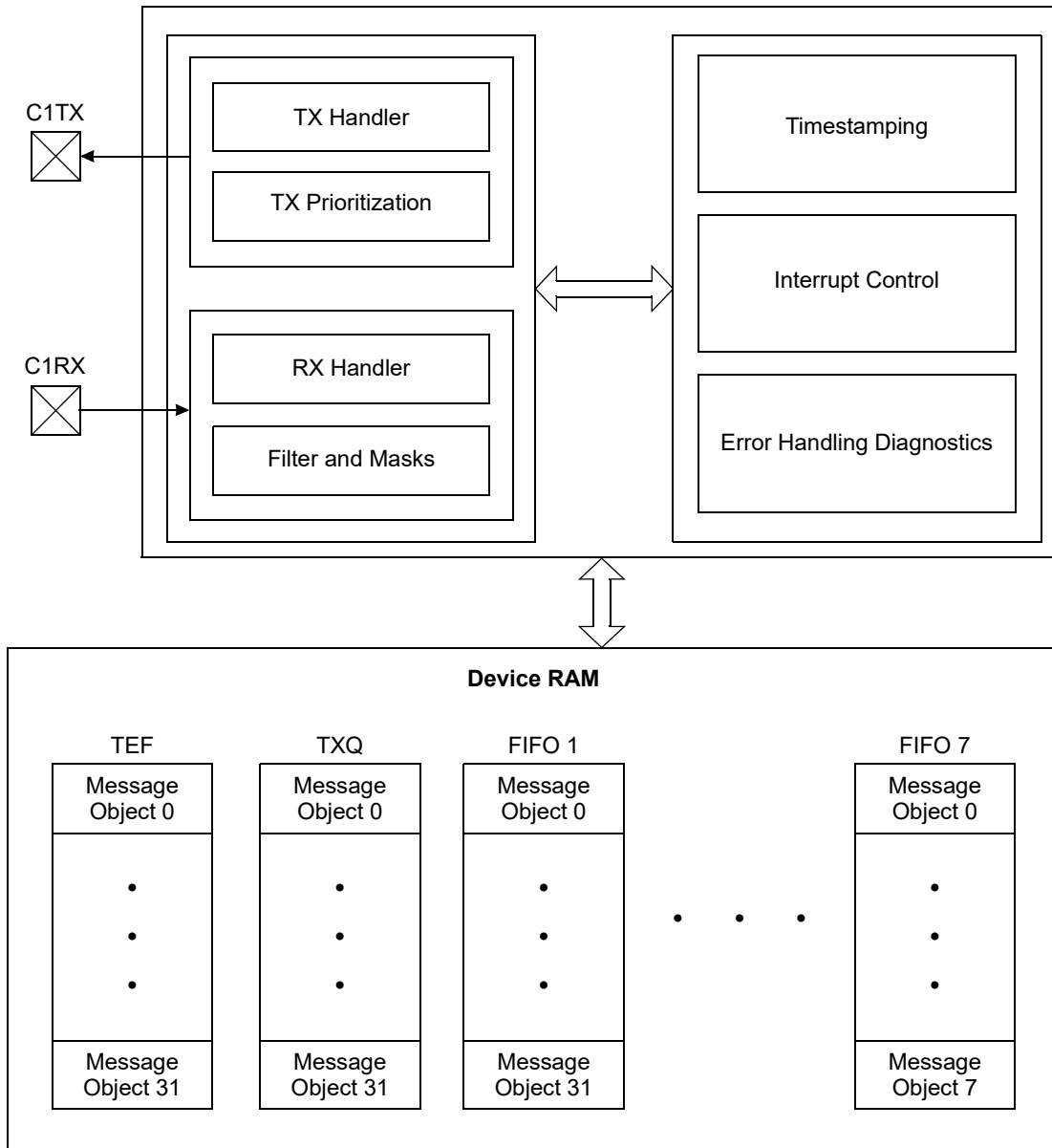
Message Reception

- 16 Flexible Filter and Mask Objects
- Each Object can be Configured to Filter either:
 - Standard ID + first 18 data bits or
 - Extended ID
- 32-Bit Timestamp
- The CAN FD Bit Stream Processor (BSP):
Implements the Medium Access Control of the CAN FD Protocol Described in ISO11898-1:2015. It serializes and deserializes the bit stream, encodes and decodes the CAN FD frames, manages the medium access, Acknowledges frames, and detects and signals errors.
- The TX Handler: Prioritizes the Messages that are Requested for Transmission by the Transmit FIFOs. It uses the RAM interface to fetch the transmit data from RAM and provides them to the BSP for transmission.
- The BSP: Provides Received Messages to the RX Handler. The RX handler uses acceptance filters to filter out messages that shall be stored in the Receive FIFOs. It uses the RAM interface to store received data into RAM.
- Each FIFO can be Configured either as a
Transmit or Receive FIFO: The FIFO control keeps track of the FIFO head and tail, and calculates the user address. In a TX FIFO, the user address points to the address in RAM where the data for the next transmit message shall be stored. In an RX FIFO, the user address points to the address in RAM where the data of the next receive message shall be read. The user notifies the FIFO that a message was written to or read from RAM by incrementing the head/tail of the FIFO.
- The Transmit Queue (TXQ): A Special Transmit FIFO that Transmits the Messages based on the ID of the Messages Stored in the Queue.

- The Transmit Event FIFO (TEF): Stores the Message IDs of the Transmitted Messages.
- A Free-Running Time Base Counter: Used to Timestamp Received Messages. Messages in the TEF can also be timestamped.
- The CAN FD Controller Modules: Generate Interrupts when New Messages are Received or when Messages were Transmitted Successfully.

Figure 12-1 shows the CAN FD system block diagram.

Figure 12-1. CAN FD Module Block Diagram



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Controller Area Network Flexible Data-Rate (...)

12.2 CAN Control/Status Registers

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x00 ... 0x05DB	Reserved									
0x05DC	C2FIFOCON1L	15:8						FRESET	TXREQ	UINC
		7:0	TXEN	RTREN	RXTSEN	TXATIE	RXOVIE	TFERFFIE	TFHRFHIE	TFNRFNIE
0x05DE	C2FIFOCON1H	15:8	PLSIZE[2:0]					FSIZE[4:0]		
		7:0	TXAT[1:0]					TXPRI[4:0]		
0x05E0	C2FIFOSTA1	15:8						FIFOC[4:0]		
		7:0	TXABT	TXLARB	TXERR	TXATIF	RXOVIF	TFERFFIF	TFHRFHIF	TFNRFNIF
0x05E2	C2FIFOSTA1H	15:8						FIFOC[4:0]		
		7:0	TXABT	TXLARB	TXERR	TXATIF	RXOVIF	TFERFFIF	TFHRFHIF	TFNRFNIF
0x05E4	C2FIFOUA1L(1)	15:8	FIFOUA[15:8]							
		7:0	FIFOUA[7:0]							
0x05E6	C2FIFOUA1H(1)	15:8	FIFOUA[31:24]							
		7:0	FIFOUA[23:16]							
0x05E8	C2FIFOCON2L	15:8						FRESET	TXREQ	UINC
		7:0	TXEN	RTREN	RXTSEN	TXATIE	RXOVIE	TFERFFIE	TFHRFHIE	TFNRFNIE
0x05EA	C2FIFOCON2H	15:8	PLSIZE[2:0]					FSIZE[4:0]		
		7:0	TXAT[1:0]					TXPRI[4:0]		
0x05EC	C2FIFOSTA2	15:8						FIFOC[4:0]		
		7:0	TXABT	TXLARB	TXERR	TXATIF	RXOVIF	TFERFFIF	TFHRFHIF	TFNRFNIF
0x05EE	C2FIFOSTA2H	15:8						FIFOC[4:0]		
		7:0	TXABT	TXLARB	TXERR	TXATIF	RXOVIF	TFERFFIF	TFHRFHIF	TFNRFNIF
0x05F0	C2FIFOUA2L(1)	15:8	FIFOUA[15:8]							
		7:0	FIFOUA[7:0]							
0x05F2	C2FIFOUA2H(1)	15:8	FIFOUA[31:24]							
		7:0	FIFOUA[23:16]							
0x05F4	C2FIFOCON3L	15:8						FRESET	TXREQ	UINC
		7:0	TXEN	RTREN	RXTSEN	TXATIE	RXOVIE	TFERFFIE	TFHRFHIE	TFNRFNIE
0x05F6	C2FIFOCON3H	15:8	PLSIZE[2:0]					FSIZE[4:0]		
		7:0	TXAT[1:0]					TXPRI[4:0]		
0x05F8	C2FIFOSTA3	15:8						FIFOC[4:0]		
		7:0	TXABT	TXLARB	TXERR	TXATIF	RXOVIF	TFERFFIF	TFHRFHIF	TFNRFNIF
0x05FA	C2FIFOSTA3H	15:8						FIFOC[4:0]		
		7:0	TXABT	TXLARB	TXERR	TXATIF	RXOVIF	TFERFFIF	TFHRFHIF	TFNRFNIF
0x05FC	C2FIFOUA3L(1)	15:8	FIFOUA[15:8]							
		7:0	FIFOUA[7:0]							
0x05FE	C2FIFOUA3H(1)	15:8	FIFOUA[31:24]							
		7:0	FIFOUA[23:16]							
0x0600	C2FIFOCON4L	15:8						FRESET	TXREQ	UINC
		7:0	TXEN	RTREN	RXTSEN	TXATIE	RXOVIE	TFERFFIE	TFHRFHIE	TFNRFNIE
0x0602	C2FIFOCON4H	15:8	PLSIZE[2:0]					FSIZE[4:0]		
		7:0	TXAT[1:0]					TXPRI[4:0]		
0x0604	C2FIFOSTA4	15:8						FIFOC[4:0]		
		7:0	TXABT	TXLARB	TXERR	TXATIF	RXOVIF	TFERFFIF	TFHRFHIF	TFNRFNIF
0x0606 ... 0x0607	Reserved									
0x0608	C2FIFOSTA4H	15:8						FIFOC[4:0]		
		7:0	TXABT	TXLARB	TXERR	TXATIF	RXOVIF	TFERFFIF	TFHRFHIF	TFNRFNIF
0x0608	C2FIFOUA4L(1)	15:8	FIFOUA[15:8]							
		7:0	FIFOUA[7:0]							
0x060A	C2FIFOUA4H(1)	15:8	FIFOUA[31:24]							
		7:0	FIFOUA[23:16]							
0x060C	C2FIFOCON5L	15:8						FRESET	TXREQ	UINC
		7:0	TXEN	RTREN	RXTSEN	TXATIE	RXOVIE	TFERFFIE	TFHRFHIE	TFNRFNIE

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Controller Area Network Flexible Data-Rate (...)

.....continued										
Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x060E	C2FIFOCON5H	15:8	PLSIZE[2:0]			FSIZE[4:0]				
		7:0	TXAT[1:0]			TXPRI[4:0]				
0x0610	C2FIFOSTA5	15:8	FIFOCI[4:0]							
		7:0	TXABT	TXLARB	TXERR	TXATIF	RXOVIF	TFERFFIF	TFHRFHIF	TFNRNFNIF
0x0612	Reserved									
...										
0x0613	C2FIFOSTA5H	15:8	FIFOCI[4:0]							
		7:0	TXABT	TXLARB	TXERR	TXATIF	RXOVIF	TFERFFIF	TFHRFHIF	TFNRNFNIF
0x0614	C2FIFOUA5L(1)	15:8	FIFOUA[15:8]							
		7:0	FIFOUA[7:0]							
0x0616	C2FIFOUA5H(1)	15:8	FIFOUA[31:24]							
		7:0	FIFOUA[23:16]							
0x0618	C2FIFOCON6L	15:8						FRESET	TXREQ	UINC
		7:0	TXEN	RTREN	RXTSEN	TXATIE	RXOVIE	TFERFFIE	TFHRFHIE	TFNRNFNIE
0x061A	C2FIFOCON6H	15:8	PLSIZE[2:0]			FSIZE[4:0]				
		7:0	TXAT[1:0]			TXPRI[4:0]				
0x061C	C2FIFOSTA6	15:8	FIFOCI[4:0]							
		7:0	TXABT	TXLARB	TXERR	TXATIF	RXOVIF	TFERFFIF	TFHRFHIF	TFNRNFNIF
0x061E	Reserved									
...										
0x061F	C2FIFOSTA6H	15:8	FIFOCI[4:0]							
		7:0	TXABT	TXLARB	TXERR	TXATIF	RXOVIF	TFERFFIF	TFHRFHIF	TFNRNFNIF
0x0620	C2FIFOUA6L(1)	15:8	FIFOUA[15:8]							
		7:0	FIFOUA[7:0]							
0x0622	C2FIFOUA6H(1)	15:8	FIFOUA[31:24]							
		7:0	FIFOUA[23:16]							
0x0624	C2FIFOCON7L	15:8						FRESET	TXREQ	UINC
		7:0	TXEN	RTREN	RXTSEN	TXATIE	RXOVIE	TFERFFIE	TFHRFHIE	TFNRNFNIE
0x0626	C2FIFOCON7H	15:8	PLSIZE[2:0]			FSIZE[4:0]				
		7:0	TXAT[1:0]			TXPRI[4:0]				
0x0628	C2FIFOSTA7	15:8	FIFOCI[4:0]							
		7:0	TXABT	TXLARB	TXERR	TXATIF	RXOVIF	TFERFFIF	TFHRFHIF	TFNRNFNIF
0x062A	Reserved									
...										
0x062B	C2FIFOUA7L(1)	15:8	FIFOUA[15:8]							
		7:0	FIFOUA[7:0]							
0x062E	C2FIFOSTA7H	15:8	FIFOCI[4:0]							
		7:0	TXABT	TXLARB	TXERR	TXATIF	RXOVIF	TFERFFIF	TFHRFHIF	TFNRNFNIF
0x062E	C2FIFOUA7H(1)	15:8	FIFOUA[31:24]							
		7:0	FIFOUA[23:16]							
0x0630	C2FLTCON0L	15:8	FLTENb			FbBP[4:0]				
		7:0	FLTENa			FaBP[4:0]				
0x0632	C2FLTCON0H	15:8	FLTEND			FdBP[4:0]				
		7:0	FLTENC			FcBP[4:0]				
0x0634	C2FLTCON1L	15:8	FLTENb			FbBP[4:0]				
		7:0	FLTENa			FaBP[4:0]				
0x0636	C2FLTCON1H	15:8	FLTEND			FdBP[4:0]				
		7:0	FLTENC			FcBP[4:0]				
0x0638	C2FLTCON2L	15:8	FLTENb			FbBP[4:0]				
		7:0	FLTENa			FaBP[4:0]				
0x063A	C2FLTCON2H	15:8	FLTEND			FdBP[4:0]				
		7:0	FLTENC			FcBP[4:0]				
0x063C	C2FLTCON3L	15:8	FLTENb			FbBP[4:0]				
		7:0	FLTENa			FaBP[4:0]				
0x063E	C2FLTCON3H	15:8	FLTEND			FdBP[4:0]				
		7:0	FLTENC			FcBP[4:0]				

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Controller Area Network Flexible Data-Rate (...)

.....continued										
Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x0640 ... 0x0641	Reserved									
0x0642	C2FLTOBJ0L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							
0x0642	C2FLTOBJ0H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x0644	C2MASK0L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x0646	C2MASK0H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							
0x0648	C2FLTOBJ1L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							
0x064A	C2FLTOBJ1H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x064C	C2MASK1L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x064E	C2MASK1H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							
0x0650	C2FLTOBJ2L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							
0x0652	C2FLTOBJ2H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x0654	C2MASK2L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x0656	C2MASK2H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							
0x0658	C2FLTOBJ3L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							
0x065A	C2FLTOBJ3H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x065C	C2MASK3L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x065E	C2MASK3H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							
0x0660	C2FLTOBJ4L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							
0x0662	C2FLTOBJ4H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x0664	C2MASK4L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x0666	C2MASK4H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							
0x0668	C2FLTOBJ5L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							
0x066A	C2FLTOBJ5H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x066C	C2MASK5L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x066E	C2MASK5H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							
0x0670	C2FLTOBJ6L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							
0x0672	C2FLTOBJ6H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x0674	C2MASK6L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x0676	C2MASK6H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							

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Controller Area Network Flexible Data-Rate (...)

.....continued

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x0678	C2FLTOBJ7L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							
0x067A	C2FLTOBJ7H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x067C	C2MASK7L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x067E	C2MASK7H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							
0x0680	C2FLTOBJ8L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							
0x0682	C2FLTOBJ8H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x0684	C2MASK8L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x0686	C2MASK8H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							
0x0688	C2FLTOBJ9L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							
0x068A	C2FLTOBJ9H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x068C	C2MASK9L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x068E	C2MASK9H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							
0x0690	C2FLTOBJ10L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							
0x0692	C2FLTOBJ10H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x0694	C2MASK10L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x0696	C2MASK10H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							
0x0698	C2FLTOBJ11L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							
0x069A	C2FLTOBJ11H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x069C	C2MASK11L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x069E	C2MASK11H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							
0x06A0	C2FLTOBJ12L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							
0x06A2	C2FLTOBJ12H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x06A4	C2MASK12L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x06A6	C2MASK12H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							
0x06A8	C2FLTOBJ13L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							
0x06AA	C2FLTOBJ13H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x06AC	C2MASK13L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x06AE	C2MASK13H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							
0x06B0	C2FLTOBJ14L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							

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Controller Area Network Flexible Data-Rate (...)

.....continued

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x06B2	C2FLTOBJ14H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x06B4	C2MASK14L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x06B6	C2MASK14H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							
0x06B8	C2FLTOBJ15L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							
0x06BA	C2FLTOBJ15H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x06BC	C2MASK15L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x06BE	C2MASK15H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							
0x06C0	C1CONL	15:8	CON		SIDL	BRSDIS	BUSY	WFT[1:0]		WAKFIL
		7:0	CLKSEL	PXEDIS	ISOCRCEN	DNCNT[4:0]				
0x06C2	C1CONH	15:8	TXBWS[3:0]				ABAT	REQOP[2:0]		
		7:0	OPMOD[2:0]			TXQEN	STEF	SERRLOM	ESIGM	RTXAT
0x06C4	C1NBTCFGL(1)	15:8		TSEG2[6:0]						
		7:0		SJW[6:0]						
0x06C6	C1NBTCFGH(1)	15:8	BRP[7:0]					TSEG1[7:0]		
		7:0	TSEG1[7:0]							
0x06C8	C1DBTCFGL(1)	15:8					TSEG2[3:0]			
		7:0					SJW[3:0]			
0x06CA	C1DBTCFGH(1)	15:8	BRP[7:0]					TSEG1[4:0]		
		7:0					TSEG1[4:0]			
0x06CC	C1TDCL(1)	15:8		TDCO[6:0]						
		7:0			TDCV[5:0]					
0x06CE	C1TDCH(1)	15:8							EDGFLTEN	SID11EN
		7:0							TDCMOD[1:0]	
0x06D0	C1TBCL(1,2)	15:8	TBC[15:8]							
		7:0	TBC[7:0]							
0x06D2	C1TBCH(1,2)	15:8	TBC[31:24]							
		7:0	TBC[23:16]							
0x06D4	C1TSCONL	15:8							TBCPRE[9:8]	
		7:0	TBCPRE[7:0]							
0x06D6	C1TSCONH	15:8								
		7:0						TSRES	TSEOF	TBCEN
0x06D8	C1VECL	15:8				FILHIT[4:0]				
		7:0		ICODE[6:0]						
0x06DA	C1VECH	15:8	RXCODE[6:0]							
		7:0		TXCODE[6:0]						
0x06DC	C1INTL	15:8	IVMIF	WAKIF	CERRIF	SERRIF	RXOVIF	TXATIF		
		7:0				TEFIF	MODIF	TBCIF	RXIF	TXIF
0x06DE	C1INTH	15:8	IVMIE	WAKIE	CERRIE	SERRIE	RXOVIE	TXATIE		
		7:0				TEFIE	MODIE	TBCIE	RXIE	TXIE
0x06E0	C1RXIFL(1)	15:8	RFIF[15:8]							
		7:0	RFIF[7:1]							
0x06E2	C1RXIFH(1)	15:8	RFIF[31:24]							
		7:0	RFIF[23:16]							
0x06E4	C1TXIFL(1)	15:8	TFIF[15:8]							
		7:0	TFIF[7:0]							
0x06E6	C1TXIFH(1)	15:8	TFIF[31:24]							
		7:0	TFIF[23:16]							
0x06E8	C1RXOVIFL(1)	15:8	RFOVIF[15:8]							
		7:0	RFOVIF[7:1]							
0x06EA	C1RXOVIFH(1)	15:8	RFOVIF[31:24]							
		7:0	RFOVIF[23:16]							

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Controller Area Network Flexible Data-Rate (...)

.....continued										
Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x06EC	C1TXATIFL(1)	15:8	TFATIF[15:8]							
		7:0	TFATIF[7:0]							
0x06EE	C1TXATIFH(1)	15:8	TFATIF[31:24]							
		7:0	TFATIF[23:16]							
0x06F0	C1TXREQL	15:8	TXREQ[15:8]							
		7:0	TXREQ[7:1]							TXREQ0
0x06F2	C1TXREQH	15:8	TXREQ[31:24]							
		7:0	TXREQ[23:16]							
0x06F4	C1TRECL	15:8	TERRCNT[7:0]							
		7:0	RERRCNT[7:0]							
0x06F6	C1TRECH	15:8								
		7:0			TXBO	TXBP	RXBP	TXWARN	RXWARN	EWARN
0x06F8	C1BDIAG0L	15:8	NTERRCNT[7:0]							
		7:0	NRERRCNT[7:0]							
0x06FA	C1BDIAG0H	15:8	DTERRCNT[7:0]							
		7:0	DRERRCNT[7:0]							
0x06FC	C1BDIAG1L	15:8	EFMSGCNT[15:8]							
		7:0	EFMSGCNT[7:0]							
0x06FE	C1BDIAG1H	15:8	DLCMM	ESI	DCRCERR	DSTUFERR	DFORMERR		DBIT1ERR	DBIT0ERR
		7:0	TXBOERR		NCRCERR	NSTUFERR	NFORMERR	NACKERR	NBIT1ERR	NBIT0ERR
0x0700	C1TEFCONL	15:8						FRESET		UINC
		7:0			TEFTSEN		TEFOVIE	TEFFIE	TEFHIE	TEFNEIE
0x0702	C1TEFCONH	15:8								
		7:0								
0x0704	C1TEFSTA	15:8								
		7:0					TEFOVIF	TEFFIF	TEFHIF	TEFNEIF
0x0706	Reserved									
...										
0x0707										
0x0708	C1TEFUAL(1)	15:8	TEFUA[15:8]							
		7:0	TEFUA[7:0]							
0x070A	C1TEFUAH(1)	15:8	TEFUA[15:8]							
		7:0	TEFUA[7:0]							
0x070C	C1FIFOBAL	15:8	FIFOBA[15:8]							
		7:0	FIFOBA[7:0]							
0x070E	C1FIFOBALH	15:8	FIFOBA[31:24]							
		7:0	FIFOBA[23:18]							FIFOBA[17:16]
0x0710	C1TXQCONL	15:8						FRESET	TXREQ	UINC
		7:0	TXEN			TXATIE		TXQEIE		TXQNIE
0x0712	C1TXQCONH	15:8	PLSIZE[2:0]				FSIZE[4:0]			
		7:0	TXAT[1:0]				TXPRI[4:0]			
0x0714	C1TXQSTAL	15:8					TXQCI[4:0]			
		7:0	TXABT	TXLARB	TXERR	TXATIF		TXQEIF		TXQNIF
0x0716	Reserved									
...										
0x0717										
0x0718	C1TXQUAL(1)	15:8	TXQUA[15:8]							
		7:0	TXQUA[7:0]							
0x071A	C1TXQUAH(1)	15:8	TXQUA[31:24]							
		7:0	TXQUA[23:16]							
0x071C	C1FIFOCON0L	15:8						FRESET	TXREQ	UINC
		7:0	TXEN	RTREN	RXTSEN	TXATIE	RXOVIE	TFERFFIE	TFHRFHIE	TFNRFNIE
0x071E	C1FIFOCON0H	15:8	PLSIZE[2:0]				FSIZE[4:0]			
		7:0	TXAT[1:0]				TXPRI[4:0]			
0x0720	C1FIFOSTA0	15:8					FIFOCI[4:0]			
		7:0	TXABT	TXLARB	TXERR	TXATIF	RXOVIF	TFERFFIF	TFHRFHIF	TFNRFNIF
0x0722	Reserved									
...										
0x0723										

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.....continued										
Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x0724	C1FIFOUA0L(1)	15:8	FIFOUA[15:8]							
		7:0	FIFOUA[7:0]							
0x0726	C1FIFOUA0H(1)	15:8	FIFOUA[31:24]							
		7:0	FIFOUA[23:16]							
0x0728	C1FIFOCON1L	15:8						FRESET	TXREQ	UINC
		7:0	TXEN	RTREN	RXTSEN	TXATIE	RXOVIE	TFERFFIE	TFHRFHIE	TFNRFNIE
0x072A	C1FIFOCON1H	15:8	PLSIZE[2:0]				FSIZE[4:0]			
		7:0	TXAT[1:0]				TXPRI[4:0]			
0x072C	C1FIFOSTA1	15:8						FIFOC[4:0]		
		7:0	TXABT	TXLARB	TXERR	TXATIF	RXOVIF	TFERFFIF	TFHRFHIF	TFNRFNIF
0x072E	Reserved									
...										
0x072F										
0x0730	C1FIFOUA1L(1)	15:8	FIFOUA[15:8]							
		7:0	FIFOUA[7:0]							
0x0732	C1FIFOUA1H(1)	15:8	FIFOUA[31:24]							
		7:0	FIFOUA[23:16]							
0x0734	C1FIFOCON2L	15:8						FRESET	TXREQ	UINC
		7:0	TXEN	RTREN	RXTSEN	TXATIE	RXOVIE	TFERFFIE	TFHRFHIE	TFNRFNIE
0x0736	C1FIFOCON2H	15:8	PLSIZE[2:0]				FSIZE[4:0]			
		7:0	TXAT[1:0]				TXPRI[4:0]			
0x0738	C1FIFOSTA2	15:8						FIFOC[4:0]		
		7:0	TXABT	TXLARB	TXERR	TXATIF	RXOVIF	TFERFFIF	TFHRFHIF	TFNRFNIF
0x073A	Reserved									
...										
0x073B										
0x073C	C1FIFOUA2L(1)	15:8	FIFOUA[15:8]							
		7:0	FIFOUA[7:0]							
0x073E	C1FIFOUA2H(1)	15:8	FIFOUA[31:24]							
		7:0	FIFOUA[23:16]							
0x0740	C1FIFOCON3L	15:8						FRESET	TXREQ	UINC
		7:0	TXEN	RTREN	RXTSEN	TXATIE	RXOVIE	TFERFFIE	TFHRFHIE	TFNRFNIE
0x0742	C1FIFOCON3H	15:8	PLSIZE[2:0]				FSIZE[4:0]			
		7:0	TXAT[1:0]				TXPRI[4:0]			
0x0744	C1FIFOSTA3	15:8						FIFOC[4:0]		
		7:0	TXABT	TXLARB	TXERR	TXATIF	RXOVIF	TFERFFIF	TFHRFHIF	TFNRFNIF
0x0746	Reserved									
...										
0x0747										
0x0748	C1FIFOUA3L(1)	15:8	FIFOUA[15:8]							
		7:0	FIFOUA[7:0]							
0x074A	C1FIFOUA3H(1)	15:8	FIFOUA[31:24]							
		7:0	FIFOUA[23:16]							
0x074C	C1FIFOCON4L	15:8						FRESET	TXREQ	UINC
		7:0	TXEN	RTREN	RXTSEN	TXATIE	RXOVIE	TFERFFIE	TFHRFHIE	TFNRFNIE
0x074C	C1FIFOCON4H	15:8	PLSIZE[2:0]				FSIZE[4:0]			
		7:0	TXAT[1:0]				TXPRI[4:0]			
0x074E	Reserved									
...										
0x074F										
0x0750	C1FIFOSTA4	15:8						FIFOC[4:0]		
		7:0	TXABT	TXLARB	TXERR	TXATIF	RXOVIF	TFERFFIF	TFHRFHIF	TFNRFNIF
0x0752	Reserved									
...										
0x0753										
0x0754	C1FIFOUA4L(1)	15:8	FIFOUA[15:8]							
		7:0	FIFOUA[7:0]							
0x0756	C1FIFOUA4H(1)	15:8	FIFOUA[31:24]							
		7:0	FIFOUA[23:16]							

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.....continued										
Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x0758	C1FIFOCON5L	15:8						FRESET	TXREQ	UINC
		7:0	TXEN	RTREN	RXTSEN	TXATIE	RXOVIE	TFERFFIE	TFHRFHIE	TFNRFNIE
0x075A	C1FIFOCON5H	15:8	PLSIZE[2:0]			FSIZE[4:0]				
		7:0		TXAT[1:0]		TXPRI[4:0]				
0x075C	C1FIFOSTA5	15:8				FIFOC[4:0]				
		7:0	TXABT	TXLARB	TXERR	TXATIF	RXOVIF	TFERFFIF	TFHRFHIF	TFNRFNIF
0x075E ... 0x075F	Reserved									
0x0760	C1FIFOUA5L(1)	15:8	FIFOUA[15:8]							
		7:0	FIFOUA[7:0]							
0x0762	C1FIFOUA5H(1)	15:8	FIFOUA[31:24]							
		7:0	FIFOUA[23:16]							
0x0764	C1FIFOCON6L	15:8						FRESET	TXREQ	UINC
		7:0	TXEN	RTREN	RXTSEN	TXATIE	RXOVIE	TFERFFIE	TFHRFHIE	TFNRFNIE
0x0766	C1FIFOCON6H	15:8	PLSIZE[2:0]			FSIZE[4:0]				
		7:0		TXAT[1:0]		TXPRI[4:0]				
0x0768	C1FIFOSTA6	15:8				FIFOC[4:0]				
		7:0	TXABT	TXLARB	TXERR	TXATIF	RXOVIF	TFERFFIF	TFHRFHIF	TFNRFNIF
0x076A ... 0x076B	Reserved									
0x076C	C1FIFOUA6L(1)	15:8	FIFOUA[15:8]							
		7:0	FIFOUA[7:0]							
0x076E	C1FIFOUA6H(1)	15:8	FIFOUA[31:24]							
		7:0	FIFOUA[23:16]							
0x0770	C1FLTCON0L	15:8	FLTENb			FbBP[4:0]				
		7:0	FLTENa			FaBP[4:0]				
0x0772	C1FLTCON0H	15:8	FLTEND			FdBP[4:0]				
		7:0	FLTENC			FcBP[4:0]				
0x0774	C1FLTCON1L	15:8	FLTENb			FbBP[4:0]				
		7:0	FLTENa			FaBP[4:0]				
0x0776	C1FLTCON1H	15:8	FLTEND			FdBP[4:0]				
		7:0	FLTENC			FcBP[4:0]				
0x0778	C1FLTCON2L	15:8	FLTENb			FbBP[4:0]				
		7:0	FLTENa			FaBP[4:0]				
0x077A	C1FLTCON2H	15:8	FLTEND			FdBP[4:0]				
		7:0	FLTENC			FcBP[4:0]				
0x077C	C1FLTCON3L	15:8	FLTENb			FbBP[4:0]				
		7:0	FLTENa			FaBP[4:0]				
0x077E	C1FLTCON3H	15:8	FLTEND			FdBP[4:0]				
		7:0	FLTENC			FcBP[4:0]				
0x0780	C1FLTOBJ0L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							
0x0782	C1FLTOBJ0H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x0784	C1MASK0L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x0786	C1MASK0H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							
0x0788	C1FLTOBJ1L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							
0x078A	C1FLTOBJ1H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x078C	C1MASK1L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x078E	C1MASK1H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							

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.....continued

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x0790	C1FLTOBJ2L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							
0x0792	C1FLTOBJ2H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x0794	C1MASK2L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x0796	C1MASK2H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							
0x0798	C1FLTOBJ3L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							
0x079A	C1FLTOBJ3H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x079C	C1MASK3L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x079E	C1MASK3H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							
0x07A0	C1FLTOBJ4L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							
0x07A2	C1FLTOBJ4H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x07A4	C1MASK4L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x07A6	C1MASK4H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							
0x07A8	C1FLTOBJ5L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							
0x07AA	C1FLTOBJ5H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x07AC	C1MASK5L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x07AE	C1MASK5H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							
0x07B0	C1FLTOBJ6L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							
0x07B2	C1FLTOBJ6H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x07B4	C1MASK6L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x07B6	C1MASK6H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							
0x07B8	C1FLTOBJ7L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							
0x07BA	C1FLTOBJ7H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x07BC	C1MASK7L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x07BE	C1MASK7H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							
0x07C0	C1FLTOBJ8L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							
0x07C2	C1FLTOBJ8H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x07C4	C1MASK8L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x07C6	C1MASK8H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							
0x07C8	C1FLTOBJ9L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							

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.....continued										
Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x07CA	C1FLTOBJ9H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x07CC	C1MASK9L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x07CE	C1MASK9H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							
0x07D0	C1FLTOBJ10L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							
0x07D2	C1FLTOBJ10H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x07D4	C1MASK10L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x07D6	C1MASK10H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							
0x07D8	C1FLTOBJ11L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							
0x07DA	C1FLTOBJ11H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x07DC	C1MASK11L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x07DE	C1MASK11H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							
0x07E0	C1FLTOBJ12L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							
0x07E2	C1FLTOBJ12H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x07E4	C1MASK12L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x07E6	C1MASK12H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							
0x07E8	C1FLTOBJ13L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							
0x07EA	C1FLTOBJ13H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x07EC	C1MASK13L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x07EE	C1MASK13H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							
0x07F0	C1FLTOBJ14L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							
0x07F2	C1FLTOBJ14H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x07F4	C1MASK14L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x07F6	C1MASK14H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							
0x07F8	C1FLTOBJ15L	15:8	EID[4:0]					SID[10:8]		
		7:0	SID[7:0]							
0x07FA	C1FLTOBJ15H	15:8		EXIDE	SID11	EID[17:13]				
		7:0	EID[12:5]							
0x07FC	C1MASK15L	15:8	MEID[4:0]					MSID[10:8]		
		7:0	MSID[7:0]							
0x07FE	C1MASK15H	15:8		MIDE	MSID11	MEID[17:13]				
		7:0	MEID[12:5]							

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12.2.1 CAN2 FIFO Control Register x Low (x = 1 to 7)

Name: C2FIFOCONxL
Offset: 0x5DC, 0x5E8, 0x5F4, 0x600, 0x60C, 0x618, 0x624

Note:

1. This bit can only be modified in Configuration mode (OPMOD[2:0] = 100).

Legend: S = Settable bit; HC = Hardware Clearable bit

Bit	15	14	13	12	11	10	9	8
						FRESET	TXREQ	UINC
Access						S/HC	R/W/HC	S/HC
Reset						1	0	0

Bit	7	6	5	4	3	2	1	0
	TXEN	RTREN	RXTSEN	TXATIE	RXOVIE	TFERFFIE	TFHRFHIE	TFNRFNIE
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 10 – FRESET FIFO Reset bit

Value	Description
1	FIFO will be reset when bit is set, cleared by hardware when FIFO is reset; user should poll whether this bit is clear before taking any action
0	No effect

Bit 9 – TXREQ Message Send Request bit

TXEN = 0 (FIFO configured as a receive FIFO):

This bit has no effect.

TXEN = 1 (FIFO configured as a transmit FIFO):

Value	Description
1	Requests sending a message; the bit will automatically clear when all the messages queued in the FIFO are successfully sent
0	Clearing the bit to '0' while set ('1') will request a message abort

Bit 8 – UINC Increment Head/Tail bit

TXEN = 1 (FIFO configured as a transmit FIFO):

When this bit is set, the FIFO head will increment by a single message.

TXEN = 0 (FIFO configured as a receive FIFO):

When this bit is set, the FIFO tail will increment by a single message.

Bit 7 – TXEN TX/RX Buffer Selection bit

Value	Description
1	Transmits message object
0	Receives message object

Bit 6 – RTREN Auto-Remote Transmit (RTR) Enable bit

Value	Description
1	When a Remote Transmit is received, TXREQ will be set
0	When a Remote Transmit is received, TXREQ will be unaffected

Bit 5 – RXTSEN Received Message Timestamp Enable bit⁽¹⁾

Value	Description
1	Captures timestamp in received message object in RAM
0	Does not capture timestamp

Bit 4 – TXATIE Transmit Attempts Exhausted Interrupt Enable bit

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Value	Description
1	Enables interrupt
0	Disables interrupt

Bit 3 – RXOVIE Overflow Interrupt Enable bit

Value	Description
1	Interrupt is enabled for overflow event
0	Interrupt is disabled for overflow event

Bit 2 – TFERFFIE Transmit/Receive FIFO Empty/Full Interrupt Enable bit

TXEN = 1 (FIFO configured as a transmit FIFO):

Transmit FIFO Empty Interrupt Enable.

1 = Interrupt is enabled for FIFO empty

0 = Interrupt is disabled for FIFO empty

TXEN = 0 (FIFO configured as a receive FIFO):

Receive FIFO Full Interrupt Enable.

1 = Interrupt is enabled for FIFO full

0 = Interrupt is disabled for FIFO full

Bit 1 – TFHRFHIE Transmit/Receive FIFO Half Empty/Half Full Interrupt Enable bit

TXEN = 1 (FIFO configured as a transmit FIFO):

Transmit FIFO Half Empty Interrupt Enable.

1 = Interrupt is enabled for FIFO half empty

0 = Interrupt is disabled for FIFO half empty

TXEN = 0 (FIFO configured as a receive FIFO):

Receive FIFO Half Full Interrupt Enable.

1 = Interrupt is enabled for FIFO half full

0 = Interrupt is disabled for FIFO half full

Bit 0 – TFNRFNIE Transmit/Receive FIFO Not Full/Not Empty Interrupt Enable bit

TXEN = 1 (FIFO configured as a transmit FIFO):

Transmit FIFO Not Full Interrupt Enable.

1 = Interrupt is enabled for FIFO not full

0 = Interrupt is disabled for FIFO not full

TXEN = 0 (FIFO configured as a receive FIFO):

Receive FIFO Not Empty Interrupt Enable.

1 = Interrupt is enabled for FIFO not empty

0 = Interrupt is disabled for FIFO not empty

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Controller Area Network Flexible Data-Rate (...)

12.2.2 CAN2 FIFO Control Register x High (x = 1 to 7)

Name: C2FIFOCONxH
Offset: 0x5DE, 0x5EA, 0x5F6, 0x602, 0x60E, 0x61A, 0x626

Note:

- These bits can only be modified in Configuration mode (OPMOD[2:0] = 100).

Bit	15	14	13	12	11	10	9	8
	PLSIZE[2:0]				FSIZE[4:0]			
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
		TXAT[1:0]			TXPRI[4:0]			
Access		R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset		1	1	0	0	0	0	0

Bits 15:13 – PLSIZE[2:0] Payload Size bits⁽¹⁾

Value	Description
111	64 data bytes
110	48 data bytes
101	32 data bytes
100	24 data bytes
011	20 data bytes
010	16 data bytes
001	12 data bytes
000	8 data bytes

Bits 12:8 – FSIZE[4:0] FIFO Size bits⁽¹⁾

Value	Description
11111	FIFO is 32 messages deep
. . .	
00010	FIFO is 3 messages deep
00001	FIFO is 2 messages deep
00000	FIFO is 1 message deep

Bits 6:5 – TXAT[1:0] Retransmission Attempts bits

This feature is enabled when RTXAT (C1CONH[0]) is set.

Value	Description
11	Unlimited number of retransmission attempts
10	Unlimited number of retransmission attempts
01	Three retransmission attempts
00	Disables retransmission attempts

Bits 4:0 – TXPRI[4:0] Message Transmit Priority bits

Value	Description
11111	Highest message priority
. . .	
00000	Lowest message priority

12.2.3 CAN2 FIFO Status Register x (x = 1 to 7)

Name: C2FIFOSTAx
Offset: 0x5E0, 0x5EC, 0x5F8, 0x604, 0x610, 0x61C, 0x628

Notes:

1. FIFOCI[4:0] bits give a zero-indexed value to the message in the FIFO. If the FIFO is four messages deep (FSIZE[4:0] = 3), FIFOCIx will take on a value of 0 to 3, depending on the state of the FIFO.
2. These bits are updated when a message completes (or aborts) or when the FIFO is reset.
3. This bit is reset on any read of this register or when the TXQ is reset. The bits are cleared when TXREQ is set or using an SPI write.

Legend: HS = Hardware Settable bit; C = Clearable bit

Bit	15	14	13	12	11	10	9	8
				FIFOCI[4:0]				
Access				R	R	R	R	R
Reset				0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	TXABT	TXLARB	TXERR	TXATIF	RXOVIF	TFERFFIF	TFHRFHIF	TFNRFNIF
Access	R	R	R	HS/C	HS/C	R	R	R
Reset	0	0	0	0	0	0	0	0

Bits 12:8 – FIFOCI[4:0] FIFO Message Index bits⁽¹⁾

TXEN = 1 (FIFO configured as a transmit buffer):

A read of this register will return an index to the message that the FIFO will next attempt to transmit.

TXEN = 0 (FIFO configured as a receive buffer):

A read of this register will return an index to the message that the FIFO will use to save the next message.

Bit 7 – TXABT Message Aborted Status bit⁽³⁾

Value	Description
1	Message was aborted
0	Message completed successfully

Bit 6 – TXLARB Message Lost Arbitration Status bit⁽²⁾

Value	Description
1	Message lost arbitration while being sent
0	Message did not lose arbitration while being sent

Bit 5 – TXERR Error Detected During Transmission bit⁽²⁾

Value	Description
1	A bus error occurred while the message was being sent
0	A bus error did not occur while the message was being sent

Bit 4 – TXATIF Transmit Attempts Exhausted Interrupt Pending bit

TXEN = 0 (FIFO configured as a receive buffer):

Unused, read as '0'.

TXEN = 1 (FIFO configured as a transmit buffer):

Value	Description
1	Interrupt is pending
0	Interrupt is not pending

Bit 3 – RXOVIF Receive FIFO Overflow Interrupt Flag bit

TXEN = 1 (FIFO configured as a transmit buffer):

Unused, read as '0'.

TXEN = 0 (FIFO configured as a receive buffer):

Value	Description
1	Overflow event has occurred
0	No overflow event has occurred

Bit 2 – TFERFFIF Transmit/Receive FIFO Empty/Full Interrupt Flag bit

TXEN = 1 (FIFO configured as a transmit FIFO):

Transmit FIFO Empty Interrupt Flag.

1 = FIFO is empty

0 = FIFO is not empty, at least one message is queued to be transmitted

TXEN = 0 (FIFO configured as a receive FIFO):

Receive FIFO Full Interrupt Flag.

1 = FIFO is full

0 = FIFO is not full

Bit 1 – TFHRFHIF Transmit/Receive FIFO Half Empty/Half Full Interrupt Flag bit

TXEN = 1 (FIFO configured as a transmit FIFO):

Transmit FIFO Half Empty Interrupt Flag.

1 = FIFO is ≤ half full

0 = FIFO is > half full

TXEN = 0 (FIFO configured as a receive FIFO):

Receive FIFO Half Full Interrupt Flag.

1 = FIFO is ≥ half full

0 = FIFO is < half full

Bit 0 – TFNRFNIF Transmit/Receive FIFO Not Full/Not Empty Interrupt Flag bit

TXEN = 1 (FIFO configured as a transmit FIFO):

Transmit FIFO Not Full Interrupt Flag.

1 = FIFO is not full

0 = FIFO is full

TXEN = 0 (FIFO configured as a receive FIFO):

Receive FIFO Not Empty Interrupt Flag.

1 = FIFO is not empty, has at least one message

0 = FIFO is empty

12.2.4 CAN2 FIFO Status High Register x (x = 1 to 7)

Name: C2FIFOSTAxH
Offset: 0x5E2, 0x5EE, 0x5FA, 0x608, 0x614, 0x620, 0x62E

Notes:

1. FIFOCI[4:0] gives a zero-indexed value to the message in the FIFO. If the FIFO is four messages deep (FSIZE[4:0] = 3), FIFOCIx will take on a value of 0 to 3, depending on the state of the FIFO.
2. These bits are updated when a message completes (or aborts) or when the FIFO is reset.
3. This bit is reset on any read of this register or when the TXQ is reset. The bits are cleared when TXREQ is set or using an SPI write.

Legend: HS = Hardware Settable bit; C = Clearable bit

Bit	15	14	13	12	11	10	9	8
				FIFOCI[4:0]				
Access				R	R	R	R	R
Reset				0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	TXABT	TXLARB	TXERR	TXATIF	RXOVIF	TFERFFIF	TFHRFHIF	TFNRFNIF
Access	R	R	R	HS/C	HS/C	R	R	R
Reset	0	0	0	0	0	0	0	0

Bits 12:8 – FIFOCI[4:0] FIFO Message Index bits⁽¹⁾

TXEN = 1 (FIFO configured as a transmit buffer):

A read of this register will return an index to the message that the FIFO will next attempt to transmit.

TXEN = 0 (FIFO configured as a receive buffer):

A read of this register will return an index to the message that the FIFO will use to save the next message.

Bit 7 – TXABT Message Aborted Status bit⁽³⁾

Value	Description
1	Message was aborted
0	Message completed successfully

Bit 6 – TXLARB Message Lost Arbitration Status bit⁽²⁾

Value	Description
1	Message lost arbitration while being sent
0	Message did not lose arbitration while being sent

Bit 5 – TXERR Error Detected During Transmission bit⁽²⁾

Value	Description
1	A bus error occurred while the message was being sent
0	A bus error did not occur while the message was being sent

Bit 4 – TXATIF Transmit Attempts Exhausted Interrupt Pending bit

TXEN = 0 (FIFO configured as a receive buffer):

Unused, read as '0'.

TXEN = 1 (FIFO configured as a transmit buffer):

Value	Description
1	Interrupt is pending
0	Interrupt is not pending

Bit 3 – RXOVIF Receive FIFO Overflow Interrupt Flag bit

TXEN = 1 (FIFO configured as a transmit buffer):

Unused, read as '0'.

TXEN = 0 (FIFO configured as a receive buffer):

Value	Description
1	Overflow event has occurred
0	No overflow event has occurred

Bit 2 – TFERFFIF Transmit/Receive FIFO Empty/Full Interrupt Flag bit

TXEN = 1 (FIFO configured as a transmit FIFO):

Transmit FIFO Empty Interrupt Flag.

1 = FIFO is empty

0 = FIFO is not empty, at least one message is queued to be transmitted

TXEN = 0 (FIFO configured as a receive FIFO):

Receive FIFO Full Interrupt Flag.

1 = FIFO is full

0 = FIFO is not full

Bit 1 – TFHRFHIF Transmit/Receive FIFO Half Empty/Half Full Interrupt Flag bit

TXEN = 1 (FIFO configured as a transmit FIFO):

Transmit FIFO Half Empty Interrupt Flag.

1 = FIFO is $\leq$ half full

0 = FIFO is $>$ half full

TXEN = 0 (FIFO configured as a receive FIFO):

Receive FIFO Half Full Interrupt Flag.

1 = FIFO is $\geq$ half full

0 = FIFO is $<$ half full

Bit 0 – TFNRFNIF Transmit/Receive FIFO Not Full/Not Empty Interrupt Flag bit

TXEN = 1 (FIFO configured as a transmit FIFO):

Transmit FIFO Not Full Interrupt Flag.

1 = FIFO is not full

0 = FIFO is full

TXEN = 0 (FIFO configured as a receive FIFO):

Receive FIFO Not Empty Interrupt Flag.

1 = FIFO is not empty, has at least one message

0 = FIFO is empty

12.2.5 CAN2 FIFO User Address Register x Low (x = 1 to 7)

Name: C2FIFOUAxL⁽¹⁾

Offset: 0x5E4, 0x5F0, 0x5FC, 0x608, 0x614, 0x620, 0x62C

Note:

1. This register is not ensured to read correctly in Configuration mode and should only be accessed when the module is not in Configuration mode.

Legend: x = Bit is unknown

Bit	15	14	13	12	11	10	9	8
	FIFOUA[15:8]							
Access	R	R	R	R	R	R	R	R
Reset	x	x	x	x	x	x	x	x
Bit	7	6	5	4	3	2	1	0
	FIFOUA[7:0]							
Access	R	R	R	R	R	R	R	R
Reset	x	x	x	x	x	x	x	x

Bits 15:0 – FIFOUA[15:0] FIFO User Address bits

TXEN = 1 (FIFO configured as a transmit buffer):

A read of this register will return the address where the next message is to be written (FIFO head).

TXEN = 0 (FIFO configured as a receive buffer):

A read of this register will return the address where the next message is to be read (FIFO tail).

12.2.6 CAN2 FIFO User Address Register x High (x = 1 to 7)

Name: C2FIFOUAxH⁽¹⁾

Offset: 0x5E6, 0x5F2, 0x5FE, 0x60A, 0x616, 0x622, 0x62E

Note:

1. This register is not ensured to read correctly in Configuration mode and should only be accessed when the module is not in Configuration mode.

Legend: x = Bit is unknown

Bit	15	14	13	12	11	10	9	8
	FIFOUA[31:24]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	FIFOUA[23:16]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	x

Bits 15:0 – FIFOUA[31:16] FIFO User Address bits

TXEN = 1 (FIFO configured as a transmit buffer):

A read of this register will return the address where the next message is to be written (FIFO head).

TXEN = 0 (FIFO configured as a receive buffer):

A read of this register will return the address where the next message is to be read (FIFO tail).

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12.2.7 CAN2 Filter Control Register x Low (x = 0 to 3; a = 0, 4, 8, 12; b = 1, 5, 9, 13)

Name: C2FLTCONxL
Offset: 0x630, 0x634, 0x638, 0x63C

Bit	15	14	13	12	11	10	9	8
	FLTENb					FbBP[4:0]		
Access	R/W			R/W	R/W	R/W	R/W	R/W
Reset	0			0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	FLTENa					FaBP[4:0]		
Access	R/W			R/W	R/W	R/W	R/W	R/W
Reset	0			0	0	0	0	0

Bit 15 – FLTENb Enable Filter b to Accept Messages bit

Value	Description
1	Filter is enabled
0	Filter is disabled

Bits 12:8 – FbBP[4:0] Pointer to Object When Filter b Hits bits

Value	Description
11111-11000	Reserved
00111	Message matching filter is stored in Object 7
00110	Message matching filter is stored in Object 6
. . .	
00010	Message matching filter is stored in Object 2
00001	Message matching filter is stored in Object 1
00000	Reserved; Object 0 is the TX Queue and can't receive messages

Bit 7 – FLTENa Enable Filter a to Accept Messages bit

Value	Description
1	Filter is enabled
0	Filter is disabled

Bits 4:0 – FaBP[4:0] Pointer to Object When Filter a Hits bits

Value	Description
11111-11000	Reserved
00111	Message matching filter is stored in Object 7
00110	Message matching filter is stored in Object 6
. . .	
00010	Message matching filter is stored in Object 2
00001	Message matching filter is stored in Object 1
00000	Reserved; Object 0 is the TX Queue and can't receive messages

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12.2.8 CAN2 Filter Control Register x High (x = 0 to 3; c = 2, 6, 10, 14; d = 3, 7, 11, 15)

Name: C2FLTCONxH
Offset: 0x632, 0x636, 0x63A, 0x63E

Bit	15	14	13	12	11	10	9	8
	FLTEND					FdBP[4:0]		
Access	R/W			R/W	R/W	R/W	R/W	R/W
Reset	0			0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	FLTENC					FcBP[4:0]		
Access	R/W			R/W	R/W	R/W	R/W	R/W
Reset	0			0	0	0	0	0

Bit 15 – FLTEND Enable Filter d to Accept Messages bit

Value	Description
1	Filter is enabled
0	Filter is disabled

Bits 12:8 – FdBP[4:0] Pointer to Object When Filter d Hits bits

Value	Description
11111-11000	Reserved
00111	Message matching filter is stored in Object 7
00110	Message matching filter is stored in Object 6
. . .	
00010	Message matching filter is stored in Object 2
00001	Message matching filter is stored in Object 1
00000	Reserved; Object 0 is the TX Queue and can't receive messages

Bit 7 – FLTENC Enable Filter c to Accept Messages bit

Value	Description
1	Filter is enabled
0	Filter is disabled

Bits 4:0 – FcBP[4:0] Pointer to Object When Filter c Hits bits

Value	Description
11111-11000	Reserved
00111	Message matching filter is stored in Object 7
00110	Message matching filter is stored in Object 6
. . .	
00010	Message matching filter is stored in Object 2
00001	Message matching filter is stored in Object 1
00000	Reserved; Object 0 is the TX Queue and can't receive messages

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12.2.9 CAN2 Filter Object Register x Low (x = 0 to 15)

Name: C2FLTOBJxL

Offset: 0x642, 0x648, 0x650, 0x658, 0x660, 0x668, 0x670, 0x678, 0x680, 0x688, 0x690, 0x698, 0x6A0, 0x6A8, 0x6B0, 0x6B8

Bit	15	14	13	12	11	10	9	8
	EID[4:0]					SID[10:8]		
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	SID[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:11 – EID[4:0] Extended Identifier Filter bits

In DeviceNet™ mode, these are the filter bits for the first two data bytes.

Bits 10:0 – SID[10:0] Standard Identifier Filter bits

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12.2.10 CAN2 Filter Object Register x High (x = 0 to 15)

Name: C2FLTOBJxH

Offset: 0x642, 0x64A, 0x652, 0x65A, 0x662, 0x66A, 0x672, 0x67A, 0x682, 0x68A, 0x692, 0x69A, 0x6A2, 0x6AA, 0x6B2, 0x6BA

Bit	15	14	13	12	11	10	9	8
		EXIDE	SID11	EID[17:13]				
Access		R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset		0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	EID[12:5]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 14 – EXIDE Extended Identifier Enable bit

If MIDE = 1:

Value	Description
1	Matches only messages with Extended Identifier addresses
0	Matches only messages with Standard Identifier addresses

Bit 13 – SID11 Standard Identifier Filter bit

Bits 12:0 – EID[17:5] Extended Identifier Filter bits

In DeviceNet™ mode, these are the filter bits for the first two data bytes.

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12.2.11 CAN2 Mask Register x Low (x = 0 to 15)

Name: C2MASKxL
Offset: 0x644, 0x64C, 0x654, 0x65C, 0x664, 0x66C, 0x674, 0x67C, 0x684, 0x68C, 0x694, 0x69C, 0x6A4, 0x6AC, 0x6B4, 0x6BC

Bit	15	14	13	12	11	10	9	8
	MEID[4:0]					MSID[10:8]		
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	MSID[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:11 – MEID[4:0] Extended Identifier Mask bits
 In DeviceNet™ mode, these are the mask bits for the first two data bytes.

Bits 10:0 – MSID[10:0] Standard Identifier Mask bits

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12.2.12 CAN2 Mask Register x High (x = 0 to 15)

Name: C2MASKxH
Offset: 0x646, 0x64E, 0x656, 0x65E, 0x666, 0x66E, 0x676, 0x67E, 0x686, 0x68E, 0x696, 0x69E, 0x6A6, 0x6AE, 0x6B6, 0x6BE

Bit	15	14	13	12	11	10	9	8
		MIDE	MSID11	MEID[17:13]				
Access		R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset		0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	MEID[12:5]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 14 – MIDE Identifier Receive Mode bit

Value	Description
1	Matches only message types (standard or extended address) that correspond to the EXIDE bit in the filter
0	Matches either standard or extended address message if filters match (i.e., if (Filter SID) = (Message SID) or if (Filter SID/EID) = (Message SID/EID))

Bit 13 – MSID11 Standard Identifier Mask bit

Bits 12:0 – MEID[17:5] Extended Identifier Mask bits

In DeviceNet™ mode, these are the mask bits for the first two data bytes.

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Controller Area Network Flexible Data-Rate (...)

12.2.13 CAN Control Register Low

Name: C1CONL
Offset: 0x6C0

Note:

- These bits can only be modified in Configuration mode (OPMOD[2:0] = 100).

Bit	15	14	13	12	11	10	9	8
	CON		SIDL	BRSDIS	BUSY	WFT[1:0]		WAKFIL
Access	R/W		R/W	R/W	R/W	R/W	R/W	R/W
Reset	0		0	0	0	1	1	1

Bit	7	6	5	4	3	2	1	0
	CLKSEL	PXEDIS	ISOCRCEN			DNCNT[4:0]		
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	1	1	0	0	0	0	0

Bit 15 – CON CAN Enable bit

Value	Description
1	CAN module is enabled
0	CAN module is disabled

Bit 13 – SIDL CAN Stop in Idle Control bit

Value	Description
1	Stops module operation in Idle mode
0	Does not stop module operation in Idle mode

Bit 12 – BRSDIS Bit Rate Switching (BRS) Disable bit

Value	Description
1	Bit Rate Switching is disabled, regardless of BRS in the transmit message object
0	Bit Rate Switching depends on BRS in the transmit message object

Bit 11 – BUSY CAN Module is Busy bit

Value	Description
1	The CAN module is active
0	The CAN module is inactive

Bits 10:9 – WFT[1:0] Selectable Wake-up Filter Time bits

Value	Description
11	T11 _{FILTER}
10	T10 _{FILTER}
01	T01 _{FILTER}
00	T00 _{FILTER}

Bit 8 – WAKFIL Enable CAN Bus Line Wake-up Filter bit⁽¹⁾

Value	Description
1	Uses CAN bus line filter for wake-up
0	CAN bus line filter is not used for wake-up

Bit 7 – CLKSEL Module Clock Source Select bit⁽¹⁾

Value	Description
1	Auxiliary clock is active when module is enabled
0	CAN clock is not active when module is enabled

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Controller Area Network Flexible Data-Rate (...)

Bit 6 – PXEDIS Protocol Exception Event Detection Disabled bit⁽¹⁾

A recessive “reserved bit” following a recessive FDF bit is called a Protocol Exception.

Value	Description
1	Protocol Exception is treated as a form error
0	If a Protocol Exception is detected, CAN will enter the Bus Integrating state

Bit 5 – ISOCRCEN Enable ISO CRC in CAN FD Frames bit⁽¹⁾

Value	Description
1	Includes stuff bit count in CRC field and uses non-zero CRC initialization vector
0	Does not include stuff bit count in CRC field and uses CRC initialization vector with all zeros

Bits 4:0 – DNCNT[4:0] DeviceNet™ Filter Bit Number bits

Value	Description
10011–11111	Invalid selection (compares up to 18 bits of data with EID)
10010	Compares up to Data Byte 2, bit 6 with EID17
. . .	
00001	Compares up to Data Byte 0, bit 7 with EID0
00000	Does not compare data bytes

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12.2.14 CAN Control Register High

Name: C1CONH
Offset: 0x6C2

Note:

- These bits can only be modified in Configuration mode (OPMOD[2:0] = 100).

Legend: S = Settable bit; HC = Hardware Clearable bit

Bit	15	14	13	12	11	10	9	8
	TXBWS[3:0]				ABAT	REQOP[2:0]		
Access	R/W	R/W	R/W	R/W	S/HC	R/W	R/W	R/W
Reset	0	0	0	0	0	1	0	0

Bit	7	6	5	4	3	2	1	0
	OPMOD[2:0]			TXQEN	STEF	SERRLOM	ESIGM	RTXAT
Access	R	R	R	R/W	R/W	R/W	R/W	R/W
Reset	1	0	0	1	1	0	0	0

Bits 15:12 – TXBWS[3:0] Transmit Bandwidth Sharing bits

Value	Description
1111–1100	4096
1011	2048
1010	1024
1001	512
1000	256
0111	128
0110	64
0101	32
0100	16
0011	8
0010	4
0001	2
0000	No delay

Bit 11 – ABAT Abort All Pending Transmissions bit

Value	Description
1	Signals all transmit buffers to abort transmission
0	Module will clear this bit when all transmissions are aborted

Bits 10:8 – REQOP[2:0] Request Operation Mode bits

Value	Description
111	Sets Restricted Operation mode
110	Sets Normal CAN 2.0 mode; error frames on CAN FD frames
101	Sets External Loopback mode
100	Sets Configuration mode
011	Sets Listen Only mode
010	Sets Internal Loopback mode
001	Sets Disable mode
000	Sets Normal CAN FD mode; supports mixing of full CAN FD and classic CAN 2.0 frames

Bits 7:5 – OPMOD[2:0] Operation Mode Status bits

Value	Description
111	Module is in Restricted Operation mode
110	Module is in Normal CAN 2.0 mode; error frames on CAN FD frames
101	Module is in External Loopback mode

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Controller Area Network Flexible Data-Rate (...)

Value	Description
100	Module is in Configuration mode
011	Module is in Listen Only mode
010	Module is in Internal Loopback mode
001	Module is in Disable mode
000	Module is in Normal CAN FD mode; supports mixing of full CAN FD and classic CAN 2.0 frames

Bit 4 – TXQEN Enable Transmit Queue bit⁽¹⁾

Value	Description
1	Enables Transmit Message Queue (TXQ) and reserves space in RAM
0	Does not reserve space in RAM for TXQ

Bit 3 – STEF Store in Transmit Event FIFO bit⁽¹⁾

Value	Description
1	Saves transmitted messages in TEF
0	Does not save transmitted messages in TEF

Bit 2 – SERRLOM Transition to Listen Only Mode on System Error bit⁽¹⁾

Value	Description
1	Transitions to Listen Only mode
0	Transitions to Restricted Operation mode

Bit 1 – ESIGM Transmit ESI in Gateway Mode bit⁽¹⁾

Value	Description
1	ESI is transmitted as recessive when ESI of the message is high or CAN controller is error passive
0	ESI reflects error status of CAN controller

Bit 0 – RTXAT Restrict Retransmission Attempts bit⁽¹⁾

Value	Description
1	Restricted retransmission attempts, uses TXAT[1:0] bits (C1TXQCONH[6:5])
0	Unlimited number of retransmission attempts, TXAT[1:0] bits will be ignored

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Controller Area Network Flexible Data-Rate (...)

12.2.15 CAN Nominal Bit Time Configuration Register Low

Name: C1NBTCFGL⁽¹⁾

Offset: 0x6C4

Note:

- These bits can only be modified in Configuration mode (OPMOD[2:0] = 100).

Bit	15	14	13	12	11	10	9	8
	TSEG2[6:0]							
Access		R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset		0	0	0	1	1	1	1
Bit	7	6	5	4	3	2	1	0
	SJW[6:0]							
Access		R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset		0	0	0	1	1	1	1

Bits 14:8 – TSEG2[6:0] Time Segment 2 bits (Phase Segment 2)

Value	Description
111 1111	Length is 128 x T _Q
. . .	
000 0000	Length is 1 x T _Q

Bits 6:0 – SJW[6:0] Synchronization Jump Width bits

Value	Description
111 1111	Length is 128 x T _Q
. . .	
000 0000	Length is 1 x T _Q

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Controller Area Network Flexible Data-Rate (...)

12.2.16 CAN Nominal Bit Time Configuration Register High

Name: C1NBTCFGH⁽¹⁾
Offset: 0x6C6

Note:

1. This register can only be modified in Configuration mode (OPMOD[2:0] = 100).

Bit	15	14	13	12	11	10	9	8
	BRP[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	TSEG1[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	1	1	1	1	0	0

Bits 15:8 – BRP[7:0] Baud Rate Prescaler bits

Value	Description
1111 1111	$T_Q = 256/F_{SYS}$
. . .	
0000 0000	$T_Q = 1/F_{SYS}$

Bits 7:0 – TSEG1[7:0] Time Segment 1 bits (Propagation Segment + Phase Segment 1)

Value	Description
1111 1111	Length is $256 \times T_Q$
. . .	
0000 0000	Length is $1 \times T_Q$

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Controller Area Network Flexible Data-Rate (...)

12.2.17 CAN Data Bit Time Configuration Register Low

Name: C1DBTCFGL⁽¹⁾

Offset: 0x6C8

Note:

1. This register can only be modified in Configuration mode (OPMOD[2:0] = 100).

Bit	15	14	13	12	11	10	9	8
					TSEG2[3:0]			
Access					R/W	R/W	R/W	R/W
Reset					0	0	1	1

Bit	7	6	5	4	3	2	1	0
					SJW[3:0]			
Access					R/W	R/W	R/W	R/W
Reset					0	0	1	1

Bits 11:8 – TSEG2[3:0] Time Segment 2 bits (Phase Segment 2)

Value	Description
1111	Length is 16 x T _Q
. . .	
0000	Length is 1 x T _Q

Bits 3:0 – SJW[3:0] Synchronization Jump Width bits

Value	Description
1111	Length is 16 x T _Q
. . .	
0000	Length is 1 x T _Q

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Controller Area Network Flexible Data-Rate (...)

12.2.18 CAN Data Bit Time Configuration Register High

Name: C1DBTCFGH⁽¹⁾

Offset: 0x6CA

Note:

1. This register can only be modified in Configuration mode (OPMOD[2:0] = 100).

Bit	15	14	13	12	11	10	9	8
	BRP[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	TSEG1[4:0]							
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	1	1	1	0

Bits 15:8 – BRP[7:0] Baud Rate Prescaler bits

Value	Description
1111 1111	$T_Q = 256/F_{SYS}$
. . .	
0000 0000	$T_Q = 1/F_{SYS}$

Bits 4:0 – TSEG1[4:0] Time Segment 1 bits (Propagation Segment + Phase Segment 1)

Value	Description
1 1111	Length is 32 x T_Q
. . .	
0 0000	Length is 1 x T_Q

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Controller Area Network Flexible Data-Rate (...)

12.2.19 CAN Transmitter Delay Compensation Register Low

Name: C1TDCL⁽¹⁾

Offset: 0x6CC

Note:

1. This register can only be modified in Configuration mode (OPMOD[2:0] = 100).

Bit	15	14	13	12	11	10	9	8
	TDCO[6:0]							
Access		R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset		0	0	1	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	TDCV[5:0]							
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 14:8 – TDCO[6:0] Transmitter Delay Compensation Offset bits (Secondary Sample Point (SSP))

Value	Description
111 1111	-64 x T _{CY}
. . .	
011 1111	63 x T _{CY}
. . .	
000 0000	0 x T _{CY}

Bits 5:0 – TDCV[5:0] Transmitter Delay Compensation Value bits (Secondary Sample Point (SSP))

Value	Description
11 1111	F _P
. . .	
00 0000	0 x F _P

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Controller Area Network Flexible Data-Rate (...)

12.2.20 CAN Transmitter Delay Compensation Register High

Name: C1TDCH⁽¹⁾

Offset: 0x6CE

Note:

1. This register can only be modified in Configuration mode (OPMOD[2:0] = 100).

Bit	15	14	13	12	11	10	9	8
							EDGFLTEN	SID11EN
Access							R/W	R/W
Reset							0	0

Bit	7	6	5	4	3	2	1	0
							TDCMOD[1:0]	
Access							R/W	R/W
Reset							1	0

Bit 9 – EDGFLTEN Enable Edge Filtering During Bus Integration State bit

Value	Description
1	Edge filtering is enabled according to ISO11898-1:2015
0	Edge filtering is disabled

Bit 8 – SID11EN Enable 12-Bit SID in CAN FD Base Format Messages bit

Value	Description
1	RRS is used as SID11 in CAN FD base format messages: SID[11:0] = {SID[10:0], SID11}
0	Does not use RRS; SID[10:0]

Bits 1:0 – TDCMOD[1:0] Transmitter Delay Compensation Mode bits (Secondary Sample Point (SSP))

Value	Description
10–11	Auto: Measures delay and adds TSEG1[4:0] (C1DBTCFGH[4:0]), adds TDCO[6:0]
01	Manual: Does not measure, uses TDCV[5:0] + TDCO[6:0] from register
00	Disabled

12.2.21 CAN Time Base Counter Register Low

Name: C1TBCL^(1,2)
Offset: 0x6D0

Notes:

1. The Time Base Counter (TBC) will be stopped and reset when TBCEN = 0 to save power.
2. The TBC prescaler count will be reset on any write to C1TBCH/L (TBCPREx will be unaffected).

Bit	15	14	13	12	11	10	9	8
	TBC[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	TBC[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – TBC[15:0] CAN Time Base Counter bits

This is a free-running timer that increments every TBCPREx clock when TBCEN is set.

12.2.22 CAN Time Base Counter Register High

Name: C1TBCH^(1,2)
Offset: 0x6D2

Notes:

1. The Time Base Counter (TBC) will be stopped and reset when TBCEN = 0 to save power.
2. The TBC prescaler count will be reset on any write to C1TBCH/L (TBCPREx will be unaffected).

Bit	15	14	13	12	11	10	9	8
	TBC[31:24]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	TBC[23:16]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – TBC[31:16] CAN Time Base Counter bits

This is a free-running timer that increments every TBCPREx clock when TBCEN is set.

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12.2.23 CAN Timestamp Control Register Low

Name: C1TSCONL
Offset: 0x6D4

Bit	15	14	13	12	11	10	9	8
							TBCPRE[9:8]	
Access							R/W	R/W
Reset							0	0

Bit	7	6	5	4	3	2	1	0
	TBCPRE[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 9:0 – TBCPRE[9:0] CAN Time Base Counter Prescaler bits

Value	Description
1023	TBC increments every 1024 clocks
. . .	
0	TBC increments every 1 clock

12.2.24 CAN Timestamp Control Register High

Name: C1TSCONH
Offset: 0x6D6

Bit	15	14	13	12	11	10	9	8
Access								
Reset								

Bit	7	6	5	4	3	2	1	0
						TSRES	TSEOF	TBCEN
Access						R/W	R/W	R/W
Reset						0	0	0

Bit 2 – TSRES Timestamp Reset bit (CAN FD frames only)

Value	Description
1	At sample point of the bit following the FDF bit
0	At sample point of Start-of-Frame (SOF)

Bit 1 – TSEOF Timestamp End-of-Frame (EOF) bit

Value	Description
1	Timestamp when frame is taken valid (11898-1 10.7): - RX no error until last, but one bit of EOF - TX no error until the end of EOF
0	Timestamp at “beginning” of frame: - Classical Frame: At sample point of SOF - FD Frame: see TSRES bit

Bit 0 – TBCEN Time Base Counter Enable bit

Value	Description
1	Enables TBC
0	Stops and resets TBC

12.2.25 CAN Interrupt Code Register Low

Name: C1VECL
Offset: 0x6D8

Bit	15	14	13	12	11	10	9	8
						FILHIT[4:0]		
Access				R	R	R	R	R
Reset				0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
					ICODE[6:0]			
Access		R	R	R	R	R	R	R
Reset		1	0	0	0	0	0	0

Bits 12:8 – FILHIT[4:0] Filter Hit Number bits

Value	Description
01111	Filter 15
01110	Filter 14
. . .	
00001	Filter 1
00000	Filter 0

Bits 6:0 – ICODE[6:0] Interrupt Flag Code bits

Value	Description
1001011-1111111	Reserved
1001010	Transmit attempt interrupt (any bit in C1TXATIF is set)
1001001	Transmit event FIFO interrupt (any bit in C1TEFSTA is set)
1001000	Invalid message occurred (IVMIF/IE)
1000111	CAN module mode change occurred (MODIF/IE)
1000110	CAN timer overflow (TBCIF/IE)
1000101	RX/TX MAB overflow/underflow (RX: Message received before previous message was saved to memory; TX: Can't feed TX MAB fast enough to transmit consistent data)
1000100	Address error interrupt (illegal FIFO address presented to system)
1000011	Receive FIFO overflow interrupt (any bit in C1RXOVIF is set)
1000010	Wake-up interrupt (WAKIF/WAKIE)
1000001	Error interrupt (CERRIF/IE)
1000000	No interrupt
0001000-0111111	Reserved
0000111	FIFO 7 interrupt (TFIF7 or RFIF7 is set)
. . .	
0000001	FIFO 1 interrupt (TFIF1 or RFIF1 is set)
0000000	FIFO 0 interrupt (TFIF0 is set)

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12.2.26 CAN Interrupt Code Register High

Name: C1VECH
Offset: 0x6DA

Bit	15	14	13	12	11	10	9	8
	RXCODE[6:0]							
Access		R	R	R	R	R	R	R
Reset		1	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	TXCODE[6:0]							
Access		R	R	R	R	R	R	R
Reset		1	0	0	0	0	0	0

Bits 14:8 – RXCODE[6:0] Receive Interrupt Flag Code bits

Value	Description
1000001-1111111	Reserved
1000000	No interrupt
0001000-0111111	Reserved
0000111	FIFO 7 interrupt (RFIF7 is set)
. . .	
0000010	FIFO 2 interrupt (RFIF2 is set)
0000001	FIFO 1 interrupt (RFIF1 is set)
0000000	Reserved; FIFO 0 cannot receive

Bits 6:0 – TXCODE[6:0] Transmit Interrupt Flag Code bits

Value	Description
1000001-1111111	Reserved
1000000	No interrupt
0001000-0111111	Reserved
0000111	FIFO 7 interrupt (TFIF7 is set)
. . .	
0000001	FIFO 1 interrupt (TFIF1 is set)
0000000	FIFO 0 interrupt (TFIF0 is set)

12.2.27 CAN Interrupt Register Low

Name: C1INTL
Offset: 0x6DC

Note:

1. C1INTL: Flags are set by hardware and cleared by application.

Legend: C = Clearable bit; HS = Hardware Settable bit

Bit	15	14	13	12	11	10	9	8
	IVMIF	WAKIF	CERRIF	SERRIF	RXOVIF	TXATIF		
Access	HS/C	HS/C	HS/C	HS/C	R	R		
Reset	0	0	0	0	0	0		

Bit	7	6	5	4	3	2	1	0
				TEFIF	MODIF	TBCIF	RXIF	TXIF
Access				R	HS/C	HS/C	R	R
Reset				0	0	0	0	0

Bit 15 – IVMIF Invalid Message Interrupt Flag bit⁽¹⁾

Value	Description
1	Invalid message interrupt occurred
0	No invalid message interrupt

Bit 14 – WAKIF Bus Wake-up Activity Interrupt Flag bit⁽¹⁾

Value	Description
1	Wake-up activity interrupt occurred
0	No wake-up activity interrupt

Bit 13 – CERRIF CAN Bus Error Interrupt Flag bit⁽¹⁾

Value	Description
1	CAN bus error interrupt occurred
0	No CAN bus error interrupt

Bit 12 – SERRIF System Error Interrupt Flag bit⁽¹⁾

Value	Description
1	System error interrupt occurred
0	No system error interrupt

Bit 11 – RXOVIF Receive Buffer Overflow Interrupt Flag bit

Value	Description
1	Receive buffer overflow interrupt occurred
0	No receive buffer overflow interrupt

Bit 10 – TXATIF Transmit Attempt Interrupt Flag bit

Value	Description
1	Transmit attempt interrupt occurred
0	No transmit attempt interrupt

Bit 4 – TEFIF Transmit Event FIFO Interrupt Flag bit

Value	Description
1	Transmit event FIFO interrupt occurred
0	No transmit event FIFO interrupt

Bit 3 – MODIF CAN Mode Change Interrupt Flag bit⁽¹⁾

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Controller Area Network Flexible Data-Rate (...)

Value	Description
1	CAN module mode change occurred (OPMOD[2:0] have changed to reflect REQOP[2:0])
0	No mode change occurred

Bit 2 – TBCIF CAN Timer Overflow Interrupt Flag bit⁽¹⁾

Value	Description
1	TBC has overflowed
0	TBC has not overflowed

Bit 1 – RXIF Receive Object Interrupt Flag bit

Value	Description
1	Receive object interrupt is pending
0	No receive object interrupts are pending

Bit 0 – TXIF Transmit Object Interrupt Flag bit

Value	Description
1	Transmit object interrupt is pending
0	No transmit object interrupts are pending

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Controller Area Network Flexible Data-Rate (...)

12.2.28 CAN Interrupt Register High

Name: C1INTH
Offset: 0x6DE

Bit	15	14	13	12	11	10	9	8
	IVMIE	WAKIE	CERRIE	SERRIE	RXOVIE	TXATIE		
Access	R/W	R/W	R/W	R/W	R/W	R/W		
Reset	0	0	0	0	0	0		

Bit	7	6	5	4	3	2	1	0
				TEFIE	MODIE	TBCIE	RXIE	TXIE
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0

Bit 15 – IVMIE Invalid Message Interrupt Enable bit

Value	Description
1	Invalid message interrupt is enabled
0	Invalid message interrupt is disabled

Bit 14 – WAKIE Bus Wake-up Activity Interrupt Enable bit

Value	Description
1	Wake-up activity interrupt is enabled
0	Wake-up Activity Interrupt is disabled

Bit 13 – CERRIE CAN Bus Error Interrupt Enable bit

Value	Description
1	CAN bus error interrupt is enabled
0	CAN bus error interrupt is disabled

Bit 12 – SERRIE System Error Interrupt Enable bit

Value	Description
1	System error interrupt is enabled
0	System error interrupt is disabled

Bit 11 – RXOVIE Receive Buffer Overflow Interrupt Enable bit

Value	Description
1	Receive buffer overflow interrupt is enabled
0	Receive buffer overflow interrupt is disabled

Bit 10 – TXATIE Transmit Attempt Interrupt Enable bit

Value	Description
1	Transmit attempt interrupt is enabled
0	Transmit attempt interrupt is disabled

Bit 4 – TEFIE Transmit Event FIFO Interrupt Enable bit

Value	Description
1	Transmit event FIFO interrupt is enabled
0	Transmit event FIFO interrupt is disabled

Bit 3 – MODIE Mode Change Interrupt Enable bit

Value	Description
1	Mode change interrupt is enabled
0	Mode change interrupt is disabled

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Controller Area Network Flexible Data-Rate (...)

Bit 2 – TBCIE CAN Timer Interrupt Enable bit

Value	Description
1	CAN timer interrupt is enabled
0	CAN timer interrupt is disabled

Bit 1 – RXIE Receive Object Interrupt Enable bit

Value	Description
1	Receive object interrupt is enabled
0	Receive object interrupt is disabled

Bit 0 – TXIE Transmit Object Interrupt Enable bit

Value	Description
1	Transmit object interrupt is enabled
0	Transmit object interrupt is disabled

12.2.29 CAN Receive Interrupt Status Register Low

Name: C1RXIFL⁽¹⁾

Offset: 0x6E0

Note:

1. C1RXIFL: FIFO: RFIFx = 'or' of enabled RX FIFO flags (flags need to be cleared in the FIFO register).

Bit	15	14	13	12	11	10	9	8
	RFIF[15:8]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	RFIF[7:1]							
Access	R	R	R	R	R	R	R	
Reset	0	0	0	0	0	0	0	

Bits 15:8 – RFIF[15:8] Unimplemented

Bits 7:1 – RFIF[7:1] Receive FIFO Interrupt Pending bits

Value	Description
1	One or more enabled receive FIFO interrupts are pending
0	No enabled receive FIFO interrupts are pending

12.2.30 CAN Receive Interrupt Status Register High

Name: C1RXIFH⁽¹⁾
Offset: 0x6E2

Note:

1. C1RXIFH: FIFO: RFIFx = 'or' of enabled RX FIFO flags (flags need to be cleared in the FIFO register).

Bit	15	14	13	12	11	10	9	8
	RFIF[31:24]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	RFIF[23:16]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – RFIF[31:16] Unimplemented

12.2.31 CAN Transmit Interrupt Status Register Low

Name: C1TXIFL⁽¹⁾
Offset: 0x6E4

Notes:

1. C1TXIFL: FIFO: TFIFx = 'or' of the enabled TX FIFO flags (flags need to be cleared in the FIFO register).
2. TFIF0 is for the Transmit Queue.

Bit	15	14	13	12	11	10	9	8
	TFIF[15:8]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	TFIF[7:0]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – TFIF[15:8] Unimplemented

Bits 7:0 – TFIF[7:0] Transmit FIFO/TXQ Attempt Interrupt Pending bits⁽²⁾

Value	Description
1	One or more enabled transmit FIFO/TXQ interrupts are pending
0	No enabled transmit FIFO/TXQ interrupts are pending

12.2.32 CAN Transmit Interrupt Status Register High

Name: C1TXIFH⁽¹⁾
Offset: 0x6E6

Note:

1. C1TXIFH: FIFO: TFIFx = 'or' of the enabled TX FIFO flags (flags need to be cleared in the FIFO register).

Bit	15	14	13	12	11	10	9	8
	TFIF[31:24]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	TFIF[23:16]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – TFIF[31:16] Unimplemented

12.2.33 CAN Receive Overflow Interrupt Status Register Low

Name: C1RXOVIFL⁽¹⁾

Offset: 0x6E8

Note:

1. C1RXOVIFL: FIFO: RFOVIFx (flag needs to be cleared in the FIFO register).

Bit	15	14	13	12	11	10	9	8
	RFOVIF[15:8]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	RFOVIF[7:1]							
Access	R	R	R	R	R	R	R	
Reset	0	0	0	0	0	0	0	

Bits 15:8 – RFOVIF[15:8] Unimplemented

Bits 7:1 – RFOVIF[7:1] Receive FIFO Overflow Interrupt Pending bits

Value	Description
1	Interrupt is pending
0	Interrupt is not pending

12.2.34 CAN Receive Overflow Interrupt Status Register High

Name: C1RXOVIFH⁽¹⁾
Offset: 0x6EA

Note:

1. C1RXOVIFH: FIFO: RFOVIFx (flag needs to be cleared in the FIFO register).

Bit	15	14	13	12	11	10	9	8
	RFOVIF[31:24]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	RFOVIF[23:16]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – RFOVIF[31:16] Unimplemented

12.2.35 CAN Transmit Attempt Interrupt Status Register Low

Name: C1TXATIFL⁽¹⁾
Offset: 0x6EC

Notes:

1. C1TXATIFL: FIFO: TFATIFx (flag needs to be cleared in the FIFO register).
2. TFATIF0 is for the Transmit Queue.

Bit	15	14	13	12	11	10	9	8
	TFATIF[15:8]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	TFATIF[7:0]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – TFATIF[15:8] Unimplemented

Bits 7:0 – TFATIF[7:0] Transmit FIFO/TXQ Attempt Interrupt Pending bits⁽²⁾

Value	Description
1	Interrupt is pending
0	Interrupt is not pending

12.2.36 CAN Transmit Attempt Interrupt Status Register High

Name: C1TXATIFH⁽¹⁾

Offset: 0x6EE

Note:

1. C1TXATIFH: FIFO: TFATIFx (flag needs to be cleared in the FIFO register).

Bit	15	14	13	12	11	10	9	8
	TFATIF[31:24]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	TFATIF[23:16]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – TFATIF[31:16] Unimplemented

12.2.37 CAN Transmit Request Register Low

Name: C1TXREQL
Offset: 0x6F0

Legend: S = Settable bit; HC = Hardware Clearable bit

Bit	15	14	13	12	11	10	9	8
	TXREQ[15:8]							
Access	S/HC	S/HC	S/HC	S/HC	S/HC	S/HC	S/HC	S/HC
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	TXREQ[7:1]							TXREQ0
Access	S/HC	S/HC	S/HC	S/HC	S/HC	S/HC	S/HC	S/HC
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – TXREQ[15:8] Unimplemented

Bits 7:1 – TXREQ[7:1] Message Send Request bits

TXEN = 1 (object configured as a transmit object):

Setting this bit to '1' requests sending a message. The bit will automatically clear when the message(s) queued in the object is (are) successfully sent. This bit can NOT be used for aborting a transmission.

TXEN = 0 (object configured as a receive object):

This bit has no effect.

Bit 0 – TXREQ0 Transmit Queue Message Send Request bit

Setting this bit to '1' requests sending a message. The bit will automatically clear when the message(s) queued in the object is (are) successfully sent. This bit can NOT be used for aborting a transmission.

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12.2.38 CAN Transmit Request Register High

Name: C1TXREQH

Offset: 0x6F2

Legend: S = Settable bit; HC = Hardware Clearable bit

Bit	15	14	13	12	11	10	9	8
	TXREQ[31:24]							
Access	S/HC	S/HC	S/HC	S/HC	S/HC	S/HC	S/HC	S/HC
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	TXREQ[23:16]							
Access	S/HC	S/HC	S/HC	S/HC	S/HC	S/HC	S/HC	S/HC
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – TXREQ[31:16] Unimplemented

dsPIC33CK512MPT608 Family

Controller Area Network Flexible Data-Rate (...)

12.2.39 CAN Transmit/Receive Error Count Register Low

Name: C1TRECL
Offset: 0x06F4
Property: [registerAccess]

Bit	15	14	13	12	11	10	9	8
	TERRCNT[7:0]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	RERRCNT[7:0]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – TERRCNT[7:0] Transmit Error Counter bits

Bits 7:0 – RERRCNT[7:0] Receive Error Counter bits

dsPIC33CK512MPT608 Family

Controller Area Network Flexible Data-Rate (...)

12.2.40 CAN Transmit/Receive Error Count Register High

Name: C1TRECH

Offset: 0x06F6

Bit	15	14	13	12	11	10	9	8
Access								
Reset								

Bit	7	6	5	4	3	2	1	0
			TXBO	TXBP	RXBP	TXWARN	RXWARN	EWARN
Access			R	R	R	R	R	R
Reset			1	0	0	0	0	0

Bit 5 – TXBO Transmitter in Bus Off Error State bit (TERRCNT[7:0] > 255)
In Configuration mode, TXBO is set since the module is not on the bus.

Bit 4 – TXBP Transmitter in Bus Passive Error State bit (TERRCNT[7:0] > 127)

Bit 3 – RXBP Receiver in Bus Passive Error State bit (RERRCNT[7:0] > 127)

Bit 2 – TXWARN Transmitter in Warning State bit (128 > TERRCNT[7:0] > 95)

Bit 1 – RXWARN Receiver in Warning State bit (128 > RERRCNT[7:0] > 95)

Bit 0 – EWARN Transmitter or Receiver in Warning State bit

dsPIC33CK512MPT608 Family

Controller Area Network Flexible Data-Rate (...)

12.2.41 CAN Bus Diagnostics Register 0 Low

Name: C1BDIAG0L
Offset: 0x6F8

Bit	15	14	13	12	11	10	9	8
	NTERRCNT[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	NRERRCNT[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – NTERRCNT[7:0] Nominal Bit Rate Transmit Error Counter bits

Bits 7:0 – NRERRCNT[7:0] Nominal Bit Rate Receive Error Counter bits

dsPIC33CK512MPT608 Family

Controller Area Network Flexible Data-Rate (...)

12.2.42 CAN Bus Diagnostics Register 0 High

Name: C1BDIAG0H
Offset: 0x6FA

Bit	15	14	13	12	11	10	9	8
	DTERRCNT[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	DRERRCNT[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – DTERRCNT[7:0] Data Bit Rate Transmit Error Counter bits

Bits 7:0 – DRERRCNT[7:0] Data Bit Rate Receive Error Counter bits

dsPIC33CK512MPT608 Family

Controller Area Network Flexible Data-Rate (...)

12.2.43 CAN Bus Diagnostics Register 1 Low

Name: C1BDIAG1L
Offset: 0x6FC

Bit	15	14	13	12	11	10	9	8
	EFMSGCNT[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	EFMSGCNT[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – EFMSGCNT[15:0] Error-Free Message Counter bits

dsPIC33CK512MPT608 Family

Controller Area Network Flexible Data-Rate (...)

12.2.44 CAN Bus Diagnostics Register 1 High

Name: C1BDIAG1H
Offset: 0x6FE

Bit	15	14	13	12	11	10	9	8
	DLCMM	ESI	DCRCERR	DSTUFERR	DFORMERR		DBIT1ERR	DBIT0ERR
Access	R/W	R/W	R/W	R/W	R/W		R/W	R/W
Reset	0	0	0	0	0		0	0

Bit	7	6	5	4	3	2	1	0
	TXBOERR		NCRCERR	NSTUFERR	NFORMERR	NACKERR	NBIT1ERR	NBIT0ERR
Access	R/W		R/W	R/W	R/W	R/W	R/W	R/W
Reset	0		0	0	0	0	0	0

Bit 15 – DLCMM DLC Mismatch bit

During a transmission or reception, the specified DLC is larger than the PLSIZE[2:0] of the FIFO element.

Bit 14 – ESI ESI Flag of a Received CAN FD Message Set bit

Bit 13 – DCRCERR Same as for nominal bit rate

Bit 12 – DSTUFERR Same as for nominal bit rate

Bit 11 – DFORMERR Same as for nominal bit rate

Bit 9 – DBIT1ERR Same as for nominal bit rate

Bit 8 – DBIT0ERR Same as for nominal bit rate

Bit 7 – TXBOERR Device Went to Bus Off bit (and auto-recovered)

Bit 5 – NCRCERR Received Message with CRC Incorrect Checksum bit

The CRC checksum of a received message was incorrect. The CRC of an incoming message does not match with the CRC calculated from the received data.

Bit 4 – NSTUFERR Received Message with Illegal Sequence bit

More than five equal bits in a sequence have occurred in a part of a received message where this is not allowed.

Bit 3 – NFORMERR Received Frame Fixed Format bit

A fixed format part of a received frame has the wrong format.

Bit 2 – NACKERR Transmitted Message Not Acknowledged bit

Transmitted message was not Acknowledged.

Bit 1 – NBIT1ERR Transmitted Message Recessive Level bit

During the transmission of a message (with the exception of the arbitration field), the device wanted to send a recessive level (bit of logical value '1'), but the monitored bus value was dominant.

Bit 0 – NBIT0ERR Transmitted Message Dominant Level bit

During the transmission of a message (or Acknowledge bit, active error flag or overload flag), the device wanted to send a dominant level (data or identifier bit of logical value '0'), but the monitored bus value was recessive. During bus off recovery, this status is set each time a sequence of 11 recessive bits has been monitored. This enables the CPU to monitor the proceeding of the bus off recovery sequence (indicating the bus is not stuck at dominant or continuously disturbed).

dsPIC33CK512MPT608 Family

Controller Area Network Flexible Data-Rate (...)

12.2.45 CAN Transmit Event FIFO Control Register Low

Name: C1TEFCONL
Offset: 0x700

Note:

1. This bit can only be modified in Configuration mode (OPMOD[2:0] = 100).

Legend: S = Settable bit; HC = Hardware Clearable bit

Bit	15	14	13	12	11	10	9	8
						FRESET		UINC
Access						S/HC		S/HC
Reset						0		0

Bit	7	6	5	4	3	2	1	0
			TEFTSEN		TEFOVIE	TEFFIE	TEFHIE	TEFNEIE
Access			R/W		R/W	R/W	R/W	R/W
Reset			0		0	0	0	0

Bit 10 – FRESET FIFO Reset bit

Value	Description
1	FIFO will be reset when bit is set, cleared by hardware when FIFO is reset; the user should poll whether this bit is clear before taking any action
0	No effect

Bit 8 – UINC Increment Tail bit

Value	Description
1	When this bit is set, the FIFO tail will increment by a single message
0	FIFO tail will not increment

Bit 5 – TEFTSEN Transmit Event FIFO Timestamp Enable bit⁽¹⁾

Value	Description
1	Timestamps elements in TEF
0	Does not timestamp elements in TEF

Bit 3 – TEFOVIE Transmit Event FIFO Overflow Interrupt Enable bit

Value	Description
1	Interrupt is enabled for overflow event
0	Interrupt is disabled for overflow event

Bit 2 – TEFFIE Transmit Event FIFO Full Interrupt Enable bit

Value	Description
1	Interrupt is enabled for FIFO full
0	Interrupt is disabled for FIFO full

Bit 1 – TEFHIE Transmit Event FIFO Half Full Interrupt Enable bit

Value	Description
1	Interrupt is enabled for FIFO half full
0	Interrupt is disabled for FIFO half full

Bit 0 – TEFNEIE Transmit Event FIFO Not Empty Interrupt Enable bit

Value	Description
1	Interrupt is enabled for FIFO not empty
0	Interrupt is disabled for FIFO not empty

dsPIC33CK512MPT608 Family

Controller Area Network Flexible Data-Rate (...)

12.2.46 CAN Transmit Event FIFO Control Register High

Name: C1TEFCONH

Offset: 0x702

Note:

1. These bits can only be modified in Configuration mode (OPMOD[2:0] = 100).

Bit	15	14	13	12	11	10	9	8
				FSIZE[4:0]				
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Access								
Reset								

Bits 12:8 – FSIZE[4:0] FIFO Size bits⁽¹⁾

Value	Description
11111	FIFO is 32 messages deep
. . .	
00010	FIFO is 3 messages deep
00001	FIFO is 2 messages deep
00000	FIFO is 1 message deep

12.2.47 CAN Transmit Event FIFO Status Register

Name: C1TEFSTA
Offset: 0x704

Note:

- These bits are read-only and reflect the status of the FIFO.

Legend: HC = Hardware Clearable bit; S = Settable bit can Set by '1'

Bit	15	14	13	12	11	10	9	8
Access								
Reset								

Bit	7	6	5	4	3	2	1	0
					TEFOVIF	TEFFIF	TEFHIF	TEFNEIF
Access					S/HC	R	R	R
Reset					0	0	0	0

Bit 3 – TEFOVIF Transmit Event FIFO Overflow Interrupt Flag bit

Value	Description
1	Overflow event has occurred
0	No overflow event has occurred

Bit 2 – TEFFIF Transmit Event FIFO Full Interrupt Flag bit⁽¹⁾

Value	Description
1	FIFO is full
0	FIFO is not full

Bit 1 – TEFHIF Transmit Event FIFO Half Full Interrupt Flag bit⁽¹⁾

Value	Description
1	FIFO is $\geq$ half full
0	FIFO is $<$ half full

Bit 0 – TEFNEIF Transmit Event FIFO Not Empty Interrupt Flag bit⁽¹⁾

Value	Description
1	FIFO is not empty
0	FIFO is empty

12.2.48 CAN Transmit Event FIFO User Address Register Low

Name: C1TEFUAL⁽¹⁾

Offset: 0x708

Note:

1. This register is not ensured to read correctly in Configuration mode and should only be accessed when the module is not in Configuration mode.

Legend: x = Bit is unknown

Bit	15	14	13	12	11	10	9	8
	TEFUA[15:8]							
Access	R	R	R	R	R	R	R	R
Reset	x	x	x	x	x	x	x	x
Bit	7	6	5	4	3	2	1	0
	TEFUA[7:0]							
Access	R	R	R	R	R	R	R	R
Reset	x	x	x	x	x	x	x	x

Bits 15:0 – TEFUA[15:0] Transmit Event FIFO User Address bits

A read of this register will return the address where the next event is to be read (FIFO tail).

12.2.49 CAN Transmit Event FIFO User Address Register High

Name: C1TEFUAH⁽¹⁾

Offset: 0x70A

Note:

1. This register is not ensured to read correctly in Configuration mode and should only be accessed when the module is not in Configuration mode.

Legend: x = Bit is unknown

Bit	15	14	13	12	11	10	9	8
	TEFUA[15:8]							
Access	R	R	R	R	R	R	R	R
Reset	x	x	x	x	x	x	x	x
Bit	7	6	5	4	3	2	1	0
	TEFUA[7:0]							
Access	R	R	R	R	R	R	R	R
Reset	x	x	x	x	x	x	x	x

Bits 15:0 – TEFUA[15:0] Transmit Event FIFO User Address bits

A read of this register will return the address where the next event is to be read (FIFO tail).

dsPIC33CK512MPT608 Family

Controller Area Network Flexible Data-Rate (...)

12.2.50 CAN Message Memory Base Address Register Low

Name: C1FIFOBAL
Offset: 0x70C

Bit	15	14	13	12	11	10	9	8
	FIFOBAL[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	FIFOBAL[7:0]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – FIFOBAL[15:8] Message Memory Base Address bits
Defines the base address for the transmit event FIFO followed by the message objects.

Bits 7:0 – FIFOBAL[7:0] Message Memory Base Address bits

dsPIC33CK512MPT608 Family

Controller Area Network Flexible Data-Rate (...)

12.2.51 CAN Message Memory Base Address Register High

Name: C1FIFOBABH
Offset: 0x70E

Bit	15	14	13	12	11	10	9	8
	FIFOBAB[31:24]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	FIFOBAB[23:18]						FIFOBAB[17:16]	
Access	R/W	R/W	R/W	R/W	R/W	R/W	R	R
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – FIFOBAB[31:24] Message Memory Base Address bits
Defines the base address for the transmit event FIFO followed by the message objects.

Bits 7:2 – FIFOBAB[23:18] Message Memory Base Address bits

Bits 1:0 – FIFOBAB[17:16] Message Memory Base Address bits

12.2.52 CAN Transmit Queue Control Register Low

Name: C1TXQCONL
Offset: 0x710

Legend: HS = Hardware Settable bit; C = Clearable bit

Bit	15	14	13	12	11	10	9	8
						FRESET	TXREQ	UINC
Access						R/W	R/W	R/W
Reset						0	0	0

Bit	7	6	5	4	3	2	1	0
	TXEN			TXATIE		TXQEIE		TXQNIE
Access	R			HS/C		R/W		R/W
Reset	0			0		0		0

Bit 10 – FRESET FIFO Reset bit

Value	Description
1	FIFO will be reset when bit is set, cleared by hardware when FIFO is reset; user should poll whether this bit is clear before taking any action
0	No effect

Bit 9 – TXREQ Message Send Request bit

Value	Description
1	Requests sending a message; the bit will automatically clear when all the messages queued in the TXQ are successfully sent
0	Clearing the bit to '0' while set ('1') will request a message abort

Bit 8 – UINC Increment Head/Tail bit

When this bit is set, the FIFO head will increment by a single message.

Bit 7 – TXEN TX Enable bit

Bit 4 – TXATIE Transmit Attempts Exhausted Interrupt Enable bit

Value	Description
1	Enables interrupt
0	Disables interrupt

Bit 2 – TXQEIE Transmit Queue Empty Interrupt Enable bit

Value	Description
1	Interrupt is enabled for TXQ empty
0	Interrupt is disabled for TXQ empty

Bit 0 – TXQNIE Transmit Queue Not Full Interrupt Enable bit

Value	Description
1	Interrupt is enabled for TXQ not full
0	Interrupt is disabled for TXQ not full

12.2.53 CAN Transmit Queue Control Register High

Name: C1TXQCONH
Offset: 0x712

Note:

- These bits can only be modified in Configuration mode (OPMOD[2:0] = 100).

Bit	15	14	13	12	11	10	9	8
	PLSIZE[2:0]			FSIZE[4:0]				
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
		TXAT[1:0]			TXPRI[4:0]			
Access		R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset		1	1	0	0	0	0	0

Bits 15:13 – PLSIZE[2:0] Payload Size bits⁽¹⁾

Value	Description
111	64 data bytes
110	48 data bytes
101	32 data bytes
100	24 data bytes
011	20 data bytes
010	16 data bytes
001	12 data bytes
000	8 data bytes

Bits 12:8 – FSIZE[4:0] FIFO Size bits⁽¹⁾

Value	Description
11111	FIFO is 32 messages deep
. . .	
00010	FIFO is 3 messages deep
00001	FIFO is 2 messages deep
00000	FIFO is 1 message deep

Bits 6:5 – TXAT[1:0] Retransmission Attempts bits

This feature is enabled when RTXAT (C1CONH[0]) is set.

Value	Description
11	Unlimited number of retransmission attempts
10	Unlimited number of retransmission attempts
01	Three retransmission attempts
00	Disables retransmission attempts

Bits 4:0 – TXPRI[4:0] Message Transmit Priority bits

Value	Description
11111	Highest message priority
. . .	
00000	Lowest message priority

12.2.54 CAN Transmit Queue Status Register

Name: C1TXQSTAL
Offset: 0x714

Notes:

1. The TXQCI[4:0] bits give a zero-indexed value to the message in the TXQ. If the TXQ is four messages deep (FSIZE[4:0] = 3), TXQCIx will take on a value of 0 to 3, depending on the state of the TXQ.
2. This bit is updated when a message completes (or aborts) or when the TXQ is reset.

Legend: HS = Hardware Settable bit; C = Clearable bit

Bit	15	14	13	12	11	10	9	8
						TXQCI[4:0]		
Access				R	R	R	R	R
Reset				0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	TXABT	TXLARB	TXERR	TXATIF		TXQEIF		TXQNIF
Access	R	R	R	HS/C		R		R
Reset	0	0	0	0		1		1

Bits 12:8 – TXQCI[4:0] Transmit Message Queue Index bits⁽¹⁾

A read of this register will return an index to the message that the FIFO will next attempt to transmit.

Bit 7 – TXABT Message Aborted Status bit⁽²⁾

Value	Description
1	Message was aborted
0	Message completed successfully

Bit 6 – TXLARB Message Lost Arbitration Status bit

Value	Description
1	Message lost arbitration while being sent
0	Message did not lose arbitration while being sent

Bit 5 – TXERR Error Detected During Transmission bit

Value	Description
1	A bus error occurred while the message was being sent
0	A bus error did not occur while the message was being sent

Bit 4 – TXATIF Transmit Attempts Exhausted Interrupt Pending bit

Value	Description
1	Interrupt is pending
0	Interrupt is not pending

Bit 2 – TXQEIF Transmit Queue Empty Interrupt Flag bit

Value	Description
1	TXQ is empty
0	TXQ is not empty, at least one message is queued to be transmitted

Bit 0 – TXQNIF Transmit Queue Not Full Interrupt Flag bit

Value	Description
1	TXQ is not full
0	TXQ is full

12.2.55 CAN Transmit Queue User Address Register Low

Name: C1TXQUAL⁽¹⁾

Offset: 0x718

Note:

1. This register is not ensured to read correctly in Configuration mode and should only be accessed when the module is not in Configuration mode.

Legend: x = Bit is unknown

Bit	15	14	13	12	11	10	9	8
	TXQUA[15:8]							
Access	R	R	R	R	R	R	R	R
Reset	x	x	x	x	x	x	x	x
Bit	7	6	5	4	3	2	1	0
	TXQUA[7:0]							
Access	R	R	R	R	R	R	R	R
Reset	x	x	x	x	x	x	x	x

Bits 15:0 – TXQUA[15:0] TXQ User Address bits

A read of this register will return the address where the next message is to be written (TXQ head).

12.2.56 CAN Transmit Queue User Address Register High

Name: C1TXQUAH⁽¹⁾
Offset: 0x71A

Note:

1. This register is not ensured to read correctly in Configuration mode and should only be accessed when the module is not in Configuration mode.

Legend: x = Bit is unknown

Bit	15	14	13	12	11	10	9	8
	TXQUA[31:24]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	TXQUA[23:16]							
Access	R	R	R	R	R	R	R	R
Reset	x	x	x	x	x	x	x	x

Bits 15:0 – TXQUA[31:16] TXQ User Address bits

A read of this register will return the address where the next message is to be written (TXQ head).

dsPIC33CK512MPT608 Family

Controller Area Network Flexible Data-Rate (...)

12.2.57 CAN FIFO Control Register x Low (x = 1 to 7)

Name: C1FIFOCONxL
Offset: 0x71C, 0x728, 0x734, 0x740, 0x74C, 0x758, 0x764

Note:

1. This bit can only be modified in Configuration mode (OPMOD[2:0] = 100).

Legend: S = Settable bit; HC = Hardware Clearable bit

Bit	15	14	13	12	11	10	9	8
						FRESET	TXREQ	UINC
Access						S/HC	R/W/HC	S/HC
Reset						1	0	0

Bit	7	6	5	4	3	2	1	0
	TXEN	RTREN	RXTSEN	TXATIE	RXOVIE	TFERFFIE	TFHRFHIE	TFNRFNIE
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 10 – FRESET FIFO Reset bit

Value	Description
1	FIFO will be reset when bit is set, cleared by hardware when FIFO is reset; user should poll whether this bit is clear before taking any action
0	No effect

Bit 9 – TXREQ Message Send Request bit

TXEN = 0 (FIFO configured as a receive FIFO):

This bit has no effect.

TXEN = 1 (FIFO configured as a transmit FIFO):

Value	Description
1	Requests sending a message; the bit will automatically clear when all the messages queued in the FIFO are successfully sent
0	Clearing the bit to '0' while set ('1') will request a message abort

Bit 8 – UINC Increment Head/Tail bit

TXEN = 1 (FIFO configured as a transmit FIFO):

When this bit is set, the FIFO head will increment by a single message.

TXEN = 0 (FIFO configured as a receive FIFO):

When this bit is set, the FIFO tail will increment by a single message.

Bit 7 – TXEN TX/RX Buffer Selection bit

Value	Description
1	Transmits message object
0	Receives message object

Bit 6 – RTREN Auto-Remote Transmit (RTR) Enable bit

Value	Description
1	When a Remote Transmit is received, TXREQ will be set
0	When a Remote Transmit is received, TXREQ will be unaffected

Bit 5 – RXTSEN Received Message Timestamp Enable bit⁽¹⁾

Value	Description
1	Captures timestamp in received message object in RAM
0	Does not capture timestamp

Bit 4 – TXATIE Transmit Attempts Exhausted Interrupt Enable bit

dsPIC33CK512MPT608 Family

Controller Area Network Flexible Data-Rate (...)

Value	Description
1	Enables interrupt
0	Disables interrupt

Bit 3 – RXOVIE Overflow Interrupt Enable bit

Value	Description
1	Interrupt is enabled for overflow event
0	Interrupt is disabled for overflow event

Bit 2 – TFERFFIE Transmit/Receive FIFO Empty/Full Interrupt Enable bit

TXEN = 1 (FIFO configured as a transmit FIFO):

Transmit FIFO Empty Interrupt Enable.

1 = Interrupt is enabled for FIFO empty

0 = Interrupt is disabled for FIFO empty

TXEN = 0 (FIFO configured as a receive FIFO):

Receive FIFO Full Interrupt Enable.

1 = Interrupt is enabled for FIFO full

0 = Interrupt is disabled for FIFO full

Bit 1 – TFHRFHIE Transmit/Receive FIFO Half Empty/Half Full Interrupt Enable bit

TXEN = 1 (FIFO configured as a transmit FIFO):

Transmit FIFO Half Empty Interrupt Enable.

1 = Interrupt is enabled for FIFO half empty

0 = Interrupt is disabled for FIFO half empty

TXEN = 0 (FIFO configured as a receive FIFO):

Receive FIFO Half Full Interrupt Enable.

1 = Interrupt is enabled for FIFO half full

0 = Interrupt is disabled for FIFO half full

Bit 0 – TFNRFNIE Transmit/Receive FIFO Not Full/Not Empty Interrupt Enable bit

TXEN = 1 (FIFO configured as a transmit FIFO):

Transmit FIFO Not Full Interrupt Enable.

1 = Interrupt is enabled for FIFO not full

0 = Interrupt is disabled for FIFO not full

TXEN = 0 (FIFO configured as a receive FIFO):

Receive FIFO Not Empty Interrupt Enable.

1 = Interrupt is enabled for FIFO not empty

0 = Interrupt is disabled for FIFO not empty

dsPIC33CK512MPT608 Family

Controller Area Network Flexible Data-Rate (...)

12.2.58 CAN FIFO Control Register x High (x = 1 to 7)

Name: C1FIFOCONxH
Offset: 0x71E, 0x72A, 0x736, 0x742, 0x74C, 0x75A, 0x766

Note:

- These bits can only be modified in Configuration mode (OPMOD[2:0] = 100).

Bit	15	14	13	12	11	10	9	8
	PLSIZE[2:0]				FSIZE[4:0]			
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
		TXAT[1:0]			TXPRI[4:0]			
Access		R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset		1	1	0	0	0	0	0

Bits 15:13 – PLSIZE[2:0] Payload Size bits⁽¹⁾

Value	Description
111	64 data bytes
110	48 data bytes
101	32 data bytes
100	24 data bytes
011	20 data bytes
010	16 data bytes
001	12 data bytes
000	8 data bytes

Bits 12:8 – FSIZE[4:0] FIFO Size bits⁽¹⁾

Value	Description
11111	FIFO is 32 messages deep
. . .	
00010	FIFO is 3 messages deep
00001	FIFO is 2 messages deep
00000	FIFO is 1 message deep

Bits 6:5 – TXAT[1:0] Retransmission Attempts bits

This feature is enabled when RTXAT (C1CONH[0]) is set.

Value	Description
11	Unlimited number of retransmission attempts
10	Unlimited number of retransmission attempts
01	Three retransmission attempts
00	Disables retransmission attempts

Bits 4:0 – TXPRI[4:0] Message Transmit Priority bits

Value	Description
11111	Highest message priority
. . .	
00000	Lowest message priority

12.2.59 CAN FIFO Status Register x (x = 1 to 7)

Name: C1FIFOSTA
Offset: 0x720, 0x72C, 0x738, 0x744, 0x750, 0x75C, 0x768

Notes:

1. FIFOCI[4:0] bits give a zero-indexed value to the message in the FIFO. If the FIFO is four messages deep (FSIZE[4:0] = 3), FIFOCIx will take on a value of 0 to 3, depending on the state of the FIFO.
2. These bits are updated when a message completes (or aborts) or when the FIFO is reset.
3. This bit is reset on any read of this register or when the TXQ is reset. The bits are cleared when TXREQ is set or using an SPI write.

Legend: HS = Hardware Settable bit; C = Clearable bit

Bit	15	14	13	12	11	10	9	8
				FIFOCI[4:0]				
Access				R	R	R	R	R
Reset				0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	TXABT	TXLARB	TXERR	TXATIF	RXOVIF	TFERFFIF	TFHRFHIF	TFNRFNIF
Access	R	R	R	HS/C	HS/C	R	R	R
Reset	0	0	0	0	0	0	0	0

Bits 12:8 – FIFOCI[4:0] FIFO Message Index bits⁽¹⁾

TXEN = 1 (FIFO configured as a transmit buffer):

A read of this register will return an index to the message that the FIFO will next attempt to transmit.

TXEN = 0 (FIFO configured as a receive buffer):

A read of this register will return an index to the message that the FIFO will use to save the next message.

Bit 7 – TXABT Message Aborted Status bit⁽³⁾

Value	Description
1	Message was aborted
0	Message completed successfully

Bit 6 – TXLARB Message Lost Arbitration Status bit⁽²⁾

Value	Description
1	Message lost arbitration while being sent
0	Message did not lose arbitration while being sent

Bit 5 – TXERR Error Detected During Transmission bit⁽²⁾

Value	Description
1	A bus error occurred while the message was being sent
0	A bus error did not occur while the message was being sent

Bit 4 – TXATIF Transmit Attempts Exhausted Interrupt Pending bit

TXEN = 0 (FIFO configured as a receive buffer):

Unused, read as '0'.

TXEN = 1 (FIFO configured as a transmit buffer):

Value	Description
1	Interrupt is pending
0	Interrupt is not pending

Bit 3 – RXOVIF Receive FIFO Overflow Interrupt Flag bit

TXEN = 1 (FIFO configured as a transmit buffer):

Unused, read as '0'.

TXEN = 0 (FIFO configured as a receive buffer):

Value	Description
1	Overflow event has occurred
0	No overflow event has occurred

Bit 2 – TFERFFIF Transmit/Receive FIFO Empty/Full Interrupt Flag bit

TXEN = 1 (FIFO configured as a transmit FIFO):

Transmit FIFO Empty Interrupt Flag.

1 = FIFO is empty

0 = FIFO is not empty, at least 1 message is queued to be transmitted

TXEN = 0 (FIFO configured as a receive FIFO):

Receive FIFO Full Interrupt Flag.

1 = FIFO is full

0 = FIFO is not full

Bit 1 – TFHRFHIF Transmit/Receive FIFO Half Empty/Half Full Interrupt Flag bit

TXEN = 1 (FIFO configured as a transmit FIFO):

Transmit FIFO Half Empty Interrupt Flag.

1 = FIFO is ≤ half full

0 = FIFO is > half full

TXEN = 0 (FIFO configured as a receive FIFO):

Receive FIFO Half Full Interrupt Flag.

1 = FIFO is ≥ half full

0 = FIFO is < half full

Bit 0 – TFNRFNIF Transmit/Receive FIFO Not Full/Not Empty Interrupt Flag bit

TXEN = 1 (FIFO configured as a transmit FIFO):

Transmit FIFO Not Full Interrupt Flag.

1 = FIFO is not full

0 = FIFO is full

TXEN = 0 (FIFO configured as a receive FIFO):

Receive FIFO Not Empty Interrupt Flag.

1 = FIFO is not empty, has at least one message

0 = FIFO is empty

12.2.60 CAN FIFO User Address Register x Low (x = 1 to 7)

Name: C1FIFOUA_{xL}⁽¹⁾

Offset: 0x724, 0x730, 0x73C, 0x748, 0x754, 0x760, 0x76C

Note:

1. This register is not ensured to read correctly in Configuration mode and should only be accessed when the module is not in Configuration mode.

Legend: x = Bit is unknown

Bit	15	14	13	12	11	10	9	8
	FIFOUA[15:8]							
Access	R	R	R	R	R	R	R	R
Reset	x	x	x	x	x	x	x	x
Bit	7	6	5	4	3	2	1	0
	FIFOUA[7:0]							
Access	R	R	R	R	R	R	R	R
Reset	x	x	x	x	x	x	x	x

Bits 15:0 – FIFOUA[15:0] FIFO User Address bits

TXEN = 1 (FIFO configured as a transmit buffer):

A read of this register will return the address where the next message is to be written (FIFO head).

TXEN = 0 (FIFO configured as a receive buffer):

A read of this register will return the address where the next message is to be read (FIFO tail).

12.2.61 CAN FIFO User Address Register x High (x = 1 to 7)

Name: C1FIFOUA_{xH}⁽¹⁾
Offset: 0x726, 0x732, 0x73E, 0x74A, 0x756, 0x762, 0x76E

Note:

1. This register is not ensured to read correctly in Configuration mode and should only be accessed when the module is not in Configuration mode.

Legend: x = Bit is unknown

Bit	15	14	13	12	11	10	9	8
	FIFOUA[31:24]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	FIFOUA[23:16]							
Access	R	R	R	R	R	R	R	R
Reset	x	x	x	x	x	x	x	x

Bits 15:0 – FIFOUA[31:16] FIFO User Address bits

TXEN = 1 (FIFO configured as a transmit buffer):

A read of this register will return the address where the next message is to be written (FIFO head).

TXEN = 0 (FIFO configured as a receive buffer):

A read of this register will return the address where the next message is to be read (FIFO tail).

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Controller Area Network Flexible Data-Rate (...)

12.2.62 CAN Filter Control Register x Low (x = 0 to 3; a = 0, 4, 8, 12; b = 1, 5, 9, 13)

Name: C1FLTCONxL
Offset: 0x770, 0x774, 0x778, 0x77C

Bit	15	14	13	12	11	10	9	8
	FLTENb					FbBP[4:0]		
Access	R/W			R/W	R/W	R/W	R/W	R/W
Reset	0			0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	FLTENA					FaBP[4:0]		
Access	R/W			R/W	R/W	R/W	R/W	R/W
Reset	0			0	0	0	0	0

Bit 15 – FLTENb Enable Filter b to Accept Messages bit

Value	Description
1	Filter is enabled
0	Filter is disabled

Bits 12:8 – FbBP[4:0] Pointer to Object When Filter b Hits bits

Value	Description
11111-11000	Reserved
00111	Message matching filter is stored in Object 7
00110	Message matching filter is stored in Object 6
. . .	
00010	Message matching filter is stored in Object 2
00001	Message matching filter is stored in Object 1
00000	Reserved; Object 0 is the TX Queue and can't receive messages

Bit 7 – FLTENA Enable Filter a to Accept Messages bit

Value	Description
1	Filter is enabled
0	Filter is disabled

Bits 4:0 – FaBP[4:0] Pointer to Object When Filter a Hits bits

Value	Description
11111-11000	Reserved
00111	Message matching filter is stored in Object 7
00110	Message matching filter is stored in Object 6
. . .	
00010	Message matching filter is stored in Object 2
00001	Message matching filter is stored in Object 1
00000	Reserved; Object 0 is the TX Queue and can't receive messages

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Controller Area Network Flexible Data-Rate (...)

12.2.63 CAN Filter Control Register x High (x = 0 to 3; c = 2, 6, 10, 14; d = 3, 7, 11, 15)

Name: C1FLTCONxH
Offset: 0x772, 0x776, 0x77A, 0x77E

Bit	15	14	13	12	11	10	9	8
	FLTEND					FdBP[4:0]		
Access	R/W			R/W	R/W	R/W	R/W	R/W
Reset	0			0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	FLTENC					FcBP[4:0]		
Access	R/W			R/W	R/W	R/W	R/W	R/W
Reset	0			0	0	0	0	0

Bit 15 – FLTEND Enable Filter d to Accept Messages bit

Value	Description
1	Filter is enabled
0	Filter is disabled

Bits 12:8 – FdBP[4:0] Pointer to Object When Filter d Hits bits

Value	Description
11111-11000	Reserved
00111	Message matching filter is stored in Object 7
00110	Message matching filter is stored in Object 6
. . .	
00010	Message matching filter is stored in Object 2
00001	Message matching filter is stored in Object 1
00000	Reserved; Object 0 is the TX Queue and can't receive messages

Bit 7 – FLTENC Enable Filter c to Accept Messages bit

Value	Description
1	Filter is enabled
0	Filter is disabled

Bits 4:0 – FcBP[4:0] Pointer to Object When Filter c Hits bits

Value	Description
11111-11000	Reserved
00111	Message matching filter is stored in Object 7
00110	Message matching filter is stored in Object 6
. . .	
00010	Message matching filter is stored in Object 2
00001	Message matching filter is stored in Object 1
00000	Reserved; Object 0 is the TX Queue and can't receive messages

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Controller Area Network Flexible Data-Rate (...)

12.2.64 CAN Filter Object Register x Low (x = 0 to 15)

Name: C1FLTOBJxL

Offset: 0x780, 0x788, 0x790, 0x798, 0x7A0, 0x7A8, 0x7B0, 0x7B8, 0x7C0, 0x7C8, 0x7D0, 0x7D8, 0x7E0, 0x7E8, 0x7F0, 0x7F8

Bit	15	14	13	12	11	10	9	8
	EID[4:0]					SID[10:8]		
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	SID[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:11 – EID[4:0] Extended Identifier Filter bits

In DeviceNet™ mode, these are the filter bits for the first two data bytes.

Bits 10:0 – SID[10:0] Standard Identifier Filter bits

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Controller Area Network Flexible Data-Rate (...)

12.2.65 CAN Filter Object Register x High (x = 0 to 15)

Name: C1FLTOBJxH

Offset: 0x782, 0x78A, 0x792, 0x79A, 0x7A2, 0x7AA, 0x7B2, 0x7BA, 0x7C2, 0x7CA, 0x7D2, 0x7DA, 0x7E2, 0x7EA, 0x7F2, 0x7FA

Bit	15	14	13	12	11	10	9	8
		EXIDE	SID11	EID[17:13]				
Access		R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset		0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	EID[12:5]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 14 – EXIDE Extended Identifier Enable bit

If MIDE = 1:

Value	Description
1	Matches only messages with Extended Identifier addresses
0	Matches only messages with Standard Identifier addresses

Bit 13 – SID11 Standard Identifier Filter bit

Bits 12:0 – EID[17:5] Extended Identifier Filter bits

In DeviceNet™ mode, these are the filter bits for the first two data bytes.

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Controller Area Network Flexible Data-Rate (...)

12.2.66 CAN Mask Register x Low (x = 0 to 15)

Name: C1MASKxL
Offset: 0x784, 0x78C, 0x794, 0x79C, 0x7A4, 0x7AC, 0x7B4, 0x7BC, 0x7C4, 0x7CC, 0x7D4, 0x7DC, 0x7E4, 0x7EC, 0x7F4, 0x7FC

Bit	15	14	13	12	11	10	9	8
	MEID[4:0]					MSID[10:8]		
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	MSID[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:11 – MEID[4:0] Extended Identifier Mask bits
 In DeviceNet™ mode, these are the mask bits for the first two data bytes.

Bits 10:0 – MSID[10:0] Standard Identifier Mask bits

12.2.67 CAN Mask Register x High (x = 0 to 15)

Name: C1MASKxH
Offset: 0x786, 0x78E, 0x796, 0x79E, 0x7A6, 0x7AE, 0x7B6, 0x7BE, 0x7C6, 0x7CE, 0x7D6, 0x7DE, 0x7E6, 0x7EE, 0x7F6, 0x7FE

Bit	15	14	13	12	11	10	9	8
		MIDE	MSID11	MEID[17:13]				
Access		R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset		0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	MEID[12:5]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 14 – MIDE Identifier Receive Mode bit

Value	Description
1	Matches only message types (standard or extended address) that correspond to the EXIDE bit in the filter
0	Matches either standard or extended address message if filters match (i.e., if (Filter SID) = (Message SID) or if (Filter SID/EID) = (Message SID/EID))

Bit 13 – MSID11 Standard Identifier Mask bit

Bits 12:0 – MEID[17:5] Extended Identifier Mask bits

In DeviceNet™ mode, these are the mask bits for the first two data bytes.

13. High-Resolution PWM with Fine Edge Placement

Note: This data sheet summarizes the features of the dsPIC33CK512MPT608 family of devices. It is not intended to be a comprehensive reference source. To complement the information in this data sheet, refer to “**High-Resolution PWM with Fine Edge Placement**” (www.microchip.com/DS70005320) in the “*dsPIC33/PIC24 Family Reference Manual*”.

The High-Speed PWM (HSPWM) module is a Pulse-Width Modulated (PWM) module to support both motor control and power supply applications. This flexible module provides features to support many types of Motor Control (MC) and Power Control (PC) applications, including:

- AC-to-DC Converters
- DC-to-DC Converters
- AC and DC Motors: BLDC, PMSM, ACIM, SRM, etc.
- Inverters
- Battery Chargers
- Digital Lighting
- Power Factor Correction (PFC)

Table 13-1. PWM Output Availability

Package Type	Total PGx Instances	Dedicated Outputs	Dedicated + PPS Outputs
100-pin	6	6 pairs	6 pairs

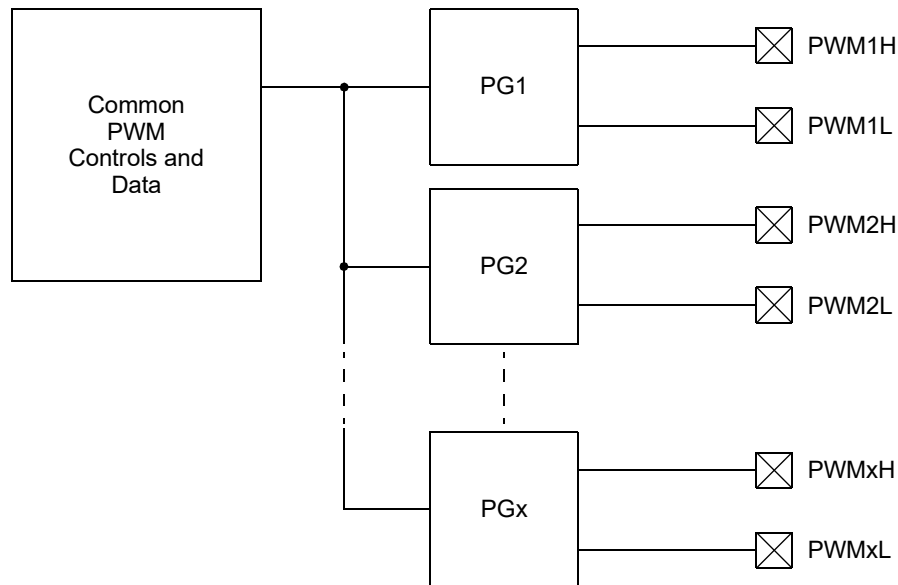
13.1 Features

- Operating modes:
 - Independent Edge mode
 - Variable Phase PWM mode
 - Center-Aligned mode
 - Double-Update Center-Aligned mode
 - Dual Edge Center-Aligned mode
 - Dual PWM mode
- Output modes:
 - Complementary
 - Independent
 - Push-Pull
- Dead-Time Generator
- Leading-Edge Blanking (LEB)
- Output Override for Fault Handling
- Flexible Period/Duty Cycle Updating Options
- Programmable Control Inputs (PCI)
- Advanced Triggering Options
- Six Combinatorial Logic Outputs
- Six PWM Event Outputs

13.2 Architecture Overview

The PWM module consists of a common set of controls and features, and multiple instantiations of PWM Generators (PGs). Each PWM Generator can be independently configured or multiple PWM Generators can be used to achieve complex multiphase systems. PWM Generators can also be used to implement sophisticated triggering, protection and logic functions. A high-level block diagram is shown in [Figure 13-1](#). In addition, this device family has another identical PWM module to accommodate additional PWM Generators.

Figure 13-1. PWM High-Level Block Diagram



13.3 Lock and Write Restrictions

The LOCK bit (PCLKCON[8]) may be set in software to block writes to certain registers. For more information, refer to “**High-Resolution PWM with Fine Edge Placement**” (www.microchip.com/DS70005320) in the “dsPIC33/PIC24 Family Reference Manual.”

The following lock/unlock sequence is required to set or clear the LOCK bit.

1. Write 0x55 to NVMKEY.
2. Write 0xAA to NVMKEY.
3. Clear (or set) the LOCK bit (PCLKCON[8]) as a single operation.

In general, modifications to configuration controls should not be done while the module is running, as indicated by the ON bit (PGxCONL[15]) being set.

13.4 PWM4H/L Output on Peripheral Pin Select

All devices support the capability to output PWM4H and PWM4L signals via Peripheral Pin Select (PPS) on to any “RPn” pin. This feature is intended for lower pin count devices that do not have PWM4H/L on dedicated pins.

Configuration bit, DUPWM (FDEVOP1[12]), provides the option to disable the fixed pin PWM4L/H functions when using the PPS. Clearing the DUPWM bit will disable PWM4 function and allow the pin to be used for another purpose. Leaving the DUPWM set (default) will output PWM4 on both fixed pin and PPS outputs. The output port enable bits, PENH and PENL (PGxIOCONH[3:2]), control output function for both dedicated and PPS pins outputs

If PWM4H/L PPS output functions are used on devices that also have fixed PWM4H/L pins, the output signal will be present on both dedicated and “RPn” pins. The output port enable bits, PENH and PENL (PGxIOCONH[3:2]), control both dedicated and PPS pins together; it is not possible to disable the dedicated pins and use only PPS.

Given the natural priority of the “RPn” functions above that of the PWM, it is possible to use the PPS output functions on the dedicated PWM4H/L pins, while the PWM4 signals are routed to other pins via PPS. Any of the peripheral outputs listed in Table 9-5, with the exception of ‘Default Port’, can be used. Input functions, including the ports and peripherals listed in 9.7. Virtual Connections, cannot be used through the “RPn” function on dedicated PWM4H/L pins when PWM4 is active.

13.5 PWM Control/Status Registers

There are two categories of Special Function Registers (SFRs) used to control the operation of the PWM module:

- Common, shared by all PWM Generators
- PWM Generator-specific

An 'x' in the register name denotes an instance of a PWM Generator.

A 'y' in the register name denotes an instance of the common function.

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13.6 Control Registers

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
00	PG6FPCIL	15:8	TSYNCDIS	TERM[2:0]			AQPS	AQSS[2:0]		
		7:0	SWTERM	PSYNC	PPS	PSS[4:0]				
0x02 ... 0x02FF	Reserved									
0x0300	PCLKCON	15:8	HRRDY	HRERR						LOCK
		7:0			DIVSEL[1:0]				MCLKSEL[1:0]	
0x0302	FSCL	15:8	FSCL[15:8]							
		7:0	FSCL[7:0]							
0x0304	FSMINPER	15:8	FSMINPER[15:8]							
		7:0	FSMINPER[7:0]							
0x0306	MPHASE	15:8	MPHASE[15:8]							
		7:0	MPHASE[7:0]							
0x0308	MDC	15:8	MDC[15:8]							
		7:0	MDC[7:0]							
0x030A	MPER	15:8	MPER[15:8]							
		7:0	MPER[7:0]							
0x030C	LFSR	15:8		LFSR[14:8]						
		7:0	LFSR[7:0]							
0x030E	CMBTRIGL	15:8								
		7:0	CTA8EN	CTA7EN	CTA6EN	CTA5EN	CTA4EN	CTA3EN	CTA2EN	CTA1EN
0x0310	CMBTRIGH	15:8								
		7:0	CTB8EN	CTB7EN	CTB6EN	CTB5EN	CTB4EN	CTB3EN	CTB2EN	CTB1EN
0x0312	LOGCONA(2)	15:8	PWMS1A[3:0]				PWMS2A[3:0]			
		7:0	S1APOL	S2APOL	PWMLFA[1:0]			PWMLFAD[2:0]		
0x0314	LOGCONB(2)	15:8	PWMS1B[3:0]				PWMS2B[3:0]			
		7:0	S1BPOL	S2BPOL	PWMLFB[1:0]			PWMLFBD[2:0]		
0x0316	LOGCONC(2)	15:8	PWMS1C[3:0]				PWMS2C[3:0]			
		7:0	S1CPOL	S2CPOL	PWMLFC[1:0]			PWMLFCD[2:0]		
0x0318	LOGCOND(2)	15:8	PWMS1D[3:0]				PWMS2D[3:0]			
		7:0	S1DPOL	S2DPOL	PWMLFD[1:0]			PWMLFDD[2:0]		
0x031A	LOGCONE(2)	15:8	PWMS1E[3:0]				PWMS2E[3:0]			
		7:0	S1EPOL	S2EPOL	PWMLFE[1:0]			PWMLFED[2:0]		
0x031C	LOGCONF(2)	15:8	PWMS1F[3:0]				PWMS2F[3:0]			
		7:0	S1FPOL	S2FPOL	PWMLFF[1:0]			PWMLFFD[2:0]		
0x031E	PWMEVTA(2)	15:8	EVTAOEN	EVTAPOL	EVTASTRD	EVTASYNC				
		7:0	EVTASEL[3:0]					EVTAPGS[2:0]		
0x0320	5)	15:8	EVTBOEN	EVTBPOL	EVTBSTRD	EVTBSYNC				
		7:0	EVTBSEL[3:0]					EVTBPGS[2:0]		
0x0322	PWMEVTB(2)	15:8	EVTBOEN	EVTBPOL	EVTBSTRD	EVTBSYNC				
		7:0	EVTBSEL[3:0]					EVTBPGS[2:0]		
0x0324	5)	15:8	EVTDOEN	EVTDPOL	EVTDSTRD	EVTDSYNC				
		7:0	EVTDSEL[3:0]					EVTDPGS[2:0]		
0x0326	PWMEVTC(2)	15:8	EVTCOEN	EVTCPOL	EVTCTSTRD	EVTCSYNC				
		7:0	EVTCSSEL[3:0]					EVTCPGS[2:0]		
0x0328	5)	15:8	EVTFOEN	EVTFPOL	EVTFSTRD	EVTFSYNC				
		7:0	EVTFSEL[3:0]					EVTFPGS[2:0]		
0x032A	PG1CONL	15:8	ON	Reserved				TRGCNT[2:0]		
		7:0	HREN			CLKSEL[1:0]		MODESEL[2:0]		
0x032C	PG1CONH	15:8	MDCSEL	MPERSEL	MPHSEL		MSTEN	UPMOD[2:0]		
		7:0	Reserved	TRGMOD			SOCS[3:0]			
0x032E	PG1STAT	15:8	SEVT	FLTEVT	CLEVT	FFEVT	SACT	FLTACT	CLACT	FFACT
		7:0	TRSET	TRCLR	CAP	UPDATE	UPDREQ	STEER	CAHALF	TRIG
0x0330	PG1IOCONL	15:8	CLMOD	SWAP	OVRENH	OVRENL	OVRDAT[1:0]		OSYNC[1:0]	
		7:0	FLTDAT[1:0]			CLDAT[1:0]		FFDAT[1:0]		DBDAT[1:0]

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High-Resolution PWM with Fine Edge Placement

.....continued										
Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x0332	PG1IOCONH	15:8		CAPSRC[2:0]						DTCMPSEL
		7:0			PMD[1:0]		PENH	PENL	POLH	POLL
0x0334	PG1EVTL	15:8	ADTR1PS[4:0]					ADTR1EN3	ADTR1EN2	ADTR1EN1
		7:0				UPDTRG[1:0]		PGTRGSEL[2:0]		
0x0336	PG1EVTH	15:8	FLTEN	CLIEN	FFIEN	SIEN			IEVTSEL[1:0]	
		7:0	ADTR2EN3	ADTR2EN2	ADTR2EN1	ADTR1OFS[4:0]				
0x0338	PG1FPCIL	15:8	TSYNCDIS	TERM[2:0]			AQPS	AQSS[2:0]		
		7:0	SWTERM	PSYNC	PPS	PSS[4:0]				
0x033A	PG1FPCIH	15:8	BPEN	BPSEL[2:0]				ACP[2:0]		
		7:0	SWPCI	SWPCIM[1:0]		LATMOD	TQPS	TQSS[2:0]		
0x033C	PG1CLPCIL	15:8	TSYNCDIS	TERM[2:0]			AQPS	AQSS[2:0]		
		7:0	SWTERM	PSYNC	PPS	PSS[4:0]				
0x033E	PG1CLPCIH	15:8	BPEN	BPSEL[2:0]				ACP[2:0]		
		7:0	SWPCI	SWPCIM[1:0]		LATMOD	TQPS	TQSS[2:0]		
0x0340	PG1FFPCIL	15:8	TSYNCDIS	TERM[2:0]			AQPS	AQSS[2:0]		
		7:0	SWTERM	PSYNC	PPS	PSS[4:0]				
0x0342	PG1FFPCIH	15:8	BPEN	BPSEL[2:0]				ACP[2:0]		
		7:0	SWPCI	SWPCIM[1:0]		LATMOD	TQPS	TQSS[2:0]		
0x0344	PG1SPCIL	15:8	TSYNCDIS	TERM[2:0]			AQPS	AQSS[2:0]		
		7:0	SWTERM	PSYNC	PPS	PSS[4:0]				
0x0346	PG1SPCIH	15:8	BPEN	BPSEL[2:0]				ACP[2:0]		
		7:0	SWPCI	SWPCIM[1:0]		LATMOD	TQPS	TQSS[2:0]		
0x0348	PG1LEBL	15:8	LEB[15:8]							
		7:0	LEB[7:0]							
0x034A	PG1LEBH	15:8						PWMPC[2:0]		
		7:0					PHR	PHF	PLR	PLF
0x034C	PG1PHASE	15:8	PG1PHASE[15:8]							
		7:0	PG1PHASE[7:0]							
0x034E	PG1DC	15:8	PG1DC[15:8]							
		7:0	PG1DC[7:0]							
0x0350	PG1DCA	15:8								
		7:0	PG1DCA[7:0]							
0x0352	PG1PER	15:8	PG1PER[15:8]							
		7:0	PG1PER[7:0]							
0x0354	PG1TRIGA	15:8	PG1TRIGA[15:8]							
		7:0	PG1TRIGA[7:0]							
0x0356	PG1TRIGB	15:8	PG1TRIGB[15:8]							
		7:0	PG1TRIGB[7:0]							
0x0358	PG1TRIGC	15:8	PG1TRIGC[15:8]							
		7:0	PG1TRIGC[7:0]							
0x035A	PG1DTL	15:8				DTL[13:8]				
		7:0	DTL[7:0]							
0x035C	PG1DTH	15:8				DTH[13:8]				
		7:0	DTH[7:0]							
0x035E	PG1CAP	15:8	PG1CAP[15:8]							
		7:0	PG1CAP[7:0]							
0x0360	PG2CONL	15:8	ON	Reserved				TRGCNT[2:0]		
		7:0	HREN			CLKSEL[1:0]		MODSEL[2:0]		
0x0362	PG2CONH	15:8	MDCSEL	MPERSEL	MPHSEL		MSTEN	UPMOD[2:0]		
		7:0	Reserved	TRGMOD		SOCS[3:0]				
0x0364	PG2STAT	15:8	SEVT	FLTEVT	CLEVT	FFEVT	SACT	FLTACT	CLACT	FFACT
		7:0	TRSET	TRCLR	CAP	UPDATE	UPDREQ	STEER	CAHALF	TRIG
0x0366	PG2IOCONL	15:8	CLMOD	SWAP	OVRENH	OVRENL	OVRDAT[1:0]		OSYNC[1:0]	
		7:0	FLTDAT[1:0]		CLDAT[1:0]		FFDAT[1:0]		DBDAT[1:0]	
0x0368	PG2IOCONH	15:8			CAPSRC[2:0]					DTCMPSEL
		7:0			PMD[1:0]		PENH	PENL	POLH	POLL
0x036A	PG2EVTL	15:8	ADTR1PS[4:0]					ADTR1EN3	ADTR1EN2	ADTR1EN1
		7:0				UPDTRG[1:0]		PGTRGSEL[2:0]		

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High-Resolution PWM with Fine Edge Placement

.....continued										
Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x036C	PG2EVTH	15:8	FLTIE	CLIE	FFIE	SIEN			IEVTSEL[1:0]	
		7:0	ADTR2EN3	ADTR2EN2	ADTR2EN1	ADTR1OFS[4:0]				
0x036E	PG2FPCIL	15:8	TSYNCDIS		TERM[2:0]		AQPS		AQSS[2:0]	
		7:0	SWTERM	PSYNC	PPS	PSS[4:0]				
0x0370	PG2FPCIH	15:8	BPEN	BPSEL[2:0]					ACP[2:0]	
		7:0	SWPCI	SWPCIM[1:0]		LATMOD	TQPS		TQSS[2:0]	
0x0372	PG2CLPCIL	15:8	TSYNCDIS		TERM[2:0]		AQPS		AQSS[2:0]	
		7:0	SWTERM	PSYNC	PPS	PSS[4:0]				
0x0374	PG2CLPCIH	15:8	BPEN	BPSEL[2:0]					ACP[2:0]	
		7:0	SWPCI	SWPCIM[1:0]		LATMOD	TQPS		TQSS[2:0]	
0x0376	PG2FFPCIL	15:8	TSYNCDIS		TERM[2:0]		AQPS		AQSS[2:0]	
		7:0	SWTERM	PSYNC	PPS	PSS[4:0]				
0x0378	PG2FFPCIH	15:8	BPEN	BPSEL[2:0]					ACP[2:0]	
		7:0	SWPCI	SWPCIM[1:0]		LATMOD	TQPS		TQSS[2:0]	
0x037A	PG2SPCIL	15:8	TSYNCDIS		TERM[2:0]		AQPS		AQSS[2:0]	
		7:0	SWTERM	PSYNC	PPS	PSS[4:0]				
0x037C	PG2SPCIH	15:8	BPEN	BPSEL[2:0]					ACP[2:0]	
		7:0	SWPCI	SWPCIM[1:0]		LATMOD	TQPS		TQSS[2:0]	
0x037E	PG2LEBL	15:8	LEB[15:8]							
		7:0	LEB[7:0]							
0x0380	PG2LEBH	15:8						PWMPCI[2:0]		
		7:0					PHR	PHF	PLR	PLF
0x0382	PG2PHASE	15:8	PG2PHASE[15:8]							
		7:0	PG2PHASE[7:0]							
0x0384	PG2DC	15:8	PG2DC[15:8]							
		7:0	PG2DC[7:0]							
0x0386	PG2DCA	15:8								
		7:0	PG2DCA[7:0]							
0x0388	PG2PER	15:8	PG2PER[15:8]							
		7:0	PG2PER[7:0]							
0x038A	PG2TRIGA	15:8	PG2TRIGA[15:8]							
		7:0	PG2TRIGA[7:0]							
0x038C	PG2TRIGB	15:8	PG2TRIGB[15:8]							
		7:0	PG2TRIGB[7:0]							
0x038E	PG2TRIGC	15:8	PG2TRIGC[15:8]							
		7:0	PG2TRIGC[7:0]							
0x0390	PG2DTL	15:8				DTL[13:8]				
		7:0	DTL[7:0]							
0x0392	PG2DTH	15:8				DTH[13:8]				
		7:0	DTH[7:0]							
0x0394	PG2CAP	15:8	PG2CAP[15:8]							
		7:0	PG2CAP[7:0]							
0x0396	PG3CONL	15:8	ON	Reserved				TRGCNT[2:0]		
		7:0	HREN			CLKSEL[1:0]		MODSEL[2:0]		
0x0398	PG3CONH	15:8	MDCSEL	MPERSEL	MPHSEL		MSTEN	UPMOD[2:0]		
		7:0	Reserved	TRGMOD			SOCS[3:0]			
0x039A	PG3STAT	15:8	SEVT	FLTEVT	CLEVT	FFEVT	SACT	FLTACT	CLACT	FFACT
		7:0	TRSET	TRCLR	CAP	UPDATE	UPDREQ	STEER	CAHALF	TRIG
0x039C	PG3IOCONL	15:8	CLMOD	SWAP	OVRENH	OVRENL	OVRDAT[1:0]		OSYNC[1:0]	
		7:0	FLTDAT[1:0]		CLDAT[1:0]		FFDAT[1:0]		DBDAT[1:0]	
0x039E	PG3IOCONH	15:8			CAPSRC[2:0]					DTCMPSEL
		7:0			PMOD[1:0]		PENH	PENL	POLH	POLL
0x03A0	PG3EVTL	15:8	ADTR1PS[4:0]					ADTR1EN3	ADTR1EN2	ADTR1EN1
		7:0				UPDTRG[1:0]		PGTRGSEL[2:0]		
0x03A2	PG3EVTH	15:8	FLTIE	CLIE	FFIE	SIEN			IEVTSEL[1:0]	
		7:0	ADTR2EN3	ADTR2EN2	ADTR2EN1	ADTR1OFS[4:0]				
0x03A4	PG3FPCIL	15:8	TSYNCDIS		TERM[2:0]		AQPS		AQSS[2:0]	
		7:0	SWTERM	PSYNC	PPS	PSS[4:0]				

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High-Resolution PWM with Fine Edge Placement

.....continued

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x03A6	PG3FPCIH	15:8	BPEN	BPSEL[2:0]				ACP[2:0]		
		7:0	SWPCI	SWPCIM[1:0]		LATMOD	TQPS	TQSS[2:0]		
0x03A8	PG3CLPCIL	15:8	TSYNCDIS	TERM[2:0]			AQPS	AQSS[2:0]		
		7:0	SWTERM	PSYNC	PPS	PSS[4:0]				
0x03AA	PG3CLPCIH	15:8	BPEN	BPSEL[2:0]				ACP[2:0]		
		7:0	SWPCI	SWPCIM[1:0]		LATMOD	TQPS	TQSS[2:0]		
0x03AC	PG3FFPCIL	15:8	TSYNCDIS	TERM[2:0]			AQPS	AQSS[2:0]		
		7:0	SWTERM	PSYNC	PPS	PSS[4:0]				
0x03AE	PG3FFPCIH	15:8	BPEN	BPSEL[2:0]				ACP[2:0]		
		7:0	SWPCI	SWPCIM[1:0]		LATMOD	TQPS	TQSS[2:0]		
0x03B0	PG3SPCIL	15:8	TSYNCDIS	TERM[2:0]			AQPS	AQSS[2:0]		
		7:0	SWTERM	PSYNC	PPS	PSS[4:0]				
0x03B2	PG3SPCIH	15:8	BPEN	BPSEL[2:0]				ACP[2:0]		
		7:0	SWPCI	SWPCIM[1:0]		LATMOD	TQPS	TQSS[2:0]		
0x03B4	PG3LEBL	15:8	LEB[15:8]							
		7:0	LEB[7:0]							
0x03B6	PG3LEBH	15:8						PWMPC[2:0]		
		7:0					PHR	PHF	PLR	PLF
0x03B8	PG3PHASE	15:8	PG3PHASE[15:8]							
		7:0	PG3PHASE[7:0]							
0x03BA	PG3DC	15:8	PG3DC[15:8]							
		7:0	PG3DC[7:0]							
0x03BC	PG3DCA	15:8								
		7:0	PG3DCA[7:0]							
0x03BE	PG3PER	15:8	PG3PER[15:8]							
		7:0	PG3PER[7:0]							
0x03C0	PG3TRIGA	15:8	PG3TRIGA[15:8]							
		7:0	PG3TRIGA[7:0]							
0x03C2	PG3TRIGB	15:8	PG3TRIGB[15:8]							
		7:0	PG3TRIGB[7:0]							
0x03C4	PG3TRIGC	15:8	PG3TRIGC[15:8]							
		7:0	PG3TRIGC[7:0]							
0x03C6	PG3DTL	15:8				DTL[13:8]				
		7:0	DTL[7:0]							
0x03C8	PG3DTH	15:8				DTH[13:8]				
		7:0	DTH[7:0]							
0x03CA	PG3CAP	15:8	PG3CAP[15:8]							
		7:0	PG3CAP[7:0]							
0x03CC	PG4CONL	15:8	ON	Reserved				TRGCNT[2:0]		
		7:0	HREN			CLKSEL[1:0]		MODSEL[2:0]		
0x03CE	PG4CONH	15:8	MDCSEL	MPERSEL	MPHSEL		MSTEN	UPMOD[2:0]		
		7:0	Reserved	TRGMOD			SOCS[3:0]			
0x03D0	PG4STAT	15:8	SEVT	FLTEVT	CLEVT	FFEVT	SACT	FLTACT	CLACT	FFACT
		7:0	TRSET	TRCLR	CAP	UPDATE	UPDREQ	STEER	CAHALF	TRIG
0x03D2	PG4IOCONL	15:8	CLMOD	SWAP	OVRENH	OVRENL	OVRDAT[1:0]		OSYNC[1:0]	
		7:0	FLTDAT[1:0]			CLDAT[1:0]		FFDAT[1:0]		DBDAT[1:0]
0x03D4	PG4IOCONH	15:8			CAPSRC[2:0]					DTCMPSEL
		7:0			PMOD[1:0]		PENH	PENL	POLH	POLL
0x03D6	PG4EVTL	15:8	ADTR1PS[4:0]					ADTR1EN3	ADTR1EN2	ADTR1EN1
		7:0			UPDTRG[1:0]			PGTRGSEL[2:0]		
0x03D8	PG4EVTH	15:8	FLTIEH	CLIEH	FFIEH	SIEN			IEVTSEL[1:0]	
		7:0	ADTR2EN3	ADTR2EN2	ADTR2EN1	ADTR1OFS[4:0]				
0x03DA	PG4FPCIL	15:8	TSYNCDIS	TERM[2:0]			AQPS	AQSS[2:0]		
		7:0	SWTERM	PSYNC	PPS	PSS[4:0]				
0x03DC	PG4FPCIH	15:8	BPEN	BPSEL[2:0]				ACP[2:0]		
		7:0	SWPCI	SWPCIM[1:0]		LATMOD	TQPS	TQSS[2:0]		
0x03DE	PG4CLPCIL	15:8	TSYNCDIS	TERM[2:0]			AQPS	AQSS[2:0]		
		7:0	SWTERM	PSYNC	PPS	PSS[4:0]				

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High-Resolution PWM with Fine Edge Placement

.....continued

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x03E0	PG4CLPCIH	15:8	BPEN	BPSEL[2:0]				ACP[2:0]		
		7:0	SWPCI	SWPCIM[1:0]		LATMOD	TQPS	TQSS[2:0]		
0x03E2	PG4FFPCIL	15:8	TSYNCDIS	TERM[2:0]			AQPS	AQSS[2:0]		
		7:0	SWTERM	PSYNC	PPS	PSS[4:0]				
0x03E4	PG4FFPCIH	15:8	BPEN	BPSEL[2:0]				ACP[2:0]		
		7:0	SWPCI	SWPCIM[1:0]		LATMOD	TQPS	TQSS[2:0]		
0x03E6	PG4SPCIL	15:8	TSYNCDIS	TERM[2:0]			AQPS	AQSS[2:0]		
		7:0	SWTERM	PSYNC	PPS	PSS[4:0]				
0x03E8	PG4SPCIH	15:8	BPEN	BPSEL[2:0]				ACP[2:0]		
		7:0	SWPCI	SWPCIM[1:0]		LATMOD	TQPS	TQSS[2:0]		
0x03EA	PG4LEBL	15:8	LEB[15:8]							
		7:0	LEB[7:0]							
0x03EC	PG4LEBH	15:8						PWMPCI[2:0]		
		7:0					PHR	PHF	PLR	PLF
0x03EE	PG4PHASE	15:8	PG4PHASE[15:8]							
		7:0	PG4PHASE[7:0]							
0x03F0	PG4DC	15:8	PG4DC[15:8]							
		7:0	PG4DC[7:0]							
0x03F2	PG4DCA	15:8								
		7:0	PG4DCA[7:0]							
0x03F4	PG4PER	15:8	PG4PER[15:8]							
		7:0	PG4PER[7:0]							
0x03F6	PG4TRIGA	15:8	PG4TRIGA[15:8]							
		7:0	PG4TRIGA[7:0]							
0x03F8	PG4TRIGB	15:8	PG4TRIGB[15:8]							
		7:0	PG4TRIGB[7:0]							
0x03FA	PG4TRIGC	15:8	PG4TRIGC[15:8]							
		7:0	PG4TRIGC[7:0]							
0x03FC	PG4DTL	15:8				DTL[13:8]				
		7:0	DTL[7:0]							
0x03FE	PG4DTH	15:8				DTH[13:8]				
		7:0	DTH[7:0]							
0x0400	PG4CAP	15:8	PG4CAP[15:8]							
		7:0	PG4CAP[7:0]							
0x0402 ... 0x0437	Reserved									
0x0438	PG6CONL	15:8	ON	Reserved				TRGCNT[2:0]		
		7:0	HREN			CLKSEL[1:0]		MODSEL[2:0]		
0x043A	PG6CONH	15:8	MDCSEL	MPERSEL	MPHSEL		MSTEN	UPMOD[2:0]		
		7:0	Reserved	TRGMOD			SOCS[3:0]			
0x043C	PG6STAT	15:8	SEVT	FLTEVT	CLEVT	FFEVT	SACT	FLTACT	CLACT	FFACT
		7:0	TRSET	TRCLR	CAP	UPDATE	UPDREQ	STEER	CAHALF	TRIG
0x043E	PG6IOCONL	15:8	CLMOD	SWAP	OVRENH	OVRENL	OVRDAT[1:0]		OSYNC[1:0]	
		7:0	FLTDAT[1:0]			CLDAT[1:0]		FFDAT[1:0]		DBDAT[1:0]
0x0440	PG6IOCONH	15:8			CAPSRC[2:0]					DTCMPSEL
		7:0			PMOD[1:0]		PENH	PENL	POLH	POLL
0x0442	PG6EVTL	15:8	ADTR1PS[4:0]					ADTR1EN3	ADTR1EN2	ADTR1EN1
		7:0				UPDTRG[1:0]		PGTRGSEL[2:0]		
0x0444	PG6EVTH	15:8	FLTIEH	CLIEH	FFIEH	SIEN			IEVTSEL[1:0]	
		7:0	ADTR2EN3	ADTR2EN2	ADTR2EN1	ADTR1OFS[4:0]				
0x0446	PG6FPCIL	15:8	TSYNCDIS	TERM[2:0]			AQPS	AQSS[2:0]		
		7:0	SWTERM	PSYNC	PPS	PSS[4:0]				
0x0448	PG6FPCIH	15:8	BPEN	BPSEL[2:0]				ACP[2:0]		
		7:0	SWPCI	SWPCIM[1:0]		LATMOD	TQPS	TQSS[2:0]		
0x044A	PG6CLPCIL	15:8	TSYNCDIS	TERM[2:0]			AQPS	AQSS[2:0]		
		7:0	SWTERM	PSYNC	PPS	PSS[4:0]				
0x044C	PG6CLPCIH	15:8	BPEN	BPSEL[2:0]				ACP[2:0]		
		7:0	SWPCI	SWPCIM[1:0]		LATMOD	TQPS	TQSS[2:0]		

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High-Resolution PWM with Fine Edge Placement

.....continued

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x044E	PG6FFPCIL	15:8	TSYNCDIS	TERM[2:0]			AQPS	AQSS[2:0]		
		7:0	SWTERM	PSYNC	PPS	PSS[4:0]				
0x0450	PG6FFPCIH	15:8	BPEN	BPSEL[2:0]				ACP[2:0]		
		7:0	SWPCI	SWPCIM[1:0]		LATMOD	TQPS	TQSS[2:0]		
0x0452	PG6SPCIL	15:8	TSYNCDIS	TERM[2:0]			AQPS	AQSS[2:0]		
		7:0	SWTERM	PSYNC	PPS	PSS[4:0]				
0x0454	PG6SPCIH	15:8	BPEN	BPSEL[2:0]				ACP[2:0]		
		7:0	SWPCI	SWPCIM[1:0]		LATMOD	TQPS	TQSS[2:0]		
0x0456	PG6LEBL	15:8	LEB[15:8]							
		7:0	LEB[7:0]							
0x0458	PG6LEBH	15:8						PWMPCI[2:0]		
		7:0					PHR	PHF	PLR	PLF
0x045A	PG6PHASE	15:8	PG6PHASE[15:8]							
		7:0	PG6PHASE[7:0]							
0x045C	PG6DC	15:8	PG6DC[15:8]							
		7:0	PG6DC[7:0]							
0x045E	PG6DCA	15:8								
		7:0	PG6DCA[7:0]							
0x0460	PG6PER	15:8	PG6PER[15:8]							
		7:0	PG6PER[7:0]							
0x0462	PG6TRIGA	15:8	PG6TRIGA[15:8]							
		7:0	PG6TRIGA[7:0]							
0x0464	PG6TRIGB	15:8	PG6TRIGB[15:8]							
		7:0	PG6TRIGB[7:0]							
0x0466	PG6TRIGC	15:8	PG6TRIGC[15:8]							
		7:0	PG6TRIGC[7:0]							
0x0468	PG6DTL	15:8			DTL[13:8]					
		7:0	DTL[7:0]							
0x046A	PG6DTH	15:8			DTH[13:8]					
		7:0	DTH[7:0]							
0x046C	PG6CAP	15:8	PG6CAP[15:8]							
		7:0	PG6CAP[7:0]							
0x046E ... 0x04A3	Reserved									
0x04A4	PG8CONL	15:8	ON	Reserved				TRGCNT[2:0]		
		7:0	HREN			CLKSEL[1:0]		MODSEL[2:0]		
0x04A6	PG8CONH	15:8	MDCSEL	MPERSEL	MPHSEL		MSTEN	UPMOD[2:0]		
		7:0	Reserved	TRGMOD			SOCS[3:0]			
0x04A8	PG8STAT	15:8	SEVT	FLTEVT	CLEVT	FFEVT	SACT	FLTACT	CLACT	FFACT
		7:0	TRSET	TRCLR	CAP	UPDATE	UPDREQ	STEER	CAHALF	TRIG
0x04AA	PG8IOCONL	15:8	CLMOD	SWAP	OVRENH	OVRENL	OVRDAT[1:0]		OSYNC[1:0]	
		7:0	FLTDAT[1:0]		CLDAT[1:0]		FFDAT[1:0]		DBDAT[1:0]	
0x04AC	PG8IOCONH	15:8			CAPSRC[2:0]					DTCMPSEL
		7:0			PMOD[1:0]		PENH	PENL	POLH	POLL
0x04AE	PG8EVTL	15:8	ADTR1PS[4:0]					ADTR1EN3	ADTR1EN2	ADTR1EN1
		7:0				UPDTRG[1:0]		PGTRGSEL[2:0]		
0x04B0	PG8EVTH	15:8	FLTEN	CLIEN	FFIEN	SIEN			IEVTSEL[1:0]	
		7:0	ADTR2EN3	ADTR2EN2	ADTR2EN1	ADTR1OFS[4:0]				
0x04B2	PG8FPCIL	15:8	TSYNCDIS	TERM[2:0]			AQPS	AQSS[2:0]		
		7:0	SWTERM	PSYNC	PPS	PSS[4:0]				
0x04B4	PG8FPCIH	15:8	BPEN	BPSEL[2:0]				ACP[2:0]		
		7:0	SWPCI	SWPCIM[1:0]		LATMOD	TQPS	TQSS[2:0]		
0x04B6	PG8CLPCIL	15:8	TSYNCDIS	TERM[2:0]			AQPS	AQSS[2:0]		
		7:0	SWTERM	PSYNC	PPS	PSS[4:0]				
0x04B8	PG8CLPCIH	15:8	BPEN	BPSEL[2:0]				ACP[2:0]		
		7:0	SWPCI	SWPCIM[1:0]		LATMOD	TQPS	TQSS[2:0]		
0x04BA	PG8FFPCIL	15:8	TSYNCDIS	TERM[2:0]			AQPS	AQSS[2:0]		
		7:0	SWTERM	PSYNC	PPS	PSS[4:0]				

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.....continued

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x04BC	PG8FFPCIH	15:8	BPEN	BPSEL[2:0]				ACP[2:0]		
		7:0	SWPCI	SWPCIM[1:0]		LATMOD	TQPS	TQSS[2:0]		
0x04BE	PG8SPCIL	15:8	TSYNCDIS	TERM[2:0]			AQPS	AQSS[2:0]		
		7:0	SWTERM	PSYNC	PPS	PSS[4:0]				
0x04C0	PG8SPCIH	15:8	BPEN	BPSEL[2:0]				ACP[2:0]		
		7:0	SWPCI	SWPCIM[1:0]		LATMOD	TQPS	TQSS[2:0]		
0x04C2	PG8LEBL	15:8	LEB[15:8]							
		7:0	LEB[7:0]							
0x04C4	PG8LEBH	15:8						PWMPCI[2:0]		
		7:0					PHR	PHF	PLR	PLF
0x04C6	PG8PHASE	15:8	PG8PHASE[15:8]							
		7:0	PG8PHASE[7:0]							
0x04C8	PG8DC	15:8	PG8DC[15:8]							
		7:0	PG8DC[7:0]							
0x04CA	PG8DCA	15:8								
		7:0	PG8DCA[7:0]							
0x04CC	PG8PER	15:8	PG8PER[15:8]							
		7:0	PG8PER[7:0]							
0x04CE	PG8TRIGA	15:8	PG8TRIGA[15:8]							
		7:0	PG8TRIGA[7:0]							
0x04D0	PG8TRIGB	15:8	PG8TRIGB[15:8]							
		7:0	PG8TRIGB[7:0]							
0x04D2	PG8TRIGC	15:8	PG8TRIGC[15:8]							
		7:0	PG8TRIGC[7:0]							
0x04D4	PG8DTL	15:8			DTL[13:8]					
		7:0	DTL[7:0]							
0x04D6 ... 0x04D7	Reserved									
0x04D8	PG8CAP	15:8	PG8CAP[15:8]							
		7:0	PG8CAP[7:0]							
0x04DA	PG8DTH	15:8			DTH[13:8]					
		7:0	DTH[7:0]							

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13.6.1 PWM Clock Control Register

Name: PCLKCON
Offset: 0x300

Notes:

1. A device-specific unlock sequence must be performed before this bit can be cleared.
2. Changing the MCLKSEL[1:0] bits while ON (PGxCONL[15]) = 1 is not recommended.

Bit	15	14	13	12	11	10	9	8
	HRRDY	HRERR						LOCK
Access	R/W	R/W						R/W
Reset	0	0						0

Bit	7	6	5	4	3	2	1	0
			DIVSEL[1:0]				MCLKSEL[1:0]	
Access			R/W	R/W			R/W	R/W
Reset			0	0			0	0

Bit 15 – HRRDY High-Resolution Ready bit

Value	Description
1	The high-resolution circuitry is ready
0	The high-resolution circuitry is not ready

Bit 14 – HRERR High-Resolution Error bit

Value	Description
1	An error has occurred; PWM signals will have limited resolution
0	No error has occurred; PWM signals will have full resolution when HRRDY = 1

Bit 8 – LOCK Lock bit⁽¹⁾

Value	Description
1	Write-protected registers and bits are locked
0	Write-protected registers and bits are unlocked

Bits 5:4 – DIVSEL[1:0] PWM Clock Divider Selection bits

Value	Description
11	Divide ratio is 1:16
10	Divide ratio is 1:8
01	Divide ratio is 1:4
00	Divide ratio is 1:2

Bits 1:0 – MCLKSEL[1:0] PWM Master Clock Selection bits⁽²⁾

Value	Description
11	AF _{PLLO} – Auxiliary PLL post-divider output
10	F _{PLLO} – Primary PLL post-divider output
01	AF _{VCO/2} – Auxiliary VCO/2
00	F _{OSC}

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13.6.2 Frequency Scale Register

Name: FSCL
Offset: 0x302

Bit	15	14	13	12	11	10	9	8
	FSCL[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	FSCL[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – FSCL[15:0] Frequency Scale Register bits

The value in this register is added to the frequency scaling accumulator at each pwm_master_clk. When the accumulated value exceeds the value of FSMINPER, a clock pulse is produced.

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13.6.3 Frequency Scaling Minimum Period Register

Name: FSMINPER

Offset: 0x304

Bit	15	14	13	12	11	10	9	8
	FSMINPER[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	FSMINPER[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – FSMINPER[15:0] Frequency Scaling Minimum Period Register bits

This register holds the minimum clock period (maximum clock frequency) that can be produced by the frequency scaling circuit.

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13.6.4 Master Phase Register

Name: MPHASE
Offset: 0x306

Bit	15	14	13	12	11	10	9	8
	MPHASE[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	MPHASE[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – MPHASE[15:0] Main Phase Register bits

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13.6.5 Master Duty Cycle Register

Name: MDC
Offset: 0x308

Bit	15	14	13	12	11	10	9	8
	MDC[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	MDC[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – MDC[15:0] Main Duty Cycle Register bits

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13.6.6 Master Period Register

Name: MPER

Offset: 0x30A

Note:

1. Period values less than '0x0010' should not be selected.

Bit	15	14	13	12	11	10	9	8
	MPER[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	MPER[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – MPER[15:0] Main Period Register bits⁽¹⁾

This register holds the period value that can be shared by multiple PWM Generators.

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13.6.7 Combinational Trigger Register Low

Name: CMBTRIGL
Offset: 0x30E

Bit	15	14	13	12	11	10	9	8
Access								
Reset								

Bit	7	6	5	4	3	2	1	0
	CTA8EN	CTA7EN	CTA6EN	CTA5EN	CTA4EN	CTA3EN	CTA2EN	CTA1EN
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 7 – CTA8EN Enable Trigger Output from PWM Generator #8 as Source for Comb. Trigger A bit

Value	Description
1	Enables specified trigger signal to be OR'd into the Combinatorial Trigger A signal
0	Disabled

Bit 6 – CTA7EN Enable Trigger Output from PWM Generator #7 as Source for Comb. Trigger A bit

Value	Description
1	Enables specified trigger signal to be OR'd into the Combinatorial Trigger A signal
0	Disabled

Bit 5 – CTA6EN Enable Trigger Output from PWM Generator #6 as Source for Comb. Trigger A bit

Value	Description
1	Enables specified trigger signal to be OR'd into the Combinatorial Trigger A signal
0	Disabled

Bit 4 – CTA5EN Enable Trigger Output from PWM Generator #5 as Source for Comb. Trigger A bit

Value	Description
1	Enables specified trigger signal to be OR'd into the Combinatorial Trigger A signal
0	Disabled

Bit 3 – CTA4EN Enable Trigger Output from PWM Generator #4 as Source for Comb. Trigger A bit

Value	Description
1	Enables specified trigger signal to be OR'd into the Combinatorial Trigger A signal
0	Disabled

Bit 2 – CTA3EN Enable Trigger Output from PWM Generator #3 as Source for Comb. Trigger A bit

Value	Description
1	Enables specified trigger signal to be OR'd into the Combinatorial Trigger A signal
0	Disabled

Bit 1 – CTA2EN Enable Trigger Output from PWM Generator #2 as Source for Comb. Trigger A bit

Value	Description
1	Enables specified trigger signal to be OR'd into the Combinatorial Trigger A signal
0	Disabled

Bit 0 – CTA1EN Enable Trigger Output from PWM Generator #1 as Source for Comb. Trigger A bit

Value	Description
1	Enables specified trigger signal to be OR'd into the Combinatorial Trigger A signal
0	Disabled

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13.6.8 Combinational Trigger Register High

Name: CMBTRIGH
Offset: 0x310

Bit	15	14	13	12	11	10	9	8
Access								
Reset								

Bit	7	6	5	4	3	2	1	0
	CTB8EN	CTB7EN	CTB6EN	CTB5EN	CTB4EN	CTB3EN	CTB2EN	CTB1EN
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 7 – CTB8EN Enable Trigger Output from PWM Generator #8 as Source for Comb. Trigger B bit

Value	Description
1	Enables specified trigger signal to be OR'd into the Combinatorial Trigger B signal
0	Disabled

Bit 6 – CTB7EN Enable Trigger Output from PWM Generator #7 as Source for Comb. Trigger B bit

Value	Description
1	Enables specified trigger signal to be OR'd into the Combinatorial Trigger B signal
0	Disabled

Bit 5 – CTB6EN Enable Trigger Output from PWM Generator #6 as Source for Comb. Trigger B bit

Value	Description
1	Enables specified trigger signal to be OR'd into the Combinatorial Trigger B signal
0	Disabled

Bit 4 – CTB5EN Enable Trigger Output from PWM Generator #5 as Source for Comb. Trigger B bit

Value	Description
1	Enables specified trigger signal to be OR'd into the Combinatorial Trigger B signal
0	Disabled

Bit 3 – CTB4EN Enable Trigger Output from PWM Generator #4 as Source for Comb. Trigger B bit

Value	Description
1	Enables specified trigger signal to be OR'd into the Combinatorial Trigger B signal
0	Disabled

Bit 2 – CTB3EN Enable Trigger Output from PWM Generator #3 as Source for Comb. Trigger B bit

Value	Description
1	Enables specified trigger signal to be OR'd into the Combinatorial Trigger B signal
0	Disabled

Bit 1 – CTB2EN Enable Trigger Output from PWM Generator #2 as Source for Comb. Trigger B bit

Value	Description
1	Enables specified trigger signal to be OR'd into the Combinatorial Trigger B signal
0	Disabled

Bit 0 – CTB1EN Enable Trigger Output from PWM Generator #1 as Source for Comb. Trigger B bit

Value	Description
1	Enables specified trigger signal to be OR'd into the Combinatorial Trigger B signal
0	Disabled

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13.6.9 Combinatorial PWM Logic Control Register y

Name: LOGCONy⁽²⁾
Offset: 0x0312, 0x0314, 0x0316, 0x0318, 0x031A, 0x031C

Notes:

1. Logic function input will be connected to '0' if the PWM channel is not present.
2. 'y' denotes a common instance (A-F).

Bit	15	14	13	12	11	10	9	8
	PWMS1y[3:0]				PWMS2y[3:0]			
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	S1yPOL	S2yPOL	PWMLFy[1:0]			PWMLFyD[2:0]		
Access	R/W	R/W	R/W	R/W		R/W	R/W	R/W
Reset	0	0	0	0		0	0	0

Bits 15:12 – PWMS1y[3:0] Combinatorial PWM Logic Source #1 Selection bits⁽¹⁾

Value	Description
1111	PWM8L
1110	PWM8H
1101–1100	Reserved
1011	PWM6L
1010	PWM6H
1001–1000	Reserved
0111	PWM4L
0110	PWM4H
0101	PWM3L
0100	PWM3H
0011	PWM2L
0010	PWM2H
0001	PWM1L
0000	PWM1H

Bits 11:8 – PWMS2y[3:0] Combinatorial PWM Logic Source #2 Selection bits⁽¹⁾

Value	Description
1111	PWM8L
1110	PWM8H
1101–1100	Reserved
1011	PWM6L
1010	PWM6H
1001–1000	Reserved
0111	PWM4L
0110	PWM4H
0101	PWM3L
0100	PWM3H
0011	PWM2L
0010	PWM2H
0001	PWM1L
0000	PWM1H

Bit 7 – S1yPOL Combinatorial PWM Logic Source #1 Polarity bit

Value	Description
1	Input is inverted

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Value	Description
0	Input is positive logic

Bit 6 – S2yPOL Combinatorial PWM Logic Source #2 Polarity bit

Value	Description
1	Input is inverted
0	Input is positive logic

Bits 5:4 – PWMLFy[1:0] Combinatorial PWM Logic Function Selection bits

Value	Description
11	Reserved
10	PWMS1 ^ PWMS2 (XOR)
01	PWMS1 & PWMS2 (AND)
00	PWMS1 PWMS2 (OR)

Bits 2:0 – PWMLFyD[2:0] Combinatorial PWM Logic Destination Selection bits

Value	Description
111	Logic function is assigned to the PWM8H or PWM8L pin
110	Reserved
101	Logic function is assigned to the PWM6H or PWM6L pin
100	Reserved
011	Logic function is assigned to the PWM4H or PWM4L pin
010	Logic function is assigned to the PWM3H or PWM3L pin
001	Logic function is assigned to the PWM2H or PWM2L pin
000	No assignment, combinatorial PWM logic function is disabled

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13.6.10 PWM Event Output Control Register y

Name: PWMEVTy^(2,5)
Offset: 0x031E, 0x0320, 0x0322, 0x0324, 0x0326, 0x0328

Notes:

1. The event signal is stretched using the peripheral clock because different PGs may be operating from different clock sources. The leading edge of the event pulse is produced in the clock domain of the PWM Generator. The trailing edge of the stretched event pulse is produced in the peripheral clock domain.
2. No event will be produced if the selected PWM Generator is not present.
3. This is the PWM Generator output signal prior to Output mode logic and any output override logic.
4. This signal should be the PGx_clk domain signal prior to any synchronization into the system clock domain.
5. 'y' denotes a common instance (A-F).

Bit	15	14	13	12	11	10	9	8
	EVTyOEN	EVTyPOL	EVTySTRD	EVTySYNC				
Access	R/W	R/W	R/W	R/W				
Reset	0	0	0	0				

Bit	7	6	5	4	3	2	1	0
	EVTySEL[3:0]					EVTyPGS[2:0]		
Access	R/W	R/W	R/W	R/W		R/W	R/W	R/W
Reset	0	0	0	0		0	0	0

Bit 15 – EVTyOEN PWM Event Output Enable bit

Value	Description
1	Event output signal is output on PWMEVTy pin
0	Event output signal is internal only

Bit 14 – EVTyPOL PWM Event Output Polarity bit

Value	Description
1	Event output signal is active-low
0	Event output signal is active-high

Bit 13 – EVTySTRD PWM Event Output Stretch Disable bit

Value	Description
1	Event output signal pulse width is not stretched
0	Event output signal is stretched to eight PWM clock cycles minimum ⁽¹⁾

Bit 12 – EVTySYNC PWM Event Output Sync bit

Event output signal pulse will be two system clocks when this bit is set and EVTySTRD = 1.

Value	Description
1	Event output signal is synchronized to the system clock
0	Event output is not synchronized to the system clock

Bits 7:4 – EVTySEL[3:0] PWM Event Selection bits

Value	Description
1111	High-resolution error event signal
1110–1010	Reserved
1001	ADC Trigger 2 signal
1000	ADC Trigger 1 signal
0111	STEER signal (available in Push-Pull Output modes only) ⁽⁴⁾
0110	CAHALF signal (available in Center-Aligned modes only) ⁽⁴⁾
0101	PCI Fault active output signal
0100	PCI current-limit active output signal
0011	PCI feed-forward active output signal

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Value	Description
0010	PCI Sync active output signal
0001	PWM Generator output signal ⁽³⁾
0000	Source is selected by the PGTRGSEL[2:0] bits

Bits 2:0 – EVTyPGS[2:0] Combinatorial PWM Logic Destination Selection bits

Value	Description
111	Logic function is assigned to the PWM8H or PWM8L pin
110	Reserved
101	Logic function is assigned to the PWM6H or PWM6L pin
100	Reserved
011	Logic function is assigned to the PWM4H or PWM4L pin
010	Logic function is assigned to the PWM3H or PWM3L pin
001	Logic function is assigned to the PWM2H or PWM2L pin
000	No assignment, combinatorial PWM logic function is disabled

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13.6.11 Linear Feedback Shift Register

Name: LFSR
Offset: 0x30C

Bit	15	14	13	12	11	10	9	8
	LFSR[14:8]							
Access		R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset		0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	LFSR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 14:0 – LFSR[14:0] Linear Feedback Shift Register bits
A read of this register will provide a 15-bit pseudorandom value.

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13.6.12 PWM Generator x Control Register Low

Name: PGxCONL
Offset: 0x32A, 0x360, 0x396, 0x3CC, 0x438, 0x4A4

Note:

1. The PWM Generator time base operates from the frequency scaling circuit clock, effectively scaling the duty cycle and period of the PWM Generator output.

Legend: r = Reserved bit

Bit	15	14	13	12	11	10	9	8
	ON	Reserved					TRGCNT[2:0]	
Access	R/W	r				R/W	R/W	R/W
Reset	0	0				0	0	0

Bit	7	6	5	4	3	2	1	0
	HREN			CLKSEL[1:0]			MODSEL[2:0]	
Access	R/W			R/W	R/W	R/W	R/W	R/W
Reset	0			0	0	0	0	0

Bit 15 – ON Enable bit

Value	Description
1	PWM Generator is enabled
0	PWM Generator is not enabled

Bit 14 – Reserved Maintain as '0'

Bits 10:8 – TRGCNT[2:0] Trigger Count Select bits

Value	Description
111	PWM Generator produces eight PWM cycles after triggered
110	PWM Generator produces seven PWM cycles after triggered
101	PWM Generator produces six PWM cycles after triggered
100	PWM Generator produces five PWM cycles after triggered
011	PWM Generator produces four PWM cycles after triggered
010	PWM Generator produces three PWM cycles after triggered
001	PWM Generator produces two PWM cycles after triggered
000	PWM Generator produces one PWM cycle after triggered

Bit 7 – HREN PWM Generator x High-Resolution Enable bit

Value	Description
1	PWM Generator x operates in High-Resolution mode
0	PWM Generator x operates in Standard Resolution mode

Bits 4:3 – CLKSEL[1:0] Clock Selection bits

Value	Description
11	PWM Generator uses Host clock scaled by frequency scaling circuit ⁽¹⁾
10	PWM Generator uses Host clock divided by clock divider circuit ⁽¹⁾
01	PWM Generator uses Host clock selected by the MCLKSEL[1:0] (PCLKCON[1:0]) control bits
00	No clock selected, PWM Generator is in Lowest Power state (default)

Bits 2:0 – MODSEL[2:0] Mode Selection bits

Value	Description
111	Dual Edge Center-Aligned PWM mode (interrupt/register update twice per cycle)
110	Dual Edge Center-Aligned PWM mode (interrupt/register update once per cycle)
101	Double-Update Center-Aligned PWM mode
100	Center-Aligned PWM mode

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Value	Description
011	Reserved
010	Independent Edge PWM mode, dual output
001	Variable Phase PWM mode
000	Independent Edge PWM mode

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13.6.13 PWM Generator x Control Register High

Name: PGxCONH
Offset: 0x32C, 0x362, 0x398, 0x3CE, 0x43A, 0x4A6

Notes:

1. The PCI selected Sync signal is always available to be OR'd with the selected SOC signal per the SOCS[3:0] bits if the PCI Sync function is enabled.
2. The source selected by the SOCS[3:0] bits MUST operate from the same clock source as the local PWM Generator. If not, the source must be routed through the PCI Sync logic so the trigger signal may be synchronized to the PWM Generator clock domain.
3. PWM Generators are grouped into groups of four: PG1-PG4 and PG5-PG8, if available. Any generator within a group of four may be used to trigger another generator within the same group.

Legend: r = Reserved bit

Bit	15	14	13	12	11	10	9	8
	MDCSEL	MPERSEL	MPHSEL		MSTEN		UPMOD[2:0]	
Access	R/W	R/W	R/W		R/W	R/W	R/W	R/W
Reset	0	0	0		0	0	0	0

Bit	7	6	5	4	3	2	1	0
	Reserved	TRGMOD					SOCS[3:0]	
Access	r	R/W			R/W	R/W	R/W	R/W
Reset	0	0			0	0	0	0

Bit 15 – MDCSEL Main Duty Cycle Register Select bit

Value	Description
1	PWM Generator uses the MDC register instead of PGxDC
0	PWM Generator uses the PGxDC register

Bit 14 – MPERSEL Main Period Register Select bit

Value	Description
1	PWM Generator uses the MPER register instead of PGxPER
0	PWM Generator uses the PGxPER register

Bit 13 – MPHSEL Main Phase Register Select bit

Value	Description
1	PWM Generator uses the MPHASE register instead of PGxPHASE
0	PWM Generator uses the PGxPHASE register

Bit 11 – MSTEN Main Update Enable bit

Value	Description
1	PWM Generator broadcasts software set/clear of the UPDATE status bit and EOC signal to other PWM Generators
0	PWM Generator does not broadcast the UPDATE status bit state or EOC signal

Bits 10:8 – UPMOD[2:0] PWM Buffer Update Mode Selection bits

Bit 7 – Reserved Maintain as '0'

Bit 6 – TRGMOD PWM Generator Trigger Mode Selection bit

Value	Description
1	PWM Generator operates in Retriggerable mode
0	PWM Generator operates in Single Trigger mode

Bits 3:0 – SOCS[3:0] Start-of-Cycle Selection bits^(1,2,3)

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Value	Description
1111	TRIG bit or PCI Sync function only (no hardware trigger source is selected)
1110-0101	Reserved
0100	PWM4(8) PG1 or PG5 trigger output selected by PGTRGSEL[2:0] (PGxEVT[2:0])
0011	PWM3 PG1 or PG5 trigger output selected by PGTRGSEL[2:0] (PGxEVT[2:0])
0010	PWM2(6) PG1 or PG5 trigger output selected by PGTRGSEL[2:0] (PGxEVT[2:0])
0001	PWM1 PG1 or PG5 trigger output selected by PGTRGSEL[2:0] (PGxEVT[2:0])
0000	Local EOC – PWM Generator is self-triggered

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13.6.14 PWM Generator x Status Register

Name: PGxSTAT
Offset: 0x32E, 0x364, 0x39A, 0x3D0, 0x43C, 0x4A8

Note:

1. User software may write a '1' to CAP as a request to initiate a software capture. The CAP status bit will be set when the capture event has occurred. No further captures will occur until CAP is cleared by software.

Legend: C = Clearable bit; HS = Hardware Settable bit

Bit	15	14	13	12	11	10	9	8
	SEVT	FLTEVT	CLEVT	FFEVT	SACT	FLTACT	CLACT	FFACT
Access	HS/C	HS/C	HS/C	HS/C	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	TRSET	TRCLR	CAP	UPDATE	UPDREQ	STEER	CAHALF	TRIG
Access	W	W	R/W/HS	R	W	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit 15 – SEVT PCI Sync Event bit

Value	Description
1	A PCI Sync event has occurred (rising edge on PCI Sync output or PCI Sync output is high when module is enabled)
0	No PCI Sync event has occurred

Bit 14 – FLTEVT PCI Fault Active Status bit

Value	Description
1	A Fault event has occurred (rising edge on PCI Fault output or PCI Fault output is high when module is enabled)
0	No Fault event has occurred

Bit 13 – CLEVT PCI Current Limit Status bit

Value	Description
1	A PCI current limit event has occurred (rising edge on PCI current-limit output or PCI current-limit output is high when module is enabled)
0	No PCI current limit event has occurred

Bit 12 – FFEVT PCI Feed-Forward Active Status bit

Value	Description
1	A PCI feed-forward event has occurred (rising edge on PCI feed-forward output or PCI feed-forward output is high when module is enabled)
0	No PCI feed-forward event has occurred

Bit 11 – SACT PCI Sync Status bit

Value	Description
1	PCI Sync output is active
0	PCI Sync output is inactive

Bit 10 – FLTACT PCI Fault Active Status bit

Value	Description
1	PCI Fault output is active
0	PCI Fault output is inactive

Bit 9 – CLACT PCI Current Limit Status bit

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Value	Description
1	PCI current limit output is active
0	PCI current limit output is inactive

Bit 8 – FFACT PCI Feed-Forward Active Status bit

Value	Description
1	PCI feed-forward output is active
0	PCI feed-forward output is inactive

Bit 7 – TRSET PWM Generator Software Trigger Set bit

User software writes a '1' to this bit location to trigger a PWM Generator cycle. The bit location always reads as '0'. The TRIG bit will indicate '1' when the PWM Generator is triggered.

Bit 6 – TRCLR PWM Generator Software Trigger Clear bit

User software writes a '1' to this bit location to stop a PWM Generator cycle. The bit location always reads as '0'. The TRIG bit will indicate '0' when the PWM Generator is not triggered.

Bit 5 – CAP Capture Status bit⁽¹⁾

Value	Description
1	PWM Generator time base value has been captured in PGxCAP
0	No capture has occurred

Bit 4 – UPDATE PWM Data Register Update Status/Control bit

Value	Description
1	PWM Data register update is pending – user Data registers are not writable
0	No PWM Data register update is pending

Bit 3 – UPDREQ PWM Data Register Update Request bit

User software writes a '1' to this bit location to request a PWM Data register update. The bit location always reads as '0'. The UPDATE status bit will indicate '1' when an update is pending.

Bit 2 – STEER Output Steering Status bit (Push-Pull Output mode only)

Value	Description
1	PWM Generator is in 2nd cycle of Push-Pull mode
0	PWM Generator is in 1st cycle of Push-Pull mode

Bit 1 – CAHALF Half Cycle Status bit (Center-Aligned modes only)

Value	Description
1	PWM Generator is in 2nd half of time base cycle
0	PWM Generator is in 1st half of time base cycle

Bit 0 – TRIG PWM Trigger Status bit

Value	Description
1	PWM Generator is triggered and PWM cycle is in progress
0	No PWM cycle is in progress

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13.6.15 PWM Generator x I/O Control Register Low

Name: PGxIOCONL
Offset: 0x330, 0x366, 0x39C, 0x3D2, 0x43E, 0x4AA

Bit	15	14	13	12	11	10	9	8
	CLMOD	SWAP	OVRENH	OVRENH	OVRDAT[1:0]		OSYNC[1:0]	
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	FLTDAT[1:0]		CLDAT[1:0]		FFDAT[1:0]		DBDAT[1:0]	
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – CLMOD Current Limit Mode Select bit

Value	Description
1	If PCI current limit is active, then the PWMxH and PWMxL output signals are inverted (bit flipping), and the CLDAT[1:0] bits are not used
0	If PCI current limit is active, then the CLDAT[1:0] bits define the PWM output levels

Bit 14 – SWAP Swap PWM Signals to PWMxH and PWMxL Device Pins bit

Value	Description
1	The PWMxH signal is connected to the PWMxL pin and the PWMxL signal is connected to the PWMxH pin
0	PWMxH/L signals are mapped to their respective pins

Bit 13 – OVRENH User Override Enable for PWMxH Pin bit

Value	Description
1	OVRDAT1 provides data for output on the PWMxH pin
0	PWM Generator provides data for the PWMxH pin

Bit 12 – OVRENH User Override Enable for PWMxL Pin bit

Value	Description
1	OVRDAT0 provides data for output on the PWMxL pin
0	PWM Generator provides data for the PWMxL pin

Bits 11:10 – OVRDAT[1:0] Data for PWMxH/PWMxL Pins if Override is Enabled bits

If OVERENH = 1, then OVRDAT1 provides data for PWMxH.

If OVERENL = 1, then OVRDAT0 provides data for PWMxL.

Bits 9:8 – OSYNC[1:0] User Output Override Synchronization Control bits

Value	Description
11	Reserved
10	User output overrides via the OVRENH/L and OVRDAT[1:0] bits occur when specified by the UPDMOD[2:0] bits in the PGxCONH register
01	User output overrides via the OVRENH/L and OVRDAT[1:0] bits occur immediately (as soon as possible)
00	User output overrides via the OVRENH/L and OVRDAT[1:0] bits are synchronized to the local PWM time base (next Start-of-Cycle)

Bits 7:6 – FLTDAT[1:0] Data for PWMxH/PWMxL Pins if Fault Event is Active bits

If Fault is active, then FLTDAT1 provides data for PWMxH.

If Fault is active, then FLTDAT0 provides data for PWMxL.

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Bits 5:4 – CLDAT[1:0] Data for PWMxH/PWMxL Pins if Current Limit Event is Active bits
If current limit is active, then CLDAT1 provides data for PWMxH.
If current limit is active, then CLDAT0 provides data for PWMxL.

Bits 3:2 – FFDAT[1:0] Data for PWMxH/PWMxL Pins if Feed-Forward Event is Active bits
If feed-forward is active, then FFDAT1 provides data for PWMxH.
If feed-forward is active, then FFDAT0 provides data for PWMxL.

Bits 1:0 – DBDAT[1:0] Data for PWMxH/PWMxL Pins if Debug Mode is Active and PTFRZ = 1 bits
If Debug mode is active and PTFRZ = 1, then DBDAT1 provides data for PWMxH.
If Debug mode is active and PTFRZ = 1, then DBDAT0 provides data for PWMxL.

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13.6.16 PWM Generator x I/O Control Register High

Name: PGxIOCONH
Offset: 0x332, 0x368, 0x39E, 0x3D4, 0x440, 0x4AC

Note:

1. A capture may be initiated in software at any time by writing a '1' to CAP (PGxSTAT[5]).

Bit	15	14	13	12	11	10	9	8
		CAPSRC[2:0]						DTCMPSEL
Access		R/W	R/W	R/W				R/W
Reset		0	0	0				0

Bit	7	6	5	4	3	2	1	0
			PMOD[1:0]		PENH	PENL	POLH	POLL
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bits 14:12 – CAPSRC[2:0] Time Base Capture Source Selection bits⁽¹⁾

Value	Description
111	Reserved
110	Reserved
101	Reserved
100	Capture time base value at assertion of selected PCI Fault signal
011	Capture time base value at assertion of selected PCI current limit signal
010	Capture time base value at assertion of selected PCI feed-forward signal
001	Capture time base value at assertion of selected PCI Sync signal
000	No hardware source selected for time base capture – software only

Bit 8 – DTCMPSEL Dead-Time Compensation Select bit

Value	Description
1	Dead-time compensation is controlled by PCI feed-forward limit logic
0	Dead-time compensation is controlled by PCI Sync logic

Bits 5:4 – PMOD[1:0] PWM Generator Output Mode Selection bits

Value	Description
11	Reserved
10	PWM Generator outputs operate in Push-Pull mode
01	PWM Generator outputs operate in Independent mode
00	PWM Generator outputs operate in Complementary mode

Bit 3 – PENH PWMxH Output Port Enable bit

Value	Description
1	PWM Generator controls the PWMxH output pin
0	PWM Generator does not control the PWMxH output pin

Bit 2 – PENL PWMxL Output Port Enable bit

Value	Description
1	PWM Generator controls the PWMxL output pin
0	PWM Generator does not control the PWMxL output pin

Bit 1 – POLH PWMxH Output Polarity bit

Value	Description
1	Output pin is active-low
0	Output pin is active-high

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Bit 0 – POLL PWMxL Output Polarity bit

Value	Description
1	Output pin is active-low
0	Output pin is active-high

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13.6.17 PWM Generator x Event Register Low

Name: PGxEVTL
Offset: 0x334, 0x36A, 0x3A0, 0x3D6, 0x442, 0x4AE

Note:

- These events are derived from the internal PWM Generator time base comparison events.

Bit	15	14	13	12	11	10	9	8
	ADTR1PS[4:0]					ADTR1EN3	ADTR1EN2	ADTR1EN1
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
				UPDTRG[1:0]		PGTRGSEL[2:0]		
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0

Bits 15:11 – ADTR1PS[4:0] ADC Trigger 1 Postscaler Selection bits

Value	Description
11111	1:32
. . .	
00010	1:3
00001	1:2
00000	1:1

Bit 10 – ADTR1EN3 ADC Trigger 1 Source is PGxTRIGC Compare Event Enable bit

Value	Description
1	PGxTRIGC register compare event is enabled as trigger source for ADC Trigger 1
0	PGxTRIGC register compare event is disabled as trigger source for ADC Trigger 1

Bit 9 – ADTR1EN2 ADC Trigger 1 Source is PGxTRIGB Compare Event Enable bit

Value	Description
1	PGxTRIGB register compare event is enabled as trigger source for ADC Trigger 1
0	PGxTRIGB register compare event is disabled as trigger source for ADC Trigger 1

Bit 8 – ADTR1EN1 ADC Trigger 1 Source is PGxTRIGA Compare Event Enable bit

Value	Description
1	PGxTRIGA register compare event is enabled as trigger source for ADC Trigger 1
0	PGxTRIGA register compare event is disabled as trigger source for ADC Trigger 1

Bits 4:3 – UPDTRG[1:0] Update Trigger Select bits

Value	Description
11	A write of the PGxTRIGA register automatically sets the UPDATE bit
10	A write of the PGxPHASE register automatically sets the UPDATE bit
01	A write of the PGxDC register automatically sets the UPDATE bit
00	User must set the UPDATE bit (PGxSTAT[4]) manually

Bits 2:0 – PGTRGSEL[2:0] PWM Generator Trigger Output Selection bits⁽¹⁾

Value	Description
111	Reserved
110	Reserved
101	Reserved
100	Reserved
011	PGxTRIGC compare event is the PWM Generator trigger
010	PGxTRIGB compare event is the PWM Generator trigger

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Value	Description
001	PGxTRIGA compare event is the PWM Generator trigger
000	EOC event is the PWM Generator trigger

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13.6.18 PWM Generator x Event Register High

Name: PGxEVTH
Offset: 0x336, 0x36C, 0x3A2, 0x3D8, 0x444, 0x4B0

Notes:

1. An interrupt is only generated on the rising edge of the PCI Fault active signal.
2. An interrupt is only generated on the rising edge of the PCI current limit active signal.
3. An interrupt is only generated on the rising edge of the PCI feed-forward active signal.
4. An interrupt is only generated on the rising edge of the PCI Sync active signal.

Bit	15	14	13	12	11	10	9	8
	FLTIE	CLIE	FFIE	SIEN			IEVTSEL[1:0]	
Access	R/W	R/W	R/W	R/W			R/W	R/W
Reset	0	0	0	0			0	0

Bit	7	6	5	4	3	2	1	0
	ADTR2EN3	ADTR2EN2	ADTR2EN1			ADTR1OFS[4:0]		
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – FLTIE PCI Fault Interrupt Enable bit⁽¹⁾

Value	Description
1	Fault interrupt is enabled
0	Fault interrupt is disabled

Bit 14 – CLIE PCI Current-Limit Interrupt Enable bit⁽²⁾

Value	Description
1	Current limit interrupt is enabled
0	Current limit interrupt is disabled

Bit 13 – FFIE PCI Feed-Forward Interrupt Enable bit⁽³⁾

Value	Description
1	Feed-forward interrupt is enabled
0	Feed-forward interrupt is disabled

Bit 12 – SIEN PCI Sync Interrupt Enable bit⁽⁴⁾

Value	Description
1	Sync interrupt is enabled
0	Sync interrupt is disabled

Bits 9:8 – IEVTSEL[1:0] Interrupt Event Selection bits

Value	Description
11	Time base interrupts are disabled (Sync, Fault, current limit and feed-forward events can be independently enabled)
10	Interrupts CPU at ADC Trigger 1 event
01	Interrupts CPU at TRIGA compare event
00	Interrupts CPU at EOC

Bit 7 – ADTR2EN3 ADC Trigger 2 Source is PGxTRIGC Compare Event Enable bit

Value	Description
1	PGxTRIGC register compare event is enabled as trigger source for ADC Trigger 2
0	PGxTRIGC register compare event is disabled as trigger source for ADC Trigger 2

Bit 6 – ADTR2EN2 ADC Trigger 2 Source is PGxTRIGB Compare Event Enable bit

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Value	Description
1	PGxTRIGB register compare event is enabled as trigger source for ADC Trigger 2
0	PGxTRIGB register compare event is disabled as trigger source for ADC Trigger 2

Bit 5 – ADTR2EN1 ADC Trigger 2 Source is PGxTRIGA Compare Event Enable bit

Value	Description
1	PGxTRIGA register compare event is enabled as trigger source for ADC Trigger 2
0	PGxTRIGA register compare event is disabled as trigger source for ADC Trigger 2

Bits 4:0 – ADTR1OFS[4:0] ADC Trigger 1 Offset Selection bits

Value	Description
11111	Offset by 31 trigger events
. . .	
00010	Offset by 2 trigger events
00001	Offset by 1 trigger event
00000	No offset

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13.6.19 PWM Generator x PCI Register Low

Name: PGxFPCIL
Offset: 0x338, 0x36E, 0x3A4, 0x3DA, 0x446, 0x4B2,

Bit	15	14	13	12	11	10	9	8
	TSYNCDIS		TERM[2:0]			AQPS	AQSS[2:0]	
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	SWTERM	PSYNC	PPS			PSS[4:0]		
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – TSYNCDIS Termination Synchronization Disable bit

Value	Description
1	Termination of latched PCI occurs immediately
0	Termination of latched PCI occurs at PWM EOC

Bits 14:12 – TERM[2:0] Termination Event Selection bits

Value	Description
111	Selects PCI Source #9
110	Selects PCI Source #8
101	Selects PCI Source #1 (PWM Generator output selected by the PWMPCI[2:0] bits)
100	PGxTRIGC trigger event
011	PGxTRIGB trigger event
010	PGxTRIGA trigger event
001	Auto-Terminate: Terminates when PCI source transitions from active to inactive
000	Manual Terminate: Terminates on a write of '1' to the SWTERM bit location

Bit 11 – AQPS Acceptance Qualifier Polarity Select bit

Value	Description
1	Inverted
0	Not inverted

Bits 10:8 – AQSS[2:0] Acceptance Qualifier Source Selection bits

Value	Description
111	SWPCI control bit only (qualifier forced to '0')
110	Selects PCI Source #9
101	Selects PCI Source #8
100	Selects PCI Source #1 (PWM Generator output selected by the PWMPCI[2:0] bits)
011	PWM Generator is triggered
010	LEB is active
001	Duty cycle is active (base PWM Generator signal)
000	No acceptance qualifier is used (qualifier forced to '1')

Bit 7 – SWTERM PCI Software Termination bit

A write of '1' to this location will produce a termination event. This bit location always reads as '0'.

Bit 6 – PSYNC PCI Synchronization Control bit

Value	Description
1	PCI source is synchronized to PWM EOC
0	PCI source is not synchronized to PWM EOC

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Bit 5 – PPS PCI Polarity Select bit

Value	Description
1	Inverted
0	Not inverted

Bits 4:0 – PSS[4:0] PCI Source Selection bits

Value	Description
11111	CLC1
11110	Reserved
11101	Comparator 3 output
11100	Comparator 2 output
11011	Comparator 1 output
11010	PWM Event D
11001	PWM Event C
11000	PWM Event B
10111	PWM Event A
10110	Device pin, PCI[22]
10101	Device pin, PCI[21]
10100	Device pin, PCI[20]
10011	Device pin, PCI[19]
10010	RPn input, PCI18R
10001	RPn input, PCI17R
10000	RPn input, PCI16R
01111	RPn input, PCI15R
01110	RPn input, PCI14R
01101	RPn input, PCI13R
01100	RPn input, PCI12R
01011	RPn input, PCI11R
01010	RPn input, PCI10R
01001	RPn input, PCI9R
01000	RPn input, PCI8R
00111	Reserved
00110	Reserved
00101	Reserved
00100	Reserved
00011	Internally connected to Combo Trigger B
00010	Internally connected to Combo Trigger A
00001	Internally connected to the output of PWMPCI[2:0] MUX
00000	Tied to '0'

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High-Resolution PWM with Fine Edge Placement

13.6.20 PWM Generator x PCI Register High

Name: PGxFPCIH
Offset: 0x33A, 0x370, 0x3A6, 0x3DC, 0x448, 0x4B4

Note:

1. Selects '0' if selected PWM Generator is not present.

Bit	15	14	13	12	11	10	9	8
	BPEN		BPSEL[2:0]				ACP[2:0]	
Access	R/W	R/W	R/W	R/W		R/W	R/W	R/W
Reset	0	0	0	0		0	0	0

Bit	7	6	5	4	3	2	1	0
	SWPCI	SWPCIM[1:0]		LATMOD	TQPS		TQSS[2:0]	
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – BPEN PCI Bypass Enable bit

Value	Description
1	PCI function is enabled and local PCI logic is bypassed; PWM Generator will be controlled by PCI function in the PWM Generator selected by the BPSEL[2:0] bits
0	PCI function is not bypassed

Bits 14:12 – BPSEL[2:0] PCI Bypass Source Selection bits⁽¹⁾

Value	Description
111	PCI control is sourced from PWM Generator 8 PCI logic when BPEN = 1
110	Reserved
101	PCI control is sourced from PWM Generator 6 PCI logic when BPEN = 1
100	Reserved
011	PCI control is sourced from PWM Generator 4 PCI logic when BPEN = 1
010	PCI control is sourced from PWM Generator 3 PCI logic when BPEN = 1
001	PCI control is sourced from PWM Generator 2 PCI logic when BPEN = 1
000	PCI control is sourced from PWM Generator 1 PCI logic when BPEN = 1

Bits 10:8 – ACP[2:0] PCI Acceptance Criteria Selection bits

Value	Description
111	Reserved
110	Reserved
101	Latched any edge
100	Latched rising edge
011	Latched
010	Any edge
001	Rising edge
000	Level-sensitive

Bit 7 – SWPCI Software PCI Control bit

Value	Description
1	Drives a '1' to PCI logic assigned to by the SWPCIM[1:0] control bits
0	Drives a '0' to PCI logic assigned to by the SWPCIM[1:0] control bits

Bits 6:5 – SWPCIM[1:0] Software PCI Control Mode bits

Value	Description
11	Reserved
10	SWPCI bit is assigned to termination qualifier logic
01	SWPCI bit is assigned to acceptance qualifier logic

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Value	Description
00	SWPCI bit is assigned to PCI acceptance logic

Bit 4 – LATMOD PCI SR Latch Mode bit

Value	Description
1	SR latch is Reset-dominant in Latched Acceptance modes
0	SR latch is set-dominant in Latched Acceptance modes

Bit 3 – TQPS Termination Qualifier Polarity Select bit

Value	Description
1	Inverted
0	Not inverted

Bits 2:0 – TQSS[2:0] Termination Qualifier Source Selection bits

Value	Description
111	SWPCI control bit only (qualifier forced to '0')
110	Selects PCI Source #9
101	Selects PCI Source #8
100	Selects PCI Source #1 (PWM Generator output selected by the PWMPCI[2:0] bits)
011	PWM Generator is triggered
010	LEB is active
001	Duty cycle is active (base PWM Generator signal)
000	No termination qualifier used (qualifier forced to '1')

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13.6.21 PWM Generator x PCI Register Low

Name: PGxCLPCIL
Offset: 0x33C, 0x372, 0x3A8, 0x3DE, 0x44A, 0x4B6

Bit	15	14	13	12	11	10	9	8
	TSYNCDIS		TERM[2:0]			AQPS	AQSS[2:0]	
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	SWTERM	PSYNC	PPS			PSS[4:0]		
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – TSYNCDIS Termination Synchronization Disable bit

Value	Description
1	Termination of latched PCI occurs immediately
0	Termination of latched PCI occurs at PWM EOC

Bits 14:12 – TERM[2:0] Termination Event Selection bits

Value	Description
111	Selects PCI Source #9
110	Selects PCI Source #8
101	Selects PCI Source #1 (PWM Generator output selected by the PWMPCI[2:0] bits)
100	PGxTRIGC trigger event
011	PGxTRIGB trigger event
010	PGxTRIGA trigger event
001	Auto-Terminate: Terminates when PCI source transitions from active to inactive
000	Manual Terminate: Terminates on a write of '1' to the SWTERM bit location

Bit 11 – AQPS Acceptance Qualifier Polarity Select bit

Value	Description
1	Inverted
0	Not inverted

Bits 10:8 – AQSS[2:0] Acceptance Qualifier Source Selection bits

Value	Description
111	SWPCI control bit only (qualifier forced to '0')
110	Selects PCI Source #9
101	Selects PCI Source #8
100	Selects PCI Source #1 (PWM Generator output selected by the PWMPCI[2:0] bits)
011	PWM Generator is triggered
010	LEB is active
001	Duty cycle is active (base PWM Generator signal)
000	No acceptance qualifier is used (qualifier forced to '1')

Bit 7 – SWTERM PCI Software Termination bit

A write of '1' to this location will produce a termination event. This bit location always reads as '0'.

Bit 6 – PSYNC PCI Synchronization Control bit

Value	Description
1	PCI source is synchronized to PWM EOC
0	PCI source is not synchronized to PWM EOC

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Bit 5 – PPS PCI Polarity Select bit

Value	Description
1	Inverted
0	Not inverted

Bits 4:0 – PSS[4:0] PCI Source Selection bits

Value	Description
11111	CLC1
11110	Reserved
11101	Comparator 3 output
11100	Comparator 2 output
11011	Comparator 1 output
11010	PWM Event D
11001	PWM Event C
11000	PWM Event B
10111	PWM Event A
10110	Device pin, PCI[22]
10101	Device pin, PCI[21]
10100	Device pin, PCI[20]
10011	Device pin, PCI[19]
10010	RPn input, PCI18R
10001	RPn input, PCI17R
10000	RPn input, PCI16R
01111	RPn input, PCI15R
01110	RPn input, PCI14R
01101	RPn input, PCI13R
01100	RPn input, PCI12R
01011	RPn input, PCI11R
01010	RPn input, PCI10R
01001	RPn input, PCI9R
01000	RPn input, PCI8R
00111	Reserved
00110	Reserved
00101	Reserved
00100	Reserved
00011	Internally connected to Combo Trigger B
00010	Internally connected to Combo Trigger A
00001	Internally connected to the output of PWMPCI[2:0] MUX
00000	Tied to '0'

dsPIC33CK512MPT608 Family

High-Resolution PWM with Fine Edge Placement

13.6.22 PWM Generator x PCI Register High

Name: PGxCLPCIH
Offset: 0x33E, 0x374, 0x3AA, 0x3E0, 0x44C, 0x4B8

Note:

1. Selects '0' if selected PWM Generator is not present.

Bit	15	14	13	12	11	10	9	8
	BPEN		BPSEL[2:0]				ACP[2:0]	
Access	R/W	R/W	R/W	R/W		R/W	R/W	R/W
Reset	0	0	0	0		0	0	0

Bit	7	6	5	4	3	2	1	0
	SWPCI	SWPCIM[1:0]		LATMOD	TQPS		TQSS[2:0]	
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – BPEN PCI Bypass Enable bit

Value	Description
1	PCI function is enabled and local PCI logic is bypassed; PWM Generator will be controlled by PCI function in the PWM Generator selected by the BPSEL[2:0] bits
0	PCI function is not bypassed

Bits 14:12 – BPSEL[2:0] PCI Bypass Source Selection bits⁽¹⁾

Value	Description
111	PCI control is sourced from PWM Generator 8 PCI logic when BPEN = 1
110	Reserved
101	PCI control is sourced from PWM Generator 6 PCI logic when BPEN = 1
100	Reserved
011	PCI control is sourced from PWM Generator 4 PCI logic when BPEN = 1
010	PCI control is sourced from PWM Generator 3 PCI logic when BPEN = 1
001	PCI control is sourced from PWM Generator 2 PCI logic when BPEN = 1
000	PCI control is sourced from PWM Generator 1 PCI logic when BPEN = 1

Bits 10:8 – ACP[2:0] PCI Acceptance Criteria Selection bits

Value	Description
111	Reserved
110	Reserved
101	Latched any edge
100	Latched rising edge
011	Latched
010	Any edge
001	Rising edge
000	Level-sensitive

Bit 7 – SWPCI Software PCI Control bit

Value	Description
1	Drives a '1' to PCI logic assigned to by the SWPCIM[1:0] control bits
0	Drives a '0' to PCI logic assigned to by the SWPCIM[1:0] control bits

Bits 6:5 – SWPCIM[1:0] Software PCI Control Mode bits

Value	Description
11	Reserved
10	SWPCI bit is assigned to termination qualifier logic
01	SWPCI bit is assigned to acceptance qualifier logic

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High-Resolution PWM with Fine Edge Placement

Value	Description
00	SWPCI bit is assigned to PCI acceptance logic

Bit 4 – LATMOD PCI SR Latch Mode bit

Value	Description
1	SR latch is Reset-dominant in Latched Acceptance modes
0	SR latch is set-dominant in Latched Acceptance modes

Bit 3 – TQPS Termination Qualifier Polarity Select bit

Value	Description
1	Inverted
0	Not inverted

Bits 2:0 – TQSS[2:0] Termination Qualifier Source Selection bits

Value	Description
111	SWPCI control bit only (qualifier forced to '0')
110	Selects PCI Source #9
101	Selects PCI Source #8
100	Selects PCI Source #1 (PWM Generator output selected by the PWMPCI[2:0] bits)
011	PWM Generator is triggered
010	LEB is active
001	Duty cycle is active (base PWM Generator signal)
000	No termination qualifier used (qualifier forced to '1')

dsPIC33CK512MPT608 Family

High-Resolution PWM with Fine Edge Placement

13.6.23 PWM Generator x PCI Register Low

Name: PGxFFPCIL
Offset: 0x340, 0x376, 0x3AC, 0x3E2, 0x44E, 0x4BA

Bit	15	14	13	12	11	10	9	8
	TSYNCDIS		TERM[2:0]			AQPS	AQSS[2:0]	
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	SWTERM	PSYNC	PPS			PSS[4:0]		
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – TSYNCDIS Termination Synchronization Disable bit

Value	Description
1	Termination of latched PCI occurs immediately
0	Termination of latched PCI occurs at PWM EOC

Bits 14:12 – TERM[2:0] Termination Event Selection bits

Value	Description
111	Selects PCI Source #9
110	Selects PCI Source #8
101	Selects PCI Source #1 (PWM Generator output selected by the PWMPCI[2:0] bits)
100	PGxTRIGC trigger event
011	PGxTRIGB trigger event
010	PGxTRIGA trigger event
001	Auto-Terminate: Terminates when PCI source transitions from active to inactive
000	Manual Terminate: Terminates on a write of '1' to the SWTERM bit location

Bit 11 – AQPS Acceptance Qualifier Polarity Select bit

Value	Description
1	Inverted
0	Not inverted

Bits 10:8 – AQSS[2:0] Acceptance Qualifier Source Selection bits

Value	Description
111	SWPCI control bit only (qualifier forced to '0')
110	Selects PCI Source #9
101	Selects PCI Source #8
100	Selects PCI Source #1 (PWM Generator output selected by the PWMPCI[2:0] bits)
011	PWM Generator is triggered
010	LEB is active
001	Duty cycle is active (base PWM Generator signal)
000	No acceptance qualifier is used (qualifier forced to '1')

Bit 7 – SWTERM PCI Software Termination bit

A write of '1' to this location will produce a termination event. This bit location always reads as '0'.

Bit 6 – PSYNC PCI Synchronization Control bit

Value	Description
1	PCI source is synchronized to PWM EOC
0	PCI source is not synchronized to PWM EOC

dsPIC33CK512MPT608 Family

High-Resolution PWM with Fine Edge Placement

Bit 5 – PPS PCI Polarity Select bit

Value	Description
1	Inverted
0	Not inverted

Bits 4:0 – PSS[4:0] PCI Source Selection bits

Value	Description
11111	CLC1
11110	Reserved
11101	Comparator 3 output
11100	Comparator 2 output
11011	Comparator 1 output
11010	PWM Event D
11001	PWM Event C
11000	PWM Event B
10111	PWM Event A
10110	Device pin, PCI[22]
10101	Device pin, PCI[21]
10100	Device pin, PCI[20]
10011	Device pin, PCI[19]
10010	RPn input, PCI18R
10001	RPn input, PCI17R
10000	RPn input, PCI16R
01111	RPn input, PCI15R
01110	RPn input, PCI14R
01101	RPn input, PCI13R
01100	RPn input, PCI12R
01011	RPn input, PCI11R
01010	RPn input, PCI10R
01001	RPn input, PCI9R
01000	RPn input, PCI8R
00111	Reserved
00110	Reserved
00101	Reserved
00100	Reserved
00011	Internally connected to Combo Trigger B
00010	Internally connected to Combo Trigger A
00001	Internally connected to the output of PWMPCI[2:0] MUX
00000	Tied to '0'

dsPIC33CK512MPT608 Family

High-Resolution PWM with Fine Edge Placement

13.6.24 PWM Generator x PCI Register High

Name: PGxFFPCIH
Offset: 0x342, 0x378, 0x3AE, 0x3E4, 0x450, 0x4BC

Note:

1. Selects '0' if selected PWM Generator is not present.

Bit	15	14	13	12	11	10	9	8
	BPEN		BPSEL[2:0]				ACP[2:0]	
Access	R/W	R/W	R/W	R/W		R/W	R/W	R/W
Reset	0	0	0	0		0	0	0

Bit	7	6	5	4	3	2	1	0
	SWPCI	SWPCIM[1:0]		LATMOD	TQPS		TQSS[2:0]	
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – BPEN PCI Bypass Enable bit

Value	Description
1	PCI function is enabled and local PCI logic is bypassed; PWM Generator will be controlled by PCI function in the PWM Generator selected by the BPSEL[2:0] bits
0	PCI function is not bypassed

Bits 14:12 – BPSEL[2:0] PCI Bypass Source Selection bits⁽¹⁾

Value	Description
111	PCI control is sourced from PWM Generator 8 PCI logic when BPEN = 1
110	Reserved
101	PCI control is sourced from PWM Generator 6 PCI logic when BPEN = 1
100	Reserved
011	PCI control is sourced from PWM Generator 4 PCI logic when BPEN = 1
010	PCI control is sourced from PWM Generator 3 PCI logic when BPEN = 1
001	PCI control is sourced from PWM Generator 2 PCI logic when BPEN = 1
000	PCI control is sourced from PWM Generator 1 PCI logic when BPEN = 1

Bits 10:8 – ACP[2:0] PCI Acceptance Criteria Selection bits

Value	Description
111	Reserved
110	Reserved
101	Latched any edge
100	Latched rising edge
011	Latched
010	Any edge
001	Rising edge
000	Level-sensitive

Bit 7 – SWPCI Software PCI Control bit

Value	Description
1	Drives a '1' to PCI logic assigned to by the SWPCIM[1:0] control bits
0	Drives a '0' to PCI logic assigned to by the SWPCIM[1:0] control bits

Bits 6:5 – SWPCIM[1:0] Software PCI Control Mode bits

Value	Description
11	Reserved
10	SWPCI bit is assigned to termination qualifier logic
01	SWPCI bit is assigned to acceptance qualifier logic

dsPIC33CK512MPT608 Family

High-Resolution PWM with Fine Edge Placement

Value	Description
00	SWPCI bit is assigned to PCI acceptance logic

Bit 4 – LATMOD PCI SR Latch Mode bit

Value	Description
1	SR latch is Reset-dominant in Latched Acceptance modes
0	SR latch is Set-dominant in Latched Acceptance modes

Bit 3 – TQPS Termination Qualifier Polarity Select bit

Value	Description
1	Inverted
0	Not inverted

Bits 2:0 – TQSS[2:0] Termination Qualifier Source Selection bits

Value	Description
111	SWPCI control bit only (qualifier forced to '0')
110	Selects PCI Source #9
101	Selects PCI Source #8
100	Selects PCI Source #1 (PWM Generator output selected by the PWMPCI[2:0] bits)
011	PWM Generator is triggered
010	LEB is active
001	Duty cycle is active (base PWM Generator signal)
000	No termination qualifier used (qualifier forced to '1')

dsPIC33CK512MPT608 Family

High-Resolution PWM with Fine Edge Placement

13.6.25 PWM Generator x PCI Register Low

Name: PGxSPCIL
Offset: 0x344, 0x37A, 0x3B0, 0x3E6, 0x452, 0x4BE

Bit	15	14	13	12	11	10	9	8
	TSYNCDIS		TERM[2:0]			AQPS	AQSS[2:0]	
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	SWTERM	PSYNC	PPS			PSS[4:0]		
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – TSYNCDIS Termination Synchronization Disable bit

Value	Description
1	Termination of latched PCI occurs immediately
0	Termination of latched PCI occurs at PWM EOC

Bits 14:12 – TERM[2:0] Termination Event Selection bits

Value	Description
111	Selects PCI Source #9
110	Selects PCI Source #8
101	Selects PCI Source #1 (PWM Generator output selected by the PWMPCI[2:0] bits)
100	PGxTRIGC trigger event
011	PGxTRIGB trigger event
010	PGxTRIGA trigger event
001	Auto-Terminate: Terminates when PCI source transitions from active to inactive
000	Manual Terminate: Terminates on a write of '1' to the SWTERM bit location

Bit 11 – AQPS Acceptance Qualifier Polarity Select bit

Value	Description
1	Inverted
0	Not inverted

Bits 10:8 – AQSS[2:0] Acceptance Qualifier Source Selection bits

Value	Description
111	SWPCI control bit only (qualifier forced to '0')
110	Selects PCI Source #9
101	Selects PCI Source #8
100	Selects PCI Source #1 (PWM Generator output selected by the PWMPCI[2:0] bits)
011	PWM Generator is triggered
010	LEB is active
001	Duty cycle is active (base PWM Generator signal)
000	No acceptance qualifier is used (qualifier forced to '1')

Bit 7 – SWTERM PCI Software Termination bit

A write of '1' to this location will produce a termination event. This bit location always reads as '0'.

Bit 6 – PSYNC PCI Synchronization Control bit

Value	Description
1	PCI source is synchronized to PWM EOC
0	PCI source is not synchronized to PWM EOC

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High-Resolution PWM with Fine Edge Placement

Bit 5 – PPS PCI Polarity Select bit

Value	Description
1	Inverted
0	Not inverted

Bits 4:0 – PSS[4:0] PCI Source Selection bits

Value	Description
11111	CLC1
11110	Reserved
11101	Comparator 3 output
11100	Comparator 2 output
11011	Comparator 1 output
11010	PWM Event D
11001	PWM Event C
11000	PWM Event B
10111	PWM Event A
10110	Device pin, PCI[22]
10101	Device pin, PCI[21]
10100	Device pin, PCI[20]
10011	Device pin, PCI[19]
10010	RPn input, PCI18R
10001	RPn input, PCI17R
10000	RPn input, PCI16R
01111	RPn input, PCI15R
01110	RPn input, PCI14R
01101	RPn input, PCI13R
01100	RPn input, PCI12R
01011	RPn input, PCI11R
01010	RPn input, PCI10R
01001	RPn input, PCI9R
01000	RPn input, PCI8R
00111	Reserved
00110	Reserved
00101	Reserved
00100	Reserved
00011	Internally connected to Combo Trigger B
00010	Internally connected to Combo Trigger A
00001	Internally connected to the output of PWMPCI[2:0] MUX
00000	Tied to '0'

dsPIC33CK512MPT608 Family

High-Resolution PWM with Fine Edge Placement

13.6.26 PWM Generator x PCI Register High

Name: PGxSPCIH
Offset: 0x346, 0x37C, 0x3B2, 0x3E8, 0x454, 0x4C0

Note:

1. Selects '0' if selected PWM Generator is not present.

Bit	15	14	13	12	11	10	9	8
	BPEN		BPSEL[2:0]				ACP[2:0]	
Access	R/W	R/W	R/W	R/W		R/W	R/W	R/W
Reset	0	0	0	0		0	0	0

Bit	7	6	5	4	3	2	1	0
	SWPCI	SWPCIM[1:0]		LATMOD	TQPS		TQSS[2:0]	
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – BPEN PCI Bypass Enable bit

Value	Description
1	PCI function is enabled and local PCI logic is bypassed; PWM Generator will be controlled by PCI function in the PWM Generator selected by the BPSEL[2:0] bits
0	PCI function is not bypassed

Bits 14:12 – BPSEL[2:0] PCI Bypass Source Selection bits⁽¹⁾

Value	Description
111	PCI control is sourced from PWM Generator 8 PCI logic when BPEN = 1
110	Reserved
101	PCI control is sourced from PWM Generator 6 PCI logic when BPEN = 1
100	Reserved
011	PCI control is sourced from PWM Generator 4 PCI logic when BPEN = 1
010	PCI control is sourced from PWM Generator 3 PCI logic when BPEN = 1
001	PCI control is sourced from PWM Generator 2 PCI logic when BPEN = 1
000	PCI control is sourced from PWM Generator 1 PCI logic when BPEN = 1

Bits 10:8 – ACP[2:0] PCI Acceptance Criteria Selection bits

Value	Description
111	Reserved
110	Reserved
101	Latched any edge
100	Latched rising edge
011	Latched
010	Any edge
001	Rising edge
000	Level-sensitive

Bit 7 – SWPCI Software PCI Control bit

Value	Description
1	Drives a '1' to PCI logic assigned to by the SWPCIM[1:0] control bits
0	Drives a '0' to PCI logic assigned to by the SWPCIM[1:0] control bits

Bits 6:5 – SWPCIM[1:0] Software PCI Control Mode bits

Value	Description
11	Reserved
10	SWPCI bit is assigned to termination qualifier logic
01	SWPCI bit is assigned to acceptance qualifier logic

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High-Resolution PWM with Fine Edge Placement

Value	Description
00	SWPCI bit is assigned to PCI acceptance logic

Bit 4 – LATMOD PCI SR Latch Mode bit

Value	Description
1	SR latch is Reset-dominant in Latched Acceptance modes
0	SR latch is set-dominant in Latched Acceptance modes

Bit 3 – TQPS Termination Qualifier Polarity Select bit

Value	Description
1	Inverted
0	Not inverted

Bits 2:0 – TQSS[2:0] Termination Qualifier Source Selection bits

Value	Description
111	SWPCI control bit only (qualifier forced to '0')
110	Selects PCI Source #9
101	Selects PCI Source #8
100	Selects PCI Source #1 (PWM Generator output selected by the PWMPCI[2:0] bits)
011	PWM Generator is triggered
010	LEB is active
001	Duty cycle is active (base PWM Generator signal)
000	No termination qualifier used (qualifier forced to '1')

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High-Resolution PWM with Fine Edge Placement

13.6.27 PWM Generator x Leading-Edge Blanking Register Low

Name: PGxLEBL
Offset: 0x348, 0x37E, 0x3B4, 0x3EA, 0x456, 0x4C2

Note:

1. Bits[2:0] are read-only and always remain as '0'.

Bit	15	14	13	12	11	10	9	8
	LEB[15:8]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	LEB[7:0]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – LEB[15:0] Leading-Edge Blanking Period bits⁽¹⁾

Leading-Edge Blanking period. The three LSBs of the blanking time are not used, providing a blanking resolution of eight PGx_clks. The minimum blanking period is eight PGx_clks which occurs when LEB[15:3] = 0.

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High-Resolution PWM with Fine Edge Placement

13.6.28 PWM Generator x Leading-Edge Blanking Register High

Name: PGxLEBH
Offset: 0x34A, 0x380, 0x3B6, 0x3EC, 0x458, 0x4C4

Note:

1. The selected PWM Generator source does not affect the LEB counter. This source can be optionally used as a PCI input, PCI qualifier, PCI terminator or PCI terminator qualifier.

Bit	15	14	13	12	11	10	9	8
						PWMPCI[2:0]		
Access						R/W	R/W	R/W
Reset						0	0	0

Bit	7	6	5	4	3	2	1	0
					PHR	PHF	PLR	PLF
Access					R/W	R/W	R/W	R/W
Reset					0	0	0	0

Bits 10:8 – PWMPCI[2:0] PWM Source for PCI Selection bits⁽¹⁾

Value	Description
111	PWM Generator #8 output is made available to PCI logic
110	Reserved
101	PWM Generator #6 output is made available to PCI logic
100	Reserved
011	PWM Generator #4 output is made available to PCI logic
010	PWM Generator #3 output is made available to PCI logic
001	PWM Generator #2 output is made available to PCI logic
000	PWM Generator #1 output is made available to PCI logic

Bit 3 – PHR PWMxH Rising bit

Value	Description
1	Rising edge of PWMxH will trigger the LEB duration counter
0	LEB ignores the rising edge of PWMxH

Bit 2 – PHF PWMxH Falling bit

Value	Description
1	Falling edge of PWMxH will trigger the LEB duration counter
0	LEB ignores the falling edge of PWMxH

Bit 1 – PLR PWMxL Rising bit

Value	Description
1	Rising edge of PWMxL will trigger the LEB duration counter
0	LEB ignores the rising edge of PWMxL

Bit 0 – PLF PWMxL Falling bit

Value	Description
1	Falling edge of PWMxL will trigger the LEB duration counter
0	LEB ignores the falling edge of PWMxL

dsPIC33CK512MPT608 Family

High-Resolution PWM with Fine Edge Placement

13.6.29 PWM Generator x Phase Register

Name: PGxPHASE
Offset: 0x34C, 0x382, 0x3B8, 0x3EE, 0x45A, 0x4C6

Bit	15	14	13	12	11	10	9	8
	PGxPHASE[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PGxPHASE[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – PGxPHASE[15:0] PWM Generator x Phase Register bits

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High-Resolution PWM with Fine Edge Placement

13.6.30 PWM Generator x Duty Cycle Register

Name: PGxDC
Offset: 0x34E, 0x384, 0x3BA, 0x3F0, 0x45C, 0x4C8

Bit	15	14	13	12	11	10	9	8
	PGxDC[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PGxDC[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – PGxDC[15:0] PWM Generator x Duty Cycle Register bits

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High-Resolution PWM with Fine Edge Placement

13.6.31 PWM Generator x Duty Cycle Adjustment Register

Name: PGxDCA
Offset: 0x350, 0x386, 0x3BC, 0x3F2, 0x45E, 0x4CA

Bit	15	14	13	12	11	10	9	8
Access								
Reset								
Bit	7	6	5	4	3	2	1	0
	PGxDCA[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 7:0 – PGxDCA[7:0] PWM Generator x Duty Cycle Adjustment Value bits

Depending on the state of the selected PCI source, the PGxDCA value will be added to the value in the PGxDC register to create the effective duty cycle. When the PCI source is active, PGxDCA is added. When the PCI source is inactive, no adjustment is made. Duty cycle adjustment is disabled when PGxDCA[7:0] = 0. The PCI source is selected using the DTCMPSEL bit.

dsPIC33CK512MPT608 Family

High-Resolution PWM with Fine Edge Placement

13.6.32 PWM Generator x Period Register

Name: PGxPER
Offset: 0x352, 0x388, 0x3BE, 0x3F4, 0x460, 0x4CC

Note:

1. Period values less than '0x0010' should not be selected.

Bit	15	14	13	12	11	10	9	8
	PGxPER[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PGxPER[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – PGxPER[15:0] PWM Generator x Period Register bits⁽¹⁾

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High-Resolution PWM with Fine Edge Placement

13.6.33 PWM Generator x Trigger A Register

Name: PGxTRIGA
Offset: 0x354, 0x38A, 0x3C0, 0x3F6, 0x462, 0x4CE

Bit	15	14	13	12	11	10	9	8
	PGxTRIGA[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PGxTRIGA[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – PGxTRIGA[15:0] PWM Generator x Trigger A Register bits

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High-Resolution PWM with Fine Edge Placement

13.6.34 PWM Generator x Trigger B Register

Name: PGxTRIGB
Offset: 0x356, 0x38C, 0x3C2, 0x3F8, 0x464, 0x4D0

Bit	15	14	13	12	11	10	9	8
	PGxTRIGB[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PGxTRIGB[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – PGxTRIGB[15:0] PWM Generator x Trigger B Register bits

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High-Resolution PWM with Fine Edge Placement

13.6.35 PWM Generator x Trigger C Register

Name: PGxTRIGC
Offset: 0x358, 0x38E, 0x3C4, 0x3FA, 0x466, 0x4D2

Bit	15	14	13	12	11	10	9	8
	PGxTRIGC[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PGxTRIGC[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – PGxTRIGC[15:0] PWM Generator x Trigger C Register bits

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High-Resolution PWM with Fine Edge Placement

13.6.36 PWM Generator x Dead-Time Register Low

Name: PGxDTL
Offset: 0x35A, 0x390, 0x3C6, 0x3FC, 0x468, 0x4D4

Note:

- DTL[13:11] bits are not available when HREN (PGxCONL[7]) = 0.

Bit	15	14	13	12	11	10	9	8
			DTL[13:8]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	DTL[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 13:0 – DTL[13:0] PWMxL Dead-Time Delay bits⁽¹⁾

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High-Resolution PWM with Fine Edge Placement

13.6.37 PWM Generator x Dead-Time Register High

Name: PGxDTH
Offset: 0x35C, 0x392, 0x3C8, 0x3FE, 0x46A, 0x4DA

Bit	15	14	13	12	11	10	9	8
			DTH[13:8]					
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	DTH[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 13:0 – DTH[13:0] PWMxH Dead-Time Delay bits

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High-Resolution PWM with Fine Edge Placement

13.6.38 PWM Generator x Capture Register

Name: PGxCAP

Offset: 0x35E, 0x394, 0x3CA, 0x400, 0x46C, 0x4D8

Note:

1. PGxCAP[1:0] will read as '00' in Standard Resolution mode. PGxCAP[4:0] will read as '00000' in High-Resolution mode.

Bit	15	14	13	12	11	10	9	8
	PGxCAP[15:8]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PGxCAP[7:0]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – PGxCAP[15:0] PGx Time Base Capture bits⁽¹⁾

14. High-Speed, 12-Bit Analog-to-Digital Converter

Notes:

1. This data sheet summarizes the features of the dsPIC33CK512MPT608 family of devices. It is not intended to be a comprehensive reference source. To complement the information in this data sheet, refer to “**12-Bit High-Speed, Multiple SARs A/D Converter (ADC)**” (www.microchip.com/DS70005213) in the “dsPIC33/PIC24 Family Reference Manual”.
2. Some registers and associated bits described in this section may not be available on all devices due to the number of implemented ADC channels. Refer to [dsPIC33CK512MPT608 Product Families](#) for ADC channel availability on device variants.

dsPIC33CK512MPT608 devices have a high-speed, 12-bit Analog-to-Digital Converter (ADC) that features a low conversion latency, high resolution and oversampling capabilities to improve performance in AC/DC and DC/DC power converters. The devices implement the ADC with five SAR cores, four dedicated and one shared.

The number of available channels and negative inputs is dependent on package size, as shown in the table below.

Table 14-1. ADC External Input Availability

Package Type	External Inputs	Negative Inputs
100-Pin	AN0-AN23, AN26, AN27	ANN0-ANN4

14.1 ADC Features Overview

The high-speed, 12-bit multiple SARs Analog-to-Digital Converter (ADC) includes the following features:

- One Shared (common) Core
- User-Configurable Resolution of Up to 12 Bits
- Up to 3.5 Msps Conversion Rate per Channel at 12-Bit Resolution
- Low Latency Conversion
- Up to 28 Analog Input Channels with a Separate 16-Bit Conversion Result Register for each Input Channel
- Conversion Result can be Formatted as Unsigned or Signed Data, on a per Channel Basis, for All Channels
- Channel Scan Capability
- Multiple Conversion Trigger Options, Including:
 - PWM triggers from CPU core
 - SCCP modules triggers
 - CLC modules triggers
 - External pin trigger event (ADTRG31)
 - Software trigger
- Four Integrated Digital Comparators with Dedicated Interrupts:
 - Multiple comparison options
 - Assignable to specific analog inputs
- Four Oversampling Filters with Dedicated Interrupts:
 - Provide increased resolution
 - Assignable to a specific analog input

Simplified block diagrams of the 12-bit ADC are shown in [Figure 14-1](#) and [Figure 14-2](#).

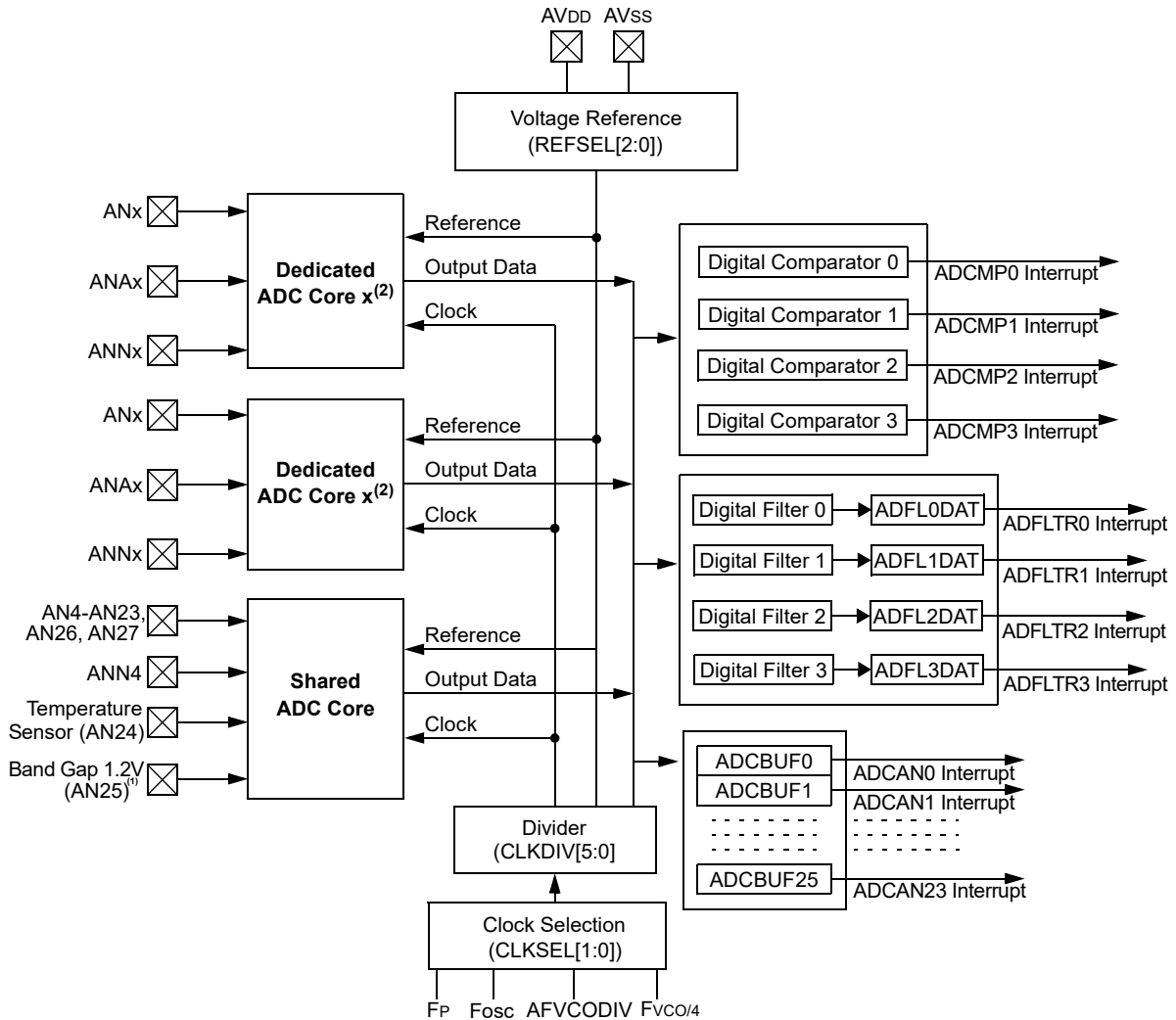
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The analog inputs (channels) are connected through multiplexers and switches to the Sample-and-Hold (S&H) circuit of the ADC core. The core uses the channel information (the output format, the Measurement mode and the input number) to process the analog sample. When conversion is complete, the result is stored in the result buffer for the specific analog input, and passed to the digital filter and digital comparator if they were configured to use data from this particular channel. If multiple ADC inputs request conversion on the shared core, the module will convert them in a sequential manner, starting with the lowest order input.

The ADC provides each analog input the ability to specify its own trigger source. This capability allows the ADC to sample and convert analog inputs that are associated with PWM Generators operating on independent time bases.

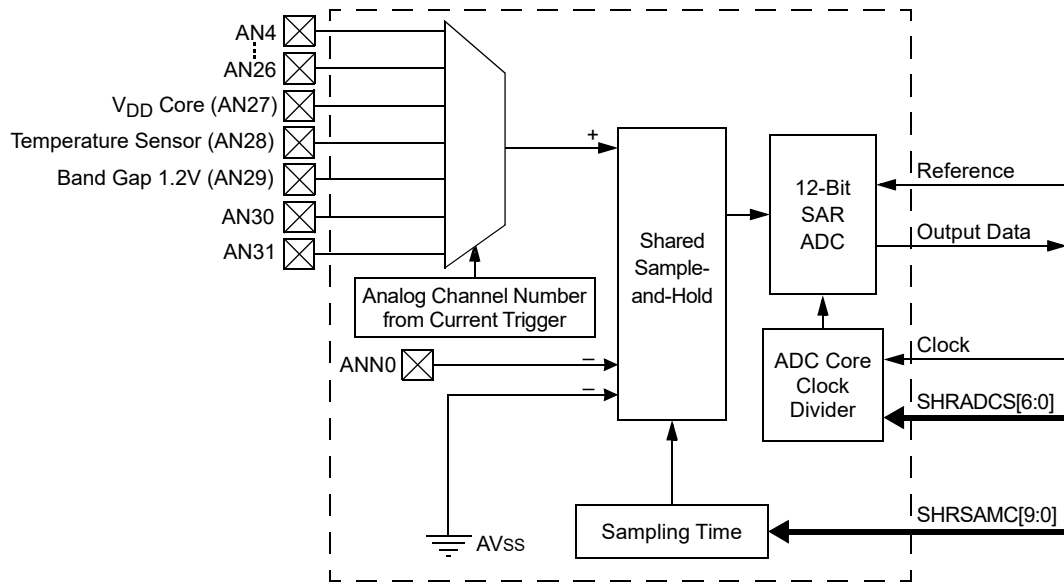
Figure 14-1. ADC Module Block Diagram



Notes:

1. Band Gap Reference (V_{BG}) is an internal analog input and is not available on device pins.
2. Your particular device may have a different number of dedicated cores; see the device-specific data sheet, pinout figures or [Table 1-1](#).

Figure 14-2. Shared Core Block Diagram⁽¹⁾



Note:

1. Check the device pinout diagram to verify if the pin is available on the specific device.

14.2 Temperature Sensor

The ADC channel, AN24, is connected to a forward biased diode. It can be used to measure die temperature. This diode provides a voltage output that can be monitored by the ADC.

The temperature coefficient is listed in [Table 33-42](#). To get the exact gain and offset numbers, the two temperature points calibration is recommended.

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14.3 ADC Control Registers

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x00 ... 0x0AFF	Reserved									
0x0B00	ADCON1L	15:8	ADON		ADSIDL					
		7:0								
0x0B02	ADCON1H	15:8								
		7:0	FORM	SHRRES[1:0]						
0x0B04	ADCON2L	15:8	REFCIE	REFERCIE		EIEN	PTGEN	SHREISEL[2:0]		
		7:0					SHRADCS[6:0]			
0x0B06	ADCON2H	15:8	REFRDY	REFERR				SHRSAMC[9:8]		
		7:0					SHRSAMC[7:0]			
0x0B08	ADCON3L	15:8	REFSEL[2:0]			SUSPEND	SUSPCIE	SUSPRDY	SHRSAMP	CNVRTCH
		7:0	SWLCTRG	SWCTRG				CNVCHSEL[5:0]		
0x0B0A	ADCON3H	15:8	CLKSEL[1:0]					CLKDIV[5:0]		
		7:0	SHREN				C3EN	C2EN	C1EN	C0EN
0x0B0C	ADCON4L	15:8						Reserved[3:0]		
		7:0					SAMC3EN	SAMC2EN	SAMC1EN	SAMC0EN
0x0B0E	ADCON4H	15:8								
		7:0	C3CHS[1:0]	C2CHS[1:0]			C1CHS[1:0]	C0CHS[1:0]		
0x0B10	ADMOD0L	15:8	DIFF7	SIGN7	DIFF6	SIGN6	DIFF5	SIGN5	DIFF4	SIGN4
		7:0	DIFF3	SIGN3	DIFF2	SIGN2	DIFF1	SIGN1	DIFF0	SIGN0
0x0B12	ADMOD0H	15:8	DIFF15	SIGN15	DIFF14	SIGN14	DIFF13	SIGN13	DIFF12	SIGN12
		7:0	DIFF11	SIGN11	DIFF10	SIGN10	DIFF9	SIGN9	DIFF8	SIGN8
0x0B14	ADMOD1L	15:8	DIFF23	SIGN23	DIFF22	SIGN22	DIFF21	SIGN21	DIFF20	SIGN20
		7:0	DIFF19	SIGN19	DIFF18	SIGN18	DIFF17	SIGN17	DIFF16	SIGN16
0x0B16	ADMOD1H	15:8								
		7:0	DIFF27	SIGN27	DIFF26	SIGN26	DIFF25	SIGN25	DIFF24	SIGN24
0x0B18 ... 0x0B1F	Reserved									
0x0B20	ADIEL	15:8					IE[15:8]			
		7:0					IE[7:0]			
0x0B22	ADIEH	15:8						IE[27:24]		
		7:0					IE[23:16]			
0x0B24 ... 0x0B2F	Reserved									
0x0B30	ADSTATL	15:8					AN[15:0]RDY			
		7:0					AN[15:0]RDY			
0x0B32	ADSTATH	15:8						AN[27:24]RDY		
		7:0					AN[23:16]RDY			
0x0B34 ... 0x0B37	Reserved									
0x0B38	ADCMP0ENL	15:8					COMPEN[15:8]			
		7:0					COMPEN[7:0]			
0x0B3A	ADCMP0ENH	15:8						COMPEN[27:24]		
		7:0					COMPEN[23:16]			
0x0B3C	ADCMP0LO	15:8					COMPLO[15:8]			
		7:0					COMPLO[7:0]			
0x0B3E	ADCMP0HI	15:8					CMPHI[15:8]			
		7:0					CMPHI[7:0]			
0x0B40	ADCMP1ENL	15:8					COMPEN[15:8]			
		7:0					COMPEN[7:0]			
0x0B42	ADCMP1ENH	15:8						COMPEN[27:24]		
		7:0					COMPEN[23:16]			

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.....continued

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x0B44	ADCMP1LO	15:8					CMPLO[15:8]			
		7:0					CMPLO[7:0]			
0x0B46	ADCMP1HI	15:8					CMPHI[15:8]			
		7:0					CMPHI[7:0]			
0x0B48	ADCMP2ENL	15:8					CMPEN[15:8]			
		7:0					CMPEN[7:0]			
0x0B4A	ADCMP2ENH	15:8					CMPEN[27:24]			
		7:0					CMPEN[23:16]			
0x0B4C	ADCMP2LO	15:8					CMPLO[15:8]			
		7:0					CMPLO[7:0]			
0x0B4E	ADCMP2HI	15:8					CMPHI[15:8]			
		7:0					CMPHI[7:0]			
0x0B50	ADCMP3ENL	15:8					CMPEN[15:8]			
		7:0					CMPEN[7:0]			
0x0B52	ADCMP3ENH	15:8					CMPEN[27:24]			
		7:0					CMPEN[23:16]			
0x0B54	ADCMP3LO	15:8					CMPLO[15:8]			
		7:0					CMPLO[7:0]			
0x0B56	ADCMP3HI	15:8					CMPHI[15:8]			
		7:0					CMPHI[7:0]			
0x0B58 ... 0x0B67	Reserved									
0x0B68	ADFL0DAT	15:8					FLDATA[15:8]			
		7:0					FLDATA[7:0]			
0x0B6A	ADFL0CON	15:8	FLEN	MODE[1:0]		OVRSAM[2:0]		IE	RDY	
		7:0				FLCHSEL[4:0]				
0x0B6C	ADFL1DAT	15:8					FLDATA[15:8]			
		7:0					FLDATA[7:0]			
0x0B6E	ADFL1CON	15:8	FLEN	MODE[1:0]		OVRSAM[2:0]		IE	RDY	
		7:0				FLCHSEL[4:0]				
0x0B70	ADFL2DAT	15:8					FLDATA[15:8]			
		7:0					FLDATA[7:0]			
0x0B72	ADFL2CON	15:8	FLEN	MODE[1:0]		OVRSAM[2:0]		IE	RDY	
		7:0				FLCHSEL[4:0]				
0x0B74	ADFL3DAT	15:8					FLDATA[15:8]			
		7:0					FLDATA[7:0]			
0x0B76	ADFL3CON	15:8	FLEN	MODE[1:0]		OVRSAM[2:0]		IE	RDY	
		7:0				FLCHSEL[4:0]				
0x0B78 ... 0x0B7F	Reserved									
0x0B80	ADTRIG0L	15:8				TRGSRC1[4:0]				
		7:0				TRGSRC0[4:0]				
0x0B82	ADTRIG0H	15:8				TRGSRC3[4:0]				
		7:0				TRGSRC2[4:0]				
0x0B84	ADTRIG1L	15:8				TRGSRC5[4:0]				
		7:0				TRGSRC4[4:0]				
0x0B86	ADTRIG1H	15:8				TRGSRC7[4:0]				
		7:0				TRGSRC6[4:0]				
0x0B88	ADTRIG2L	15:8				TRGSRC9[4:0]				
		7:0				TRGSRC8[4:0]				
0x0B8A	ADTRIG2H	15:8				TRGSRC11[4:0]				
		7:0				TRGSRC10[4:0]				
0x0B8C	ADTRIG3L	15:8				TRGSRC13[4:0]				
		7:0				TRGSRC12[4:0]				
0x0B8E	ADTRIG3H	15:8				TRGSRC15[4:0]				
		7:0				TRGSRC14[4:0]				

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.....continued										
Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x0B90	ADTRIG4L	15:8				TRGSRC17[4:0]				
		7:0				TRGSRC16[4:0]				
0x0B92	ADTRIG4H	15:8				TRGSRC19[4:0]				
		7:0				TRGSRC18[4:0]				
0x0B94	ADTRIG5L	15:8				TRGSRC21[4:0]				
		7:0				TRGSRC20[4:0]				
0x0B96 ... 0x0B9F	Reserved									
0x0BA0	ADCMP0CON	15:8				CHNL[4:0]				
		7:0	CMPEN	IE	STAT	BTWN	HIHI	HILO	LOHI	LOLO
0x0BA2 ... 0x0BA3	Reserved									
0x0BA4	ADCMP1CON	15:8				CHNL[4:0]				
		7:0	CMPEN	IE	STAT	BTWN	HIHI	HILO	LOHI	LOLO
0x0BA6 ... 0x0BA7	Reserved									
0x0BA8	ADCMP2CON	15:8				CHNL[4:0]				
		7:0	CMPEN	IE	STAT	BTWN	HIHI	HILO	LOHI	LOLO
0x0BAA ... 0x0BAB	Reserved									
0x0BAC	ADCMP3CON	15:8				CHNL[4:0]				
		7:0	CMPEN	IE	STAT	BTWN	HIHI	HILO	LOHI	LOLO
0x0BAE ... 0x0BCF	Reserved									
0x0BD0	ADLVLTRGL	15:8	LVLEN[15:8]							
		7:0	LVLEN[7:0]							
0x0BD2	ADLVLTRGH	15:8	LVLEN[27:24]							
		7:0	LVLEN[23:16]							
0x0BD4	ADCORE0L	15:8	SAMC[9:8]							
		7:0	SAMC[7:0]							
0x0BD6	ADCORE0H	15:8	EISEL[2:0]							
		7:0	ADCS[6:0]							
0x0BD8	ADCORE1L	15:8	SAMC[9:8]							
		7:0	SAMC[7:0]							
0x0BDA	ADCORE1H	15:8	EISEL[2:0]							
		7:0	ADCS[6:0]							
0x0BDC	ADCORE2L	15:8	SAMC[9:8]							
		7:0	SAMC[7:0]							
0x0BDE	ADCORE2H	15:8	EISEL[2:0]							
		7:0	ADCS[6:0]							
0x0BE0	ADCORE3L	15:8	SAMC[9:8]							
		7:0	SAMC[7:0]							
0x0BE2	ADCORE3H	15:8	EISEL[2:0]							
		7:0	ADCS[6:0]							
0x0BE4 ... 0x0BEF	Reserved									
0x0BF0	ADEIEL	15:8	EIEN[15:8]							
		7:0	EIEN[7:0]							
0x0BF2	ADEIEH	15:8	EIEN[27:24]							
		7:0	EIEN[23:16]							
0x0BF4 ... 0x0BF7	Reserved									

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High-Speed, 12-Bit Analog-to-Digital Convert...

.....continued

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x0BF8	ADEISTATL	15:8	EISTAT[15:8]							
		7:0	EISTAT[7:0]							
0x0BFA	ADEISTATH	15:8	EISTAT[27:24]							
		7:0	EISTAT[23:16]							
0x0BFC ... 0x0BFF	Reserved									
0x0C00	ADCON5L	15:8	SHRRDY				C3RDY	C2RDY	C1RDY	C0RDY
		7:0	SHRPWR				C3PWR	C2PWR	C1PWR	C0PWR
0x0C02	ADCON5H	15:8	WARMTIME[3:0]							
		7:0	SHRCIE				C3CIE	C2CIE	C1CIE	C0CIE
0x0C04 ... 0x0C0B	Reserved									
0x0C0C	ADCBUF0	15:8	ADCBUF0[15:8]							
		7:0	ADCBUF0[7:0]							
0x0C0E	ADCBUF1	15:8	ADCBUF1[15:8]							
		7:0	ADCBUF1[7:0]							
0x0C10	ADCBUF2	15:8	ADCBUF2[15:8]							
		7:0	ADCBUF2[7:0]							
0x0C12	ADCBUF3	15:8	ADCBUF3[15:8]							
		7:0	ADCBUF3[7:0]							
0x0C14	ADCBUF4	15:8	ADCBUF4[15:8]							
		7:0	ADCBUF4[7:0]							
0x0C16	ADCBUF5	15:8	ADCBUF5[15:8]							
		7:0	ADCBUF5[7:0]							
0x0C18	ADCBUF6	15:8	ADCBUF6[15:8]							
		7:0	ADCBUF6[7:0]							
0x0C1A	ADCBUF7	15:8	ADCBUF7[15:8]							
		7:0	ADCBUF7[7:0]							
0x0C1C	ADCBUF8	15:8	ADCBUF8[15:8]							
		7:0	ADCBUF8[7:0]							
0x0C1E	ADCBUF9	15:8	ADCBUF9[15:8]							
		7:0	ADCBUF9[7:0]							
0x0C20	ADCBUF10	15:8	ADCBUF10[15:8]							
		7:0	ADCBUF10[7:0]							
0x0C22	ADCBUF11	15:8	ADCBUF11[15:8]							
		7:0	ADCBUF11[7:0]							
0x0C24	ADCBUF12	15:8	ADCBUF12[15:8]							
		7:0	ADCBUF12[7:0]							
0x0C26	ADCBUF13	15:8	ADCBUF13[15:8]							
		7:0	ADCBUF13[7:0]							
0x0C28	ADCBUF14	15:8	ADCBUF14[15:8]							
		7:0	ADCBUF14[7:0]							
0x0C2A	ADCBUF15	15:8	ADCBUF15[15:8]							
		7:0	ADCBUF15[7:0]							
0x0C2C	ADCBUF16	15:8	ADCBUF16[15:8]							
		7:0	ADCBUF16[7:0]							
0x0C2E	ADCBUF17	15:8	ADCBUF17[15:8]							
		7:0	ADCBUF17[7:0]							
0x0C30	ADCBUF18	15:8	ADCBUF18[15:8]							
		7:0	ADCBUF18[7:0]							
0x0C32	ADCBUF19	15:8	ADCBUF19[15:8]							
		7:0	ADCBUF19[7:0]							
0x0C34	ADCBUF20	15:8	ADCBUF20[15:8]							
		7:0	ADCBUF20[7:0]							
0x0C36	ADCBUF21	15:8	ADCBUF21[15:8]							
		7:0	ADCBUF21[7:0]							

14.3.1 ADC Control Register 1 Low

Name: ADCON1L

Offset: 0xB00

Note:

1. Set the ADON bit only after the ADC module has been configured. Changing ADC Configuration bits when ADON = 1 will result in unpredictable behavior.

Bit	15	14	13	12	11	10	9	8
	ADON		ADSIDL					
Access	R/W		R/W					
Reset	0		0					

Bit	7	6	5	4	3	2	1	0
Access								
Reset								

Bit 15 – ADON ADC Enable bit⁽¹⁾

Value	Description
1	ADC module is enabled
0	ADC module is off

Bit 13 – ADSIDL ADC Stop in Idle Mode bit

Value	Description
1	Discontinues module operation when device enters Idle mode
0	Continues module operation in Idle mode

14.3.2 ADC Control Register 1 High

Name: ADCON1H
Offset: 0xB02

Bit	15	14	13	12	11	10	9	8
Access								
Reset								

Bit	7	6	5	4	3	2	1	0
	FORM	SHRRES[1:0]						
Access	R/W	R/W	R/W					
Reset	0	1	1					

Bit 7 – FORM Fractional Data Output Format bit

Value	Description
1	Fractional
0	Integer

Bits 6:5 – SHRRES[1:0] Shared ADC Core Resolution Selection bits

Value	Description
11	12-bit resolution
10	10-bit resolution
01	8-bit resolution
00	6-bit resolution

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14.3.3 ADC Control Register 2 Low

Name: ADCON2L

Offset: 0xB04

Note:

- For the 6-bit shared ADC core resolution (SHRRES[1:0] = 00), the SHREISEL[2:0] settings, from '100' to '111', are not valid and should not be used. For the 8-bit shared ADC core resolution (SHRRES[1:0] = 01), the SHREISEL[2:0] settings, '110' and '111', are not valid and should not be used.

Bit	15	14	13	12	11	10	9	8
	REFCIE	REFERCIE		EIEN	PTGEN		SHREISEL[2:0]	
Access	R/W	R/W		R/W	R/W	R/W	R/W	R/W
Reset	0	0		0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
					SHRADCS[6:0]			
Access		R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset		0	0	0	0	0	0	0

Bit 15 – REFCIE Band Gap and Reference Voltage Ready Common Interrupt Enable bit

Value	Description
1	Common interrupt will be generated when the band gap becomes ready
0	Common interrupt is disabled for the band gap ready event

Bit 14 – REPERCIE Band Gap or Reference Voltage Error Common Interrupt Enable bit

Value	Description
1	Common interrupt will be generated when a band gap or reference voltage error is detected
0	Common interrupt is disabled for the band gap and reference voltage error event

Bit 12 – EIEN Early Interrupts Enable bit

Value	Description
1	Early interrupt feature is enabled for input channel interrupts (when EISTATx flag is set)
0	Individual interrupts are generated when conversion is done (when ANxRDY flag is set)

Bit 11 – PTGEN External Conversion Request Interface bit

Setting this bit will enable the PTG to request conversion of an ADC input.

Bits 10:8 – SHREISEL[2:0] Shared Core Early Interrupt Time Selection bits⁽¹⁾

Value	Description
111	Early interrupt is set, interrupt is generated eight T_{ADCORE} clocks prior to when data are ready
110	Early interrupt is set, interrupt is generated seven T_{ADCORE} clocks prior to when data are ready
101	Early interrupt is set, interrupt is generated six T_{ADCORE} clocks prior to when data are ready
100	Early interrupt is set, interrupt is generated five T_{ADCORE} clocks prior to when the data are ready
011	Early interrupt is set, interrupt is generated four T_{ADCORE} clocks prior to when data are ready
010	Early interrupt is set, interrupt is generated three T_{ADCORE} clocks prior to when data are ready
001	Early interrupt is set, interrupt is generated two T_{ADCORE} clocks prior to when data are ready
000	Early interrupt is set, interrupt is generated one T_{ADCORE} clock prior to when data are ready

Bits 6:0 – SHRADCS[6:0] Shared ADC Core Input Clock Divider bits

These bits determine the number of $T_{Coresrc}$ (Source Clock Periods) for one shared T_{ADCORE} (Core Clock Period).

Value	Description
1111111	254 Source Clock Periods
. . .	
0000011	6 Source Clock Periods
0000010	4 Source Clock Periods

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Value	Description
0000001	2 Source Clock Periods
0000000	2 Source Clock Periods

14.3.4 ADC Control Register 2 High

Name: ADCON2H
Offset: 0xB06

Legend: HSC = Hardware Settable/Clearable bit

Bit	15	14	13	12	11	10	9	8
	REFRDY	REFERR					SHRSAMC[9:8]	
Access	HSC/R	HSC/R					R/W	R/W
Reset	0	0					0	0

Bit	7	6	5	4	3	2	1	0
	SHRSAMC[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – REFRDY Band Gap and Reference Voltage Ready Flag bit

Value	Description
1	Band gap is ready
0	Band gap is not ready

Bit 14 – REFERR Band Gap or Reference Voltage Error Flag bit

Value	Description
1	Band gap was removed after the ADC module was enabled (ADON = 1)
0	No band gap error was detected

Bits 9:0 – SHRSAMC[9:0] Shared ADC Core Sample Time Selection bits

These bits specify the number of shared ADC Core Clock Periods (T_{ADCORE}) for the shared ADC core sample time (Sample Time = (SHRSAMC[9:0] + 2) * T_{ADCORE}).

Value	Description
1111111111	1025 T_{ADCORE}
. . .	
0000000001	3 T_{ADCORE}
0000000000	2 T_{ADCORE}

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14.3.5 ADC Control Register 3 Low

Name: ADCON3L

Offset: 0xB08

Legend: HSC = Hardware Settable/Clearable bit

Bit	15	14	13	12	11	10	9	8
	REFSEL[2:0]			SUSPEND	SUSPCIE	SUSPRDY	SHRSAMP	CNVRTCH
Access	R/W	R/W	R/W	R/W	R/W	HSC/R	R/W	HSC/R
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	SWLCTRG	SWCTRG	CNVCHSEL[5:0]					
Access	R/W	HSC/R	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:13 – REFSEL[2:0] ADC Reference Voltage Selection bits

Value	V_{REFH}	V_{REFL}
001–111	Unimplemented: Do not use	
000	AV_{DD}	AV_{SS}

Bit 12 – SUSPEND All ADC Core Triggers Disable bit

Value	Description
1	All new trigger events for the ADC core are disabled
0	The ADC core can be triggered

Bit 11 – SUSPCIE Suspend All ADC Cores Common Interrupt Enable bit

Value	Description
1	Common interrupt will be generated when ADC core triggers are suspended (SUSPEND bit = 1) and all previous conversions are finished (SUSPRDY bit becomes set)
0	Common interrupt is not generated for suspend ADC cores event

Bit 10 – SUSPRDY ADC Core Suspended Flag bit

Value	Description
1	The ADC core is suspended (SUSPEND bit = 1) and has no conversions in progress
0	The ADC core has previous conversions in progress

Bit 9 – SHRSAMP Shared ADC Core Sampling Direct Control bit

This bit should be used with the individual channel conversion trigger controlled by the CNVRTCH bit. It connects an analog input, specified by the CNVCHSEL[5:0] bits, to the shared ADC core and allows extending the sampling time. This bit is not controlled by hardware and must be cleared before the conversion starts (setting CNVRTCH to '1').

Value	Description
1	Shared ADC core samples an analog input specified by the CNVCHSEL[5:0] bits
0	Sampling is controlled by the shared ADC core hardware

Bit 8 – CNVRTCH Software Individual Channel Conversion Trigger bit

Value	Description
1	Single trigger is generated for an analog input specified by the CNVCHSEL[5:0] bits; when the bit is set, it is automatically cleared by hardware on the next instruction cycle
0	Next individual channel conversion trigger can be generated

Bit 7 – SWLCTRG Software Level-Sensitive Common Trigger bit

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Value	Description
1	Triggers are continuously generated for all channels with the software; level-sensitive common trigger selected as a source in the ADTRIGnL and ADTRIGxH registers
0	No software, level-sensitive common triggers are generated

Bit 6 – SWCTRG Software Common Trigger bit

Value	Description
1	Single trigger is generated for all channels with the software; common trigger selected as a source in the ADTRIGnL and ADTRIGxH registers; when the bit is set, it is automatically cleared by hardware on the next instruction cycle
0	Ready to generate the next software common trigger

Bits 5:0 – CNVCHSEL[5:0] Channel No. Selection for Software Individual Channel Conv. Trigger bits

These bits define a channel to be converted when the CNVRTCH bit is set.

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14.3.6 ADC Control Register 3 High

Name: ADCON3H

Offset: 0xB0A

Notes:

1. The ADC input clock frequency, selected by the CLKSEL[1:0] bits, must not exceed AD67 (see [Table 33-40](#)).
2. The ADC clock frequency, after the first divider selected by the CLKDIV[5:0] bits, must not exceed AD67 (see [Table 33-40](#)).

Bit	15	14	13	12	11	10	9	8
	CLKSEL[1:0]		CLKDIV[5:0]					
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	SHREN				C3EN	C2EN	C1EN	C0EN
Access	R/W				R/W	R/W	R/W	R/W
Reset	0				0	0	0	0

Bits 15:14 – CLKSEL[1:0] ADC Module Clock Source Selection bits⁽¹⁾

Value	Description
11	F _{VCO} /4
10	AF _{VCO} DIV
01	F _{OSC}
00	F _P (Peripheral Clock)

Bits 13:8 – CLKDIV[5:0] ADC Module Clock Source Divider bits⁽²⁾

The divider forms a T_{CORESRC} clock used by all ADC cores (shared and dedicated), from the T_{SRC} ADC module clock source, selected by the CLKSEL[1:0] bits. Then, each ADC core individually divides the T_{CORESRC} clock to get a core-specific T_{ADCORE} clock using the ADCS[6:0] bits in the ADCORExH register or the SHRADCS[6:0] bits in the ADCON2L register.

Value	Description
111111	64 Source Clock Periods
. . .	
000011	4 Source Clock Periods
000010	3 Source Clock Periods
000001	2 Source Clock Periods
000000	1 Source Clock Period

Bit 7 – SHREN Shared ADC Core Enable bit

Value	Description
1	Shared ADC core is enabled
0	Shared ADC core is disabled

Bit 3 – C3EN Dedicated ADC Core 3 Enable bit

Value	Description
1	Dedicated ADC Core 3 is enabled
0	Dedicated ADC Core 3 is disabled

Bit 2 – C2EN Dedicated ADC Core 2 Enable bit

Value	Description
1	Dedicated ADC Core 2 is enabled
0	Dedicated ADC Core 2 is disabled

Bit 1 – C1EN Dedicated ADC Core 1 Enable bit

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Value	Description
1	Dedicated ADC Core 1 is enabled
0	Dedicated ADC Core 1 is disabled

Bit 0 – C0EN Dedicated ADC Core 0 Enable bit

Value	Description
1	Dedicated ADC Core 0 is enabled
0	Dedicated ADC Core 0 is disabled

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14.3.7 ADC Control Register 4 Low

Name: ADCON4L

Offset: 0xB0C

Legend: r = Reserved bit

Bit	15	14	13	12	11	10	9	8
						Reserved[3:0]		
Access					r	r	r	r
Reset					0	0	0	0

Bit	7	6	5	4	3	2	1	0
					SAMC3EN	SAMC2EN	SAMC1EN	SAMC0EN
Access					R/W	R/W	R/W	R/W
Reset					0	0	0	0

Bits 11:8 – Reserved[3:0] Must be written as '0'

Bit 3 – SAMC3EN Dedicated ADC Core 3 Conversion Delay Enable bit

Value	Description
1	After trigger, the conversion will be delayed and the ADC core will continue sampling during the time specified by the SAMC[9:0] bits in the ADCORE2L register
0	After trigger, the sampling will be stopped immediately and the conversion will be started on the next core clock cycle

Bit 2 – SAMC2EN Dedicated ADC Core 2 Conversion Delay Enable bit

Value	Description
1	After trigger, the conversion will be delayed and the ADC core will continue sampling during the time specified by the SAMC[9:0] bits in the ADCORE2L register
0	After trigger, the sampling will be stopped immediately and the conversion will be started on the next core clock cycle

Bit 1 – SAMC1EN Dedicated ADC Core 1 Conversion Delay Enable bit

Value	Description
1	After trigger, the conversion will be delayed and the ADC core will continue sampling during the time specified by the SAMC[9:0] bits in the ADCORE1L register
0	After trigger, the sampling will be stopped immediately and the conversion will be started on the next core clock cycle

Bit 0 – SAMC0EN Dedicated ADC Core 0 Conversion Delay Enable bit

Value	Description
1	After trigger, the conversion will be delayed and the ADC core will continue sampling during the time specified by the SAMC[9:0] bits in the ADCORE0L register
0	After trigger, the sampling will be stopped immediately and the conversion will be started on the next core clock cycle

14.3.8 ADC Control Register 4 High

Name: ADCON4H
Offset: 0xB0E

Bit	15	14	13	12	11	10	9	8
Access								
Reset								

Bit	7	6	5	4	3	2	1	0
	C3CHS[1:0]		C2CHS[1:0]		C1CHS[1:0]		C0CHS[1:0]	
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 7:6 – C3CHS[1:0] Dedicated ADC Core 3 Input Channel Selection bits

Value	Description
10	ANB3
01	ANA3
00	AN3

Bits 5:4 – C2CHS[1:0] Dedicated ADC Core 2 Input Channel Selection bits

Value	Description
11	ANC2
10	ANB2
01	ANA2
00	AN2

Bits 3:2 – C1CHS[1:0] Dedicated ADC Core 1 Input Channel Selection bits

Value	Description
11	ANC1
10	ANB1
01	ANA1
00	AN1

Bits 1:0 – C0CHS[1:0] Dedicated ADC Core 0 Input Channel Selection bits

Value	Description
11	ANC0
10	ANB0
01	ANA0
00	AN0

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14.3.9 ADC Input Mode Control Register 0 Low

Name: ADMOD0L
Offset: 0xB10

Bit	15	14	13	12	11	10	9	8
	DIFF7	SIGN7	DIFF6	SIGN6	DIFF5	SIGN5	DIFF4	SIGN4
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	DIFF3	SIGN3	DIFF2	SIGN2	DIFF1	SIGN1	DIFF0	SIGN0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – DIFF7 Differential-Mode for Corresponding Analog Inputs bits

Value	Description
1	Channel is differential
0	Channel is single-ended

Bit 14 – SIGN7 Output Data Sign for Corresponding Analog Input bit

Value	Description
1	Channel output data are signed
0	Channel output data are unsigned

Bit 13 – DIFF6 Differential-Mode for Corresponding Analog Inputs bits

Value	Description
1	Channel is differential
0	Channel is single-ended

Bit 12 – SIGN6 Output Data Sign for Corresponding Analog Input bit

Value	Description
1	Channel output data are signed
0	Channel output data are unsigned

Bit 11 – DIFF5 Differential-Mode for Corresponding Analog Inputs bits

Value	Description
1	Channel is differential
0	Channel is single-ended

Bit 10 – SIGN5 Output Data Sign for Corresponding Analog Input bit

Value	Description
1	Channel output data are signed
0	Channel output data are unsigned

Bit 9 – DIFF4 Differential-Mode for Corresponding Analog Inputs bits

Value	Description
1	Channel is differential
0	Channel is single-ended

Bit 8 – SIGN4 Output Data Sign for Corresponding Analog Input bit

Value	Description
1	Channel output data are signed
0	Channel output data are unsigned

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Bit 7 – DIFF3 Differential-Mode for Corresponding Analog Inputs bits

Value	Description
1	Channel is differential
0	Channel is single-ended

Bit 6 – SIGN3 Output Data Sign for Corresponding Analog Input bit

Value	Description
1	Channel output data are signed
0	Channel output data are unsigned

Bit 5 – DIFF2 Differential-Mode for Corresponding Analog Inputs bits

Value	Description
1	Channel is differential
0	Channel is single-ended

Bit 4 – SIGN2 Output Data Sign for Corresponding Analog Input bit

Value	Description
1	Channel output data are signed
0	Channel output data are unsigned

Bit 3 – DIFF1 Differential-Mode for Corresponding Analog Inputs bits

Value	Description
1	Channel is differential
0	Channel is single-ended

Bit 2 – SIGN1 Output Data Sign for Corresponding Analog Input bit

Value	Description
1	Channel output data are signed
0	Channel output data are unsigned

Bit 1 – DIFF0 Differential-Mode for Corresponding Analog Inputs bits

Value	Description
1	Channel is differential
0	Channel is single-ended

Bit 0 – SIGN0 Output Data Sign for Corresponding Analog Input bit

Value	Description
1	Channel output data are signed
0	Channel output data are unsigned

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14.3.10 ADC Input Mode Control Register 0 High

Name: ADMOD0H
Offset: 0xB12

Bit	15	14	13	12	11	10	9	8
	DIFF15	SIGN15	DIFF14	SIGN14	DIFF13	SIGN13	DIFF12	SIGN12
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	DIFF11	SIGN11	DIFF10	SIGN10	DIFF9	SIGN9	DIFF8	SIGN8
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – DIFF15 Differential-Mode for Corresponding Analog Inputs bits

Value	Description
1	Channel is differential
0	Channel is single-ended

Bit 14 – SIGN15 Output Data Sign for Corresponding Analog Input bit

Value	Description
1	Channel output data are signed
0	Channel output data are unsigned

Bit 13 – DIFF14 Differential-Mode for Corresponding Analog Inputs bits

Value	Description
1	Channel is differential
0	Channel is single-ended

Bit 12 – SIGN14 Output Data Sign for Corresponding Analog Input bit

Value	Description
1	Channel output data are signed
0	Channel output data are unsigned

Bit 11 – DIFF13 Differential-Mode for Corresponding Analog Inputs bits

Value	Description
1	Channel is differential
0	Channel is single-ended

Bit 10 – SIGN13 Output Data Sign for Corresponding Analog Input bit

Value	Description
1	Channel output data are signed
0	Channel output data are unsigned

Bit 9 – DIFF12 Differential-Mode for Corresponding Analog Inputs bits

Value	Description
1	Channel is differential
0	Channel is single-ended

Bit 8 – SIGN12 Output Data Sign for Corresponding Analog Input bit

Value	Description
1	Channel output data are signed
0	Channel output data are unsigned

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Bit 7 – DIFF11 Differential-Mode for Corresponding Analog Inputs bits

Value	Description
1	Channel is differential
0	Channel is single-ended

Bit 6 – SIGN11 Output Data Sign for Corresponding Analog Input bit

Value	Description
1	Channel output data are signed
0	Channel output data are unsigned

Bit 5 – DIFF10 Differential-Mode for Corresponding Analog Inputs bits

Value	Description
1	Channel is differential
0	Channel is single-ended

Bit 4 – SIGN10 Output Data Sign for Corresponding Analog Input bit

Value	Description
1	Channel output data are signed
0	Channel output data are unsigned

Bit 3 – DIFF9 Differential-Mode for Corresponding Analog Inputs bits

Value	Description
1	Channel is differential
0	Channel is single-ended

Bit 2 – SIGN9 Output Data Sign for Corresponding Analog Input bit

Value	Description
1	Channel output data are signed
0	Channel output data are unsigned

Bit 1 – DIFF8 Differential-Mode for Corresponding Analog Inputs bits

Value	Description
1	Channel is differential
0	Channel is single-ended

Bit 0 – SIGN8 Output Data Sign for Corresponding Analog Input bit

Value	Description
1	Channel output data are signed
0	Channel output data are unsigned

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14.3.11 ADC Input Mode Control Register 1 Low

Name: ADMOD1L
Offset: 0xB14

Bit	15	14	13	12	11	10	9	8
	DIFF23	SIGN23	DIFF22	SIGN22	DIFF21	SIGN21	DIFF20	SIGN20
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	DIFF19	SIGN19	DIFF18	SIGN18	DIFF17	SIGN17	DIFF16	SIGN16
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – DIFF23 Differential-Mode for Corresponding Analog Inputs bits

Value	Description
1	Channel is differential
0	Channel is single-ended

Bit 14 – SIGN23 Output Data Sign for Corresponding Analog Input bit

Value	Description
1	Channel output data are signed
0	Channel output data are unsigned

Bit 13 – DIFF22 Differential-Mode for Corresponding Analog Inputs bits

Value	Description
1	Channel is differential
0	Channel is single-ended

Bit 12 – SIGN22 Output Data Sign for Corresponding Analog Input bit

Value	Description
1	Channel output data are signed
0	Channel output data are unsigned

Bit 11 – DIFF21 Differential-Mode for Corresponding Analog Inputs bits

Value	Description
1	Channel is differential
0	Channel is single-ended

Bit 10 – SIGN21 Output Data Sign for Corresponding Analog Input bit

Value	Description
1	Channel output data are signed
0	Channel output data are unsigned

Bit 9 – DIFF20 Differential-Mode for Corresponding Analog Inputs bits

Value	Description
1	Channel is differential
0	Channel is single-ended

Bit 8 – SIGN20 Output Data Sign for Corresponding Analog Input bit

Value	Description
1	Channel output data are signed
0	Channel output data are unsigned

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Bit 7 – DIFF19 Differential-Mode for Corresponding Analog Inputs bits

Value	Description
1	Channel is differential
0	Channel is single-ended

Bit 6 – SIGN19 Output Data Sign for Corresponding Analog Input bit

Value	Description
1	Channel output data are signed
0	Channel output data are unsigned

Bit 5 – DIFF18 Differential-Mode for Corresponding Analog Inputs bits

Value	Description
1	Channel is differential
0	Channel is single-ended

Bit 4 – SIGN18 Output Data Sign for Corresponding Analog Input bit

Value	Description
1	Channel output data are signed
0	Channel output data are unsigned

Bit 3 – DIFF17 Differential-Mode for Corresponding Analog Inputs bits

Value	Description
1	Channel is differential
0	Channel is single-ended

Bit 2 – SIGN17 Output Data Sign for Corresponding Analog Input bit

Value	Description
1	Channel output data are signed
0	Channel output data are unsigned

Bit 1 – DIFF16 Differential-Mode for Corresponding Analog Inputs bits

Value	Description
1	Channel is differential
0	Channel is single-ended

Bit 0 – SIGN16 Output Data Sign for Corresponding Analog Input bit

Value	Description
1	Channel output data are signed
0	Channel output data are unsigned

14.3.12 ADC Input Mode Control Register 1 High

Name: ADMOD1H
Offset: 0xB16

Bit	15	14	13	12	11	10	9	8
Access								
Reset								

Bit	7	6	5	4	3	2	1	0
	DIFF27	SIGN27	DIFF26	SIGN26	DIFF25	SIGN25	DIFF24	SIGN24
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 7 – DIFF27 Differential-Mode for Corresponding Analog Inputs bit

Value	Description
1	Channel is differential
0	Channel is single-ended

Bit 6 – SIGN27 Output Data Sign for Corresponding Analog Inputs bit

Value	Description
1	Channel output data is signed
0	Channel output data is unsigned

Bit 5 – DIFF26 Differential-Mode for Corresponding Analog Inputs bit

Value	Description
1	Channel is differential
0	Channel is single-ended

Bit 4 – SIGN26 Output Data Sign for Corresponding Analog Inputs bit

Value	Description
1	Channel output data is signed
0	Channel output data is unsigned

Bit 3 – DIFF25 Differential-Mode for Corresponding Analog Inputs bit

Value	Description
1	Channel is differential
0	Channel is single-ended

Bit 2 – SIGN25 Output Data Sign for Corresponding Analog Inputs bit

Value	Description
1	Channel output data is signed
0	Channel output data is unsigned

Bit 1 – DIFF24 Differential-Mode for Corresponding Analog Inputs bit

Value	Description
1	Channel is differential
0	Channel is single-ended

Bit 0 – SIGN24 Output Data Sign for Corresponding Analog Inputs bit

Value	Description
1	Channel output data is signed
0	Channel output data is unsigned

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14.3.13 ADC Interrupt Enable Register Low

Name: ADIEL

Offset: 0xB20

Bit	15	14	13	12	11	10	9	8
	IE[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	IE[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – IE[15:0] Common Interrupt Enable bits

Value	Description
1	Common and individual interrupts are enabled for the corresponding channel
0	Common and individual interrupts are disabled for the corresponding channel

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14.3.14 ADC Interrupt Enable Register High

Name: ADIEH
Offset: 0xB22

Bit	15	14	13	12	11	10	9	8
					IE[27:24]			
Access					R/W	R/W	R/W	R/W
Reset					0	0	0	0
Bit	7	6	5	4	3	2	1	0
	IE[23:16]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 11:0 – IE[27:16] Common Interrupt Enable bits

Value	Description
1	Common and individual interrupts are enabled for the corresponding channel
0	Common and individual interrupts are disabled for the corresponding channel

14.3.15 ADC Data Ready Status Register Low

Name: ADSTATL

Offset: 0xB30

Legend: HSC = Hardware Settable/Clearable bit

Bit	15	14	13	12	11	10	9	8
	AN[15:0]RDY							
Access	HSC/R	HSC/R	HSC/R	HSC/R	HSC/R	HSC/R	HSC/R	HSC/R
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	AN[15:0]RDY							
Access	HSC/R	HSC/R	HSC/R	HSC/R	HSC/R	HSC/R	HSC/R	HSC/R
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – AN[15:0]RDY Common Interrupt Enable for Corresponding Analog Inputs bits

Value	Description
1	Channel conversion result is ready in the corresponding ADCBUFx register
0	Channel conversion result is not ready

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High-Speed, 12-Bit Analog-to-Digital Convert...

14.3.16 ADC Data Ready Status Register High

Name: ADSTATH

Offset: 0xB32

Legend: HSC = Hardware Settable/Clearable bit

Bit	15	14	13	12	11	10	9	8
					AN[27:24]RDY			
Access					HSC/R	HSC/R	HSC/R	HSC/R
Reset					0	0	0	0

Bit	7	6	5	4	3	2	1	0
	AN[23:16]RDY							
Access	HSC/R	HSC/R	HSC/R	HSC/R	HSC/R	HSC/R	HSC/R	HSC/R
Reset	0	0	0	0	0	0	0	0

Bits 11:8 – AN[27:24]RDY Common Interrupt Enable for Corresponding Analog Input bits

Value	Description
1	Channel conversion result is ready in the corresponding ADCBUFx register
0	Channel conversion result is not ready

Bits 7:0 – AN[23:16]RDY Common Interrupt Enable for Corresponding Analog Input bits

Value	Description
1	Channel conversion result is ready in the corresponding ADCBUFx register
0	Channel conversion result is not ready

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High-Speed, 12-Bit Analog-to-Digital Convert...

14.3.17 ADC Digital Comparator x Channel Enable Register Low (x = 0, 1, 2, 3)

Name: ADCMPxENL
Offset: 0xB38, 0xB40, 0xB48, 0xB50

Bit	15	14	13	12	11	10	9	8
	CMPEN[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	CMPEN[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – CMPEN[15:0] Comparator Enable for Corresponding Input Channel bits

Value	Description
1	Conversion result for corresponding channel is used by the comparator
0	Conversion result for corresponding channel is not used by the comparator

dsPIC33CK512MPT608 Family

High-Speed, 12-Bit Analog-to-Digital Convert...

14.3.18 ADC Digital Comparator x Channel Enable Register High (x = 0, 1, 2, 3)

Name: ADCMPxENH
Offset: 0xB3A, 0xB42, 0xB4A, 0xB52

Bit	15	14	13	12	11	10	9	8
					CMPEN[27:24]			
Access					R/W	R/W	R/W	R/W
Reset					0	0	0	0
Bit	7	6	5	4	3	2	1	0
	CMPEN[23:16]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 11:0 – CMPEN[27:16] Comparator Enable for Corresponding Input Channel bits

Value	Description
1	Conversion result for corresponding channel is used by the comparator
0	Conversion result for corresponding channel is not used by the comparator

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High-Speed, 12-Bit Analog-to-Digital Convert...

14.3.19 ADC Comparator x Threshold Low Register

Name: ADCMPxLO
Offset: 0xB3C, 0xB44, 0xB4C, 0xB54

Bit	15	14	13	12	11	10	9	8
	CMPLO[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	CMPLO[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – CMPLO[15:0] ADC Comparator Lower Threshold bits

The register stores the 16-bit low digital comparison values for use by the digital comparators.

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High-Speed, 12-Bit Analog-to-Digital Convert...

14.3.20 ADC Comparator x Threshold High Register

Name: ADCMPxHI
Offset: 0xB3E, 0xB46, 0xB4E, 0xB56

Bit	15	14	13	12	11	10	9	8
	CMPHI[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	CMPHI[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – CMPHI[15:0] ADC Comparator Upper Threshold bits

The register stores the 16-bit upper digital comparison values for use by the digital comparators.

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High-Speed, 12-Bit Analog-to-Digital Convert...

14.3.21 Oversampling Filter (x = 0, 1, 2, 3) Output Register

Name: ADFLxDAT
Offset: 0xB68, 0xB6C, 0xB70, 0xB74

Bit	15	14	13	12	11	10	9	8
	FLDATA[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	FLDATA[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – FLDATA[15:0] 16-Bit Output Data from Oversampling Filters bits

dsPIC33CK512MPT608 Family

High-Speed, 12-Bit Analog-to-Digital Convert...

14.3.22 ADC Digital Filter x Control Register (x = 0, 1, 2, 3)

Name: ADFLxCON
Offset: 0xB6A, 0xB6E, 0xB72, 0xB76

Legend: HSC = Hardware Settable/Clearable bit

Bit	15	14	13	12	11	10	9	8
	FLEN	MODE[1:0]		OVSAM[2:0]			IE	RDY
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	HSC/R
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
				FLCHSEL[4:0]				
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0

Bit 15 – FLEN Filter Enable bit

Value	Description
1	Filter is enabled
0	Filter is disabled and the RDY bit is cleared

Bits 14:13 – MODE[1:0] Filter Mode bits

Value	Description
11	Averaging mode
10	Reserved
01	Reserved
00	Oversampling mode

Bits 12:10 – OVSAM[2:0] Filter Averaging/Oversampling Ratio bits

If MODE[1:0] = 00:

111 = 128x (16-bit result in the ADFLxDAT register is in 12.4 format)
 110 = 32x (15-bit result in the ADFLxDAT register is in 12.3 format)
 101 = 8x (14-bit result in the ADFLxDAT register is in 12.2 format)
 100 = 2x (13-bit result in the ADFLxDAT register is in 12.1 format)
 011 = 256x (16-bit result in the ADFLxDAT register is in 12.4 format)
 010 = 64x (15-bit result in the ADFLxDAT register is in 12.3 format)
 001 = 16x (14-bit result in the ADFLxDAT register is in 12.2 format)
 000 = 4x (13-bit result in the ADFLxDAT register is in 12.1 format)

If MODE[1:0] = 11 (12-bit result in the ADFLxDAT register in all instances):

111 = 256x
 110 = 128x
 101 = 64x
 100 = 32x
 011 = 16x
 010 = 8x
 001 = 4x
 000 = 2x

Bit 9 – IE Filter Common ADC Interrupt Enable bit

Value	Description
1	Common ADC interrupt will be generated when the filter result will be ready
0	Common ADC interrupt will not be generated for the filter

Bit 8 – RDY Oversampling Filter Data Ready Flag bit

This bit is cleared by hardware when the result is read from the ADFLxDAT register.

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High-Speed, 12-Bit Analog-to-Digital Convert...

Value	Description
1	Data in the ADFLxDAT register are ready
0	The ADFLxDAT register has been read and new data in the ADFLxDAT register are not ready

Bits 4:0 – FLCHSEL[4:0] Oversampling Filter Input Channel Selection bits

Value	Description
11111–11100	Reserved
11011	AN27
11010	AN26
11001	Band Gap, 1.2V (AN25)
11000	Temperature Sensor (AN24)
10111	AN23
. . .	
00010	AN2
00001	AN1
00000	AN0

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High-Speed, 12-Bit Analog-to-Digital Convert...

14.3.23 ADC Channel Trigger 0 Selection Register Low

Name: ADTRIG0L
Offset: 0xB80

Bit	15	14	13	12	11	10	9	8
						TRGSRC1[4:0]		
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
						TRGSRC0[4:0]		
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0

Bits 12:8 – TRGSRC1[4:0] Trigger Source Selection for Corresponding Analog Inputs bits

Value	Description
11111	ADTRG31 (PPS input)
11110	PTG
11101	CLC2
11100	CLC1
11011	Input Capture/Output Compare 9
11010	Input Capture/Output Compare 7
11001	Input Capture/Output Compare 6
11000	Input Capture/Output Compare 5
10111	Input Capture/Output Compare 4
10110	Input Capture/Output Compare 3
10101	Input Capture/Output Compare 2
10100	Input Capture/Output Compare 1
10011	PWM8 Trigger 2
10010	PWM8 Trigger 1
10001-10000	Reserved
01111	PWM6 Trigger 2
01110	PWM6 Trigger 1
01101-01100	Reserved
01011	PWM4 Trigger 2
01010	PWM4 Trigger 1
01001	PWM3 Trigger 2
01000	PWM3 Trigger 1
00111	PWM2 Trigger 2
00110	PWM2 Trigger 1
00101	PWM1 Trigger 2
00100	PWM1 Trigger 1
00011	Reserved
00010	Level software trigger
00001	Common software trigger
00000	No trigger is enabled

Bits 4:0 – TRGSRC0[4:0] Common Interrupt Enable for Corresponding Analog Inputs bits

Value	Description
11111	ADTRG31 (PPS input)
11110	PTG
11101	CLC2
11100	CLC1
11011	Input Capture/Output Compare 9

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High-Speed, 12-Bit Analog-to-Digital Convert...

Value	Description
11010	Input Capture/Output Compare 7
11001	Input Capture/Output Compare 6
11000	Input Capture/Output Compare 5
10111	Input Capture/Output Compare 4
10110	Input Capture/Output Compare 3
10101	Input Capture/Output Compare 2
10100	Input Capture/Output Compare 1
10011	PWM8 Trigger 2
10010	PWM8 Trigger 1
10001-10000	Reserved
01111	PWM6 Trigger 2
01110	PWM6 Trigger 1
01101-01100	Reserved
01011	PWM4 Trigger 2
01010	PWM4 Trigger 1
01001	PWM3 Trigger 2
01000	PWM3 Trigger 1
00111	PWM2 Trigger 2
00110	PWM2 Trigger 1
00101	PWM1 Trigger 2
00100	PWM1 Trigger 1
00011	Reserved
00010	Level software trigger
00001	Common software trigger
00000	No trigger is enabled

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High-Speed, 12-Bit Analog-to-Digital Convert...

14.3.24 ADC Channel Trigger 0 Selection Register High

Name: ADTRIG0H
Offset: 0xB82

Bit	15	14	13	12	11	10	9	8
						TRGSRC3[4:0]		
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
						TRGSRC2[4:0]		
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0

Bits 12:8 – TRGSRC3[4:0] Trigger Source Selection for Corresponding Analog Inputs bits

Value	Description
11111	ADTRG31 (PPS input)
11110	PTG
11101	CLC2
11100	CLC1
11011	Input Capture/Output Compare 9
11010	Input Capture/Output Compare 7
11001	Input Capture/Output Compare 6
11000	Input Capture/Output Compare 5
10111	Input Capture/Output Compare 4
10110	Input Capture/Output Compare 3
10101	Input Capture/Output Compare 2
10100	Input Capture/Output Compare 1
10011	PWM8 Trigger 2
10010	PWM8 Trigger 1
10001-10000	Reserved
01111	PWM6 Trigger 2
01110	PWM6 Trigger 1
01101-01100	Reserved
01011	PWM4 Trigger 2
01010	PWM4 Trigger 1
01001	PWM3 Trigger 2
01000	PWM3 Trigger 1
00111	PWM2 Trigger 2
00110	PWM2 Trigger 1
00101	PWM1 Trigger 2
00100	PWM1 Trigger 1
00011	Reserved
00010	Level software trigger
00001	Common software trigger
00000	No trigger is enabled

Bits 4:0 – TRGSRC2[4:0] Common Interrupt Enable for Corresponding Analog Inputs bits

Value	Description
11111	ADTRG31 (PPS input)
11110	PTG
11101	CLC2
11100	CLC1
11011	Input Capture/Output Compare 9

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High-Speed, 12-Bit Analog-to-Digital Convert...

Value	Description
11010	Input Capture/Output Compare 7
11001	Input Capture/Output Compare 6
11000	Input Capture/Output Compare 5
10111	Input Capture/Output Compare 4
10110	Input Capture/Output Compare 3
10101	Input Capture/Output Compare 2
10100	Input Capture/Output Compare 1
10011	PWM8 Trigger 2
10010	PWM8 Trigger 1
10001-10000	Reserved
01111	PWM6 Trigger 2
01110	PWM6 Trigger 1
01101-01100	Reserved
01011	PWM4 Trigger 2
01010	PWM4 Trigger 1
01001	PWM3 Trigger 2
01000	PWM3 Trigger 1
00111	PWM2 Trigger 2
00110	PWM2 Trigger 1
00101	PWM1 Trigger 2
00100	PWM1 Trigger 1
00011	Reserved
00010	Level software trigger
00001	Common software trigger
00000	No trigger is enabled

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High-Speed, 12-Bit Analog-to-Digital Convert...

14.3.25 ADC Channel Trigger 1 Selection Register Low

Name: ADTRIG1L
Offset: 0xB84

Bit	15	14	13	12	11	10	9	8
						TRGSRC5[4:0]		
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
						TRGSRC4[4:0]		
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0

Bits 12:8 – TRGSRC5[4:0] Trigger Source Selection for Corresponding Analog Inputs bits

Value	Description
11111	ADTRG31 (PPS input)
11110	PTG
11101	CLC2
11100	CLC1
11011	Input Capture/Output Compare 9
11010	Input Capture/Output Compare 7
11001	Input Capture/Output Compare 6
11000	Input Capture/Output Compare 5
10111	Input Capture/Output Compare 4
10110	Input Capture/Output Compare 3
10101	Input Capture/Output Compare 2
10100	Input Capture/Output Compare 1
10011	PWM8 Trigger 2
10010	PWM8 Trigger 1
10001-10000	Reserved
01111	PWM6 Trigger 2
01110	PWM6 Trigger 1
01101-01100	Reserved
01011	PWM4 Trigger 2
01010	PWM4 Trigger 1
01001	PWM3 Trigger 2
01000	PWM3 Trigger 1
00111	PWM2 Trigger 2
00110	PWM2 Trigger 1
00101	PWM1 Trigger 2
00100	PWM1 Trigger 1
00011	Reserved
00010	Level software trigger
00001	Common software trigger
00000	No trigger is enabled

Bits 4:0 – TRGSRC4[4:0] Common Interrupt Enable for Corresponding Analog Inputs bits

Value	Description
11111	ADTRG31 (PPS input)
11110	PTG
11101	CLC2
11100	CLC1
11011	Input Capture/Output Compare 9

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High-Speed, 12-Bit Analog-to-Digital Convert...

Value	Description
11010	Input Capture/Output Compare 7
11001	Input Capture/Output Compare 6
11000	Input Capture/Output Compare 5
10111	Input Capture/Output Compare 4
10110	Input Capture/Output Compare 3
10101	Input Capture/Output Compare 2
10100	Input Capture/Output Compare 1
10011	PWM8 Trigger 2
10010	PWM8 Trigger 1
10001-10000	Reserved
01111	PWM6 Trigger 2
01110	PWM6 Trigger 1
01101-01100	Reserved
01011	PWM4 Trigger 2
01010	PWM4 Trigger 1
01001	PWM3 Trigger 2
01000	PWM3 Trigger 1
00111	PWM2 Trigger 2
00110	PWM2 Trigger 1
00101	PWM1 Trigger 2
00100	PWM1 Trigger 1
00011	Reserved
00010	Level software trigger
00001	Common software trigger
00000	No trigger is enabled

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High-Speed, 12-Bit Analog-to-Digital Convert...

14.3.26 ADC Channel Trigger 1 Selection Register High

Name: ADTRIG1H
Offset: 0xB86

Bit	15	14	13	12	11	10	9	8
						TRGSRC7[4:0]		
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
						TRGSRC6[4:0]		
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0

Bits 12:8 – TRGSRC7[4:0] Trigger Source Selection for Corresponding Analog Inputs bits

Value	Description
11111	ADTRG31 (PPS input)
11110	PTG
11101	CLC2
11100	CLC1
11011	Input Capture/Output Compare 9
11010	Input Capture/Output Compare 7
11001	Input Capture/Output Compare 6
11000	Input Capture/Output Compare 5
10111	Input Capture/Output Compare 4
10110	Input Capture/Output Compare 3
10101	Input Capture/Output Compare 2
10100	Input Capture/Output Compare 1
10011	PWM8 Trigger 2
10010	PWM8 Trigger 1
10001-10000	Reserved
01111	PWM6 Trigger 2
01110	PWM6 Trigger 1
01101-01100	Reserved
01011	PWM4 Trigger 2
01010	PWM4 Trigger 1
01001	PWM3 Trigger 2
01000	PWM3 Trigger 1
00111	PWM2 Trigger 2
00110	PWM2 Trigger 1
00101	PWM1 Trigger 2
00100	PWM1 Trigger 1
00011	Reserved
00010	Level software trigger
00001	Common software trigger
00000	No trigger is enabled

Bits 4:0 – TRGSRC6[4:0] Common Interrupt Enable for Corresponding Analog Inputs bits

Value	Description
11111	ADTRG31 (PPS input)
11110	PTG
11101	CLC2
11100	CLC1
11011	Input Capture/Output Compare 9

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High-Speed, 12-Bit Analog-to-Digital Convert...

Value	Description
11010	Input Capture/Output Compare 7
11001	Input Capture/Output Compare 6
11000	Input Capture/Output Compare 5
10111	Input Capture/Output Compare 4
10110	Input Capture/Output Compare 3
10101	Input Capture/Output Compare 2
10100	Input Capture/Output Compare 1
10011	PWM8 Trigger 2
10010	PWM8 Trigger 1
10001-10000	Reserved
01111	PWM6 Trigger 2
01110	PWM6 Trigger 1
01101-01100	Reserved
01011	PWM4 Trigger 2
01010	PWM4 Trigger 1
01001	PWM3 Trigger 2
01000	PWM3 Trigger 1
00111	PWM2 Trigger 2
00110	PWM2 Trigger 1
00101	PWM1 Trigger 2
00100	PWM1 Trigger 1
00011	Reserved
00010	Level software trigger
00001	Common software trigger
00000	No trigger is enabled

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14.3.27 ADC Channel Trigger 2 Selection Register Low

Name: ADTRIG2L

Offset: 0xB88

Bit	15	14	13	12	11	10	9	8
						TRGSRC9[4:0]		
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
						TRGSRC8[4:0]		
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0

Bits 12:8 – TRGSRC9[4:0] Trigger Source Selection for Corresponding Analog Inputs bits

Value	Description
11111	ADTRG31 (PPS input)
11110	PTG
11101	CLC2
11100	CLC1
11011	Input Capture/Output Compare 9
11010	Input Capture/Output Compare 7
11001	Input Capture/Output Compare 6
11000	Input Capture/Output Compare 5
10111	Input Capture/Output Compare 4
10110	Input Capture/Output Compare 3
10101	Input Capture/Output Compare 2
10100	Input Capture/Output Compare 1
10011	PWM8 Trigger 2
10010	PWM8 Trigger 1
10001-10000	Reserved
01111	PWM6 Trigger 2
01110	PWM6 Trigger 1
01101-01100	Reserved
01011	PWM4 Trigger 2
01010	PWM4 Trigger 1
01001	PWM3 Trigger 2
01000	PWM3 Trigger 1
00111	PWM2 Trigger 2
00110	PWM2 Trigger 1
00101	PWM1 Trigger 2
00100	PWM1 Trigger 1
00011	Reserved
00010	Level software trigger
00001	Common software trigger
00000	No trigger is enabled

Bits 4:0 – TRGSRC8[4:0] Common Interrupt Enable for Corresponding Analog Inputs bits

Value	Description
11111	ADTRG31 (PPS input)
11110	PTG
11101	CLC2
11100	CLC1
11011	Input Capture/Output Compare 9

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High-Speed, 12-Bit Analog-to-Digital Convert...

Value	Description
11010	Input Capture/Output Compare 7
11001	Input Capture/Output Compare 6
11000	Input Capture/Output Compare 5
10111	Input Capture/Output Compare 4
10110	Input Capture/Output Compare 3
10101	Input Capture/Output Compare 2
10100	Input Capture/Output Compare 1
10011	PWM8 Trigger 2
10010	PWM8 Trigger 1
10001-10000	Reserved
01111	PWM6 Trigger 2
01110	PWM6 Trigger 1
01101-01100	Reserved
01011	PWM4 Trigger 2
01010	PWM4 Trigger 1
01001	PWM3 Trigger 2
01000	PWM3 Trigger 1
00111	PWM2 Trigger 2
00110	PWM2 Trigger 1
00101	PWM1 Trigger 2
00100	PWM1 Trigger 1
00011	Reserved
00010	Level software trigger
00001	Common software trigger
00000	No trigger is enabled

dsPIC33CK512MPT608 Family

High-Speed, 12-Bit Analog-to-Digital Convert...

14.3.28 ADC Channel Trigger 2 Selection Register High

Name: ADTRIG2H
Offset: 0xB8A

Bit	15	14	13	12	11	10	9	8
				TRGSRC11[4:0]				
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
				TRGSRC10[4:0]				
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0

Bits 12:8 – TRGSRC11[4:0] Trigger Source Selection for Corresponding Analog Inputs bits

Value	Description
11111	ADTRG31 (PPS input)
11110	PTG
11101	CLC2
11100	CLC1
11011	Input Capture/Output Compare 9
11010	Input Capture/Output Compare 7
11001	Input Capture/Output Compare 6
11000	Input Capture/Output Compare 5
10111	Input Capture/Output Compare 4
10110	Input Capture/Output Compare 3
10101	Input Capture/Output Compare 2
10100	Input Capture/Output Compare 1
10011	PWM8 Trigger 2
10010	PWM8 Trigger 1
10001-10000	Reserved
01111	PWM6 Trigger 2
01110	PWM6 Trigger 1
01101-01100	Reserved
01011	PWM4 Trigger 2
01010	PWM4 Trigger 1
01001	PWM3 Trigger 2
01000	PWM3 Trigger 1
00111	PWM2 Trigger 2
00110	PWM2 Trigger 1
00101	PWM1 Trigger 2
00100	PWM1 Trigger 1
00011	Reserved
00010	Level software trigger
00001	Common software trigger
00000	No trigger is enabled

Bits 4:0 – TRGSRC10[4:0] Common Interrupt Enable for Corresponding Analog Inputs bits

Value	Description
11111	ADTRG31 (PPS input)
11110	PTG
11101	CLC2
11100	CLC1
11011	Input Capture/Output Compare 9

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High-Speed, 12-Bit Analog-to-Digital Convert...

Value	Description
11010	Input Capture/Output Compare 7
11001	Input Capture/Output Compare 6
11000	Input Capture/Output Compare 5
10111	Input Capture/Output Compare 4
10110	Input Capture/Output Compare 3
10101	Input Capture/Output Compare 2
10100	Input Capture/Output Compare 1
10011	PWM8 Trigger 2
10010	PWM8 Trigger 1
10001-10000	Reserved
01111	PWM6 Trigger 2
01110	PWM6 Trigger 1
01101-01100	Reserved
01011	PWM4 Trigger 2
01010	PWM4 Trigger 1
01001	PWM3 Trigger 2
01000	PWM3 Trigger 1
00111	PWM2 Trigger 2
00110	PWM2 Trigger 1
00101	PWM1 Trigger 2
00100	PWM1 Trigger 1
00011	Reserved
00010	Level software trigger
00001	Common software trigger
00000	No trigger is enabled

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High-Speed, 12-Bit Analog-to-Digital Convert...

14.3.29 ADC Channel Trigger 3 Selection Register Low

Name: ADTRIG3L
Offset: 0xB8C

Bit	15	14	13	12	11	10	9	8
				TRGSRC13[4:0]				
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
				TRGSRC12[4:0]				
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0

Bits 12:8 – TRGSRC13[4:0] Trigger Source Selection for Corresponding Analog Inputs bits

Value	Description
11111	ADTRG31 (PPS input)
11110	PTG
11101	CLC2
11100	CLC1
11011	Input Capture/Output Compare 9
11010	Input Capture/Output Compare 7
11001	Input Capture/Output Compare 6
11000	Input Capture/Output Compare 5
10111	Input Capture/Output Compare 4
10110	Input Capture/Output Compare 3
10101	Input Capture/Output Compare 2
10100	Input Capture/Output Compare 1
10011	PWM8 Trigger 2
10010	PWM8 Trigger 1
10001-10000	Reserved
01111	PWM6 Trigger 2
01110	PWM6 Trigger 1
01101-01100	Reserved
01011	PWM4 Trigger 2
01010	PWM4 Trigger 1
01001	PWM3 Trigger 2
01000	PWM3 Trigger 1
00111	PWM2 Trigger 2
00110	PWM2 Trigger 1
00101	PWM1 Trigger 2
00100	PWM1 Trigger 1
00011	Reserved
00010	Level software trigger
00001	Common software trigger
00000	No trigger is enabled

Bits 4:0 – TRGSRC12[4:0] Common Interrupt Enable for Corresponding Analog Inputs bits

Value	Description
11111	ADTRG31 (PPS input)
11110	PTG
11101	CLC2
11100	CLC1
11011	Input Capture/Output Compare 9

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High-Speed, 12-Bit Analog-to-Digital Convert...

Value	Description
11010	Input Capture/Output Compare 7
11001	Input Capture/Output Compare 6
11000	Input Capture/Output Compare 5
10111	Input Capture/Output Compare 4
10110	Input Capture/Output Compare 3
10101	Input Capture/Output Compare 2
10100	Input Capture/Output Compare 1
10011	PWM8 Trigger 2
10010	PWM8 Trigger 1
10001-10000	Reserved
01111	PWM6 Trigger 2
01110	PWM6 Trigger 1
01101-01100	Reserved
01011	PWM4 Trigger 2
01010	PWM4 Trigger 1
01001	PWM3 Trigger 2
01000	PWM3 Trigger 1
00111	PWM2 Trigger 2
00110	PWM2 Trigger 1
00101	PWM1 Trigger 2
00100	PWM1 Trigger 1
00011	Reserved
00010	Level software trigger
00001	Common software trigger
00000	No trigger is enabled

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High-Speed, 12-Bit Analog-to-Digital Convert...

14.3.30 ADC Channel Trigger 3 Selection Register High

Name: ADTRIG3H
Offset: 0xB8E

Bit	15	14	13	12	11	10	9	8
				TRGSRC15[4:0]				
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
				TRGSRC14[4:0]				
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0

Bits 12:8 – TRGSRC15[4:0] Trigger Source Selection for Corresponding Analog Inputs bits

Value	Description
11111	ADTRG31 (PPS input)
11110	PTG
11101	CLC2
11100	CLC1
11011	Input Capture/Output Compare 9
11010	Input Capture/Output Compare 7
11001	Input Capture/Output Compare 6
11000	Input Capture/Output Compare 5
10111	Input Capture/Output Compare 4
10110	Input Capture/Output Compare 3
10101	Input Capture/Output Compare 2
10100	Input Capture/Output Compare 1
10011	PWM8 Trigger 2
10010	PWM8 Trigger 1
10001-10000	Reserved
01111	PWM6 Trigger 2
01110	PWM6 Trigger 1
01101-01011	Reserved
01011	PWM4 Trigger 2
01010	PWM4 Trigger 1
01001	PWM3 Trigger 2
01000	PWM3 Trigger 1
00111	PWM2 Trigger 2
00110	PWM2 Trigger 1
00101	PWM1 Trigger 2
00100	PWM1 Trigger 1
00011	Reserved
00010	Level software trigger
00001	Common software trigger
00000	No trigger is enabled

Bits 4:0 – TRGSRC14[4:0] Common Interrupt Enable for Corresponding Analog Inputs bits

Value	Description
11111	ADTRG31 (PPS input)
11110	PTG
11101	CLC2
11100	CLC1
11011	Input Capture/Output Compare 9

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High-Speed, 12-Bit Analog-to-Digital Convert...

Value	Description
11010	Input Capture/Output Compare 7
11001	Input Capture/Output Compare 6
11000	Input Capture/Output Compare 5
10111	Input Capture/Output Compare 4
10110	Input Capture/Output Compare 3
10101	Input Capture/Output Compare 2
10100	Input Capture/Output Compare 1
10011	PWM8 Trigger 2
10010	PWM8 Trigger 1
10001-10000	Reserved
01111	PWM6 Trigger 2
01110	PWM6 Trigger 1
01101-01100	Reserved
01011	PWM4 Trigger 2
01010	PWM4 Trigger 1
01001	PWM3 Trigger 2
01000	PWM3 Trigger 1
00111	PWM2 Trigger 2
00110	PWM2 Trigger 1
00101	PWM1 Trigger 2
00100	PWM1 Trigger 1
00011	Reserved
00010	Level software trigger
00001	Common software trigger
00000	No trigger is enabled

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High-Speed, 12-Bit Analog-to-Digital Convert...

14.3.31 ADC Channel Trigger 4 Selection Register Low

Name: ADTRIG4L
Offset: 0xB90

Bit	15	14	13	12	11	10	9	8
				TRGSRC17[4:0]				
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
				TRGSRC16[4:0]				
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0

Bits 12:8 – TRGSRC17[4:0] Trigger Source Selection for Corresponding Analog Inputs bits

Value	Description
11111	ADTRG31 (PPS input)
11110	PTG
11101	CLC2
11100	CLC1
11011	Input Capture/Output Compare 9
11010	Input Capture/Output Compare 7
11001	Input Capture/Output Compare 6
11000	Input Capture/Output Compare 5
10111	Input Capture/Output Compare 4
10110	Input Capture/Output Compare 3
10101	Input Capture/Output Compare 2
10100	Input Capture/Output Compare 1
10011	PWM8 Trigger 2
10010	PWM8 Trigger 1
10001-10000	Reserved
01111	PWM6 Trigger 2
01110	PWM6 Trigger 1
01101-01100	Reserved
01011	PWM4 Trigger 2
01010	PWM4 Trigger 1
01001	PWM3 Trigger 2
01000	PWM3 Trigger 1
00111	PWM2 Trigger 2
00110	PWM2 Trigger 1
00101	PWM1 Trigger 2
00100	PWM1 Trigger 1
00011	Reserved
00010	Level software trigger
00001	Common software trigger
00000	No trigger is enabled

Bits 4:0 – TRGSRC16[4:0] Common Interrupt Enable for Corresponding Analog Inputs bits

Value	Description
11111	ADTRG31 (PPS input)
11110	PTG
11101	CLC2
11100	CLC1
11011	Input Capture/Output Compare 9

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High-Speed, 12-Bit Analog-to-Digital Convert...

Value	Description
11010	Input Capture/Output Compare 7
11001	Input Capture/Output Compare 6
11000	Input Capture/Output Compare 5
10111	Input Capture/Output Compare 4
10110	Input Capture/Output Compare 3
10101	Input Capture/Output Compare 2
10100	Input Capture/Output Compare 1
10011	PWM8 Trigger 2
10010	PWM8 Trigger 1
10001-10000	Reserved
01111	PWM6 Trigger 2
01110	PWM6 Trigger 1
01101-01100	Reserved
01011	PWM4 Trigger 2
01010	PWM4 Trigger 1
01001	PWM3 Trigger 2
01000	PWM3 Trigger 1
00111	PWM2 Trigger 2
00110	PWM2 Trigger 1
00101	PWM1 Trigger 2
00100	PWM1 Trigger 1
00011	Reserved
00010	Level software trigger
00001	Common software trigger
00000	No trigger is enabled

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High-Speed, 12-Bit Analog-to-Digital Convert...

14.3.32 ADC Channel Trigger 4 Selection Register High

Name: ADTRIG4H
Offset: 0xB92

Bit	15	14	13	12	11	10	9	8
						TRGSRC19[4:0]		
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
						TRGSRC18[4:0]		
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0

Bits 12:8 – TRGSRC19[4:0] Trigger Source Selection for Corresponding Analog Inputs bits

Value	Description
11111	ADTRG31 (PPS input)
11110	PTG
11101	CLC2
11100	CLC1
11011	Input Capture/Output Compare 9
11010	Input Capture/Output Compare 7
11001	Input Capture/Output Compare 6
11000	Input Capture/Output Compare 5
10111	Input Capture/Output Compare 4
10110	Input Capture/Output Compare 3
10101	Input Capture/Output Compare 2
10100	Input Capture/Output Compare 1
10011	PWM8 Trigger 2
10010	PWM8 Trigger 1
10001-10000	Reserved
01111	PWM6 Trigger 2
01110	PWM6 Trigger 1
01101-01100	Reserved
01011	PWM4 Trigger 2
01010	PWM4 Trigger 1
01001	PWM3 Trigger 2
01000	PWM3 Trigger 1
00111	PWM2 Trigger 2
00110	PWM2 Trigger 1
00101	PWM1 Trigger 2
00100	PWM1 Trigger 1
00011	Reserved
00010	Level software trigger
00001	Common software trigger
00000	No trigger is enabled

Bits 4:0 – TRGSRC18[4:0] Common Interrupt Enable for Corresponding Analog Inputs bits

Value	Description
11111	ADTRG31 (PPS input)
11110	PTG
11101	CLC2
11100	CLC1
11011	Input Capture/Output Compare 9

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High-Speed, 12-Bit Analog-to-Digital Convert...

Value	Description
11010	Input Capture/Output Compare 7
11001	Input Capture/Output Compare 6
11000	Input Capture/Output Compare 5
10111	Input Capture/Output Compare 4
10110	Input Capture/Output Compare 3
10101	Input Capture/Output Compare 2
10100	Input Capture/Output Compare 1
10011	PWM8 Trigger 2
10010	PWM8 Trigger 1
10001-10000	Reserved
01111	PWM6 Trigger 2
01110	PWM6 Trigger 1
01101-01100	Reserved
01011	PWM4 Trigger 2
01010	PWM4 Trigger 1
01001	PWM3 Trigger 2
01000	PWM3 Trigger 1
00111	PWM2 Trigger 2
00110	PWM2 Trigger 1
00101	PWM1 Trigger 2
00100	PWM1 Trigger 1
00011	Reserved
00010	Level software trigger
00001	Common software trigger
00000	No trigger is enabled

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High-Speed, 12-Bit Analog-to-Digital Convert...

14.3.33 ADC Channel Trigger 5 Selection Register Low

Name: ADTRIG5L
Offset: 0xB94

Bit	15	14	13	12	11	10	9	8
				TRGSRC21[4:0]				
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
				TRGSRC20[4:0]				
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0

Bits 12:8 – TRGSRC21[4:0] Trigger Source Selection for Corresponding Analog Inputs bits

Value	Description
11111	ADTRG31 (PPS input)
11110	PTG
11101	CLC2
11100	CLC1
11011	Input Capture/Output Compare 9
11010	Input Capture/Output Compare 7
11001	Input Capture/Output Compare 6
11000	Input Capture/Output Compare 5
10111	Input Capture/Output Compare 4
10110	Input Capture/Output Compare 3
10101	Input Capture/Output Compare 2
10100	Input Capture/Output Compare 1
10011	PWM8 Trigger 2
10010	PWM8 Trigger 1
10001-10000	Reserved
01111	PWM6 Trigger 2
01110	PWM6 Trigger 1
01101-01100	Reserved
01011	PWM4 Trigger 2
01010	PWM4 Trigger 1
01001	PWM3 Trigger 2
01000	PWM3 Trigger 1
00111	PWM2 Trigger 2
00110	PWM2 Trigger 1
00101	PWM1 Trigger 2
00100	PWM1 Trigger 1
00011	Reserved
00010	Level software trigger
00001	Common software trigger
00000	No trigger is enabled

Bits 4:0 – TRGSRC20[4:0] Common Interrupt Enable for Corresponding Analog Inputs bits

Value	Description
11111	ADTRG31 (PPS input)
11110	PTG
11101	CLC2
11100	CLC1
11011	Input Capture/Output Compare 9

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High-Speed, 12-Bit Analog-to-Digital Convert...

Value	Description
11010	Input Capture/Output Compare 7
11001	Input Capture/Output Compare 6
11000	Input Capture/Output Compare 5
10111	Input Capture/Output Compare 4
10110	Input Capture/Output Compare 3
10101	Input Capture/Output Compare 2
10100	Input Capture/Output Compare 1
10011	PWM8 Trigger 2
10010	PWM8 Trigger 1
10001-10000	Reserved
01111	PWM6 Trigger 2
01110	PWM6 Trigger 1
01101-01100	Reserved
01011	PWM4 Trigger 2
01010	PWM4 Trigger 1
01001	PWM3 Trigger 2
01000	PWM3 Trigger 1
00111	PWM2 Trigger 2
00110	PWM2 Trigger 1
00101	PWM1 Trigger 2
00100	PWM1 Trigger 1
00011	Reserved
00010	Level software trigger
00001	Common software trigger
00000	No trigger is enabled

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High-Speed, 12-Bit Analog-to-Digital Convert...

14.3.34 ADC Digital Comparator x Control Register (x = 0, 1, 2, 3)

Name: ADCMPxCON
Offset: 0xBA0, 0xBA4, 0xBA8, 0xBAC

Legend: HC = Hardware Clearable bit; HS = Hardware Settable bit; HSC = Hardware Settable/Clearable bit

Bit	15	14	13	12	11	10	9	8
				CHNL[4:0]				
Access				HSC/R	HSC/R	HSC/R	HSC/R	HSC/R
Reset				0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	CMPEN	IE	STAT	BTWN	HIHI	HILO	LOHI	LOLO
Access	R/W	R/W	HS/HC/R	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 12:8 – CHNL[4:0] Input Channel Number bits

Value	Description
11111–11100	Reserved
11011	AN27
11010	AN26
11001	Band gap, 1.2V (AN25)
11000	Temperature sensor (AN24)
10111	AN23
. . .	
00010	AN2
00001	AN1
00000	AN0

Bit 7 – CMPEN Comparator Enable bit

Value	Description
1	Comparator is enabled
0	Comparator is disabled and the STAT status bit is cleared

Bit 6 – IE Comparator Common ADC Interrupt Enable bit

Value	Description
1	Common ADC interrupt will be generated if the comparator detects a comparison event instead of digital comparator interrupt
0	ADC interrupt will not be generated for the comparator

Bit 5 – STAT Comparator Event Status bit

This bit is cleared by hardware when the channel number is read from the CHNL[4:0] bits.

Value	Description
1	A comparison event has been detected since the last read of the CHNL[4:0] bits
0	A comparison event has not been detected since the last read of the CHNL[4:0] bits

Bit 4 – BTWN Between Low/High Comparator Event bit

Value	Description
1	Generates a comparator event when $ADCMPxLO \leq ADCBUFx < ADCMPxHI$
0	Does not generate a digital comparator event when $ADCMPxLO \leq ADCBUFx < ADCMPxHI$

Bit 3 – HIHI High/High Comparator Event bit

Value	Description
1	Generates a digital comparator event when $ADCBUFx \geq ADCMPxHI$
0	Does not generate a digital comparator event when $ADCBUFx \geq ADCMPxHI$

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Bit 2 – HILO High/Low Comparator Event bit

Value	Description
1	Generates a digital comparator event when $ADCBUFx < ADCMPxHI$
0	Does not generate a digital comparator event when $ADCBUFx < ADCMPxHI$

Bit 1 – LOHI Low/High Comparator Event bit

Value	Description
1	Generates a digital comparator event when $ADCBUFx \geq ADCMPxLO$
0	Does not generate a digital comparator event when $ADCBUFx \geq ADCMPxLO$

Bit 0 – LOLO Low/Low Comparator Event bit

Value	Description
1	Generates a digital comparator event when $ADCBUFx < ADCMPxLO$
0	Does not generate a digital comparator event when $ADCBUFx < ADCMPxLO$

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High-Speed, 12-Bit Analog-to-Digital Convert...

14.3.35 ADC Level-Sensitive Trigger Control Register Low

Name: ADLVLTRGL
Offset: 0xBD0

Bit	15	14	13	12	11	10	9	8
	LVLEN[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	LVLEN[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – LVLEN[15:0] Level Trigger for Corresponding Analog Input Enable bits

Value	Description
1	Input trigger is level-sensitive
0	Input trigger is edge-sensitive

dsPIC33CK512MPT608 Family

High-Speed, 12-Bit Analog-to-Digital Convert...

14.3.36 ADC Level-Sensitive Trigger Control Register High

Name: ADLVLTRGH
Offset: 0xBD2

Bit	15	14	13	12	11	10	9	8
					LVLEN[27:24]			
Access					R/W	R/W	R/W	R/W
Reset					0	0	0	0
Bit	7	6	5	4	3	2	1	0
	LVLEN[23:16]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 11:0 – LVLEN[27:16] Level Trigger for Corresponding Analog Input Enable bits

Value	Description
1	Input trigger is level-sensitive
0	Input trigger is edge-sensitive

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High-Speed, 12-Bit Analog-to-Digital Convert...

14.3.37 Dedicated ADC Core x Control Register Low (x = 0 to 3)

Name: ADCORExL
Offset: 0xBD4, 0xBD8, 0xBDC, 0xBE0

Bit	15	14	13	12	11	10	9	8
							SAMC[9:8]	
Access							R/W	R/W
Reset							0	0
Bit	7	6	5	4	3	2	1	0
	SAMC[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 9:0 – SAMC[9:0] Dedicated ADC Core x Conversion Delay Selection bits

These bits determine the time between the trigger event and the start of conversion in the number of the Core Clock Periods (T_{ADCORE}). During this time, the ADC Core x still continues sampling. This feature is enabled by the SAMCxEN bits in the ADCON4L register.

Value	Description
1111111111	1025 T_{ADCORE}
. . .	
0000000001	3 T_{ADCORE}
0000000000	2 T_{ADCORE}

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High-Speed, 12-Bit Analog-to-Digital Convert...

14.3.38 Dedicated ADC Core x Control Register High (x = 0 to 3)

Name: ADCORExH
Offset: 0xBD6, 0xBDA, 0xBDE, 0xBE2

Note:

- For the 6-bit ADC core resolution (RES[1:0] = 00), the EISEL[2:0] bits settings, from '100' to '111', are not valid and should not be used. For the 8-bit ADC core resolution (RES[1:0] = 01), the EISEL[2:0] bits settings, '110' and '111', are not valid and should not be used.

Bit	15	14	13	12	11	10	9	8
				EISEL[2:0]			RES[1:0]	
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
		ADCS[6:0]						
Access		R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset		0	0	0	0	0	0	0

Bits 12:10 – EISEL[2:0] ADC Core x Early Interrupt Time Selection bits

Value	Description
111	Early interrupt is set and interrupt is generated 8 T_{ADCORE} clocks prior to when data are ready
110	Early interrupt is set and interrupt is generated 7 T_{ADCORE} clocks prior to when data are ready
101	Early interrupt is set and interrupt is generated 6 T_{ADCORE} clocks prior to when data are ready
100	Early interrupt is set and interrupt is generated 5 T_{ADCORE} clocks prior to when data are ready
011	Early interrupt is set and interrupt is generated 4 T_{ADCORE} clocks prior to when data are ready
010	Early interrupt is set and interrupt is generated 3 T_{ADCORE} clocks prior to when data are ready
001	Early interrupt is set and interrupt is generated 2 T_{ADCORE} clocks prior to when data are ready
000	Early interrupt is set and interrupt is generated 1 T_{ADCORE} clock prior to when data are ready

Bits 9:8 – RES[1:0] ADC Core x Resolution Selection bits

Value	Description
11	12-bit resolution
10	10-bit resolution
01	8-bit resolution ⁽¹⁾
00	6-bit resolution ⁽¹⁾

Bits 6:0 – ADCS[6:0] ADC Core x Input Clock Divider bits

These bits determine the number of Source Clock Periods ($T_{CORESRC}$) for one Core Clock Period (T_{ADCORE}).

Value	Description
1111111	254 Source Clock Periods
. . .	
0000011	6 Source Clock Periods
0000010	4 Source Clock Periods
0000001	2 Source Clock Periods
0000000	2 Source Clock Periods

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High-Speed, 12-Bit Analog-to-Digital Convert...

14.3.39 ADC Early Interrupt Enable Register Low

Name: ADEIEL

Offset: 0xBF0

Bit	15	14	13	12	11	10	9	8
	EIEN[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	EIEN[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – EIEN[15:0] Early Interrupt Enable for Corresponding Analog Input bits

Value	Description
1	Early interrupt is enabled for the channel
0	Early interrupt is disabled for the channel

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High-Speed, 12-Bit Analog-to-Digital Convert...

14.3.40 ADC Early Interrupt Enable Register High

Name: ADEIEH
Offset: 0xBF2

Bit	15	14	13	12	11	10	9	8
					EIEN[27:24]			
Access					R/W	R/W	R/W	R/W
Reset					0	0	0	0
Bit	7	6	5	4	3	2	1	0
	EIEN[23:16]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 11:0 – EIEN[27:16] Early Interrupt Enable for Corresponding Analog Input bits

Value	Description
1	Early interrupt is enabled for the channel
0	Early interrupt is disabled for the channel

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High-Speed, 12-Bit Analog-to-Digital Convert...

14.3.41 ADC Early Interrupt Status Register Low

Name: ADEISTATL
Offset: 0xBF8

Bit	15	14	13	12	11	10	9	8
	EISTAT[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	EISTAT[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – EISTAT[15:0] Early Interrupt Status for Corresponding Analog Input bits

Value	Description
1	Early interrupt was generated
0	Early interrupt was not generated since the last ADCBUFx read

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High-Speed, 12-Bit Analog-to-Digital Convert...

14.3.42 ADC Early Interrupt Status Register High

Name: ADEISTATH
Offset: 0xBFA

Bit	15	14	13	12	11	10	9	8
	EISTAT[27:24]							
Access					R/W	R/W	R/W	R/W
Reset					0	0	0	0
Bit	7	6	5	4	3	2	1	0
	EISTAT[23:16]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 11:0 – EISTAT[27:16] Early Interrupt Status for Corresponding Analog Input bits

Value	Description
1	Early interrupt was generated
0	Early interrupt was not generated since the last ADCBUFx read

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High-Speed, 12-Bit Analog-to-Digital Convert...

14.3.43 ADC Control Register 5 Low

Name: ADCON5L

Offset: 0xC00

Legend: HSC = Hardware Settable/Clearable bit

Bit	15	14	13	12	11	10	9	8
	SHRRDY				C3RDY	C2RDY	C1RDY	C0RDY
Access	R/HSC				R/W	R/W	R/W	R/W
Reset	0				0	0	0	0

Bit	7	6	5	4	3	2	1	0
	SHRPWR				C3PWR	C2PWR	C1PWR	C0PWR
Access	R/W				R/W	R/W	R/W	R/W
Reset	0				0	0	0	0

Bit 15 – SHRRDY Shared ADC Core Ready Flag bit

Value	Description
1	ADC core is powered and ready for operation
0	ADC core is not ready for operation

Bit 11 – C3RDY Dedicated ADC Core 3 Ready Flag bit

Value	Description
1	ADC Core 3 is powered and ready for operation
0	ADC Core 3 is not ready for operation

Bit 10 – C2RDY Dedicated ADC Core 2 Ready Flag bit

Value	Description
1	ADC Core 2 is powered and ready for operation
0	ADC Core 2 is not ready for operation

Bit 9 – C1RDY Dedicated ADC Core 1 Ready Flag bit

Value	Description
1	ADC Core 1 is powered and ready for operation
0	ADC Core 1 is not ready for operation

Bit 8 – C0RDY Dedicated ADC Core 0 Ready Flag bit

Value	Description
1	ADC Core 0 is powered and ready for operation
0	ADC Core 0 is not ready for operation

Bit 7 – SHRPWR Shared ADC Core Power Enable bit

Value	Description
1	ADC core is powered
0	ADC core is off

Bit 3 – C3PWR Dedicated ADC Core 3 Power Enable bit

Value	Description
1	ADC Core 3 is powered
0	ADC Core 3 off

Bit 2 – C2PWR Dedicated ADC Core 2 Power Enable bit

Value	Description
1	ADC Core 2 is powered
0	ADC Core 2 is off

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Bit 1 – C1PWR Dedicated ADC Core 1 Power Enable bit

Value	Description
1	ADC Core 1 is powered
0	ADC Core 1 is off

Bit 0 – C0PWR Dedicated ADC Core 0 Power Enable bit

Value	Description
1	ADC Core 0 is powered
0	ADC Core 0 is off

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14.3.44 ADC Control Register 5 High

Name: ADCON5H
Offset: 0xC02

Bit	15	14	13	12	11	10	9	8
					WARMTIME[3:0]			
Access					R/W	R/W	R/W	R/W
Reset					0	0	0	0

Bit	7	6	5	4	3	2	1	0
	SHRCIE				C3CIE	C2CIE	C1CIE	C0CIE
Access	R/W				R/W	R/W	R/W	R/W
Reset	0				0	0	0	0

Bits 11:8 – WARMTIME[3:0] ADC Dedicated Core Power-up Delay bits

These bits determine the power-up delay in the number of the Core Source Clock Periods ($T_{CORESRC}$) for all ADC cores.

Value	Description
1111	32768 Source Clock Periods
1110	16384 Source Clock Periods
1101	8192 Source Clock Periods
1100	4096 Source Clock Periods
1011	2048 Source Clock Periods
1010	1024 Source Clock Periods
1001	512 Source Clock Periods
1000	256 Source Clock Periods
0111	128 Source Clock Periods
0110	64 Source Clock Periods
0101	32 Source Clock Periods
0100	16 Source Clock Periods
00xx	16 Source Clock Periods

Bit 7 – SHRCIE Shared ADC Core Ready Common Interrupt Enable bit

Value	Description
1	Common interrupt will be generated when ADC core is powered and ready for operation
0	Common interrupt is disabled for an ADC core ready event

Bit 3 – C3CIE Dedicated ADC Core 3 Ready Common Interrupt Enable bit

Value	Description
1	Common interrupt will be generated when ADC Core 3 is powered and ready for operation
0	Common interrupt is disabled for an ADC Core 3 ready event

Bit 2 – C2CIE Dedicated ADC Core 2 Ready Common Interrupt Enable bit

Value	Description
1	Common interrupt will be generated when ADC Core 2 is powered and ready for operation
0	Common interrupt is disabled for an ADC Core 2 ready event

Bit 1 – C1CIE Dedicated ADC Core 1 Ready Common Interrupt Enable bit

Value	Description
1	Common interrupt will be generated when ADC Core 1 is powered and ready for operation
0	Common interrupt is disabled for an ADC Core 1 ready event

Bit 0 – C0CIE Dedicated ADC Core 0 Ready Common Interrupt Enable bit

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High-Speed, 12-Bit Analog-to-Digital Convert...

Value	Description
1	Common interrupt will be generated when ADC Core 0 is powered and ready for operation
0	Common interrupt is disabled for an ADC Core 0 ready event

14.3.45 ADC Buffer x Register

Name: ADCBUFx
Offset: 0xC0C, 0xC0E, 0xC10, 0xC12, 0xC14, 0xC16, 0xC18, 0xC1A, 0xC1C, 0xC1E, 0xC20, 0xC22, 0xC24, 0xC26, 0xC28, 0xC2A, 0xC2C, 0xC2E, 0xC30, 0xC32, 0xC34, 0xC36

Bit	15	14	13	12	11	10	9	8
	ADCBUFx[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	ADCBUFx[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – ADCBUFx[15:0] Buffer Data bits

14.4 Analog-to-Digital Converter Resources

Many useful resources are provided on the main product page of the Microchip website for the devices listed in this data sheet. This product page contains the latest updates and additional information.

14.4.1 Differential-Mode

ANNx negative external inputs are used for Differential-mode as shown in [Figure 14-2](#). To enable Differential-mode, the DIFF bit (in the ADMODxL or ADMODxH register) is set for the corresponding channel.

14.4.2 Key Resources

- “12-Bit High-Speed, Multiple SARs A/D Converter (ADC)” (www.microchip.com/DS70005213) in the “dsPIC33/PIC24 Family Reference Manual”
- Code Samples
- Application Notes
- Software Libraries
- Webinars
- All Related “dsPIC33/PIC24 Family Reference Manual” Sections
- Development Tools

15. High-Speed Analog Comparator with Slope Compensation DAC

Notes:

1. This data sheet summarizes the features of the dsPIC33CK512MPT608 family of devices. It is not intended to be a comprehensive reference source. To complement the information in this data sheet, refer to “**High-Speed Analog Comparator Module**” (www.microchip.com/DS70005280) in the “*dsPIC33/PIC24 Family Reference Manual*.”
2. Some registers and associated bits described in this section may not be available on all devices. Refer to [4. Memory Organization](#) in this data sheet for device-specific register and bit information.

The high-speed analog comparator module provides a method to monitor voltage, current and other critical signals in a power conversion application that may be too fast for the CPU and ADC to capture. The analog comparator module can be used to implement Peak Current mode control, Critical Conduction mode (variable frequency) and Hysteretic Control mode.

15.1 Overview

The high-speed analog comparator module is comprised of a high-speed comparator, Pulse Density Modulation (PDM) DAC and a slope compensation unit. The slope compensation unit provides a user-defined slope which can be used to alter the DAC output. This feature is useful in applications, such as Peak Current mode control, where slope compensation is required to maintain the stability of the power supply. The user simply specifies the direction and rate of change for the slope compensation, and the output of the DAC is modified accordingly.

The DAC consists of a PDM unit, followed by a digitally controlled multiphase RC filter. The PDM unit uses a phase accumulator circuit to generate an output stream of pulses. The density of the pulse stream is proportional to the input data value, relative to the maximum value supported by the bit width of the accumulator. The output pulse density is representative of the desired output voltage. The pulse stream is filtered with an RC filter to yield an analog voltage. The output of the DAC is connected to the negative input of the comparator. The positive input of the comparator can be selected using a MUX from the input pins. The comparator provides a high-speed operation with a typical delay of 15 ns.

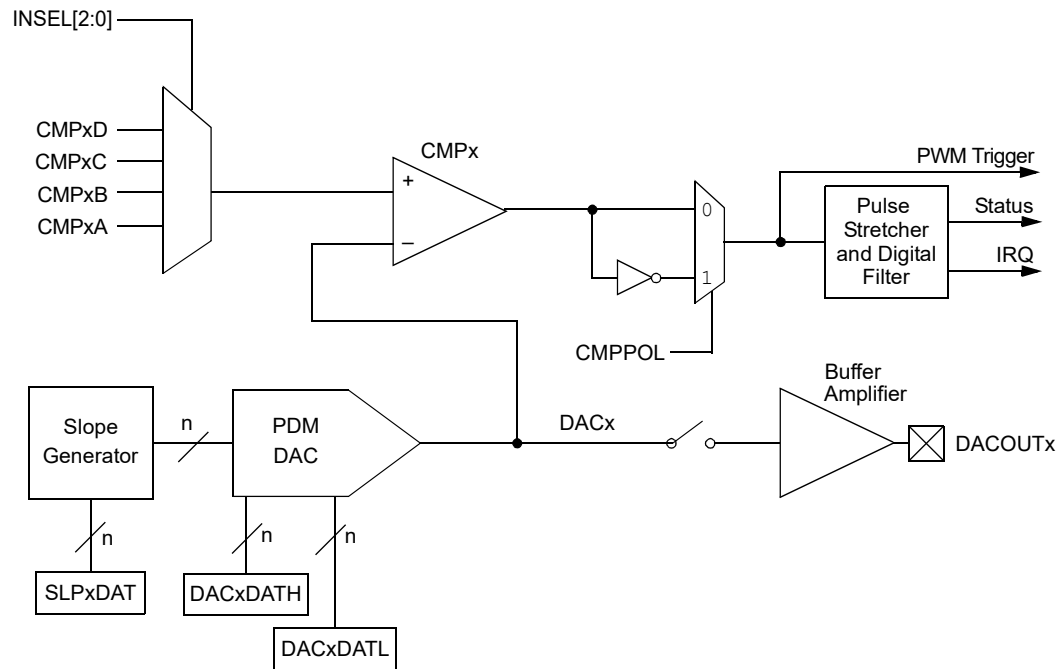
The output of the comparator is processed by the pulse stretcher and the digital filter blocks, which prevent comparator response to unintended fast transients in the inputs. [Figure 15-1](#) shows a block diagram of the high-speed analog comparator module. The DAC module can be operated in one of three modes: Slope Generation mode, Hysteretic mode and Triangle Wave mode. Each of these modes can be used in a variety of power supply applications.

Note: This device supports two DACOUT pins, DACOUT1 and DACOUT2. DAC instances, DAC1, DAC2 and DAC3, are associated with DACOUT1. DAC instances, DAC4, DAC5 and DAC6, are associated with DACOUT2. The DACOUTx pin can only be associated with a single DAC output at any given time. If more than one DACOEN bit is set, the DACOUTx pin will be a combination of the signals.

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High-Speed Analog Comparator with Slope Comp...

Figure 15-1. High-Speed Analog Comparator Module Block Diagram



Note: n = 16

15.2 Features Overview

- Four Rail-to-Rail Analog Comparators
- Up to Five Selectable Input Sources per Comparator:
 - Three external inputs
- Programmable Comparator Hysteresis
- Programmable Output Polarity
- Interrupt Generation Capability
- Dedicated Pulse Density Modulation DAC for each Analog Comparator:
 - PDM unit followed by a digitally controlled multimode multipole RC filter
- Multimode Multipole RC Output Filter:
 - Transition mode: Provides the fastest response
 - Fast mode: For tracking DAC slopes
 - Steady-State mode: Provides 12-bit resolution
- Slope Compensation along with each DAC:
 - Slope Generation mode
 - Hysteretic Control mode
 - Triangle Wave mode
- Functional Support for the High-Speed PWM module which Includes:
 - PWM duty cycle control
 - PWM period control
 - PWM Fault detect

15.3 DAC Control Registers

The DACCTRL1L and DACCTRL2H/L registers are common configuration registers for DAC modules. The DACxCON, DACxDAT, SLPxCON and SLPxDAT registers specify the operation of individual modules.

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High-Speed Analog Comparator with Slope Comp...

15.4 DAC Control Registers

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x00 ... 0x0C47	Reserved									
0x0C48	DACCTRL1L	15:8	DACON		DACSIDL					
		7:0	CLKSEL[1:0]		CLKDIV[1:0]			FCLKDIV[2:0]		
0x0C4A ... 0x0C4B	Reserved									
0x0C4C	DACCTRL2L	15:8							TMODTIME[9:0]	
		7:0	TMODTIME[9:0]							
0x0C4E	DACCTRL2H	15:8							SSTIME[9:0]	
		7:0	SSTIME[9:0]							
0x0C50	DAC1CONL	15:8	DACEN	IRQM[1:0]				CBE	DACOEN	FLTREN
		7:0	CMPSTAT	CMPPOL	INSEL[2:0]			HYSPOL	HYSSEL[1:0]	
0x0C52	DAC1CONH	15:8							TMCB[9:0]	
		7:0	TMCB[9:0]							
0x0C54	DAC1DATL	15:8					DACLOW[11:0]			
		7:0	DACLOW[11:0]							
0x0C56	DAC1DATH	15:8					DACDAT[11:0]			
		7:0	DACDAT[11:0]							
0x0C58	SLP1CONL	15:8	HCFSEL[3:0]				SLPSTOPA[3:0]			
		7:0	SLPSTOPB[3:0]				SLPSTRT[3:0]			
0x0C5A	SLP1CONH	15:8	SLOPEN				HME	TWME	PSE	
		7:0								
0x0C5C	SLP1DAT	15:8	SLPDAT[15:8]							
		7:0	SLPDAT[7:0]							
0x0C5E ... 0x0C5F	Reserved									
0x0C60	DAC2CONL	15:8	DACEN	IRQM[1:0]				CBE	DACOEN	FLTREN
		7:0	CMPSTAT	CMPPOL	INSEL[2:0]			HYSPOL	HYSSEL[1:0]	
0x0C62	DAC2CONH	15:8							TMCB[9:0]	
		7:0	TMCB[9:0]							
0x0C64	DAC2DATL	15:8					DACLOW[11:0]			
		7:0	DACLOW[11:0]							
0x0C66	DAC2DATH	15:8					DACDAT[11:0]			
		7:0	DACDAT[11:0]							
0x0C68	SLP2CONL	15:8	HCFSEL[3:0]				SLPSTOPA[3:0]			
		7:0	SLPSTOPB[3:0]				SLPSTRT[3:0]			
0x0C6A	SLP2CONH	15:8	SLOPEN				HME	TWME	PSE	
		7:0								
0x0C6C	SLP2DAT	15:8	SLPDAT[15:8]							
		7:0	SLPDAT[7:0]							
0x0C6E ... 0x0C6F	Reserved									
0x0C70	DAC3CONL	15:8	DACEN	IRQM[1:0]				CBE	DACOEN	FLTREN
		7:0	CMPSTAT	CMPPOL	INSEL[2:0]			HYSPOL	HYSSEL[1:0]	
0x0C72	DAC3CONH	15:8							TMCB[9:0]	
		7:0	TMCB[9:0]							
0x0C74	DAC3DATL	15:8					DACLOW[11:0]			
		7:0	DACLOW[11:0]							
0x0C76	DAC3DATH	15:8					DACDAT[11:0]			
		7:0	DACDAT[11:0]							
0x0C78	SLP3CONL	15:8	HCFSEL[3:0]				SLPSTOPA[3:0]			
		7:0	SLPSTOPB[3:0]				SLPSTRT[3:0]			

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High-Speed Analog Comparator with Slope Comp...

.....continued

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x0C7A	SLP3CONH	15:8	SLOPEN				HME	TWME	PSE	
		7:0								
0x0C7C	SLP3DAT	15:8	SLPDAT[15:8]							
		7:0	SLPDAT[7:0]							
0x0C7E ... 0x0C7F	Reserved									
0x0C80	DAC4CONL	15:8	DACEN	IRQM[1:0]				CBE	DACOEN	FLTREN
		7:0	CMPSTAT	CMPPOL	INSEL[2:0]			HYSPOL	HYSSEL[1:0]	
0x0C82	DAC4CONH	15:8							TMCB[9:0]	
		7:0	TMCB[9:0]							
0x0C84	DAC4DATL	15:8					DACLOW[11:0]			
		7:0	DACLOW[11:0]							
0x0C86	DAC4DATH	15:8					DACDAT[11:0]			
		7:0	DACDAT[11:0]							
0x0C88	SLP4CONL	15:8	HCFSEL[3:0]				SLPSTOPA[3:0]			
		7:0	SLPSTOPB[3:0]				SLPSTRT[3:0]			
0x0C8A	SLP4CONH	15:8	SLOPEN				HME	TWME	PSE	
		7:0								
0x0C8C	SLP4DAT	15:8	SLPDAT[15:8]							
		7:0	SLPDAT[7:0]							
0x0C8E ... 0x0C8F	Reserved									
0x0C90	DAC5CONL	15:8	DACEN	IRQM[1:0]				CBE	DACOEN	FLTREN
		7:0	CMPSTAT	CMPPOL	INSEL[2:0]			HYSPOL	HYSSEL[1:0]	
0x0C92	DAC5CONH	15:8							TMCB[9:0]	
		7:0	TMCB[9:0]							
0x0C94	DAC5DATL	15:8					DACLOW[11:0]			
		7:0	DACLOW[11:0]							
0x0C96	DAC5DATH	15:8					DACDAT[11:0]			
		7:0	DACDAT[11:0]							
0x0C98	SLP5CONL	15:8	HCFSEL[3:0]				SLPSTOPA[3:0]			
		7:0	SLPSTOPB[3:0]				SLPSTRT[3:0]			
0x0C9A	SLP5CONH	15:8	SLOPEN				HME	TWME	PSE	
		7:0								
0x0C9C	SLP5DAT	15:8	SLPDAT[15:8]							
		7:0	SLPDAT[7:0]							
0x0C9E ... 0x0C9F	Reserved									
0x0CA0	DAC6CONL	15:8	DACEN	IRQM[1:0]				CBE	DACOEN	FLTREN
		7:0	CMPSTAT	CMPPOL	INSEL[2:0]			HYSPOL	HYSSEL[1:0]	
0x0CA2	DAC6CONH	15:8							TMCB[9:0]	
		7:0	TMCB[9:0]							
0x0CA4	DAC6DATL	15:8					DACLOW[11:0]			
		7:0	DACLOW[11:0]							
0x0CA6	DAC6DATH	15:8					DACDAT[11:0]			
		7:0	DACDAT[11:0]							
0x0CA8	SLP6CONL	15:8	HCFSEL[3:0]				SLPSTOPA[3:0]			
		7:0	SLPSTOPB[3:0]				SLPSTRT[3:0]			
0x0CAA	SLP6CONH	15:8	SLOPEN				HME	TWME	PSE	
		7:0								
0x0CAC	SLP6DAT	15:8	SLPDAT[15:8]							
		7:0	SLPDAT[7:0]							

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High-Speed Analog Comparator with Slope Comp...

15.4.1 DAC Control 1 Low Register

Name: DACCTRL1L
Offset: 0xC48

Note:

- These bits should only be changed when DACON = 0 to avoid unpredictable behavior.

Bit	15	14	13	12	11	10	9	8
	DACON		DACSIDL					
Access	R/W		R/W					
Reset	0		0					

Bit	7	6	5	4	3	2	1	0
	CLKSEL[1:0]		CLKDIV[1:0]			FCLKDIV[2:0]		
Access	R/W	R/W	R/W	R/W		R/W	R/W	R/W
Reset	0	0	0	0		0	0	0

Bit 15 – DACON Common DAC Module Enable bit

Value	Description
1	Enables DAC modules
0	Disables DAC modules and disables FSCM clocks to reduce power consumption; any pending Slope mode and/or Underflow conditions are cleared

Bit 13 – DACSIDL DAC Stop in Idle Mode bit

Value	Description
1	Discontinues module operation when device enters Idle mode
0	Continues module operation in Idle mode

Bits 7:6 – CLKSEL[1:0] DAC Clock Source Select bits⁽¹⁾

Value	Description
11	F _{PLLO}
10	AF _{PLLO}
01	F _{VCO/2}
00	AF _{VCO/2}

Bits 5:4 – CLKDIV[1:0] DAC Clock Divider bits⁽¹⁾

Value	Description
11	Divide-by-4
10	Divide-by-3 (non-uniform duty cycle)
01	Divide-by-2
00	1x

Bits 2:0 – FCLKDIV[2:0] Comparator Filter Clock Divider bits

Value	Description
111	Divide-by-8
110	Divide-by-7
101	Divide-by-6
100	Divide-by-5
011	Divide-by-4
010	Divide-by-3
001	Divide-by-2
000	1x

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15.4.2 DAC Control 2 Low Register

Name: DACCTRL2L
Offset: 0xC4C

Bit	15	14	13	12	11	10	9	8
							TMODTIME[9:0]	
Access							R/W	R/W
Reset							0	0
Bit	7	6	5	4	3	2	1	0
	TMODTIME[9:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 9:0 – TMODTIME[9:0] Transition Mode Duration bits

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15.4.3 DAC Control 2 High Register

Name: DACCTRL2H

Offset: 0xC4E

Note:

1. The value for SSTIME[9:0] should be greater than the TMODTIME[9:0] value.

Bit	15	14	13	12	11	10	9	8
							SSTIME[9:0]	
Access							R/W	R/W
Reset							0	0
Bit	7	6	5	4	3	2	1	0
	SSTIME[9:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 9:0 – SSTIME[9:0] Time from Start of Transition Mode until Steady-State Filter is Enabled bits⁽¹⁾

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High-Speed Analog Comparator with Slope Comp...

15.4.4 DACx Control Low Register

Name: DACxCONL
Offset: 0xC50, 0xC60, 0xC70, 0xC80, 0xC90, 0xCA0

Bit	15	14	13	12	11	10	9	8
	DACEN	IRQM[1:0]				CBE	DACOEN	FLTREN
Access	R/W	R/W	R/W			R/W	R/W	R/W
Reset	0	0	0			0	0	0

Bit	7	6	5	4	3	2	1	0
	CMPSTAT	CMPPOL	INSEL[2:0]			HYSPOL	HYSSEL[1:0]	
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – DACEN Individual DACx Module Enable bit

Value	Description
1	Enables DACx module
0	Disables DACx module to reduce power consumption; any pending Slope mode and/or Underflow conditions are cleared

Bits 14:13 – IRQM[1:0] Interrupt Mode select bits

Value	Description
11	Generates an interrupt on either a rising or falling edge detect
10	Generates an interrupt on a falling edge detect
01	Generates an interrupt on a rising edge detect
00	Interrupts are disabled

Bit 10 – CBE Comparator Blank Enable bit

Value	Description
1	Enables the analog comparator output to be blanked (gated off) during the recovery transition following the completion of a slope operation
0	Disables the blanking signal to the analog comparator; therefore, the analog comparator output is always active

Bit 9 – DACOEN DACx Output Buffer Enable bit

Value	Description
1	DACx analog voltage is connected to the DACOUTx pin
0	DACx analog voltage is not connected to the DACOUTx pin

Bit 8 – FLTREN Comparator Digital Filter Enable bit

Value	Description
1	Digital filter is enabled
0	Digital filter is disabled

Bit 7 – CMPSTAT Comparator Status bits

Bit 6 – CMPPOL Comparator Output Polarity Control bit

Value	Description
1	Output is inverted
0	Output is noninverted

Bits 5:3 – INSEL[2:0] Comparator Input Source Select bits

Value	Description
111	Reserved

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Value	Description
110	Reserved
101	Reserved
100	Reserved
011	CMPxD input pin
010	CMPxC input pin
001	CMPxB input pin
000	CMPxA input pin

Bit 2 – HYPOL Comparator Hysteresis Polarity Select bit

Value	Description
1	Hysteresis is applied to the falling edge of the comparator output
0	Hysteresis is applied to the rising edge of the comparator output

Bits 1:0 – HYSSEL[1:0] Comparator Hysteresis Select bits

Value	Description
11	45 mv hysteresis
10	30 mv hysteresis
01	15 mv hysteresis
00	No hysteresis is selected

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High-Speed Analog Comparator with Slope Comp...

15.4.5 DACx Control High Register

Name: DACxCONH
Offset: 0xC52, 0xC62, 0xC72, 0xC82, 0xC92, 0xCA2

Bit	15	14	13	12	11	10	9	8
							TMCB[9:0]	
Access							R/W	R/W
Reset							0	0
Bit	7	6	5	4	3	2	1	0
	TMCB[9:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 9:0 – TMCB[9:0] DACx Leading-Edge Blanking bits

These register bits specify the blanking period for the comparator, following changes to the DAC output during Change-of-State (COS), for the input signal selected by the HCFSEL[3:0] bits in [15.4.8. SLPxCONL](#).

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15.4.6 DACx Data Low Register

Name: DACxDATL
Offset: 0xC54, 0xC64, 0xC74, 0xC84, 0xC94, 0xCA4

Bit	15	14	13	12	11	10	9	8
	DACLOW[11:0]							
Access					R/W	R/W	R/W	R/W
Reset					0	0	0	0
Bit	7	6	5	4	3	2	1	0
	DACLOW[11:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 11:0 – DACLOW[11:0] DACx Low Data bits

In Hysteretic mode, Slope Generator mode and Triangle mode, this register specifies the low data value and/or limit for the DACx module. Valid values are from 205 to 3890.

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15.4.7 DACx Data High Register

Name: DACxDATH
Offset: 0xC56, 0xC66, 0xC76, 0xC86, 0xC96, 0xCA6

Bit	15	14	13	12	11	10	9	8
	DACDAT[11:0]							
Access					R/W	R/W	R/W	R/W
Reset					0	0	0	0
Bit	7	6	5	4	3	2	1	0
	DACDAT[11:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 11:0 – DACDAT[11:0] DACx High Data bits

This register specifies the high DACx data value. Valid values are from 205 to 3890.

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15.4.8 DAC Slope x Control Low Register

Name: SLPxCONL
Offset: 0xC58, 0xC68, 0xC78, 0xC88, 0xC98, 0xCA8

Bit	15	14	13	12	11	10	9	8
	HCFSEL[3:0]				SLPSTOPA[3:0]			
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	SLPSTOPB[3:0]				SLPSTRT[3:0]			
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:12 – HCFSEL[3:0] Hysteretic Comparator Function Input Select bits

The selected input signal controls the switching between the DACx high limit (DACxDATH) and the DACx low limit (DACxDATL) as the data source for the PDM DAC. It modifies the polarity of the comparator, and the rising and falling edges initiate the start of the LEB counter (TMCB[9:0] bits in [15.4.5. DACxCONH](#)).

Input Selection	Source
1111	1
1100	0
1011	0
1010	0
1001	0
1000	PWM8H
0111	Reserved
0110	PWM6H
0101	Reserved
0100	PWM4H
0011	PWM3H
0010	PWM2H
0001	PWM1H
0000	0

Bits 11:8 – SLPSTOPA[3:0] Slope Stop A Signal Select bits

The selected Slope Stop A signal is logically OR'd with the selected Slope Stop B signal to terminate the slope function.

Slope Stop A Signal Selection	Source
1101–1111	1
1000	PWM8 Trigger 2
0111	Reserved
0110	PWM6 Trigger 2
0101	Reserved
0100	PWM4 Trigger 2
0011	PWM3 Trigger 2
0010	PWM2 Trigger 2
0001	PWM1 Trigger 2
0000	0

Bits 7:4 – SLPSTOPB[3:0] Slope Stop B Signal Select bits

The selected Slope Stop B signal is logically OR'd with the selected Slope Stop A signal to terminate the slope function.

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Slope Stop B Signal Selection	Source
0100–0111	1
0110	CMP6 out
0101	CMP5 out
0100	CMP4 out
0011	CMP3 out
0010	CMP2 out
0001	CMP1 out
0000	0

Bits 3:0 – SLPSTRT[3:0] Slope Start Signal Select bits

Slope Start Signal Selection	Source
1101–1111	1
1000	PWM8 Trigger 1
0111	Reserved
0110	PWM6 Trigger 1
0101	Reserved
0100	PWM4 Trigger 1
0011	PWM3 Trigger 1
0010	PWM2 Trigger 1
0001	PWM1 Trigger 1
0000	0

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15.4.9 DAC Slope x Control High Register

Name: SLPxCONH
Offset: 0xC5A, 0xC6A, 0xC7A, 0xC8A, 0xC9A, 0xCA

Notes:

1. HME mode requires the user to disable the slope function (SLOPEN = 0).
2. TWME mode requires the user to enable the slope function (SLOPEN = 1).

Bit	15	14	13	12	11	10	9	8
	SLOPEN				HME	TWME	PSE	
Access	R/W				R/W	R/W	R/W	
Reset	0				0	0	0	

Bit	7	6	5	4	3	2	1	0
Access								
Reset								

Bit 15 – SLOPEN Slope Function Enable/On bit

Value	Description
1	Enables slope function
0	Disables slope function; slope accumulator is disabled to reduce power consumption

Bit 11 – HME Hysteretic Mode Enable bit⁽¹⁾

Value	Description
1	Enables Hysteretic mode for DACx
0	Disables Hysteretic mode for DACx

Bit 10 – TWME Triangle Wave Mode Enable bit⁽²⁾

Value	Description
1	Enables Triangle Wave mode for DACx
0	Disables Triangle Wave mode for DACx

Bit 9 – PSE Positive Slope Mode Enable bit

Value	Description
1	Slope mode is positive (increasing)
0	Slope mode is negative (decreasing)

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15.4.10 DAC Slope x Data Register

Name: SLPxDAT
Offset: 0xC5C, 0xC6C, 0xC7C, 0xC8C, 0xC9C, 0xCAC

Bit	15	14	13	12	11	10	9	8
	SLPDAT[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	SLPDAT[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – SLPDAT[15:0] Slope Ramp Rate Value bits
The SLPDATx value is in 12.4 format.

16. Quadrature Encoder Interface (QEI)

Notes:

- 1. This data sheet summarizes the features of the dsPIC33CK512MPT608 family of devices. It is not intended to be a comprehensive resource. For more information, refer to the “**Quadrature Encoder Interface (QEI)**” (www.microchip.com/DS70000601) in the “dsPIC33/PIC24 Family Reference Manual.”
- 2. Some registers and associated bits described in this section may not be available on all devices. Refer to [4. Memory Organization](#) in this data sheet for device-specific register and bit information.

The Quadrature Encoder Interface (QEI) module provides the interface to incremental encoders for obtaining mechanical position data. Quadrature Encoders, also known as incremental encoders or optical encoders, detect position and speed of rotating motion systems. Quadrature Encoders enable closed-loop control of motor control applications, such as Switched Reluctance (SR) and AC Induction Motors (ACIM).

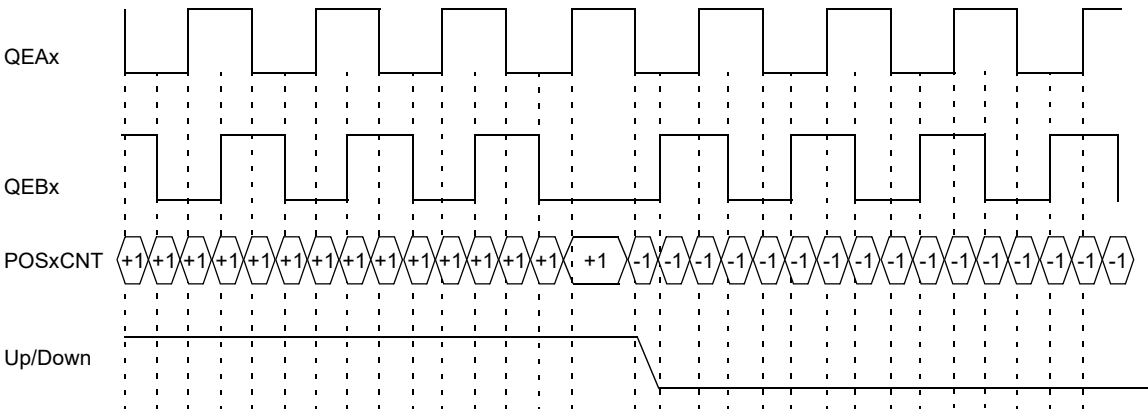
A typical Quadrature Encoder includes a slotted wheel attached to the shaft of the motor and an emitter/detector module that senses the slots in the wheel. Typically, three output channels, Phase A (QEAx), Phase B (QEBx) and Index (INDXx), provide information on the movement of the motor shaft, including distance and direction.

The two channels, Phase A (QEAx) and Phase B (QEBx), are typically 90 degrees out of phase with respect to each other. The Phase A and Phase B channels have a unique relationship. If Phase A leads Phase B, the direction of the motor is deemed positive or forward. If Phase A lags Phase B, the direction of the motor is deemed negative or reverse. The Index pulse occurs once per mechanical revolution and is used as a reference to indicate an absolute position. [Figure 16-1](#) illustrates the Quadrature Encoder Interface signals.

The Quadrature signals from the encoder can have four unique states (‘01’, ‘00’, ‘10’ and ‘11’) that reflect the relationship between QEAx and QEBx. [Figure 16-1](#) illustrates these states for one count cycle. The order of the states get reversed when the direction of travel changes.

The Quadrature Decoder increments or decrements the 32-bit up/down Position x Counter (POSxCNTH/L) registers for each Change-of-State (COS). The counter increments when QEAx leads QEBx and decrements when QEBx leads QEAx. [Table 16-1](#) shows an overview of the QEI module.

Figure 16-1. Quadrature Encoder Interface Signals



[Table 16-1](#) shows the truth table that describes how the Quadrature signals are decoded.

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Table 16-1. Truth Table for Quadrature Encoder

Current Quadrature State		Previous Quadrature State		Action
QA	QB	QA	QB	
1	1	1	1	No count or direction change
1	1	1	0	Count up
1	1	0	1	Count down
1	1	0	0	Invalid state change; ignore
1	0	1	1	Count down
1	0	1	0	No count or direction change
1	0	0	1	Invalid state change; ignore
1	0	0	0	Count up
0	1	1	1	Count up
0	1	1	0	Invalid state change; ignore
0	1	0	1	No count or direction change
0	1	0	0	Count down
0	0	1	1	Invalid state change; ignore
0	0	1	0	Count down
0	0	0	1	Count up
0	0	0	0	No count or direction change

Figure 16-2 illustrates the simplified block diagram of the QEI module. The QEI module consists of decoder logic to interpret the Phase A (QEAx) and Phase B (QEBx) signals, and an up/down counter to accumulate the count. The counter pulses are generated when the Quadrature state changes. The count direction information must be maintained in a register until a direction change is detected. The module also includes digital noise filters, which condition the input signal.

The QEI module consists of the following major features:

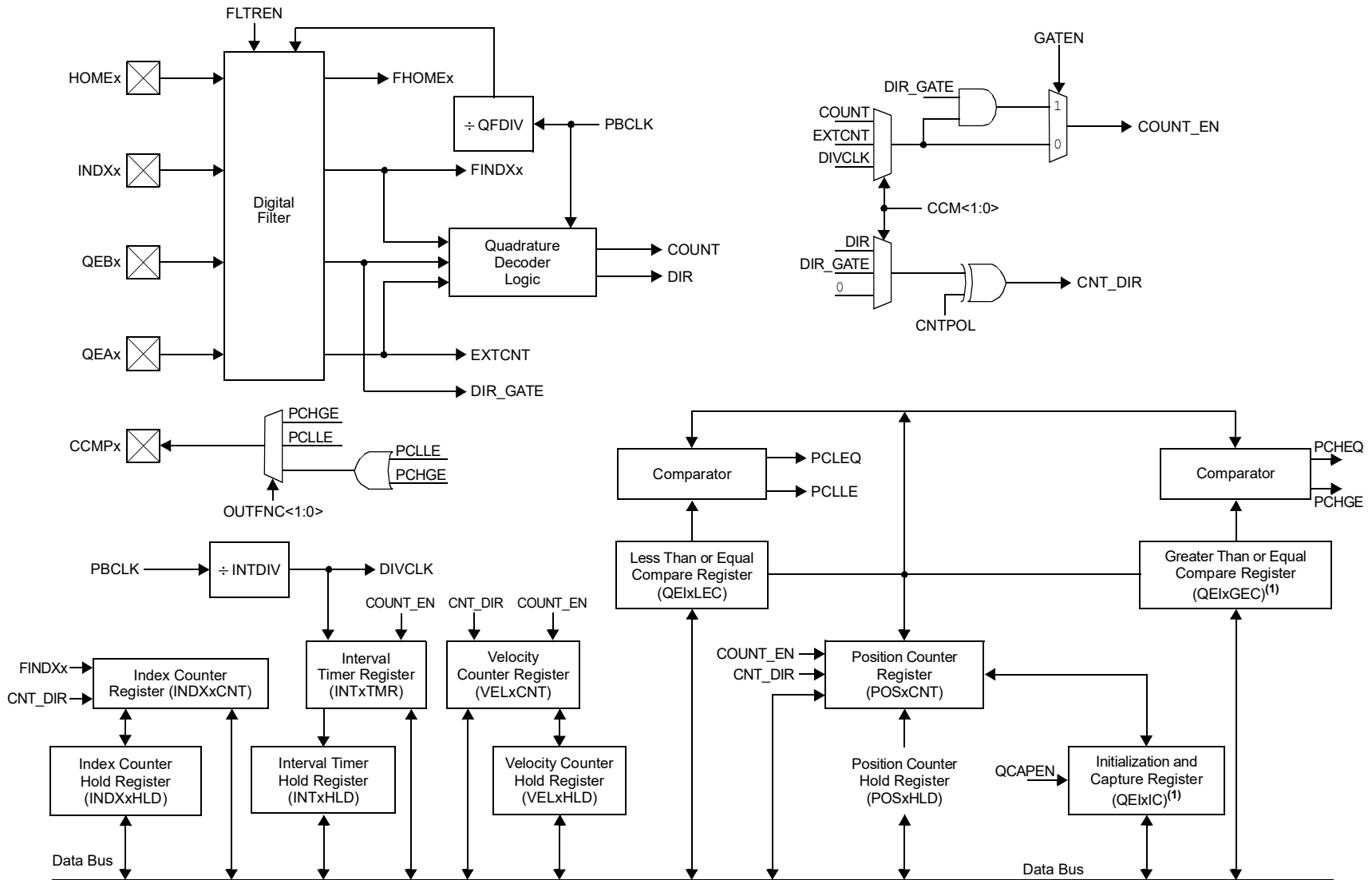
- Four Input Pins: Two Phase Signals, an Index Pulse and a Home Pulse
- Programmable Digital Noise Filters on Inputs
- Quadrature Decoder providing Counter Pulses and Count Direction
- Count Direction Status
- 4x Count Resolution
- Index (INDXx) Pulse to Reset the Position Counter
- General Purpose 32-Bit Timer/Counter mode
- Interrupts generated by QEI or Counter Events
- 32-Bit Velocity Counter
- 32-Bit Position Counter
- 32-Bit Index Pulse Counter
- 32-Bit Interval Timer
- 32-Bit Position Initialization/Capture Register
- 32-Bit Compare Less Than and Greater Than Registers
- External Up/Down Count mode
- External Gated Count mode
- External Gated Timer mode

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- Interval Timer mode

Figure 16-2. Quadrature Encoder Interface (QEI) Module Block Diagram



Note:

1. These registers map to the same memory location.

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Quadrature Encoder Interface (QEI)

16.1 QEI Control/Status Registers

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x00 ... 0x013F	Reserved									
0x0140	QE1CON	15:8 7:0	QE1EN		QE1SIDL	PIMOD[2:0]			IMV[1:0]	
0x0142 ... 0x0143	Reserved									
0x0144	QE1IOCL	15:8 7:0	QCAPEN HOMPOL	FLTREN IDXPOL		QFDIV[2:0] QEAPOL	HOME	OUTFNC[1:0] INDEX		SWPAB QEA
0x0146	QE1IOCH	15:8 7:0								HCAPEN
0x0148	QE1STAT	15:8 7:0			PCHEQIRQ VELOVIRQ	PCHEQIEN VELOVIEN	PCLEQIRQ HOMIRQ	PCLEQIEN HOMIEN	POSOVIRQ IDXIRQ	POSOVIEN IDXIEN
0x014A ... 0x014B	Reserved									
0x014C	POS1CNTL	15:8 7:0	POSCNT[15:8] POSCNT[7:0]							
0x014E	POS1CNTH	15:8 7:0	POSCNT[31:24] POSCNT[23:16]							
0x0150 ... 0x0151	Reserved									
0x0152	POS1HLD	15:8 7:0	POSHLD[31:24] POSHLD[23:16]							
0x0154	VEL1CNTL	15:8 7:0	VELCNT[15:8] VELCNT[7:0]							
0x0156	VEL1CNTH	15:8 7:0	VELCNT[31:24] VELCNT[23:16]							
0x0158 ... 0x0159	Reserved									
0x015A	VEL1HLD	15:8 7:0	VELHLD[31:24] VELHLD[23:16]							
0x015C	INT1TMRL	15:8 7:0	INTTMR[15:8] INTTMR[7:0]							
0x015E	INT1TMRH	15:8 7:0	INTTMR[31:24] INTTMR[23:16]							
0x0160	INT1HLDL	15:8 7:0	INTXHLD[15:8] INTXHLD[7:0]							
0x0162	INT1HLDH	15:8 7:0	INTHLD[31:24] INTHLD[23:16]							
0x0164	INDX1CNTL	15:8 7:0	INDXCNT[15:8] INDXCNT[7:0]							
0x0166	INDX1CNTH	15:8 7:0	INDXCNT[31:24] INDXCNT[23:16]							
0x0168 ... 0x0169	Reserved									
0x016A	INDX1HLD	15:8 7:0	INDXHLD[31:24] INDXHLD[23:16]							
0x016C	QE1GECL	15:8 7:0	QEIGEC[15:8] QEIGEC[7:0]							
0x016E	QE1GECH	15:8 7:0	QEIGEC[31:24] QEIGEC[23:16]							

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.....continued

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x0170	QE1LECL	15:8	QEILEC[15:8]							
		7:0	QEILEC[7:0]							
0x0172	QE1LECH	15:8	QEILEC[31:24]							
		7:0	QEILEC[23:16]							
0x0174	QE2CON	15:8	QEIEN		QEISIDL	PIMOD[2:0]			IMV[1:0]	
		7:0		INTDIV[2:0]			CNTPOL	GATEN	CCM[1:0]	
0x0176	Reserved									
0x0177										
0x0178	QE2IOCL	15:8	QCAPEN	FLTREN	QFDIV[2:0]			OUTFNC[1:0]		SWPAB
		7:0	HOMPOL	IDXPOL	QEBPOL	QEAPOL	HOME	INDEX	QEB	QEA
0x017A	QE2IOCH	15:8								
		7:0								HCAPEN
0x017C	QE2STAT	15:8			PCHEQIRQ	PCHEQIEN	PCLEQIRQ	PCLEQIEN	POSOVIRQ	POSOVIEN
		7:0	PCIIRQ	PCIIEN	VELOVIRQ	VELOVIEN	HOMIRQ	HOMIEN	IDXIRQ	IDXIEN
0x017E	Reserved									
0x017F										
0x0180	POS2CNTL	15:8	POSCNT[15:8]							
		7:0	POSCNT[7:0]							
0x0182	POS2CNTH	15:8	POSCNT[31:24]							
		7:0	POSCNT[23:16]							
0x0183	VEL2HLD	15:8	VELHLD[31:24]							
		7:0	VELHLD[23:16]							
0x0185	Reserved									
0x0186	POS2HLD	15:8	POSHLD[31:24]							
		7:0	POSHLD[23:16]							
0x0188	VEL2CNTL	15:8	VELCNT[15:8]							
		7:0	VELCNT[7:0]							
0x018A	VEL2CNTH	15:8	VELCNT[31:24]							
		7:0	VELCNT[23:16]							
0x018C	Reserved									
0x018F										
0x0190	INT2TMRL	15:8	INTTMR[15:8]							
		7:0	INTTMR[7:0]							
0x0192	INT2TMRH	15:8	INTTMR[31:24]							
		7:0	INTTMR[23:16]							
0x0194	INT2HLDL	15:8	INTXHLD[15:8]							
		7:0	INTXHLD[7:0]							
0x0196	INT2HLDH	15:8	INTHLD[31:24]							
		7:0	INTHLD[23:16]							
0x0198	INDX2CNTL	15:8	INDXCNT[15:8]							
		7:0	INDXCNT[7:0]							
0x019A	INDX2CNTH	15:8	INDXCNT[31:24]							
		7:0	INDXCNT[23:16]							
0x019C	Reserved									
0x019D										
0x019E	INDX2HLD	15:8	INDXHLD[31:24]							
		7:0	INDXHLD[23:16]							
0x01A0	QE2GECL	15:8	QEIGEC[15:8]							
		7:0	QEIGEC[7:0]							
0x01A2	QE2GECH	15:8	QEIGEC[31:24]							
		7:0	QEIGEC[23:16]							
0x01A4	QE2LECL	15:8	QEILEC[15:8]							
		7:0	QEILEC[7:0]							
0x01A6	QE2LECH	15:8	QEILEC[31:24]							
		7:0	QEILEC[23:16]							

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.....continued

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x01A8 ... 0x053F	Reserved									
0x0540	QEI3CON	15:8	QEIEN		QEISIDL	PIMOD[2:0]			IMV[1:0]	
		7:0		INTDIV[2:0]			CNTPOL	GATEN	CCM[1:0]	
0x0542 ... 0x0543	Reserved									
0x0544	QEI3IOCL	15:8	QCAPEN	FLTREN	QFDIV[2:0]			OUTFNC[1:0]		SWPAB
		7:0	HOMPOL	IDXPOL	QEBPOL	QEAPOL	HOME	INDEX	QEB	QEA
0x0546	QEI3IOCH	15:8								
		7:0								HCAPEN
0x0548	QEI3STAT	15:8			PCHEQIRQ	PCHEQIEN	PCLEQIRQ	PCLEQIEN	POSOVIRQ	POSOVIEN
		7:0	PCIIRQ	PCIIEN	VELOVIRQ	VELOVIEN	HOMIRQ	HOMIEN	IDXIRQ	IDXIEN
0x054A ... 0x054B	Reserved									
0x054C	POS3CNTL	15:8	POSCNT[15:8]							
		7:0	POSCNT[7:0]							
0x054E	POS3CNTH	15:8	POSCNT[31:24]							
		7:0	POSCNT[23:16]							
0x0550 ... 0x0551	Reserved									
0x0552	POS3HLD	15:8	POSHLD[31:24]							
		7:0	POSHLD[23:16]							
0x0554	VEL3CNTL	15:8	VELCNT[15:8]							
		7:0	VELCNT[7:0]							
0x0556	VEL3CNTH	15:8	VELCNT[31:24]							
		7:0	VELCNT[23:16]							
0x0558 ... 0x0559	Reserved									
0x055A	VEL3HLD	15:8	VELHLD[31:24]							
		7:0	VELHLD[23:16]							
0x055C	INT3TMRL	15:8	INTTMR[15:8]							
		7:0	INTTMR[7:0]							
0x055E	INT3TMRH	15:8	INTTMR[31:24]							
		7:0	INTTMR[23:16]							
0x0560	INT3HLDL	15:8	INTXHLD[15:8]							
		7:0	INTXHLD[7:0]							
0x0562	INT3HLDH	15:8	INTHLD[31:24]							
		7:0	INTHLD[23:16]							
0x0564	INDX3CNTL	15:8	INDXCNT[15:8]							
		7:0	INDXCNT[7:0]							
0x0566	INDX3CNTH	15:8	INDXCNT[31:24]							
		7:0	INDXCNT[23:16]							
0x0568 ... 0x0569	Reserved									
0x056A	INDX3HLD	15:8	INDXHLD[31:24]							
		7:0	INDXHLD[23:16]							
0x056C	QEI3GECL	15:8	QEIGEC[15:8]							
		7:0	QEIGEC[7:0]							
0x056E	QEI3GECH	15:8	QEIGEC[31:24]							
		7:0	QEIGEC[23:16]							
0x0570	QEI3LECL	15:8	QEILEC[15:8]							
		7:0	QEILEC[7:0]							

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Quadrature Encoder Interface (QEI)

.....continued

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x0572	QEI3LECH	15:8	QEILEC[31:24]							
		7:0	QEILEC[23:16]							

16.1.1 QEIx Control Register

Name: QEIXCON
Offset: 0x140, 0x174, 0x540

Notes:

1. When CCMx = 10 or CCMx = 11, all of the QEI counters operate as timers and the PIMOD[2:0] bits are ignored.
2. When CCMx = 00, and QEAX and QEBx values match the Index Match Value (IMV), the POSxCNTL and POSxCNTL registers are reset.
3. The selected clock rate should be at least twice the expected maximum quadrature count rate.
4. Not all devices support this mode.
5. The QCAPEN and HCAPEN bits must be cleared during PIMODx Modes 2 through 7 to ensure proper functionality. Not all devices support HCAPEN.

Bit	15	14	13	12	11	10	9	8
	QEIEN		QEISIDL		PIMOD[2:0]		IMV[1:0]	
Access	R/W		R/W	R/W	R/W	R/W	R/W	R/W
Reset	0		0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
			INTDIV[2:0]		CNTPOL	GATEN	CCM[1:0]	
Access		R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset		0	0	0	0	0	0	0

Bit 15 – QEIEN Quadrature Encoder Interface Module Enable bit

Value	Description
1	Module counters are enabled
0	Module counters are disabled, but SFRs can be read or written

Bit 13 – QEISIDL QEI Stop in Idle Mode bit

Value	Description
1	Discontinues module operation when device enters Idle mode
0	Continues module operation in Idle mode

Bits 12:10 – PIMOD[2:0] Position Counter Initialization Mode Select bits^(1,5)

Value	Description
111	Modulo Count mode for position counter and every Index event resets the position counter ⁽⁴⁾
110	Modulo Count mode for position counter
101	Resets the position counter when the position counter equals the QEIXGEC register
100	Second Index event after Home event initializes the position counter with the contents of the QEIXIC register
011	First Index event after Home event initializes the position counter with the contents of the QEIXIC register
010	Next Index input event initializes the position counter with the contents of the QEIXIC register
001	Every Index input event resets the position counter
000	Index input event does not affect the position counter

Bits 9:8 – IMV[1:0] Index Match Value bits⁽²⁾

Value	Description
11	Index match occurs when QEBx = 1 and QEAX = 1
10	Index match occurs when QEBx = 1 and QEAX = 0
01	Index match occurs when QEBx = 0 and QEAX = 1
00	Index match occurs when QEBx = 0 and QEAX = 0

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Quadrature Encoder Interface (QEI)

Bits 6:4 – INTDIV[2:0] Timer Input Clock Prescale Select bits⁽³⁾ (interval timer, main timer (position counter), velocity counter and Index counter internal clock divider select)

Value	Description
111	1:256 prescale value
110	1:64 prescale value
101	1:32 prescale value
100	1:16 prescale value
011	1:8 prescale value
010	1:4 prescale value
001	1:2 prescale value
000	1:1 prescale value

Bit 3 – CNTPOL Position and Index Counter/Timer Direction Select bit

Value	Description
1	Counter direction is negative unless modified by an external up/down signal
0	Counter direction is positive unless modified by an external up/down signal

Bit 2 – GATEN External Count Gate Enable bit

Value	Description
1	External gate signal controls position counter operation
0	External gate signal does not affect position counter operation

Bits 1:0 – CCM[1:0] Counter Control Mode Selection bits

Value	Description
11	Internal Timer mode
10	External Clock Count with External Gate mode
01	External Clock Count with External Up/Down mode
00	Quadrature Encoder mode

16.1.2 QEIx I/O Control Register

Name: QEIXIOCL
Offset: 0x144, 0x178, 0x544

Legend: x = Bit is unknown

Bit	15	14	13	12	11	10	9	8
	QCAPEN	FLTREN	QFDIV[2:0]			OUTFNC[1:0]		SWPAB
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	HOMPOL	IDXPOL	QEBPOL	QEAPOL	HOME	INDEX	QEB	QEA
Access	R/W	R/W	R/W	R/W	R	R	R	R
Reset	0	0	0	0	x	x	x	x

Bit 15 – QCAPEN QEIX Position Counter Input Capture Enable bit

Value	Description
1	HOMEx input event (positive edge) triggers a position capture event (HCAPEN must be cleared)
0	HOMEx input event (positive edge) does not trigger a position capture event

Bit 14 – FLTREN QEAX/QEBX/INDXx/HOMEx Digital Filter Enable bit

Value	Description
1	Input pin digital filter is enabled
0	Input pin digital filter is disabled (bypassed)

Bits 13:11 – QFDIV[2:0] QEAX/QEBX/INDXx/HOMEx Digital Input Filter Clock Divide Select bits

Value	Description
111	1:256 clock divide
110	1:64 clock divide
101	1:32 clock divide
100	1:16 clock divide
011	1:8 clock divide
010	1:4 clock divide
001	1:2 clock divide
000	1:1 clock divide

Bits 10:9 – OUTFNC[1:0] QEIX Module Output Function Mode Select bits

Value	Description
11	The CNTCMPx pin goes high when POSxCNT ≤ QEIXLEC or POSxCNT ≥ QEIXGEC
10	The CNTCMPx pin goes high when POSxCNT ≤ QEIXLEC
01	The CNTCMPx pin goes high when POSxCNT ≥ QEIXGEC
00	Output is disabled

Bit 8 – SWPAB Swap QEAX and QEBX Inputs bit

Value	Description
1	QEAX and QEBX are swapped prior to Quadrature Decoder logic
0	QEAX and QEBX are not swapped

Bit 7 – HOMPOL HOMEx Input Polarity Select bit

Value	Description
1	Input is inverted
0	Input is not inverted

Bit 6 – IDXPOL INDXx Input Polarity Select bit

Value	Description
1	Input is inverted
0	Input is not inverted

dsPIC33CK512MPT608 Family

Quadrature Encoder Interface (QEI)

Value	Description
1	Input is inverted
0	Input is not inverted

Bit 5 – QEBPOL QEBx Input Polarity Select bit

Value	Description
1	Input is inverted
0	Input is not inverted

Bit 4 – QEAPOL QEAx Input Polarity Select bit

Value	Description
1	Input is inverted
0	Input is not inverted

Bit 3 – HOME Status of HOMEx Input Pin After Polarity Control bit (read-only)

Value	Description
1	Pin is at logic '1' if HOMPOL bit is set to '0'; pin is at logic '0' if HOMPOL bit is set to '1'
0	Pin is at logic '0' if HOMPOL bit is set to '0'; pin is at logic '1' if HOMPOL bit is set to '1'

Bit 2 – INDEX Status of INDXX Input Pin After Polarity Control bit (read-only)

Value	Description
1	Pin is at logic '1' if the IDXPOL bit is set to '0'; pin is at logic '0' if the IDXPOL bit is set to '1'
0	Pin is at logic '0' if the IDXPOL bit is set to '0'; pin is at logic '1' if the IDXPOL bit is set to '1'

Bit 1 – QEB Status of QEBx Input Pin After Polarity Control and SWPAB Pin Swapping bit (read-only)

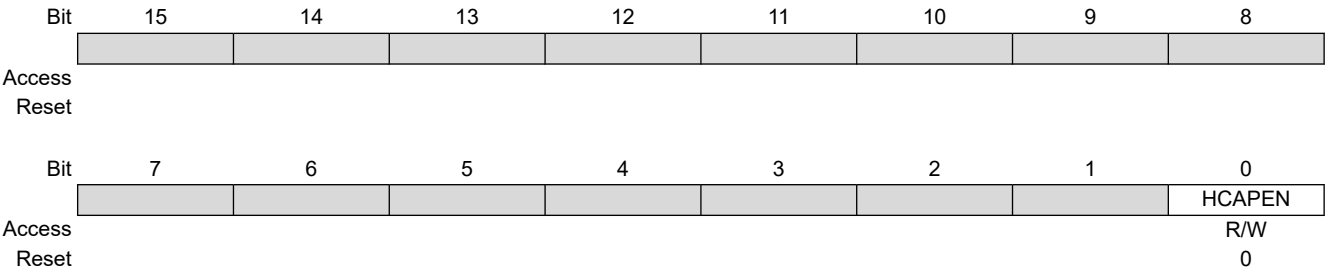
Value	Description
1	Physical pin, QEBx, is at logic '1' if QEBPOL bit is set to '0' and SWPAB bit is set to '0'; physical pin, QEBx, is at logic '0' if QEBPOL bit is set to '1' and SWPAB bit is set to '0'; physical pin, QEAx, is at logic '1' if QEBPOL bit is set to '0' and SWPAB bit is set to '1'; physical pin, QEAx, is at logic '0' if QEBPOL bit is set to '1' and SWPAB bit is set to '1'
0	Physical pin, QEBx, is at logic '0' if QEBPOL bit is set to '0' and SWPAB bit is set to '0'; physical pin, QEBx, is at logic '1' if QEBPOL bit is set to '1' and SWPAB bit is set to '0'; physical pin, QEAx, is at logic '0' if QEBPOL bit is set to '0' and SWPAB bit is set to '1'; physical pin, QEAx, is at logic '1' if QEBPOL bit is set to '1' and SWPAB bit is set to '1'

Bit 0 – QEA Status of QEAx Input Pin After Polarity Control and SWPAB Pin Swapping bit (read-only)

Value	Description
1	Physical pin, QEAx, is at logic '1' if QEAPOL bit is set to '0' and SWPAB bit is set to '0'; physical pin, QEAx, is at logic '0' if QEAPOL bit is set to '1' and SWPAB bit is set to '0'; physical pin, QEBx, is at logic '1' if QEAPOL bit is set to '0' and SWPAB bit is set to '1'; physical pin, QEBx, is at logic '0' if QEAPOL bit is set to '1' and SWPAB bit is set to '1'
0	Physical pin, QEAx, is at logic '0' if QEAPOL bit is set to '0' and SWPAB bit is set to '0'; physical pin, QEAx, is at logic '1' if QEAPOL bit is set to '1' and SWPAB bit is set to '0'; physical pin, QEBx, is at logic '0' if QEAPOL bit is set to '0' and SWPAB bit is set to '1'; physical pin, QEBx, is at logic '1' if the QEAPOL bit is set to '1' and the SWPAB bit is set to '1'

16.1.3 QEIx I/O Control High Register

Name: QEIXIOCH
Offset: 0x146, 0x17A, 0x546



Bit 0 – HCAPEN Position Counter Input Capture by Home Event Enable bit

Value	Description
1	HOMEx input event (positive edge) triggers a position capture event
0	HOMEx input event (positive edge) does not trigger a position capture event

16.1.4 QEIx Status Register

Name: QEIXSTAT
Offset: 0x148, 0x17C, 0x548

Note:

- This status bit is only applicable to PIMOD[2:0] modes, '011' and '100'.

Legend: C = Clearable bit, HS = Hardware Settable bit

Bit	15	14	13	12	11	10	9	8
			PCHEQIRQ	PCHEQIEN	PCLEQIRQ	PCLEQIEN	POSOVIRQ	POSOVIEN
Access			R/C/HS	R/W	R/C/HS	R/W	R/C/HS	R/W
Reset			0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	PCIIRQ	PCIEN	VELOVIRQ	VELOVIEN	HOMIRQ	HOMIEN	IDXIRQ	IDXIEN
Access	R/C/HS	R/W	R/C/HS	R/W	R/C/HS	R/W	R/C/HS	R/W
Reset	0	0	0	0	0	0	0	0

Bit 13 – PCHEQIRQ Position Counter Greater Than Compare Status bit

Value	Description
1	POSxCNT > QEIXGEC
0	POSxCNT < QEIXGEC

Bit 12 – PCHEQIEN Position Counter Greater Than Compare Interrupt Enable bit

Value	Description
1	Interrupt is enabled
0	Interrupt is disabled

Bit 11 – PCLEQIRQ Position Counter Less Than Compare Status bit

Value	Description
1	POSxCNT < QEIXLEC
0	POSxCNT > QEIXLEC

Bit 10 – PCLEQIEN Position Counter Less Than Compare Interrupt Enable bit

Value	Description
1	Interrupt is enabled
0	Interrupt is disabled

Bit 9 – POSOVIRQ Position Counter Overflow Status bit

Value	Description
1	Overflow has occurred
0	No overflow has occurred

Bit 8 – POSOVIEN Position Counter Overflow Interrupt Enable bit

Value	Description
1	Interrupt is enabled
0	Interrupt is disabled

Bit 7 – PCIIRQ Position Counter (Homing) Initialization Process Complete Status bit⁽¹⁾

Value	Description
1	POSxCNT was reinitialized
0	POSxCNT was not reinitialized

Bit 6 – PCIEN Position Counter (Homing) Initialization Process Complete Interrupt Enable bit

dsPIC33CK512MPT608 Family

Quadrature Encoder Interface (QEI)

Value	Description
1	Interrupt is enabled
0	Interrupt is disabled

Bit 5 – VELOVIRQ Velocity Counter Overflow Status bit

Value	Description
1	Overflow has occurred
0	No overflow has occurred

Bit 4 – VELOVIEN Velocity Counter Overflow Interrupt Enable bit

Value	Description
1	Interrupt is enabled
0	Interrupt is disabled

Bit 3 – HOMIRQ Status Flag for Home Event Status bit

Value	Description
1	Home event has occurred
0	No Home event has occurred

Bit 2 – HOMIEN Home Input Event Interrupt Enable bit

Value	Description
1	Interrupt is enabled
0	Interrupt is disabled

Bit 1 – IDXIRQ Status Flag for Index Event Status bit

Value	Description
1	Index event has occurred
0	No Index event has occurred

Bit 0 – IDXIEN Index Input Event Interrupt Enable bit

Value	Description
1	Interrupt is enabled
0	Interrupt is disabled

dsPIC33CK512MPT608 Family

Quadrature Encoder Interface (QEI)

16.1.5 Position x Counter Register Low

Name: POSxCNTL
Offset: 0x14C, 0x180, 0x54C

Bit	15	14	13	12	11	10	9	8
	POSCNT[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	POSCNT[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – POSCNT[15:0] Low Word Used to Form 32-Bit Position Counter Register (POSxCNT) bits

dsPIC33CK512MPT608 Family

Quadrature Encoder Interface (QEI)

16.1.6 Position x Counter Register High

Name: POSxCNTH
Offset: 0x14E, 0x182, 0x54E

Bit	15	14	13	12	11	10	9	8
	POSCNT[31:24]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	POSCNT[23:16]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – POSCNT[31:16] High Word Used to Form 32-Bit Position Counter Register (POSxCNT) bits

dsPIC33CK512MPT608 Family

Quadrature Encoder Interface (QEI)

16.1.7 Position x Counter Hold Register

Name: POSxHLD
Offset: 0x152, 0x186, 0x552

Bit	15	14	13	12	11	10	9	8
	POSHLD[31:24]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	POSHLD[23:16]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – POSHLD[31:16] Hold for Reading/Writing Position x Counter High Word Register POSxCNTH (POSxCNT[31:16]) bits

dsPIC33CK512MPT608 Family

Quadrature Encoder Interface (QEI)

16.1.8 Velocity x Counter Register Low

Name: VELxCNTL
Offset: 0x154, 0x188, 0x554

Bit	15	14	13	12	11	10	9	8
	VELCNT[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	VELCNT[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – VELCNT[15:0] Velocity Counter bits

dsPIC33CK512MPT608 Family

Quadrature Encoder Interface (QEI)

16.1.9 Velocity x Counter Register High

Name: VELxCNTH
Offset: 0x156, 0x18A, 0x556

Bit	15	14	13	12	11	10	9	8
	VELCNT[31:24]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	VELCNT[23:16]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – VELCNT[31:16] Velocity Counter bits

dsPIC33CK512MPT608 Family

Quadrature Encoder Interface (QEI)

16.1.10 Velocity x Counter Hold Register

Name: VELxHLD
Offset: 0x15A, 0x183, 0x55A

Bit	15	14	13	12	11	10	9	8
	VELHLD[31:24]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	VELHLD[23:16]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – VELHLD[31:16] Hold for Reading/Writing Velocity x Counter High Word Register VELxCNTH (VELxCNT[31:16]) bits

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Quadrature Encoder Interface (QEI)

16.1.11 Interval x Timer Register Low

Name: INTxTMRL
Offset: 0x15C, 0x190, 0x55C

Bit	15	14	13	12	11	10	9	8
	INTTMR[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	INTTMR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – INTTMR[15:0] Low Word Used to Form 32-Bit Interval Timer Register (INTxTMR) bits

dsPIC33CK512MPT608 Family

Quadrature Encoder Interface (QEI)

16.1.12 Interval x Timer Register High

Name: INTxTMRH
Offset: 0x15E, 0x192, 0x55E

Bit	15	14	13	12	11	10	9	8
	INTTMR[31:24]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	INTTMR[23:16]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – INTTMR[31:16] High Word Used to Form 32-Bit Interval Timer Register (INTxTMR) bits

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Quadrature Encoder Interface (QEI)

16.1.13 Interval x Timer Hold Register Low

Name: INTxHLDL
Offset: 0x160, 0x194, 0x560

Bit	15	14	13	12	11	10	9	8
	INTXHLD[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	INTXHLD[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – INTXHLD[15:0] Hold for Reading/Writing Interval x Counter Register (INTxTMR) bits

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Quadrature Encoder Interface (QEI)

16.1.14 Interval x Timer Hold Register High

Name: INTxHLDH
Offset: 0x162, 0x196, 0x562

Bit	15	14	13	12	11	10	9	8
	INTHLD[31:24]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	INTHLD[23:16]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – INTHLD[31:16] Hold for Reading/Writing Interval x Counter Register (INTxTMR) bits

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Quadrature Encoder Interface (QEI)

16.1.15 Index x Counter Register Low

Name: INDXXCNTL
Offset: 0x164, 0x198, 0x564

Bit	15	14	13	12	11	10	9	8
	INDXCNT[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	INDXCNT[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – INDXCNT[15:0] Low Word Used to Form 32-Bit Index x Counter Register (INDXXCNT) bits

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Quadrature Encoder Interface (QEI)

16.1.16 Index x Counter Register High

Name: INDXXCNTH
Offset: 0x166, 0x19A, 0x566

Bit	15	14	13	12	11	10	9	8
	INDXCNT[31:24]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	INDXCNT[23:16]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – INDXCNT[31:16] High Word Used to Form 32-Bit Index x Counter Register (INDXXCNT) bits

dsPIC33CK512MPT608 Family

Quadrature Encoder Interface (QEI)

16.1.17 Index x Counter Hold Register

Name: INDXxHLD
Offset: 0x16A, 0x19E, 0x56A

Bit	15	14	13	12	11	10	9	8
	INDXHLD[31:24]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	INDXHLD[23:16]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – INDXHLD[31:16] Hold Register for Reading/Writing Index 1 Counter High Word Register (INDX1CNTH) bits

dsPIC33CK512MPT608 Family

Quadrature Encoder Interface (QEI)

16.1.18 QEIx Greater Than or Equal Compare Register Low

Name: QEIxGECL
Offset: 0x16C, 0x1A0, 0x56C

Bit	15	14	13	12	11	10	9	8
	QEIGEC[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	QEIGEC[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – QEIGEC[15:0] Low Word Used to Form 32-Bit Greater Than or Equal Compare Register (QEIGEC) bits

dsPIC33CK512MPT608 Family

Quadrature Encoder Interface (QEI)

16.1.19 QEIx Greater Than or Equal Compare Register High

Name: QEIxGECH
Offset: 0x16E, 0x1A2, 0x56E

Bit	15	14	13	12	11	10	9	8
	QEIGEC[31:24]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	QEIGEC[23:16]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – QEIGEC[31:16] High Word Used to Form 32-Bit Greater Than or Equal Compare Register (QEIGEC) bits

dsPIC33CK512MPT608 Family

Quadrature Encoder Interface (QEI)

16.1.20 QEIx Less Than or Equal Compare Register Low

Name: QEIXLECL
Offset: 0x170, 0x1A4, 0x570

Bit	15	14	13	12	11	10	9	8
	QEILEC[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	QEILEC[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – QEILEC[15:0] Low Word Used to Form 32-Bit Less Than or Equal Compare Register (QEIXLEC) bits

dsPIC33CK512MPT608 Family

Quadrature Encoder Interface (QEI)

16.1.21 QEIx Less Than or Equal Compare Register High

Name: QEIXLECH
Offset: 0x172, 0x1A6, 0x572

Bit	15	14	13	12	11	10	9	8
	QEILEC[31:24]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	QEILEC[23:16]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – QEILEC[31:16] High Word Used to Form 32-Bit Less Than or Equal Compare Register (QEIXLEC) bits

17. Universal Asynchronous Receiver Transmitter (UART)

Note: This data sheet summarizes the features of the dsPIC33CK512MPT608 family of devices. It is not intended to be a comprehensive reference source. To complement the information in this data sheet, refer to “**Multiprotocol Universal Asynchronous Receiver Transmitter (UART) Module**” (www.microchip.com/DS70005288) in the “dsPIC33/PIC24 Family Reference Manual”.

The Universal Asynchronous Receiver Transmitter (UART) is a flexible serial communication peripheral used to interface dsPIC® microcontrollers with other equipment, including computers and peripherals. The UART is a full-duplex, asynchronous communication channel that can be used to implement protocols, such as RS-232 and RS-485. The UART also supports the following hardware extensions:

- LIN/J2602
- Digital Multiplex (DMX)
- Smart Card

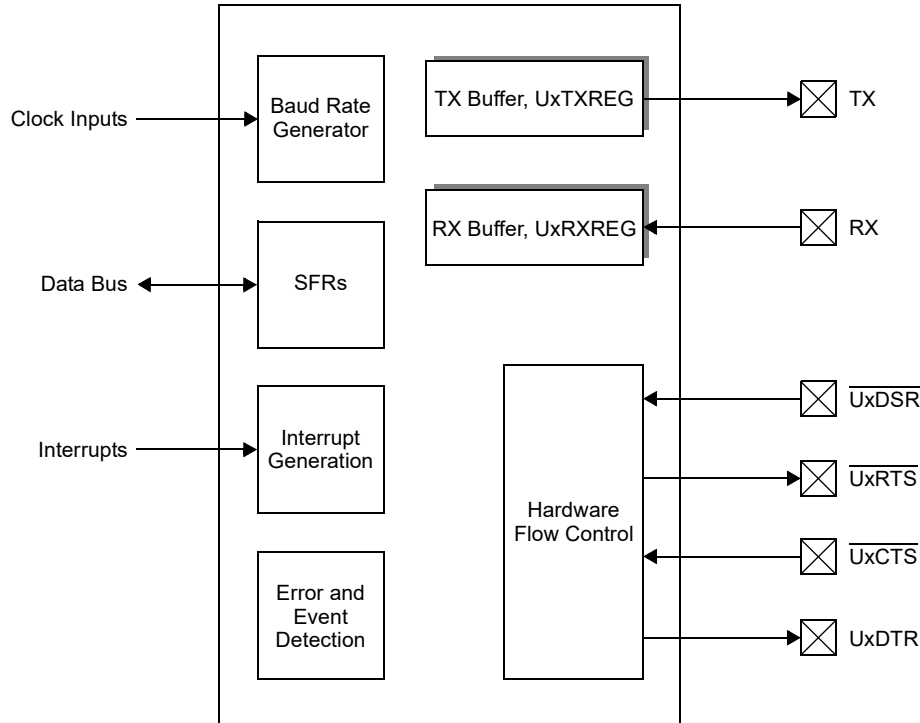
The primary features of the UART are:

- Full or Half-Duplex Operation
- Up to 8-Deep TX and RX First-In First-Out (FIFO) Buffers
- 8-Bit or 9-Bit Data Width
- Configurable Stop Bit Length
- Flow Control
- Auto-Baud Calibration
- Parity, Framing and Buffer Overrun Error Detection
- Address Detect
- Break Transmission
- Transmit and Receive Polarity Control
- Manchester Encoder/Decoder
- Operation in Sleep mode
- Wake from Sleep on Sync Break Received Interrupt

17.1 Architectural Overview

The UART transfers bytes of data, to and from device pins, using First-In First-Out (FIFO) buffers up to eight bytes deep. The status of the buffers and data is made available to user software through Special Function Registers (SFRs). The UART implements multiple interrupt channels for handling transmit, receive and error events. A simplified block diagram of the UART is shown in [Figure 17-1](#).

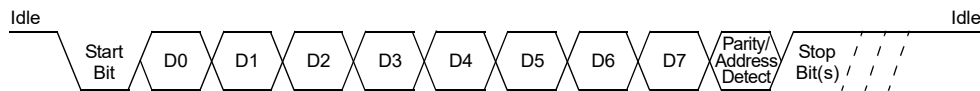
Figure 17-1. Simplified UARTx Block Diagram



17.2 Character Frame

A typical UART character frame is shown in Figure 17-2. The Idle state is high with a 'Start' condition indicated by a falling edge. The Start bit is followed by the number of data, parity/address detect and Stop bits defined by the MOD[3:0] (UxMODE[3:0]) bits selected.

Figure 17-2. UART Character Frame



17.3 Data Buffers

Both transmit and receive functions use buffers to store data shifted to/from the pins. These buffers are FIFOs and are accessed by reading the SFRs, UxTXREG and UxRXREG, respectively. Each data buffer has multiple flags associated with its operation to allow software to read the status. Interrupts can also be configured based on the space available in the buffers. The transmit and receive buffers can be cleared and their pointers reset using the associated TX/RX Buffer Empty Status bits, UTXBE (UxSTAH[5]) and URXBE (UxSTAH[1]).

17.4 Protocol Extensions

The UART provides hardware support for LIN/J2602, DMX and smart card protocol extensions to reduce software overhead. A protocol extension is enabled by writing a value to the MOD[3:0] (UxMODE[3:0]) selection bits and further configured using the UARTx Timing Parameter registers, UxP1, UxP2, UxP3 and UxP3H. Details regarding operation and usage are discussed in their respective chapters. Not all protocols are available on all devices.

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17.5 UART Control/Status Registers

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x00 ... 0x0237	Reserved									
0x0238	U1MODE	15:8	UARTEN		USIDL	WAKE	RXBIMD		BRKOVRR	UTXBRK
		7:0	BRGH	ABAUD	UTXEN	URXEN	MOD[3:0]			
0x023A	U1MODEH	15:8	SLPEN	ACTIVE			BCLKMOD	BCLKSEL[1:0]	HALFDPLX	
		7:0	RUNOVF	URXINV	STSEL[1:0]		C0EN	UTXINV	FLO[1:0]	
0x023C	U1STA	15:8	TXMTIE	PERIE	ABDOVF	CERIE	FERIE	RXBKIE	OERIE	TXCIE
		7:0	TRMT	PERR	ABDOVF	CERIF	FERR	RXBKIF	OERR	TXCIF
0x023E	U1STAH	15:8			UTXISEL[2:0]				URXISEL[2:0]	
		7:0	TXWRE	STPMD	UTXBE	UTXBF	RIDLE	XON	URXBE	URXBF
0x0240	U1BRG	15:8	BRG[15:8]							
		7:0	BRG[7:0]							
0x0242	U1BRGH	15:8								
		7:0					BRG[19:16]			
0x0244	U1RXREG	15:8								
		7:0	RXREG[7:0]							
0x0246 ... 0x0247	Reserved									
0x0248	U1TXREG	15:8	LAST							
		7:0	TXREG[7:0]							
0x024A ... 0x024B	Reserved									
0x024C	U1P1	15:8								P1[8]
		7:0	P1[7:0]							
0x024E	U1P2	15:8								P2[8]
		7:0	P2[7:0]							
0x0250	U1P3	15:8								
		7:0	P3[7:0]							
0x0252	U1P3H	15:8								
		7:0	P3[23:16]							
0x0254	U1TXCHK	15:8								
		7:0	TXCHK[7:0]							
0x0256	U1RXCHK	15:8								
		7:0	RXCHK[7:0]							
0x0258	U1SCCON	15:8								
		7:0			TXRPT[1:0]		CONV	T0PD	PRTCL	
0x025A	U1SCINT	15:8			RXRPTIF	TXRPTIF		BTCIF	WTCIF	GTCIF
		7:0			RXRPTIE	TXRPTIE		BTCIE	WTCIE	GTCIE
0x025C	U1INT	15:8								
		7:0	WUIF	ABDIF				ABDIE		
0x025E ... 0x025F	Reserved									
0x0260	U2MODE	15:8	UARTEN		USIDL	WAKE	RXBIMD		BRKOVRR	UTXBRK
		7:0	BRGH	ABAUD	UTXEN	URXEN	MOD[3:0]			
0x0262	U2MODEH	15:8	SLPEN	ACTIVE			BCLKMOD	BCLKSEL[1:0]	HALFDPLX	
		7:0	RUNOVF	URXINV	STSEL[1:0]		C0EN	UTXINV	FLO[1:0]	
0x0264	U2STA	15:8	TXMTIE	PERIE	ABDOVF	CERIE	FERIE	RXBKIE	OERIE	TXCIE
		7:0	TRMT	PERR	ABDOVF	CERIF	FERR	RXBKIF	OERR	TXCIF
0x0266	U2STAH	15:8			UTXISEL[2:0]				URXISEL[2:0]	
		7:0	TXWRE	STPMD	UTXBE	UTXBF	RIDLE	XON	URXBE	URXBF
0x0268	U2BRG	15:8	BRG[15:8]							
		7:0	BRG[7:0]							

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.....continued

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x026A	U2BRGH	15:8								
		7:0					BRG[19:16]			
0x026C	U2RXREG	15:8								
		7:0	RXREG[7:0]							
0x026E ... 0x026F	Reserved									
0x0270	U2TXREG	15:8	LAST							
		7:0	TXREG[7:0]							
0x0272 ... 0x0273	Reserved									
0x0274	U2P1	15:8								P1[8]
		7:0	P1[7:0]							
0x0276	U2P2	15:8								P2[8]
		7:0	P2[7:0]							
0x0278	U2P3	15:8								
		7:0	P3[7:0]							
0x027A	U2P3H	15:8								
		7:0	P3[23:16]							
0x027C	U2TXCHK	15:8								
		7:0	TXCHK[7:0]							
0x027E	U2RXCHK	15:8								
		7:0	RXCHK[7:0]							
0x0280	U2SCCON	15:8								
		7:0			TXRPT[1:0]		CONV	T0PD	PRTCL	
0x0282	U2SCINT	15:8			RXRPTIF	TXRPTIF		BTCIF	WTCIF	GTCIF
		7:0			RXRPTIE	TXRPTIE		BTCIE	WTCIE	GTCIE
0x0284	U2INT	15:8								
		7:0	WUIF	ABDIF				ABDIE		
0x0286 ... 0x0EFF	Reserved									
0x0F00	U3MODE	15:8	UARTEN		USIDL	WAKE	RXBIMD		BRKOV	UTXBRK
		7:0	BRGH	ABAUD	UTXEN	URXEN	MOD[3:0]			
0x0F02	U3MODEH	15:8	SLPEN	ACTIVE			BCLKMOD	BCLKSEL[1:0]		HALFDPLX
		7:0	RUNOVF	URXINV	STSEL[1:0]		C0EN	UTXINV	FLO[1:0]	
0x0F04	U3STA	15:8	TXMTIE	PERIE	ABDOVF	CERIE	FERIE	RXBKIE	OERIE	TXCIE
		7:0	TRMT	PERR	ABDOVF	CERIF	FERR	RXBKIF	OERR	TXCIF
0x0F06	U3STAH	15:8			UTXISEL[2:0]			URXISEL[2:0]		
		7:0	TXWRE	STPMD	UTXBE	UTXBF	RIDLE	XON	URXBE	URXBF
0x0F08	U3BRG	15:8	BRG[15:8]							
		7:0	BRG[7:0]							
0x0F0A	U3BRGH	15:8								
		7:0					BRG[19:16]			
0x0F0C	U3RXREG	15:8								
		7:0	RXREG[7:0]							
0x0F0E ... 0x0F0F	Reserved									
0x0F10	U3TXREG	15:8	LAST							
		7:0	TXREG[7:0]							
0x0F12 ... 0x0F13	Reserved									
0x0F14	U3P1	15:8								P1[8]
		7:0	P1[7:0]							
0x0F16	U3P2	15:8								P2[8]
		7:0	P2[7:0]							

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.....continued

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x0F18	U3P3	15:8	P3[15:8]							
		7:0	P3[7:0]							
0x0F1A	U3P3H	15:8								
		7:0	P3[23:16]							
0x0F1C	U3TXCHK	15:8								
		7:0	TXCHK[7:0]							
0x0F1E	U3RXCHK	15:8								
		7:0	RXCHK[7:0]							
0x0F20	U3SCCON	15:8								
		7:0			TXRPT[1:0]		CONV	T0PD	PRTCL	
0x0F22	U3SCINT	15:8			RXRPTIF	TXRPTIF		BTCIF	WTCIF	GTCIF
		7:0			RXRPTIE	TXRPTIE		BTCIE	WTCIE	GTCIE
0x0F24	U3INT	15:8								
		7:0	WUIF	ABDIF				ABDIE		

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17.5.1 UARTx Configuration Register

Name: UxMODE
Offset: 0x238, 0x260, 0xF00

Note:

1. R/HS/HC in DMX and LIN mode.

Legend: HC = Hardware Clearable bit; HS = Hardware Settable bit

Bit	15	14	13	12	11	10	9	8
	UARTEN		USIDL	WAKE	RXBIMD		BRKOVVR	UTXBRK
Access	R/W		R/W	R/W	R/W		R/W	R/W/HC
Reset	0		0	0	0		0	0

Bit	7	6	5	4	3	2	1	0
	BRGH	ABAUD	UTXEN	URXEN	MOD[3:0]			
Access	R/W	R/W/HC	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – UARTEN UART Enable bit

Value	Description
1	UART is ready to transmit and receive
0	UART state machine, FIFO Buffer Pointers and counters are reset; registers are readable and writable

Bit 13 – USIDL UART Stop in Idle Mode bit

Value	Description
1	Discontinues module operation when device enters Idle mode
0	Continues module operation in Idle mode

Bit 12 – WAKE Wake-up Enable bit

Value	Description
1	Module will continue to sample the RX pin – interrupt generated on falling edge, bit cleared in hardware on following rising edge; if ABAUD is set, Auto-Baud Detection (ABD) will begin immediately
0	RX pin is not monitored nor rising edge detected

Bit 11 – RXBIMD Receive Break Interrupt Mode bit

Value	Description
1	RXBKIF flag when a minimum of 23 (DMX)/11 (asynchronous or LIN/J2602) low bit periods are detected
0	RXBKIF flag when the Break makes a low-to-high transition after being low for at least 23/11-bit periods

Bit 9 – BRKOVVR Send Break Software Override bit

Overrides the TX Data Line:

Value	Description
1	Makes the TX line active (Output 0 when UTXINV = 0, Output 1 when UTXINV = 1)
0	TX line is driven by the shifter

Bit 8 – UTXBRK UART Transmit Break bit⁽¹⁾

Value	Description
1	Sends Sync Break on next transmission; cleared by hardware upon completion
0	Sync Break transmission is disabled or has completed

Bit 7 – BRGH High Baud Rate Select bit

Value	Description
1	High Speed: Baud rate is baudclk/4

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Value	Description
0	Low Speed: Baud rate is baudclk/16

Bit 6 – ABAUD Auto-Baud Detect Enable bit (read-only when MOD[3:0] = 1xxx)

Value	Description
1	Enables baud rate measurement on the next character – requires reception of a Sync field (55h); cleared in hardware upon completion
0	Baud rate measurement is disabled or has completed

Bit 5 – UTXEN UART Transmit Enable bit

Value	Description
1	Transmit enabled – except during Auto-Baud Detection
0	Transmit disabled – all transmit counters, pointers and state machines are reset; TX buffer is not flushed, status bits are not reset

Bit 4 – URXEN UART Receive Enable bit

Value	Description
1	Receive enabled – except during Auto-Baud Detection
0	Receive disabled – all receive counters, pointers and state machines are reset; RX buffer is not flushed, status bits are not reset

Bits 3:0 – MOD[3:0] UART Mode bits

Value	Description
Other	Reserved
1111	Smart card
1110	IrDA [®]
1101	Reserved
1100	LIN Commander/Responder
1011	LIN Responder only
1010	DMX
1001	Reserved
1000	Reserved
0111	Reserved
0110	Reserved
0101	Reserved
0100	Asynchronous 9-bit UART with address detect, ninth bit = 1 signals address
0011	Asynchronous 8-bit UART without address detect, ninth bit is used as an even parity bit
0010	Asynchronous 8-bit UART without address detect, ninth bit is used as an odd parity bit
0001	Asynchronous 7-bit UART
0000	Asynchronous 8-bit UART

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17.5.2 UARTx Configuration Register High

Name: UxMODEH
Offset: 0x23A, 0x262, 0xF02

Bit	15	14	13	12	11	10	9	8
	SLPEN	ACTIVE			BCLKMOD	BCLKSEL[1:0]		HALFDPLX
Access	R/W	R			R/W	R/W	R/W	R/W
Reset	0	0			0	0	0	0

Bit	7	6	5	4	3	2	1	0
	RUNOVF	URXINV	STSEL[1:0]		C0EN	UTXINV	FLO[1:0]	
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – SLPEN Run During Sleep Enable bit

Value	Description
1	UART BRG clock runs during Sleep
0	UART BRG clock is turned off during Sleep

Bit 14 – ACTIVE UART Running Status bit

Value	Description
1	UART clock request is active (user can not update the UxMODE/UxMODEH registers)
0	UART clock request is not active (user can update the UxMODE/UxMODEH registers)

Bit 11 – BCLKMOD Baud Clock Generation Mode Select bit

Value	Description
1	Uses fractional Baud Rate Generation
0	Uses legacy divide-by-x counter for baud clock generation (x = 4 or 16 depending on the BRGH bit)

Bits 10:9 – BCLKSEL[1:0] Baud Clock Source Selection bits

Value	Description
11	$AF_{VCO}/3$
10	F_{OSC}
01	Reserved
00	$F_{OSC}/2$ (F_P)

Bit 8 – HALFDPLX UART Half-Duplex Selection Mode bit

Value	Description
1	Half-Duplex mode: UxTX is driven as an output when transmitting and tri-stated when TX is Idle
0	Full-Duplex mode: UxTX is driven as an output at all times when both UxRTEN and UxTXEN are set

Bit 7 – RUNOVF Run During Overflow Condition Mode bit

Value	Description
1	When an Overflow Error (OERR) condition is detected, the RX shifter continues to run so as to remain synchronized with incoming RX data; data are not transferred to UxRXREG when it is full (i.e., no UxRXREG data are overwritten)
0	When an Overflow Error (OERR) condition is detected, the RX shifter stops accepting new data (Legacy mode)

Bit 6 – URXINV UART Receive Polarity bit

Value	Description
1	Inverts RX polarity; Idle state is low
0	Input is not inverted; Idle state is high

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Bits 5:4 – STSEL[1:0] Number of Stop Bits Selection bits

Value	Description
11	2 Stop bits sent, 1 checked at receive
10	2 Stop bits sent, 2 checked at receive
01	1.5 Stop bits sent, 1.5 checked at receive
00	1 Stop bit sent, 1 checked at receive

Bit 3 – C0EN Enable Legacy Checksum (C0) Transmit and Receive bit

Value	Description
1	Checksum Mode 1 (enhanced LIN checksum in LIN mode; add all TX/RX words in all other modes)
0	Checksum Mode 0 (legacy LIN checksum in LIN mode; not used in all other modes)

Bit 2 – UTXINV UART Transmit Polarity bit

Value	Description
1	Inverts TX polarity; TX is low in Idle state
0	Output data are not inverted; TX output is high in Idle state

Bits 1:0 – FLO[1:0] Flow Control Enable bits (only valid when MOD[3:0] = 0xxx)

Value	Description
11	Reserved
10	RTS-DSR (for TX side)/CTS-DTR (for RX side) hardware flow control
01	XON/XOFF software flow control
00	Flow control off

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17.5.3 UARTx Status Register

Name: UxSTA
Offset: 0x23C, 0x264, 0xF04

Legend: HC = Hardware Clearable bit; HS = Hardware Settable bit

Bit	15	14	13	12	11	10	9	8
	TXMTIE	PERIE	ABDOVE	CERIE	FERIE	RXBKIE	OERIE	TXCIE
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	TRMT	PERR	ABDOVF	CERIF	FERR	RXBKIF	OERR	TXCIF
Access	R	R	R/W/HS	R/W/HC	R	R/W/HC	R/W/HC	R/W/HC
Reset	1	0	0	0	0	0	0	0

Bit 15 – TXMTIE Transmit Shifter Empty Interrupt Enable bit

Value	Description
1	Interrupt is enabled
0	Interrupt is disabled

Bit 14 – PERIE Parity Error Interrupt Enable bit

Value	Description
1	Interrupt is enabled
0	Interrupt is disabled

Bit 13 – ABDOVE Auto-Baud Rate Acquisition Interrupt Enable bit

Value	Description
1	Interrupt is enabled
0	Interrupt is disabled

Bit 12 – CERIE Checksum Error Interrupt Enable bit

Value	Description
1	Interrupt is enabled
0	Interrupt is disabled

Bit 11 – FERR Framing Error Interrupt Enable bit

Value	Description
1	Interrupt is enabled
0	Interrupt is disabled

Bit 10 – RXBKIE Receive Break Interrupt Enable bit

Value	Description
1	Interrupt is enabled
0	Interrupt is disabled

Bit 9 – OERR Receive Buffer Overflow Interrupt Enable bit

Value	Description
1	Interrupt is enabled
0	Interrupt is disabled

Bit 8 – TXCIE Transmit Collision Interrupt Enable bit

Value	Description
1	Interrupt is enabled
0	Interrupt is disabled

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Bit 7 – TRMT Transmit Shifter Empty Interrupt Flag bit (read-only)

Value	Description
1	Transmit Shift Register (TSR) is empty (end of last Stop bit when STPMD = 1 or middle of first Stop bit when STPMD = 0)
0	Transmit Shift Register is not empty

Bit 6 – PERR Parity Error/Address Received/Forward Frame Interrupt Flag bit

LIN and Parity Modes:

1 = Parity error detected

0 = No parity error detected

Address Mode:

1 = Address received

0 = No address detected

All Other Modes:

Not used.

Bit 5 – ABDOVF Auto-Baud Rate Acquisition Interrupt Flag bit (must be cleared by software)

Value	Description
1	BRG rolled over during the auto-baud rate acquisition sequence (must be cleared in software)
0	BRG has not rolled over during the auto-baud rate acquisition sequence

Bit 4 – CERIF Checksum Error Interrupt Flag bit (must be cleared by software)

Value	Description
1	Checksum error
0	No checksum error

Bit 3 – FERR Framing Error Interrupt Flag bit

Value	Description
1	Framing Error: Inverted level of the Stop bit corresponding to the topmost character in the buffer; propagates through the buffer with the received character
0	No framing error

Bit 2 – RXBKIF Receive Break Interrupt Flag bit (must be cleared by software)

Value	Description
1	A Break was received
0	No Break was detected

Bit 1 – OERR Receive Buffer Overflow Interrupt Flag bit (must be cleared by software)

Value	Description
1	Receive buffer has overflowed
0	Receive buffer has not overflowed

Bit 0 – TXCIF Transmit Collision Interrupt Flag bit (must be cleared by software)

Value	Description
1	Transmitted word is not equal to the received word
0	Transmitted word is equal to the received word

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17.5.4 UARTx Status Register High

Name: UxSTAH
Offset: 0x23E, 0x266, 0xF06

Note:

- The receive watermark interrupt is not set if PERIF or FERIF is set and the corresponding IE bit is set.

Legend: S = Settable bit, HS = Hardware Settable bit

Bit	15	14	13	12	11	10	9	8
		UTXISEL[2:0]				URXISEL[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		0	0	0		0	0	0

Bit	7	6	5	4	3	2	1	0
	TXWRE	STPMD	UTXBE	UTXBF	RIDLE	XON	URXBE	URXBF
Access	R/W/HS	R/W	R/S	R	R	R	R/S	R
Reset	0	0	1	0	1	1	1	0

Bits 14:12 – UTXISEL[2:0] UART Transmit Interrupt Select bits

Value	Description
111	Sets transmit interrupt when there is one empty slot left in the buffer
. . .	
010	Sets transmit interrupt when there are six empty slots or more in the buffer
001	Sets transmit interrupt when there are seven empty slots or more in the buffer
000	Sets transmit interrupt when there are eight empty slots in the buffer; TX buffer is empty

Bits 10:8 – URXISEL[2:0] UART Receive Interrupt Select bits⁽¹⁾

Value	Description
111	Triggers receive interrupt when there are eight words in the buffer; RX buffer is full
. . .	
001	Triggers receive interrupt when there are two words or more in the buffer
000	Triggers receive interrupt when there is one word or more in the buffer

Bit 7 – TXWRE TX Write Transmit Error Status bit

LIN and Parity Modes:

- 1 = A new byte was written when buffer was full or when P2[8:0] = 0 (must be cleared by software)
 0 = No error

Address Detect Mode:

- 1 = A new byte was written when buffer was full or to P1[8:0] when P1x was full (must be cleared by software)
 0 = No error

Other Modes:

- 1 = A new byte was written when buffer was full (must be cleared by software)
 0 = No error

Bit 6 – STPMD Stop Bit Detection Mode bit

Value	Description
1	Triggers RXIF at the end of the last Stop bit
0	Triggers RXIF in the middle of the first (or second, depending on the STSEL[1:0] setting) Stop bit

Bit 5 – UTXBE UART TX Buffer Empty Status bit

Value	Description
1	Transmit buffer is empty; writing '1' when UTXEN = 0 will reset the TX FIFO Pointers and counters
0	Transmit buffer is not empty

Bit 4 – UTXBF UART TX Buffer Full Status bit

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Value	Description
1	Transmit buffer is full
0	Transmit buffer is not full

Bit 3 – RIDLE Receive Idle bit

Value	Description
1	UART RX line is in the Idle state
0	UART RX line is receiving something

Bit 2 – XON UART in XON Mode bit

Only valid when FLO[1:0] control bits are set to XON/XOFF mode.

Value	Description
1	UART has received XON
0	UART has not received XON or XOFF was received

Bit 1 – URXBE UART RX Buffer Empty Status bit

Value	Description
1	Receive buffer is empty; writing '1' when URXEN = 0 will reset the RX FIFO Pointers and counters
0	Receive buffer is not empty

Bit 0 – URXBF UART RX Buffer Full Status bit

Value	Description
1	Receive buffer is full
0	Receive buffer is not full

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17.5.5 UARTx Baud Rate Register Low

Name: UxBRG
Offset: 0x240, 0x268, 0xF08

Bit	15	14	13	12	11	10	9	8
	BRG[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	BRG[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

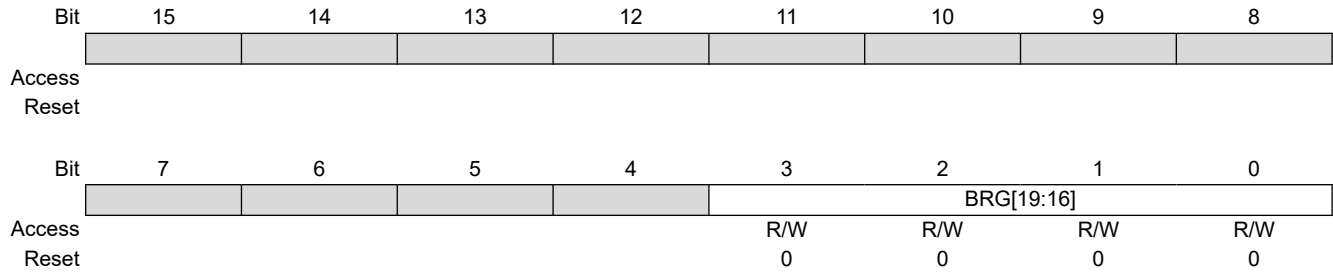
Bits 15:0 – BRG[15:0] Baud Rate Divisor bits

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17.5.6 UARTx Baud Rate Register High

Name: UxBRGH
Offset: 0x242, 0x26A, 0xF0A



Bits 3:0 – BRG[19:16] Baud Rate Divisor bits

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17.5.7 UARTx Receive Buffer Register

Name: UxRXREG
Offset: 0x244, 0x26C, 0xF0C

Legend: x = Bit is unknown

Bit	15	14	13	12	11	10	9	8
Access								
Reset								

Bit	7	6	5	4	3	2	1	0
	RXREG[7:0]							
Access	R	R	R	R	R	R	R	R
Reset	x	x	x	x	x	x	x	x

Bits 7:0 – RXREG[7:0] Received Character Data bits 7-0

dsPIC33CK512MPT608 Family

Universal Asynchronous Receiver Transmitter ...

17.5.8 UARTx Transmit Buffer Register

Name: UxTXREG
Offset: 0x248, 0x270, 0xF10

Legend: x = Bit is unknown

Bit	15	14	13	12	11	10	9	8
	LAST							
Access	W							
Reset	x							

Bit	7	6	5	4	3	2	1	0
	TXREG[7:0]							
Access	W	W	W	W	W	W	W	W
Reset	x	x	x	x	x	x	x	x

Bit 15 – LAST Last Byte Indicator for Smart Card Support bit

Bits 7:0 – TXREG[7:0] Transmitted Character Data bits 7-0
If the buffer is full, further writes to the buffer are ignored.

dsPIC33CK512MPT608 Family

Universal Asynchronous Receiver Transmitter ...

17.5.9 UARTx Timing Parameter 1 Register

Name: UxP1
Offset: 0x24C, 0x274, 0xF14

Bit	15	14	13	12	11	10	9	8
								P1[8]
Access								R/W
Reset								0

Bit	7	6	5	4	3	2	1	0
								P1[7:0]
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 8:0 – P1[8:0] Parameter 1 bits

DMX TX:

Number of Bytes to Transmit – 1 (not including Start code).

LIN TX:

PID to transmit (bits[5:0]).

Asynchronous TX with Address Detect:

Address to transmit. A '1' is automatically inserted into bit 9 (bits[7:0]).

Smart Card Mode:

Guard Time Counter bits. This counter is operated on the bit clock whose period is always equal to one ETU (bits[8:0]).

Other Modes:

Not used.

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Universal Asynchronous Receiver Transmitter ...

17.5.10 UARTx Timing Parameter 2 Register

Name: UxP2
Offset: 0x24E, 0x276, 0xF16

Bit	15	14	13	12	11	10	9	8
								P2[8]
Access								R/W
Reset								0

Bit	7	6	5	4	3	2	1	0
								P2[7:0]
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 8:0 – P2[8:0] Parameter 2 bits

DMX RX:

The first byte number to receive – 1, not including Start code (bits[8:0]).

LIN Responder TX:

Number of bytes to transmit (bits[7:0]).

Asynchronous RX with Address Detect:

Address to start matching (bits[7:0]).

Smart Card Mode:

Block Time Counter bits. This counter is operated on the bit clock whose period is always equal to one ETU (bits[8:0]).

Other Modes:

Not used.

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Universal Asynchronous Receiver Transmitter ...

17.5.11 UARTx Timing Parameter 3 Register Low

Name: UxP3
Offset: 0x250, 0x278, 0xF18

Bit	15	14	13	12	11	10	9	8
	P3[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	P3[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – P3[15:0] Parameter 3 bits

DMX RX:

The last byte number to receive – 1, not including Start code (bits[8:0]).

LIN Responder RX:

Number of bytes to receive (bits[7:0]).

Asynchronous RX:

Used to mask the UxP2 address bits; 1 = P2 address bit is used, 0 = P2 address bit is masked off (bits[7:0]).

Smart Card Mode:

Waiting Time Counter bits (bits[15:0]).

Other Modes:

Not used.

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Universal Asynchronous Receiver Transmitter ...

17.5.12 UARTx Timing Parameter 3 Register High

Name: UxP3H
Offset: 0x252, 0x27A, 0xF1A

Bit	15	14	13	12	11	10	9	8
Access								
Reset								
Bit	7	6	5	4	3	2	1	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 7:0 – P3[23:16] Parameter 3 High bits

Smart Card Mode:

Waiting Time Counter bits (bits[23:16]).

Other Modes:

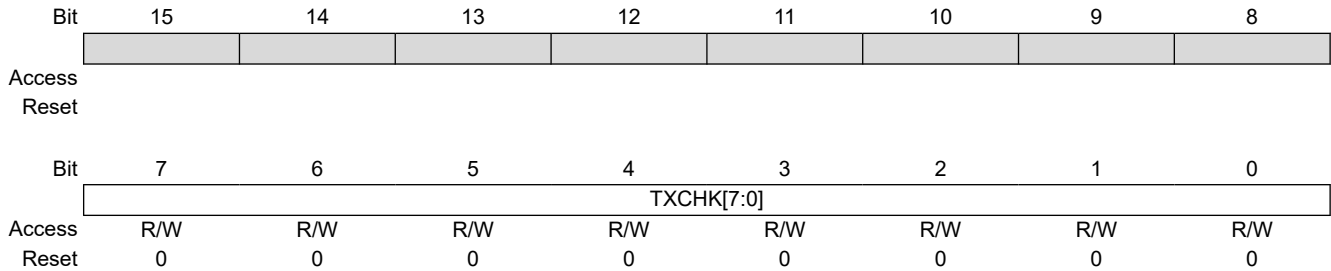
Not used.

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Universal Asynchronous Receiver Transmitter ...

17.5.13 UARTx Transmit Checksum Register

Name: UxTXCHK
Offset: 0x254, 0x27C, 0xF1C



Bits 7:0 – TXCHK[7:0] Transmit Checksum bits (calculated from TX words)

LIN Modes:

C0EN = 1: Sum of all transmitted data + addition carries, including PID.

C0EN = 0: Sum of all transmitted data + addition carries, excluding PID.

LIN Responder:

Cleared when Break is detected.

LIN Commander/Responder:

Cleared when Break is detected.

Other Modes:

C0EN = 1: Sum of every byte transmitted + addition carries.

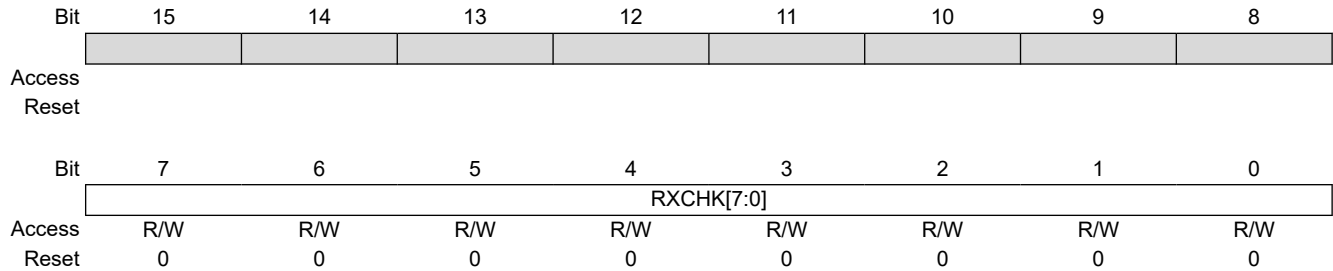
C0EN = 0: Value remains unchanged.

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Universal Asynchronous Receiver Transmitter ...

17.5.14 UARTx Receive Checksum Register

Name: UxRXCHK
Offset: 0x256, 0x27E, 0xF1E



Bits 7:0 – RXCHK[7:0] Receive Checksum bits (calculated from RX words)

LIN Modes:

C0EN = 1: Sum of all received data + addition carries, including PID.

C0EN = 0: Sum of all received data + addition carries, excluding PID.

LIN Responder:

Cleared when Break is detected.

LIN Commander/Responder:

Cleared when Break is detected.

Other Modes:

C0EN = 1: Sum of every byte received + addition carries.

C0EN = 0: Value remains unchanged.

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17.5.15 UARTx Smart Card Configuration Register

Name: UxSCCON
Offset: 0x258, 0x280, 0xF20

Bit	15	14	13	12	11	10	9	8
Access								
Reset								

Bit	7	6	5	4	3	2	1	0
			TXRPT[1:0]		CONV	T0PD	PRTCL	
Access			R/W	R/W	R/W	R/W	R/W	
Reset			0	0	0	0	0	

Bits 5:4 – TXRPT[1:0] Transmit Repeat Selection bits

Value	Description
11	Retransmits the error byte four times
10	Retransmits the error byte three times
01	Retransmits the error byte twice
00	Retransmits the error byte once

Bit 3 – CONV Logic Convention Selection bit

Value	Description
1	Inverse logic convention
0	Direct logic convention

Bit 2 – T0PD Pull-Down Duration for T = 0 Error Handling bit

Value	Description
1	2 ETU
0	1 ETU

Bit 1 – PRTCL Smart Card Protocol Selection bit

Value	Description
1	T = 1
0	T = 0

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Universal Asynchronous Receiver Transmitter ...

17.5.16 UARTx Smart Card Interrupt Register

Name: UxSCINT
Offset: 0x25A, 0x282, 0xF22

Legend: HS = Hardware Settable bit

Bit	15	14	13	12	11	10	9	8
			RXRPTIF	TXRPTIF		BTCIF	WTCIF	GTCIF
Access			R/W/HS	R/W/HS		R/W/HS	R/W/HS	R/W/HS
Reset			0	0		0	0	0

Bit	7	6	5	4	3	2	1	0
			RXRPTIE	TXRPTIE		BTCIE	WTCIE	GTCIE
Access			R/W	R/W		R/W	R/W	R/W
Reset			0	0		0	0	0

Bit 13 – RXRPTIF Receive Repeat Interrupt Flag bit

Value	Description
1	Parity error has persisted after the same character has been received five times (four retransmits)
0	Flag is cleared

Bit 12 – TXRPTIF Transmit Repeat Interrupt Flag bit

Value	Description
1	Line error has been detected after the last retransmit per TXRPT[1:0]
0	Flag is cleared

Bit 10 – BTCIF Block Time Counter Interrupt Flag bit

Value	Description
1	Block Time Counter has reached 0
0	Block Time Counter has not reached 0

Bit 9 – WTCIF Waiting Time Counter Interrupt Flag bit

Value	Description
1	Waiting Time Counter has reached 0
0	Waiting Time Counter has not reached 0

Bit 8 – GTCIF Guard Time Counter Interrupt Flag bit

Value	Description
1	Guard Time Counter has reached 0
0	Guard Time Counter has not reached 0

Bit 5 – RXRPTIE Receive Repeat Interrupt Enable bit

Value	Description
1	An interrupt is invoked when a parity error has persisted after the same character has been received five times (four retransmits)
0	Interrupt is disabled

Bit 4 – TXRPTIE Transmit Repeat Interrupt Enable bit

Value	Description
1	An interrupt is invoked when a line error is detected after the last retransmit per TXRPT[1:0] has been completed
0	Interrupt is disabled

Bit 2 – BTCIE Block Time Counter Interrupt Enable bit

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Universal Asynchronous Receiver Transmitter ...

Value	Description
1	Block Time Counter interrupt is enabled
0	Block Time Counter interrupt is disabled

Bit 1 – WTCIE Waiting Time Counter Interrupt Enable bit

Value	Description
1	Waiting Time Counter interrupt is enabled
0	Waiting Time Counter interrupt is disabled

Bit 0 – GTCIE Guard Time Counter Interrupt Enable bit

Value	Description
1	Guard Time Counter interrupt is enabled
0	Guard Time Counter interrupt is disabled

17.5.17 UARTx Interrupt Register

Name: UxINT
Offset: 0x25C, 0x284, 0xF24

Legend: HS = Hardware Settable bit

Bit	15	14	13	12	11	10	9	8
Access								
Reset								

Bit	7	6	5	4	3	2	1	0
	WUIF	ABDIF				ABDIE		
Access	R/W/HS	R/W/HS				R/W		
Reset	0	0				0		

Bit 7 – WUIF Wake-up Interrupt Flag bit

Value	Description
1	Sets when WAKE = 1 and RX makes a 1-to-0 transition; triggers event interrupt (must be cleared by software)
0	WAKE is not enabled or WAKE is enabled, but no wake-up event has occurred

Bit 6 – ABDIF Auto-Baud Completed Interrupt Flag bit

Value	Description
1	Sets when ABD sequence makes the final 1-to-0 transition; triggers event interrupt (must be cleared by software)
0	ABAUD is not enabled or ABAUD is enabled but auto-baud has not completed

Bit 2 – ABDIE Auto-Baud Completed Interrupt Enable Flag bit

Value	Description
1	Allows ABDIF to set an event interrupt
0	ABDIF does not set an event interrupt

18. Serial Peripheral Interface (SPI)

Note: This data sheet summarizes the features of the dsPIC33CK512MPT608 family of devices. It is not intended to be a comprehensive reference source. To complement the information in this data sheet, refer to “**Serial Peripheral Interface (SPI) with Audio Codec Support**” (www.microchip.com/DS70005136) in the “dsPIC33/PIC24 Family Reference Manual”.

The Serial Peripheral Interface (SPI) module is a synchronous serial interface, useful for communicating with other peripheral or microcontroller devices. These peripheral devices may be serial EEPROMs, shift registers, display drivers, A/D Converters, etc. The SPI module is compatible with the Motorola® SPI and SIOP interfaces. All devices in the dsPIC33CK512MPT608 family include three SPI modules.

Note: This family of devices has three SPIs. However, one of them is used for communication with the Secure Subsystem. Only two are available for general purpose.

The module supports operation in two Buffer modes. In Standard mode, data are shifted through a single serial buffer. In Enhanced Buffer mode, data are shifted through a FIFO buffer. The FIFO level depends on the configured mode.

Note: FIFO depth for this device is four (in 8-Bit Data mode).

Variable length data can be transmitted and received, from 2 to 32 bits.

Note: Do not perform Read-Modify-Write operations (such as bit-oriented instructions) on the SPIxBUF register in either Standard or Enhanced Buffer mode.

The module also supports Audio modes. Four different Audio modes are available.

- I²S mode
- Left Justified mode
- Right Justified mode
- PCM/DSP mode

In each of these modes, the serial clock is free-running and audio data are always transferred.

The SPI serial interface consists of four pins:

- SDIx: Serial Data Input
- SDOx: Serial Data Output
- SCKx: Shift Clock Input or Output

The SPI module can be configured to operate using two, three or four pins.

The SPI module has the ability to generate three interrupts, reflecting the events that occur during the data communication. The following types of interrupts can be generated:

1. Receive interrupts are signaled by SPIxRXIF. This event occurs when:
 - RX watermark interrupt
 - SPIROV = 1
 - SPIRBF = 1
 - SPIRBE = 1

provided the respective mask bits are enabled in SPIxIMSKL/H.

2. Transmit interrupts are signalled by SPIxTXIF. This event occurs when:
 - TX watermark interrupt
 - SPITUR = 1
 - SPITBF = 1
 - SPITBE = 1

provided the respective mask bits are enabled in SPIxIMSKL/H.

3. General interrupts are signalled by SPIxGIF. This event occurs when:
 - FRMERR = 1

- SPIBUSY = 1
- SRMT = 1

provided the respective mask bits are enabled in SPIxIMSKL/H.

Block diagrams of the module in Standard and Enhanced modes are shown in [Figure 18-1](#) and [Figure 18-2](#).

Note: In this section, the SPI modules are referred to together as SPIx, or separately as SPI1, SPI2 or SPI3. Special Function Registers will follow a similar notation. For example, SPIxCON1 and SPIxCON2 refer to the control registers for any of the three SPI modules.

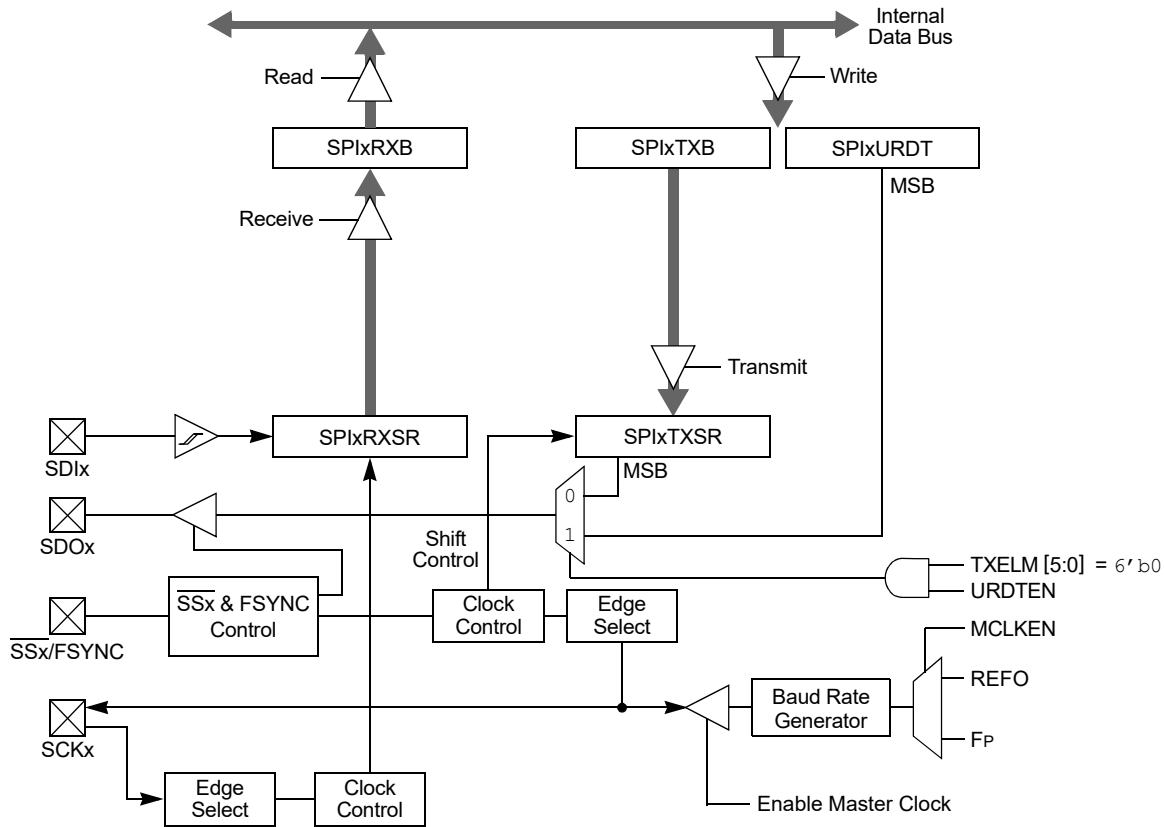
To set up the SPIx module for the Standard Host mode of operation:

1. If using interrupts:
 - a. Clear the interrupt flag bits in the respective IFSx register.
 - b. Set the interrupt enable bits in the respective IECx register.
 - c. Write the SPIxIP bits in the respective IPCx register to set the interrupt priority.
2. Write the desired settings to the SPIxCON1L and SPIxCON1H registers with the MSTEN bit (SPIxCON1L[5]) = 1.
3. Clear the SPIROV bit (SPIxSTATL[6]).
4. Enable SPIx operation by setting the SPIEN bit (SPIxCON1L[15]).
5. Write the data to be transmitted to the SPIxBUFL and SPIxBUFH registers. Transmission (and reception) will start as soon as data are written to the SPIxBUFL and SPIxBUFH registers.

To set up the SPIx module for the Standard Client mode of operation:

1. Clear the SPIxBUF registers.
2. If using interrupts:
 - a. Clear the SPIxBUFL and SPIxBUFH registers.
 - b. Set the interrupt enable bits in the respective IECx register.
 - c. Write the SPIxIP bits in the respective IPCx register to set the interrupt priority.
3. Write the desired settings to the SPIxCON1L, SPIxCON1H and SPIxCON2L registers with the MSTEN bit (SPIxCON1L[5]) = 0.
4. Clear the SMP bit.
5. If the CKE bit (SPIxCON1L[8]) is set, then the SSEN bit (SPIxCON1L[7]) must be set to enable the $\overline{SS}_x$ pin.
6. Clear the SPIROV bit (SPIxSTATL[6]).
7. Enable SPIx operation by setting the SPIEN bit (SPIxCON1L[15]).

Figure 18-1. SPIx Module Block Diagram (Standard Mode)



To set up the SPIx module for the Enhanced Buffer Host mode of operation:

1. If using interrupts:
 - a. Clear the interrupt flag bits in the respective IFSx register.
 - b. Set the interrupt enable bits in the respective IECx register.
 - c. Write the SPIxIP bits in the respective IPCx register.
2. Write the desired settings to the SPIxCON1L, SPIxCON1H and SPIxCON2L registers with MSTEN (SPIxCON1L[5]) = 1.
3. Clear the SPIROV bit (SPIxSTATL[6]).
4. Select Enhanced Buffer mode by setting the ENHBUF bit (SPIxCON1L[0]).
5. Enable SPIx operation by setting the SPIEN bit (SPIxCON1L[15]).
6. Write the data to be transmitted to the SPIxBUFL and SPIxBUFH registers. Transmission (and reception) will start as soon as data are written to the SPIxBUFL and SPIxBUFH registers.

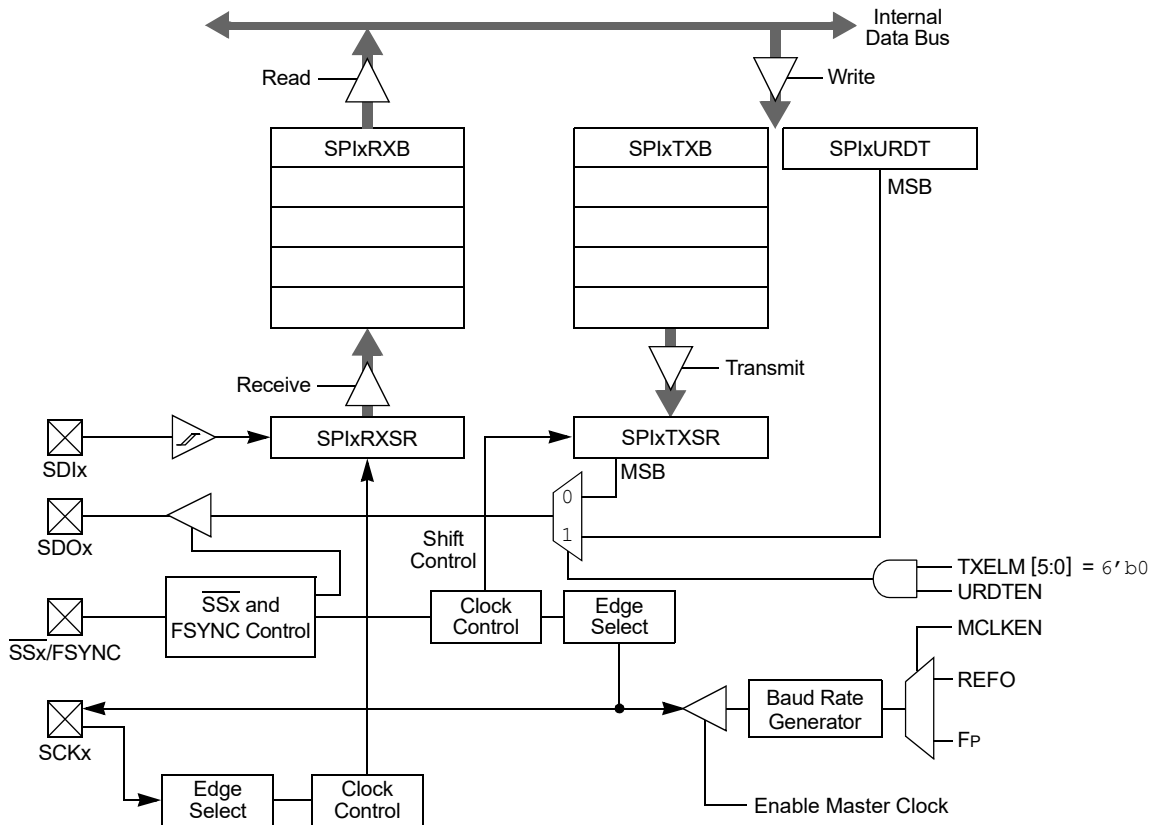
To set up the SPIx module for the Enhanced Buffer Client mode of operation:

1. Clear the SPIxBUFL and SPIxBUFH registers.
2. If using interrupts:
 - a. Clear the interrupt flag bits in the respective IFSx register.
 - b. Set the interrupt enable bits in the respective IECx register.
 - c. Write the SPIxIP bits in the respective IPCx register to set the interrupt priority.
3. Write the desired settings to the SPIxCON1L, SPIxCON1H and SPIxCON2L registers with the MSTEN bit (SPIxCON1L[5]) = 0.
4. Clear the SMP bit.
5. If the CKE bit is set, then the SSx pin must be set, thus enabling the SSx pin.
6. Clear the SPIROV bit (SPIxSTATL[6]).

Serial Peripheral Interface (SPI)

7. Select Enhanced Buffer mode by setting the ENHBUF bit (SPIxCON1L[0]).
8. Enable SPIx operation by setting the SPIEN bit (SPIxCON1L[15]).

Figure 18-2. SPIx Module Block Diagram (Enhanced Mode)



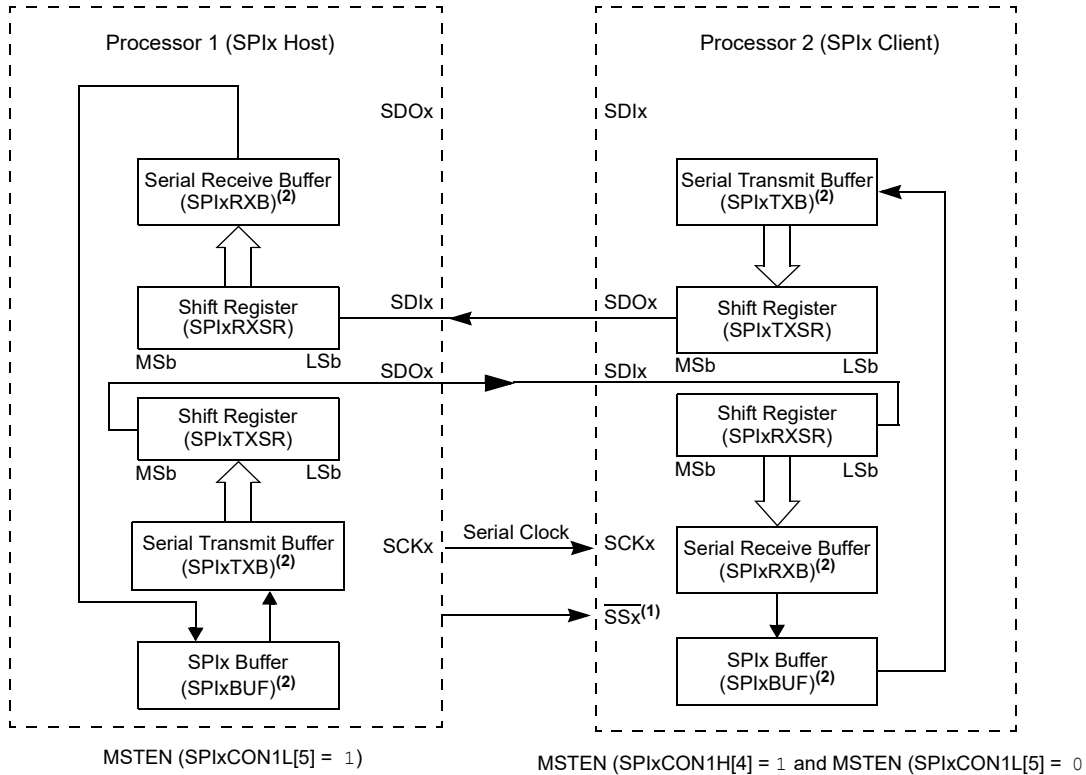
To set up the SPIx module for Audio mode:

1. Clear the SPIxBUFL and SPIxBUFH registers.
2. If using interrupts:
 - a. Clear the interrupt flag bits in the respective IFSx register.
 - b. Set the interrupt enable bits in the respective IECx register.
 - c. Write the SPIxIP bits in the respective IPCx register to set the interrupt priority.
3. Write the desired settings to the SPIxCON1L, SPIxCON1H and SPIxCON2L registers with AUDEN (SPIxCON1H[15]) = 1.
4. Clear the SPIROV bit (SPIxSTATL[6]).
5. Enable SPIx operation by setting the SPIEN bit (SPIxCON1L[15]).
6. Write the data to be transmitted to the SPIxBUFL and SPIxBUFH registers. Transmission (and reception) will start as soon as data are written to the SPIxBUFL and SPIxBUFH registers.

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Serial Peripheral Interface (SPI)

Figure 18-3. SPIx Main Connection (Standard Mode)



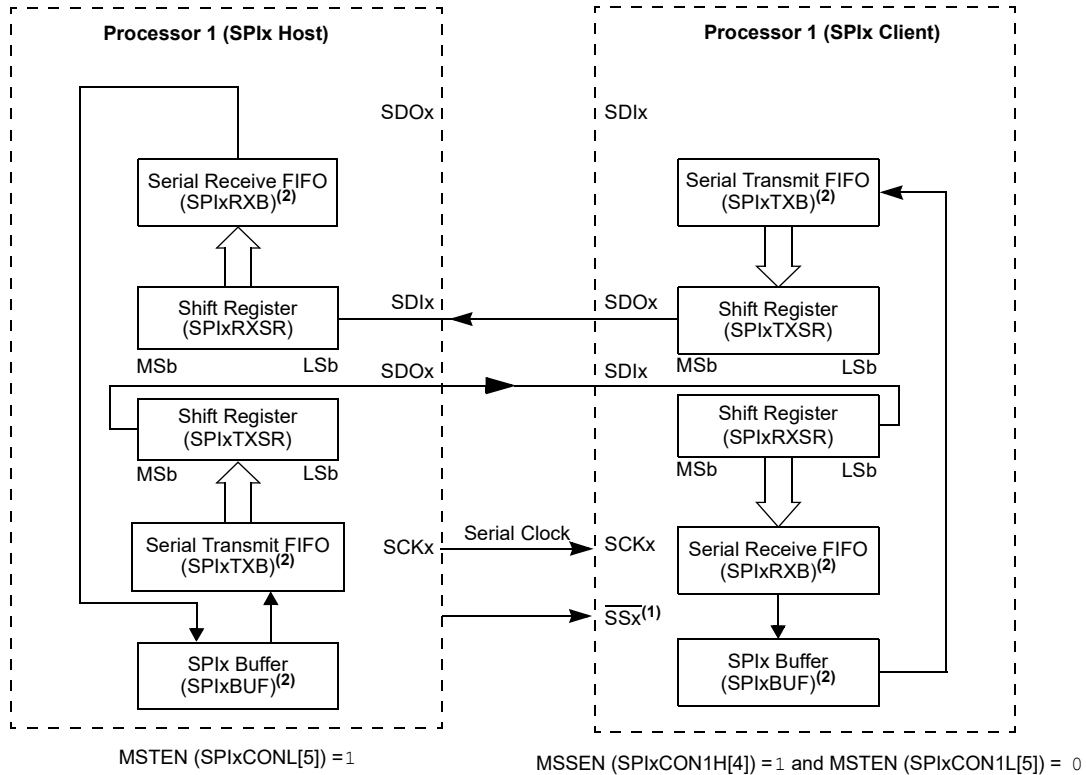
Note 1: Using the $\overline{SSx}$ pin in Client mode of operation is optional.

Note 2: User must write transmit data to read the received data from SPIxBUF. The SPIxTXB and SPIxRXB registers are memory-mapped to SPIxBUF.

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Serial Peripheral Interface (SPI)

Figure 18-4. SPIx Main Connection (Enhanced Buffer Modes)



- Note 1:** Using the $\overline{SSx}$ pin in Client mode is optional.
- Note 2:** User must write transmit data to read the received data from SPIxBUF. The SPIxTXB and SPIxRXB registers are memory-mapped to SPIxBUF.

Equation 18-1. Relationship Between Device and SPIx Clock Speed

$$\text{Baud Rate} = \frac{F_P}{(2 * (\text{SPIxBRG} + 1))}$$

Where:

F_P is the Peripheral Bus Clock Frequency.

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Serial Peripheral Interface (SPI)

18.1 SPI Control/Status Registers

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x00 ... 0x02AB	Reserved									
0x02AC	SPI1CON1L	15:8	SPIEN		SPISIDL	DISSDO	MODE32 and	MODE16[1:0]	SMP	CKE
		7:0	SSEN	CKP	MSTEN	DISSDI	DISSCK	MCLKEN	SPIFE	ENHBUF
0x02AE	SPI1CON1H	15:8	AUDEN	SPISGNEXT	IGNROV	IGNTUR	AUDMONO	URDTEN	AUDMOD[1:0]	
		7:0	FRMEN	FRMSYNC	FRMPOL	MSEN	FRMSYPW	FRMCNT[2:0]		
0x02B0	SPI1CON2L	15:8								
		7:0				WLENGTH[4:0]				
0x02B2 ... 0x02B3	Reserved									
0x02B4	SPI1STATL	15:8				FRMERR	SPIBUSY			SPITUR
		7:0	SRMT	SPIROV	SPIRBE		SPITBE		SPITBF	SPIRBF
0x02B6	SPI1STATH	15:8			RXELM[5:0]					
		7:0			TXELM[5:0]					
0x02B8	SPI1BUFL	15:8	DATA[15:8]							
		7:0	DATA[7:0]							
0x02BA	SPI1BUFH	15:8	DATA[31:24]							
		7:0	DATA[23:16]							
0x02BC	SPI1BRGL	15:8				BRG[12:8]				
		7:0	BRG[7:0]							
0x02BE ... 0x02BF	Reserved									
0x02C0	SPI1MSKL	15:8				FRMERREN	BUSYEN			SPITUREN
		7:0	SRMTEN	SPIROVEN	SPIRBEN		SPITBEN		SPITBFEN	SPIRBFEN
0x02C2	SPI1MSKH	15:8	RXWIEN		RXMSK[5:0]					
		7:0	TXWIEN		TXMSK[5:0]					
0x02C4	SPI1URDTL	15:8	SPI1URDTL[15:8]							
		7:0	SPI1URDTL[7:0]							
0x02C6	SPI1URDTH	15:8	SPI1URDTH[31:24]							
		7:0	SPI1URDTH[23:16]							
0x02C8	SPI2CON1L	15:8	SPIEN		SPISIDL	DISSDO	MODE32 and	MODE16[1:0]	SMP	CKE
		7:0	SSEN	CKP	MSTEN	DISSDI	DISSCK	MCLKEN	SPIFE	ENHBUF
0x02CA	SPI2CON1H	15:8	AUDEN	SPISGNEXT	IGNROV	IGNTUR	AUDMONO	URDTEN	AUDMOD[1:0]	
		7:0	FRMEN	FRMSYNC	FRMPOL	MSEN	FRMSYPW	FRMCNT[2:0]		
0x02CC	SPI2CON2L	15:8								
		7:0				WLENGTH[4:0]				
0x02CE ... 0x02CF	Reserved									
0x02D0	SPI2STATL	15:8				FRMERR	SPIBUSY			SPITUR
		7:0	SRMT	SPIROV	SPIRBE		SPITBE		SPITBF	SPIRBF
0x02D2	SPI2STATH	15:8			RXELM[5:0]					
		7:0			TXELM[5:0]					
0x02D4	SPI2BUFL	15:8	DATA[15:8]							
		7:0	DATA[7:0]							
0x02D6	SPI2BUFH	15:8	DATA[31:24]							
		7:0	DATA[23:16]							
0x02D8	SPI2BRGL	15:8				BRG[12:8]				
		7:0	BRG[7:0]							
0x02DA ... 0x02DB	Reserved									

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Serial Peripheral Interface (SPI)

.....continued

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x02DC	SPI2IMSKL	15:8				FRMERREN	BUSYEN			SPITUREN
		7:0	SRMTEN	SPIROVEN	SPIRBEN		SPITBEN		SPITBFEN	SPIRBFEN
0x02DE	SPI2IMSKH	15:8	RXWIEN		RXMSK[5:0]					
		7:0	TXWIEN		TXMSK[5:0]					
0x02E0	SPI2URDTL	15:8	SPI2URDTL[15:8]							
		7:0	SPI2URDTL[7:0]							
0x02E2	SPI2URDTH	15:8	SPI2URDTH[31:24]							
		7:0	SPI2URDTH[23:16]							
0x02E4	SPI3CON1L	15:8	SPIEN		SPISIDL	DISSDO	MODE32 and MODE16[1:0]		SMP	CKE
		7:0	SSEN	CKP	MSTEN	DISSDI	DISSCK	MCLKEN	SPIFE	ENHBUF
0x02E6	SPI3CON1H	15:8	AUDEN	SPISGNEXT	IGNROV	IGNTUR	AUDMONO	URDTEN	AUDMOD[1:0]	
		7:0	FRMEN	FRMSYNC	FRMPOL	MSEN	FRMSYPW	FRMCNT[2:0]		
0x02E8	SPI3CON2L	15:8								
		7:0				WLENGTH[4:0]				
0x02EA ... 0x02EB	Reserved									
0x02EC	SPI3STATL	15:8				FRMERR	SPIBUSY			SPITUR
		7:0	SRMT	SPIROV	SPIRBE		SPITBE		SPITBF	SPIRBF
0x02EE	SPI3STATH	15:8			RXELM[5:0]					
		7:0			TXELM[5:0]					
0x02F0 ... 0x02F7	Reserved									
0x02F8	SPI3IMSKL	15:8				FRMERREN	BUSYEN			SPITUREN
		7:0	SRMTEN	SPIROVEN	SPIRBEN		SPITBEN		SPITBFEN	SPIRBFEN
0x02FA	SPI3IMSKH	15:8	RXWIEN		RXMSK[5:0]					
		7:0	TXWIEN		TXMSK[5:0]					

18.1.1 SPIx Control Register 1 Low

Name: SPIxCON1L
Offset: 0x2AC, 0x2C8, 0x2E4

Notes:

1. When AUDEN = 1, this module functions as if CKE = 0, regardless of its actual value.
2. When FRMEN = 1, SSEN is not used.
3. MCLKEN can only be written when the SPIEN bit = 0.
4. This channel is not meaningful for DSP/PCM mode as LRC follows FRMSYPW.

Bit	15	14	13	12	11	10	9	8
	SPIEN		SPISIDL	DISSDO	MODE32 and MODE16[1:0]		SMP	CKE
Access	R/W		R/W	R/W	R/W	R/W	R/W	R/W
Reset	0		0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	SSEN	CKP	MSTEN	DISSDI	DISSCK	MCLKEN	SPIFE	ENHBUF
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – SPIEN SPIx On bit

Value	Description
1	Enables module
0	Turns off and resets module, disables clocks, disables interrupt event generation, allows SFR modifications

Bit 13 – SPISIDL SPIx Stop in Idle Mode bit

Value	Description
1	Halts in CPU Idle mode
0	Continues to operate in CPU Idle mode

Bit 12 – DISSDO Disable SDOx Output Port bit

Value	Description
1	SDOx pin is not used by the module; pin is controlled by port function
0	SDOx pin is controlled by the module

Bits 11:10 – MODE32 and MODE16[1:0] Serial Word Length Select bits^(1,4)

MODE32	MODE16	AUDEN	Communication
1	x	0	32-Bit
0	1		16-Bit
0	0		8-Bit
1	1	1	24-Bit Data, 32-Bit FIFO, 32-Bit Channel/64-Bit Frame
1	0		32-Bit Data, 32-Bit FIFO, 32-Bit Channel/64-Bit Frame
0	1		16-Bit Data, 16-Bit FIFO, 32-Bit Channel/64-Bit Frame
0	0		16-Bit FIFO, 16-Bit Channel/32-Bit Frame

Bit 9 – SMP SPIx Data Input Sample Phase bitClient Mode:

Input data are always sampled at the middle of data output time, regardless of the SMP setting.

Host Mode:

Value	Description
1	Input data are sampled at the end of data output time
0	Input data are sampled at the middle of data output time

Bit 8 – CKE SPIx Clock Edge Select bit⁽¹⁾

Value	Description
1	Transmit happens on transition from Active Clock state to Idle Clock state
0	Transmit happens on transition from Idle Clock state to Active Clock state

Bit 7 – SSEN Client Select Enable bit (Client mode)⁽²⁾

Value	Description
1	SSx pin is used by the macro in Client mode; SSx pin is used as the Client select input
0	SSx pin is not used by the macro (SSx pin will be controlled by the port I/O)

Bit 6 – CKP Clock Polarity Select bit

Value	Description
1	Idle state for clock is a high level; Active state is a low level
0	Idle state for clock is a low level; Active state is a high level

Bit 5 – MSTEN Host Mode Enable bit

Value	Description
1	Host mode
0	Client mode

Bit 4 – DISSDI Disable SDIx Input Port bit

Value	Description
1	SDIx pin is not used by the module; pin is controlled by port function
0	SDIx pin is controlled by the module

Bit 3 – DISSCK Disable SCKx Output Port bit

Value	Description
1	SCKx pin is not used by the module; pin is controlled by port function
0	SCKx pin is controlled by the module

Bit 2 – MCLKEN Host Clock Enable bit⁽³⁾

Value	Description
1	REFO is used by the BRG
0	PBCLK is used by the BRG

Bit 1 – SPIFE Frame Sync Pulse Edge Select bit

Value	Description
1	Frame Sync pulse (Idle-to-active edge) coincides with the first bit clock
0	Frame Sync pulse (Idle-to-active edge) precedes the first bit clock

Bit 0 – ENHBUF Enhanced Buffer Enable bit

Value	Description
1	Enhanced Buffer mode is enabled
0	Enhanced Buffer mode is disabled

18.1.2 SPIx Control Register 1 High

Name: SPIxCON1H
Offset: 0x2AE, 0x2CA, 0x2E6

Notes:

1. AUDEN can only be written when the SPIEN bit = 0.
2. AUDMONO can only be written when the SPIEN bit = 0 and is only valid for AUDEN = 1.
3. URDTEN is only valid when IGNTUR = 1.
4. AUDMOD[1:0] can only be written when the SPIEN bit = 0 and is only valid when AUDEN = 1. When NOT in PCM/DSP mode, this module functions as if FRMSYPW = 1, regardless of its actual value.

Bit	15	14	13	12	11	10	9	8
	AUDEN	SPISGNEXT	IGNROV	IGNTUR	AUDMONO	URDTEN	AUDMOD[1:0]	
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	FRMEN	FRMSYNC	FRMPOL	MSEN	FRMSYPW	FRMCNT[2:0]		
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – AUDEN Audio Codec Support Enable bit⁽¹⁾

Value	Description
1	Audio protocol is enabled; MSTEN controls the direction of both SCKx and frame (a.k.a. LRC), and this module functions as if FRMEN = 1, FRMSYNC = MSTEN, FRMCNT[2:0] = 001 and SMP = 0, regardless of their actual values
0	Audio protocol is disabled

Bit 14 – SPISGNEXT SPIx Sign-Extend RX FIFO Read Data Enable bit

Value	Description
1	Data from RX FIFO are sign-extended
0	Data from RX FIFO are not sign-extended

Bit 13 – IGNROV Ignore Receive Overflow bit

Value	Description
1	A Receive Overflow (ROV) is NOT a critical error; during ROV, data in the FIFO are not overwritten by the receive data
0	A ROV is a critical error that stops SPI operation

Bit 12 – IGNTUR Ignore Transmit Underrun bit

Value	Description
1	A Transmit Underrun (TUR) is NOT a critical error and data indicated by URDTEN are transmitted until the SPIxTXB is not empty
0	A TUR is a critical error that stops SPI operation

Bit 11 – AUDMONO Audio Data Format Transmit bit⁽²⁾

Value	Description
1	Audio data are mono (i.e., each data word is transmitted on both left and right channels)
0	Audio data are stereo

Bit 10 – URDTEN Transmit Underrun Data Enable bit⁽³⁾

Value	Description
1	Transmits data out of SPIxURDT register during Transmit Underrun conditions
0	Transmits the last received data during Transmit Underrun conditions

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Serial Peripheral Interface (SPI)

Bits 9:8 – AUDMOD[1:0] Audio Protocol Mode Selection bits⁽⁴⁾

Value	Description
11	PCM/DSP mode
10	Right Justified mode: This module functions as if SPIFE = 1, regardless of its actual value
01	Left Justified mode: This module functions as if SPIFE = 1, regardless of its actual value
00	I ² S mode: This module functions as if SPIFE = 0, regardless of its actual value

Bit 7 – FRMEN Framed SPIx Support bit

Value	Description
1	Framed SPIx support is enabled (SSx pin is used as the FSYNC input/output)
0	Framed SPIx support is disabled

Bit 6 – FRMSYNC Frame Sync Pulse Direction Control bit

Value	Description
1	Frame Sync pulse input (Client)
0	Frame Sync pulse output (Host)

Bit 5 – FRMPOL Frame Sync/Client Select Polarity bit

Value	Description
1	Frame Sync pulse/Client select is active-high
0	Frame Sync pulse/Client select is active-low

Bit 4 – MSSEN Host Mode Client Select Enable bit

Value	Description
1	SPIx Client select support is enabled with polarity determined by FRMPOL (SSx pin is automatically driven during transmission in Host mode)
0	Client select SPIx support is disabled (SSx pin will be controlled by port I/O)

Bit 3 – FRMSYPW Frame Sync Pulse-Width bit

Value	Description
1	Frame Sync pulse is one serial word length wide (as defined by MODE[32,16]/WLENGTH[4:0])
0	Frame Sync pulse is one clock (SCKx) wide

Bits 2:0 – FRMCNT[2:0] Frame Sync Pulse Counter bits

Controls the number of serial words transmitted per Sync pulse.

Value	Description
111	Reserved
110	Reserved
101	Generates a Frame Sync pulse on every 32 serial words
100	Generates a Frame Sync pulse on every 16 serial words
011	Generates a Frame Sync pulse on every 8 serial words
010	Generates a Frame Sync pulse on every 4 serial words
001	Generates a Frame Sync pulse on every 2 serial words (value used by audio protocols)
000	Generates a Frame Sync pulse on each serial word

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Serial Peripheral Interface (SPI)

18.1.3 SPIx Control Register 2 Low

Name: SPIxCON2L
Offset: 0x2B0, 0x2CC, 0x2E8

Notes:

- These bits are effective when AUDEN = 0 only.
- Varying the length by changing these bits does not affect the depth of the TX/RX FIFO.

Bit	15	14	13	12	11	10	9	8
Access								
Reset								

Bit	7	6	5	4	3	2	1	0
						WLENGTH[4:0]		
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0

Bits 4:0 – WLENGTH[4:0] Variable Word Length bits^(1,2)

Value	Description
11111	32-bit data
11110	31-bit data
11101	30-bit data
11100	29-bit data
11011	28-bit data
11010	27-bit data
11001	26-bit data
11000	25-bit data
10111	24-bit data
10110	23-bit data
10101	22-bit data
10100	21-bit data
10011	20-bit data
10010	19-bit data
10001	18-bit data
10000	17-bit data
01111	16-bit data
01110	15-bit data
01101	14-bit data
01100	13-bit data
01011	12-bit data
01010	11-bit data
01001	10-bit data
01000	9-bit data
00111	8-bit data
00110	7-bit data
00101	6-bit data
00100	5-bit data
00011	4-bit data
00010	3-bit data
00001	2-bit data
00000	See MODE[32,16] bits in SPIxCON1L[11:10]

18.1.4 SPIx Status Register Low

Name: SPIxSTATL
Offset: 0x2B4, 0x2D0, 0x2EC

Note:

1. SPITUR is cleared when SPIEN = 0. When IGNTUR = 1, SPITUR provides dynamic status of the Transmit Underrun condition, but does not stop RX/TX operation and does not need to be cleared by software.

Legend: C = Clearable bit; HS = Hardware Settable bit; HSC = Hardware Settable/Clearable bit

Bit	15	14	13	12	11	10	9	8
				FRMERR	SPIBUSY			SPITUR
Access				R/C/HS	R/HSC			R/HSC
Reset				0	0			0

Bit	7	6	5	4	3	2	1	0
	SRMT	SPIROV	SPIRBE		SPITBE		SPITBF	SPIRBF
Access	R/HSC	R/C/HS	R/HSC		R/HSC		R/HSC	R/HSC
Reset	0	0	1		1		0	0

Bit 12 – FRMERR SPIx Frame Error Status bit

Value	Description
1	Frame error is detected
0	No frame error is detected

Bit 11 – SPIBUSY SPIx Activity Status bit

Value	Description
1	Module is currently busy with some transactions
0	No ongoing transactions (at time of read)

Bit 8 – SPITUR SPIx Transmit Underrun Status bit⁽¹⁾

Value	Description
1	Transmit buffer has encountered a Transmit Underrun condition
0	Transmit buffer does not have a Transmit Underrun condition

Bit 7 – SRMT Shift Register Empty Status bit

Value	Description
1	No current or pending transactions (i.e., neither SPIxTXB or SPIxTXSR contains data to transmit)
0	Current or pending transactions

Bit 6 – SPIROV SPIx Receive Overflow Status bit

Value	Description
1	A new byte/half-word/word has been completely received when the SPIxRXB was full
0	No overflow

Bit 5 – SPIRBE SPIx RX Buffer Empty Status bit

Standard Buffer Mode:

Automatically set in hardware when SPIxBUF is read from, reading SPIxRXB. Automatically cleared in hardware when SPIx transfers data from SPIxRXSR to SPIxRXB.

Enhanced Buffer Mode:

Indicates RXELM[5:0] = 000000.

Value	Description
1	RX buffer is empty
0	RX buffer is not empty

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Serial Peripheral Interface (SPI)

Bit 3 – SPITBE SPIx Transmit Buffer Empty Status bit

Standard Buffer Mode:

Automatically set in hardware when SPIx transfers data from SPIxTXB to SPIxTXSR. Automatically cleared in hardware when SPIxBUF is written, loading SPIxTXB.

Enhanced Buffer Mode:

Indicates TXELM[5:0] = 000000.

Value	Description
1	SPIxTXB is empty
0	SPIxTXB is not empty

Bit 1 – SPITBF SPIx Transmit Buffer Full Status bit

Standard Buffer Mode:

Automatically set in hardware when SPIxBUF is written, loading SPIxTXB. Automatically cleared in hardware when SPIx transfers data from SPIxTXB to SPIxTXSR.

Enhanced Buffer Mode:

Indicates TXELM[5:0] = 111111.

Value	Description
1	SPIxTXB is full
0	SPIxTXB not full

Bit 0 – SPIRBF SPIx Receive Buffer Full Status bit

Standard Buffer Mode:

Automatically set in hardware when SPIx transfers data from SPIxRXSR to SPIxRXB. Automatically cleared in hardware when SPIxBUF is read from, reading SPIxRXB.

Enhanced Buffer Mode:

Indicates RXELM[5:0] = 111111.

Value	Description
1	SPIxRXB is full
0	SPIxRXB is not full

18.1.5 SPIx Status Register High

Name: SPIxSTATH
Offset: 0x2B6, 0x2D2, 0x2EE

Notes:

1. RXELM3 and TXELM3 bits are only present when FIFODEPTH = 8 or higher.
2. RXELM4 and TXELM4 bits are only present when FIFODEPTH = 16 or higher.
3. RXELM5 and TXELM5 bits are only present when FIFODEPTH = 32.

Legend: HSC = Hardware Settable/Clearable bit

Bit	15	14	13	12	11	10	9	8
			RXELM[5:0]					
Access			R/HSC	R/HSC	R/HSC	R/HSC	R/HSC	R/HSC
Reset			0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
			TXELM[5:0]					
Access			R/HSC	R/HSC	R/HSC	R/HSC	R/HSC	R/HSC
Reset			0	0	0	0	0	0

Bits 13:8 – RXELM[5:0] Receive Buffer Element Count bits (valid in Enhanced Buffer mode)^(1,2,3)

Bits 5:0 – TXELM[5:0] Transmit Buffer Element Count bits (valid in Enhanced Buffer mode)^(1,2,3)

18.1.6 SPIx Buffer Register Low

Name: SPIxBUFL
Offset: 0x2B8, 0x2D4

Bit	15	14	13	12	11	10	9	8
	DATA[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	DATA[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – DATA[15:0] SPI FIFO Data bits

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Serial Peripheral Interface (SPI)

18.1.7 SPIx Buffer Register High

Name: SPIxBUFH
Offset: 0x2BA, 0x2D6

Bit	15	14	13	12	11	10	9	8
	DATA[31:24]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	DATA[23:16]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – DATA[31:16] SPI FIFO Data bits

18.1.8 SPIx Baud Rate Generator Register Low

Name: SPIxBRGL
Offset: 0x2BC, 0x2D8

Note:
1. Changing the BRG value when SPIEN = 1 causes undefined behavior.

Bit	15	14	13	12	11	10	9	8
	BRG[12:8]							
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	BRG[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 12:0 – BRG[12:0] SPI Baud Rate Generator Divisor bits⁽¹⁾

18.1.9 SPIx Interrupt Mask Register Low

Name: SPIxIMSKL
Offset: 0x2C0, 0x2DC, 0x2F8

Bit	15	14	13	12	11	10	9	8
				FRMERREN	BUSYEN			SPITUREN
Access				R/W	R/W			R/W
Reset				0	0			0

Bit	7	6	5	4	3	2	1	0
	SRMTEN	SPIROVEN	SPIRBEN		SPITBEN		SPITBFEN	SPIRBFEN
Access	R/W	R/W	R/W		R/W		R/W	R/W
Reset	0	0	0		0		0	0

Bit 12 – FRMERREN Enable Interrupt Events via FRMERR bit

Value	Description
1	Frame error generates an interrupt event
0	Frame error does not generate an interrupt event

Bit 11 – BUSYEN Enable Interrupt Events via SPIBUSY bit

Value	Description
1	SPIBUSY generates an interrupt event
0	SPIBUSY does not generate an interrupt event

Bit 8 – SPITUREN Enable Interrupt Events via SPITUR bit

Value	Description
1	Transmit Underrun (TUR) generates an interrupt event
0	Transmit Underrun does not generate an interrupt event

Bit 7 – SRMTEN Enable Interrupt Events via SRMT bit

Value	Description
1	Shift Register Empty (SRMT) generates interrupt events
0	Shift Register Empty does not generate interrupt events

Bit 6 – SPIROVEN Enable Interrupt Events via SPIROV bit

Value	Description
1	SPIx Receive Overflow (ROV) generates an interrupt event
0	SPIx Receive Overflow does not generate an interrupt event

Bit 5 – SPIRBEN Enable Interrupt Events via SPIRBE bit

Value	Description
1	SPIx RX buffer empty generates an interrupt event
0	SPIx RX buffer empty does not generate an interrupt event

Bit 3 – SPITBEN Enable Interrupt Events via SPITBE bit

Value	Description
1	SPIx transmit buffer empty generates an interrupt event
0	SPIx transmit buffer empty does not generate an interrupt event

Bit 1 – SPITBFEN Enable Interrupt Events via SPITBF bit

Value	Description
1	SPIx transmit buffer full generates an interrupt event
0	SPIx transmit buffer full does not generate an interrupt event

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Serial Peripheral Interface (SPI)

Bit 0 – SPIRBFEN Enable Interrupt Events via SPIRBF bit

Value	Description
1	SPIx receive buffer full generates an interrupt event
0	SPIx receive buffer full does not generate an interrupt event

18.1.10 SPIx Interrupt Mask Register High

Name: SPIxIMSKH
Offset: 0x2C2, 0x2DE, 0x2FA

Notes:

1. Mask values higher than FIFODEPTH are not valid. The module will not trigger a match for any value in this case.
2. RXMSK2 and TXMSK2 bits are only present when FIFODEPTH = 8 or higher.
3. RXMSK3 and TXMSK3 bits are only present when FIFODEPTH = 16 or higher.
4. RXMSK4 and TXMSK4 bits are only present when FIFODEPTH = 32.

Bit	15	14	13	12	11	10	9	8
	RXWIEN		RXMSK[5:0]					
Access	R/W		R/W	R/W	R/W	R/W	R/W	R/W
Reset	0		0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	TXWIEN		TXMSK[5:0]					
Access	R/W		R/W	R/W	R/W	R/W	R/W	R/W
Reset	0		0	0	0	0	0	0

Bit 15 – RXWIEN Receive Watermark Interrupt Enable bit

Value	Description
1	Triggers receive buffer element watermark interrupt when RXMSK[5:0] ≤ RXELM[5:0]
0	Disables receive buffer element watermark interrupt

Bits 13:8 – RXMSK[5:0] RX Buffer Mask bits^(1,2,3,4)

RX mask bits; used in conjunction with the RXWIEN bit.

Bit 7 – TXWIEN Transmit Watermark Interrupt Enable bit

Value	Description
1	Triggers transmit buffer element watermark interrupt when TXMSK[5:0] = TXELM[5:0]
0	Disables transmit buffer element watermark interrupt

Bits 5:0 – TXMSK[5:0] TX Buffer Mask bits^(1,2,3,4)

TX mask bits; used in conjunction with the TXWIEN bit.

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Serial Peripheral Interface (SPI)

18.1.11 SPIx Underrun Data Register Low

Name: SPIxURDTL
Offset: 0x2C4, 0x2E0

Bit	15	14	13	12	11	10	9	8
	SPIxURDTL[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	SPIxURDTL[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – SPIxURDTL[15:0] SPI Underrun Data bits

These bits are only used when URDTEN = 1. This register holds the data to transmit when a Transmit Underrun condition occurs.

When the MODE[32:16] or WLENGTH[4:0] bits select 16 to 9-bit data, the SPIx only uses URDATA[15:0]. When the MODE[32:16] or WLENGTH[4:0] bits select 8 to 2-bit data, the SPIx only uses URDATA[7:0].

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Serial Peripheral Interface (SPI)

18.1.12 SPIx Underrun Data Register Low

Name: SPIxURDTH
Offset: 0x2C6, 0x2E2

Bit	15	14	13	12	11	10	9	8
	SPIxURDTH[31:24]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	SPIxURDTH[23:16]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – SPIxURDTH[31:16] SPI Underrun Data bits

These bits are only used when URDTEN = 1. This register holds the data to transmit when a Transmit Underrun condition occurs.

When the MODE[32:16] or WLENGTH[4:0] bits select 32 to 25-bit data, the SPIx only uses URDATA[31:16]. When the MODE[32:16] or WLENGTH[4:0] bits select 24 to 17-bit data, the SPIx only uses URDATA[23:16].

19. Inter-Integrated Circuit (I²C)

Note: This data sheet summarizes the features of the dsPIC33CK512MPT608 family of devices. It is not intended to be a comprehensive reference source. For more information, refer to “Inter-Integrated Circuit (I²C)” (www.microchip.com/DS70000195) in the “dsPIC33/PIC24 Family Reference Manual”.

The Inter-Integrated Circuit (I²C) module is a serial interface useful for communicating with other peripheral or microcontroller devices. These peripheral devices may be serial EEPROMs, display drivers, A/D Converters, etc.

The I²C module supports these features:

- 7-Bit and 10-Bit Device Addresses
- General Call Address as Defined in the I²C Protocol
- Both 100 kHz and 400 kHz Bus Specifications
- Configurable Address Masking
- Multi-Host modes to Prevent Loss of Messages in Arbitration
- Bus Repeater mode, Allowing the Acceptance of All Messages as a Client, regardless of the Address
- Automatic SCL

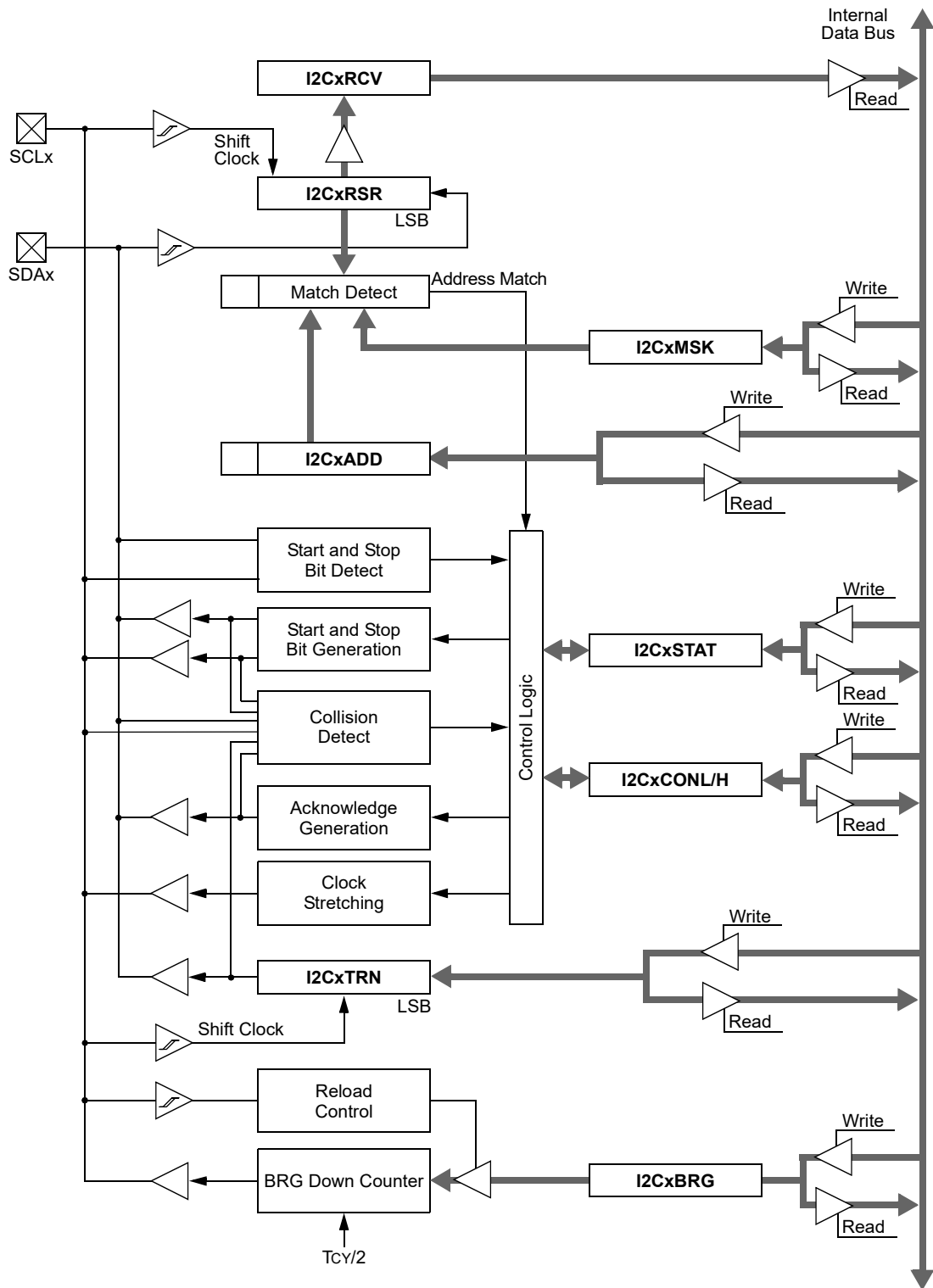
A block diagram of the module is shown in [Figure 19-1](#).

19.1 Communicating as a Host in a Single Host Environment

The details of sending a message in Host mode depends on the communication protocol for the device being communicated with. Typically, the sequence of events is as follows:

1. Assert a Start condition on SDAx and SCLx.
2. Send the I²C device address byte to the Client with a write indication.
3. Wait for and verify an Acknowledge from the Client.
4. Send the first data byte (sometimes known as the command) to the Client.
5. Wait for and verify an Acknowledge from the Client.
6. Send the serial memory address low byte to the Client.
7. Repeat Steps 4 and 5 until all data bytes are sent.
8. Assert a Repeated Start condition on SDAx and SCLx.
9. Send the device address byte to the Client with a read indication.
10. Wait for and verify an Acknowledge from the Client.
11. Enable Host reception to receive serial memory data.
12. Generate an ACK or NACK condition at the end of a received byte of data.
13. Generate a Stop condition on SDAx and SCLx.

Figure 19-1. I2Cx Block Diagram



19.2 Setting Baud Rate When Operating as a Bus Main

To compute the Baud Rate Generator reload value, use [Equation 19-1](#).

Equation 19-1. Computing Baud Rate Reload Value^(1,2,3)

$$I2CxBRG = ((1/F_{SCL} - Delay) \cdot F_P/2) - 2$$

Notes:

1. These clock rate values are for guidance only. The actual clock rate should be measured in its intended application.
2. Typical value of delay varies from 110 ns to 150 ns.
3. I2CxBRG values of 0 to 3 are expressly forbidden. The user should never program the I2CxBRG with a value of 0x0, 0x1, 0x2 or 0x3 as indeterminate results may occur.

19.3 Client Address Masking

The I2CxMSK register ([I2CxMASK](#)) designates address bit positions as “don’t care” for both 7-Bit and 10-Bit Addressing modes. Setting a particular bit location (= 1) in the I2CxMSK register causes the Client module to respond, whether the corresponding address bit value is a ‘0’ or a ‘1’. For example, when I2CxMSK is set to ‘0010000000’, the Client module will detect both addresses, ‘0000000000’ and ‘0010000000’.

To enable address masking, the Intelligent Peripheral Management Interface (IPMI) must be disabled by clearing the STRICT bit (I2CxCONL[11]).

Note: As a result of changes in the I²C protocol, the addresses in [Table 19-2](#) are reserved and will not be Acknowledged in Client mode. This includes any address mask settings that include any of these addresses.

Table 19-1. I2Cx Clock Rates^(1,2)

F _{CY}	F _{SCL}	I2CxBRG Value	
		Decimal	Hexadecimal
100 MHz	1 MHz	41	29
100 MHz	400 kHz	116	74
100 MHz	100 kHz	491	1EB
80 MHz	1 MHz	32	20
80 MHz	400 kHz	92	5C
80 MHz	100 kHz	392	188
60 MHz	1 MHz	24	18
60 MHz	400 kHz	69	45
60 MHz	100 kHz	294	126
40 MHz	1 MHz	15	0F
40 MHz	400 kHz	45	2D
40 MHz	100 kHz	195	C3
20 MHz	1 MHz	7	7

Notes:

1. Based on F_{CY} = F_{OSC}/2; Doze mode and PLL are disabled.
2. These clock rate values are for guidance only. The actual clock rate can be affected by various system-level parameters. The actual clock rate should be measured in its intended application.

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.....continued

F _{CY}	F _{SCL}	I2CxBRG Value	
		Decimal	Hexadecimal
20 MHz	400 kHz	22	16
20 MHz	100 kHz	97	61

Notes:

1. Based on $F_{CY} = F_{OSC}/2$; Doze mode and PLL are disabled.
2. These clock rate values are for guidance only. The actual clock rate can be affected by various system-level parameters. The actual clock rate should be measured in its intended application.

Table 19-2. I2Cx Reserved Addresses⁽¹⁾

Client Address	R/W Bit	Description
0000 000	0	General Call Address ⁽²⁾
0000 000	1	Start Byte
0000 001	x	Cbus Address
0000 01x	x	Reserved
0000 1xx	x	HS Mode Host Code
1111 0xx	x	10-Bit Client Upper Byte ⁽³⁾
1111 1xx	x	Reserved

Note:

1. The address bits listed here will never cause an address match independent of address mask settings.
2. This address will be Acknowledged only if GCEN = 1.
3. A match on this address can only occur on the upper byte in 10-Bit Addressing mode.

19.4 SMBus Support

The dsPIC33CK512MPT608 family devices have support for SMBus through options in the input voltage thresholds. There are two control bits to select one of three options: SMEN (I2CxCONL[8]) and Configuration bit, SMBEN (FDEVOP[10]). [I2CxCONL](#) details the setting of these control bits.

Table 19-3. I²C Pin Voltage Threshold

	SMEN SFR Bit (I2CxCONL[8])	SMBEN Configuration Bit (FDEVOP[10])
I ² C (default)	0	x
SMBus 2.0	1	0
SMBus 3.0	1	1

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19.5 I2C Control/Status Registers

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x00 ... 0x01FF	Reserved									
0x0200	I2C1CONL	15:8	I2CEN		I2CSIDL	SCLREL	STRICT	A10M	DISSLW	SMEN
		7:0	GCEN	STREN	ACKDT	ACKEN	RCEN	PEN	RSEN	SEN
0x0202	I2C1CONH	15:8								
		7:0		PCIE	SCIE	BOEN	SDAHT	SBCDE	AHEN	DHEN
0x0204	I2C1STAT	15:8	ACKSTAT	TRSTAT	ACKTIM			BCL	GCSTAT	ADD10
		7:0	IWCOL	I2COV	D/A	P	S	R/W	RBF	TBF
0x0206 ... 0x020B	Reserved									
0x020C	I2C1MSK	15:8							MSK[9:8]	
		7:0	MSK[7:0]							
0x020E ... 0x021B	Reserved									
0x021C	I2C2CONL	15:8	I2CEN		I2CSIDL	SCLREL	STRICT	A10M	DISSLW	SMEN
		7:0	GCEN	STREN	ACKDT	ACKEN	RCEN	PEN	RSEN	SEN
0x021E	I2C2CONH	15:8								
		7:0		PCIE	SCIE	BOEN	SDAHT	SBCDE	AHEN	DHEN
0x0220	I2C2STAT	15:8	ACKSTAT	TRSTAT	ACKTIM			BCL	GCSTAT	ADD10
		7:0	IWCOL	I2COV	D/A	P	S	R/W	RBF	TBF
0x0222 ... 0x0227	Reserved									
0x0228	I2C2MSK	15:8							MSK[9:8]	
		7:0	MSK[7:0]							
0x022A ... 0x0F5B	Reserved									
0x0F5C	I2C3CONL	15:8	I2CEN		I2CSIDL	SCLREL	STRICT	A10M	DISSLW	SMEN
		7:0	GCEN	STREN	ACKDT	ACKEN	RCEN	PEN	RSEN	SEN
0x0F5E	I2C3CONH	15:8								
		7:0		PCIE	SCIE	BOEN	SDAHT	SBCDE	AHEN	DHEN
0x0F60	I2C3STAT	15:8	ACKSTAT	TRSTAT	ACKTIM			BCL	GCSTAT	ADD10
		7:0	IWCOL	I2COV	D/A	P	S	R/W	RBF	TBF
0x0F62 ... 0x0F67	Reserved									
0x0F68	I2C3MSK	15:8							MSK[9:8]	
		7:0	MSK[7:0]							

19.5.1 I2Cx Control Register Low

Name: I2CxCONL
Offset: 0x200, 0x21C, 0xF5C

Notes:

1. Automatically cleared to '0' at the beginning of Client transmission; automatically cleared to '0' at the end of Client reception.
2. Automatically cleared to '0' at the beginning of Client transmission.

Legend: HC = Hardware Clearable bit

Bit	15	14	13	12	11	10	9	8
	I2CEN		I2CSIDL	SCLREL	STRICT	A10M	DISSLW	SMEN
Access	R/W		R/W/HC	R/W	R/W	R/W	R/W	R/W
Reset	0		0	1	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	GCEN	STREN	ACKDT	ACKEN	RCEN	PEN	RSEN	SEN
Access	R/W	R/W	R/W	R/W/HC	R/W/HC	R/W/HC	R/W/HC	R/W/HC
Reset	0	0	0	0	0	0	0	0

Bit 15 – I2CEN I2Cx Enable bit (writable from software only)

Value	Description
1	Enables the I2Cx module, and configures the SDAx and SCLx pins as serial port pins
0	Disables the I2Cx module; all I ² C pins are controlled by port functions

Bit 13 – I2CSIDL I2Cx Stop in Idle Mode bit

Value	Description
1	Discontinues module operation when device enters Idle mode
0	Continues module operation in Idle mode

Bit 12 – SCLREL SCLx Release Control bit (I²C Client mode only)⁽¹⁾

If STREN = 1:⁽²⁾

User software may write '0' to initiate a clock stretch and write '1' to release the clock. Hardware clears at the beginning of every Client data byte transmission. Hardware clears at the end of every Client address byte reception. Hardware clears at the end of every Client data byte reception.

If STREN = 0:

User software may only write '1' to release the clock. Hardware clears at the beginning of every Client data byte transmission. Hardware clears at the end of every Client address byte reception.

Value	Description
1	Releases the SCLx clock
0	Holds the SCLx clock low (clock stretch)

Bit 11 – STRICT I2Cx Strict Reserved Address Rule Enable bit

Value	Description
1	Strict reserved addressing is enforced; for reserved addresses. (In Client Mode) – The device doesn't respond to reserved address space and addresses falling in that category are NACKed. (In Host Mode) – The device is allowed to generate addresses with reserved address space.
0	Reserved addressing would be Acknowledged. (In Client Mode) – The device will respond to an address falling in the reserved address space. When there is a match with any of the reserved addresses, the device will generate an ACK. (In Host Mode) – Reserved.

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Bit 10 – A10M 10-Bit Client Address Flag bit

Value	Description
1	I2CxADD is a 10-bit Client address
0	I2CxADD is a 7-bit Client address

Bit 9 – DISSLW Slew Rate Control Disable bit

Value	Description
1	Slew rate control is disabled for Standard Speed mode (100 kHz, also disabled for 1 MHz mode)
0	Slew rate control is enabled for High-Speed mode (400 kHz)

Bit 8 – SMEN SMBus Input Levels Enable bit

Value	Description
1	Enables input logic so thresholds are compliant with the SMBus specification
0	Disables SMBus-specific inputs

Bit 7 – GCEN General Call Enable bit (in I²C Client mode only)

Value	Description
1	Enables interrupt when a general call address is received in I2CxRSR; module is enabled for reception
0	General call address is disabled.

Bit 6 – STREN SCLx Clock Stretch Enable bit

In I²C Client mode only; used in conjunction with the SCLREL bit.

Value	Description
1	Enables clock stretching
0	Disables clock stretching

Bit 5 – ACKDT Acknowledge Data bit

In I²C Host mode during Host Receive mode. The value that will be transmitted when the user initiates an Acknowledge sequence at the end of a receive.

In I²C Client mode when AHEN = 1 or DHEN = 1. The value that the Client will transmit when it initiates an Acknowledge sequence at the end of an address or data reception.

Value	Description
1	NACK is sent
0	ACK is sent

Bit 4 – ACKEN Acknowledge Sequence Enable bit

In I²C Host mode only; applicable during Host Receive mode.

Value	Description
1	Initiates Acknowledge sequence on SDAx and SCLx pins, and transmits ACKDT data bit
0	Acknowledge sequence is Idle

Bit 3 – RCEN Receive Enable bit (in I²C Host mode only)

Value	Description
1	Enables Receive mode for I ² C; automatically cleared by hardware at end of 8-bit receive data byte
0	Receive sequence is not in progress

Bit 2 – PEN Stop Condition Enable bit (in I²C Host mode only)

Value	Description
1	Initiates Stop condition on SDAx and SCLx pins
0	Stop condition is Idle

Bit 1 – RSEN Restart Condition Enable bit (in I²C Host mode only)

Value	Description
1	Initiates Restart condition on SDAx and SCLx pins
0	Restart condition is Idle

Bit 0 – SEN Start Condition Enable bit (in I²C Host mode only)

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Value	Description
1	Initiates Start condition on SDAx and SCLx pins
0	Start condition is Idle

19.5.2 I2Cx Control Register High

Name: I2CxCONH
Offset: 0x202, 0x21E, 0xF5E

Bit	15	14	13	12	11	10	9	8
Access								
Reset								

Bit	7	6	5	4	3	2	1	0
Access		PCIE	SCIE	BOEN	SDAHT	SBCDE	AHEN	DHEN
Reset		R/W	R/W	R/W	R/W	R/W	R/W	R/W
		0	0	0	0	0	0	0

Bit 6 – PCIE Stop Condition Interrupt Enable bit

Value	Description
1	Enables interrupt on detection of Stop condition
0	Stop detection interrupts are disabled

Bit 5 – SCIE Start Condition Interrupt Enable bit

Value	Description
1	Enables interrupt on detection of Start or Restart conditions
0	Start detection interrupts are disabled

Bit 4 – BOEN Buffer Overwrite Enable bit

Value	Description
1	I2CxRCV is updated and an ACK is generated for a received address/data byte, ignoring the state of the I2COV bit only if RBF bit = 0
0	I2CxRCV is only updated when I2COV is clear

Bit 3 – SDAHT SDAx Hold Time Selection bit

Value	Description
1	Minimum of 300 ns hold time on SDAx after the falling edge of SCLx
0	Minimum of 100 ns hold time on SDAx after the falling edge of SCLx

Bit 2 – SBCDE Client Mode Bus Collision Detect Enable bit

If, on the rising edge of SCLx, SDAx is sampled low when the module is outputting a High state, the BCL bit is set and the bus goes Idle. This Detection mode is only valid during data and ACK transmit sequences.

Value	Description
1	Enables Client bus collision interrupts
0	Client bus collision interrupts are disabled

Bit 1 – AHEN Address Hold Enable bit

Value	Description
1	Following the 8th falling edge of SCLx for a matching received address byte; SCLREL bit (I2CxCONL[12]) will be cleared and the SCLx will be held low
0	Address holding is disabled

Bit 0 – DHEN Data Hold Enable bit

Value	Description
1	Following the 8th falling edge of SCLx for a received data byte; Client hardware clears the SCLREL bit (I2CxCONL[12]) and SCLx is held low
0	Data holding is disabled

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Inter-Integrated Circuit (I2C)

19.5.3 I2Cx Status Register

Name: I2CxSTAT
Offset: 0x204, 0x220, 0xF60

Legend: C = Clearable bit; HS = Hardware Settable bit; HSC = Hardware Settable/Clearable bit

Bit	15	14	13	12	11	10	9	8
	ACKSTAT	TRSTAT	ACKTIM			BCL	GCSTAT	ADD10
Access	R/HSC	R/HSC	R/HSC			R/C/HSC	R/HSC	R/HSC
Reset	0	0	0			0	0	0

Bit	7	6	5	4	3	2	1	0
	IWCOL	I2COV	D/A	P	S	R/W	RBF	TBF
Access	R/C/HS	R/C/HS	R/HSC	R/HSC	R/HSC	R/HSC	R/HSC	R/HSC
Reset	0	0	0	0	0	0	0	0

Bit 15 – ACKSTAT Acknowledge Status bit (updated in all Host and Client modes)

Value	Description
1	Acknowledge was not received from Client
0	Acknowledge was received from Client

Bit 14 – TRSTAT Transmit Status bit (when operating as I²C Host; applicable to Host transmit operation)

Value	Description
1	Host transmit is in progress (eight bits + ACK)
0	Host transmit is not in progress

Bit 13 – ACKTIM Acknowledge Time Status bit (valid in I²C Client mode only)

Value	Description
1	Indicates I ² C bus is in an Acknowledge sequence, set on 8th falling edge of SCLx clock
0	Not an Acknowledge sequence, cleared on 9th rising edge of SCLx clock

Bit 10 – BCL Bus Collision Detect bit
 (Host/Client mode; cleared when I²C module is disabled, I2CEN = 0)

Value	Description
1	A bus collision has been detected during a Host or Client transmit operation
0	No bus collision has been detected

Bit 9 – GCSTAT General Call Status bit (cleared after Stop detection)

Value	Description
1	General call address was received
0	General call address was not received

Bit 8 – ADD10 10-Bit Address Status bit (cleared after Stop detection)

Value	Description
1	10-bit address was matched
0	10-bit address was not matched

Bit 7 – IWCOL I2Cx Write Collision Detect bit

Value	Description
1	An attempt to write to the I2CxTRN register failed because the I ² C module is busy; must be cleared in software
0	No collision

Bit 6 – I2COV I2Cx Receive Overflow Flag bit

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Value	Description
1	A byte was received while the I2CxRCV register is still holding the previous byte; I2COV is a “don’t care” in Transmit mode, must be cleared in software
0	No overflow

Bit 5 – D/A Data/Address bit (when operating as I²C Client)

Value	Description
1	Indicates that the last byte received was data
0	Indicates that the last byte received or transmitted was an address

Bit 4 – P I2Cx Stop bit

Updated when Start, Reset or Stop is detected; cleared when the I²C module is disabled, I2CEN = 0.

Value	Description
1	Indicates that a Stop bit has been detected last
0	Stop bit was not detected last

Bit 3 – S I2Cx Start bit

Updated when Start, Reset or Stop is detected; cleared when the I²C module is disabled, I2CEN = 0.

Value	Description
1	Indicates that a Start (or Repeated Start) bit has been detected last
0	Start bit was not detected last

Bit 2 – R/W Read/Write Information bit (when operating as I²C Client)

Value	Description
1	Read: Indicates the data transfer is output from the Client
0	Write: Indicates the data transfer is input to the Client

Bit 1 – RBF Receive Buffer Full Status bit

Value	Description
1	Receive is complete, I2CxRCV is full
0	Receive is not complete, I2CxRCV is empty

Bit 0 – TBF Transmit Buffer Full Status bit

Value	Description
1	Transmit is in progress, I2CxTRN is full (eight bits of data)
0	Transmit is complete, I2CxTRN is empty

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Inter-Integrated Circuit (I2C)

19.5.4 I2Cx Client Mode Address Mask Register

Name: I2CxMSK
Offset: 0x20C, 0x228, 0xF68

Bit	15	14	13	12	11	10	9	8
							MSK[9:8]	
Access							R/W	R/W
Reset							0	0

Bit	7	6	5	4	3	2	1	0
	MSK[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 9:0 – MSK[9:0] I2Cx Mask for Address Bit x Select bits

Value	Description
1	Enables masking for bit x of the incoming message address; bit match is not required in this position
0	Disables masking for bit x; bit match is required in this position

20. Single-Edge Nibble Transmission (SENT)

Note: This data sheet summarizes the features of this group of dsPIC33CK512MPT608 family devices. It is not intended to be a comprehensive reference source. To complement the information in this data sheet, refer to “Single-Edge Nibble Transmission (SENT) Module” (www.microchip.com/DS70005145) in the “dsPIC33/PIC24 Family Reference Manual”.

The Single-Edge Nibble Transmission (SENT) module is based on the SAE J2716, “SENT – Single-Edge Nibble Transmission for Automotive Applications”. The SENT protocol is a one-way, single wire, time-modulated serial communication based on successive falling edges. It is intended for use in applications where high-resolution sensor data need to be communicated from a sensor to an Engine Control Unit (ECU).

The SENTx module has the following major features:

- Selectable Transmit or Receive mode
- Synchronous or Asynchronous Transmit modes
- Automatic Data Rate Synchronization
- Optional Automatic Detection of CRC Errors in Receive mode
- Optional Hardware Calculation of CRC in Transmit mode
- Support for Optional Pause Pulse Period
- Data Buffering for One Message Frame
- Selectable Data Length for Transmit/Receive, Up to Six Nibbles
- Automatic Detection of Framing Errors

SENT protocol timing is based on a predetermined time unit, T_{TICK} . Both the transmitter and receiver must be preconfigured for T_{TICK} , which can vary from 3 to 90 μs . A SENT message frame starts with a Sync pulse. The purpose of the Sync pulse is to allow the receiver to calculate the data rate of the message encoded by the transmitter. The SENT specification allows messages to be validated with up to a 20% variation in T_{TICK} . This allows for the transmitter and receiver to run from different clocks that may be inaccurate, and drift with time and temperature. The data nibbles are four bits in length and are encoded as the data value + 12 ticks. This yields a 0 value of 12 ticks and the maximum value, 0xF, of 27 ticks.

A SENT message consists of the following:

- A synchronization/calibration period of 56 tick times
- A status nibble of 12-27 tick times
- Up to six data nibbles of 12-27 tick times
- A CRC nibble of 12-27 tick times
- An optional pause pulse period of 12-768 tick times

Figure 20-1 shows a block diagram of the SENTx module.

Figure 20-2 shows the construction of a typical 6-nibble data frame, with the numbers representing the minimum or maximum number of tick times for each section.

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Figure 20-1. SENTx Module Block Diagram

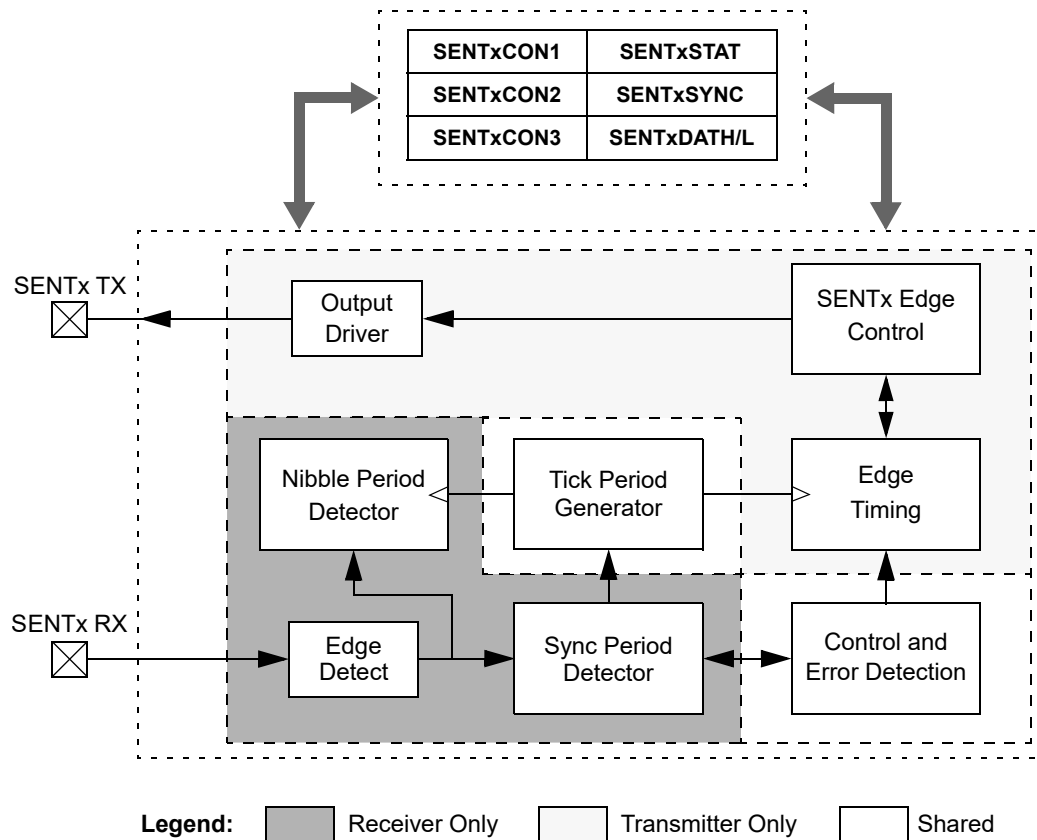
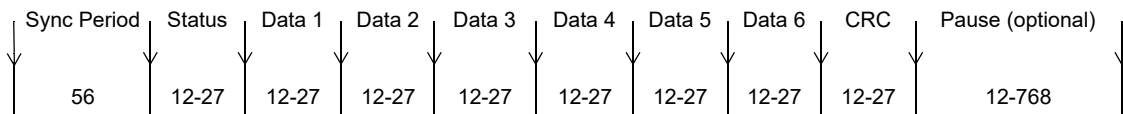


Figure 20-2. SENTx Protocol Data Frames



20.1 Transmit Mode

By default, the SENTx module is configured for transmit operation. The module can be configured for continuous asynchronous message frame transmission, or alternatively, for Synchronous mode triggered by software. When enabled, the transmitter will send a Sync, followed by the appropriate number of data nibbles, an optional CRC and optional pause pulse. The tick period used by the SENTx transmitter is set by writing a value to the TICKTIME[15:0] (SENTxCON2[15:0]) bits. The tick period calculations are shown in [Equation 20-1](#).

Equation 20-1. Tick Period Calculation

$$TICKTIME[15:0] = \frac{T_{TICK}}{T_{CLK}} - 1$$

An optional pause pulse can be used in Asynchronous mode to provide a fixed message frame time period. The frame period used by the SENTx transmitter is set by writing a value to the FRAMETIME[15:0] (SENTxCON3[15:0]) bits. The formulas used to calculate the value of frame time are shown in [Equation 20-2](#).

Equation 20-2. Frame Time Calculations

$$FRAMETIME[15:0] = TTICK/TFRAME$$

$$FRAMETIME[15:0] \geq 122 + 27N$$

$$FRAMETIME[15:0] \geq 848 + 12N$$

Where:

$TFRAME$ = Total time of the message from ms

N = The number of data nibbles in message, 1-6

Note: The module will not produce a pause period with less than 12 ticks, regardless of the $FRAMETIME[15:0]$ value. $FRAMETIME[15:0]$ values beyond 2047 will have no effect on the length of a data frame.

20.1.1 Transmit Mode Configuration

20.1.1.1 Initializing the SENTx Module

Perform the following steps to initialize the module:

1. Write RCVEN (SENTxCON1[11]) = 0 for Transmit mode.
2. Write TXM (SENTxCON1[10]) = 0 for Asynchronous Transmit mode or TXM = 1 for Synchronous mode.
3. Write NIBCNT[2:0] (SENTxCON1[2:0]) for the desired data frame length.
4. Write CRCEN (SENTxCON1[8]) for hardware or software CRC calculation.
5. Write PPP (SENTxCON1[7]) for optional pause pulse.
6. If PPP = 1, write TFRAME to SENTxCON3.
7. Write SENTxCON2 with the appropriate value for the desired tick period.
8. Enable interrupts and set interrupt priority.
9. Write initial status and data values to SENTxDATH/L.
10. If CRCEN = 0, calculate CRC and write the value to CRC[3:0] (SENTxDATL[3:0]).
11. Set the SNTEN (SENTxCON1[15]) bit to enable the module.

User software updates to SENTxDATH/L must be performed after the completion of the CRC and before the next message frame's status nibble. The recommended method is to use the message frame completion interrupt to trigger data writes. The data should be read from the SENTxDATL/H registers after the completion of the CRC and before the next message frame's status nibble. The recommended method is to use the message frame completion interrupt trigger.

20.2 Receive Mode

The module can be configured for receive operation by setting the RCVEN (SENTxCON1[11]) bit. The time between each falling edge is compared to SYNCMIN[15:0] (SENTxCON3[15:0]) and SYNCMAX[15:0] (SENTxCON2[15:0]), and if the measured time lies between the minimum and maximum limits, the module begins to receive data. The validated Sync time is captured in the SENTxSYNC register and the tick time is calculated. Subsequent falling edges are verified to be within the valid data width and the data are stored in the SENTxDATL/H registers. An interrupt event is generated at the completion of the message and the user software should read the SENTx Data registers before the reception of the next nibble. The equation for SYNCMIN[15:0] and SYNCMAX[15:0] is shown in [Equation 20-3](#).

Equation 20-3. SYNCMIN[15:0] and SYNCMAX[15:0] Calculations

$$T_{TICK} = T_{CLK} \cdot (TICKTIME[15:0] + 1)$$

$$FRAME_{TIME}[15:0] = T_{TICK} / T_{FRAME}$$

$$SyncCount = 8 \times FRCV \times T_{TICK}$$

$$SYNC_{MIN}[15:0] = 0.8 \times SyncCount$$

$$SYNC_{MAX}[15:0] = 1.2 \times SyncCount$$

$$FRAME_{TIME}[15:0] \geq 122 + 27N$$

$$FRAME_{TIME}[15:0] \geq 848 + 12N$$

Where:

T_{FRAME} = Total time of the message from ms

N = The number of data nibbles in message, 1-6

$FRCV$ = $FCY \times \text{Prescaler}$

T_{CLK} = $FCY / \text{Prescaler}$

For $T_{TICK} = 3.0 \mu s$ and $F_{CLK} = 4 \text{ MHz}$, $SYNC_{MIN}[15:0] = 76$.

Note: To ensure a Sync period can be identified, the value written to $SYNC_{MIN}[15:0]$ must be less than the value written to $SYNC_{MAX}[15:0]$.

20.2.1 Initializing the SENTx Module

Perform the following steps to initialize the module:

1. Write RCVEN (SENTxCON1[11]) = 1 for Receive mode.
2. Write NIBCNT[2:0] (SENTxCON1[2:0]) for the desired data frame length.
3. Write CRCEN (SENTxCON1[8]) for hardware or software CRC validation.
4. Write PPP (SENTxCON1[7]) = 1 if pause pulse is present.
5. Write SENTxCON2 with the value of SYNCMAXx (Nominal Sync Period + 20%).
6. Write SENTxCON3 with the value of SYNCMINx (Nominal Sync Period – 20%).
7. Enable interrupts and set interrupt priority.
8. Set the SNTEN (SENTxCON1[15]) bit to enable the module.

The data should be read from the SENTxDATL/H registers after the completion of the CRC and before the next message frame's status nibble. The recommended method is to use the message frame completion interrupt trigger.

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20.3 SENT Control/Status Registers

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x00 ... 0x7F	Reserved									
0x80	SENT1CON1	15:8	SNTEN		SNTSIDL		RCVEN	TXM	TXPOL	CRCEN
		7:0	PPP	SPCEN		PS		NIBCNT[2:0]		
0x82 ... 0x83	Reserved									
0x84	SENT2CON1	15:8	SNTEN		SNTSIDL		RCVEN	TXM	TXPOL	CRCEN
		7:0	PPP	SPCEN		PS		NIBCNT[2:0]		
0x86 ... 0x87	Reserved									
0x88	SENT3CON1	15:8	SNTEN		SNTSIDL		RCVEN	TXM	TXPOL	CRCEN
		7:0	PPP	SPCEN		PS		NIBCNT[2:0]		
0x8A ... 0x8B	Reserved									
0x8C	SENT1STAT	15:8								
		7:0	PAUSE	NIB[2:0]			CRCERR	FRMERR	RXIDLE	SYNCTXEN
0x8E ... 0x93	Reserved									
0x94	SENT1DATAL(1)	15:8	DATA4[3:0]				DATA5[3:0]			
		7:0	DATA6[3:0]				CRC[3:0]			
0x96	SENT1DATAH(1)	15:8	STAT[3:0]				DATA1[3:0]			
		7:0	DATA2[3:0]				DATA3[3:0]			
0x98 ... 0xA3	Reserved									
0xA4	SENT2STAT	15:8								
		7:0	PAUSE	NIB[2:0]			CRCERR	FRMERR	RXIDLE	SYNCTXEN
0xA6 ... 0xAB	Reserved									
0xAC	SENT2DATAL(1)	15:8	DATA4[3:0]				DATA5[3:0]			
		7:0	DATA6[3:0]				CRC[3:0]			
0xAE	SENT2DATAH(1)	15:8	STAT[3:0]				DATA1[3:0]			
		7:0	DATA2[3:0]				DATA3[3:0]			

dsPIC33CK512MPT608 Family

Single-Edge Nibble Transmission (SENT)

20.3.1 SENTx Control Register 1

Name: SENTxCON1
Offset: 0x080, 0x084, 0x088

Notes:

1. This bit has no function in Receive mode (RCVEN = 1).
2. This bit has no function in Transmit mode (RCVEN = 0).

Bit	15	14	13	12	11	10	9	8
	SNTEN		SNTSIDL		RCVEN	TXM	TXPOL	CRCEN
Access	R/W		R/W		R/W	R/W	R/W	R/W
Reset	0		0		0	0	0	0

Bit	7	6	5	4	3	2	1	0
	PPP	SPCEN		PS		NIBCNT[2:0]		
Access	R/W	R/W		R/W		R/W	R/W	R/W
Reset	0	0		0		0	0	0

Bit 15 – SNTEN SENTx Enable bit

Value	Description
1	SENTx is enabled
0	SENTx is disabled

Bit 13 – SNTSIDL SENTx Stop in Idle Mode bit

Value	Description
1	Discontinues module operation when the device enters Idle mode
0	Continues module operation in Idle mode

Bit 11 – RCVEN SENTx Receive Enable bit

Value	Description
1	SENTx operates as a receiver
0	SENTx operates as a transmitter (sensor)

Bit 10 – TXM SENTx Transmit Mode bit⁽¹⁾

Value	Description
1	SENTx transmits data frame only when triggered using the SYNCTXEN status bit
0	SENTx transmits data frames continuously while SNTEN = 1

Bit 9 – TXPOL SENTx Transmit Polarity bit⁽¹⁾

Value	Description
1	SENTx data output pin is low in the Idle state
0	SENTx data output pin is high in the Idle state

Bit 8 – CRCEN CRC Enable bit

Module in Receive Mode (RCVEN = 1):

- 1 = SENTx performs CRC verification on received data using the preferred J2716 method
- 0 = SENTx does not perform CRC verification on received data

Module in Transmit Mode (RCVEN = 0):

- 1 = SENTx automatically calculates CRC using the preferred J2716 method
- 0 = SENTx does not calculate CRC

Bit 7 – PPP Pause Pulse Present bit

Value	Description
1	SENTx is configured to transmit/receive SENT messages with pause pulse
0	SENTx is configured to transmit/receive SENT messages without pause pulse

dsPIC33CK512MPT608 Family

Single-Edge Nibble Transmission (SENT)

Bit 6 – SPCEN Short PWM Code Enable bit⁽²⁾

Value	Description
1	SPC control from external source is enabled
0	SPC control from external source is disabled

Bit 4 – PS SENTx Module Clock Prescaler (divider) bit

Value	Description
1	Divide-by-4
0	Divide-by-1

Bits 2:0 – NIBCNT[2:0] Nibble Count Control bits

Value	Description
111	Reserved; do not use
110	Module transmits/receives six data nibbles in a SENT data pocket
101	Module transmits/receives five data nibbles in a SENT data pocket
100	Module transmits/receives four data nibbles in a SENT data pocket
011	Module transmits/receives three data nibbles in a SENT data pocket
010	Module transmits/receives two data nibbles in a SENT data pocket
001	Module transmits/receives one data nibble in a SENT data pocket
000	Reserved; do not use

dsPIC33CK512MPT608 Family

Single-Edge Nibble Transmission (SENT)

20.3.2 SENTx Status Register

Name: SENTxSTAT
Offset: 0x08C, 0x0A4

Note:

1. In Receive mode (RCVEN = 1), the SYNCTXEN bit is read-only.

Legend: C = Clearable bit, HC = Hardware Clearable bit

Bit	15	14	13	12	11	10	9	8
Access								
Reset								

Bit	7	6	5	4	3	2	1	0
Access	R	R	R	R	R	R/C	R	R/W/HC
Reset	0	0	0	0	0	0	0	0

Bit 7 – PAUSE Pause Period Status bit

Value	Description
1	The module is transmitting/receiving a pause period
0	The module is not transmitting/receiving a pause period

Bits 6:4 – NIB[2:0] Nibble Status bits

Module in Transmit Mode (RCVEN = 0):

- 111 = Module is transmitting a CRC nibble
- 110 = Module is transmitting Data Nibble 6
- 101 = Module is transmitting Data Nibble 5
- 100 = Module is transmitting Data Nibble 4
- 011 = Module is transmitting Data Nibble 3
- 010 = Module is transmitting Data Nibble 2
- 001 = Module is transmitting Data Nibble 1
- 000 = Module is transmitting a status nibble or pause period, or is not transmitting

Module in Receive Mode (RCVEN = 1):

- 111 = Module is receiving a CRC nibble or was receiving this nibble when an error occurred
- 110 = Module is receiving Data Nibble 6 or was receiving this nibble when an error occurred
- 101 = Module is receiving Data Nibble 5 or was receiving this nibble when an error occurred
- 100 = Module is receiving Data Nibble 4 or was receiving this nibble when an error occurred
- 011 = Module is receiving Data Nibble 3 or was receiving this nibble when an error occurred
- 010 = Module is receiving Data Nibble 2 or was receiving this nibble when an error occurred
- 001 = Module is receiving Data Nibble 1 or was receiving this nibble when an error occurred
- 000 = Module is receiving a status nibble or waiting for Sync

Bit 3 – CRCERR CRC Status bit (Receive mode only)

Value	Description
1	A CRC error has occurred for the 1-6 data nibbles in SENTxDATL/H
0	A CRC error has not occurred

Bit 2 – FRMERR Framing Error Status bit (Receive mode only)

Value	Description
1	A data nibble was received with less than 12 tick periods or greater than 27 tick periods
0	Framing error has not occurred

Bit 1 – RXIDLE SENTx Receiver Idle Status bit (Receive mode only)

dsPIC33CK512MPT608 Family

Single-Edge Nibble Transmission (SENT)

Value	Description
1	The SENTx data bus has been Idle (high) for a period of SYNCMAX[15:0] or greater
0	The SENTx data bus is not Idle

Bit 0 – SYNCTXEN SENTx Synchronization Period Status/Transmit Enable bit⁽¹⁾

Module in Receive Mode (RCVEN = 1):

1 = A valid synchronization period was detected; the module is receiving nibble data

0 = No synchronization period has been detected; the module is not receiving nibble data

Module in Asynchronous Transmit Mode (RCVEN = 0, TXM = 0):

The bit always reads as '1' when the module is enabled, indicating the module transmits SENTx data frames continuously. The bit reads '0' when the module is disabled.

Module in Synchronous Transmit Mode (RCVEN = 0, TXM = 1):

1 = The module is transmitting a SENTx data frame

0 = The module is not transmitting a data frame, user software may set SYNCTXEN to start another data frame transmission

dsPIC33CK512MPT608 Family

Single-Edge Nibble Transmission (SENT)

20.3.3 SENTx Receive Data Register Low

Name: SENTxDATAL⁽¹⁾

Offset: 0x094, 0x0AC

Note:

1. Register bits are read-only in Receive mode (RCVEN = 1). In Transmit mode, the CRC[3:0] bits are read-only when automatic CRC calculation is enabled (RCVEN = 0, CRCEN = 1).

Bit	15	14	13	12	11	10	9	8
	DATA4[3:0]				DATA5[3:0]			
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	DATA6[3:0]				CRC[3:0]			
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:12 – DATA4[3:0] Data Nibble 4 Data bits

Bits 11:8 – DATA5[3:0] Data Nibble 5 Data bits

Bits 7:4 – DATA6[3:0] Data Nibble 6 Data bits

Bits 3:0 – CRC[3:0] CRC Nibble Data bits

dsPIC33CK512MPT608 Family

Single-Edge Nibble Transmission (SENT)

20.3.4 SENTx Receive Data Register High

Name: SENTxDATAH⁽¹⁾

Offset: 0x096, 0x0AE

Note:

1. Register bits are read-only in Receive mode (RCVEN = 1). In Transmit mode, the CRC[3:0] bits are read-only when automatic CRC calculation is enabled (RCVEN = 0, CRCEN = 1).

Bit	15	14	13	12	11	10	9	8
	STAT[3:0]				DATA1[3:0]			
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	DATA2[3:0]				DATA3[3:0]			
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:12 – STAT[3:0] Status Nibble Data bits

Bits 11:8 – DATA1[3:0] Data Nibble 1 Data bits

Bits 7:4 – DATA2[3:0] Data Nibble 2 Data bits

Bits 3:0 – DATA3[3:0] Data Nibble 3 Data bits

21. Timer1

Notes:

1. This data sheet summarizes the features of the dsPIC33CK512MPT608 family of devices. It is not intended to be a comprehensive reference source. To complement the information in this data sheet, refer to “**Timers**” (www.microchip.com/DS70362) in the “*dsPIC33/PIC24 Family Reference Manual*”, which is available from the Microchip website (www.microchip.com/).
2. Some registers and associated bits described in this section may not be available on all devices. Refer to [4. Memory Organization](#) in this data sheet for device-specific register and bit information.

The Timer1 module is a 16-bit timer that can operate as a free-running interval timer/counter.

The Timer1 module has the following unique features over other timers:

- Can be Operated in Asynchronous Counter mode
- Asynchronous Timer
- Operational during CPU Sleep mode
- Software Selectable Prescalers 1:1, 1:8, 1:64 and 1:256
- External Clock Selection Control
- The Timer1 External Clock Input (T1CK) can Optionally be Synchronized to the Internal Device Clock and the Clock Synchronization is Performed after the Prescaler

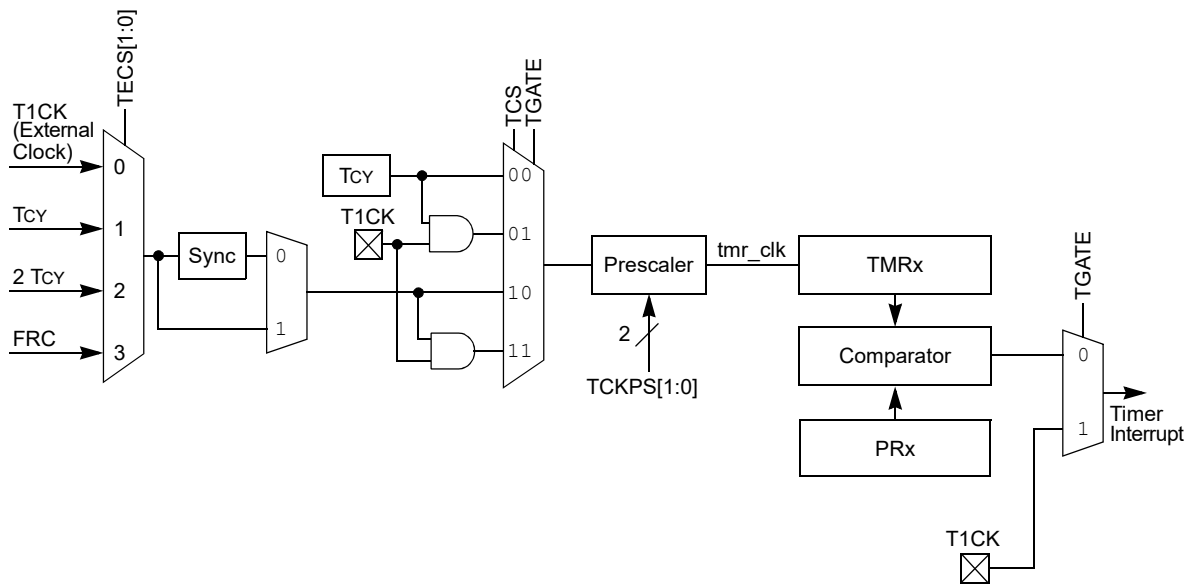
If Timer1 is used for SCCP, the timer should be running in Synchronous mode.

The Timer1 module can operate in one of the following modes:

- Timer mode
- Gated Timer mode
- Synchronous Counter mode
- Asynchronous Counter mode

A block diagram of Timer1 is shown in [Figure 21-1](#).

Figure 21-1. 16-Bit Timer1 Module Block Diagram



21.1 Timer1 Control Register

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x00 ... 0xFF	Reserved									
0x0100	T1CON	15:8	TON		SIDL	TMWDIS	TMWIP	PRWIP	TECS[1:0]	
		7:0	TGATE		TCKPS[1:0]			TSYNC	TCS	
0x0102 ... 0x0103	Reserved									
0x0104	TMR1	15:8	TMR[15:8]							
		7:0	TMR[7:0]							
0x0106 ... 0x0107	Reserved									
0x0108	PR1	15:8	PR[15:8]							
		7:0	PR[7:0]							

21.1.1 Timer1 Control Register

Name: T1CON

Offset: 0x100

Note:

- When Timer1 is enabled in External Synchronous Counter mode ($TCS = 1$, $TSYNC = 1$, $TON = 1$), any attempts by user software to write to the TMR1 register are ignored.

Bit	15	14	13	12	11	10	9	8
	TON		SIDL	TMWDIS	TMWIP	PRWIP	TECS[1:0]	
Access	R/W		R/W	R/W	R	R	R/W	R/W
Reset	0		0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	TGATE		TCKPS[1:0]			TSYNC	TCS	
Access	R/W		R/W	R/W		R/W	R/W	
Reset	0		0	0		0	0	

Bit 15 – TON Timer1 On bit⁽¹⁾

Value	Description
1	Starts 16-bit Timer1
0	Stops 16-bit Timer1

Bit 13 – SIDL Timer1 Stop in Idle Mode bit

Value	Description
1	Discontinues module operation when device enters Idle mode
0	Continues module operation in Idle mode

Bit 12 – TMWDIS Asynchronous Timer1 Write Disable bit

Value	Description
1	Timer writes are ignored while a posted write to TMR1 or PR1 is synchronized to the asynchronous clock domain
0	Back-to-back writes are enabled in Asynchronous mode

Bit 11 – TMWIP Asynchronous Timer1 Write in Progress bit

Value	Description
1	Write to the timer in Asynchronous mode is pending
0	Write to the timer in Asynchronous mode is complete

Bit 10 – PRWIP Asynchronous Period Write in Progress bit

Value	Description
1	Write to the Period register in Asynchronous mode is pending
0	Write to the Period register in Asynchronous mode is complete

Bits 9:8 – TECS[1:0] Timer1 Extended Clock Select bits

Value	Description
11	FRC clock
10	$2 T_{CY}$
01	T_{CY}
00	External Clock comes from the T1CK pin

Bit 7 – TGATE Timer1 Gated Time Accumulation Enable bit

When $TCS = 1$:

This bit is ignored.

When $TCS = 0$:

Value	Description
1	Gated time accumulation is enabled
0	Gated time accumulation is disabled

Bits 5:4 – TCKPS[1:0] Timer1 Input Clock Prescale Select bits

Value	Description
11	1:256
10	1:64
01	1:8
00	1:1

Bit 2 – TSYNC Timer1 External Clock Input Synchronization Select bit⁽¹⁾

When TCS = 0:

This bit is ignored.

When TCS = 1:

Value	Description
1	Synchronizes the External Clock input
0	Does not synchronize the External Clock input

Bit 1 – TCS Timer1 Clock Source Select bit⁽¹⁾

Value	Description
1	External Clock source selected by TECS[1:0]
0	Internal peripheral clock (F _P)

21.1.2 Timer1 Counter Register**Name:** TMR1**Offset:** 0x104

Bit	15	14	13	12	11	10	9	8
	TMR[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	TMR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – TMR[15:0] Timer1 Value bits

21.1.3 Period Register 1

Name: PR1
Offset: 0x108

Bit	15	14	13	12	11	10	9	8
	PR[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PR[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – PR[15:0] Period Register bits

22. Capture/Compare/PWM/Timer Modules (SCCP)

Note: This data sheet summarizes the features of the dsPIC33CK512MPT608 family of devices. It is not intended to be a comprehensive reference source. To complement the information in this data sheet, refer to “**Capture/Compare/PWM/Timer (MCCP and SCCP)**” (www.microchip.com/DS30003035) in the “dsPIC33/PIC24 Family Reference Manual”.

dsPIC33CK512MPT608 family devices include several Capture/Compare/PWM/Timer base modules, which provide the functionality of three different peripherals from earlier PIC24F devices. The module can operate in one of three major modes:

- General Purpose Timer
- Input Capture
- Output Compare/PWM

Single CCP output modules (SCCPs) provide only one PWM output.

The SCCP module can be operated only in one of the three major modes at any time. The other modes are not available unless the module is reconfigured for the new mode.

A conceptual block diagram for the module is shown in [Figure 22-1](#). All three modes share a time base generator and a common Timer register pair (CCPxTMRH/L); other shared hardware components are added as a particular mode requires.

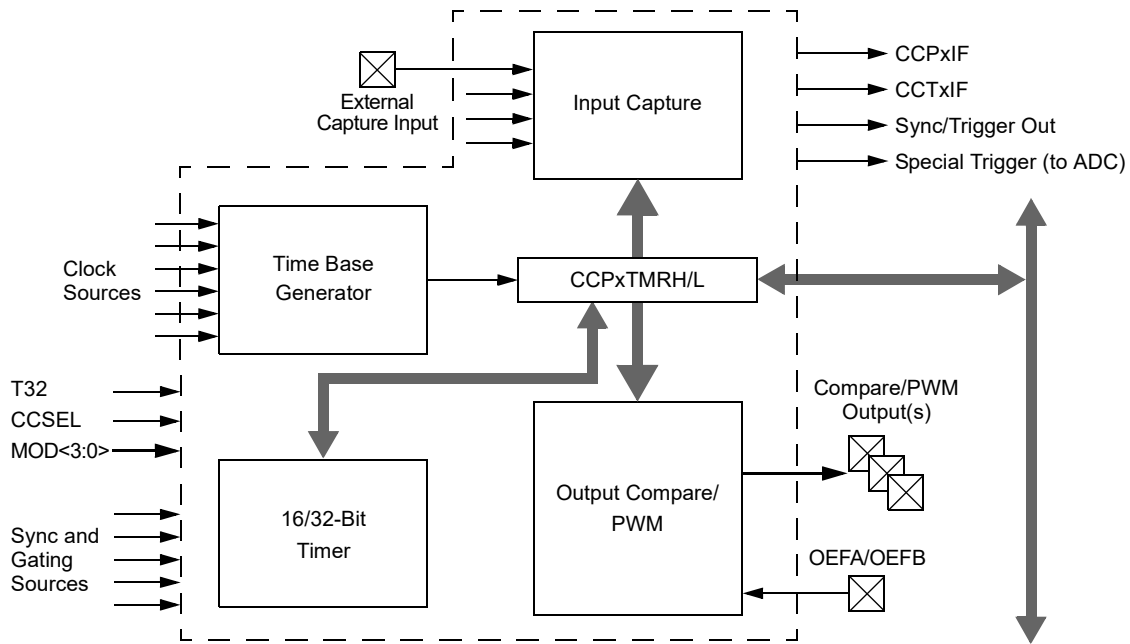
Each module has a total of six control and status registers:

- CCPxCON1L
- CCPxCON1H
- CCPxCON2L
- CCPxCON2H
- CCPxCON3H
- CCPxSTATL

Each module also includes eight buffer/counter registers that serve as Timer Value registers or data holding buffers:

- CCPxTMRH/CCPxTMRL (CCPx Timer High/Low Counters)
- CCPxPRH/CCPxPRL (CCPx Timer Period High/Low)
- CCPxRA (CCPx Primary Output Compare Data Buffer)
- CCPxRB (CCPx Secondary Output Compare Data Buffer)
- CCPxBUFH/CCPxBUFL (CCPx Input Capture High/Low Buffers)

Figure 22-1. SCCPx Conceptual Block Diagram

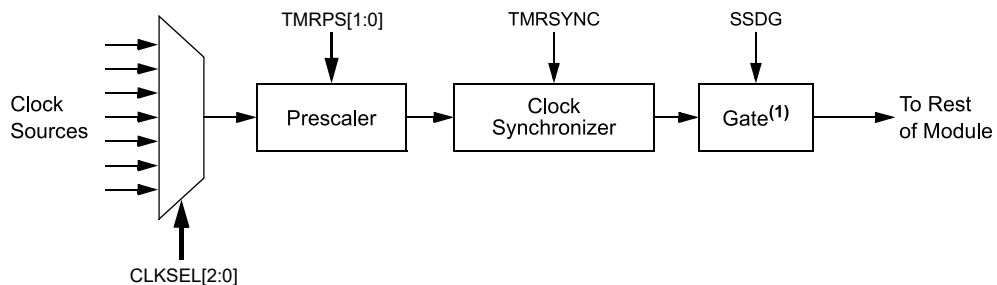


22.1 Time Base Generator

The Timer Clock Generator (TCG) generates a clock for the module's internal time base, using one of the clock signals already available on the microcontroller. This is used as the time reference for the module in its three major modes. The internal time base is shown in Figure 22-2.

There are eight inputs available to the clock generator, which are selected using the CLKSEL[2:0] bits (CCPxCON1L[10:8]). Available sources include the FRC, the Secondary Oscillator and the TCLKI External Clock inputs. The system clock is the default source (CLKSEL[2:0] = 000).

Figure 22-2. Timer Clock Generator



Note 1: Gating is available in Timer modes only.

22.2 General Purpose Timer

Timer mode is selected when CCSEL = 0 and MOD[3:0] = 0000. The timer can function as a 32-bit timer or a dual 16-bit timer, depending on the setting of the T32 bit (Table 22-1).

dsPIC33CK512MPT608 Family

Capture/Compare/PWM/Timer Modules (SCCP)

Table 22-1. Timer Operation Mode

T32 (CCPxCON1L[5])	Operating Mode
0	Dual Timer Mode (16-bit)
1	Timer Mode (32-bit)

Dual 16-Bit Timer mode provides a simple timer function with two independent 16-bit timer/counters. The primary timer uses CCPxTMRL and CCPxPRL. Only the primary timer can interact with other modules on the device. It generates the SCCPx Sync out signals for use by other SCCP modules. It can also use the SYNC[4:0] bits' signal generated by other modules.

The secondary timer uses CCPxTMRH and CCPxPRH. It is intended to be used only as a periodic interrupt source for scheduling CPU events. It does not generate an output Sync/trigger signal like the primary time base. In Dual Timer mode, the CCPx Secondary Timer Period register, CCPxPRH, generates the SCCP compare event (CCPxIF) used by many other modules on the device.

The 32-Bit Timer mode uses the CCPxTMRL and CCPxTMRH registers, together, as a single 32-bit timer. When CCPxTMRL overflows, CCPxTMRH increments by one. This mode provides a simple timer function when it is important to track long time periods. Note that the T32 bit (CCPxCON1L[5]) should be set before the CCPxTMRL or CCPxPRH registers are written to initialize the 32-bit timer.

22.3 Output Compare Mode

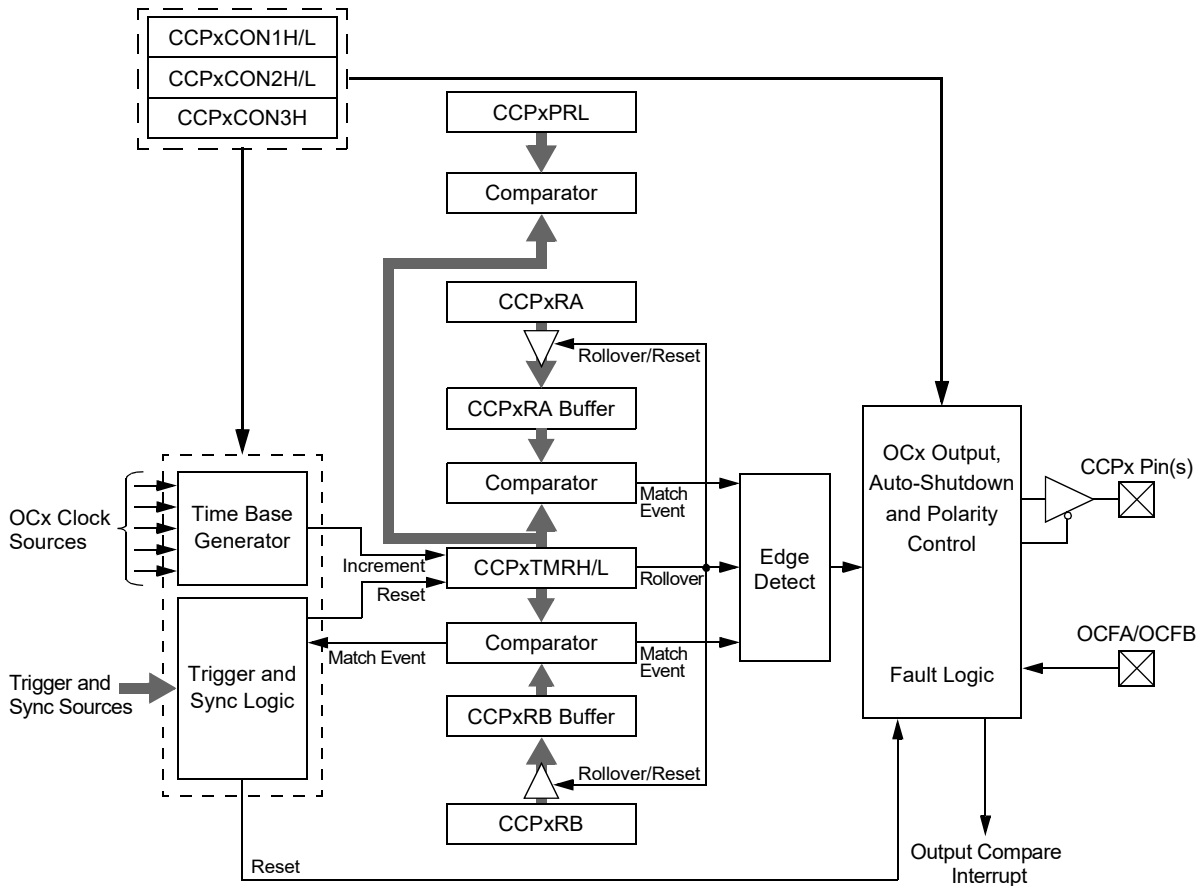
Output Compare mode compares the Timer register value with the value of one or two Compare registers, depending on its mode of operation. The Output Compare x module, on compare match events, has the ability to generate a single output transition or a train of output pulses. Like most PIC[®] MCU peripherals, the Output Compare x module can also generate interrupts on a compare match event.

Table 22-2 shows the various modes available in Output Compare modes.

Table 22-2. Output Compare x/PWMx Modes

MOD[3:0] (CCPxCON1L[3:0])	T32 (CCPxCON1L[5])	Operating Mode	
0001	0	Output High on Compare (16-bit)	Single Edge Mode
0001	1	Output High on Compare (32-bit)	
0010	0	Output Low on Compare (16-bit)	
0010	1	Output Low on Compare (32-bit)	
0011	0	Output Toggle on Compare (16-bit)	
0011	1	Output Toggle on Compare (32-bit)	
0100	0	Dual Edge Compare (16-bit)	Dual Edge Mode
0101	0	Dual Edge Compare (16-bit buffered)	PWM Mode

Figure 22-3. Output Compare x Block Diagram



22.4 Input Capture Mode

Input Capture mode is used to capture a timer value from an independent timer base, upon an event, on an input pin or other internal trigger source. The input capture features are useful in applications requiring frequency (time period) and pulse measurement. [Figure 22-4](#) depicts a simplified block diagram of Input Capture mode.

Input Capture mode uses a dedicated 16/32-bit, synchronous, up counting timer for the capture function. The timer value is written to the FIFO when a capture event occurs. The internal value may be read (with a synchronization delay) using the CCPxTMRH/L registers.

To use Input Capture mode, the CCSEL bit (CCPxCON1L[4]) must be set. The T32 and the MOD[3:0] bits are used to select the proper Capture mode, as shown in [Table 22-3](#).

Table 22-3. Input Capture x Modes

MOD[3:0] (CCPxCON1L[3:0])	T32 (CCPxCON1L[5])	Operating Mode
0000	0	Edge Detect (16-bit capture)
0000	1	Edge Detect (32-bit capture)
0001	0	Every Rising (16-bit capture)
0001	1	Every Rising (32-bit capture)
0010	0	Every Falling (16-bit capture)

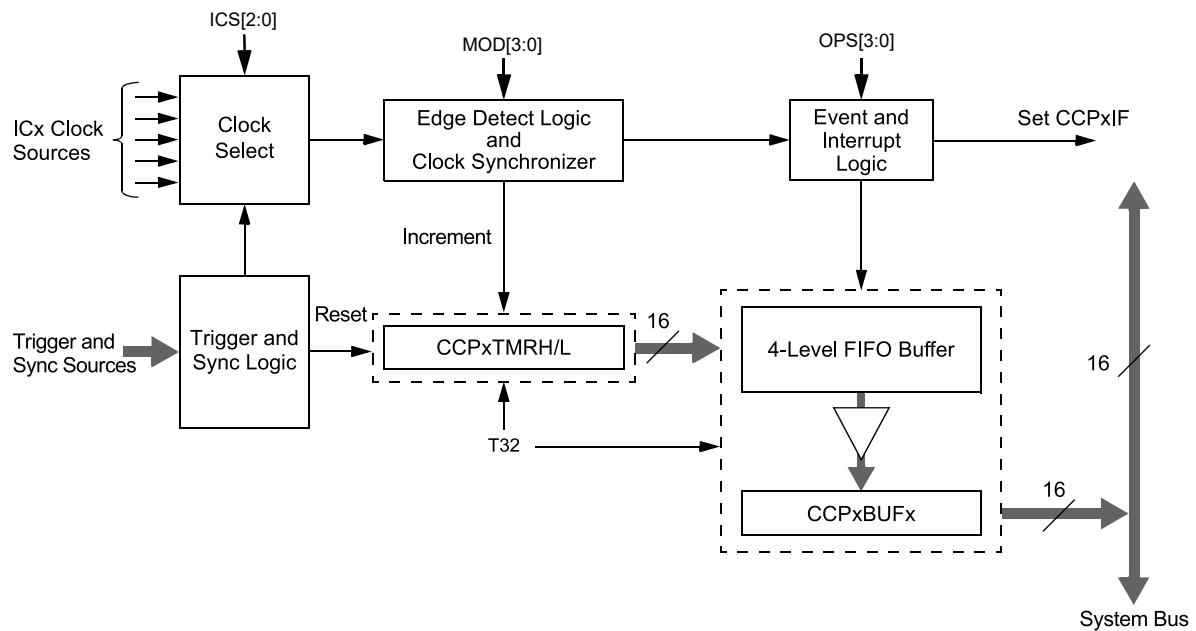
dsPIC33CK512MPT608 Family

Capture/Compare/PWM/Timer Modules (SCCP)

.....continued

MOD[3:0] (CCPxCON1L[3:0])	T32 (CCPxCON1L[5])	Operating Mode
0010	1	Every Falling (32-bit capture)
0011	0	Every Rising/Falling (16-bit capture)
0011	1	Every Rising/Falling (32-bit capture)
0100	0	Every 4th Rising (16-bit capture)
0100	1	Every 4th Rising (32-bit capture)
0101	0	Every 16th Rising (16-bit capture)
0101	1	Every 16th Rising (32-bit capture)

Figure 22-4. Input Capture x Block Diagram



22.5 Auxiliary Output

The SCCPx modules have an auxiliary (secondary) output that provides other peripherals access to internal module signals. The auxiliary output is intended to connect to other SCCP modules, or other digital peripherals, to provide these types of functions:

- Time Base Synchronization
- Peripheral Trigger and Clock Inputs
- Signal Gating

The type of output signal is selected using the AUXOUT[1:0] control bits (CCPxCON2H[4:3]). The type of output signal is also dependent on the module operating mode.

Table 22-4. Auxiliary Output

AUXOUT[1:0]	CCSEL	MOD[3:0]	Comments	Signal Description
00	x	xxxx	Auxiliary output disabled	No Output
01	0	0000	Time Base modes	Time Base Period Reset or Rollover
10				Special Event Trigger Output
11				No Output
01	0	0001 through 1111	Output Compare modes	Time Base Period Reset or Rollover
10				Output Compare Event Signal
11				Output Compare Signal
01	1	xxxx	Input Capture modes	Time Base Period Reset or Rollover
10				Reflects the Value of the ICDIS bit
11				Input Capture Event Signal

dsPIC33CK512MPT608 Family

Capture/Compare/PWM/Timer Modules (SCCP)

22.6 SCCP Control/Status Registers

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x00 ... 0x094F	Reserved									
0x0950	CCP1CON1L	15:8	CCPON		CCPSIDL	CCPSLP	TMRSYNC	CLKSEL[2:0]		
		7:0	TMRPS[1:0]		T32	CCSEL	MOD[3:0]			
0x0952	CCP1CON1H	15:8	OPSSRC	RTRGEN			OPS3[3:0]			
		7:0	TRIGEN	ONESHOT	ALTSYNC	SYNC[4:0]				
0x0954	CCP1CON2L	15:8	PWMRSEN	ASDGM		SSDG				
		7:0	ASDG[7:0]							
0x0956	CCP1CON2H	15:8	OENSYNC							OCAEN
		7:0	ICGSM[1:0]			AUXOUT[1:0]		ICS[2:0]		
0x0958 ... 0x0959	Reserved									
0x095A	CCP1CON3H	15:8	OETRIG	OSCNT[2:0]						
		7:0			POLACE		PSSACE[1:0]		PSSBDF[1:0]	
0x095C	CCP1STATL	15:8						ICGARM		
		7:0	CCPTRIG	TRSET	TRCLR	ASEVT	SCEVT	ICDIS	ICOV	ICBNE
0x095E ... 0x095F	Reserved									
0x0960	CCP1TMRL	15:8	TMRL[15:8]							
		7:0	TMRL[7:0]							
0x0962	CCP1TMRH	15:8	TMRH[31:24]							
		7:0	TMRH[23:16]							
0x0964	CCP1PRL	15:8	PRL[15:8]							
		7:0	PRL[7:0]							
0x0966	CCP1PRH	15:8	PRH[31:24]							
		7:0	PRH[23:16]							
0x0968	CCP1RA	15:8	CMP[15:8]							
		7:0	CMP[7:0]							
0x096A ... 0x096B	Reserved									
0x096C	CCP1RB	15:8	CMP[15:8]							
		7:0	CMP[7:0]							
0x096E ... 0x096F	Reserved									
0x0970	CCP1BUFL	15:8	BUF[15:8]							
		7:0	BUF[7:0]							
0x0972	CCP1BUFH	15:8	BUF[31:24]							
		7:0	BUF[23:16]							
0x0974	CCP2CON1L	15:8	CCPON		CCPSIDL	CCPSLP	TMRSYNC	CLKSEL[2:0]		
		7:0	TMRPS[1:0]		T32	CCSEL	MOD[3:0]			
0x0974	CCP2CON2L	15:8	PWMRSEN	ASDGM		SSDG				
		7:0	ASDG[7:0]							
0x0976	CCP2CON1H	15:8	OPSSRC	RTRGEN			OPS3[3:0]			
		7:0	TRIGEN	ONESHOT	ALTSYNC	SYNC[4:0]				
0x0976	CCP2CON2H	15:8	OENSYNC							OCAEN
		7:0	ICGSM[1:0]			AUXOUT[1:0]		ICS[2:0]		
0x0978 ... 0x097D	Reserved									
0x097E	CCP2CON3H	15:8	OETRIG	OSCNT[2:0]						
		7:0			POLACE		PSSACE[1:0]		PSSBDF[1:0]	

dsPIC33CK512MPT608 Family

Capture/Compare/PWM/Timer Modules (SCCP)

.....continued

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x0980	CCP2STATL	15:8						ICGARM		
		7:0	CCPTRIG	TRSET	TRCLR	ASEVT	SCEVT	ICDIS	ICOV	ICBNE
0x0982 ... 0x0983	Reserved									
0x0984	CCP2TMRL	15:8	TMRL[15:8]							
		7:0	TMRL[7:0]							
0x0986	CCP2TMRH	15:8	TMRH[31:24]							
		7:0	TMRH[23:16]							
0x0988	CCP2PRL	15:8	PRL[15:8]							
		7:0	PRL[7:0]							
0x098A	CCP2PRH	15:8	PRH[31:24]							
		7:0	PRH[23:16]							
0x098C	CCP2RA	15:8	CMP[15:8]							
		7:0	CMP[7:0]							
0x098E ... 0x098F	Reserved									
0x0990	CCP2RB	15:8	CMP[15:8]							
		7:0	CMP[7:0]							
0x0992 ... 0x0993	Reserved									
0x0994	CCP2BUFL	15:8	BUF[15:8]							
		7:0	BUF[7:0]							
0x0996	CCP2BUFH	15:8	BUF[31:24]							
		7:0	BUF[23:16]							
0x0998	CCP3CON1L	15:8	CCPON		CCPSIDL	CCPSLP	TMRSYNC	CLKSEL[2:0]		
		7:0	TMRPS[1:0]		T32	CCSEL	MOD[3:0]			
0x099A	CCP3CON1H	15:8	OPSSRC	RTRGEN			OPS3[3:0]			
		7:0	TRIGEN	ONESHOT	ALTSYNC	SYNC[4:0]				
0x099C	CCP3CON2L	15:8	PWMRSEN	ASDGM		SSDG				
		7:0	ASDG[7:0]							
0x099E	CCP3CON2H	15:8	OENSYNC							OCAEN
		7:0	ICGSM[1:0]			AUXOUT[1:0]		ICS[2:0]		
0x09A0 ... 0x09A1	Reserved									
0x09A2	CCP3CON3H	15:8	OETRIG	OSCNT[2:0]						
		7:0			POLACE		PSSACE[1:0]		PSSBDF[1:0]	
0x09A4	CCP3STATL	15:8						ICGARM		
		7:0	CCPTRIG	TRSET	TRCLR	ASEVT	SCEVT	ICDIS	ICOV	ICBNE
0x09A6 ... 0x09A7	Reserved									
0x09A8	CCP3TMRL	15:8	TMRL[15:8]							
		7:0	TMRL[7:0]							
0x09AA	CCP3TMRH	15:8	TMRH[31:24]							
		7:0	TMRH[23:16]							
0x09AC	CCP3PRL	15:8	PRL[15:8]							
		7:0	PRL[7:0]							
0x09AE	CCP3PRH	15:8	PRH[31:24]							
		7:0	PRH[23:16]							
0x09B0	CCP3RA	15:8	CMP[15:8]							
		7:0	CMP[7:0]							
0x09B2 ... 0x09B3	Reserved									

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Capture/Compare/PWM/Timer Modules (SCCP)

.....continued

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x09B4	CCP3RB	15:8	CMP[15:8]							
		7:0	CMP[7:0]							
0x09B6 ... 0x09B7	Reserved									
0x09B8	CCP3BUFL	15:8	BUF[15:8]							
		7:0	BUF[7:0]							
0x09BA	CCP3BUFH	15:8	BUF[31:24]							
		7:0	BUF[23:16]							
0x09BC	CCP4CON1L	15:8	CCPON		CCPSIDL	CCPSLP	TMRSYNC	CLKSEL[2:0]		
		7:0	TMRPS[1:0]		T32	CCSEL	MOD[3:0]			
0x09BE	CCP4CON1H	15:8	OPSSRC	RTRGEN			OPS3[3:0]			
		7:0	TRIGEN	ONESHOT	ALTSYNC	SYNC[4:0]				
0x09C0	CCP4CON2L	15:8	PWMRSEN	ASDGM		SSDG				
		7:0	ASDG[7:0]							
0x09C2	CCP4CON2H	15:8	OENSYNC							OCAEN
		7:0	ICGSM[1:0]			AUXOUT[1:0]		ICS[2:0]		
0x09C4 ... 0x09C5	Reserved									
0x09C6	CCP4CON3H	15:8	OETRIG	OSCNT[2:0]						
		7:0			POLACE		PSSACE[1:0]		PSSBDF[1:0]	
0x09C8	CCP4STATL	15:8						ICGARM		
		7:0	CCPTRIG	TRSET	TRCLR	ASEVT	SCEVT	ICDIS	ICOV	ICBNE
0x09CA ... 0x09CB	Reserved									
0x09CC	CCP4TMRL	15:8	TMRL[15:8]							
		7:0	TMRL[7:0]							
0x09CE	CCP4TMRH	15:8	TMRH[31:24]							
		7:0	TMRH[23:16]							
0x09D0	CCP4PRL	15:8	PRL[15:8]							
		7:0	PRL[7:0]							
0x09D2	CCP4PRH	15:8	PRH[31:24]							
		7:0	PRH[23:16]							
0x09D4	CCP4RA	15:8	CMP[15:8]							
		7:0	CMP[7:0]							
0x09D6 ... 0x09D7	Reserved									
0x09D8	CCP4RB	15:8	CMP[15:8]							
		7:0	CMP[7:0]							
0x09DA ... 0x09DB	Reserved									
0x09DC	CCP4BUFL	15:8	BUF[15:8]							
		7:0	BUF[7:0]							
0x09DE	CCP4BUFH	15:8	BUF[31:24]							
		7:0	BUF[23:16]							
0x09E0	CCP5CON1L	15:8	CCPON		CCPSIDL	CCPSLP	TMRSYNC	CLKSEL[2:0]		
		7:0	TMRPS[1:0]		T32	CCSEL	MOD[3:0]			
0x09E2	CCP5CON1H	15:8	OPSSRC	RTRGEN			OPS3[3:0]			
		7:0	TRIGEN	ONESHOT	ALTSYNC	SYNC[4:0]				
0x09E4	CCP5CON2L	15:8	PWMRSEN	ASDGM		SSDG				
		7:0	ASDG[7:0]							
0x09E6	CCP5CON2H	15:8	OENSYNC							OCAEN
		7:0	ICGSM[1:0]			AUXOUT[1:0]		ICS[2:0]		

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Capture/Compare/PWM/Timer Modules (SCCP)

.....continued

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x09E8 ... 0x09E9	Reserved									
0x09EA	CCP5CON3H	15:8	OETRIG	OSCNT[2:0]						
		7:0			POLACE		PSSACE[1:0]		PSSBDF[1:0]	
0x09EC	CCP5STATL	15:8						ICGARM		
		7:0	CCPTRIG	TRSET	TRCLR	ASEVT	SCEVT	ICDIS	ICOV	ICBNE
0x09EE ... 0x09EF	Reserved									
0x09F0	CCP5TMRL	15:8	TMRL[15:8]							
		7:0	TMRL[7:0]							
0x09F2	CCP5TMRH	15:8	TMRH[31:24]							
		7:0	TMRH[23:16]							
0x09F4	CCP5PRL	15:8	PRL[15:8]							
		7:0	PRL[7:0]							
0x09F6	CCP5PRH	15:8	PRH[31:24]							
		7:0	PRH[23:16]							
0x09F8	CCP5RA	15:8	CMP[15:8]							
		7:0	CMP[7:0]							
0x09FA ... 0x09FB	Reserved									
0x09FC	CCP5RB	15:8	CMP[15:8]							
		7:0	CMP[7:0]							
0x09FE ... 0x09FF	Reserved									
0x0A00	CCP5BUFL	15:8	BUF[15:8]							
		7:0	BUF[7:0]							
0x0A02	CCP5BUFH	15:8	BUF[31:24]							
		7:0	BUF[23:16]							
0x0A04	CCP6CON1L	15:8	CCPON		CCPSIDL	CCPSLP	TMRSYNC	CLKSEL[2:0]		
		7:0	TMRPS[1:0]		T32	CCSEL	MOD[3:0]			
0x0A06	CCP6CON1H	15:8	OPSSRC	RTRGEN			OPS3[3:0]			
		7:0	TRIGEN	ONESHOT	ALTSYNC	SYNC[4:0]				
0x0A08	CCP6CON2L	15:8	PWMRSEN	ASDGM		SSDG				
		7:0	ASDG[7:0]							
0x0A0A	CCP6CON2H	15:8	OENSYNC							OCAEN
		7:0	ICGSM[1:0]			AUXOUT[1:0]		ICS[2:0]		
0x0A0C ... 0x0A0D	Reserved									
0x0A0E	CCP6CON3H	15:8	OETRIG	OSCNT[2:0]						
		7:0			POLACE		PSSACE[1:0]		PSSBDF[1:0]	
0x0A10	CCP6STATL	15:8						ICGARM		
		7:0	CCPTRIG	TRSET	TRCLR	ASEVT	SCEVT	ICDIS	ICOV	ICBNE
0x0A12 ... 0x0A13	Reserved									
0x0A14	CCP6TMRL	15:8	TMRL[15:8]							
		7:0	TMRL[7:0]							
0x0A16	CCP6TMRH	15:8	TMRH[31:24]							
		7:0	TMRH[23:16]							
0x0A18	CCP6PRL	15:8	PRL[15:8]							
		7:0	PRL[7:0]							
0x0A1A	CCP6PRH	15:8	PRH[31:24]							
		7:0	PRH[23:16]							

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Capture/Compare/PWM/Timer Modules (SCCP)

.....continued

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x0A1C	CCP6RA	15:8	CMP[15:8]							
		7:0	CMP[7:0]							
0x0A1E ... 0x0A1F	Reserved									
0x0A20	CCP6RB	15:8	CMP[15:8]							
		7:0	CMP[7:0]							
0x0A22 ... 0x0A23	Reserved									
0x0A24	CCP6BUFL	15:8	BUF[15:8]							
		7:0	BUF[7:0]							
0x0A26	CCP6BUFH	15:8	BUF[31:24]							
		7:0	BUF[23:16]							
0x0A28	CCP7CON1L	15:8	CCPON		CCPSIDL	CCPSLP	TMRSYNC	CLKSEL[2:0]		
		7:0	TMRPS[1:0]		T32	CCSEL	MOD[3:0]			
0x0A2A	CCP7CON1H	15:8	OPSSRC	RTRGEN			OPS3[3:0]			
		7:0	TRIGEN	ONESHOT	ALTSYNC	SYNC[4:0]				
0x0A2C	CCP7CON2L	15:8	PWMRSEN	ASDGM		SSDG				
		7:0	ASDG[7:0]							
0x0A2E	CCP7CON2H	15:8	OENSYNC							OCAEN
		7:0	ICGSM[1:0]			AUXOUT[1:0]		ICS[2:0]		
0x0A30 ... 0x0A31	Reserved									
0x0A32	CCP7CON3H	15:8	OETRIG	OSCNT[2:0]						
		7:0			POLACE		PSSACE[1:0]		PSSBDF[1:0]	
0x0A34	CCP7STATL	15:8						ICGARM		
		7:0	CCPTRIG	TRSET	TRCLR	ASEVT	SCEVT	ICDIS	ICOV	ICBNE
0x0A36 ... 0x0A37	Reserved									
0x0A38	CCP7TMRL	15:8	TMRL[15:8]							
		7:0	TMRL[7:0]							
0x0A3A	CCP7TMRH	15:8	TMRH[31:24]							
		7:0	TMRH[23:16]							
0x0A3C	CCP7PRL	15:8	PRL[15:8]							
		7:0	PRL[7:0]							
0x0A3E	CCP7PRH	15:8	PRH[31:24]							
		7:0	PRH[23:16]							
0x0A40	CCP7RA	15:8	CMP[15:8]							
		7:0	CMP[7:0]							
0x0A42 ... 0x0A43	Reserved									
0x0A44	CCP7RB	15:8	CMP[15:8]							
		7:0	CMP[7:0]							
0x0A46 ... 0x0A47	Reserved									
0x0A48	CCP7BUFL	15:8	BUF[15:8]							
		7:0	BUF[7:0]							
0x0A4A	CCP7BUFH	15:8	BUF[31:24]							
		7:0	BUF[23:16]							
0x0A4C	CCP8CON1L	15:8	CCPON		CCPSIDL	CCPSLP	TMRSYNC	CLKSEL[2:0]		
		7:0	TMRPS[1:0]		T32	CCSEL	MOD[3:0]			
0x0A4E	CCP8CON1H	15:8	OPSSRC	RTRGEN			OPS3[3:0]			
		7:0	TRIGEN	ONESHOT	ALTSYNC	SYNC[4:0]				

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Capture/Compare/PWM/Timer Modules (SCCP)

.....continued

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x0A50	CCP8CON2L	15:8	PWMRSEN	ASDGM		SSDG				
		7:0	ASDG[7:0]							
0x0A52	CCP8CON2H	15:8	OENSYNC							OCAEN
		7:0	ICGSM[1:0]			AUXOUT[1:0]		ICS[2:0]		
0x0A54 ... 0x0A55	Reserved									
0x0A56	CCP8CON3H	15:8	OETRIG	OSCNT[2:0]						
		7:0			POLACE		PSSACE[1:0]		PSSBDF[1:0]	
0x0A58	CCP8STATL	15:8						ICGARM		
		7:0	CCPTRIG	TRSET	TRCLR	ASEVT	SCEVT	ICDIS	ICOV	ICBNE
0x0A5A ... 0x0A5F	Reserved									
0x0A60	CCP8PRL	15:8	PRL[15:8]							
		7:0	PRL[7:0]							
0x0A62	CCP8PRH	15:8	PRH[31:24]							
		7:0	PRH[23:16]							
0x0A64	CCP8RA	15:8	CMP[15:8]							
		7:0	CMP[7:0]							
0x0A66 ... 0x0A67	Reserved									
0x0A68	CCP8RB	15:8	CMP[15:8]							
		7:0	CMP[7:0]							
0x0A6A ... 0x0A6B	Reserved									
0x0A6C	CCP8BUFL	15:8	BUF[15:8]							
		7:0	BUF[7:0]							
0x0A6E	CCP8BUFH	15:8	BUF[31:24]							
		7:0	BUF[23:16]							
0x0A70	CCP9CON1L	15:8	CCPON		CCPSIDL	CCPSLP	TMRSYNC	CLKSEL[2:0]		
		7:0	TMRPS[1:0]		T32	CCSEL	MOD[3:0]			
0x0A72	CCP9CON1H	15:8	OPSSRC	RTRGEN			OPS3[3:0]			
		7:0	TRIGEN	ONESHOT	ALTSYNC	SYNC[4:0]				
0x0A74	CCP9CON2L	15:8	PWMRSEN	ASDGM		SSDG				
		7:0	ASDG[7:0]							
0x0A74	CCP9CON3H	15:8	OETRIG	OSCNT[2:0]						
		7:0			POLACE		PSSACE[1:0]		PSSBDF[1:0]	
0x0A76	CCP9CON2H	15:8	OENSYNC							OCAEN
		7:0	ICGSM[1:0]			AUXOUT[1:0]		ICS[2:0]		
0x0A78 ... 0x0A7B	Reserved									
0x0A7C	CCP9STATL	15:8						ICGARM		
		7:0	CCPTRIG	TRSET	TRCLR	ASEVT	SCEVT	ICDIS	ICOV	ICBNE
0x0A7E ... 0x0A83	Reserved									
0x0A84	CCP9PRL	15:8	PRL[15:8]							
		7:0	PRL[7:0]							
0x0A86	CCP9PRH	15:8	PRH[31:24]							
		7:0	PRH[23:16]							
0x0A88	CCP9RA	15:8	CMP[15:8]							
		7:0	CMP[7:0]							
0x0A8A ... 0x0A8B	Reserved									

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Capture/Compare/PWM/Timer Modules (SCCP)

.....continued

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x0A8C	CCP9RB	15:8	CMP[15:8]							
		7:0	CMP[7:0]							
0x0A8E ... 0x0A8F	Reserved									
0x0A90	CCP9BUFL	15:8	BUF[15:8]							
		7:0	BUF[7:0]							
0x0A92	CCP9BUFH	15:8	BUF[31:24]							
		7:0	BUF[23:16]							

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Capture/Compare/PWM/Timer Modules (SCCP)

22.6.1 CCPx Control 1 Low Register

Name: CCPXCON1L
Offset: 0x950, 0x974, 0x998, 0x9BC, 0x9E0, 0xA04, 0xA28, 0xA4C, 0xA70

Note:

- Only available on the MCCP.

Bit	15	14	13	12	11	10	9	8
	CCPON			CCPSIDL	CCPSLP	TMRSYNC	CLKSEL[2:0]	
Access	R/W			R/W	R/W	R/W	R/W	R/W
Reset	0			0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	TMRPS[1:0]		T32	CCSEL		MOD[3:0]		
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – CCPON CCPx Module Enable bit

Value	Description
1	Module is enabled with an operating mode specified by the MOD[3:0] control bits
0	Module is disabled

Bit 13 – CCPSIDL CCPx Stop in Idle Mode Bit

Value	Description
1	Discontinues module operation when device enters Idle mode
0	Continues module operation in Idle mode

Bit 12 – CCPSLP CCPx Sleep Mode Enable bit

Value	Description
1	Module continues to operate in Sleep modes
0	Module does not operate in Sleep modes

Bit 11 – TMRSYNC Time Base Clock Synchronization bit

Value	Description
1	Asynchronous module time base clock is selected and synchronized to the internal system clocks (CLKSEL[2:0] ≠ 000)
0	Synchronous module time base clock is selected and does not require synchronization (CLKSEL[2:0] = 000)

Bits 10:8 – CLKSEL[2:0] CCPx Time Base Clock Select bits⁽¹⁾

Value	Description
111	External CCP TCKIx
110	CLC4
101	CLC3
100	CLC2
011	CLC1
010	F _{OSC}
001	Reference Clock (REFCLKO)
000	F _{OSC} /2 (F _P)

Bits 7:6 – TMRPS[1:0] Time Base Prescale Select bits

Value	Description
11	1:64 Prescaler
10	1:16 Prescaler
01	1:4 Prescaler

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Capture/Compare/PWM/Timer Modules (SCCP)

Value	Description
00	1:1 Prescaler

Bit 5 – T32 32-Bit Time Base Select bit

Value	Description
1	Uses 32-bit time base for timer, single edge output compare or input capture function
0	Uses 16-bit time base for timer, single edge output compare or input capture function

Bit 4 – CCSEL Capture/Compare Mode Select bit

Value	Description
1	Input Capture peripheral
0	Output Compare/PWM/Timer peripheral (exact function is selected by the MOD[3:0] bits)

Bits 3:0 – MOD[3:0] CCPx Mode Select bits

For CCSEL = 1 (Input Capture modes):

Value	Description
1xxx	Reserved
011x	Reserved
0101	Capture every 16th rising edge
0100	Capture every 4th rising edge
0011	Capture every rising and falling edge
0010	Capture every falling edge
0001	Capture every rising edge
0000	Capture every rising and falling edge (Edge Detect mode)

For CCSEL = 0 (Output Compare/Timer modes):

Value	Description
1111	External Input mode: Pulse generator is disabled, source is selected by ICS[2:0]
1110	Reserved
110x	Reserved
10xx	Reserved
0111	Variable Frequency Pulse mode ⁽¹⁾
0110	Center Aligned Pulse Compare mode, buffered ⁽¹⁾
0101	Dual Edge Compare mode, buffered
0100	Dual Edge Compare mode
0011	16-Bit/32-Bit Single Edge mode, toggles output on compare match
0010	16-Bit/32-Bit Single Edge mode, drives output low on compare match
0001	16-Bit/32-Bit Single Edge mode, drives output high on compare match
0000	16-Bit/32-Bit Timer mode, output functions are disabled

dsPIC33CK512MPT608 Family

Capture/Compare/PWM/Timer Modules (SCCP)

22.6.2 CCPx Control 1 High Register

Name: CCPxCON1H
Offset: 0x952, 0x976, 0x99A, 0x9BE, 0x9E2, 0xA06, 0xA2A, 0xA4E, 0xA72

Notes:

1. This control bit has no function in Input Capture modes.
2. This control bit has no function when TRIGEN = 0.
3. Output postscale settings, from 1:5 to 1:16 (0100–1111), will result in a FIFO buffer overflow for Input Capture modes.

Bit	15	14	13	12	11	10	9	8
	OPSSRC	RTRGEN			OPS3[3:0]			
Access	R/W	R/W			R/W	R/W	R/W	R/W
Reset	0	0			0	0	0	0

Bit	7	6	5	4	3	2	1	0
	TRIGEN	ONESHOT	ALTSYNC		SYNC[4:0]			
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – OPSSRC Output Postscaler Source Select bit⁽¹⁾

Value	Description
1	Output postscaler scales module trigger output events
0	Output postscaler scales time base interrupt events

Bit 14 – RTRGEN Retrigger Enable bit⁽²⁾

Value	Description
1	Time base can be retriggered when TRIGEN bit = 1
0	Time base may not be retriggered when TRIGEN bit = 1

Bits 11:8 – OPS3[3:0] CCPx Interrupt Output Postscale Select bits⁽³⁾

Value	Description
1111	Interrupt every 16th time base period match
1110	Interrupt every 15th time base period match
. . .	
0100	Interrupt every 5th time base period match
0011	Interrupt every 4th time base period match or 4th input capture event
0010	Interrupt every 3rd time base period match or 3rd input capture event
0001	Interrupt every 2nd time base period match or 2nd input capture event
0000	Interrupt after each time base period match or input capture event

Bit 7 – TRIGEN CCPx Trigger Enable bit

Value	Description
1	Trigger operation of time base is enabled
0	Trigger operation of time base is disabled

Bit 6 – ONESHOT One-Shot Trigger Mode Enable bit

Value	Description
1	One-Shot Trigger mode is enabled; trigger duration is set by OSCNT[2:0]
0	One-Shot Trigger mode is disabled

Bit 5 – ALTSYNC CCPx Clock Select bits

Value	Description
1	An alternate signal is used as the module synchronization output signal
0	The module synchronization output signal is the Time Base Reset/rollover event

dsPIC33CK512MPT608 Family

Capture/Compare/PWM/Timer Modules (SCCP)

Bits 4:0 – SYNC[4:0] CCPx Synchronization Source Select bits
See [22.6.15. Synchronization Sources](#) for the definition of inputs.

dsPIC33CK512MPT608 Family

Capture/Compare/PWM/Timer Modules (SCCP)

22.6.3 CCPx Control 2 Low Register

Name: CCPxCON2L
Offset: 0x954, 0x974, 0x99C, 0x9C0, 0x9E4, 0xA08, 0xA2C, 0xA50, 0xA74

Bit	15	14	13	12	11	10	9	8
	PWMRSEN	ASDGM		SSDG				
Access	R/W	R/W		R/W				
Reset	0	0		0				

Bit	7	6	5	4	3	2	1	0
	ASDG[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – PWMRSEN CCPx PWM Restart Enable bit

Value	Description
1	ASEVT bit clears automatically at the beginning of the next PWM period, after the shutdown input has ended
0	ASEVT bit must be cleared in software to resume PWM activity on output pins

Bit 14 – ASDGM CCPx Auto-Shutdown Gate Mode Enable bit

Value	Description
1	Waits until the next Time Base Reset or rollover for shutdown to occur
0	Shutdown event occurs immediately

Bit 12 – SSDG CCPx Software Shutdown/Gate Control bit

Value	Description
1	Manually forces auto-shutdown, timer clock gate or input capture signal gate event (setting of ASDGM bit still applies)
0	Normal module operation

Bits 7:0 – ASDG[7:0] CCPx Auto-Shutdown/Gating Source Enable bits

Value	Description
1	ASDGx Source n is enabled (see 22.6.16. Auto-Shutdown and Gating Sources for auto-shutdown/gating sources)
0	ASDGx Source n is disabled

dsPIC33CK512MPT608 Family

Capture/Compare/PWM/Timer Modules (SCCP)

22.6.4 CCPx Control 2 High Register

Name: CCPXCON2H
Offset: 0x956, 0x976, 0x99E, 0x9C2, 0x9E6, 0xA0A, 0xA2E, 0xA52, 0xA76

Bit	15	14	13	12	11	10	9	8
	OENSYNC							OCAEN
Access	R/W							R/W
Reset	0							0

Bit	7	6	5	4	3	2	1	0
	ICGSM[1:0]			AUXOUT[1:0]			ICS[2:0]	
Access	R/W	R/W		R/W	R/W	R/W	R/W	R/W
Reset	0	0		0	0	0	0	0

Bit 15 – OENSYNC Output Enable Synchronization bit

Value	Description
1	Update by output enable bits occurs on the next Time Base Reset or rollover
0	Update by output enable bits occurs immediately

Bit 8 – OCAEN Output Enable/Steering Control bit

Value	Description
1	OCx pin is controlled by the CCPx module and produces an output compare or PWM signal
0	OCx pin is not controlled by the CCPx module; the pin is available to the port logic or another peripheral multiplexed on the pin

Bits 7:6 – ICGSM[1:0] Input Capture Gating Source Mode Control bits

Value	Description
11	Reserved
10	One-Shot mode: Falling edge from gating source disables future capture events (ICDIS = 1)
01	One-Shot mode: Rising edge from gating source enables future capture events (ICDIS = 0)
00	Level-Sensitive mode: A high level from gating source will enable future capture events; a low level will disable future capture events

Bits 4:3 – AUXOUT[1:0] Auxiliary Output Signal on Event Selection bits

Value	Description
11	Input capture or output compare event; no signal in Timer mode
10	Signal output is defined by module operating mode (see 22.5. Auxiliary Output)
01	Time base rollover event (all modes)
00	Disabled

Bits 2:0 – ICS[2:0] Input Capture Source Select bits

Value	Description
111	CLC4 Output
110	CLC3 Output
101	CLC2 Output
100	CLC1 Output
011	Comparator 3
010	Comparator 2
001	Comparator 1
000	SCCP Input Capture x (ICx) pin (PPS)

dsPIC33CK512MPT608 Family

Capture/Compare/PWM/Timer Modules (SCCP)

22.6.5 CCPx Control 3 High Register

Name: CCPXCON3H
Offset: 0x95A, 0x97E, 0x9A2, 0x9C6, 0x9EA, 0xA0E, 0xA32, 0xA56, 0xA74

Bit	15	14	13	12	11	10	9	8
	OETRIG	OSCNT[2:0]						
Access	R/W	R/W	R/W	R/W				
Reset	0	0	0	0				

Bit	7	6	5	4	3	2	1	0
			POLACE		PSSACE[1:0]		PSSBDF[1:0]	
Access			R/W		R/W	R/W	R/W	R/W
Reset			0		0	0	0	0

Bit 15 – OETRIG CCPx Dead-Time Select bit

Value	Description
1	For Triggered mode (TRIGEN = 1): Module does not drive enabled output pins until triggered
0	Normal output pin operation

Bits 14:12 – OSCNT[2:0] One-Shot Event Count bits

Value	Description
111	Extends one-shot event by 7 time base periods (8 time base periods total)
110	Extends one-shot event by 6 time base periods (7 time base periods total)
101	Extends one-shot event by 5 time base periods (6 time base periods total)
100	Extends one-shot event by 4 time base periods (5 time base periods total)
011	Extends one-shot event by 3 time base periods (4 time base periods total)
010	Extends one-shot event by 2 time base periods (3 time base periods total)
001	Extends one-shot event by 1 time base period (2 time base periods total)
000	Does not extend one-shot trigger event

Bit 5 – POLACE CCPx Output Pins, OCxA, OCxC and OCxE, Polarity Control bit

Value	Description
1	Output pin polarity is active-low
0	Output pin polarity is active-high

Bits 3:2 – PSSACE[1:0] PWMx Output Pins, OCxA, OCxC and OCxE, Shutdown State Control bits

Value	Description
11	Pins are driven active when a shutdown event occurs
10	Pins are driven inactive when a shutdown event occurs
0x	Pins are in a High-Impedance state when a shutdown event occurs

Bits 1:0 – PSSBDF[1:0] PWMx Output Pins, OCMxB, OCMxD and OCMxF, Shutdown State Control bits

Value	Description
11	Pins are driven active when a shutdown event occurs
10	Pins are driven inactive when a shutdown event occurs
0x	Pins are in a High-Impedance state when a shutdown event occurs

dsPIC33CK512MPT608 Family

Capture/Compare/PWM/Timer Modules (SCCP)

22.6.6 CCPx Status Register

Name: CCPxSTATL
Offset: 0x95C, 0x980, 0x9A4, 0x9C8, 0x9EC, 0xA10, 0xA34, 0xA58, 0xA7C

Legend: C = Clearable bit; W1 = Write '1' Only bit

Bit	15	14	13	12	11	10	9	8
						ICGARM		
Access						R		
Reset						0		

Bit	7	6	5	4	3	2	1	0
	CCPTRIG	TRSET	TRCLR	ASEVT	SCEVT	ICDIS	ICOV	ICBNE
Access	R	W1	W1	R/C	R/C	R/C	R/C	R/C
Reset	0	0	0	0	0	0	0	0

Bit 10 – ICGARM Input Capture Gate Arm bit

Value	Description
1	Input capture gating logic is armed for a one-shot gate event when ICGSM[1:0] = 01 or 10; bit always reads as '0'
0	Input capture gating logic is not armed for a one-shot gate event when ICGSM[1:0] = 01 or 10; bit always reads as '0'

Bit 7 – CCPTRIG CCPx Trigger Status bit

Value	Description
1	Timer has been triggered and is running
0	Timer has not been triggered and is held in Reset

Bit 6 – TRSET CCPx Trigger Set Request bit

Writes '1' to this location to trigger the timer when TRIGEN = 1 (location always reads as '0').

Bit 5 – TRCLR CCPx Trigger Clear Request bit

Writes '1' to this location to cancel the timer trigger when TRIGEN = 1 (location always reads as '0').

Bit 4 – ASEVT CCPx Auto-Shutdown Event Status/Control bit

Value	Description
1	A shutdown event is in progress; CCPx outputs are in the Shutdown state
0	CCPx outputs operate normally

Bit 3 – SCEVT Single Edge Compare Event Status bit

Value	Description
1	A single edge compare event has occurred
0	A single edge compare event has not occurred

Bit 2 – ICDIS Input Capture x Disable bit

Value	Description
1	Event on Input Capture x pin (ICx) does not generate a capture event
0	Event on Input Capture x pin will generate a capture event

Bit 1 – ICOV Input Capture x Buffer Overflow Status bit

Value	Description
1	The Input Capture x FIFO buffer has overflowed
0	The Input Capture x FIFO buffer has not overflowed

Bit 0 – ICBNE Input Capture x Buffer Status bit

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Capture/Compare/PWM/Timer Modules (SCCP)

Value	Description
1	Input Capture x buffer has data available
0	Input Capture x buffer is empty

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Capture/Compare/PWM/Timer Modules (SCCP)

22.6.7 CCPx Time Base Register Low

Name: CCPxTMRL
Offset: 0x960, 0x984, 0x9A8, 0x9CC, 0x9F0, 0xA14, 0xA38, A5C, A80

Bit	15	14	13	12	11	10	9	8
	TMRL[15:8]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	TMRL[7:0]							
Access	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – TMRL[15:0] CCPx 16-Bit Time Base Value bits

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Capture/Compare/PWM/Timer Modules (SCCP)

22.6.8 CCPx Time Base High Register

Name: CCPxTMRH

Offset: 0x962, 0x986, 0x9AA, 0x9CE, 0x9F2, 0xA16, 0xA3A, A5E, A82

Bit	15	14	13	12	11	10	9	8
	TMRH[31:24]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	TMRH[23:16]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – TMRH[31:16] CCPx 16-Bit Time Base Value bits

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Capture/Compare/PWM/Timer Modules (SCCP)

22.6.9 CCPx Period Low Register

Name: CCPxPRL

Offset: 0x964, 0x988, 0x9AC, 0x9D0, 0x9F4, 0xA18, 0xA3C, 0xA60, 0xA84

Bit	15	14	13	12	11	10	9	8
	PRL[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PRL[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – PRL[15:0] CCPx Period Low Register bits

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Capture/Compare/PWM/Timer Modules (SCCP)

22.6.10 CCPx Period High Register

Name: CCPxPRH

Offset: 0x966, 0x98A, 0x9AE, 0x9D2, 0x9F6, 0xA1A, 0xA3E, 0xA62, 0xA86

Bit	15	14	13	12	11	10	9	8
	PRH[31:24]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PRH[23:16]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – PRH[31:16] CCPx Period High Register bits

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Capture/Compare/PWM/Timer Modules (SCCP)

22.6.11 CCPx Primary Compare Register (Timer/Compare Modes Only)

Name: CCPxRA
Offset: 0x968, 0x98C, 0x9B0, 0x9D4, 0x9F8, 0xA1C, 0xA40, 0xA64, 0xA88

Bit	15	14	13	12	11	10	9	8
	CMP[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	CMP[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – CMP[15:0] CCPx Primary Compare Value bits
The 16-bit value to be compared against the CCP time base.

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Capture/Compare/PWM/Timer Modules (SCCP)

22.6.12 CCPx Secondary Compare Register (Timer/Compare Modes Only)

Name: CCPxRB
Offset: 0x96C, 0x990, 0x9B4, 0x9D8, 0x9FC, 0xA20, 0xA44, 0xA68, 0xA8C

Bit	15	14	13	12	11	10	9	8
	CMP[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	CMP[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – CMP[15:0] CCPx Secondary Compare Value bits
The 16-bit value to be compared against the CCP time base.

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Capture/Compare/PWM/Timer Modules (SCCP)

22.6.13 CCPx Capture Buffer Register Low (Capture Modes Only)

Name: CCPxBUFL
Offset: 0x970, 0x994, 0x9B8, 0x9DC, 0xA00, 0xA24, 0xA48, 0xA6C, 0xA90

Bit	15	14	13	12	11	10	9	8
	BUF[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	BUF[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – BUF[15:0] CCPx Compare Buffer Value bits
 Indicates the oldest captured time base value in the FIFO.

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Capture/Compare/PWM/Timer Modules (SCCP)

22.6.14 CCPx Capture Buffer High Register (Capture Modes Only)

Name: CCPxBUFH
Offset: 0x972, 0x996, 0x9BA, 0x9DE, 0xA02, 0xA26, 0xA4A, 0xA6E, 0xA92

Bit	15	14	13	12	11	10	9	8
	BUF[31:24]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	BUF[23:16]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – BUF[31:16] CCPx Compare Buffer Value bits

22.6.15 Synchronization Sources

Table 22-5. Synchronization Sources

SYNC[4:0]	Synchronization Source
00000	None; Timer with Rollover on CCPxPR Match or FFFFh
00001	Module's Own Timer Sync Out
00010	Sync Output SCCP2
00011	Sync Output SCCP3
00100	Sync Output SCCP4
00101	Sync Output SCCP5
00110	Sync Output SCCP6
00111	Sync Output SCCP7
01000	Sync Output SCCP8
01001	INT0
01010	INT1
01011	INT2
01100	UART1 RX Edge Detect
01101	UART1 TX Edge Detect
01110	UART2 RX Edge Detect
01111	UART2 TX Edge Detect
10000	CLC1 Output
10001	CLC2 Output
10010	CLC3 Output
10011	CLC4 Output
10100	UART3 RX Edge Detect
10101	UART3 TX Edge Detect

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Capture/Compare/PWM/Timer Modules (SCCP)

.....continued	
SYNC[4:0]	Synchronization Source
10110	Sync Output MCCP9
10111	Comparator 1 Output
11000	Comparator 2 Output
11001	Comparator 3 Output
11010–11110	Reserved
11111	None; Timer with Auto-Rollover (FFFFh → 0000h)

22.6.16 Auto-Shutdown and Gating Sources

Table 22-6. Auto-Shutdown and Gating Sources

ASDG[x] Bit	Auto-Shutdown/Gating Source								
	SCCP1	SCCP2	SCCP3	SCCP4	SCCP5	SCCP6	SCCP7	SCCP8	MCCP9
0	Comparator 1 Output								
1	Comparator 2 Output								
2	OCFC								
3	OCFD								
4	ICM1 ⁽¹⁾	ICM2 ⁽¹⁾	ICM3 ⁽¹⁾	ICM4 ⁽¹⁾	ICM5 ⁽¹⁾	ICM6 ⁽¹⁾	ICM7 ⁽¹⁾	ICM8 ⁽¹⁾	ICM9 ⁽¹⁾
5	CLC1 Output ⁽¹⁾								
6	OCFA ⁽¹⁾								
7	OCFB ⁽¹⁾								
Note:									
1. Selected by Peripheral Pin Select (PPS).									

23. Configurable Logic Cell (CLC)

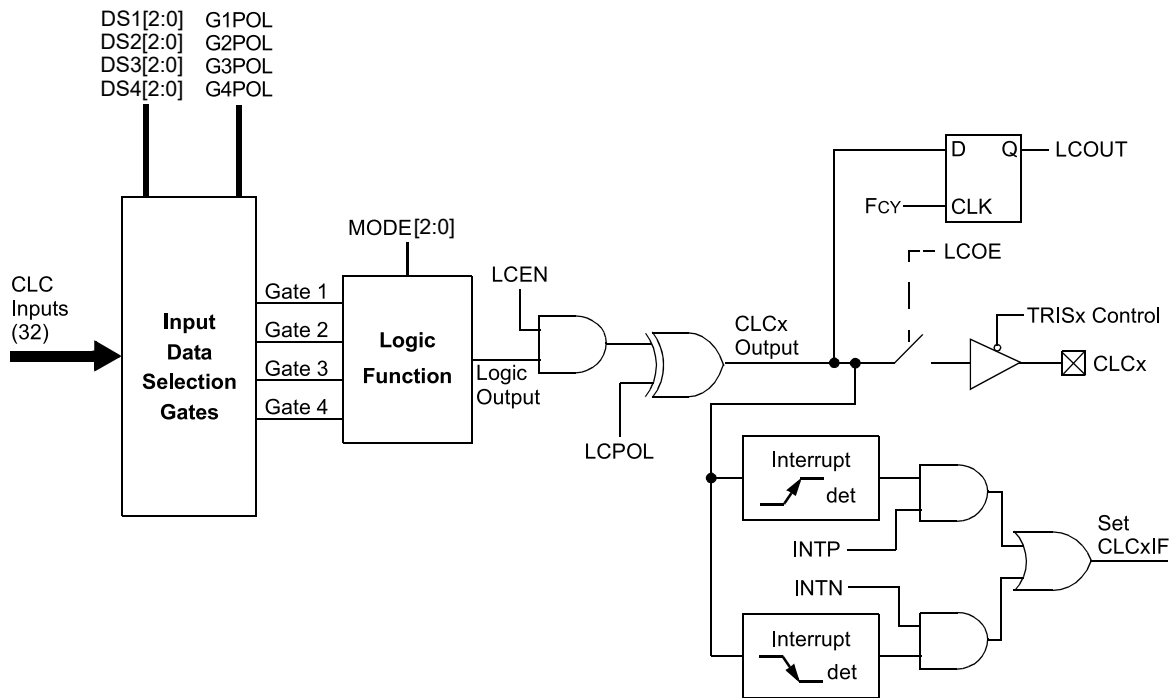
Note: This data sheet summarizes the features of the dsPIC33CK512MPT608 family of devices. It is not intended to be a comprehensive reference source. For more information, refer to “**Configurable Logic Cell (CLC)**” (www.microchip.com/DS70005298) in the “dsPIC33/PIC24 Family Reference Manual”. The information in this data sheet supersedes the information in the FRM.

The Configurable Logic Cell (CLC) module allows the user to specify combinations of signals as inputs to a logic function and to use the logic output to control other peripherals or I/O pins. This provides greater flexibility and potential in embedded designs, since the CLC module can operate outside the limitations of software execution, and supports a vast amount of output designs.

There are four input gates to the selected logic function. These four input gates select from a pool of up to 32 signals that are selected using four data source selection multiplexers.

Figure 23-1 shows the details of the data source multiplexers and Figure 23-2 shows the logic input gate connections.

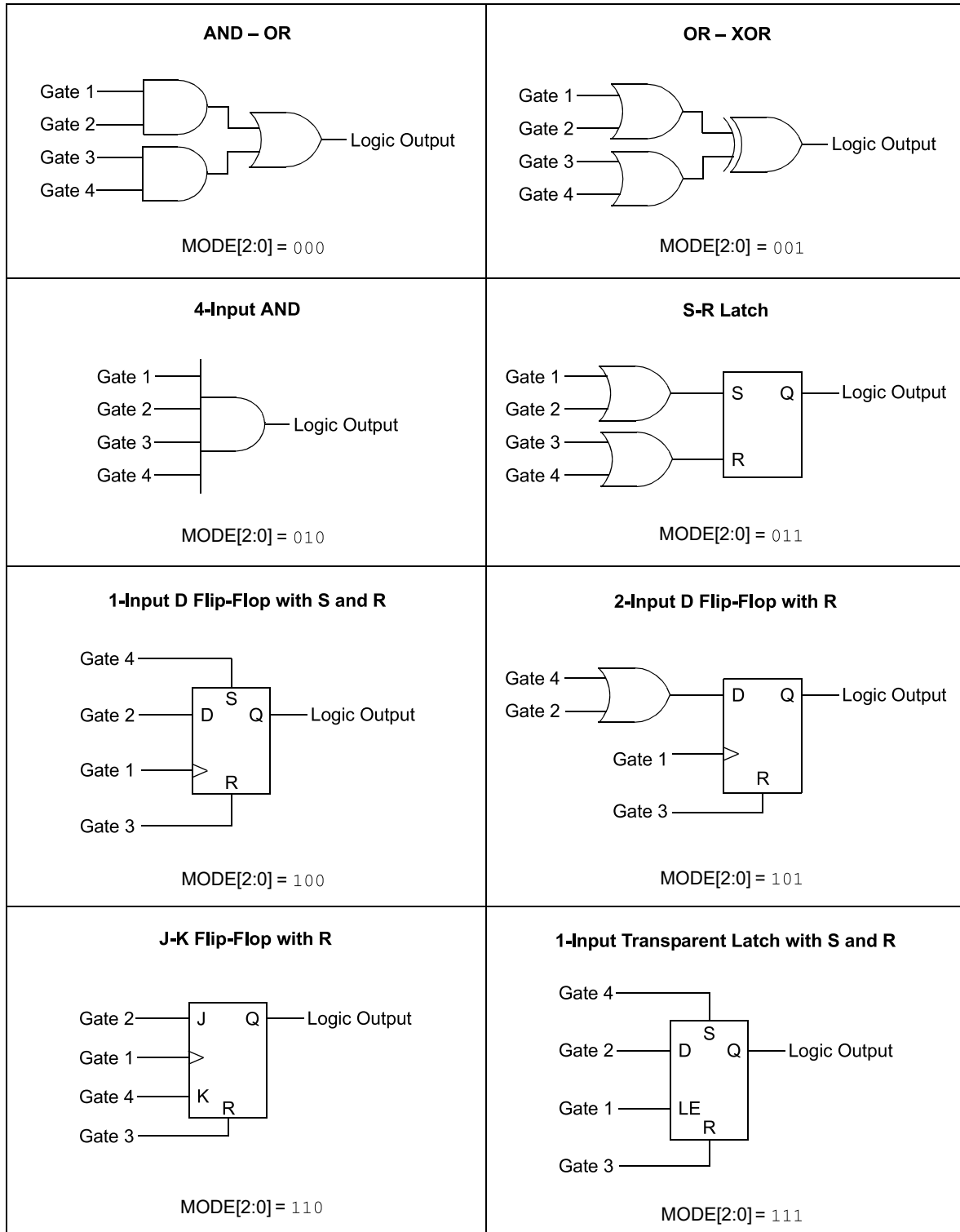
Figure 23-1. CLCx Module



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Configurable Logic Cell (CLC)

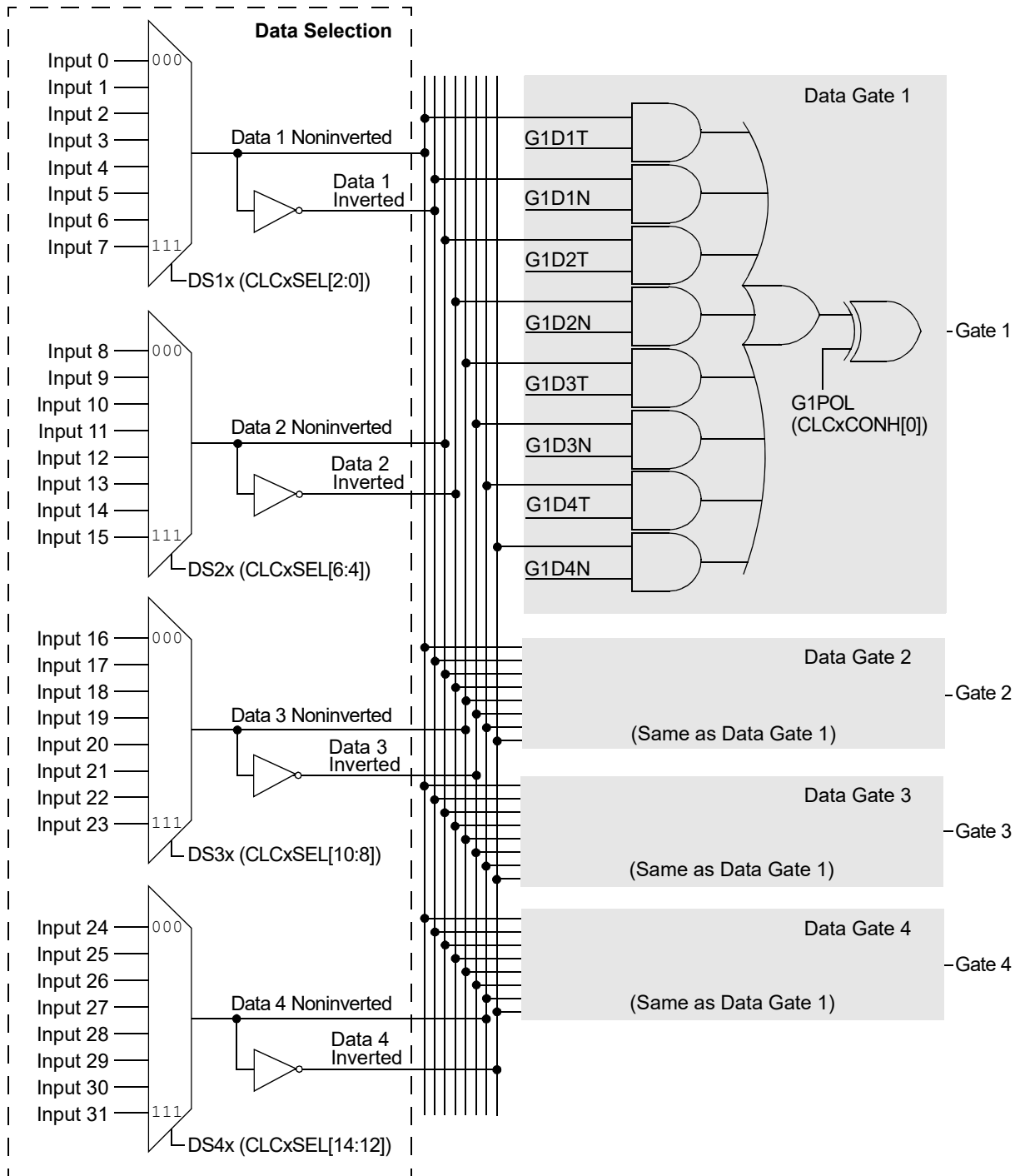
Figure 23-2. CLCx Logic Function Combinatorial Options



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Configurable Logic Cell (CLC)

Figure 23-3. CLCx Input Source Selection Diagram



Note: All controls are undefined at power-up.

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Configurable Logic Cell (CLC)

23.1 CLC Control Registers

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x00 ... 0xC1	Reserved									
0xC2	CLC1CONH	15:8 7:0					G4POL INTP	G3POL INTN	G2POL	G1POL
0xC4	CLC1CONL	15:8 7:0	LCEN LCOE	LCOUT	LCPOL			MODE[2:0]		
0xC6 ... 0xC7	Reserved									
0xC8	CLC1SEL	15:8 7:0		DS4[2:0] DS2[2:0]				DS3[2:0] DS1[2:0]		
0xC8	CLC1GLSL	15:8 7:0	G2D4T G1D4T	G2D4N G1D4N	G2D3T G1D3T	G2D3N G1D3N	G2D2T G1D2T	G2D2N G1D2N	G2D1T G1D1T	G2D1N G1D1N
0xCA	CLC1GLSH	15:8 7:0	G4D4T G3D4T	G4D4N G3D4N	G4D3T G3D3T	G4D3N G3D3N	G4D2T G3D2T	G4D2N G3D2N	G4D1T G3D1T	G4D1N G3D1N
0xCC ... 0xCD	Reserved									
0xCE	CLC2CONH	15:8 7:0					G4POL INTP	G3POL INTN	G2POL	G1POL
0xD0	CLC2CONL	15:8 7:0	LCEN LCOE	LCOUT	LCPOL			MODE[2:0]		
0xD2 ... 0xD3	Reserved									
0xD4	CLC2SEL	15:8 7:0		DS4[2:0] DS2[2:0]				DS3[2:0] DS1[2:0]		
0xD4	CLC2GLSL	15:8 7:0	G2D4T G1D4T	G2D4N G1D4N	G2D3T G1D3T	G2D3N G1D3N	G2D2T G1D2T	G2D2N G1D2N	G2D1T G1D1T	G2D1N G1D1N
0xD6	CLC2GLSH	15:8 7:0	G4D4T G3D4T	G4D4N G3D4N	G4D3T G3D3T	G4D3N G3D3N	G4D2T G3D2T	G4D2N G3D2N	G4D1T G3D1T	G4D1N G3D1N
0xD8 ... 0xD9	Reserved									
0xDA	CLC3CONH	15:8 7:0					G4POL INTP	G3POL INTN	G2POL	G1POL
0xDC	CLC3CONL	15:8 7:0	LCEN LCOE	LCOUT	LCPOL			MODE[2:0]		
0xDE ... 0xDF	Reserved									
0xE0	CLC3SEL	15:8 7:0		DS4[2:0] DS2[2:0]				DS3[2:0] DS1[2:0]		
0xE0	CLC3GLSL	15:8 7:0	G2D4T G1D4T	G2D4N G1D4N	G2D3T G1D3T	G2D3N G1D3N	G2D2T G1D2T	G2D2N G1D2N	G2D1T G1D1T	G2D1N G1D1N
0xE2	CLC3GLSH	15:8 7:0	G4D4T G3D4T	G4D4N G3D4N	G4D3T G3D3T	G4D3N G3D3N	G4D2T G3D2T	G4D2N G3D2N	G4D1T G3D1T	G4D1N G3D1N
0xE4 ... 0xE5	Reserved									
0xE6	CLC4CONH	15:8 7:0					G4POL INTP	G3POL INTN	G2POL	G1POL
0xE8	CLC4CONL	15:8 7:0	LCEN LCOE	LCOUT	LCPOL			MODE[2:0]		

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Configurable Logic Cell (CLC)

.....continued										
Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0xEA ... 0xEB	Reserved									
0xEC	CLC4SEL	15:8		DS4[2:0]				DS3[2:0]		
		7:0		DS2[2:0]				DS1[2:0]		
0xEC	CLC4GLSL	15:8	G2D4T	G2D4N	G2D3T	G2D3N	G2D2T	G2D2N	G2D1T	G2D1N
		7:0	G1D4T	G1D4N	G1D3T	G1D3N	G1D2T	G1D2N	G1D1T	G1D1N
0xEE	CLC4GLSH	15:8	G4D4T	G4D4N	G4D3T	G4D3N	G4D2T	G4D2N	G4D1T	G4D1N
		7:0	G3D4T	G3D4N	G3D3T	G3D3N	G3D2T	G3D2N	G3D1T	G3D1N

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Configurable Logic Cell (CLC)

23.1.1 CLCx Control Register Low

Name: CLCxCONL
Offset: 0x0C4, 0x0D0, 0x0DC, 0x0E8

Bit	15	14	13	12	11	10	9	8
	LCEN				INTP	INTN		
Access	R/W				R/W	R/W		
Reset	0				0	0		

Bit	7	6	5	4	3	2	1	0
	LCOE	LCOUT	LCPOL				MODE[2:0]	
Access	R/W	R	R/W			R/W	R/W	R/W
Reset	0	0	0			0	0	0

Bit 15 – LCEN CLCx Enable bit

Value	Description
1	CLCx is enabled and mixing input signals
0	CLCx is disabled and has logic zero outputs

Bit 11 – INTP CLCx Positive Edge Interrupt Enable bit

Value	Description
1	Interrupt will be generated when a rising edge occurs on LCOUT
0	Interrupt will not be generated

Bit 10 – INTN CLCx Negative Edge Interrupt Enable bit

Value	Description
1	Interrupt will be generated when a falling edge occurs on LCOUT
0	Interrupt will not be generated

Bit 7 – LCOE CLCx Port Enable bit

Value	Description
1	CLCx port pin output is enabled
0	CLCx port pin output is disabled

Bit 6 – LCOUT CLCx Data Output Status bit

Value	Description
1	CLCx output high
0	CLCx output low

Bit 5 – LCPOL CLCx Output Polarity Control bit

Value	Description
1	The output of the module is inverted
0	The output of the module is not inverted

Bits 2:0 – MODE[2:0] CLCx Mode bits

Value	Description
111	Single input transparent latch with S and R
110	JK flip-flop with R
101	Two-input D flip-flop with R
100	Single input D flip-flop with S and R
011	SR latch
010	Four-input AND
001	Four-input OR-XOR
000	Four-input AND-OR

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Configurable Logic Cell (CLC)

23.1.2 CLCx Control Register High

Name: CLCxCONH
Offset: 0x0C2, 0x0CE, 0x0DA, 0x0E6

Bit	15	14	13	12	11	10	9	8
Access								
Reset								

Bit	7	6	5	4	3	2	1	0
					G4POL	G3POL	G2POL	G1POL
Access					R/W	R/W	R/W	R/W
Reset					0	0	0	0

Bit 3 – G4POL Gate 4 Polarity Control bit

Value	Description
1	Channel 4 logic output is inverted when applied to the logic cell
0	Channel 4 logic output is not inverted

Bit 2 – G3POL Gate 3 Polarity Control bit

Value	Description
1	Channel 3 logic output is inverted when applied to the logic cell
0	Channel 3 logic output is not inverted

Bit 1 – G2POL Gate 2 Polarity Control bit

Value	Description
1	Channel 2 logic output is inverted when applied to the logic cell
0	Channel 2 logic output is not inverted

Bit 0 – G1POL Gate 1 Polarity Control bit

Value	Description
1	Channel 1 logic output is inverted when applied to the logic cell
0	Channel 1 logic output is not inverted

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Configurable Logic Cell (CLC)

23.1.3 CLCx Input MUX Select Register

Name: CLCxSEL
Offset: 0x0C8, 0x0D4, 0x0E0, 0x0EC

Bit	15	14	13	12	11	10	9	8
		DS4[2:0]				DS3[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		0	0	0		0	0	0

Bit	7	6	5	4	3	2	1	0
		DS2[2:0]				DS1[2:0]		
Access		R/W	R/W	R/W		R/W	R/W	R/W
Reset		0	0	0		0	0	0

Bits 14:12 – DS4[2:0] Data Selection MUX 4 Signal Selection bits

Value	Description
111	SCCP3 auxiliary out
110	SCCP1 auxiliary out
101	CLCIND RP pin
100	Reserved
011	SPI1 Input (SDIx)
010	Comparator 3 out
001	CLC2 output
000	PWM Event A

Bits 10:8 – DS3[2:0] Data Selection MUX 3 Signal Selection bits

Value	Description
111	SCCP4 OC out
110	SCCP3 OC out
101	CLC4 out
100	UART1 RX
011	SPI1 Output (SDOx)
010	Comparator 2 output
001	CLC1 output
000	CLCINC I/O pin

Bits 6:4 – DS2[2:0] Data Selection MUX 2 Signal Selection bits

Value	Description
111	SCCP2 OC out
110	SCCP1 OC out
101	Comparator 6 output
100	Comparator 5 output
011	UART1 TX
010	Comparator 1 output
001	Comparator 4 Output
000	CLCINB I/O pin

Bits 2:0 – DS1[2:0] Data Selection MUX 1 Signal Selection bits

Value	Description
111	SCCP4 auxiliary out
110	SCCP2 auxiliary out
101	Reserved
100	REFCLKO output
011	INTRC/LPRC clock source

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Configurable Logic Cell (CLC)

Value	Description
010	CLC3 out
001	System clock (F_{CY})
000	CLCINA I/O pin

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Configurable Logic Cell (CLC)

23.1.4 CLCx Gate Logic Input Select Low Register

Name: CLCxGLSL
Offset: 0x0C8, 0x0D4, 0x0E0, 0x0EC

Bit	15	14	13	12	11	10	9	8
	G2D4T	G2D4N	G2D3T	G2D3N	G2D2T	G2D2N	G2D1T	G2D1N
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	G1D4T	G1D4N	G1D3T	G1D3N	G1D2T	G1D2N	G1D1T	G1D1N
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – G2D4T Gate 2 Data Source 4 True Enable bit

Value	Description
1	Data Source 4 signal is enabled for Gate 2
0	Data Source 4 signal is disabled for Gate 2

Bit 14 – G2D4N Gate 2 Data Source 4 Negated Enable bit

Value	Description
1	Data Source 4 inverted signal is enabled for Gate 2
0	Data Source 4 inverted signal is disabled for Gate 2

Bit 13 – G2D3T Gate 2 Data Source 3 True Enable bit

Value	Description
1	Data Source 3 signal is enabled for Gate 2
0	Data Source 3 signal is disabled for Gate 2

Bit 12 – G2D3N Gate 2 Data Source 3 Negated Enable bit

Value	Description
1	Data Source 3 inverted signal is enabled for Gate 2
0	Data Source 3 inverted signal is disabled for Gate 2

Bit 11 – G2D2T Gate 2 Data Source 2 True Enable bit

Value	Description
1	Data Source 2 signal is enabled for Gate 2
0	Data Source 2 signal is disabled for Gate 2

Bit 10 – G2D2N Gate 2 Data Source 2 Negated Enable bit

Value	Description
1	Data Source 2 inverted signal is enabled for Gate 2
0	Data Source 2 inverted signal is disabled for Gate 2

Bit 9 – G2D1T Gate 2 Data Source 1 True Enable bit

Value	Description
1	Data Source 1 signal is enabled for Gate 2
0	Data Source 1 signal is disabled for Gate 2

Bit 8 – G2D1N Gate 2 Data Source 1 Negated Enable bit

Value	Description
1	Data Source 1 inverted signal is enabled for Gate 2
0	Data Source 1 inverted signal is disabled for Gate 2

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Bit 7 – G1D4T Gate 1 Data Source 4 True Enable bit

Value	Description
1	Data Source 4 signal is enabled for Gate 1
0	Data Source 4 signal is disabled for Gate 1

Bit 6 – G1D4N Gate 1 Data Source 4 Negated Enable bit

Value	Description
1	Data Source 4 inverted signal is enabled for Gate 1
0	Data Source 4 inverted signal is disabled for Gate 1

Bit 5 – G1D3T Gate 1 Data Source 3 True Enable bit

Value	Description
1	Data Source 3 signal is enabled for Gate 1
0	Data Source 3 signal is disabled for Gate 1

Bit 4 – G1D3N Gate 1 Data Source 3 Negated Enable bit

Value	Description
1	Data Source 3 inverted signal is enabled for Gate 1
0	Data Source 3 inverted signal is disabled for Gate 1

Bit 3 – G1D2T Gate 1 Data Source 2 True Enable bit

Value	Description
1	Data Source 2 signal is enabled for Gate 1
0	Data Source 2 signal is disabled for Gate 1

Bit 2 – G1D2N Gate 1 Data Source 2 Negated Enable bit

Value	Description
1	Data Source 2 inverted signal is enabled for Gate 1
0	Data Source 2 inverted signal is disabled for Gate 1

Bit 1 – G1D1T Gate 1 Data Source 1 True Enable bit

Value	Description
1	Data Source 1 signal is enabled for Gate 1
0	Data Source 1 signal is disabled for Gate 1

Bit 0 – G1D1N Gate 1 Data Source 1 Negated Enable bit

Value	Description
1	Data Source 1 inverted signal is enabled for Gate 1
0	Data Source 1 inverted signal is disabled for Gate 1

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Configurable Logic Cell (CLC)

23.1.5 CLCx Gate Logic Input Select High Register

Name: CLCxGLSH
Offset: 0x0CA, 0x0D6, 0x0E2, 0x0EE

Bit	15	14	13	12	11	10	9	8
	G4D4T	G4D4N	G4D3T	G4D3N	G4D2T	G4D2N	G4D1T	G4D1N
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	G3D4T	G3D4N	G3D3T	G3D3N	G3D2T	G3D2N	G3D1T	G3D1N
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 15 – G4D4T Gate 4 Data Source 4 True Enable bit

Value	Description
1	Data Source 4 signal is enabled for Gate 4
0	Data Source 4 signal is disabled for Gate 4

Bit 14 – G4D4N Gate 4 Data Source 4 Negated Enable bit

Value	Description
1	Data Source 4 inverted signal is enabled for Gate 4
0	Data Source 4 inverted signal is disabled for Gate 4

Bit 13 – G4D3T Gate 4 Data Source 3 True Enable bit

Value	Description
1	Data Source 3 signal is enabled for Gate 4
0	Data Source 3 signal is disabled for Gate 4

Bit 12 – G4D3N Gate 4 Data Source 3 Negated Enable bit

Value	Description
1	Data Source 3 inverted signal is enabled for Gate 4
0	Data Source 3 inverted signal is disabled for Gate 4

Bit 11 – G4D2T Gate 4 Data Source 2 True Enable bit

Value	Description
1	Data Source 2 signal is enabled for Gate 4
0	Data Source 2 signal is disabled for Gate 4

Bit 10 – G4D2N Gate 4 Data Source 2 Negated Enable bit

Value	Description
1	Data Source 2 inverted signal is enabled for Gate 4
0	Data Source 2 inverted signal is disabled for Gate 4

Bit 9 – G4D1T Gate 4 Data Source 1 True Enable bit

Value	Description
1	Data Source 1 signal is enabled for Gate 4
0	Data Source 1 signal is disabled for Gate 4

Bit 8 – G4D1N Gate 4 Data Source 1 Negated Enable bit

Value	Description
1	Data Source 1 inverted signal is enabled for Gate 4
0	Data Source 1 inverted signal is disabled for Gate 4

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Bit 7 – G3D4T Gate 3 Data Source 4 True Enable bit

Value	Description
1	Data Source 4 signal is enabled for Gate 3
0	Data Source 4 signal is disabled for Gate 3

Bit 6 – G3D4N Gate 3 Data Source 4 Negated Enable bit

Value	Description
1	Data Source 4 inverted signal is enabled for Gate 3
0	Data Source 4 inverted signal is disabled for Gate 3

Bit 5 – G3D3T Gate 3 Data Source 3 True Enable bit

Value	Description
1	Data Source 3 signal is enabled for Gate 3
0	Data Source 3 signal is disabled for Gate 3

Bit 4 – G3D3N Gate 3 Data Source 3 Negated Enable bit

Value	Description
1	Data Source 3 inverted signal is enabled for Gate 3
0	Data Source 3 inverted signal is disabled for Gate 3

Bit 3 – G3D2T Gate 3 Data Source 2 True Enable bit

Value	Description
1	Data Source 2 signal is enabled for Gate 3
0	Data Source 2 signal is disabled for Gate 3

Bit 2 – G3D2N Gate 3 Data Source 2 Negated Enable bit

Value	Description
1	Data Source 2 inverted signal is enabled for Gate 3
0	Data Source 2 inverted signal is disabled for Gate 3

Bit 1 – G3D1T Gate 3 Data Source 1 True Enable bit

Value	Description
1	Data Source 1 signal is enabled for Gate 3
0	Data Source 1 signal is disabled for Gate 3

Bit 0 – G3D1N Gate 3 Data Source 1 Negated Enable bit

Value	Description
1	Data Source 1 inverted signal is enabled for Gate 3
0	Data Source 1 inverted signal is disabled for Gate 3

24. Peripheral Trigger Generator (PTG)

Note: This data sheet summarizes the features of the dsPIC33CK512MPT608 family of devices. It is not intended to be a comprehensive reference source. To complement the information in this data sheet, refer to “**Peripheral Trigger Generator (PTG)**” (www.microchip.com/DS70000669) in the “*dsPIC33/PIC24 Family Reference Manual*”.

The dsPIC33CK512MPT608 family Peripheral Trigger Generator (PTG) module is a user-programmable sequencer that is capable of generating complex trigger signal sequences to coordinate the operation of other peripherals. The PTG module is designed to interface with other modules, such as an Analog-to-Digital Converter (ADC), output compare and PWM modules, timers and interrupt controllers.

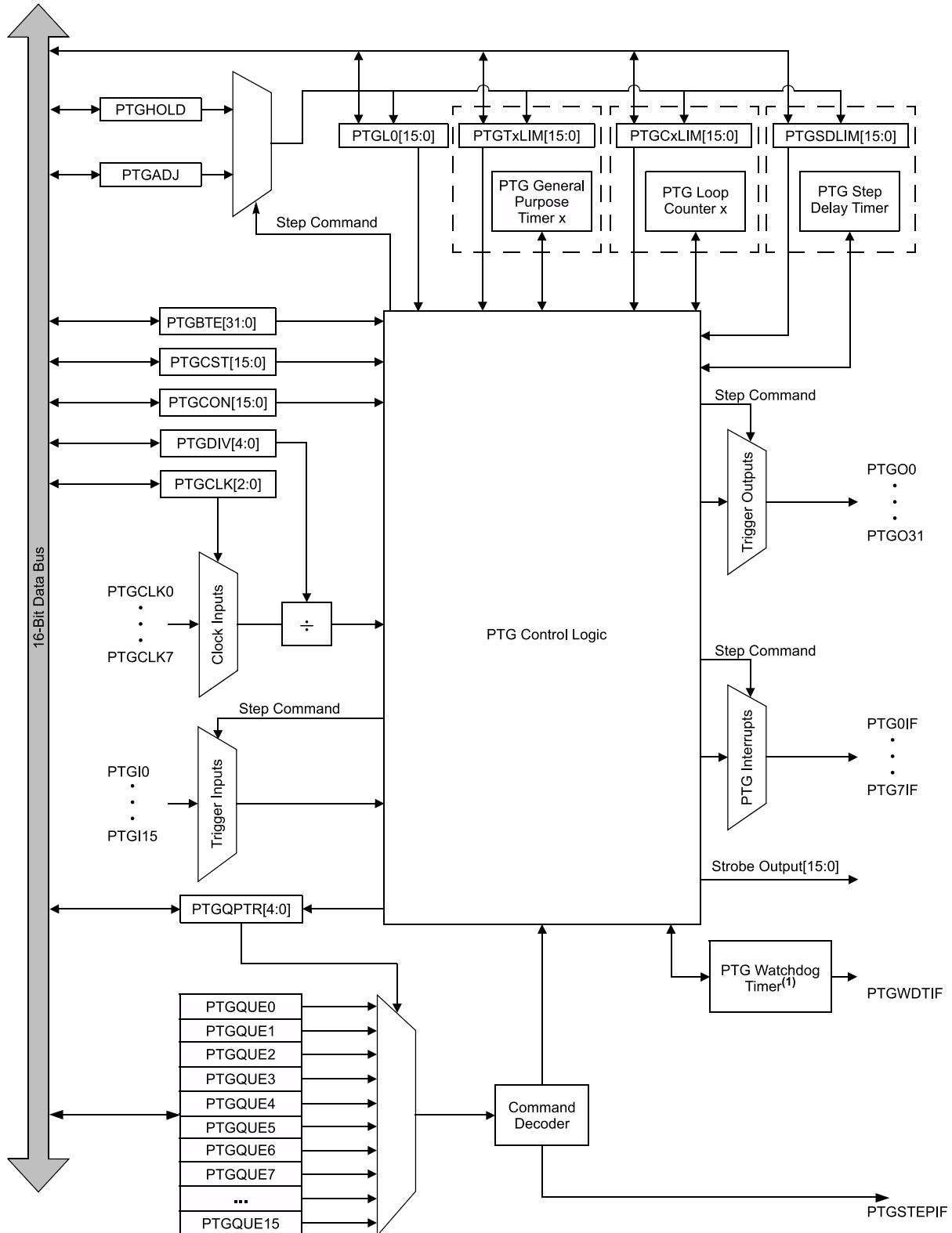
24.1 Features

- Behavior is Step Command Driven:
 - Step commands are eight bits wide
- Commands are Stored in a Step Queue:
 - Queue depth is up to 32 entries
 - Programmable Step execution time (Step delay)
- Supports the Command Sequence Loop:
 - Can be nested one-level deep
 - Conditional or unconditional loop
 - Two 16-bit loop counters
- 15 Hardware Input Triggers:
 - Sensitive to either positive or negative edges, or a high or low level
- One Software Input Trigger
- Generates up to 32 Unique Output Trigger Signals
- Generates Two Types of Trigger Outputs:
 - Individual
 - Broadcast
- Strobed Output Port for Literal Data Values:
 - 5-bit literal write (literal part of a command)
 - 16-bit literal write (literal held in the PTGL0 register)
- Generates up to Ten Unique Interrupt Signals
- Two 16-Bit General Purpose Timers
- Flexible Self-Contained Watchdog Timer (WDT) to Set an Upper Limit to Trigger Wait Time
- Single-Step Command Capability in Debug mode
- Selectable Clock (System, Pulse-Width Modulator (PWM) or ADC)
- Programmable Clock Divider

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Peripheral Trigger Generator (PTG)

Figure 24-1. PTG Block Diagram



Note:

1. This is a dedicated Watchdog Timer for the PTG module and is independent of the device Watchdog Timer.

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Peripheral Trigger Generator (PTG)

24.2 PTG Registers

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x00 ... 0x08FF	Reserved									
0x0900	PTGCST	15:8	PTGEN		PTGSIDL	PTGTOGL		PTGSWT	PTGSSEN	PTGIVIS
		7:0	PTGSTRT	PTGWDTO	PTGBUSY				PTGITM[1:0]	
0x0902	PTGCON	15:8	PTGCLK[2:0]			PTGDIV[4:0]				
		7:0	PTGPWD[3:0]					PTGWDT[2:0]		
0x0904	PTGBTEL(1)	15:8	PTGBTE[15:8]							
		7:0	PTGBTE[7:0]							
0x0906	PTGBTEH(1)	15:8	PTGBTE[31:24]							
		7:0		PTGBTE[23:17]						
0x0908	PTGHOLD(1)	15:8	PTGHOLD[15:8]							
		7:0	PTGHOLD[7:0]							
0x090A ... 0x090B	Reserved									
0x090C	PTGT0LIM(1)	15:8	PTGT0LIM[15:8]							
		7:0	PTGT0LIM[7:0]							
0x090E ... 0x090F	Reserved									
0x0910	PTGT1LIM(1)	15:8	PTGT1LIM[15:8]							
		7:0	PTGT1LIM[7:0]							
0x0912 ... 0x0913	Reserved									
0x0914	PTGSDLIM(1)	15:8	PTGSDLIM[15:8]							
		7:0	PTGSDLIM[7:0]							
0x0916 ... 0x0917	Reserved									
0x0918	PTGC0LIM(1)	15:8	PTGC0LIM[15:8]							
		7:0	PTGC0LIM[7:0]							
0x091A ... 0x091B	Reserved									
0x091C	PTGC1LIM(1)	15:8	PTGC1LIM[15:8]							
		7:0	PTGC1LIM[7:0]							
0x091E ... 0x091F	Reserved									
0x0920	PTGADJ(1)	15:8	PTGADJ[15:8]							
		7:0	PTGADJ[7:0]							
0x0922 ... 0x0923	Reserved									
0x0924	PTGL0(1,2)	15:8	PTGL0[15:8]							
		7:0	PTGL0[7:0]							
0x0926 ... 0x0927	Reserved									
0x0928	PTGQPTR(1)	15:8				PTGQPTR[4:0]				
		7:0								
0x092A ... 0x092F	Reserved									

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.....continued

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x0930	PTGQUE0(1)	15:8					STEP20+1[7:0]			
		7:0					STEP20[7:0]			
0x0932	PTGQUE1(1)	15:8					STEP21+1[7:0]			
		7:0					STEP21[7:0]			
0x0934	PTGQUE2(1)	15:8					STEP22+1[7:0]			
		7:0					STEP22[7:0]			
0x0936	PTGQUE3(1)	15:8					STEP23+1[7:0]			
		7:0					STEP23[7:0]			
0x0938	PTGQUE4(1)	15:8					STEP24+1[7:0]			
		7:0					STEP24[7:0]			
0x093A	PTGQUE5(1)	15:8					STEP25+1[7:0]			
		7:0					STEP25[7:0]			
0x093C	PTGQUE6(1)	15:8					STEP26+1[7:0]			
		7:0					STEP26[7:0]			
0x093E	PTGQUE7(1)	15:8					STEP27+1[7:0]			
		7:0					STEP27[7:0]			
0x0940	PTGQUE8(1)	15:8					STEP28+1[7:0]			
		7:0					STEP28[7:0]			
0x0942	PTGQUE9(1)	15:8					STEP29+1[7:0]			
		7:0					STEP29[7:0]			
0x0944	PTGQUE10(1)	15:8					STEP20+1[7:0]			
		7:0					STEP20[7:0]			
0x0946	PTGQUE11(1)	15:8					STEP211+1[7:0]			
		7:0					STEP211[7:0]			
0x0948	PTGQUE12(1)	15:8					STEP22+1[7:0]			
		7:0					STEP22[7:0]			
0x094A	PTGQUE13(1)	15:8					STEP23+1[7:0]			
		7:0					STEP23[7:0]			
0x094C	PTGQUE14(1)	15:8					STEP24+1[7:0]			
		7:0					STEP24[7:0]			
0x094E	PTGQUE15(1)	15:8					STEP25+1[7:0]			
		7:0					STEP25[7:0]			

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Peripheral Trigger Generator (PTG)

24.2.1 PTG Control/Status Low Register

Name: PTGCST

Offset: 0x900

Notes:

1. These bits apply to the PTGWHI and PTGWLO commands only.
2. This bit is only used with the PTGCTRL Step command software trigger option.
3. The PTGSSEN bit may only be written when in Debug mode.

Bit	15	14	13	12	11	10	9	8
	PTGEN		PTGSIDL	PTGTOGL		PTGSWT	PTGSSEN	PTGIVIS
Access	R/W		R/W	R/W		R/W	R/W	R/W
Reset	0		0	0		0	0	0

Bit	7	6	5	4	3	2	1	0
	PTGSTRT	PTGWDTO	PTGBUSY				PTGITM[1:0]	
Access	R/W	R/W	R/W				R/W	R/W
Reset	0	0	0				0	0

Bit 15 – PTGEN PTG Broadcast Trigger Enable bit

Value	Description
1	PTG is enabled
0	PTG is disabled

Bit 13 – PTGSIDL PTG Freeze in Debug Mode bit

Value	Description
1	Halts PTG operation when device is Idle
0	PTG operation continues when device is Idle

Bit 12 – PTGTOGL PTG Toggle Trigger Output bit

Value	Description
1	Toggles state of TRIG output for each execution of PTGTRIG
0	Generates a single TRIG pulse for each execution of PTGTRIG

Bit 10 – PTGSWT PTG Software Trigger bit⁽²⁾

Value	Description
1	If the PTG state machine is executing the “Wait for software trigger” Step command (OPTION[3:0] = 1010 or 1011), the command will complete and execution will continue
0	No action other than to clear the bit

Bit 9 – PTGSSEN PTG Single-Step Command bit⁽³⁾

Value	Description
1	Enables single Step when in Debug mode
0	Disables single Step

Bit 8 – PTGIVIS PTG Counter/Timer Visibility bit

Value	Description
1	Reading the PTGSDLIM, PTGCxLIM or PTGTxLIM register returns the current values of their corresponding Counter/Timer registers (PTGSDLIM, PTGCxLIM and PTGTxLIM)
0	Reading the PTGSDLIM, PTGCxLIM or PTGTxLIM register returns the value of these Limit registers

Bit 7 – PTGSTRT PTG Start Sequencer bit

Value	Description
1	Starts to sequentially execute the commands (Continuous mode)
0	Stops executing the commands

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Bit 6 – PTGWDTO PTG Watchdog Timer Time-out Status bit

Value	Description
1	PTG Watchdog Timer has timed out
0	PTG Watchdog Timer has not timed out

Bit 5 – PTGBUSY PTG State Machine Busy bit

Value	Description
1	PTG is running on the selected clock source; no SFR writes are allowed to PTGCLK[2:0] or PTGDIV[4:0]
0	PTG state machine is not running

Bits 1:0 – PTGITM[1:0] PTG Input Trigger Operation Selection bits⁽¹⁾

Value	Description
11	Single-level detect with Step delay not executed on exit of command (regardless of the PTGCTRL command) (Mode 3)
10	Single-level detect with Step delay executed on exit of command (Mode 2)
01	Continuous edge detect with Step delay not executed on exit of command (regardless of the PTGCTRL command) (Mode 1)
00	Continuous edge detect with Step delay executed on exit of command (Mode 0)

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Peripheral Trigger Generator (PTG)

24.2.2 PTG Control/Status Register

Name: PTGCON
Offset: 0x902

Bit	15	14	13	12	11	10	9	8
	PTGCLK[2:0]			PTGDIV[4:0]				
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PTGPWD[3:0]					PTGWDT[2:0]		
Access	R/W	R/W	R/W	R/W		R/W	R/W	R/W
Reset	0	0	0	0		0	0	0

Bits 15:13 – PTGCLK[2:0] PTG Module Clock Source Selection bits

Value	Description
111	CLC1 output
110	$F_{VCO}/4$
101	Reserved
100	Reserved
011	Input from Timer1 Clock pin, T1CK
010	ADC clock
001	F_{CY}
000	F_P

Bits 12:8 – PTGDIV[4:0] PTG Module Clock Prescaler (Divider) bits

Value	Description
11111	Divide-by-32
11110	Divide-by-31
. . .	
00001	Divide-by-2
00000	Divide-by-1

Bits 7:4 – PTGPWD[3:0] PTG Trigger Output Pulse-Width (in PTG clock cycles) bits

Value	Description
1111	All trigger outputs are 16 PTG clock cycles wide
1110	All trigger outputs are 15 PTG clock cycles wide
. . .	
0001	All trigger outputs are 2 PTG clock cycles wide
0000	All trigger outputs are 1 PTG clock cycle wide

Bits 2:0 – PTGWDT[2:0] PTG Watchdog Timer Time-out Selection bits

Value	Description
111	Watchdog Timer will time out after 512 PTG clocks
110	Watchdog Timer will time out after 256 PTG clocks
101	Watchdog Timer will time out after 128 PTG clocks
100	Watchdog Timer will time out after 64 PTG clocks
011	Watchdog Timer will time out after 32 PTG clocks
010	Watchdog Timer will time out after 16 PTG clocks
001	Watchdog Timer will time out after 8 PTG clocks
000	Watchdog Timer is disabled

24.2.3 PTG Broadcast Trigger Enable Register Low

Name: PTGBTEL⁽¹⁾

Offset: 0x904

Note:

- These bits are read-only when the module is executing Step commands.

Bit	15	14	13	12	11	10	9	8
	PTGBTE[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PTGBTE[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – PTGBTE[15:0] PTG Broadcast Trigger Enable bits

Value	Description
1	Generates trigger when the broadcast command is executed
0	Does not generate trigger when the broadcast command is executed

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Peripheral Trigger Generator (PTG)

24.2.4 PTG Broadcast Trigger Enable Low Register

Name: PTGBTEH⁽¹⁾

Offset: 0x906

Note:

- These bits are read-only when the module is executing Step commands.

Bit	15	14	13	12	11	10	9	8
	PTGBTE[31:24]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PTGBTE[23:17]							
Access		R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset		0	0	0	0	0	0	0

Bits 15:8 – PTGBTE[31:24] PTG Broadcast Trigger Enable bits

Value	Description
1	Generates trigger when the broadcast command is executed
0	Does not generate trigger when the broadcast command is executed

Bits 6:0 – PTGBTE[23:17] PTG Broadcast Trigger Enable bits

Value	Description
1	Generates trigger when the broadcast command is executed
0	Does not generate trigger when the broadcast command is executed

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Peripheral Trigger Generator (PTG)

24.2.5 PTG Hold Register

Name: PTGHOLD⁽¹⁾

Offset: 0x908

Note:

1. These bits are read-only when the module is executing Step commands.

Bit	15	14	13	12	11	10	9	8
	PTGHOLD[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PTGHOLD[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – PTGHOLD[15:0] PTG General Purpose Hold Register bits

This register holds the user-supplied data to be copied to the PTGTxLIM, PTGCxLIM, PTGSDLIM or PTGL0 register using the `PTGCOPY` command.

24.2.6 PTG Timer0 Limit Register

Name: PTGT0LIM⁽¹⁾

Offset: 0x90C

Note:

- These bits are read-only when the module is executing Step commands.

Bit	15	14	13	12	11	10	9	8
	PTGT0LIM[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PTGT0LIM[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – PTGT0LIM[15:0] PTG Timer0 Limit Register bits
General Purpose Timer0 Limit register.

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Peripheral Trigger Generator (PTG)

24.2.7 PTG Timer1 Limit Register

Name: PTGT1LIM⁽¹⁾

Offset: 0x910

Note:

1. These bits are read-only when the module is executing Step commands.

Bit	15	14	13	12	11	10	9	8
	PTGT1LIM[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PTGT1LIM[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – PTGT1LIM[15:0] PTG Timer1 Limit Register bits
General Purpose Timer1 Limit register.

24.2.8 PTG Step Delay Limit Register

Name: PTGSDLIM⁽¹⁾

Offset: 0x914

Note:

- These bits are read-only when the module is executing Step commands.

Bit	15	14	13	12	11	10	9	8
	PTGSDLIM[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PTGSDLIM[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – PTGSDLIM[15:0] PTG Step Delay Limit Register bits

This register holds a PTG Step delay value representing the number of additional PTG clocks between the start of a Step command and the completion of a Step command.

24.2.9 PTG Counter 0 Limit Register

Name: PTGC0LIM⁽¹⁾

Offset: 0x918

Note:

- These bits are read-only when the module is executing Step commands.

Bit	15	14	13	12	11	10	9	8
	PTGC0LIM[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PTGC0LIM[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – PTGC0LIM[15:0] PTG Counter 0 Limit Register bits

This register is used to specify the loop count for the PTGJMPC0 Step command or as a Limit register for the General Purpose Counter 0.

24.2.10 PTG Counter 1 Limit Register

Name: PTGC1LIM⁽¹⁾

Offset: 0x91C

Note:

1. These bits are read-only when the module is executing Step commands.

Bit	15	14	13	12	11	10	9	8
	PTGC1LIM[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PTGC1LIM[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – PTGC1LIM[15:0] PTG Counter 1 Limit Register bits

This register is used to specify the loop count for the PTGJMPC1 Step command or as a Limit register for the General Purpose Counter 1.

24.2.11 PTG Adjust Register

Name: PTGADJ⁽¹⁾

Offset: 0x920

Note:

- These bits are read-only when the module is executing Step commands.

Bit	15	14	13	12	11	10	9	8
	PTGADJ[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PTGADJ[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – PTGADJ[15:0] PTG Adjust Register bits

This register holds the user-supplied data to be added to the PTGTxLIM, PTGCxLIM, PTGSDLIM or PTGL0 register using the `PTGADD` command.

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24.2.12 PTG Literal 0 Register

Name: PTGL0^(1,2)

Offset: 0x924

Notes:

1. These bits are read-only when the module is executing Step commands.
2. The PTG strobe output is typically connected to the ADC Channel Select register. This allows the PTG to directly control ADC channel switching. See the specific device data sheet for connections of the PTG output.

Bit	15	14	13	12	11	10	9	8
	PTGL0[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PTGL0[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

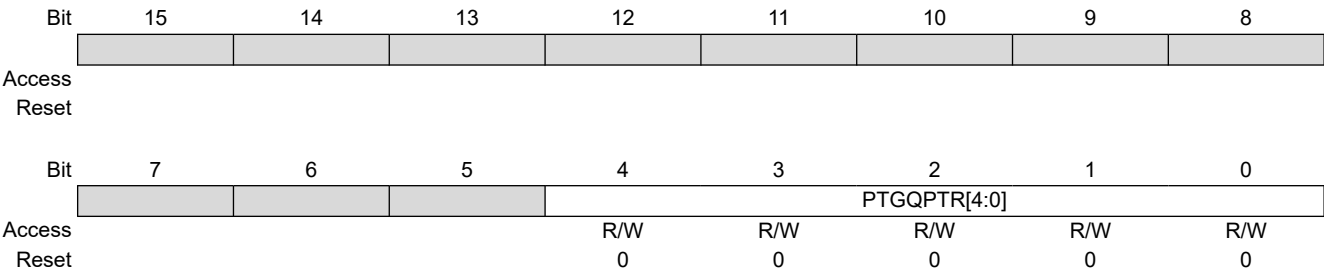
Bits 15:0 – PTGL0[15:0] PTG Literal 0 Register bits

This register holds the 6-bit value to be written to the CNVCHSEL[5:0] bits (ADCON3L[5:0]) with the `PTGCTRL` Step command.

24.2.13 PTG Step Queue Pointer Register

Name: PTGQPTR⁽¹⁾
Offset: 0x928

Note:
1. These bits are read-only when the module is executing Step commands.



Bits 4:0 – PTGQPTR[4:0] PTG Step Queue Pointer Register bits
This register points to the currently active Step command in the Step queue.

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24.2.14 PTG Step Queue n Pointer Register (n = 0-15)

Name: PTGQUEn⁽¹⁾
Offset: 0x930, 0x932, 0x934, 0x936, 0x938, 0x93A, 0x93C, 0x93E, 0x940, 0x942, 0x944, 0x946, 0x948, 0x94A, 0x94C, 0x94E

Notes:

- These bits are read-only when the module is executing Step commands.
- Refer to [24.3. PTG Step Commands](#) for the Step command encoding.

Bit	15	14	13	12	11	10	9	8
	STEP2n+1[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	STEP2n[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:8 – STEP2n+1[7:0] PTG Command 2n+1 bits⁽²⁾

A queue location for storage of the STEP2n+1 command byte, where 'n' is from PTGQUEn.

Bits 7:0 – STEP2n[7:0] PTG Command 2n bits⁽²⁾

A queue location for storage of the STEP2n command byte, where 'n' is the odd numbered Step Queue Pointers.

24.3 PTG Step Commands

Table 24-1. PTG Step Command Format and Description

Step Command Byte			
STEPx[7:0]			
CMD[3:0]		OPTION[3:0]	
bit 7	bit 4	bit 3	bit 0

Table 24-2. PTG Command Options

bit 7-4	Step Command	CMD[3:0]	Command Description
	PTGCTRL	0000	Execute the control command as described by the OPTION[3:0] bits.
	PTGADD	0001	Add contents of the PTGADJ register to the target register as described by the OPTION[3:0] bits.
	PTGCOPY		Copy contents of the PTGHOLD register to the target register as described by the OPTION[3:0] bits.
	PTGSTRB	001x	Copy the values contained in the bits, CMD[0]:OPTION[3:0], to the strobe output bits[4:0].
	PTGWHI	0100	Wait for a low-to-high edge input from a selected PTG trigger input as described by the OPTION[3:0] bits.
	PTGWLO	0101	Wait for a high-to-low edge input from a selected PTG trigger input as described by the OPTION[3:0] bits.
	—	0110	Reserved; do not use. ⁽¹⁾
	PTGIRQ	0111	Generate individual interrupt request as described by the OPTION[3:0] bits.
	PTGTRIG	100x	Generate individual trigger output as described by the bits, CMD[0]:OPTION[3:0].
	PTGJMP	101x	Copy the values contained in the bits, CMD[0]:OPTION[3:0], to the PTGQPTR register and jump to that Step queue.
	PTGJMPC0	110x	PTGC0 = PTGC0LIM: Increment the PTGQPTR register.
			PTGC0 ≠ PTGC0LIM: Increment Counter 0 (PTGC0) and copy the values contained in the bits, CMD[0]:OPTION[3:0], to the PTGQPTR register, and jump to that Step queue.
	PTGJMPC1	111x	PTGC1 = PTGC1LIM: Increment the PTGQPTR register.
			PTGC1 ≠ PTGC1LIM: Increment Counter 1 (PTGC1) and copy the values contained in the bits, CMD[0]:OPTION[3:0], to the PTGQPTR register, and jump to that Step queue.

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bit 3-0	PTGCTRL ⁽¹⁾	0000	NOP.
		0001	Reserved; do not use.
		0010	Disable Step delay timer (PTGSD).
		0011	Reserved; do not use.
		0100	Reserved; do not use.
		0101	Reserved; do not use.
		0110	Enable Step delay timer (PTGSD).
		0111	Reserved; do not use.
		1000	Start and wait for the PTG Timer0 to match the PTGT0LIM register.
		1001	Start and wait for the PTG Timer1 to match the PTGT1LIM register.
		1010	Wait for the software trigger (level, PTGSWT = 1).
		1011	Wait for the software trigger (positive edge, PTGSWT = 0 to 1).
	PTGADD ⁽¹⁾	1100	Copy the PTGC0LIM register contents to the strobe output.
		1101	Copy the PTGC1LIM register contents to the strobe output.
		1110	Copy the PTGL0 register contents to the strobe output.
		1111	Generate the triggers indicated in the PTGBTE register.
		0000	Add the PTGADJ register contents to the PTGC0LIM register.
		0001	Add the PTGADJ register contents to the PTGC1LIM register.
		0010	Add the PTGADJ register contents to the PTGT0LIM register.
		0011	Add the PTGADJ register contents to the PTGT1LIM register.
	PTGCOPY ⁽¹⁾	0100	Add the PTGADJ register contents to the PTGSDLIM register.
		0101	Add the PTGADJ register contents to the PTGL0 register.
		0110	Reserved; do not use.
		0111	Reserved; do not use.
		1000	Copy the PTGHOLD register contents to the PTGC0LIM register.
		1001	Copy the PTGHOLD register contents to the PTGC1LIM register.
		1010	Copy the PTGHOLD register contents to the PTGT0LIM register.
		1011	Copy the PTGHOLD register contents to the PTGT1LIM register.
	PTGWHI ⁽¹⁾ or PTGWLO ⁽¹⁾	1100	Copy the PTGHOLD register contents to the PTGSDLIM register.
		1101	Copy the PTGHOLD register contents to the PTGL0 register.
		1110	Reserved; do not use.
		1111	Reserved; do not use.
		0000	PTGI0 (see Table 24-3 for input assignments).
		...	...
		1111	PTGI15 (see Table 24-3 for input assignments).
	PTGIRQ ⁽¹⁾	0000	Generate PTG Interrupt 0.
		...	...
		0111	Generate PTG Interrupt 7.
		1000	Reserved; do not use.
		...	...
		1111	Reserved; do not use.
	PTGTRIG	0000	PTGO0 (see Table 24-4 for input assignments).
		0001	PTGO1 (see Table 24-4 for input assignments).
		...	...
		1110	PTGO30 (see Table 24-4 for input assignments).
	PTGWHI ⁽¹⁾ or PTGWLO ⁽¹⁾	1111	PTGO31 (see Table 24-4 for input assignments).
		0000	PTGI0 (see specific device data sheet for interrupt assignments).
		...	...
		1111	PTGI15 (see specific device data sheet for interrupt assignments).
	PTGIRQ ⁽¹⁾	0000	Generate PTG Interrupt 0 (see specific device data sheet for interrupt assignments).
		...	...
		0111	Generate PTG Interrupt 7 (see specific device data sheet for interrupt assignments).
		1000	Reserved; do not use.
		...	...
		1111	Reserved; do not use.
	PTGTRIG	0000	PTGO0 (see specific device data sheet for interrupt assignments).
		0001	PTGO1 (see specific device data sheet for interrupt assignments).

Note:

1. All reserved commands or options will execute, but they do not have any affect (i.e., execute as a NOP instruction).

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Peripheral Trigger Generator (PTG)

Table 24-3. PTG Input Descriptions

PTG Input Number	PTG Input Description
PTG Trigger Input 0	Trigger Input from PWM1 ADC Trigger 2
PTG Trigger Input 1	Trigger Input from PWM2 ADC Trigger 2
PTG Trigger Input 2	Trigger Input from PWM3 ADC Trigger 2
PTG Trigger Input 3	Trigger Input from PWM4 ADC Trigger 2
PTG Trigger Input 4	Reserved
PTG Trigger Input 5	Trigger Input from PWM6 ADC Trigger 2
PTG Trigger Input 6	Reserved
PTG Trigger Input 7	Trigger Input from SCCP4 OCMP Output
PTG Trigger Input 8	Trigger Input from SCCP5 OCMP Output
PTG Trigger Input 9	Trigger Input from Comparator 1
PTG Trigger Input 10	Trigger Input from Comparator 2
PTG Trigger Input 11	Trigger Input from Comparator 3
PTG Trigger Input 12	Trigger Input from CLC1
PTG Trigger Input 13	Trigger Input ADC Done Group Interrupt
PTG Trigger Input 14	Trigger Input from CLC2
PTG Trigger Input 15	Trigger Input from INT2 PPS

Table 24-4. PTG Output Descriptions

PTG Output Number	PTG Output Description
PTGO0 to PTGO11	Reserved
PTGO12	ADC TRGSRC[30]
PTGO13 to PTGO23	Reserved
PTGO24	PPS Output RP46
PTGO25	PPS Output RP47
PTGO26	PPS Input RP6
PTGO27	PPS Input RP7
PTGO28 to PTGO31	Reserved

25. 32-Bit Programmable Cyclic Redundancy Check (CRC) Generator

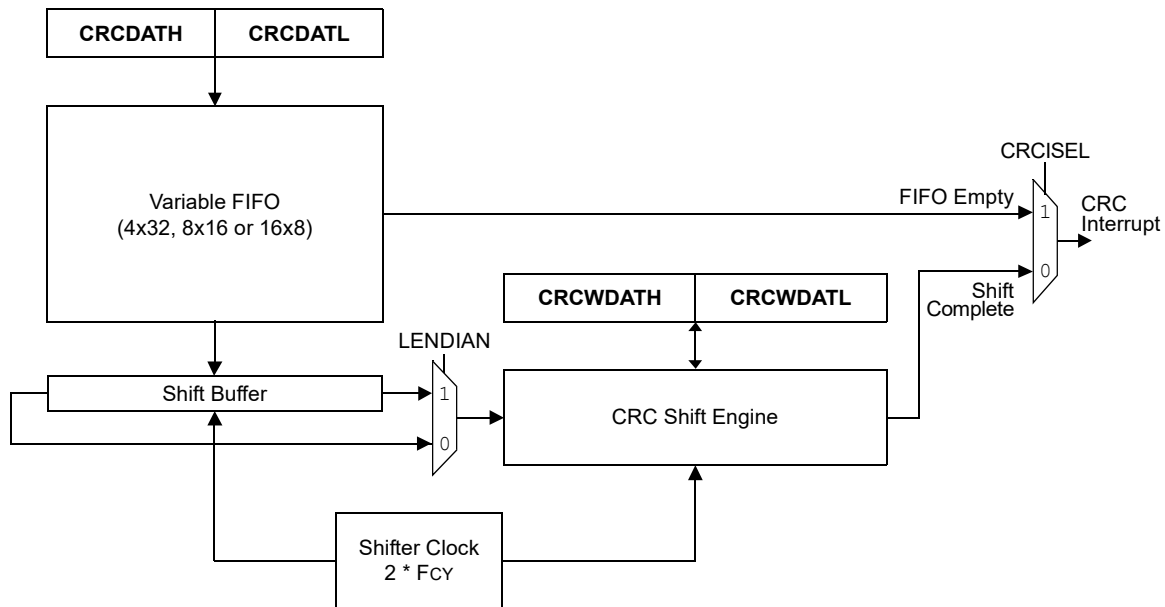
Note: This data sheet summarizes the features of the dsPIC33CK512MPT608 family of devices. It is not intended to be a comprehensive reference source. For more information, refer to “**32-Bit Programmable Cyclic Redundancy Check (CRC)**” (www.microchip.com/DS30009729) in the “dsPIC33/PIC24 Family Reference Manual”.

The 32-bit programmable CRC generator provides a hardware implemented method of quickly generating checksums for various networking and security applications. It offers the following features:

- User-Programmable CRC Polynomial Equation, Up to 32 Bits
- Programmable Shift Direction (little or big-endian)
- Independent Data and Polynomial Lengths
- Configurable Interrupt Output
- Data FIFO

Figure 25-1 displays a simplified block diagram of the CRC generator.

Figure 25-1. CRC Module Block Diagram



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32-Bit Programmable Cyclic Redundancy Check ...

25.1 CRC Control Registers

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x00 ... 0xAF	Reserved									
0xB0	CRCCONL	15:8	CRCEN		CSIDL	VWORD[4:0]				
		7:0	CRCFUL	CRCMPT	CRCISEL	CRCGO	LENDIAN	MOD		
0xB2	CRCCONH	15:8				DWIDTH[4:0]				
		7:0				PLEN[4:0]				
0xB4	CRCXORL	15:8	X[15:8]							
		7:0	X[7:1]							
0xB6	CRCXORH	15:8	X[31:24]							
		7:0	X[23:16]							

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32-Bit Programmable Cyclic Redundancy Check ...

25.1.1 CRC Control Register Low

Name: CRCCONL
Offset: 0x0B0

Legend: HC = Hardware Clearable bit, HSC = Hardware Settable/Clearable bit

Bit	15	14	13	12	11	10	9	8
	CRCEN		CSIDL	VWORD[4:0]				
Access	R/W		R/W	HSC/R	HSC/R	HSC/R	HSC/R	HSC/R
Reset	0		0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	CRCFUL	CRCMPT	CRCISEL	CRCGO	LENDIAN	MOD		
Access	HSC/R	HSC/R	R/W	HC/R/W	R/W	R/W		
Reset	0	1	0	0	0	0		

Bit 15 – CRCEN CRC Enable bit

Value	Description
1	Enables module
0	Disables module

Bit 13 – CSIDL CRC Stop in Idle Mode bit

Value	Description
1	Discontinues module operation when device enters Idle mode
0	Continues module operation in Idle mode

Bits 12:8 – VWORD[4:0] Pointer Value bits

Indicates the number of valid words in the FIFO. Has a maximum value of 8 when PLEN[4:0] ≥ 7 or 16 when PLEN[4:0] ≤ 7.

Bit 7 – CRCFUL CRC FIFO Full bit

Value	Description
1	FIFO is full
0	FIFO is not full

Bit 6 – CRCMPT CRC FIFO Empty bit

Value	Description
1	FIFO is empty
0	FIFO is not empty

Bit 5 – CRCISEL CRC Interrupt Selection bit

Value	Description
1	Interrupt on FIFO is empty; the final word of data is still shifting through the CRC
0	Interrupt on shift is complete and results are ready

Bit 4 – CRCGO CRC Start bit

Value	Description
1	Starts CRC serial shifter
0	CRC serial shifter is turned off

Bit 3 – LENDIAN Data Shift Direction Select bit

Value	Description
1	Data word is shifted into the FIFO, starting with the LSb (little-endian)
0	Data word is shifted into the FIFO, starting with the MSb (big-endian)

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32-Bit Programmable Cyclic Redundancy Check ...

Bit 2 – MOD CRC Calculation Mode bit

Value	Description
1	Alternate mode
0	Legacy mode bit

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32-Bit Programmable Cyclic Redundancy Check ...

25.1.2 CRC Control Register High

Name: CRCCONH
Offset: 0x0B2

Bit	15	14	13	12	11	10	9	8
				DWIDTH[4:0]				
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
				PLEN[4:0]				
Access				R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0

Bits 12:8 – DWIDTH[4:0] Data Word Width Configuration bits
Configures the width of the data word (Data Word Width – 1).

Bits 4:0 – PLEN[4:0] Polynomial Length Configuration bits
Configures the length of the polynomial (Polynomial Length – 1).

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32-Bit Programmable Cyclic Redundancy Check ...

25.1.3 CRC XOR Polynomial Register, Low Byte

Name: CRCXORL
Offset: 0x0B4

Bit	15	14	13	12	11	10	9	8
	X[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	X[7:1]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Reset	0	0	0	0	0	0	0	

Bits 15:8 – X[15:8] XOR of Polynomial Term x^n Enable bits

Bits 7:1 – X[7:1] XOR of Polynomial Term x^n Enable bits

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32-Bit Programmable Cyclic Redundancy Check ...

25.1.4 CRC XOR Polynomial Register, High Byte

Name: CRCXORH
Offset: 0x0B6

Bit	15	14	13	12	11	10	9	8
	X[31:24]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	X[23:16]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – X[31:16] XOR of Polynomial Term x^n Enable bits

26. Current Bias Generator (CBG)

Note: This data sheet summarizes the features of the dsPIC33CK512MPT608 family of devices. It is not intended to be a comprehensive reference source. To complement the information in this data sheet, refer to “**Current Bias Generator (CBG)**” (www.microchip.com/DS70005253) in the “*dsPIC33/PIC24 Family Reference Manual*”.

The Current Bias Generator (CBG) consists of two classes of current sources: 10 μA and 50 μA sources. The major features of each current source are:

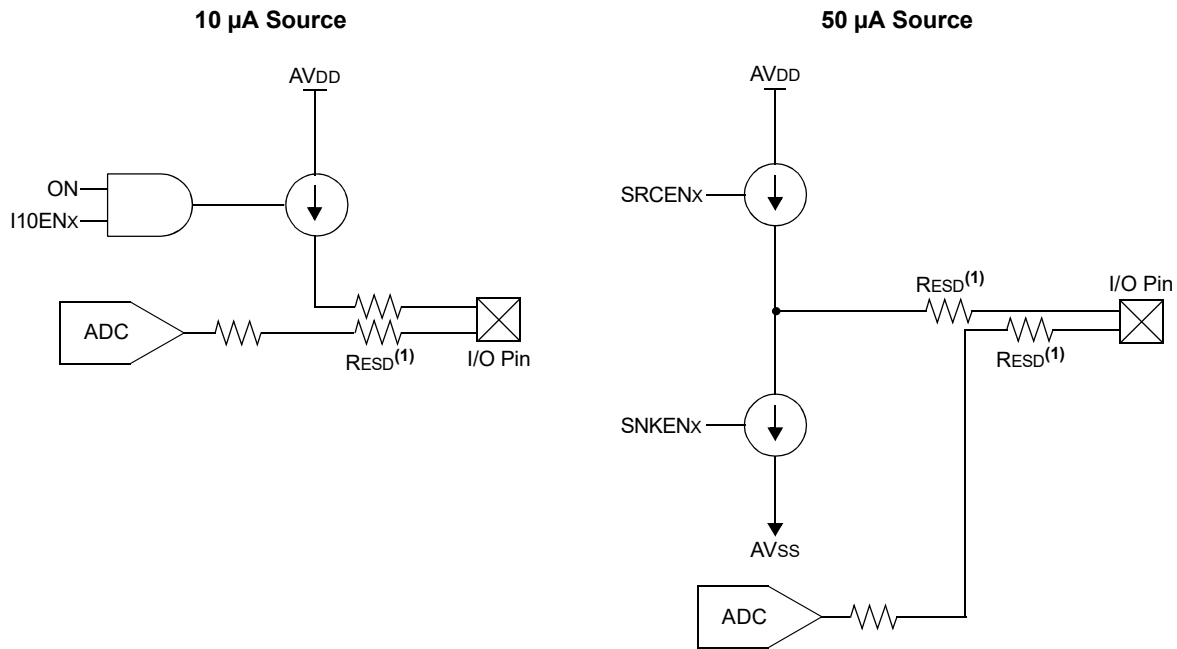
- 10 μA Current Sources:
 - Current sourcing only
 - Up to four independent sources
- 50 μA Current Sources:
 - Selectable current sourcing or sinking
 - Selectable current mirroring for sourcing and sinking

Table 26-1. CBG Channel Availability

Package Type	ISRCx	IBIASx
100-Pin	0,1,2,3	0,1,2,3

A simplified block diagram of the CBG module is shown in [Figure 26-1](#).

Figure 26-1. Constant-Current Source Module Block Diagram⁽²⁾



Notes:

1. RESD is typically 350 Ohms.
2. The ADC analog input is shown only for clarity. Each analog peripheral connected to the pin has a separate Electrostatic Discharge (ESD) resistor.

26.1 Current Bias Generator Control Registers

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x00 ... 0x08EF	Reserved									
0x08F0	BIASCON	15:8	ON							
		7:0					I10EN3	I10EN2	I10EN1	I10EN0
0x08F2 ... 0x08F3	Reserved									
0x08F4	IBIASCON0L	15:8			SHRSRCEN1	SHRSNKEN1	GENSRCEN1	GENSNKEN1	SRCEN1	SNKEN1
		7:0			SHRSRCEN0	SHRSNKEN0	GENSRCEN0	GENSNKEN0	SRCEN0	SNKEN0
0x08F6	IBIASCON0H	15:8			SHRSRCEN3	SHRSNKEN3	GENSRCEN3	GENSNKEN3	SRCEN3	SNKEN3
		7:0			SHRSRCEN2	SHRSNKEN2	GENSRCEN2	GENSNKEN2	SRCEN2	SNKEN2

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Current Bias Generator (CBG)

26.1.1 Current Bias Generator Control Register

Name: BIASCON
Offset: 0x8F0

Bit	15	14	13	12	11	10	9	8
	ON							
Access	R/W							
Reset	0							

Bit	7	6	5	4	3	2	1	0
					I10EN3	I10EN2	I10EN1	I10EN0
Access					R/W	R/W	R/W	R/W
Reset					0	0	0	0

Bit 15 – ON Current Bias Module Enable bit

Value	Description
1	Module is enabled
0	Module is disabled

Bit 3 – I10EN3 10 μ A Enable for Output 3 bit

Value	Description
1	10 μ A output is enabled
0	10 μ A output is disabled

Bit 2 – I10EN2 10 μ A Enable for Output 2 bit

Value	Description
1	10 μ A output is enabled
0	10 μ A output is disabled

Bit 1 – I10EN1 10 μ A Enable for Output 1 bit

Value	Description
1	10 μ A output is enabled
0	10 μ A output is disabled

Bit 0 – I10EN0 10 μ A Enable for Output 0 bit

Value	Description
1	10 μ A output is enabled
0	10 μ A output is disabled

26.1.2 Current Bias Generator 50 μ A Current Source Control Low Register

Name: IBIASCON0L

Offset: 0x8F4

Bit	15	14	13	12	11	10	9	8
			SHRSRCEN1	SHRSNKEN1	GENSRCEN1	GENSNKEN1	SRCEN1	SNKEN1
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
			SHRSRCEN0	SHRSNKEN0	GENSRCEN0	GENSNKEN0	SRCEN0	SNKEN0
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bit 13 – SHRSRCEN1 Share Source Enable for Output #1 bit

Value	Description
1	Sourcing Current Mirror mode is enabled (uses reference from another source)
0	Sourcing Current Mirror mode is disabled

Bit 12 – SHRSNKEN1 Share Sink Enable for Output #1 bit

Value	Description
1	Sinking Current Mirror mode is enabled (uses reference from another source)
0	Sinking Current Mirror mode is disabled

Bit 11 – GENSRCEN1 Generated Source Enable for Output #1 bit

Value	Description
1	Source generates the current source mirror reference
0	Source does not generate the current source mirror reference

Bit 10 – GENSNKEN1 Generated Sink Enable for Output #1 bit

Value	Description
1	Source generates the current source mirror reference
0	Source does not generate the current source mirror reference

Bit 9 – SRCEN1 Source Enable for Output #1 bit

Value	Description
1	Current source is enabled
0	Current source is disabled

Bit 8 – SNKEN1 Sink Enable for Output #1 bit

Value	Description
1	Current sink is enabled
0	Current sink is disabled

Bit 5 – SHRSRCEN0 Share Source Enable for Output #0 bit

Value	Description
1	Sourcing Current Mirror mode is enabled (uses reference from another source)
0	Sourcing Current Mirror mode is disabled

Bit 4 – SHRSNKEN0 Share Sink Enable for Output #0 bit

Value	Description
1	Sinking Current Mirror mode is enabled (uses reference from another source)
0	Sinking Current Mirror mode is disabled

dsPIC33CK512MPT608 Family

Current Bias Generator (CBG)

Bit 3 – GENSRCEN0 Generated Source Enable for Output #0 bit

Value	Description
1	Source generates the current source mirror reference
0	Source does not generate the current source mirror reference

Bit 2 – GENSNKEN0 Generated Sink Enable for Output #0 bit

Value	Description
1	Source generates the current source mirror reference
0	Source does not generate the current source mirror reference

Bit 1 – SRCEN0 Source Enable for Output #0 bit

Value	Description
1	Current source is enabled
0	Current source is disabled

Bit 0 – SNKEN0 Sink Enable for Output #0 bit

Value	Description
1	Current sink is enabled
0	Current sink is disabled

26.1.3 Current Bias Generator 50 μ A Current Source Control High Register**Name:** IBIASCON0H**Offset:** 0x8F6

Bit	15	14	13	12	11	10	9	8
			SHRSRCEN3	SHRSNKEN3	GENSRCEN3	GENSNKEN3	SRCEN3	SNKEN3
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
			SHRSRCEN2	SHRSNKEN2	GENSRCEN2	GENSNKEN2	SRCEN2	SNKEN2
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bit 13 – SHRSRCEN3 Share Source Enable for Output #3 bit

Value	Description
1	Sourcing Current Mirror mode is enabled (uses reference from another source)
0	Sourcing Current Mirror mode is disabled

Bit 12 – SHRSNKEN3 Share Sink Enable for Output #3 bit

Value	Description
1	Sinking Current Mirror mode is enabled (uses reference from another source)
0	Sinking Current Mirror mode is disabled

Bit 11 – GENSRCEN3 Generated Source Enable for Output #3 bit

Value	Description
1	Source generates the current source mirror reference
0	Source does not generate the current source mirror reference

Bit 10 – GENSNKEN3 Generated Sink Enable for Output #3 bit

Value	Description
1	Source generates the current source mirror reference
0	Source does not generate the current source mirror reference

Bit 9 – SRCEN3 Source Enable for Output #3 bit

Value	Description
1	Current source is enabled
0	Current source is disabled

Bit 8 – SNKEN3 Sink Enable for Output #3 bit

Value	Description
1	Current sink is enabled
0	Current sink is disabled

Bit 5 – SHRSRCEN2 Share Source Enable for Output #2 bit

Value	Description
1	Sourcing Current Mirror mode is enabled (uses reference from another source)
0	Sourcing Current Mirror mode is disabled

Bit 4 – SHRSNKEN2 Share Sink Enable for Output #2 bit

Value	Description
1	Sinking Current Mirror mode is enabled (uses reference from another source)
0	Sinking Current Mirror mode is disabled

dsPIC33CK512MPT608 Family

Current Bias Generator (CBG)

Bit 3 – GENSRCEN2 Generated Source Enable for Output #2 bit

Value	Description
1	Source generates the current source mirror reference
0	Source does not generate the current source mirror reference

Bit 2 – GENSNKEN2 Generated Sink Enable for Output #2 bit

Value	Description
1	Source generates the current source mirror reference
0	Source does not generate the current source mirror reference

Bit 1 – SRCEN2 Source Enable for Output #2 bit

Value	Description
1	Current source is enabled
0	Current source is disabled

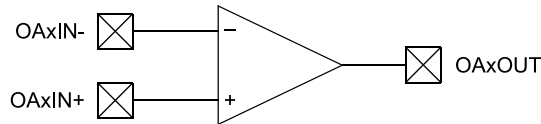
Bit 0 – SNKEN2 Sink Enable for Output #2 bit

Value	Description
1	Current sink is enabled
0	Current sink is disabled

27. Operational Amplifier

The dsPIC33CK512MPT608 family implements three instances of operational amplifiers (op amps). The op amps can be used for a wide variety of purposes, including signal conditioning and filtering. The three op amps are functionally identical. The block diagram for a single amplifier is shown in [Figure 27-1](#).

Figure 27-1. Single Operational Amplifier Block Diagram



The op amps are controlled by two SFR registers: AMPCON1L and AMPCON1H. They remain in a Low-Power state until the AMPON bit is set. Each op amp can then be enabled independently by setting the corresponding AMPENx bit (x = 1, 2, 3).

The NCHDISx bit provides some flexibility regarding input range versus Integral Nonlinearity (INL). When NCHDISx = 0 (default), the op amps have a wider input voltage range (see [33.2. AC Characteristics and Timing Parameters](#)). When NCHDISx = 1, the wider input range is traded for improved INL performance (lower INL).

27.1 Operational Amplifier Control Registers

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x00 ... 0x08DB	Reserved									
0x08DC	AMPCON1L	15:8	AMPON							
		7:0						AMPEN3	AMPEN2	AMPEN1
0x08DE	AMPCON1H	15:8								
		7:0						NCHDIS3	NCHDIS2	NCHDIS1

27.1.1 Op Amp Control Register Low

Name: AMPCON1L
Offset: 0x8DC

Bit	15	14	13	12	11	10	9	8
	AMPON							
Access	R/W							
Reset	0							

Bit	7	6	5	4	3	2	1	0
						AMPEN3	AMPEN2	AMPEN1
Access						R/W	R/W	R/W
Reset						0	0	0

Bit 15 – AMPON Op Amp Enable/On bit

Value	Description
1	Enables op amp modules if their respective AMPENx bits are also asserted
0	Disables all op amp modules

Bit 2 – AMPEN3 Op Amp #3 Enable bit

Value	Description
1	Enables Op Amp #3 if the AMPON bit is also asserted
0	Disables Op Amp #3

Bit 1 – AMPEN2 Op Amp #2 Enable bit

Value	Description
1	Enables Op Amp #2 if the AMPON bit is also asserted
0	Disables Op Amp #2

Bit 0 – AMPEN1 Op Amp #1 Enable bit

Value	Description
1	Enables Op Amp #1 if the AMPON bit is also asserted
0	Disables Op Amp #1

dsPIC33CK512MPT608 Family

Operational Amplifier

27.1.2 Op Amp Control Register High

Name: AMPCON1H

Offset: 0x8DE

Bit	15	14	13	12	11	10	9	8
Access								
Reset								

Bit	7	6	5	4	3	2	1	0
						NCHDIS3	NCHDIS2	NCHDIS1
Access						R/W	R/W	R/W
Reset						0	0	0

Bit 2 – NCHDIS3 Op Amp #3 N Channel Disable bit

Value	Description
1	Disables Op Amp #3 N channels input stage; reduced INL, but lowered input voltage range
0	Wide input range for Op Amp #3

Bit 1 – NCHDIS2 Op Amp #2 N Channel Disable bit

Value	Description
1	Disables Op Amp #2 N channels input stage; reduced INL, but lowered input voltage range
0	Wide input range for Op Amp #2

Bit 0 – NCHDIS1 Op Amp #1 N Channel Disable bit

Value	Description
1	Disables Op Amp #1 N channels input stage; reduced INL, but lowered input voltage range
0	Wide input range for Op Amp #1

28. Deadman Timer (DMT)

Note: This data sheet summarizes the features of the dsPIC33CK512MPT608 family of devices. It is not intended to be a comprehensive reference source. To complement the information in this data sheet, refer to “**Deadman Timer (DMT)**” (www.microchip.com/DS70005155) in the “*dsPIC33/PIC24 Family Reference Manual*”.

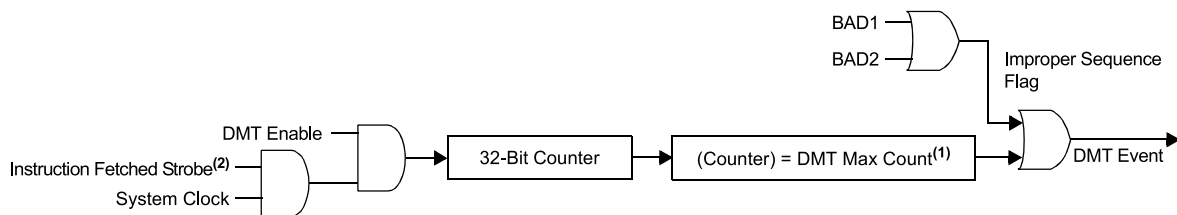
The primary function of the Deadman Timer (DMT) is to interrupt the processor in the event of a software malfunction. The DMT, which works on the system clock, is a free-running instruction fetch timer, which is clocked whenever an instruction fetch occurs, until a count match occurs. Instructions are not fetched when the processor is in Sleep mode.

DMT can be enabled in the Configuration fuse or by software in the DMTCON register by setting the ON bit. The DMT consists of a 32-bit counter with a time-out count match value, as specified by the two 16-bit Configuration Fuse registers: FDMTCNTL and FDMTCNTH.

A DMT is typically used in mission-critical and safety-critical applications, where any single failure of the software functionality and sequencing must be detected.

Figure 28-1 shows a block diagram of the Deadman Timer module.

Figure 28-1. Deadman Timer Block Diagram



Notes:

1. DMT Max Count is controlled by the initial value of the FDMTCNTL and FDMTCNTH Configuration registers.
2. DMT window interval is controlled by the value of the FDMTIVTL and FDMTIVTH Configuration registers.

28.1 Deadman Timer Control/Status Registers

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x00 ... 0x5B	Reserved									
0x5C	DMTCON	15:8 7:0	ON							
0x5E ... 0x5F	Reserved									
0x60	DMTPRECLR	15:8 7:0	STEP1[7:0]							
0x62 ... 0x63	Reserved									
0x64	DMTCLR	15:8 7:0	STEP2[7:0]							
0x66 ... 0x67	Reserved									
0x68	DMTSTAT	15:8 7:0	BAD1	BAD2	DMTEVENT					WINOPN
0x6A ... 0x6B	Reserved									
0x6C	DMTCNTL	15:8 7:0	COUNTER[15:8] COUNTER[7:0]							
0x6E	DMTCNTH	15:8 7:0	COUNTER[31:24] COUNTER[23:16]							
0x70	HOLDREG(1)	15:8 7:0	UPRCNT[15:8] UPRCNT[7:0]							
0x72 ... 0x73	Reserved									
0x74	PSCNTL	15:8 7:0	PSCNT[15:8] PSCNT[7:0]							
0x76	PSCNTH	15:8 7:0	PSCNT[31:24] PSCNT[23:16]							
0x78	PSINTVL	15:8 7:0	PSINTV[15:8] PSINTV[7:0]							
0x7A	PSINTVH	15:8 7:0	PSINTV[31:24] PSINTV[23:16]							

28.1.1 Deadman Timer Control Register**Name:** DMTCON**Offset:** 0x05C**Legend:** SO = Settable Only bit**Notes:**

1. This bit has control only when DMTDIS = 0 in the FDMT register.
2. DMT cannot be disabled in software. Writing '0' to this bit has no effect.

Bit	15	14	13	12	11	10	9	8
	ON							
Access	R/SO							
Reset	0							
Bit	7	6	5	4	3	2	1	0
Access								
Reset								

Bit 15 – ON DMT Enable bit^(1,2)

Value	Description
1	Deadman Timer module is enabled
0	Deadman Timer module is not enabled

28.1.2 Deadman Timer Preclear Register**Name:** DMTPRECLR**Offset:** 0x060**Note:**

- Bits 15:8 are cleared when the DMT Counter is reset by writing a correct sequence of STEP1 and STEP2. STEP1 is also cleared if DMTCLR[STEP2] is loaded with the correct value in the correct sequence.

Bit	15	14	13	12	11	10	9	8
	STEP1[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
Access								
Reset								

Bits 15:8 – STEP1[7:0] DMT Preclear Enable bits

Value	Description
01000000	Enables the Deadman Timer preclear (Step 1)
All Other Write Patterns	Sets the BAD1 flag ⁽¹⁾

28.1.3 Deadman Timer Clear Register**Name:** DMTCLR**Offset:** 0x064**Note:**

1. Bits 7:0 are cleared when the DMT Counter is reset by writing a correct sequence of STEP1 and STEP2.

Bit	15	14	13	12	11	10	9	8
Access								
Reset								
Bit	7	6	5	4	3	2	1	0
	STEP2[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 7:0 – STEP2[7:0] DMT Clear Timer bits

Value	Description
00001000	Clears STEP1[7:0], STEP2[7:0] and the Deadman Timer if preceded by the correct loading of the STEP1[7:0] bits in the correct sequence. The write to these bits may be verified by reading the DMTCNTL/H registers and observing the counter being reset.
All Other Write Patterns	Sets the BAD2 bit; the value of STEP1[7:0] will remain unchanged and the new value being written to STEP2[7:0] will be captured ⁽¹⁾ .

28.1.4 Deadman Timer Status Register**Name:** DMTSTAT**Offset:** 0x068**Note:**

- BAD1, BAD2 and DMTEVENT bits are cleared only on a reset.

Bit	15	14	13	12	11	10	9	8
Access								
Reset								

Bit	7	6	5	4	3	2	1	0
	BAD1	BAD2	DMTEVENT					WINOPN
Access	R/W	R/W	R/W					R/W
Reset	0	0	0					0

Bit 7 – BAD1 Deadman Timer Bad STEP1[7:0] Value Detect bit⁽¹⁾

Value	Description
1	Incorrect STEP1[7:0] value was detected
0	Incorrect STEP1[7:0] value was not detected

Bit 6 – BAD2 Deadman Timer Bad STEP2[7:0] Value Detect bit⁽¹⁾

Value	Description
1	Incorrect STEP2[7:0] value was detected
0	Incorrect STEP2[7:0] value was not detected

Bit 5 – DMTEVENT Deadman Timer Event bit⁽¹⁾

Value	Description
1	Deadman Timer event was detected (counter expired, or bad STEP1[7:0] or STEP2[7:0] value was entered prior to counter increment)
0	Deadman Timer event was not detected

Bit 0 – WINOPN Deadman Timer Clear Window bit

Value	Description
1	Deadman Timer clear window is open
0	Deadman Timer clear window is not open

28.1.5 Deadman Timer Count Register Low**Name:** DMTCNTL**Offset:** 0x06C

Bit	15	14	13	12	11	10	9	8
	COUNTER[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	COUNTER[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – COUNTER[15:0] Read Current Contents of Lower DMT Counter bits

28.1.6 Deadman Timer Count Register High**Name:** DMTCNTH**Offset:** 0x06E

Bit	15	14	13	12	11	10	9	8
	COUNTER[31:24]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	COUNTER[23:16]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – COUNTER[31:16] Read Current Contents of Higher DMT Counter bits

28.1.7 DMT Hold Register

Name: HOLDREG⁽¹⁾
Offset: 0x070

- Note:**
1. The DMTHOLDREG register is initialized to '0' on Reset, and is only loaded when the DMTCNTH and DMTCNTH registers are read.

Bit	15	14	13	12	11	10	9	8
	UPRCNT[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	UPRCNT[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – UPRCNT[15:0] DMTCNTH Register Value when DMTCNTH and DMTCNTH were Last Read bits

28.1.8 DMT Post-Configure Count Status Register Low**Name:** PSCNTL**Offset:** 0x074

Bit	15	14	13	12	11	10	9	8
	PSCNT[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PSCNT[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – PSCNT[15:0] Lower DMT Instruction Count Value Configuration Status bits

This is always the value of the FDMTCNTL Configuration register.

28.1.9 DMT Post-Configure Count Status Register High**Name:** PSCNTH**Offset:** 0x076

Bit	15	14	13	12	11	10	9	8
	PSCNT[31:24]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PSCNT[23:16]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – PSCNT[31:16] Higher DMT Instruction Count Value Configuration Status bits

This is always the value of the FDMTCNTH Configuration register.

28.1.10 DMT Post-Configure Interval Status Register Low**Name:** PSINTVL**Offset:** 0x078

Bit	15	14	13	12	11	10	9	8
	PSINTV[15:8]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PSINTV[7:0]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – PSINTV[15:0] Lower DMT Window Interval Configuration Status bits

This is always the value of the FDMTIVTL Configuration register.

28.1.11 DMT Post-Configure Interval Status Register High**Name:** PSINTVH**Offset:** 0x07A

Bit	15	14	13	12	11	10	9	8
	PSINTV[31:24]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	PSINTV[23:16]							
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – PSINTV[31:16] Higher DMT Window Interval Configuration Status bits

This is always the value of the FDMTIVTH Configuration register.

29. Power-Saving Features

Note: This data sheet summarizes the features of the dsPIC33CK512MPT608 family of devices. It is not intended to be a comprehensive reference source. To complement the information in this data sheet, refer to “**Watchdog Timer and Power-Saving Modes**” (www.microchip.com/DS70615) in the “dsPIC33/PIC24 Family Reference Manual”.

The dsPIC33CK512MPT608 family devices provide the ability to manage power consumption by selectively managing clocking to the CPU and the peripherals. In general, a lower clock frequency and a reduction in the number of peripherals being clocked constitutes lower consumed power.

dsPIC33CK512MPT608 family devices can manage power consumption in four ways:

- Clock Frequency
- Instruction-Based Sleep and Idle modes
- Software-Controlled Doze mode
- Selective Peripheral Control in Software

Combinations of these methods can be used to selectively tailor an application’s power consumption while still maintaining critical application features, such as timing-sensitive communications.

29.1 Clock Frequency and Clock Switching

The dsPIC33CK512MPT608 family devices allow a wide range of clock frequencies to be selected under application control. If the system clock configuration is not locked, users can choose low-power or high-precision oscillators by simply changing the NOSCx bits (OSCCON[10:8]). The process of changing a system clock during operation, as well as limitations to the process, are discussed in more detail in [10. Oscillator with High-Frequency PLL](#).

29.2 Instruction-Based Power-Saving Modes

The dsPIC33CK512MPT608 family devices have two special power-saving modes that are entered through the execution of a special `PWRSV` instruction. Sleep mode stops clock operation and halts all code execution. Idle mode halts the CPU and code execution, but allows peripheral modules to continue operation. The assembler syntax of the `PWRSV` instruction is shown in [Example 29-1](#) and [Example 29-2](#).

Note: `SLEEP_MODE` and `IDLE_MODE` are constants defined in the assembler include file for the selected device.

Sleep and Idle modes can be exited as a result of an enabled interrupt, WDT time-out or a device Reset. When the device exits these modes, it is said to “wake-up”.

Example 29-1. `PWRSV` Instruction Syntax in Assembly

```
PWRSV #SLEEP_MODE    ; Put the device into Sleep mode
PWRSV #IDLE_MODE     ; Put the device into Idle mode
```

Example 29-2. `PWRSV` Instruction Syntax in C Language

```
Sleep()    // Put the device into Sleep mode
Idle ()    // Put the device into Idle mode
```


29.2.1 Sleep Mode

The following occurs in Sleep mode:

- The system clock source is shut down. If an on-chip oscillator is used, it is turned off.
- The device current consumption is reduced to a minimum, provided that no I/O pin is sourcing current.
- The Fail-Safe Clock Monitor does not operate, since the system clock source is disabled.
- The WDT, if enabled, is automatically cleared prior to entering Sleep or Idle mode.
- Some device features or peripherals can continue to operate. This includes items such as the Input Change Notification on the I/O ports or peripherals that use an External Clock input.
- Any peripheral that requires the system clock source for its operation is disabled.

The device wakes up from Sleep mode on any of these events:

- Any interrupt source that is individually enabled
- Any form of device Reset
- A WDT time-out

On wake-up from Sleep mode, the processor restarts with the same clock source that was active when Sleep mode was entered.

For optimal power savings, the internal regulator and the Flash regulator can be configured to go into standby when Sleep mode is entered by clearing the VREGS (RCON[8]) bit (default configuration).

If the application requires a faster wake-up time and can accept higher current requirements, the VREGS (RCON[8]) bit can be set to keep the internal regulator and the Flash regulator active during Sleep mode. The available Low-Power Sleep modes are shown in [Table 29-1](#). Additional regulator information is available in [30.5. On-Chip Voltage Regulators](#).

Table 29-1. Low-Power Sleep Modes

Relative Power	LPWREN	VREGS	Mode
Highest	0	1	Full power, active
—	0	0	Full power, standby
—	1 ⁽¹⁾	1	Low power, active
Lowest	1 ⁽¹⁾	0	Low power, standby

Note:

1. Low-Power modes, when LPWREN = 1, can only be used in the industrial temperature range.

29.2.2 Idle Mode

The following occurs in Idle mode:

- The CPU stops executing instructions.
- The WDT is automatically cleared.
- The system clock source remains active. By default, all peripheral modules continue to operate normally from the system clock source, but can also be selectively disabled (see [29.4. Peripheral Module Disable](#)).

The device wakes up from Idle mode on any of these events:

- Any interrupt that is individually enabled
- Any device Reset
- A WDT time-out

On wake-up from Idle mode, the clock is reapplied to the CPU and instruction execution will begin (two to four clock cycles later), starting with the instruction following the `PWRSAB` instruction or the first instruction in the ISR.

All peripherals also have the option to discontinue operation when Idle mode is entered to allow for increased power savings. This option is selectable in the control register of each peripheral; for example, the SIDL bit in the Timer1 Control register (T1CON[13]).

29.2.3 Interrupts Coincident with Power Save Instructions

Any interrupt that coincides with the execution of a `PWRSAY` instruction is held off until entry into Sleep or Idle mode has completed. The device then wakes up from Sleep or Idle mode.

29.3 Doze Mode

The preferred strategies for reducing power consumption are changing clock speed and invoking one of the power-saving modes. In some circumstances, this cannot be practical. For example, it may be necessary for an application to maintain uninterrupted synchronous communication, even while it is doing nothing else. Reducing system clock speed can introduce communication errors, while using a power-saving mode can stop communications completely.

Doze mode is a simple and effective alternative method to reduce power consumption while the device is still executing code. In this mode, the system clock continues to operate from the same source and at the same speed. Peripheral modules continue to be clocked at the same speed, while the CPU clock speed is reduced. Synchronization between the two clock domains is maintained, allowing the peripherals to access the SFRs while the CPU executes code at a slower rate.

Doze mode is enabled by setting the DOZEN bit (CLKDIV[11]). The ratio between peripheral and core clock speed is determined by the DOZE[2:0] bits (CLKDIV[14:12]). There are eight possible configurations, from 1:1 to 1:128, with 1:1 being the default setting.

Programs can use Doze mode to selectively reduce power consumption in event-driven applications. This allows clock-sensitive functions, such as synchronous communications, to continue without interruption while the CPU is not in Idle, waiting for something to invoke an interrupt routine. An automatic return to full-speed CPU operation on interrupts can be enabled by setting the ROI bit (CLKDIV[15]). By default, interrupt events have no effect on Doze mode operation.

29.4 Peripheral Module Disable

The Peripheral Module Disable (PMD) registers provide a method to disable a peripheral module by stopping all clock sources supplied to that module. When a peripheral is disabled using the appropriate PMD control bit, the peripheral is in a Minimum Power Consumption state. The control and status registers associated with the peripheral are also disabled, so writes to those registers do not have any effect and read values are invalid.

A peripheral module is enabled only if both the associated bit in the PMD register is cleared and the peripheral is supported by the specific dsPIC® DSC variant. If the peripheral is present in the device, it is enabled in the PMD register by default.

Note: If a PMD bit is set, the corresponding module is disabled after a delay of one instruction cycle. Similarly, if a PMD bit is cleared, the corresponding module is enabled after a delay of one instruction cycle (assuming the module control registers are already configured to enable module operation).

29.5 Power-Saving Resources

Many useful resources are provided on the main product page of the Microchip website for the devices listed in this data sheet. This product page contains the latest updates and additional information.

29.5.1 Key Resources

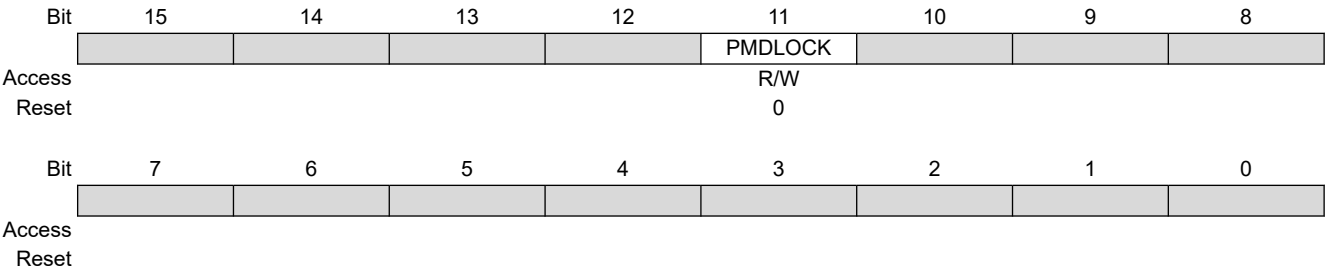
- “**Watchdog Timer and Power-Saving Modes**” (www.microchip.com/DS70615) in the “*dsPIC33/PIC24 Family Reference Manual*.”
- Code Samples
- Application Notes
- Software Libraries
- Webinars
- All related “*dsPIC33/PIC24 Family Reference Manual*” Sections
- Development Tools

29.6 Power-Saving Control Registers

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x00 ... 0x0F9F	Reserved									
0x0FA0	PMDCONL	15:8					PMDLOCK			
		7:0								
0x0FA2 ... 0x0FA3	Reserved									
0x0FA4	PMD1	15:8					TAMD	QE1MD	PWMMD	
		7:0	I2C1MD	U2MD	U1MD	SPI2MD	SPI1MD	C2MD	C1MD	ADC1MD
0x0FA6	PMD2	15:8								CCP9MD
		7:0	CCP8MD	CCP7MD	CCP6MD	CCP5MD	CCP4MD	CCP3MD	CCP2MD	CCP1MD
0x0FA8	PMD3	15:8								
		7:0	CRCMD		QE2MD		U3MD	I2C3MD	I2C2MD	
0x0FAA	PMD4	15:8								
		7:0					REFOMD			
0x0FAC ... 0x0FAD	Reserved									
0x0FAE	PMD6	15:8	DMA7MD	DMA6MD	DMA5MD	DMA4MD	DMA3MD	DMA2MD	DMA1MD	DMA0MD
		7:0	QE13MD							SPI3MD
0x0FB0	PMD7	15:8			PDC6MD	PDC5MD	PDC4MD	PDC3MD	PDC2MD	PDC1MD
		7:0					PTGMD			
0x0FB2	PMD8	15:8			OPAMPMD	SENT2MD	SENT1MD			DMTMD
		7:0			CLC4MD	CLC3MD	CLC2MD	CLC1MD	BIASMD	

29.6.1 Peripheral Module Disable Control Register Low

Name: PMDCONL
Offset: 0xFA0



Bit 11 – PMDLOCK PMD Lock bit

Value	Description
1	All PMD registers are locked and cannot be written
0	All PMD registers are unlocked and can be written

29.6.2 Peripheral Module Disable 1 Register

Name: PMD1
Offset: 0xFA4

Bit	15	14	13	12	11	10	9	8
					TAMD	QE1MD	PWMMD	
Access					R/W	R/W	R/W	
Reset					0	0	0	

Bit	7	6	5	4	3	2	1	0
	I2C1MD	U2MD	U1MD	SPI2MD	SPI1MD	C2MD	C1MD	ADC1MD
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 11 – TAMD Timer A Module Disable bit

Value	Description
1	Timer A module is disabled
0	Timer A module is enabled

Bit 10 – QE1MD QE1 Module Disable bit

Value	Description
1	QE1 module is disabled
0	QE1 module is enabled

Bit 9 – PWMMD PWM Module Disable bit

Value	Description
1	PWM module is disabled
0	PWM module is enabled

Bit 7 – I2C1MD I2C1 Module Disable bit

Value	Description
1	I2C1 module is disabled
0	I2C1 module is enabled

Bit 6 – U2MD UART2 Module Disable bit

Value	Description
1	UART2 module is disabled
0	UART2 module is enabled

Bit 5 – U1MD UART1 Module Disable bit

Value	Description
1	UART1 module is disabled
0	UART1 module is enabled

Bit 4 – SPI2MD SPI2 Module Disable bit

Value	Description
1	SPI2 module is disabled
0	SPI2 module is enabled

Bit 3 – SPI1MD SPI1 Module Disable bit

Value	Description
1	SPI1 module is disabled
0	SPI1 module is enabled

Bit 2 – C2MD CAN2 Module Disable bit

Value	Description
1	CAN2 module is disabled
0	CAN2 module is enabled

Bit 1 – C1MD CAN1 Module Disable bit

Value	Description
1	CAN1 module is disabled
0	CAN1 module is enabled

Bit 0 – ADC1MD ADC Module Disable bit

Value	Description
1	ADC module is disabled
0	ADC module is enabled

29.6.3 Peripheral Module Disable 2 Register⁽¹⁾**Name:** PMD2**Offset:** 0xFA6**Note:**

- Availability is dependent on the supported peripherals, refer to [dsPIC33CK512MPT608 Product Families](#).

Bit	15	14	13	12	11	10	9	8
								CCP9MD
Access								R/W
Reset								0

Bit	7	6	5	4	3	2	1	0
	CCP8MD	CCP7MD	CCP6MD	CCP5MD	CCP4MD	CCP3MD	CCP2MD	CCP1MD
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit 8 – CCP9MD M CCP9 Module Disable bit

Value	Description
1	M CCP9 module is disabled
0	M CCP9 module is enabled

Bit 7 – CCP8MD S CCP8 Module Disable bit

Value	Description
1	S CCP8 module is disabled
0	S CCP8 module is enabled

Bit 6 – CCP7MD S CCP7 Module Disable bit

Value	Description
1	S CCP7 module is disabled
0	S CCP7 module is enabled

Bit 5 – CCP6MD S CCP6 Module Disable bit

Value	Description
1	S CCP6 module is disabled
0	S CCP6 module is enabled

Bit 4 – CCP5MD S CCP5 Module Disable bit

Value	Description
1	S CCP5 module is disabled
0	S CCP5 module is enabled

Bit 3 – CCP4MD S CCP4 Module Disable bit

Value	Description
1	S CCP4 module is disabled
0	S CCP4 module is enabled

Bit 2 – CCP3MD S CCP3 Module Disable bit

Value	Description
1	S CCP3 module is disabled
0	S CCP3 module is enabled

Bit 1 – CCP2MD S CCP2 Module Disable bit

Value	Description
1	S CCP2 module is disabled

Value	Description
0	SCCP2 module is enabled

Bit 0 – CCP1MD SCCP1 Module Disable bit

Value	Description
1	SCCP1 module is disabled
0	SCCP1 module is enabled

29.6.4 Peripheral Module Disable 3 Register**Name:** PMD3**Offset:** 0xFA8

Bit	15	14	13	12	11	10	9	8
Access								
Reset								

Bit	7	6	5	4	3	2	1	0
	CRCMD		QEI2MD		U3MD	I2C3MD	I2C2MD	
Access	R/W		R/W		R/W	R/W	R/W	
Reset	0		0		0	0	0	

Bit 7 – CRCMD CRC Module Disable bit

Value	Description
1	CRC module is disabled
0	CRC module is enabled

Bit 5 – QEI2MD QEI Module 2 Disable bit

Value	Description
1	QEI2 module is disabled
0	QEI2 module is enabled

Bit 3 – U3MD UART3 Module Disable bit

Value	Description
1	UART3 module is disabled
0	UART3 module is enabled

Bit 2 – I2C3MD I2C3 Module Disable bit

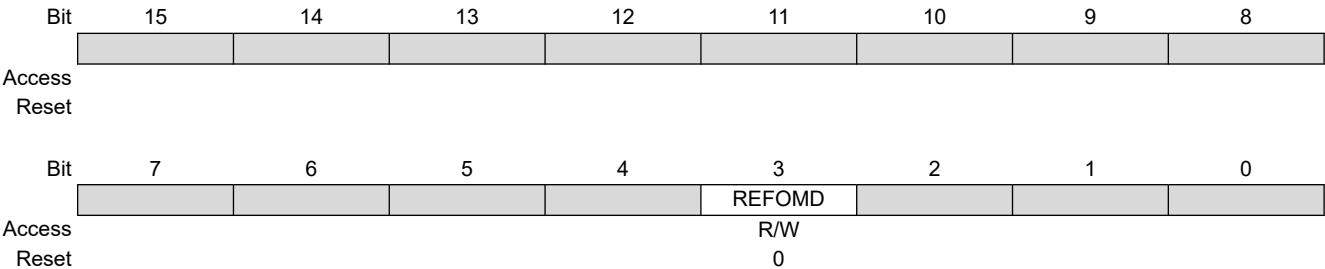
Value	Description
1	I2C3 module is disabled
0	I2C3 module is enabled

Bit 1 – I2C2MD I2C2 Module Disable bit

Value	Description
1	I2C2 module is disabled
0	I2C2 module is enabled

29.6.5 Peripheral Module Disable 4 Register

Name: PMD4
Offset: 0xFAA



Bit 3 – REFOMD Reference Clock Module Disable bit

Value	Description
1	Reference clock module is disabled
0	Reference clock module is enabled

29.6.6 Peripheral Module Disable 6 Register**Name:** PMD6**Offset:** 0xFAE

Bit	15	14	13	12	11	10	9	8
	DMA7MD	DMA6MD	DMA5MD	DMA4MD	DMA3MD	DMA2MD	DMA1MD	DMA0MD
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	QE13MD							SPI3MD
Access	R/W							R/W
Reset	0							0

Bit 15 – DMA7MD DMA7 Module Disable bit

Value	Description
1	DMA7 module is disabled
0	DMA7 module is enabled

Bit 14 – DMA6MD DMA6 Module Disable bit

Value	Description
1	DMA6 module is disabled
0	DMA6 module is enabled

Bit 13 – DMA5MD DMA5 Module Disable bit

Value	Description
1	DMA5 module is disabled
0	DMA5 module is enabled

Bit 12 – DMA4MD DMA4 Module Disable bit

Value	Description
1	DMA4 module is disabled
0	DMA4 module is enabled

Bit 11 – DMA3MD DMA3 Module Disable bit

Value	Description
1	DMA3 module is disabled
0	DMA3 module is enabled

Bit 10 – DMA2MD DMA2 Module Disable bit

Value	Description
1	DMA2 module is disabled
0	DMA2 module is enabled

Bit 9 – DMA1MD DMA1 Module Disable bit

Value	Description
1	DMA1 module is disabled
0	DMA1 module is enabled

Bit 8 – DMA0MD DMA0 Module Disable bit

Value	Description
1	DMA0 module is disabled
0	DMA0 module is enabled

Bit 7 – QEI3MD QEI3 Module Disable bit

Value	Description
1	QEI3 module is disabled
0	QEI3 module is enabled

Bit 0 – SPI3MD SPI3 Module Disable bit

Value	Description
1	SPI3 module is disabled
0	SPI3 module is enabled

29.6.7 Peripheral Module Disable 7 Register**Name:** PMD7**Offset:** 0xFB0

Bit	15	14	13	12	11	10	9	8
			PDC6MD	PDC5MD	PDC4MD	PDC3MD	PDC2MD	PDC1MD
Access			R/W	R/W	R/W	R/W	R/W	R/W
Reset			0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
					PTGMD			
Access					R/W			
Reset					0			

Bit 13 – PDC6MD Peripheral DMA Controller 6 Module Disable bit

Value	Description
1	PDC6 module is disabled
0	PDC6 module is enabled

Bit 12 – PDC5MD Peripheral DMA Controller 5 Module Disable bit

Value	Description
1	PDC5 module is disabled
0	PDC5 module is enabled

Bit 11 – PDC4MD Peripheral DMA Controller 4 Module Disable bit

Value	Description
1	PDC4 module is disabled
0	PDC4 module is enabled

Bit 10 – PDC3MD Peripheral DMA Controller 3 Module Disable bit

Value	Description
1	PDC3 module is disabled
0	PDC3 module is enabled

Bit 9 – PDC2MD Peripheral DMA Controller 2 Module Disable bit

Value	Description
1	PDC2 module is disabled
0	PDC2 module is enabled

Bit 8 – PDC1MD Peripheral DMA Controller 1 Module Disable bit

Value	Description
1	PDC1 module is disabled
0	PDC1 module is enabled

Bit 3 – PTGMD PTG Module Disable bit

Value	Description
1	PTG module is disabled
0	PTG module is enabled

29.6.8 Peripheral Module Disable 8 Register

Name: PMD8

Offset: 0xFB2

Bit	15	14	13	12	11	10	9	8
			OPAMPMD	SENT2MD	SENT1MD			DMTMD
Access			R/W	R/W	R/W			R/W
Reset			0	0	0			0

Bit	7	6	5	4	3	2	1	0
			CLC4MD	CLC3MD	CLC2MD	CLC1MD	BIASMD	
Access			R/W	R/W	R/W	R/W	R/W	
Reset			0	0	0	0	0	

Bit 13 – OPAMPMD Op Amp Module Disable bit

Value	Description
1	Op amp module is disabled
0	Op amp module is enabled

Bit 12 – SENT2MD SENT2 Module Disable bit

Value	Description
1	SENT2 module is disabled
0	SENT2 module is enabled

Bit 11 – SENT1MD SENT1 Module Disable bit

Value	Description
1	SENT1 module is disabled
0	SENT1 module is enabled

Bit 8 – DMTMD Deadman Timer Module Disable bit

Value	Description
1	Deadman Timer module is disabled
0	Deadman Timer module is enabled

Bit 5 – CLC4MD CLC4 Module Disable bit

Value	Description
1	CLC4 module is disabled
0	CLC4 module is enabled

Bit 4 – CLC3MD CLC3 Module Disable bit

Value	Description
1	CLC3 module is disabled
0	CLC3 module is enabled

Bit 3 – CLC2MD CLC2 Module Disable bit

Value	Description
1	CLC2 module is disabled
0	CLC2 module is enabled

Bit 2 – CLC1MD CLC1 Module Disable bit

Value	Description
1	CLC1 module is disabled
0	CLC1 module is enabled

Bit 1 – BIASMD Constant-Current Source Module Disable bit

Value	Description
1	Constant-current source module is disabled
0	Constant-current source module is enabled

30. Special Features

Note: This data sheet summarizes the features of the dsPIC33CK512MPT608 family of devices. It is not intended to be a comprehensive reference source. To complement the information in this data sheet, refer to the related section of the “dsPIC33/PIC24 Family Reference Manual”, which is available from the Microchip website (www.microchip.com).

The dsPIC33CK512MPT608 family devices include several features intended to maximize application flexibility and reliability, and minimize cost through elimination of external components. These are:

- Flexible Configuration
- Watchdog Timer (WDT)
- Code Protection and CodeGuard™ Security
- JTAG Boundary Scan Interface
- In-Circuit Serial Programming™ (ICSP™)
- In-Circuit Emulation
- Brown-out Reset (BOR)

30.1 Configuration Bits

In dsPIC33CK512MPT608 family devices, the Configuration Words are implemented as volatile memory. This means that configuration data will get loaded to volatile memory (from the Flash Configuration Words) each time the device is powered up. Configuration data are stored at the end of the on-chip program memory space, known as the Flash Configuration Words. Their specific locations are shown in [Table 30-1](#). The configuration data are automatically loaded from the Flash Configuration Words to the proper Configuration Shadow registers during device Resets.

When creating applications for these devices, users should always specifically allocate the location of the Flash Configuration Words for configuration data in their code for the compiler. This is to make certain that program code is not stored in this address when the code is compiled. Program code executing out of configuration space will cause a device Reset.

Note: Performing a page erase operation on the last page of program memory clears the Flash Configuration Words.

Table 30-1. dsPIC33CK512MPT608 Configuration Addresses

Register Name	Single Partition		Dual Partition, Active		Dual Partition, Inactive	
	256k Address	512k Address	256k Address	512k Address	256k Address	512k Address
FSEC ⁽¹⁾	0x02BF00	0x057F00	0x015F00	0x02BF00	0x415F00	0x42BF00
FBSLIM ⁽¹⁾	0x02BF10	0x057F10	0x015F10	0x02BF10	0x415F10	0x42BF10
FSIGN ⁽¹⁾	0x02BF14	0x057F14	0x015F14	0x02BF14	0x415F14	0x42BF14
FOSCSEL	0x02BF18	0x057F18	0x015F18	0x02BF18	0x415F18	0x42BF18
FOSC	0x02BF1C	0x057F1C	0x015F1C	0x02BF1C	0x415F1C	0x42BF1C
FWDT	0x02BF20	0x057F20	0x015F20	0x02BF20	0x415F20	0x42BF20
FPOR	0x02BF24	0x057F24	0x015F24	0x02BF24	0x415F24	0x42BF24
FICD	0x02BF28	0x057F28	0x015F28	0x02BF28	0x415F28	0x42BF28
FDMTIVTL	0x02BF2C	0x057F2C	0x015F2C	0x02BF2C	0x415F2C	0x42BF2C

Notes:

1. Changes to the Inactive Partition Configuration Words affect how the Active Partition accesses the Inactive Partition.
2. FBOOT resides in calibration memory space.

dsPIC33CK512MPT608 Family

Special Features

.....continued

Register Name	Single Partition		Dual Partition, Active		Dual Partition, Inactive	
	256k Address	512k Address	256k Address	512k Address	256k Address	512k Address
FDMTIVTH	0x02BF30	0x057F30	0x015F30	0x02BF30	0x415F30	0x42BF30
FDMTCNTL	0x02BF34	0x057F34	0x015F34	0x02BF34	0x415F34	0x42BF34
FDMTCNTH	0x02BF38	0x057F38	0x015F38	0x02BF38	0x415F38	0x42BF38
FDMT	0x02BF3C	0x057F3C	0x015F3C	0x02BF3C	0x415F3C	0x42BF3C
FDEVOPT	0x02BF40	0x057F40	0x015F40	0x02BF40	0x415F40	0x42BF40
FALTREG	0x02BF44	0x057F44	0x015F44	0x02BF44	0x415F44	0x42BF44
FALTSEQ	0x02BFFC	0x057FFC	0x015FFC	0x02BFFC	0x415FFC	0x42BFFC
FBOOT ⁽²⁾	0x801800					

Notes:

1. Changes to the Inactive Partition Configuration Words affect how the Active Partition accesses the Inactive Partition.
2. FBOOT resides in calibration memory space.

30.2 Configuration Registers

Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x00	FSEC	23:16								
		15:8	AIVTDIS					CSS[2:0]		CWRP
		7:0		GSS[1:0]	GWRP		BSEN	BSS[1:0]		BWRP
0x03 ... 0x0F0F	Reserved									
0x0F10	FBSLIM	23:16								
		15:8						BSLIM[12:8]		
		7:0						BSLIM[7:0]		
0x0F13	Reserved									
0x0F14	FSIGN	23:16								
		15:8	Reserved							
		7:0								
0x0F17	Reserved									
0x0F18	FOSCSEL	23:16								
		15:8								
		7:0	IESO					FNOSC[2:0]		
0x0F1B	Reserved									
0x0F1C	FOSC	23:16								
		15:8				XTBST	XTCFG[1:0]			PLLKEN
		7:0		FCKSM[1:0]				OSCIOFNC	POSCMD[1:0]	
0x0F1F	Reserved									
0x0F20	FWDT	23:16								
		15:8	FWDTEN			SWDTPS[4:0]			WDTWIN[1:0]	
		7:0	WINDIS		RCLKSEL[1:0]			RWDTPS[4:0]		
0x0F23	Reserved									
0x0F24	FPOR	23:16								
		15:8								
		7:0		BISTDIS	Reserved[1:0]					
0x0F27	Reserved									
0x0F28	FICD	23:16								
		15:8	NOBTSWP				Reserved[6:0]			
		7:0			JTAGEN				ICS[1:0]	
0x0F2B	Reserved									
0x0F2C	FDMTIVTL	23:16								
		15:8					DMTIVT[15:8]			
		7:0					DMTIVT[7:0]			
0x0F2F	Reserved									
0x0F30	FDMTIVTH	23:16								
		15:8					DMTIVT[31:24]			
		7:0					DMTIVT[23:16]			
0x0F33	Reserved									
0x0F34	FDMTCNTL	23:16								
		15:8					DMTCNT[15:8]			
		7:0					DMTCNT[7:0]			
0x0F37	Reserved									
0x0F38	FDMTCNTH	23:16								
		15:8					DMTCNT[31:24]			
		7:0					DMTCNT[23:16]			
0x0F3B	Reserved									
0x0F3C	FDMT	23:16								
		15:8								
		7:0								DMTDIS
0x0F3F	Reserved									

dsPIC33CK512MPT608 Family

Special Features

.....continued										
Offset	Name	Bit Pos.	7	6	5	4	3	2	1	0
0x0F40	FDEV OPT	23:16								
		15:8		ALT TMS	SPI2PIN	DUP PWM		SMBEN	Reserved[1:0]	
		7:0	Reserved				ALT I2C1	Reserved		
0x0F43	Reserved									
0x0F44	FALT REG	23:16								
		15:8		CTXT4[2:0]				CTXT3[2:0]		
		7:0		CTXT2[2:0]				CTXT1[2:0]		
0x0F47 ... 0x0FFB	Reserved									
0x0FFC	FBTSEQ	23:16	IBSEQ[11:0]							
		15:8	IBSEQ[11:0]				BSEQ[11:8]			
		7:0	BSEQ[7:0]							

30.2.1 FSEC Configuration Register

Name: FSEC
Offset: 0x000

Legend: PO = Program Once bit

Bit	23	22	21	20	19	18	17	16
Access								
Reset								

Bit	15	14	13	12	11	10	9	8
Access	AIVTDIS					CSS[2:0]		CWRP
Reset	R/PO				R/PO	R/PO	R/PO	R/PO
Reset	1				1	1	1	1

Bit	7	6	5	4	3	2	1	0
Access	GSS[1:0]		GWRP		BSEN		BSS[1:0]	BWRP
Reset	R/PO	R/PO	R/PO		R/PO	R/PO	R/PO	R/PO
Reset	1	1	1		1	1	1	1

Bit 15 – AIVTDIS Alternate Interrupt Vector Table Disable bit

Value	Description
1	Disables AIVT
0	Enables AIVT

Bits 11:9 – CSS[2:0] Configuration Segment Code Flash Protection Level bits

Value	Description
111	No protection (other than CWRP write protection)
110	Standard security
10x	Enhanced security
0xx	High security

Bit 8 – CWRP Configuration Segment Write-Protect bit

Value	Description
1	Configuration Segment is not write-protected
0	Configuration Segment is write-protected

Bits 7:6 – GSS[1:0] General Segment Code Flash Protection Level bits

Value	Description
11	No protection (other than GWRP write protection)
10	Standard security
0x	High security

Bit 5 – GWRP General Segment Write-Protect bit

Value	Description
1	User program memory is not write-protected
0	User program memory is write-protected

Bit 3 – BSEN Boot Segment Control bit

Value	Description
1	No Boot Segment
0	Boot Segment size is determined by BSLIM[12:0]

Bits 2:1 – BSS[1:0] Boot Segment Code Flash Protection Level bits

Value	Description
11	No protection (other than BWRP write protection)
10	Standard security
0x	High security

Bit 0 – BWRP Boot Segment Write-Protect bit

Value	Description
1	User program memory is not write-protected
0	User program memory is write-protected

30.2.2 FBSLIM Configuration Register

Name: FBSLIM

Offset: 0xF10

Legend: PO = Program Once bit

Bit	23	22	21	20	19	18	17	16
Access								
Reset								
Bit	15	14	13	12	11	10	9	8
				BSLIM[12:8]				
Access				R/PO	R/PO	R/PO	R/PO	R/PO
Reset				1	1	1	1	1
Bit	7	6	5	4	3	2	1	0
	BSLIM[7:0]							
Access	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO
Reset	1	1	1	1	1	1	1	1

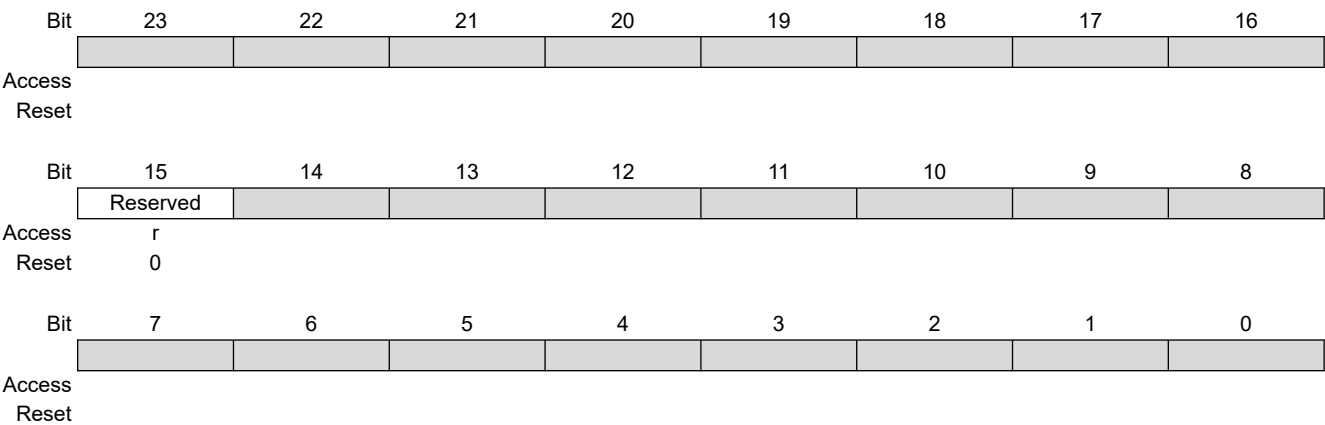
Bits 12:0 – BSLIM[12:0] Boot Segment Code Flash Page Address Limit bits

Contains the page address of the first active General Segment page. The value to be programmed is the inverted page address, such that programming additional '0's can only increase the Boot Segment size.

30.2.3 FSIGN Configuration Register

Name: FSIGN
Offset: 0xF14

Legend: PO = Program Once bit, r = Reserved bit



Bit 15 – Reserved Maintain as '0'

30.2.4 FOSCSEL Configuration Register

Name: FOSCSEL

Offset: 0xF18

Legend: PO = Program Once bit

Bit	23	22	21	20	19	18	17	16
Access								
Reset								

Bit	15	14	13	12	11	10	9	8
Access								
Reset								

Bit	7	6	5	4	3	2	1	0
	IESO					FNOSC[2:0]		
Access	R/PO					R/PO	R/PO	R/PO
Reset	1					1	1	1

Bit 7 – IESO Internal External Switchover bit

Value	Description
1	Internal External Switchover mode is enabled (Two-Speed Start-up is enabled)
0	Internal External Switchover mode is disabled (Two-Speed Start-up is disabled)

Bits 2:0 – FNOSC[2:0] Initial Oscillator Source Selection bits

Value	Description
111	Internal Fast RC (FRC) Oscillator with Postscaler
110	Backup Fast RC (BFRC)
101	LPRC Oscillator
100	Reserved
011	Primary Oscillator with PLL (XTPLL, HSPLL, ECPLL)
010	Primary (XT, HS, EC) Oscillator
001	Internal Fast RC Oscillator with PLL (FRCPLL)
000	Fast RC (FRC) Oscillator

30.2.5 FOSC Configuration Register

Name: FOSC
Offset: 0xF1C

Note:

1. A time-out period will occur when the system clock switching logic requests the PLL clock source and the PLL is not already enabled.

Legend: PO = Program Once bit,

Bit	23	22	21	20	19	18	17	16
Access								
Reset								

Bit	15	14	13	12	11	10	9	8
				XTBST	XTCFG[1:0]			PLLKEN
Access				R/PO	R/PO	R/PO		R/PO
Reset				1	1	1		1

Bit	7	6	5	4	3	2	1	0
	FCKSM[1:0]					OSCIOFNC	POSCMD[1:0]	
Access	R/PO	R/PO				R/PO	R/PO	R/PO
Reset	1	1				1	1	1

Bit 12 – XTBST Oscillator Kick-Start Programmability bit

Value	Description
1	Boosts the kick-start
0	Default kick-start

Bits 11:10 – XTCFG[1:0] Crystal Oscillator Drive Select bits

Current gain programmability for oscillator (output drive).

Value	Description
11	Gain3 (use for 24-32 MHz crystals)
10	Gain2 (use for 16-24 MHz crystals)
01	Gain1 (use for 8-16 MHz crystals)
00	Gain0 (use for 4-8 MHz crystals)

Bit 8 – PLLKEN PLL Lock Status Control bit⁽¹⁾

Value	Description
1	PLL lock signal will be used to disable PLL clock output if lock is lost
0	PLL lock signal is not used; the PLL clock output will not be disabled if lock is lost

Bits 7:6 – FCKSM[1:0] Clock Switching Mode bits

Value	Description
1x	Clock switching is disabled, Fail-Safe Clock Monitor is disabled
01	Clock switching is enabled, Fail-Safe Clock Monitor is disabled
00	Clock switching is enabled, Fail-Safe Clock Monitor is enabled

Bit 2 – OSCIOFNC OSCO Pin Function bit (except in XT and HS modes)⁽¹⁾

Value	Description
1	OSCO is the clock output
0	OSCO is the general purpose digital I/O pin

Bits 1:0 – POSCMD[1:0] Primary Oscillator Mode Select bits

Value	Description
11	Primary Oscillator is disabled
10	HS Crystal Oscillator mode (10 MHz-32 MHz)
01	XT Crystal Oscillator mode (3.5 MHz-10 MHz)
00	EC (External Clock) mode

30.2.6 FWDT Configuration Register**Name:** FWDT**Offset:** 0xF20**Legend:** PO = Program Once bit

Bit	23	22	21	20	19	18	17	16
Access								
Reset								
Bit	15	14	13	12	11	10	9	8
Access	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO
Reset	1	1	1	1	1	1	1	1
Bit	7	6	5	4	3	2	1	0
Access	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO
Reset	1	1	1	1	1	1	1	1

Bit 15 – FWDTEN Watchdog Timer Enable bit

Value	Description
1	WDT is enabled in hardware
0	WDT controller via the ON bit (WDTCONL[15])

Bits 14:10 – SWDTPS[4:0] Sleep Mode Watchdog Timer Period Select bits

Value	Description
11111	Divide by $2^{31} = 2,147,483,648$
11110	Divide by $2^{30} = 1,073,741,824$
...	
00001	Divide by $2^1, 2$
00000	Divide by $2^0, 1$

Bits 9:8 – WDTWIN[1:0] Watchdog Timer Window Select bits

Value	Description
11	WDT window is 25% of the WDT period
10	WDT window is 37.5% of the WDT period
01	WDT window is 50% of the WDT period
00	WDT Window is 75% of the WDT period

Bit 7 – WINDIS Watchdog Timer Window Enable bit

Value	Description
1	Watchdog Timer is in Non-Window mode
0	Watchdog Timer is in Window mode

Bits 6:5 – RCLKSEL[1:0] Watchdog Timer Clock Select bits

Value	Description
11	LPRC clock
10	Uses FRC when WINDIS = 0, system clock is not INTOSC/LPRC and device is not in Sleep; otherwise, uses INTOSC/LPRC
01	Uses peripheral clock when system clock is not INTOSC/LPRC and device is not in Sleep; otherwise, uses INTOSC/LPRC
00	Reserved

Bits 4:0 – RWDTPS[4:0] Run Mode Watchdog Timer Period Select bits

Value	Description
11111	Divide by $2^{31} = 2,147,483,648$
11110	Divide by $2^{30} = 1,073,741,824$
...	
00001	Divide by $2^{1, 2}$
00000	Divide by $2^{0, 1}$

30.2.7 FPOR Configuration Register

Name: FPOR

Offset: 0xF24

Legend: PO = Program Once bit, r = Reserved bit

Note:

1. Applies to a Power-on Reset (POR) only.

Bit	23	22	21	20	19	18	17	16
Access								
Reset								
Bit	15	14	13	12	11	10	9	8
Access								
Reset								
Bit	7	6	5	4	3	2	1	0
		BISTDIS	Reserved[1:0]					
Access		R/PO	r	r				
Reset		1	1	1				

Bit 6 – BISTDIS Memory BIST Feature Disable bit⁽¹⁾

Value	Description
1	Normal start-up operation after Reset; executes instruction at 0x000000
0	Optional start-up operation after Reset; executes instruction at start of boot space

Bits 5:4 – Reserved[1:0] Maintain as '1'

30.2.8 FICD Configuration Register

Name: FICD
Offset: 0xF28

Legend: PO = Program Once bit, r = Reserved bit

Bit	23	22	21	20	19	18	17	16
Access								
Reset								
Bit	15	14	13	12	11	10	9	8
Access	NOBTSWP	Reserved[6:0]						
Reset	R/PO	r	r	r	r	r	r	r
Reset	1	0	0	0	0	0	0	1
Bit	7	6	5	4	3	2	1	0
Access			JTAGEN				ICS[1:0]	
Reset			R/PO				R/PO	R/PO
Reset			1				1	1

Bit 15 – NOBTSWP BOOTSWP Instruction Disable bit

Value	Description
1	BOOTSWP instruction is disabled
0	BOOTSWP instruction is enabled

Bits 14:8 – Reserved[6:0] Maintain as '1'

Bit 5 – JTAGEN JTAG Enable bit

Value	Description
1	JTAG port is enabled
0	JTAG port is disabled

Bits 1:0 – ICS[1:0] ICD Communication Channel Select bits

Value	Description
11	Communicates on PGC1 and PGD1
10	Communicates on PGC2 and PGD2
01	Communicates on PGC3 and PGD3
00	Reserved, do not use

30.2.9 FDMTIVTL Configuration Register

Name: FDMTIVTL

Offset: 0xF2C

Legend: PO = Program Once bit

Bit	23	22	21	20	19	18	17	16
Access								
Reset								
Bit	15	14	13	12	11	10	9	8
	DMTIVT[15:8]							
Access	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO
Reset	1	1	1	1	1	1	1	1
Bit	7	6	5	4	3	2	1	0
	DMTIVT[7:0]							
Access	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO
Reset	1	1	1	1	1	1	1	1

Bits 15:0 – DMTIVT[15:0] DMT Window Interval Lower 16 bits

30.2.10 FDMTIVTH Configuration Register

Name: FDMTIVTH
Offset: 0xF30

Legend: PO = Program Once bit

Bit	23	22	21	20	19	18	17	16
Access								
Reset								
Bit	15	14	13	12	11	10	9	8
	DMTIVT[31:24]							
Access	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	DMTIVT[23:16]							
Access	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO
Reset	0	0	0	0	0	0	0	1

Bits 15:0 – DMTIVT[31:16] DMT Window Interval Higher 16 bits

30.2.11 FDMTCNTL Configuration Register

Name: FDMTCNTL

Offset: 0xF34

Legend: PO = Program Once bit

Bit	23	22	21	20	19	18	17	16
Access								
Reset								
Bit	15	14	13	12	11	10	9	8
	DMTCNT[15:8]							
Access	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO
Reset	1	1	1	1	1	1	1	1
Bit	7	6	5	4	3	2	1	0
	DMTCNT[7:0]							
Access	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO
Reset	1	1	1	1	1	1	1	1

Bits 15:0 – DMTCNT[15:0] DMT Instruction Count Time-out Value Lower 16 bits

30.2.12 FDMTCNTH Configuration Register

Name: FDMTCNTH

Offset: 0xF38

Legend: PO = Program Once bit

Bit	23	22	21	20	19	18	17	16
Access								
Reset								
Bit	15	14	13	12	11	10	9	8
	DMTCNT[31:24]							
Access	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO
Reset	1	1	1	1	1	1	1	1
Bit	7	6	5	4	3	2	1	0
	DMTCNT[23:16]							
Access	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO
Reset	1	1	1	1	1	1	1	1

Bits 15:0 – DMTCNT[31:16] DMT Instruction Count Time-out Value Upper 16 bits

30.2.13 FDMT Configuration Register

Name: FDMT
Offset: 0xF3C

Legend: PO = Program Once bit

Bit	23	22	21	20	19	18	17	16
Access								
Reset								

Bit	15	14	13	12	11	10	9	8
Access								
Reset								

Bit	7	6	5	4	3	2	1	0
								DMTDIS
Access								R/PO
Reset								1

Bit 0 – DMTDIS DMT Disable bit

Value	Description
1	Deadman Timer is disabled and can be enabled by software using the ON bit (DMTCON[15])
0	Deadman Timer is enabled and cannot be disabled by software

30.2.14 FDEVOPT Configuration Register**Name:** FDEVOPT**Offset:** 0xF40**Note:**

- Fixed pin option is only available for higher pin packages (48-pin, 64-pin and 80-pin).

Legend: PO = Program Once bit, r = Reserved bit

Bit	23	22	21	20	19	18	17	16
Access								
Reset								
Bit	15	14	13	12	11	10	9	8
		ALTTMS	SPI2PIN	DUPPWM		SMBEN	Reserved[1:0]	
Access		R/PO	R/PO	R/PO		R/PO	r	r
Reset		1	1	1		1	0	0
Bit	7	6	5	4	3	2	1	0
	Reserved				ALT12C1	Reserved		
Access	r				R/PO	r		
Reset	1				1	1		

Bit 14 – ALTTMS Alternate TMS Pin Mapping bit

Value	Description
1	Default location for TMS pin (shared with PWM pin: RB10)
0	Alternate location for TMS pin (without PWM pin: RB2)

Bit 13 – SPI2PIN Main SPI #2 Fast I/O Pad Disable bit⁽¹⁾

Value	Description
1	Host SPI2 uses PPS (I/O remap) to make connections with device pins
0	Host SPI2 uses direct connections with specified device pins

Bit 12 – DUPPWM Duplicate PWM Pin Mapping bit

Value	Description
1	Default PWM4 functions on PPS and fixed pins
0	PWM4 functions only on fixed pins

Bit 10 – SMBEN Select Input Voltage Threshold for I²C Pads to be SMBus 3.0 Compliant bit

Value	Description
1	Enables SMBus 3.0 input threshold voltage
0	Enables SMBus 2.0 input threshold voltage

Bits 9:8 – Reserved[1:0] Maintain as '0'**Bit 7 – Reserved** Maintain as '1'**Bit 3 – ALT12C1** Alternate I2C1 Pin Mapping bit

Value	Description
1	Default location for SCL1/SDA1 pins
0	Alternate location for SCL1/SDA1 pins (ASCL1/ASDA1)

Bit 2 – Reserved Maintain as '1'

30.2.15 FALTREG Configuration Register**Name:** FALTREG**Offset:** 0xF44**Legend:** PO = Program Once bit

Bit	23	22	21	20	19	18	17	16
Access								
Reset								
Bit	15	14	13	12	11	10	9	8
Access			CTXT4[2:0]				CTXT3[2:0]	
Reset		R/PO	R/PO	R/PO		R/PO	R/PO	R/PO
		1	1	1		1	1	1
Bit	7	6	5	4	3	2	1	0
Access			CTXT2[2:0]				CTXT1[2:0]	
Reset		R/PO	R/PO	R/PO		R/PO	R/PO	R/PO
		1	1	1		1	1	1

Bits 14:12 – CTXT4[2:0] Specifies Alternate Working Register Set #4 with IPLs bits

Value	Description
111	Not assigned
110	Alternate Register Set #4 is assigned to IPL Level 7
101	Alternate Register Set #4 is assigned to IPL Level 6
100	Alternate Register Set #4 is assigned to IPL Level 5
011	Alternate Register Set #4 is assigned to IPL Level 4
010	Alternate Register Set #4 is assigned to IPL Level 3
001	Alternate Register Set #4 is assigned to IPL Level 2
000	Alternate Register Set #4 is assigned to IPL Level 1

Bits 10:8 – CTXT3[2:0] Specifies Alternate Working Register Set #3 with IPLs bits

Value	Description
111	Not assigned
110	Alternate Register Set #3 is assigned to IPL Level 7
101	Alternate Register Set #3 is assigned to IPL Level 6
100	Alternate Register Set #3 is assigned to IPL Level 5
011	Alternate Register Set #3 is assigned to IPL Level 4
010	Alternate Register Set #3 is assigned to IPL Level 3
001	Alternate Register Set #3 is assigned to IPL Level 2
000	Alternate Register Set #3 is assigned to IPL Level 1

Bits 6:4 – CTXT2[2:0] Specifies Alternate Working Register Set #2 with IPLs bits

Value	Description
111	Not assigned
110	Alternate Register Set #2 is assigned to IPL Level 7
101	Alternate Register Set #2 is assigned to IPL Level 6
100	Alternate Register Set #2 is assigned to IPL Level 5
011	Alternate Register Set #2 is assigned to IPL Level 4
010	Alternate Register Set #2 is assigned to IPL Level 3
001	Alternate Register Set #2 is assigned to IPL Level 2
000	Alternate Register Set #2 is assigned to IPL Level 1

Bits 2:0 – CTXT1[2:0] Specifies Alternate Working Register Set #1 with IPLs bits

Value	Description
111	Not assigned
110	Alternate Register Set #1 is assigned to IPL Level 7
101	Alternate Register Set #1 is assigned to IPL Level 6
100	Alternate Register Set #1 is assigned to IPL Level 5
011	Alternate Register Set #1 is assigned to IPL Level 4
010	Alternate Register Set #1 is assigned to IPL Level 3
001	Alternate Register Set #1 is assigned to IPL Level 2
000	Alternate Register Set #1 is assigned to IPL Level 1

30.2.16 FBTSEQ Configuration Register

Name: FBTSEQ

Offset: 0xFFC

Legend: PO = Program Once bit

Bit	23	22	21	20	19	18	17	16
	IBSEQ[11:0]							
Access	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO
Reset	1	1	1	1	1	1	1	1
Bit	15	14	13	12	11	10	9	8
	IBSEQ[11:0]				BSEQ[11:8]			
Access	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO
Reset	1	1	1	1	1	1	1	1
Bit	7	6	5	4	3	2	1	0
	BSEQ[7:0]							
Access	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO	R/PO
Reset	1	1	1	1	1	1	1	1

Bits 23:12 – IBSEQ[11:0] Inverse Boot Sequence Number bits (Dual Partition modes only)

The one's complement of BSEQ[11:0]; must be calculated by the user and written into device programming.

Bits 11:0 – BSEQ[11:0] Boot Sequence Number bits (Dual Partition modes only)

Relative value defining which partition will be active after a device Reset; the partition containing a lower boot number will be active.

30.2.17 FBOOT Configuration Register

Name: FBOOT

Legend: PO = Program Once bit

Bit	23	22	21	20	19	18	17	16
Access								
Reset								

Bit	15	14	13	12	11	10	9	8
Access								
Reset								

Bit	7	6	5	4	3	2	1	0
							BTMODE[1:0]	
Access							R/PO	R/PO
Reset							1	1

Bits 1:0 – BTMODE[1:0] Device Partition Mode Configuration Status bits

Value	Description
11	Single Partition mode
10	Dual Partition mode
01	Protected Dual Partition mode (Partition 1 is write-protected when inactive)
00	Reserved, do not use

30.3 Device Calibration and Identification

The dsPIC33CK512MPT608 devices have two Identification registers, near the end of configuration memory space, that store the Device ID (DEVID) and Device Revision (DEVREV). These registers are used to determine the mask, variant and manufacturing information about the device. These registers are read-only and are shown in [30.3.2. DEVID](#) and [30.3.1. DEVREV](#).

30.3.1 Device Revision Register

Name: DEVREV

Legend: R = Read-only bit, r = Reserved bit

Bit	23	22	21	20	19	18	17	16
Access								
Reset								
Bit	15	14	13	12	11	10	9	8
	Reserved							
Access	r							
Reset	0							
Bit	7	6	5	4	3	2	1	0
					DEVREV[3:0]			
Access					R	R	R	R
Reset								

Bit 15 – Reserved Maintain as '0'

Bits 3:0 – DEVREV[3:0] Device Revision bits

30.3.2 Device ID Register

Name: DEVID

Note:

- See [Table 30-2](#) for the list of Device Identifier bits.

Legend: R = Read-only bit

Bit	23	22	21	20	19	18	17	16
Access								
Reset								
Bit	15	14	13	12	11	10	9	8
Access	R	R	R	R	R	R	R	R
Reset								
Bit	7	6	5	4	3	2	1	0
Access	R	R	R	R	R	R	R	R
Reset								

Bits 15:8 – FAMID[7:0] Device Family Identifier bits

Value	Description
1010 0000	dsPIC33CK512MPT608 family

Bits 7:0 – DEV[7:0] Individual Device Identifier bits⁽¹⁾

30.3.3 Device IDs

Table 30-2. Device IDs for the dsPIC33CK512MPT608 Family

Device	DEVID
dsPIC33CK512MPT608-I/PT	0x9F5C
dsPIC33CK256MPT608-I/PT	0x9F4C

30.4 User OTP Memory

The dsPIC33CK512MPT608 family devices contain 64 One-Time-Programmable (OTP) double words, located at addresses, 801700h through 8017FEh. Each 48-bit OTP double word can only be written one time. The OTP Words can be used for storing checksums, code revisions, manufacturing dates, manufacturing lot numbers or any other application-specific information.

The OTP area is not cleared by any erase command. This memory can be written only once.

30.5 On-Chip Voltage Regulators

dsPIC33CK512MPT608 family devices have a capacitorless internal voltage regulator to supply power to the core at 1.2V (typical).

The regulators have Low-Power and Standby modes for use in Sleep modes. For additional information about Sleep, see [29.2.1. Sleep Mode](#).

When the regulators are in Low-Power mode (LPWREN = 1), the power available to the core is limited. Before the LPWREN bit is set, the device should be placed into a Lower Power state by disabling peripherals and lowering CPU frequency (e.g., 8 MHz FRC without PLL).

30.5.1 Voltage Regulator Control Register**Name:** VREGCON⁽¹⁾**Offset:** 0xCFC**Note:**

1. Hardware resets this register only on a POR Reset.

Legend: r = Reserved bit

Bit	15	14	13	12	11	10	9	8
	LPWREN							
Access	R/W							
Reset	0							

Bit	7	6	5	4	3	2	1	0
			Reserved[1:0]		Reserved[1:0]		Reserved[1:0]	
Access			r	r	r	r	r	r
Reset			0	0	0	0	0	0

Bit 15 – LPWREN Low-Power Mode Enable bit

Value	Description
1	Voltage regulators are in Low-Power mode
0	Voltage regulators are in Full-Power mode

Bits 5:4 – Reserved[1:0] Read as '0'**Bits 3:2 – Reserved[1:0]** Read as '0'**Bits 1:0 – Reserved[1:0]** Read as '0'**30.6 Brown-out Reset (BOR)**

The Brown-out Reset (BOR) module is based on an internal voltage reference circuit that monitors the regulated supply voltage. The main purpose of the BOR module is to generate a device Reset when a Brown-out condition occurs. Brown-out conditions are generally caused by glitches on the AC mains (for example, missing portions of the AC cycle waveform due to bad power transmission lines or voltage sags due to excessive current draw when a large inductive load is turned on).

A BOR generates a Reset pulse which resets the device. The BOR selects the clock source based on the device Configuration bit selections.

If an Oscillator mode is selected, the BOR activates the Oscillator Start-up Timer (OST). The system clock is held until OST expires. If the PLL is used, the clock is held until the LOCK bit (OSCCON[5]) is '1'.

The BOR status bit (RCON[1]) is set to indicate that a BOR has occurred. The BOR circuit continues to operate while in Sleep or Idle mode and resets the device should V_{DD} fall below the BOR threshold voltage.

Note: When V_{DD} is between VBOR-25 mV and VBOR, a BOR reset may repeatedly occur. A tristate IO can be used to observe this behavior. The VBOR specification is listed in the [33. Electrical Characteristics](#) in [Table 33-20](#). An external voltage monitor IC can be used to avoid this behavior. MCP111-300E and similar devices are recommended for this purpose.

30.7 Dual Watchdog Timer (WDT)

Note: This data sheet summarizes the features of the dsPIC33CK512MPT608 family of devices. It is not intended to be a comprehensive reference source. To complement the information in this data sheet, refer to “**Dual Watchdog Timer**” (www.microchip.com/DS70005250) in the “dsPIC33/PIC24 Family Reference Manual”.

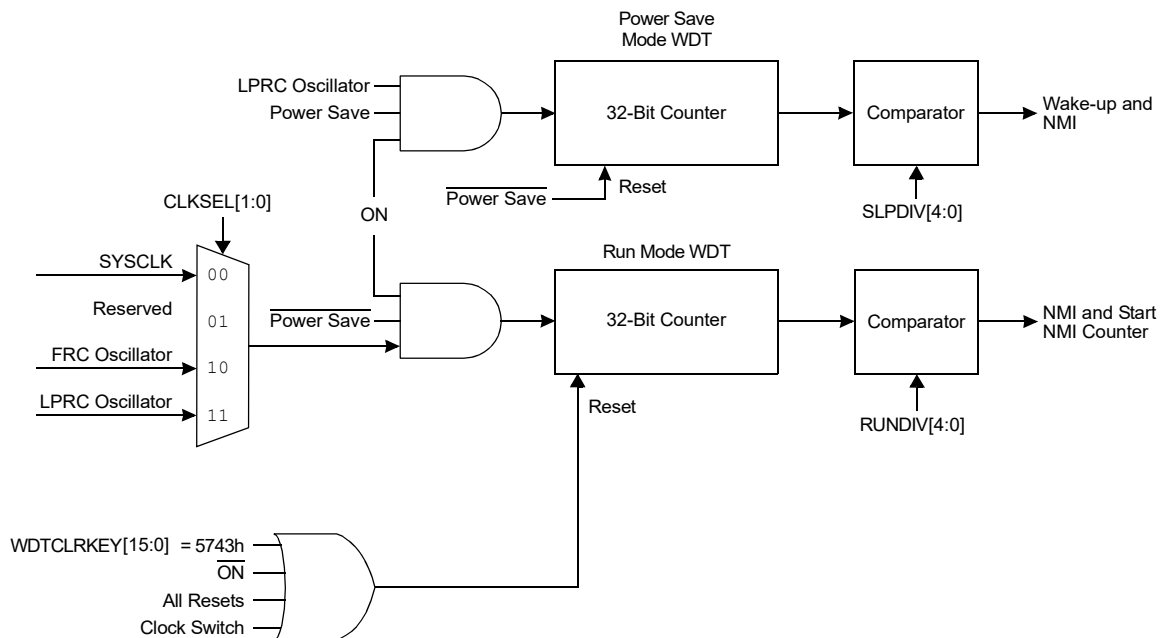
The dsPIC33 dual Watchdog Timer (WDT) is described in this section. Refer to [Figure 30-1](#) for a block diagram of the WDT.

The WDT, when enabled, operates from the internal Low-Power RC Oscillator clock source or a selectable clock source in Run mode. The WDT can be used to detect system software malfunctions by resetting the device if the WDT is not cleared periodically in software. The WDT can be configured in Windowed mode or Non-Windowed mode. Various WDT time-out periods can be selected using the WDT postscaler. The WDT can also be used to wake the device from Sleep or Idle mode (Power Save mode). If the WDT expires and issues a device Reset, the WTDO bit in the RCON register will be set.

The following are some of the key features of the WDT modules:

- Configuration or Software Controlled
- Separate User-Configurable Time-out Periods for Run and Sleep/Idle
- Can Wake the Device from Sleep or Idle
- User-Selectable Clock Source in Run mode
- Operates from LPRC in Sleep/Idle mode

Figure 30-1. Watchdog Timer Block Diagram



30.7.1 Watchdog Timer Control Register Low

Name: WDTCONL

Offset: 0xFB4

Notes:

1. A read of this bit will result in a '1' if the WDT is enabled by the device configuration or by software. The user's software should not read or write the peripheral's SFRs in the SYSCLK cycle immediately following the instruction that clears the module's ON bit.
2. The user's software should not read or write the peripheral's SFRs in the SYSCLK cycle immediately following the instruction that clears the module's ON bit.
3. These bits reflect the value of the Configuration bits.
4. The WDTWINEN bit reflects the status of the Configuration bit if the bit is set. If the bit is cleared, the value is controlled by software.
5. The available clock sources are device-dependent.

Legend: y = Value from Configuration bit on POR, HS = Hardware Settable bit

Bit	15	14	13	12	11	10	9	8
	ON			RUNDIV[4:0]				
Access	R/W			R	R	R	R	R
Reset	0			0	0	0	0	y

Bit	7	6	5	4	3	2	1	0
	CLKSEL[1:0]		SLPDIV[4:0]					WDTWINEN
Access	R	R	R	R	R	R	R	HS/R/W
Reset			0	0	0	0	y	0

Bit 15 – ON Watchdog Timer Enable bit^(1,2)

Value	Description
1	Enables the Watchdog Timer if it is not enabled by the device configuration
0	Disables the Watchdog Timer if it was enabled in software

Bits 12:8 – RUNDIV[4:0] WDT Run Mode Postscaler Status bits⁽³⁾

Value	Description
11111	Divide by $2^{31} = 2,147,483,648$
11110	Divide by $2^{30} = 1,073,741,824$
. . .	
00001	Divide by $2^1, 2$
00000	Divide by $2^0, 1$

Bits 7:6 – CLKSEL[1:0] WDT Run Mode Clock Select Status bits^(3,5)

Value	Description
11	LPRC Oscillator
10	FRC Oscillator
01	Reserved
00	SYSCLK

Bits 5:1 – SLPDIV[4:0] Sleep and Idle Mode WDT Postscaler Status bits⁽³⁾

Value	Description
11111	Divide by $2^{31} = 2,147,483,648$
11110	Divide by $2^{30} = 1,073,741,824$
. . .	
00001	Divide by $2^1, 2$
00000	Divide by $2^0, 1$

Bit 0 – WDTWINEN Watchdog Timer Window Enable bit⁽⁴⁾

Value	Description
1	Enables Window mode
0	Disables Window mode

30.7.2 Watchdog Timer Control Register High

Name: WDTCONH

Offset: 0xFB6

Bit	15	14	13	12	11	10	9	8
	WDTCLRKEY[15:8]							
Access	W	W	W	W	W	W	W	W
Reset	0	0	0	0	0	0	0	0
Bit	7	6	5	4	3	2	1	0
	WDTCLRKEY[7:0]							
Access	W	W	W	W	W	W	W	W
Reset	0	0	0	0	0	0	0	0

Bits 15:0 – WDTCLRKEY[15:0] Watchdog Timer Clear Key Bits

To clear the Watchdog Timer to prevent a time-out, software must write the value, 0x5743, to this location using a single 16-bit write.

30.8 JTAG Interface

The dsPIC33CK512MPT608 family devices implement a JTAG interface, which supports boundary scan device testing. Programming is not supported through the JTAG interface; only boundary scan is supported.

Note: Refer to “**Programming and Diagnostics**” (www.microchip.com/DS70608) in the “dsPIC33/PIC24 Family Reference Manual” for further information on usage, configuration and operation of the JTAG interface.

30.9 In-Circuit Debugger

When MPLAB® ICD 3 or the REAL ICE™ emulator is selected as a debugger, the in-circuit debugging functionality is enabled. This function allows simple debugging functions when used with MPLAB IDE. Debugging functionality is controlled through the PGCx (Emulation/Debug Clock) and PGDx (Emulation/Debug Data) pin functions.

Any of the three pairs of debugging clock/data pins can be used:

- PGC1 and PGD1 Host Debug or Client Debug.
- PGC2 and PGD2 Host Debug or Client Debug.
- PGC3 and PGD3 Host Debug or Client Debug for debugging Host and Client simultaneously; two ICD or the REAL ICE™ emulator are required. This mode of debugging, where the Host and Client are simultaneously debugged, is called the Dual Debug mode.

To use the in-circuit debugger function of the device, the design must implement ICSP connections to $\overline{\text{MCLR}}$, V_{DD} , V_{SS} and the PGCx/PGDx pin pair. In addition, when the feature is enabled, some of the resources are not available for general use. These resources include the first 80 bytes of data RAM and two or five (in Dual Debug mode) I/O pins (PGCx and PGDx).

30.10 Code Protection and CodeGuard™ Security

dsPIC33CK512MPT608 family devices offer multiple levels of security for protecting individual intellectual property. The program Flash protection can be broken up into three segments: Boot Segment (BS), General Segment (GS) and Configuration Segment (CS). Boot Segment has the highest security privilege and can be thought to have limited restrictions when accessing other segments. General Segment has the least security and is intended for the end user system code. Configuration Segment contains only the device user configuration data, which are located at the end of the program memory space.

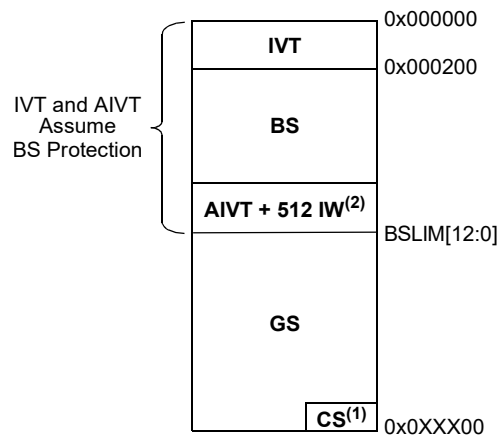
The code protection features are controlled by the Configuration registers, FSEC and FBSLIM. The FSEC register controls the code-protect level for each segment and if that segment is write-protected. The size of BS and GS will

depend on the BSLIM[12:0] bits setting and if the Alternate Interrupt Vector Table (AIVT) is enabled. The BSLIM[12:0] bits define the number of pages for BS, with each page containing 1024 IW. The smallest BS size is one page, which will consist of the Interrupt Vector Table (IVT) and 512 IW of code protection.

If the AIVT is enabled, the last page of BS will contain the AIVT and will not contain any BS code. With AIVT enabled, the smallest BS size is now two pages (2048 IW), with one page for the IVT and BS code, and the other page for the AIVT. Write protection of the BS does not cover the AIVT. The last page of BS can always be programmed or erased by BS code. The General Segment will start at the next page and will consume the rest of program Flash, except for the Flash Configuration Words. The IVT will assume GS security only if BS is not enabled. The IVT is protected from being programmed or page erased when either security segment has enabled write protection.

The different device security segments are shown in [Figure 30-2](#). Here, all three segments are shown, but are not required. If only basic code protection is required, then GS can be enabled independently or combined with CS, if desired.

Figure 30-2. Security Segments Example



Notes:

1. If CS is write-protected, the last page (GS + CS) of program memory will be protected from an Erase condition.
2. The last half (256 IW) of the last page of BS is unusable program memory.

31. Instruction Set Summary

Note: This data sheet summarizes the features of the dsPIC33CK512MPT608 family of devices. It is not intended to be a comprehensive reference source. To complement the information in this data sheet, refer to the related section in the “dsPIC33/PIC24 Family Reference Manual”, which is available from the Microchip website (www.microchip.com).

The dsPIC33CK instruction set is almost identical to that of the dsPIC30F and dsPIC33F.

Most instructions are a single program memory word (24 bits). Only three instructions require two program memory locations.

Each single-word instruction is a 24-bit word, divided into an 8-bit opcode, which specifies the instruction type and one or more operands, which further specify the operation of the instruction.

The instruction set is highly orthogonal and is grouped into five basic categories:

- Word or byte-oriented operations
- Bit-oriented operations
- Literal operations
- DSP operations
- Control operations

Table 31-1 lists the general symbols used in describing the instructions.

The dsPIC33 instruction set summary in Table 31-2 lists all the instructions, along with the status flags affected by each instruction.

Most word or byte-oriented W register instructions (including barrel shift instructions) have three operands:

- The first source operand, which is typically a register ‘Wb’ without any address modifier
- The second source operand, which is typically a register ‘Ws’ with or without an address modifier
- The destination of the result, which is typically a register ‘Wd’ with or without an address modifier

However, word or byte-oriented file register instructions have two operands:

- The file register specified by the value ‘f’
- The destination, which could be either the file register ‘f’ or the W0 register, which is denoted as ‘WREG’

Most bit-oriented instructions (including simple rotate/shift instructions) have two operands:

- The W register (with or without an address modifier) or file register (specified by the value of ‘Ws’ or ‘f’)
- The bit in the W register or file register (specified by a literal value or indirectly by the contents of register ‘Wb’)

The literal instructions that involve data movement can use some of the following operands:

- A literal value to be loaded into a W register or file register (specified by ‘k’)
- The W register or file register where the literal value is to be loaded (specified by ‘Wb’ or ‘f’)

However, literal instructions that involve arithmetic or logical operations use some of the following operands:

- The first source operand, which is a register ‘Wb’ without any address modifier
- The second source operand, which is a literal value
- The destination of the result (only if not the same as the first source operand), which is typically a register ‘Wd’ with or without an address modifier

The MAC class of DSP instructions can use some of the following operands:

- The accumulator (A or B) to be used (required operand)
- The W registers to be used as the two operands
- The X and Y address space prefetch operations
- The X and Y address space prefetch destinations
- The accumulator write-back destination

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The other DSP instructions do not involve any multiplication and can include:

- The accumulator to be used (required)
- The source or destination operand (designated as Wso or Wdo, respectively) with or without an address modifier
- The amount of shift specified by a W register 'Wn' or a literal value

The control instructions can use some of the following operands:

- A program memory address
- The mode of the Table Read and Table Write instructions

Most instructions are a single word. Certain double-word instructions are designed to provide all the required information in these 48 bits. In the second word, the eight MSBs are '0's. If this second word is executed as an instruction (by itself), it executes as a NOP.

The double-word instructions execute in two instruction cycles.

Most single-word instructions are executed in a single instruction cycle, unless a conditional test is true or the Program Counter is changed as a result of the instruction, or a PSV or Table Read is performed. In these cases, the execution takes multiple instruction cycles, with the additional instruction cycle(s) executed as a NOP. Certain instructions that involve skipping over the subsequent instruction require either two or three cycles if the skip is performed, depending on whether the instruction being skipped is a single-word or two-word instruction. Moreover, double-word moves require two cycles.

Note: For more details on the instruction set, refer to the “16-Bit MCU and DSC Programmer’s Reference Manual” (www.microchip.com/DS70000157).

Table 31-1. Symbols Used in Opcode Descriptions

Field	Description
#text	Means literal defined by “text”
(text)	Means “content of text”
[text]	Means “the location addressed by text”
{ }	Optional field or operation
$a \in \{b, c, d\}$	a is selected from the set of values b, c, d
[n:m]	Register bit field
.b	Byte mode selection
.d	Double-Word mode selection
.S	Shadow register select
.w	Word mode selection (default)
Acc	One of two accumulators {A, B}
AWB	Accumulator Write-Back Destination Address register $\in \{W13, [W13]+ = 2\}$
bit4	4-bit bit selection field (used in word-addressed instructions) $\in \{0...15\}$
C, DC, N, OV, Z	MCU Status bits: Carry, Digit Carry, Negative, Overflow, Sticky Zero
Expr	Absolute address, label or expression (resolved by the linker)
f	File register address $\in \{0x0000...0x1FFF\}$
lit1	1-bit unsigned literal $\in \{0,1\}$

Note: In dsPIC33CK512MPT608 devices, read and Read-Modify-Write (RMW) operations on non-CPU Special Function Registers require an additional cycle when compared to dsPIC30F, dsPIC33F, PIC24F and PIC24H devices.

.....continued	
Field	Description
lit4	4-bit unsigned literal $\in \{0...15\}$
lit5	5-bit unsigned literal $\in \{0...31\}$
lit8	8-bit unsigned literal $\in \{0...255\}$
lit10	10-bit unsigned literal $\in \{0...255\}$ for Byte mode, $\{0:1023\}$ for Word mode
lit14	14-bit unsigned literal $\in \{0...16384\}$
lit16	16-bit unsigned literal $\in \{0...65535\}$
lit23	23-bit unsigned literal $\in \{0...8388608\}$; LSb must be '0'
None	Field does not require an entry, can be blank
OA, OB, SA, SB	DSP Status bits: ACCA Overflow, ACCB Overflow, ACCA Saturate, ACCB Saturate
PC	Program Counter
Slit10	10-bit signed literal $\in \{-512...511\}$
Slit16	16-bit signed literal $\in \{-32768...32767\}$
Slit6	6-bit signed literal $\in \{-16...16\}$
Wb	Base W register $\in \{W0...W15\}$
Wd	Destination W register $\in \{Wd, [Wd], [Wd++] , [Wd--], [++Wd], [--Wd] \}$
Wdo	Destination W register $\in \{Wnd, [Wnd], [Wnd++] , [Wnd--], [++Wnd], [--Wnd], [Wnd+Wb] \}$
Wm,Wn	Dividend, Divisor Working register pair (direct addressing)
Wm*Wm	Multiplicand and Multiplier Working register pair for Square instructions $\in \{W4 * W4, W5 * W5, W6 * W6, W7 * W7\}$
Wm*Wn	Multiplicand and Multiplier Working register pair for DSP instructions $\in \{W4 * W5, W4 * W6, W4 * W7, W5 * W6, W5 * W7, W6 * W7\}$
Wn	One of 16 Working registers $\in \{W0...W15\}$
Wnd	One of 16 Destination Working registers $\in \{W0...W15\}$
Wns	One of 16 Source Working registers $\in \{W0...W15\}$
WREG	W0 (Working register used in file register instructions)
Ws	Source W register $\in \{Ws, [Ws], [Ws++] , [Ws--], [++Ws], [--Ws] \}$
Wso	Source W register $\in \{Wns, [Wns], [Wns++] , [Wns--], [++Wns], [--Wns], [Wns+Wb] \}$
Wx	X Data Space Prefetch Address register for DSP instructions $\in \{[W8] + = 6, [W8] + = 4, [W8] + = 2, [W8], [W8] - = 6, [W8] - = 4, [W8] - = 2,$ $[W9] + = 6, [W9] + = 4, [W9] + = 2, [W9], [W9] - = 6, [W9] - = 4, [W9] - = 2,$ $[W9 + W12], \text{none}\}$
Wxd	X Data Space Prefetch Destination register for DSP instructions $\in \{W4...W7\}$
Note: In dsPIC33CK512MPT608 devices, read and Read-Modify-Write (RMW) operations on non-CPU Special Function Registers require an additional cycle when compared to dsPIC30F, dsPIC33F, PIC24F and PIC24H devices.	

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Instruction Set Summary

.....continued	
Field	Description
Wy	Y Data Space Prefetch Address register for DSP instructions $\in \{[W10] + 6, [W10] + 4, [W10] + 2, [W10], [W10] - 6, [W10] - 4, [W10] - 2, [W11] + 6, [W11] + 4, [W11] + 2, [W11], [W11] - 6, [W11] - 4, [W11] - 2, [W11 + W12], \text{none}\}$
Wyd	Y Data Space Prefetch Destination register for DSP instructions $\in \{W4...W7\}$
Note: In dsPIC33CK512MPT608 devices, read and Read-Modify-Write (RMW) operations on non-CPU Special Function Registers require an additional cycle when compared to dsPIC30F, dsPIC33F, PIC24F and PIC24H devices.	

Table 31-2. Instruction Set Overview

Base Instr #	Assembly Mnemonic	Assembly Syntax		Description	# of Words	# of Cycles ⁽¹⁾	Status Flags Affected
1	ADD	ADD	Acc	Add Accumulators	1	1	OA,OB,SA,SB
		ADD	f	$f = f + \text{WREG}$	1	1	C,DC,N,OV,Z
		ADD	f, WREG	$\text{WREG} = f + \text{WREG}$	1	1	C,DC,N,OV,Z
		ADD	#lit10, Wn	$\text{Wd} = \text{lit10} + \text{Wd}$	1	1	C,DC,N,OV,Z
		ADD	Wb, Ws, Wd	$\text{Wd} = \text{Wb} + \text{Ws}$	1	1	C,DC,N,OV,Z
		ADD	Wb, #lit5, Wd	$\text{Wd} = \text{Wb} + \text{lit5}$	1	1	C,DC,N,OV,Z
		ADD	Wso, #Slit4, Acc	16-bit Signed Add to Accumulator	1	1	OA,OB,SA,SB
2	ADDC	ADDC	f	$f = f + \text{WREG} + (C)$	1	1	C,DC,N,OV,Z
		ADDC	f, WREG	$\text{WREG} = f + \text{WREG} + (C)$	1	1	C,DC,N,OV,Z
		ADDC	#lit10, Wn	$\text{Wd} = \text{lit10} + \text{Wd} + (C)$	1	1	C,DC,N,OV,Z
		ADDC	Wb, Ws, Wd	$\text{Wd} = \text{Wb} + \text{Ws} + (C)$	1	1	C,DC,N,OV,Z
		ADDC	Wb, #lit5, Wd	$\text{Wd} = \text{Wb} + \text{lit5} + (C)$	1	1	C,DC,N,OV,Z
3	AND	AND	f	$f = f .\text{AND.} \text{WREG}$	1	1	N,Z
		AND	f, WREG	$\text{WREG} = f .\text{AND.} \text{WREG}$	1	1	N,Z
		AND	#lit10, Wn	$\text{Wd} = \text{lit10} .\text{AND.} \text{Wd}$	1	1	N,Z
		AND	Wb, Ws, Wd	$\text{Wd} = \text{Wb} .\text{AND.} \text{Ws}$	1	1	N,Z
		AND	Wb, #lit5, Wd	$\text{Wd} = \text{Wb} .\text{AND.} \text{lit5}$	1	1	N,Z
4	ASR	ASR	f	$f = \text{Arithmetic Right Shift } f$	1	1	C,N,OV,Z
		ASR	f, WREG	$\text{WREG} = \text{Arithmetic Right Shift } f$	1	1	C,N,OV,Z
		ASR	Ws, Wd	$\text{Wd} = \text{Arithmetic Right Shift } \text{Ws}$	1	1	C,N,OV,Z
		ASR	Wb, Wns, Wnd	$\text{Wnd} = \text{Arithmetic Right Shift } \text{Wb by } \text{Wns}$	1	1	N,Z
		ASR	Wb, #lit5, Wnd	$\text{Wnd} = \text{Arithmetic Right Shift } \text{Wb by } \text{lit5}$	1	1	N,Z
5	BCLR	BCLR	f, #bit4	Bit Clear f	1	1	None
		BCLR	Ws, #bit4	Bit Clear Ws	1	1	None

Notes:

1. Read and Read-Modify-Write (e.g., bit operations and logical operations) on non-CPU SFRs incur an additional instruction cycle.
2. For dsPIC33CK512MPT608 devices, the divide instructions must be preceded with a "REPEAT #5" instruction, such that they are executed six consecutive times.

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Instruction Set Summary

.....continued							
Base Instr #	Assembly Mnemonic	Assembly Syntax		Description	# of Words	# of Cycles(1)	Status Flags Affected
6	BFEXT	BFEXT	bit4, wid5, Ws, Wb	Bit Field Extract from Ws to Wb	2	2	None
		BFEXT	bit4, wid5, f, Wb	Bit Field Extract from f to Wb	2	2	None
7	BFINS	BFINS	bit4, wid5, Wb, Ws	Bit Field Insert from Wb into Ws	2	2	None
		BFINS	bit4, wid5, Wb, f	Bit Field Insert from Wb into f	2	2	None
		BFINS	bit4, wid5, lit8, Ws	Bit Field Insert from #lit8 to Ws	2	2	None
8	BOOTSWP	BOOTSWP		Swap the Active and Inactive Program Flash Space	1	2	None
9	BRA	BRA	C, Expr	Branch if Carry	1	1 (4)/1	None
		BRA	GE, Expr	Branch if Greater Than or Equal	1	1 (4)/1	None
		BRA	GEU, Expr	Branch if unsigned Greater Than or Equal	1	1 (4)/1	None
		BRA	GT, Expr	Branch if Greater Than	1	1 (4)/1	None
		BRA	GTU, Expr	Branch if Unsigned Greater Than	1	1 (4)/1	None
		BRA	LE, Expr	Branch if Less Than or Equal	1	1 (4)/1	None
		BRA	LEU, Expr	Branch if Unsigned Less Than or Equal	1	1 (4)/1	None
		BRA	LT, Expr	Branch if Less Than	1	1 (4)/1	None
		BRA	LTU, Expr	Branch if Unsigned Less Than	1	1 (4)/1	None
		BRA	N, Expr	Branch if Negative	1	1 (4)/1	None
		BRA	NC, Expr	Branch if Not Carry	1	1 (4)/1	None
		BRA	NN, Expr	Branch if Not Negative	1	1 (4)/1	None
		BRA	NOV, Expr	Branch if Not Overflow	1	1 (4)/1	None
		BRA	NZ, Expr	Branch if Not Zero	1	1 (4)/1	None
		BRA	OA, Expr	Branch if Accumulator A Overflow	1	1 (4)/1	None
		BRA	OB, Expr	Branch if Accumulator B Overflow	1	1 (4)/1	None
		BRA	OV, Expr	Branch if Overflow	1	1 (4)/1	None
		BRA	SA, Expr	Branch if Accumulator A Saturated	1	1 (4)/1	None
		BRA	SB, Expr	Branch if Accumulator B Saturated	1	1 (4)/1	None
		BRA	Expr	Branch Unconditionally	1	4/2	None
		BRA	Z, Expr	Branch if Zero	1	1 (4)/1	None
		BRA	Wn	Computed Branch	1	4	None
10	BREAK	BREAK		Stop User Code Execution	1	1	None
11	BSET	BSET	f, #bit4	Bit Set f	1	1	None
			Ws, #bit4	Bit Set Ws	1	1	None
Notes: 1. Read and Read-Modify-Write (e.g., bit operations and logical operations) on non-CPU SFRs incur an additional instruction cycle. 2. For dsPIC33CK512MPT608 devices, the divide instructions must be preceded with a “REPEAT #5” instruction, such that they are executed six consecutive times.							

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Instruction Set Summary

.....continued

Base Instr #	Assembly Mnemonic	Assembly Syntax		Description	# of Words	# of Cycles ⁽¹⁾	Status Flags Affected
12	BSW	BSW.C	Ws,Wb	Write C Bit to Ws[Wb]	1	1	None
		BSW.Z	Ws,Wb	Write Z Bit to Ws[Wb]	1	1	None
13	BTG	BTG	f,#bit4	Bit Toggle f	1	1	None
		BTG	Ws,#bit4	Bit Toggle Ws	1	1	None
14	BTSC	BTSC	f,#bit4	Bit Test f, Skip if Clear	1	1 (2 or 3)	None
		BTSC	Ws,#bit4	Bit Test Ws, Skip if Clear	1	1 (2 or 3)	None
15	BTSS	BTSS	f,#bit4	Bit Test f, Skip if Set	1	1 (2 or 3)	None
		BTSS	Ws,#bit4	Bit Test Ws, Skip if Set	1	1 (2 or 3)	None
16	BTST	BTST	f,#bit4	Bit Test f	1	1	Z
		BTST.C	Ws,#bit4	Bit Test Ws to C	1	1	C
		BTST.Z	Ws,#bit4	Bit Test Ws to Z	1	1	Z
		BTST.C	Ws,Wb	Bit Test Ws[Wb] to C	1	1	C
		BTST.Z	Ws,Wb	Bit Test Ws[Wb] to Z	1	1	Z
17	BTSTS	BTSTS	f,#bit4	Bit Test then Set f	1	1	Z
		BTSTS.C	Ws,#bit4	Bit Test Ws to C, then Set	1	1	C
		BTSTS.Z	Ws,#bit4	Bit Test Ws to Z, then Set	1	1	Z
18	CALL	CALL	lit23	Call Subroutine	2	4(2)	SFA
		CALL	Wn	Call Indirect Subroutine	1	4(2)	SFA
		CALL.L	Wn	Call Indirect Subroutine (long address)	1	4(2)	SFA
19	CLR	CLR	f	f = 0x0000	1	1	None
		CLR	WREG	WREG = 0x0000	1	1	None
		CLR	Ws	Ws = 0x0000	1	1	None
		CLR	Acc,Wx,Wxd,Wy,Wyd,AWB	Clear Accumulator	1	1	OA,OB,SA,SB
20	CLRWDT	CLRWDT		Clear Watchdog Timer	1	1	WDTO,Sleep
21	COM	COM	f	f = $\bar{f}$	1	1	N,Z
		COM	f,WREG	WREG = $\bar{f}$	1	1	N,Z
		COM	Ws,Wd	Wd = $\overline{Ws}$	1	1	N,Z
22	CP	CP	f	Compare f with WREG	1	1	C,DC,N,OV,Z
		CP	Wb,#lit8	Compare Wb with lit8	1	1	C,DC,N,OV,Z
		CP	Wb,Ws	Compare Wb with Ws (Wb – Ws)	1	1	C,DC,N,OV,Z
23	CP0	CP0	f	Compare f with 0x0000	1	1	C,DC,N,OV,Z
		CP0	Ws	Compare Ws with 0x0000	1	1	C,DC,N,OV,Z
Notes: 1. Read and Read-Modify-Write (e.g., bit operations and logical operations) on non-CPU SFRs incur an additional instruction cycle. 2. For dsPIC33CK512MPT608 devices, the divide instructions must be preceded with a “REPEAT #5” instruction, such that they are executed six consecutive times.							

dsPIC33CK512MPT608 Family

Instruction Set Summary

.....continued							
Base Instr #	Assembly Mnemonic	Assembly Syntax		Description	# of Words	# of Cycles(1)	Status Flags Affected
24	CPB	CPB	f	Compare f with WREG, with Borrow	1	1	C,DC,N,OV,Z
		CPB	Wb, #lit8	Compare Wb with lit8, with Borrow	1	1	C,DC,N,OV,Z
		CPB	Wb, Ws	Compare Wb with Ws, with Borrow (Wb – Ws – C̅)	1	1	C,DC,N,OV,Z
25	CPSEQ	CPSEQ	Wb, Wn	Compare Wb with Wn, Skip if =	1	1 (2 or 3)	None
	CPBEQ	CPBEQ	Wb, Wn, Expr	Compare Wb with Wn, Branch if =	1	1 (5)	None
26	CPSGT	CPSGT	Wb, Wn	Compare Wb with Wn, Skip if >	1	1 (2 or 3)	None
	CPBGT	CPBGT	Wb, Wn, Expr	Compare Wb with Wn, Branch if >	1	1 (5)	None
27	CPSLT	CPSLT	Wb, Wn	Compare Wb with Wn, Skip if <	1	1 (2 or 3)	None
	CPBLT	CPBLT	Wb, Wn, Expr	Compare Wb with Wn, Branch if <	1	1 (5)	None
28	CPSNE	CPSNE	Wb, Wn	Compare Wb with Wn, Skip if ≠	1	1 (2 or 3)	None
	CPBNE	CPBNE	Wb, Wn, Expr	Compare Wb with Wn, Branch if ≠	1	1 (5)	None
29	CTXTSWP	CTXTSWP	#lit3	Switch CPU Register Context to Context Defined by lit3	1	2	None
30	CTXTSWP	CTXTSWP	Wn	Switch CPU Register Context to Context Defined by Wn	1	2	None
31	DAW.B	DAW.B	Wn	Wn = Decimal Adjust Wn	1	1	C
32	DEC	DEC	f	f = f – 1	1	1	C,DC,N,OV,Z
		DEC	f, WREG	WREG = f – 1	1	1	C,DC,N,OV,Z
		DEC	Ws, Wd	Wd = Ws – 1	1	1	C,DC,N,OV,Z
33	DEC2	DEC2	f	f = f – 2	1	1	C,DC,N,OV,Z
		DEC2	f, WREG	WREG = f – 2	1	1	C,DC,N,OV,Z
		DEC2	Ws, Wd	Wd = Ws – 2	1	1	C,DC,N,OV,Z
34	DISI	DISI	#lit14	Disable Interrupts for k Instruction Cycles	1	1	None
35	DIVF ⁽²⁾	DIVF	Wm, Wn	Signed 16/16-Bit Fractional Divide	1	6	N,Z,C,OV
36	DIV.S ⁽²⁾	DIV.S	Wm, Wn	Signed 16/16-Bit Integer Divide	1	6	N,Z,C,OV
		DIV.SD	Wm, Wn	Signed 32/16-Bit Integer Divide	1	6	N,Z,C,OV
Notes: 1. Read and Read-Modify-Write (e.g., bit operations and logical operations) on non-CPU SFRs incur an additional instruction cycle. 2. For dsPIC33CK512MPT608 devices, the divide instructions must be preceded with a “REPEAT #5” instruction, such that they are executed six consecutive times.							

dsPIC33CK512MPT608 Family

Instruction Set Summary

.....continued							
Base Instr #	Assembly Mnemonic	Assembly Syntax		Description	# of Words	# of Cycles(1)	Status Flags Affected
37	DIV.U ⁽²⁾	DIV.U	Wm, Wn	Unsigned 16/16-Bit Integer Divide	1	6	N,Z,C,OV
		DIV.UD	Wm,Wn	Unsigned 32/16-Bit Integer Divide	1	6	N,Z,C,OV
38	DIVF2 ⁽²⁾	DIVF2	Wm, Wn	Signed 16/16-Bit Fractional Divide (W1:W0 preserved)	1	6	N,Z,C,OV
39	DIV2.S ⁽²⁾	DIV2.S	Wm, Wn	Signed 16/16-Bit Integer Divide (W1:W0 preserved)	1	6	N,Z,C,OV
		DIV2.SD	Wm, Wn	Signed 32/16-Bit Integer Divide (W1:W0 preserved)	1	6	N,Z,C,OV
40	DIV2.U ⁽²⁾	DIV2.U	Wm, Wn	Unsigned 16/16-Bit Integer Divide (W1:W0 preserved)	1	6	N,Z,C,OV
		DIV2.UD	Wm, Wn	Unsigned 32/16-Bit Integer Divide (W1:W0 preserved)	1	6	N,Z,C,OV
41	DO	DO	#lit15, Expr	Do Code to PC + Expr, lit15 + 1 Times	2	2	None
		DO	Wn, Expr	Do code to PC + Expr, (Wn) + 1 Times	2	2	None
42	ED	ED	Wm*Wm, Acc, Wx, Wy, Wxd	Euclidean Distance (no accumulate)	1	1	OA,OB,OAB,SA,SB,SAB
43	EDAC	EDAC	Wm*Wm, Acc, Wx, Wy, Wxd	Euclidean Distance	1	1	OA,OB,OAB,SA,SB,SAB
44	EXCH	EXCH	Wns, Wnd	Swap Wns with Wnd	1	1	None
46	FBCL	FBCL	Ws, Wnd	Find Bit Change from Left (MSb) Side	1	1	C
47	FF1L	FF1L	Ws, Wnd	Find First One from Left (MSb) Side	1	1	C
48	FF1R	FF1R	Ws, Wnd	Find First One from Right (LSb) Side	1	1	C
49	FLIM	FLIM	Wb, Ws	Force Data (Upper and Lower) Range Limit without Limit Excess Result	1	1	N,Z,OV
		FLIM.V	Wb, Ws, Wd	Force Data (Upper and Lower) Range Limit with Limit Excess Result	1	1	N,Z,OV
50	GOTO	GOTO	Expr	Go to Address	2	4/2	None
		GOTO	Wn	Go to Indirect	1	4/2	None
		GOTO.L	Wn	Go to Indirect (long address)	1	4/2	None
51	INC	INC	f	f = f + 1	1	1	C,DC,N,OV,Z
		INC	f, WREG	WREG = f + 1	1	1	C,DC,N,OV,Z
		INC	Ws, Wd	Wd = Ws + 1	1	1	C,DC,N,OV,Z
Notes:							
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dsPIC33CK512MPT608 Family

Instruction Set Summary

.....continued

Base Instr #	Assembly Mnemonic	Assembly Syntax		Description	# of Words	# of Cycles(1)	Status Flags Affected
52	INC2	INC2	f	f = f + 2	1	1	C,DC,N,OV,Z
		INC2	f,WREG	WREG = f + 2	1	1	C,DC,N,OV,Z
		INC2	Ws,Wd	Wd = Ws + 2	1	1	C,DC,N,OV,Z
53	IOR	IOR	f	f = f .IOR. WREG	1	1	N,Z
		IOR	f,WREG	WREG = f .IOR. WREG	1	1	N,Z
		IOR	#lit10,Wn	Wd = lit10 .IOR. Wd	1	1	N,Z
		IOR	Wb,Ws,Wd	Wd = Wb .IOR. Ws	1	1	N,Z
		IOR	Wb,#lit5,Wd	Wd = Wb .IOR. lit5	1	1	N,Z
54	LAC	LAC	Wso,#Slit4,Acc	Load Accumulator	1	1	OA,OB,OAB,SA,SB,SAB
		LAC.D	Wso, #Slit4, Acc	Load Accumulator Double	1	2	OA,SA,OB,SB
56	LNK	LNK	#lit14	Link Frame Pointer	1	1	SFA
57	LSR	LSR	f	f = Logical Right Shift f	1	1	C,N,OV,Z
		LSR	f,WREG	WREG = Logical Right Shift f	1	1	C,N,OV,Z
		LSR	Ws,Wd	Wd = Logical Right Shift Ws	1	1	C,N,OV,Z
		LSR	Wb,Wns,Wnd	Wnd = Logical Right Shift Wb by Wns	1	1	N,Z
		LSR	Wb,#lit5,Wnd	Wnd = Logical Right Shift Wb by lit5	1	1	N,Z
58	MAC	MAC	Wm*Wn,Acc,Wx,Wxd,Wy,Wyd,AWB	Multiply and Accumulate	1	1	OA,OB,OAB,SA,SB,SAB
		MAC	Wm*Wm,Acc,Wx,Wxd,Wy,Wyd	Square and Accumulate	1	1	OA,OB,OAB,SA,SB,SAB
59	MAX	MAX	Acc	Force Data Maximum Range Limit	1	1	N,OV,Z
		MAX.V	Acc, Wnd	Force Data Maximum Range Limit with Result	1	1	N,OV,Z
60	MIN	MIN	Acc	If Accumulator A Less than B Load Accumulator with B or vice versa	1	1	N,OV,Z
		MIN.V	Acc, Wd	If Accumulator A Less than B Accumulator Force Minimum Data Range Limit with Limit Excess Result	1	1	N,OV,Z
		MINZ	Acc	Accumulator Force Minimum Data Range Limit	1	1	N,OV,Z
		MINZ.V	Acc, Wd	Accumulator Force Minimum Data Range Limit with Limit Excess Result	1	1	N,OV,Z

Notes:

1.

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2.

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dsPIC33CK512MPT608 Family

Instruction Set Summary

.....continued							
Base Instr #	Assembly Mnemonic	Assembly Syntax		Description	# of Words	# of Cycles ⁽¹⁾	Status Flags Affected
61	MOV	MOV	f, Wn	Move f to Wn	1	1	None
		MOV	f	Move f to f	1	1	None
		MOV	f, WREG	Move f to WREG	1	1	None
		MOV	#lit16, Wn	Move 16-Bit Literal to Wn	1	1	None
		MOV.b	#lit8, Wn	Move 8-Bit Literal to Wn	1	1	None
		MOV	Wn, f	Move Wn to f	1	1	None
		MOV	Wso, Wdo	Move Ws to Wd	1	1	None
		MOV	WREG, f	Move WREG to f	1	1	None
		MOV.D	Wns, Wd	Move Double from W(ns):W(ns + 1) to Wd	1	2	None
		MOV.D	Ws, Wnd	Move Double from Ws to W(nd + 1):W(nd)	1	2	None
62	MOVPG	MOVPG	#lit10, DSRPAG	Move 10-Bit Literal to DSRPAG	1	1	None
		MOVPG	#lit8, TBLPAG	Move 8-Bit Literal to TBLPAG	1	1	None
		MOVPG	Ws, DSRPAG	Move Ws[9:0] to DSRPAG	1	1	None
		MOVPG	Ws, TBLPAG	Move Ws[7:0] to TBLPAG	1	1	None
64	MOVSAC	MOVSAC	Acc, Wx, Wxd, Wy, Wyd, AWB	Prefetch and Store Accumulator	1	1	None
65	MPY	MPY	Wm*Wn, Acc, Wx, Wxd, Wy, Wyd	Multiply Wm by Wn to Accumulator	1	1	OA,OB,OAB, SA,SB,SAB
		MPY	Wm*Wm, Acc, Wx, Wxd, Wy, Wyd	Square Wm to Accumulator	1	1	OA,OB,OAB, SA,SB,SAB
66	MPY.N	MPY.N	Wm*Wn, Acc, Wx, Wxd, Wy, Wyd	-(Multiply Wm by Wn) to Accumulator	1	1	None
67	MSC	MSC	Wm*Wm, Acc, Wx, Wxd, Wy, Wyd, AWB	Multiply and Subtract from Accumulator	1	1	OA,OB,OAB, SA,SB,SAB
Notes: - Read and Read-Modify-Write (e.g., bit operations and logical operations) on non-CPU SFRs incur an additional instruction cycle. - For dsPIC33CK512MPT608 devices, the divide instructions must be preceded with a "REPEAT #5" instruction, such that they are executed six consecutive times.							

dsPIC33CK512MPT608 Family

Instruction Set Summary

.....continued

Base Instr #	Assembly Mnemonic	Assembly Syntax		Description	# of Words	# of Cycles ⁽¹⁾	Status Flags Affected
68	MUL	MUL.SS	Wb, Ws, Wnd	{Wnd + 1, Wnd} = Signed(Wb) * Signed(Ws)	1	1	None
		MUL.SS	Wb, Ws, Acc	Accumulator = Signed(Wb) * Signed(Ws)	1	1	None
		MUL.SU	Wb, Ws, Wnd	{Wnd + 1, Wnd} = Signed(Wb) * Unsigned(Ws)	1	1	None
		MUL.SU	Wb, Ws, Acc	Accumulator = Signed(Wb) * Unsigned(Ws)	1	1	None
		MUL.SU	Wb, #lit5, Acc	Accumulator = Signed(Wb) * Unsigned(lit5)	1	1	None
		MUL.US	Wb, Ws, Wnd	{Wnd + 1, Wnd} = Unsigned(Wb) * Signed(Ws)	1	1	None
		MUL.US	Wb, Ws, Acc	Accumulator = Unsigned(Wb) * Signed(Ws)	1	1	None
		MUL.UU	Wb, Ws, Wnd	{Wnd + 1, Wnd} = Unsigned(Wb) * Unsigned(Ws)	1	1	None
		MUL.UU	Wb, #lit5, Acc	Accumulator = Unsigned(Wb) * Unsigned(lit5)	1	1	None
		MUL.UU	Wb, Ws, Acc	Accumulator = Unsigned(Wb) * Unsigned(Ws)	1	1	None
		MULW.SS	Wb, Ws, Wnd	Wnd = Signed(Wb) * Signed(Ws)	1	1	None
		MULW.SU	Wb, Ws, Wnd	Wnd = Signed(Wb) * Unsigned(Ws)	1	1	None
		MULW.US	Wb, Ws, Wnd	Wnd = Unsigned(Wb) * Signed(Ws)	1	1	None
		MULW.UU	Wb, Ws, Wnd	Wnd = Unsigned(Wb) * Unsigned(Ws)	1	1	None
		MUL.SU	Wb, #lit5, Wnd	{Wnd + 1, Wnd} = Signed(Wb) * Unsigned(lit5)	1	1	None
		MUL.SU	Wb, #lit5, Wnd	Wnd = Signed(Wb) * Unsigned(lit5)	1	1	None
		MUL.UU	Wb, #lit5, Wnd	{Wnd + 1, Wnd} = Unsigned(Wb) * Unsigned(lit5)	1	1	None
		MUL.UU	Wb, #lit5, Wnd	Wnd = Unsigned(Wb) * Unsigned(lit5)	1	1	None
		MUL	f	W3:W2 = f * WREG	1	1	None
		69	NEG	NEG	Acc	Negate Accumulator	1
NEG	f			f = $\overline{f}$ + 1	1	1	C,DC,N,OV,Z
NEG	f, WREG			WREG = $\overline{f}$ + 1	1	1	C,DC,N,OV,Z
NEG	Ws, Wd			Wd = $\overline{Ws}$ + 1	1	1	C,DC,N,OV,Z

Notes:

1.

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2.

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dsPIC33CK512MPT608 Family

Instruction Set Summary

.....continued

Base Instr #	Assembly Mnemonic	Assembly Syntax		Description	# of Words	# of Cycles ⁽¹⁾	Status Flags Affected
70	NOP	NOP		No Operation	1	1	None
		NOPR		No Operation	1	1	None
71	NORM	NORM	Acc, Wd	Normalize Accumulator	1	1	N,OV,Z
72	POP	POP	f	Pop f from Top-of-Stack (TOS)	1	1	None
		POP	Wdo	Pop from Top-of-Stack (TOS) to Wdo	1	1	None
		POP.D	Wnd	Pop from Top-of-Stack (TOS) to W(nd):W(nd + 1)	1	2	None
		POP.S		Pop Shadow Registers	1	1	All
73	PUSH	PUSH	f	Push f to Top-of-Stack (TOS)	1	1	None
		PUSH	Wso	Push Wso to Top-of-Stack (TOS)	1	1	None
		PUSH.D	Wns	Push W(ns):W(ns + 1) to Top-of-Stack (TOS)	1	2	None
		PUSH.S		Push Shadow Registers	1	1	None
74	PWRSV	PWRSV	#lit1	Go into Sleep or Idle mode	1	1	WDTO,Sleep
75	RCALL	RCALL	Expr	Relative Call	1	4/2 ⁽²⁾	SFA
		RCALL	Wn	Computed Call	1	4/2 ⁽²⁾	SFA
76	REPEAT	REPEAT	#lit15	Repeat Next Instruction lit15 + 1 Times	1	1	None
		REPEAT	Wn	Repeat Next Instruction (Wn) + 1 Times	1	1	None
77	RESET	RESET		Software Device Reset	1	1	None
78	RETFIE	RETFIE		Return from Interrupt	1	6 (5)/3 ⁽²⁾	SFA
79	RETLW	RETLW	#lit10, Wn	Return with Literal in Wn	1	6 (5)/3 ⁽²⁾	SFA
80	RETURN	RETURN		Return from Subroutine	1	6 (5)/3 ⁽²⁾	SFA
81	RLC	RLC	f	f = Rotate Left through Carry f	1	1	C,N,Z
		RLC	f, WREG	WREG = Rotate Left through Carry f	1	1	C,N,Z
		RLC	Ws, Wd	Wd = Rotate Left through Carry Ws	1	1	C,N,Z
82	RLNC	RLNC	f	f = Rotate Left (No Carry) f	1	1	N,Z
		RLNC	f, WREG	WREG = Rotate Left (No Carry) f	1	1	N,Z
		RLNC	Ws, Wd	Wd = Rotate Left (No Carry) Ws	1	1	N,Z
83	RRC	RRC	f	f = Rotate Right through Carry f	1	1	C,N,Z
		RRC	f, WREG	WREG = Rotate Right through Carry f	1	1	C,N,Z
		RRC	Ws, Wd	Wd = Rotate Right through Carry Ws	1	1	C,N,Z

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1.

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dsPIC33CK512MPT608 Family

Instruction Set Summary

.....continued							
Base Instr #	Assembly Mnemonic	Assembly Syntax		Description	# of Words	# of Cycles ⁽¹⁾	Status Flags Affected
84	RRNC	RRNC	f	f = Rotate Right (No Carry) f	1	1	N,Z
		RRNC	f,WREG	WREG = Rotate Right (No Carry) f	1	1	N,Z
		RRNC	Ws,Wd	Wd = Rotate Right (No Carry) Ws	1	1	N,Z
85	SAC	SAC	Acc,#Slit4,Wdo	Store Accumulator	1	1	None
		SAC.R	Acc,#Slit4,Wdo	Store Rounded Accumulator	1	1	None
		SAC.D	#Slit4, Wdo	Store Accumulator Double	1	1	None
86	SE	SE	Ws,Wnd	Wnd = Sign-Extended Ws	1	1	C,N,Z
87	SETM	SETM	f	f = 0xFFFF	1	1	None
		SETM	WREG	WREG = 0xFFFF	1	1	None
		SETM	Ws	Ws = 0xFFFF	1	1	None
88	SFTAC	SFTAC	Acc,Wn	Arithmetic Shift Accumulator by (Wn)	1	1	OA,OB,OAB,SA,SB,SAB
		SFTAC	Acc,#Slit6	Arithmetic Shift Accumulator by Slit6	1	1	OA,OB,OAB,SA,SB,SAB
89	SL	SL	f	f = Left Shift f	1	1	C,N,OV,Z
		SL	f,WREG	WREG = Left Shift f	1	1	C,N,OV,Z
		SL	Ws,Wd	Wd = Left Shift Ws	1	1	C,N,OV,Z
		SL	Wb,Wns,Wnd	Wnd = Left Shift Wb by Wns	1	1	N,Z
		SL	Wb,#lit5,Wnd	Wnd = Left Shift Wb by lit5	1	1	N,Z
91	SUB	SUB	Acc	Subtract Accumulators	1	1	OA,OB,OAB,SA,SB,SAB
		SUB	f	f = f – WREG	1	1	C,DC,N,OV,Z
		SUB	f,WREG	WREG = f – WREG	1	1	C,DC,N,OV,Z
		SUB	#lit10,Wn	Wn = Wn – lit10	1	1	C,DC,N,OV,Z
		SUB	Wb,Ws,Wd	Wd = Wb – Ws	1	1	C,DC,N,OV,Z
		SUB	Wb,#lit5,Wd	Wd = Wb – lit5	1	1	C,DC,N,OV,Z
92	SUBB	SUBB	f	f = f – WREG – ($\overline{C}$)	1	1	C,DC,N,OV,Z
		SUBB	f,WREG	WREG = f – WREG – ($\overline{C}$)	1	1	C,DC,N,OV,Z
		SUBB	#lit10,Wn	Wn = Wn – lit10 – ($\overline{C}$)	1	1	C,DC,N,OV,Z
		SUBB	Wb,Ws,Wd	Wd = Wb – Ws – ($\overline{C}$)	1	1	C,DC,N,OV,Z
		SUBB	Wb,#lit5,Wd	Wd = Wb – lit5 – ($\overline{C}$)	1	1	C,DC,N,OV,Z
93	SUBR	SUBR	f	f = WREG – f	1	1	C,DC,N,OV,Z
		SUBR	f,WREG	WREG = WREG – f	1	1	C,DC,N,OV,Z
		SUBR	Wb,Ws,Wd	Wd = Ws – Wb	1	1	C,DC,N,OV,Z
		SUBR	Wb,#lit5,Wd	Wd = lit5 – Wb	1	1	C,DC,N,OV,Z
94	SUBBR	SUBBR	f	f = WREG – f – ($\overline{C}$)	1	1	C,DC,N,OV,Z
		SUBBR	f,WREG	WREG = WREG – f – ($\overline{C}$)	1	1	C,DC,N,OV,Z
		SUBBR	Wb,Ws,Wd	Wd = Ws – Wb – ($\overline{C}$)	1	1	C,DC,N,OV,Z
		SUBBR	Wb,#lit5,Wd	Wd = lit5 – Wb – ($\overline{C}$)	1	1	C,DC,N,OV,Z
95	SWAP	SWAP.b	Wn	Wn = Nibble Swap Wn	1	1	None
		SWAP	Wn	Wn = Byte Swap Wn	1	1	None
Notes:							
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dsPIC33CK512MPT608 Family

Instruction Set Summary

.....continued

Base Instr #	Assembly Mnemonic	Assembly Syntax		Description	# of Words	# of Cycles ⁽¹⁾	Status Flags Affected
96	TBLRDH	TBLRDH	Ws, Wd	Read Prog[23:16] to Wd[7:0]	1	5/3	None
97	TBLRDL	TBLRDL	Ws, Wd	Read Prog[15:0] to Wd	1	5/3	None
98	TBLWTH	TBLWTH	Ws, Wd	Write Ws[7:0] to Prog[23:16]	1	2	None
99	TBLWTL	TBLWTL	Ws, Wd	Write Ws to Prog[15:0]	1	2	None
101	ULNK	ULNK		Unlink Frame Pointer	1	1	SFA
104	XOR	XOR	f	f = f .XOR. WREG	1	1	N,Z
		XOR	f, WREG	WREG = f .XOR. WREG	1	1	N,Z
		XOR	#lit10, Wn	Wd = lit10 .XOR. Wd	1	1	N,Z
		XOR	Wb, Ws, Wd	Wd = Wb .XOR. Ws	1	1	N,Z
		XOR	Wb, #lit5, Wd	Wd = Wb .XOR. lit5	1	1	N,Z
105	ZE	ZE	Ws, Wnd	Wnd = Zero-Extend Ws	1	1	C,Z,N

Notes:

1. Read and Read-Modify-Write (e.g., bit operations and logical operations) on non-CPU SFRs incur an additional instruction cycle.
2. For dsPIC33CK512MPT608 devices, the divide instructions must be preceded with a "REPEAT #5" instruction, such that they are executed six consecutive times.

32. Development Support

Move a design from concept to production in record time with Microchip's award-winning development tools. Microchip tools work together to provide state of the art debugging for any project with easy-to-use Graphical User Interfaces (GUIs) in our free MPLAB® X and Atmel Studio Integrated Development Environments (IDEs), and our code generation tools. Providing the ultimate ease-of-use experience, Microchip's line of programmers, debuggers and emulators work seamlessly with our software tools. Microchip development boards help evaluate the best silicon device for an application, while our line of third party tools round out our comprehensive development tool solutions.

Microchip's MPLAB X and Atmel Studio ecosystems provide a variety of embedded design tools to consider, which support multiple devices, such as PIC® MCUs, AVR® MCUs, SAM MCUs and dsPIC® DSCs. MPLAB X tools are compatible with Windows®, Linux® and Mac® operating systems while Atmel Studio tools are compatible with Windows.

Go to the following website for more information and details:

www.microchip.com/development-tools/

33. Electrical Characteristics

This section provides an overview of the dsPIC33CK512MPT608 family electrical characteristics. Additional information will be provided in future revisions of this document as it becomes available.

Absolute maximum ratings for the dsPIC33CK512MPT608 family are listed below. Exposure to these Maximum Rating conditions for extended periods may affect device reliability. Functional operation of the device at these, or any other conditions above the parameters indicated in the operation listings of this specification, is not implied.

Table 33-1. Absolute Maximum Ratings⁽¹⁾

Ambient temperature under bias	-40°C to +125°C
Storage temperature	-65°C to +125°C
Voltage on V_{DD} with respect to V_{SS}	-0.3V to +4.0V
Voltage on any pin that is not 5V tolerant with respect to V_{SS} ⁽³⁾	-0.3V to ($V_{DD} + 0.3V$)
Voltage on any 5V tolerant pin with respect to V_{SS} when $V_{DD} \geq 3.0V$ ⁽³⁾	-0.3V to +5.5V
Voltage on any 5V tolerant pin with respect to V_{SS} when $V_{DD} < 3.0V$ ⁽³⁾	-0.3V to +3.6V
Maximum current out of V_{SS} pin	300 mA
Maximum current into V_{DD} pin ⁽²⁾	300 mA
Maximum current sunk/sourced by any 4x I/O pin	15 mA
Maximum current sunk/sourced by any 8x I/O pin	25 mA
Maximum current sunk by a group of I/Os between two V_{SS} pins ⁽⁴⁾	75 mA
Maximum current sourced by a group of I/Os between two V_{DD} pins ⁽⁴⁾	75 mA
Maximum current sunk by all ports ⁽²⁾	200 mA

Notes:

1. Stresses above those listed under “Absolute Maximum Ratings” may cause permanent damage to the device. This is a stress rating only and functional operation of the device at those, or any other conditions above those indicated in the operation listings of this specification, is not implied. Exposure to Maximum Rating conditions for extended periods may affect device reliability.
2. Maximum allowable current is a function of device maximum power dissipation (see [33.1. DC Characteristics](#)).
3. See the [Pin Diagram](#) section for the 5V tolerant pins.
4. Not applicable to AV_{DD} and AV_{SS} pins.

33.1 DC Characteristics

Table 33-2. Operating MIPS vs. Voltage

Characteristic	V _{DD} Range (in Volts)	Temperature Range (in °C)	Maximum MIPS dsPIC33CK512MPT608 Family
—	3.0V to 3.6V	-40°C to +85°C	100
	3.0V to 3.6V	-40°C to +125°C	100

Table 33-3. Thermal Operating Conditions

Rating	Symbol	Min.	Typ.	Max.	Unit
Industrial Temperature Devices					
Operating Junction Temperature Range	T _J	-40	—	+125	°C
Operating Ambient Temperature Range	T _A	-40	—	+85	°C
Extended Temperature Devices					
Operating Junction Temperature Range	T _J	-40	—	+140	°C
Operating Ambient Temperature Range	T _A	-40	—	+125	°C
Power Dissipation: Internal Chip Power Dissipation: $P_{INT} = V_{DD} \times (I_{DD} - \sum I_{OH})$	P _D	P _{INT} + P _{I/O}			W
I/O Pin Power Dissipation: $I/O = \sum (\{V_{DD} - V_{OH}\} \times I_{OH}) + \sum (V_{OL} \times I_{OL})$					
Maximum Allowed Power Dissipation	P _{DMAX}	(T _J - T _A)/θ _{JA}			W

Table 33-4. Thermal Packaging Characteristics

Characteristic	Symbol	Typ.	Max.	Unit	Notes
Package Thermal Resistance, 100-Pin TQFP, 12x12x1 mm	θ _{JA}	44.7	—	°C/W	1

Note:

1. Junction to ambient thermal resistance, Theta-JA (θ_{JA}) numbers are achieved by package simulations.

Table 33-5. Operating Voltage Specifications

Operating Conditions: 3.0V to 3.6V (unless otherwise stated) ⁽¹⁾ Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial $-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended							
Param No.	Symbol	Characteristic	Min.	Typ.	Max.	Units	Conditions
Operating Voltage							
DC10	V_{DD}	Supply Voltage	3.0	—	3.6	V	
DC11	AV_{DD}	Supply Voltage	Greater of: $V_{DD} - 0.3$ or 3.0	—	Lesser of: $V_{DD} + 0.3$ or 3.6	V	The difference between AV_{DD} supply and V_{DD} supply must not exceed ± 300 mV at all times, including during device power-up
DC16	$V_{POR}^{(3)}$	V_{DD} Start Voltage to Ensure Internal Power-on Reset Signal	—	—	V_{SS}	V	
DC17	SV_{DD}	V_{DD} Rise Rate to Ensure Internal Power-on Reset Signal	0.03	—	100	V/ms	0V-3V in 100 ms
BO10	$V_{BOR}^{(3)}$	BOR Event on V_{DD} Transition High-to-Low⁽²⁾	2.68	2.84	2.99	V	
Notes:							
1. Device is functional at $V_{BORMIN} < V_{DD} < V_{DDMIN}$. Analog modules (ADC and comparators) may have degraded performance.							
2. Parameters are characterized but not tested.							
3. This parameter is not applicable to the Secure Subsystem.							

Table 33-6. DC Characteristics: Operating Current (I_{DD})

DC Characteristics	Run		Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated)			
			Operating temperature -40°C ≤ T _A ≤ +85°C for Industrial -40°C ≤ T _A ≤ +125°C for Extended			
Parameter No.	Typ.	Max.	Units	Conditions		
Operating Current (I _{DD}) ^(1,2)						
DC20	7.76	10.7	mA	-40°C	3.3V	10 MIPS (N = 1, N2 = 5, N3 = 2, M = 50, F _{VCO} = 400 MHz, F _{PLLO} = 40 MHz)
	7.49	10	mA	+25°C		
	7.82	15.5	mA	+85°C		
	10.32	23.5	mA	+125°C		
DC21	10.36	13.1	mA	-40°C	3.3V	20 MIPS (N = 1, N2 = 5, N3 = 1, M = 60, F _{VCO} = 480 MHz, F _{PLLO} = 280 MHz)
	10.09	12.45	mA	+25°C		
	10.42	17.5	mA	+85°C		
	12.89	25.5	mA	+125°C		
DC22	14.54	17.45	mA	-40°C	3.3V	40 MIPS (N = 1, N2 = 3, N3 = 1, M = 60, F _{VCO} = 480 MHz, F _{PLLO} = 160 MHz)
	14.26	16.7	mA	+25°C		
	14.58	22	mA	+85°C		
	17.06	30	mA	+125°C		
DC23	22.2	25.4	mA	-40°C	3.3V	70 MIPS (N = 1, N2 = 2, N3 = 1, M = 70, F _{VCO} = 560 MHz, F _{PLLO} = 280 MHz)
	21.91	24.9	mA	+25°C		
	22.21	30.75	mA	+85°C		
	24.65	37.5	mA	+125°C		

Notes:

- I_{DD} is primarily a function of the operating voltage and frequency. Other factors, such as I/O pin loading and switching rate, oscillator type, internal code execution pattern and temperature, also have an impact on the current consumption.
Base Run current (I_{DD}) is measured as follows:
 - Oscillator is switched to EC+PLL mode in software
 - OSC1 pin is driven with external 8 MHz square wave with levels from 0.3V to $V_{DD} - 0.3V$
 - OSC2 is configured as an I/O in the Configuration Words (OSCIOFNC (FOSC[2]) = 0)
 - FSCM is disabled (FCKSM[1:0] (FOSC[7:6]) = 01)
 - Watchdog Timer is disabled (FWDT[15] = 0 and WDTCONL[15] = 0)
 - All I/O pins (except OSC1) are configured as outputs and driving low
 - No peripheral modules are operating or being clocked (defined PMDx bits are all '1's)
 - JTAG is disabled (JTAGEN (FICD[5]) = 0)
 - NOP instructions are executed in while(1) loop
- The Secure Subsystem may be configured to operate either in Sleep or Idle mode when there is no SPI communication with Host. Accordingly, add Sleep or Idle current of the Secure Subsystem for overall operating current of the device. The SPI lines used for communication must be configured as inputs and pulled up.

.....continued

DC Characteristics	Run		Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated) Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial $-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended			
	Parameter No.	Typ.	Max.	Units	Conditions	
DC24		27.36	30.7	mA	-40°C +25°C +85°C +125°C	3.3V 90 MIPS (N = 1, N2 = 2, N3 = 1, M = 90, F _{VCO} = 720 MHz, F _{PLLO} = 360 MHz)
		26.96	30.5	mA		
		26.68	35	mA		
		29.01	42	mA		
DC25		27.14	30.9	mA	-40°C +25°C +85°C +125°C	3.3V 100 MIPS (N = 1, N2 = 1, N3 = 1, M = 50, F _{VCO} = 400 MHz, F _{PLLO} = 400 MHz)
		26.54	30.1	mA		
		26.79	35	mA		
		29.23	42.5	mA		

Notes:

1. I_{DD} is primarily a function of the operating voltage and frequency. Other factors, such as I/O pin loading and switching rate, oscillator type, internal code execution pattern and temperature, also have an impact on the current consumption.
Base Run current (I_{DD}) is measured as follows:
 - Oscillator is switched to EC+PLL mode in software
 - OSC1 pin is driven with external 8 MHz square wave with levels from 0.3V to V_{DD} – 0.3V
 - OSC2 is configured as an I/O in the Configuration Words (OSCIOFNC (FOSC[2]) = 0)
 - FSCM is disabled (FCKSM[1:0] (FOSC[7:6]) = 01)
 - Watchdog Timer is disabled (FWDTC[15] = 0 and WDTCONL[15] = 0)
 - All I/O pins (except OSC1) are configured as outputs and driving low
 - No peripheral modules are operating or being clocked (defined PMDx bits are all '1's)
 - JTAG is disabled (JTAGEN (FICD[5]) = 0)
 - NOP instructions are executed in while(1) loop
2. The Secure Subsystem may be configured to operate either in Sleep or Idle mode when there is no SPI communication with Host. Accordingly, add Sleep or Idle current of the Secure Subsystem for overall operating current of the device. The SPI lines used for communication must be configured as inputs and pulled up.

Table 33-7. Idle Current (I_{IDLE})^(2,3)

Parameter No.	Typ. ⁽¹⁾	Max.	Units	Conditions		
DC40	6.41	8.47	mA	-40°C	3.3V	10 MIPS (N = 1, N2 = 5, N3 = 2, M = 50, F _{VCO} = 400 MHz, F _{PLLO} = 40 MHz)
	6.15	7.57	mA	+25°C		
	6.45	13	mA	+85°C		
	8.95	22	mA	+125°C		
DC41	7.31	10.1	mA	-40°C	3.3V	20 MIPS (N = 1, N2 = 5, N3 = 1, M = 50, F _{VCO} = 400 MHz, F _{PLLO} = 80 MHz)
	7.04	9.1	mA	+25°C		
	7.36	14.75	mA	+85°C		
	9.83	22.75	mA	+125°C		

.....continued

Parameter No.	Typ. ⁽¹⁾	Max.	Units	Conditions		
DC42	9.4	12.3	mA	-40°C	3.3V	40 MIPS (N = 1, N2 = 3, N3 = 1, M = 60, F _{VCO} = 480 MHz, F _{PLLO} = 160 MHz)
	9.13	11.2	mA	+25°C		
	9.45	16.5	mA	+85°C		
	11.92	25	mA	+125°C		
DC43	12.39	15.3	mA	-40°C	3.3V	70 MIPS (N = 1, N2 = 2, N3 = 1, M = 70, F _{VCO} = 560 MHz, F _{PLLO} = 280 MHz)
	12.11	14.3	mA	+25°C		
	12.43	19.75	mA	+85°C		
	14.89	28.25	mA	+125°C		
DC44	14.78	17.85	mA	-40°C	3.3V	90 MIPS (N = 1, N2 = 2, N3 = 1, M = 90, F _{VCO} = 720 MHz, F _{PLLO} = 360 MHz)
	14.5	16.9	mA	+25°C		
	14.81	22.5	mA	+85°C		
	17.26	29.5	mA	+125°C		
DC45	14.44	17.55	mA	-40°C	3.3V	100 MIPS (N = 1, N2 = 1, N3 = 1, M = 50, F _{VCO} = 400 MHz, F _{PLLO} = 400 MHz)
	14.15	16.5	mA	+25°C		
	14.46	22.25	mA	+85°C		
	17.5	30	mA	+125°C		

Notes:

- Data in the "Typ." column are for design guidance only and are not tested.
- Base Idle current (I_{IDLE}) is measured as follows:
 - Oscillator is switched to EC+PLL mode in software
 - OSC1 pin is driven with external 8 MHz square wave with levels from 0.3V to V_{DD} - 0.3V
 - OSC2 is configured as an I/O in the Configuration Words (OSCIOFNC (FOSC[2]) = 0)
 - FSCM is disabled (FCKSM[1:0] (FOSC[7:6]) = 01)
 - Watchdog Timer is disabled (FWDTEN (FWDT[15]) = 0)
 - All I/O pins (except OSC1) are configured as outputs and driving low
 - No peripheral modules are operating or being clocked (defined PMDx bits are all '1's)
 - JTAG is disabled (JTAGEN (FICD[5]) = 0)
 - Flash in standby with NVMSIDL (NVMCON[12]) = 1
- The Secure Subsystem may be configured to operate either in Sleep or Idle mode when there is no SPI communication with Host. Accordingly, add Sleep or Idle current of the Secure Subsystem for overall Idle current of the device. The SPI lines used for communication must be configured as inputs and pulled up.

Table 33-8. DC Characteristics: Power-Down Current (I_{PD})

DC Charateristics		Sleep		Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated) Operating temperature -40°C ≤ T _A ≤ +85°C for Industrial -40°C ≤ T _A ≤ +125°C for Extended	
Parameter No.	Typ. ⁽¹⁾	Max.	Units	Conditions	
Power-Down Current (I _{PD}) ^(2,4)					
DC60	270	650	μA	-40°C	3.3V
	525	1000	μA	+25°C	
	2150	7250	μA	+85°C	
	7.03	18.5	mA	+125°C ⁽³⁾	

Notes:

1. Data in the "Typ." column are for design guidance only and are not tested.
2. I_{PD} (Sleep) current is measured as follows:
 - CPU core is off, oscillator is configured in EC mode and External Clock is active; OSCI is driven with external square wave from rail-to-rail (EC clock overshoot/undershoot < 250 mV required)
 - CLK0 is configured as an I/O input pin in the Configuration Word
 - All I/O pins are configured as output low
 - $MCLR = V_{DD}$, WDT and FSCM are disabled
 - All peripheral modules are disabled (PMDx bits are all set)
 - The VREGS bit (RCON[8]) = 0 (i.e., core regulator is set to standby while the device is in Sleep mode)
 - JTAG is disabled
3. The regulators are in High-Power mode, LPWREN (VREGCON[15]) = 0.
4. The Secure Subsystem may be configured to operate either in Sleep or Idle mode when there is no SPI communication with Host. Accordingly, add Sleep or Idle current of the Secure Subsystem for overall Idle current of the device. The SPI lines used for communication must be configured as inputs and pulled up.

Table 33-9. Doze Current (I_{DOZE})

Parameter No.	Typ. ⁽¹⁾	Max	Doze Ratio	Units	Conditions		
DC70	18.19	20	1:2	mA	-40°C	3.3V	70 MIPS (N = 1, N2 = 2, N3 = 1, M = 70, F _{VCO} = 560 MHz, F _{PLLO} = 280 MHz)
	12.66	15	1:128	mA			
	17.54	20.15	1:2	mA	+25°C		
	12.39	14.7	1:128	mA			
	17.85	25	1:2	mA	+85°C		
	12.7	20	1:128	mA			
	20.32	32.5	1:2	mA	+125°C		
	16.05	28.5	1:128	mA			

.....continued

Parameter No.	Typ. ⁽¹⁾	Max	Doze Ratio	Units	Conditions		
DC71	22.3	25.55	1:2	mA	-40°C	3.3V	100 MIPS (N = 1, N2 = 1, N3 = 1, M = 50, F _{VCO} = 400 MHz, F _{PLLO} = 400 MHz)
	14.83	17.25	1:128	mA			
	21.86	25.05	1:2	mA	+25°C		
	14.55	16.95	1:128	mA			
	22.16	30	1:2	mA	+85°C		
	14.86	22	1:128	mA			
	24.62	36.5	1:2	mA	+125°C		
	17.45	30	1:128	mA			

Note:

1. Data in the "Typ." column are for design guidance only and are not tested.

Table 33-10. DC Characteristics: Watchdog Timer Delta Current (ΔI_{WDT})⁽¹⁾

DC Characteristics		Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated)			
		Operating temperature -40°C ≤ T _A ≤ +85°C for Industrial -40°C ≤ T _A ≤ +125°C for Extended			
Parameter No.	Typ.	Max.	Units	Conditions	
DC61c	25	125	μA	-40°C	3.3V
DC61b	110	600	μA	+25°C	
DC61a	100	600	μA	+85°C	
DC61d	100	600	μA	+125°C	

Note:

1. The ΔI_{WDT} current is the additional current consumed when the module is enabled. This includes the LPRC/ BFRC clock source current. This current should be added to the base I_{PD} current. All parameters are characterized but not tested during manufacturing.

Table 33-11. DC Characteristics: PWM Delta Current⁽¹⁾

DC Characteristics	Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated)				
	Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial $-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended				
Parameter No.	Typ.	Max.	Units	Conditions	
DC100	5.96	6.6	mA	-40°C , 3.3V	PWM Output 500 kHz, PWM Input ($\text{AF}_{\text{PLLO}} = 500 \text{ MHz}$), AVCO = 1000 MHz, PLLFBD = 125, APLLDIV = 2
	5.99	6.7	mA	$+25^{\circ}\text{C}$, 3.3V	
	5.92	6.9	mA	$+85^{\circ}\text{C}$, 3.3V	
	5.47	7	mA	$+125^{\circ}\text{C}$, 3.3V	
DC101	4.89	5.4	mA	-40°C , 3.3V	PWM Output 500 kHz, PWM Input ($\text{AF}_{\text{PLLO}} = 400 \text{ MHz}$), AVCO = 400 MHz, PLLFBD = 50, APLLDIV = 1
	4.91	5.5	mA	$+25^{\circ}\text{C}$, 3.3V	
	4.85	5.7	mA	$+85^{\circ}\text{C}$, 3.3V	
	4.42	5.7	mA	$+125^{\circ}\text{C}$, 3.3V	
DC102	2.77	3.7	mA	-40°C , 3.3V	PWM Output 500 kHz, PWM Input ($\text{AF}_{\text{PLLO}} = 200 \text{ MHz}$), AVCO = 400 MHz, PLLFBD = 50, APLLDIV = 2
	2.75	3.7	mA	$+25^{\circ}\text{C}$, 3.3V	
	2.7	3.7	mA	$+85^{\circ}\text{C}$, 3.3V	
	2.26	3.7	mA	$+125^{\circ}\text{C}$, 3.3V	
DC103	1.67	2	mA	-40°C , 3.3V	PWM Output 500 kHz, PWM Input ($\text{AF}_{\text{PLLO}} = 100 \text{ MHz}$), AVCO = 400 MHz, PLLFBD = 50, APLLDIV = 4
	1.66	2.2	mA	$+25^{\circ}\text{C}$, 3.3V	
	1.63	2.3	mA	$+85^{\circ}\text{C}$, 3.3V	
	1.17	2.3	mA	$+125^{\circ}\text{C}$, 3.3V	

Note:

1. APLL current is not included. The APLL current will be the same if more than one PWM is running. Listed delta currents are for only one PWM instance when $\text{HREN} = 0$ ($\text{PGxCONL}[7]$). All parameters are characterized but not tested during manufacturing.

Table 33-12. DC Characteristics: APLL Delta Current

DC Characteristics	Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated)				
	Operating temperature -40°C ≤ T _A ≤ +85°C for Industrial -40°C ≤ T _A ≤ +125°C for Extended				
Parameter No.	Typ.	Max.	Units	Conditions ⁽¹⁾	
DC110	5.93	6.6	mA	-40°C, 3.3V	AF _{PLLO} = 500 MHz, AVCO = 1000 MHz, PLLFBF = 125, APLLDIV = 2
	5.95	7	mA	+25°C, 3.3V	
	6.15	7.6	mA	+85°C, 3.3V	
	7.15	9	mA	+125°C, 3.3V	

Note:

1. The APLL current will be the same if more than one PWM or DAC is run to the APLL clock. All parameters are characterized but not tested during manufacturing.

.....continued

DC Characteristics	Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated)				
	Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial $-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended				
Parameter No.	Typ.	Max.	Units	Conditions ⁽¹⁾	
DC111	2.72	3.3	mA	-40°C, 3.3V	AF _{PLLO} = 400 MHz, AVCO = 400 MHz, PLLFBF = 50, APLLDIV = 1
	2.74	3.7	mA	+25°C, 3.3V	
	2.92	4.3	mA	+85°C, 3.3V	
	3.87	5.6		+125°C, 3.3V	
DC112	1.39	2.7	mA	-40°C, 3.3V	AF _{PLLO} = 200 MHz, AVCO = 400 MHz, PLLFBF = 50, APLLDIV = 2
	1.49	2.7	mA	+25°C, 3.3V	
	1.65	3	mA	+85°C, 3.3V	
	2.6	4.4	mA	+125°C, 3.3V	
DC113	0.79	1.1	mA	-40°C, 3.3V	AF _{PLLO} = 100 MHz, AVCO = 400 MHz, PLLFBF = 50, APLLDIV = 4
	0.84	1.4	mA	+25°C, 3.3V	
	0.96	2.3	mA	+85°C, 3.3V	
	1.93	3.6	mA	+125°C, 3.3V	

Note:

1. The APLL current will be the same if more than one PWM or DAC is run to the APLL clock. All parameters are characterized but not tested during manufacturing.

Table 33-13. DC Characteristics: ADC Δ Current

DC Characteristics	Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated)					
	Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial $-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended					
Parameter No.	Typ.	Max.	Units	Conditions		
DC120	3.61	4	mA	-40°C	3.3V	T _{AD} = 14.3 ns (3.5 Msps conversion rate)
	3.68	4.1	mA	+25°C		
	3.69	4.2	mA	+85°C		
	3.89	4.6	mA	+125°C		

Table 33-14. DC Characteristics: Comparator + DAC Delta Current

DC Characteristics	Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated)				
	Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial $-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended				
Parameter No.	Typ.	Max.	Units	Conditions	
DC130	1.2	1.35	mA	-40°C , 3.3V	AF _{PLLO} @ 500 MHz ⁽¹⁾
	1.23	1.65	mA	$+25^{\circ}\text{C}$, 3.3V	
	1.23	1.65	mA	$+85^{\circ}\text{C}$, 3.3V	
	1.24	1.65	mA	$+125^{\circ}\text{C}$, 3.3V	

Note:

1. APLL current is not included. Listed delta currents are for only one comparator + DAC instance. All parameters are characterized but not tested during manufacturing.

Table 33-15. Op Amp Delta Current⁽¹⁾

Parameter No.	Typ.	Max.	Units	Conditions	
DC140	0.25	1	mA	-40°C	3.3V
	0.27	1.1	mA	$+25^{\circ}\text{C}$	
	0.32	1.4	mA	$+85^{\circ}\text{C}$	
	0.4647	1.7	mA	$+125^{\circ}\text{C}$	

Note:

1. Listed delta currents are for only one op amp instance. All parameters are characterized but not tested during manufacturing.

Table 33-16. I/O Pin Input Specifications

Operating Conditions: 3.0V to 3.6V (unless otherwise stated) Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial $-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended							
Param No.	Symbol	Characteristic	Min.	Typ. ⁽¹⁾	Max.	Units	Conditions
DI10	V _{IL}	Input Low Voltage					
		Any I/O Pin and MCLR	V _{SS}	—	0.2 V _{DD}	V	
		I/O Pins with SDAx, SCLx	V _{SS}	—	0.3 V _{DD}	V	SMBus disabled
		I/O Pins with SDAx, SCLx	V _{SS}	—	0.8	V	SMBus enabled
		I/O Pins with SDAx, SCLx	V _{SS}	—	0.8	V	SMBus 3.0

Notes:

1. Data in "Typ." column are at 3.3V, $+25^{\circ}\text{C}$ unless otherwise stated.
2. Negative current is defined as current sourced by the pin.
3. See the [Pin Diagram](#) section for the 5V tolerant I/O pins.
4. All parameters are characterized but not tested during manufacturing.

.....continued

Operating Conditions: 3.0V to 3.6V (unless otherwise stated)

Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial $-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended

Param No.	Symbol	Characteristic	Min.	Typ. ⁽¹⁾	Max.	Units	Conditions
	V_{IH}	Input High Voltage					
DI20		I/O Pins Not 5V Tolerant ⁽³⁾	$0.8 V_{DD}$	—	V_{DD}	V	
		5V Tolerant I/O Pins and $\overline{MCLR}$ ⁽³⁾	$0.8 V_{DD}$	—	5.5	V	
		5V Tolerant I/O Pins with SDAx, SCLx ⁽³⁾	$0.8 V_{DD}$	—	5.5	V	SMBus disabled
		5V Tolerant I/O Pins with SDAx, SCLx ⁽³⁾	2.1	—	5.5	V	SMBus enabled
		5V Tolerant I/O Pins with SDAx, SCLx ⁽³⁾	1.35	—	5.5	V	SMBus 3.0
		I/O Pins with SDAx, SCLx Not 5V Tolerant ⁽³⁾	$0.8 V_{DD}$	—	V_{DD}	V	SMBus disabled
		I/O Pins with SDAx, SCLx Not 5V Tolerant ⁽³⁾	2.1	—	V_{DD}	V	SMBus enabled
		I/O Pins with SDAx, SCLx Not 5V Tolerant ⁽³⁾	1.35	—	V_{DD}	V	SMBus 3.0
DI30	I_{CNPU}	Input Change Notification Pull-up Current ^(2,4)	175	360	545	μA	$V_{DD} = 3.6\text{V}$, $V_{PIN} = V_{SS}$
DI31	I_{CNPD}	Input Change Notification Pull-Down Current ⁽⁴⁾	65	215	360	μA	$V_{DD} = 3.6\text{V}$, $V_{PIN} = V_{DD}$

Notes:

1. Data in "Typ." column are at 3.3V, +25°C unless otherwise stated.
2. Negative current is defined as current sourced by the pin.
3. See the [Pin Diagram](#) section for the 5V tolerant I/O pins.
4. All parameters are characterized but not tested during manufacturing.

Table 33-17. I/O Pin Input Specifications

Operating Conditions: 3.0V to 3.6V (unless otherwise stated) Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial $-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended						
Param No.	Symbol	Characteristic	Min.	Max.	Units	Conditions
DI50	I_{IL}	Input Leakage Current ⁽¹⁾				
		I/O Pins 5V Tolerant ⁽²⁾	-700	+700	nA	$V_{PIN} = V_{SS}$ or V_{DD}
		I/O Pins Not 5V Tolerant ⁽²⁾	-700	+700	nA	
		MCLR	-700	+700	nA	
		OSCI	-700	+700	nA	XT and HS modes
Notes: 1. Negative current is defined as current sourced by the pin. 2. See the Pin Diagram section for the 5V tolerant I/O pins. All parameters are characterized but not tested during manufacturing.						

Table 33-18. I/O Pin Input Injection Current Specifications

Operating Conditions: 3.0V to 3.6V (unless otherwise stated) Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial $-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended						
Param No.	Symbol	Characteristic	Min.	Max.	Units	Conditions
DI60a	I_{ICL}	Input Low Injection Current	0	-5 ^(1,4)	mA	All pins
DI60b	I_{ICH}	Input High Injection Current	0	+5 ^(2,3,4)	mA	All pins, except all 5V tolerant pins
DI60c	ΣI_{ICT}	Total Input Injection Current (sum of all I/O and control pins) ⁽⁵⁾	-20	+20	mA	Absolute instantaneous sum of all $\pm$ input injection currents from all I/O pins $(I_{ICL} + I_{ICH}) \leq \Sigma I_{ICT}$
Notes: V_{IL} Source $< (V_{SS} - 0.3)$. V_{IH} Source $> (V_{DD} + 0.3)$ for non-5V tolerant pins only. 5V tolerant pins do not have an internal high-side diode to V_{DD}, and therefore, cannot tolerate any "positive" input injection current. - Injection currents can affect the ADC results. - Any number and/or combination of I/O pins, not excluded under I_{ICL} or I_{ICH} conditions, are permitted in the sum.						

Table 33-19. I/O Pin Output Specifications

Operating Conditions: 3.0V to 3.6V (unless otherwise stated) Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial $-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended							
Param.	Symbol	Characteristic	Min.	Typ.	Max.	Units	Conditions
DO10	V_{OL}	Output Low Voltage 4x Sink Driver Pins	—	—	0.42	V	$V_{DD} = 3.6\text{V}$, $I_{OL} < 9\text{ mA}$
		Output Low Voltage 8x Sink Driver Pins ⁽¹⁾	—	—	0.4	V	$V_{DD} = 3.6\text{V}$, $I_{OL} < 11\text{ mA}$
DO20	V_{OH}	Output High Voltage 4x Source Driver Pins	2.4	—	—	V	$V_{DD} = 3.6\text{V}$, $I_{OH} > -8\text{ mA}$
		Output High Voltage 8x Source Driver Pins ⁽¹⁾	2.4	—	—	V	$V_{DD} = 3.6\text{V}$, $I_{OH} > -12\text{ mA}$
Note:							
1. The 8x sink/source pins are RB1, RC8, RC9 and RD8 pins; all other ports are 4x sink drivers.							

Table 33-20. Electrical Characteristics: BOR

DC Characteristics			Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated) ⁽¹⁾ Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial $-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended				
Param No.	Symbol	Characteristic	Min.	Typ.	Max.	Units	Conditions
BO10	V_{BOR}	BOR Event on V_{DD} Transition High-to-Low	2.68	2.84	2.99	V	V_{DD} (Note 2)
Notes:							
1. Device is functional at $V_{BORMIN} < V_{DD} < V_{DDMIN}$, but will have degraded performance. Device functionality is tested, but not characterized. Analog modules (ADC and comparators) may have degraded performance.							
2. Parameters are for design guidance only and are not tested in manufacturing.							

Table 33-21. Program Memory

Operating Conditions: 3.0V to 3.6V (unless otherwise stated) Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial $-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended						
Param No.	Symbol	Characteristic	Min.	Max.	Units	Conditions
Program Flash Memory						
D130	E_P	Cell Endurance	10,000	—	E/W	-40°C to $+125^{\circ}\text{C}$
D131	V_{PR}	V_{DD} for Read	3.0	3.6	V	
D132b	V_{PEW}	V_{DD} for Self-Timed Write	3.0	3.6	V	
D134	T_{RETD}	Characteristic Retention	20	—	Year	Provided no other specifications are violated, -40°C to $+125^{\circ}\text{C}$
D137a	T_{PE}	Page Erase Time	15.3	16.82	ms	T_{PE} = 128,454 FRC cycles (Note 1)
D138a	T_{WW}	Word Write Time	47.7	52.3	μs	T_{WW} = 400 FRC cycles (Note 1)
D139a	T_{RW}	Row Write Time	2.0	2.2	ms	T_{RW} = 16,782 FRC cycles (Note 1)
Note: 1. Other conditions: FRC = 8 MHz, $TUN[5:0] = 011111$ (for Minimum), $TUN[5:0] = 100000$ (for Maximum). This parameter depends on the FRC accuracy (see Table 33-27) and the value of the FRC Oscillator Tuning register (see 10.11.4. OSCTUN). For complete details on calculating the Minimum and Maximum time, see 33.2. AC Characteristics and Timing Parameters .						

Table 33-22. Secure Subsystem Δ Current

Operating Conditions: 3.0V to 3.6V (unless otherwise stated) Operating temperature -40°C ≤ T _A ≤ +85°C for Industrial -40°C ≤ T _A ≤ +125°C for Extended							
Param No.	Symbol	Characteristic	Min.	Typ. ⁽²⁾	Max.	Units	Conditions
D141	Iss_Active	Operating Current ⁽¹⁾	—	13.02	—	mA	-40°C, 3.3V
			—	13.89	26	mA	+25°C, 3.3V
			—	14.17	—	mA	+85°C,3.3V
			—	14.11	—	mA	+125°C,3.3V
D142	Iss_Idle	Idle Current	—	5.05	—	mA	-40°C, 3.3V
			—	5.23	10	mA	+25°C, 3.3V
			—	5.42	—	mA	+85°C,3.3V
			—	5.7	—	mA	+125°C,3.3V
D143	Iss_Sleep	Sleep Current	—	3.13	—	μA	-40°C, 3.3V
			—	3.3	15	μA	+25°C, 3.3V
			—	3.56	—	μA	+85°C,3.3V
			—	4.42	—	μA	+125°C,3.3V

Notes:

1.

This parameter can be measured by executing a command that generates a private or symmetric key using a random number generator.

2.

These values are characterized but not tested at manufacturing.

33.2 AC Characteristics and Timing Parameters

Figure 33-1. Load Conditions for Device Timing Specifications

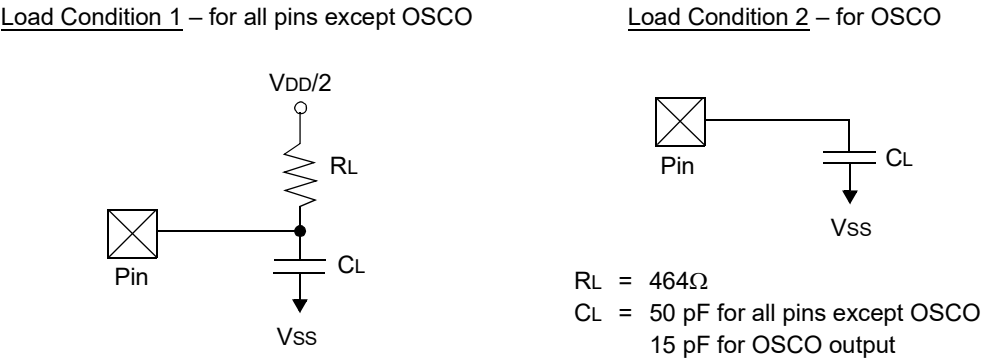


Table 33-23. Capacitive Loading Requirements on Output Pins

Param No.	Symbol	Characteristic	Min.	Typ.	Max.	Units	Conditions
DO50	C_{OSCO}	OSCO Pin	—	—	15	pF	In XT and HS modes, when External Clock is used to drive OSC1
DO56	C_{IO}	All I/O Pins and OSCO	—	—	50	pF	EC mode
DO58	C_B	SCLx, SDAx	—	—	400	pF	In I ² C mode

Figure 33-2. External Clock Timing

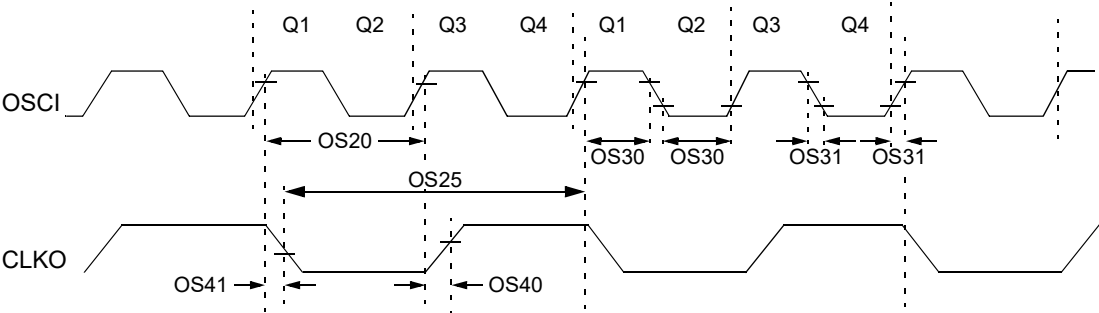


Table 33-24. External Clock Timing Requirements

Operating Conditions: 3.0V to 3.6V (unless otherwise stated)							
Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial							
$-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended							
Param No.	Sym	Characteristic	Min.	Typ. ⁽¹⁾	Max.	Units	Conditions
OS10	FIN	External CLKI Frequency (External Clocks allowed only in EC and ECPLL modes)	DC	—	64	MHz	EC
		Oscillator Crystal Frequency	3.5	—	10	MHz	XT
			10	—	32	MHz	HS
OS20	TOSC	$TOSC = 1/FOSC$	15.6	—	DC	ns	
OS25	TCY	Instruction Cycle Time ⁽²⁾	10	—	DC	ns	
OS30	TOSL, TOSH	External Clock in (OSCI) High or Low Time	$0.45 \times TOSC$	—	$0.55 \times TOSC$	ns	EC
OS31	TOSR, TOSF	External Clock in (OSCI) Rise or Fall Time	—	—	20	ns	EC
OS40	TCKR	CLKO Rise Time ^(3,4)	—	5.4	—	ns	
OS41	TCKF	CLKO Fall Time ^(3,4)	—	6.4	—	ns	
Notes:							
1. Data in "Typ." column are at 3.3V, +25°C unless otherwise stated.							
2. Instruction cycle period (TCY) equals two times the input oscillator time base period. All specified values are based on characterization data for that particular oscillator type, under Standard Operating conditions, with the device executing code. Exceeding these specified limits may result in an unstable oscillator operation and/or higher than expected current consumption. All devices are tested to operate at "Minimum" values with an External Clock applied to the OSCI pin. When an External Clock input is used, the "Maximum" cycle time limit is "DC" (no clock) for all devices.							
3. Measurements are taken in EC mode. The CLKO signal is measured on the OSCO pin.							
4. This parameter is characterized but not tested in manufacturing.							

.....continued

Operating Conditions: 3.0V to 3.6V (unless otherwise stated)

Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial $-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended

Param No.	Sym	Characteristic	Min.	Typ. ⁽¹⁾	Max.	Units	Conditions
OS42	GM	External Oscillator Transconductance ⁽³⁾	2.7	—	4	mA/V	XTCFG[1:0] = 00, XTBST = 0
			4	—	7	mA/V	XTCFG[1:0] = 00, XTBST = 1
			4.5	—	7	mA/V	XTCFG[1:0] = 01, XTBST = 0
			6	—	11.9	mA/V	XTCFG[1:0] = 01, XTBST = 1
			5.9	—	9.7	mA/V	XTCFG[1:0] = 10, XTBST = 0
			6.9	—	15.9	mA/V	XTCFG[1:0] = 10, XTBST = 1
			6.7	—	12	mA/V	XTCFG[1:0] = 11, XTBST = 0
			7.5	—	19	mA/V	XTCFG[1:0] = 11, XTBST = 1

Notes:

1. Data in "Typ." column are at 3.3V, +25°C unless otherwise stated.
2. Instruction cycle period (T_{CY}) equals two times the input oscillator time base period. All specified values are based on characterization data for that particular oscillator type, under Standard Operating conditions, with the device executing code. Exceeding these specified limits may result in an unstable oscillator operation and/or higher than expected current consumption. All devices are tested to operate at "Minimum" values with an External Clock applied to the OSCI pin. When an External Clock input is used, the "Maximum" cycle time limit is "DC" (no clock) for all devices.
3. Measurements are taken in EC mode. The CLKO signal is measured on the OSCO pin.
4. This parameter is characterized but not tested in manufacturing.

Table 33-25. PLL Clock Timing Specifications

Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated)							
Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial							
$-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended							
Param No.	Symbol	Characteristic	Min.	Typ. ⁽¹⁾	Max.	Units	Conditions
OS50	F _{PLLI}	PLL Voltage Controlled Oscillator (VCO) Input Frequency Range	8 ⁽²⁾	—	64	MHz	ECPLL, XTPLL modes
OS51	F _{VCO}	On-Chip VCO System Frequency	400	—	1600	MHz	
OS52	T _{LOCK}	PLL Start-up Time (Lock Time)	—	60	—	μs	
Notes:							
1. Data in "Typ." column are at 3.3V, +25°C unless otherwise stated. Parameters are for design guidance only and are not tested.							
2. Inclusive of FRC Tolerance Specification, Parameter F20a .							

Table 33-26. Auxiliary PLL Clock Timing Specifications

Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated)							
Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial							
$-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended							
Param No.	Symbol	Characteristic	Min.	Typ. ⁽¹⁾	Max.	Units	Conditions
OS50	F _{PLLI}	APLL Voltage Controlled Oscillator (VCO) Input Frequency Range	8 ⁽²⁾	—	64	MHz	ECPLL, XTPLL modes
OS51	F _{VCO}	On-Chip VCO System Frequency	400	—	1600	MHz	
OS52	T _{LOCK}	APLL Start-up Time (Lock Time)	—	125	—	μs	
Notes:							
1. Data in "Typ." column are at 3.3V, +25°C unless otherwise stated. Parameters are for design guidance only and are not tested.							
2. Inclusive of FRC Tolerance Specification, Parameter F20a .							

Table 33-27. Internal FRC Accuracy

Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated)					
Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial					
$-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended					
Param No.	Characteristic	Min.	Max.	Units	Conditions
Internal FRC Accuracy @ FRC Frequency = 8 MHz ⁽¹⁾					
F20a	FRC	-2 ⁽²⁾	+2	%	$-40^{\circ}\text{C} \leq T_A \leq 0^{\circ}\text{C}$
		-1.5	+1.5	%	$-5^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$
		-2	+2	%	$+85^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$
F22	BFRC/LPRC	-17	+17	%	$-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$
Notes: - Frequency is calibrated at $+25^{\circ}\text{C}$ and 3.3V. TUNx bits can be used to compensate for temperature drift. - Due to the effect of aging, this value may drift by an additional -0.5% over the lifetime of the device					

Figure 33-3. I/O Timing Characteristics

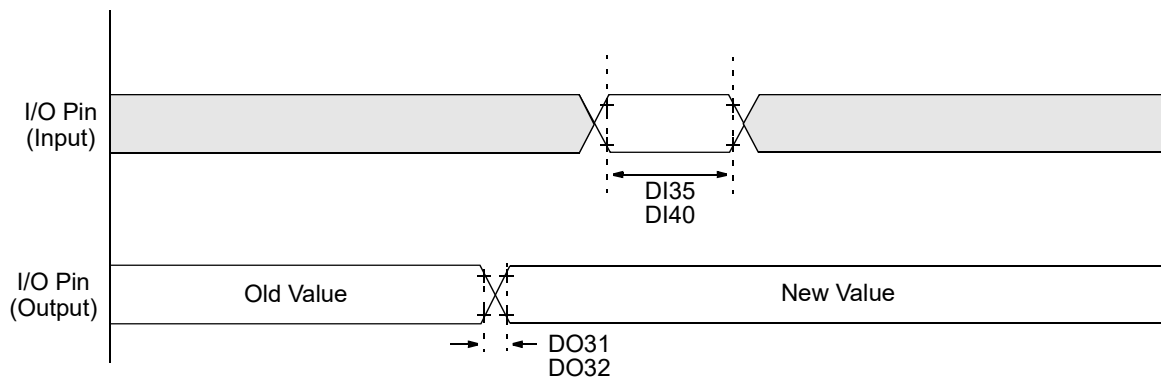


Table 33-28. I/O Timing Requirements

Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated)							
Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial							
$-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended							
Param No.	Symbol	Characteristic	Min.	Typ. ⁽¹⁾	Max.	Units	Conditions
DO31	$T_{IO\text{R}}$	Port Output Rise Time ⁽²⁾	—	6.5	9.7	ns	
DO32	$T_{IO\text{F}}$	Port Output Fall Time ⁽²⁾	—	3.2	4.2	ns	
DI35	T_{INP}	INTx Pin High or Low Time (input)	20	—	—	ns	
DI40	T_{RBP}	CNx High or Low Time (input)	2	—	—	T_{CY}	
Notes: - Data in "Typ." column are at 3.3V, $+25^{\circ}\text{C}$ unless otherwise stated. - This parameter is characterized but not tested in manufacturing.							

Figure 33-4. BOR and Master Clear Reset Timing Characteristics

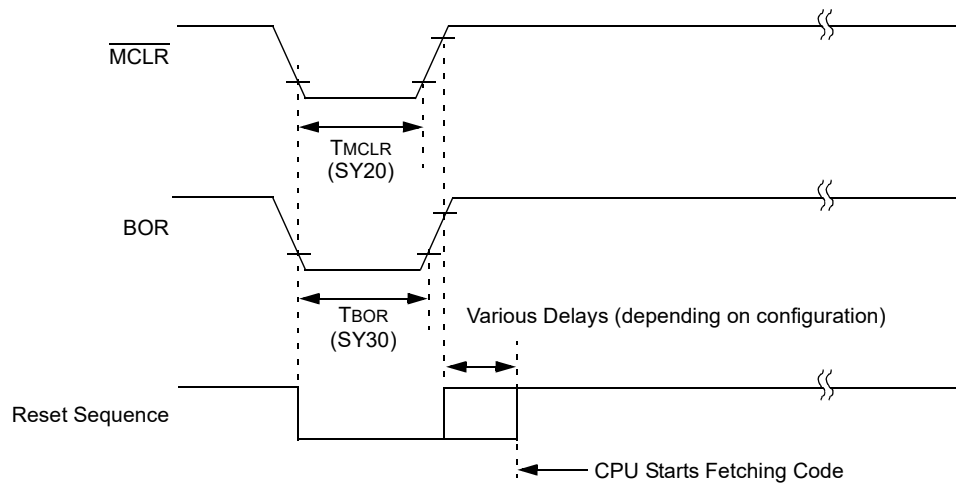


Table 33-29. Reset, Watchdog Timer, Oscillator Start-up Timer, Power-up Timer Timing Requirements

Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated)							
Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial							
$-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended							
Param No.	Symbol	Characteristic ⁽¹⁾	Min.	Typ. ⁽²⁾	Max.	Units	Conditions
SY00	T_{PU}	Power-up Period	—	200	—	μs	
SY10	T_{OST}	Oscillator Start-up Time	—	$1024 T_{OSC}$	—	—	$T_{OSC} = \text{OSCI period}$
SY13	T_{IOZ}	I/O High-Impedance from MCLR Low or Watchdog Timer Reset	—	1.5	—	μs	
SY20	T_{MCLR}	MCLR Pulse Width (low)	2	—	—	μs	
SY30	T_{BOR}	BOR Pulse Width (low)	1	—	—	μs	
SY35	T_{FSCM}	Fail-Safe Clock Monitor Delay	—	500	900	μs	-40°C to $+85^{\circ}\text{C}$
SY36	T_{VREG}	Voltage Regulator Standby-to-Active mode Transition Time	—	—	40	μs	Clock fail to BFRC switch
SY37	$T_{OSCDFRC}$	FRC Oscillator Start-up Delay	—	—	15	μs	From POR event
SY38	$T_{OSCDLPRC}$	LPRC Oscillator Start-up Delay	—	—	50	μs	From Reset event
Notes:							
1. These parameters are characterized but not tested in manufacturing.							
2. Data in "Typ." column are at 3.3V, $+25^{\circ}\text{C}$ unless otherwise stated.							

Figure 33-5. High-Speed PWMx Module Fault Timing Characteristics

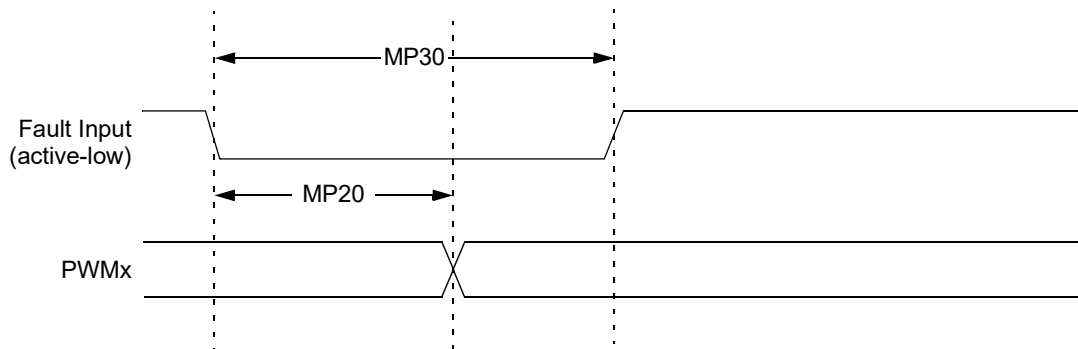


Figure 33-6. High-Speed PWMx Module Timing Characteristics

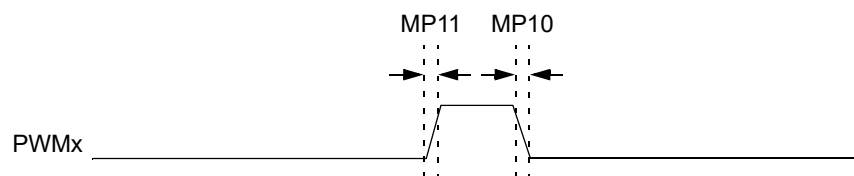


Table 33-30. High-Speed PWMx Module Timing Requirements

Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated)							
Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial							
$-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended							
Param No.	Symbol	Characteristic ⁽¹⁾	Min.	Typ.	Max.	Units	Conditions
MP00	F _{IN}	PWM Input Frequency	450	—	500	MHz	Note 2
MP10	T _{FPWM}	PWMx Output Fall Time	—	—	—	ns	See Parameter DO32
MP11	T _{RPWM}	PWMx Output Rise Time	—	—	—	ns	See Parameter DO31
MP20	T _{FD}	Fault Input ↓ to PWMx I/O Change	—	—	26	ns	PCI Inputs 19 through 22
MP30	T _{FH}	Fault Input Pulse Width	8	—	—	ns	
Notes:							
1. These parameters are characterized but not tested in manufacturing.							
2. Input frequency of 500 MHz is recommended for High-Resolution mode.							

Table 33-31. SPIx Maximum Data/Clock Rate Summary

SPI Host Transmit Only (Half-Duplex)	SPI Host Transmit/ Receive (Full-Duplex)	SPI Client Transmit/ Receive (Full-Duplex)	CKE	Maximum Data Rate (MHz)	Condition
Figure 33-7 Figure 33-8	—	—	0	15	Using PPS
				40	Dedicated Pin
Figure 33-8 Table 33-32	—	—	1	15	Using PPS
				40	Dedicated Pin
—	Figure 33-9 Table 33-33	—	0	9	Using PPS
				40	Dedicated Pin
—	Figure 33-10 Table 33-34	—	1	9	Using PPS
				40	Dedicated Pin
—	—	Figure 33-11 Table 33-35	0	15	Using PPS
				40	Dedicated Pin
—	—	Figure 33-12 Table 33-36	1	15	Using PPS
				40	Dedicated Pin

Figure 33-7. SPIx Host Mode (Half-Duplex, Transmit Only, CKE = 0)
Timing Characteristics

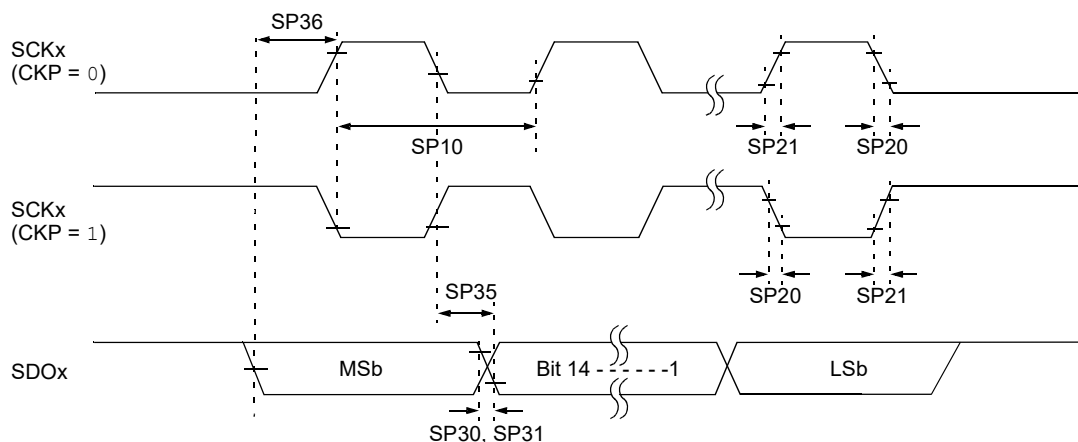


Figure 33-8. SPIx Host Mode (Half-Duplex, Transmit Only, CKE = 1)
Timing Characteristics

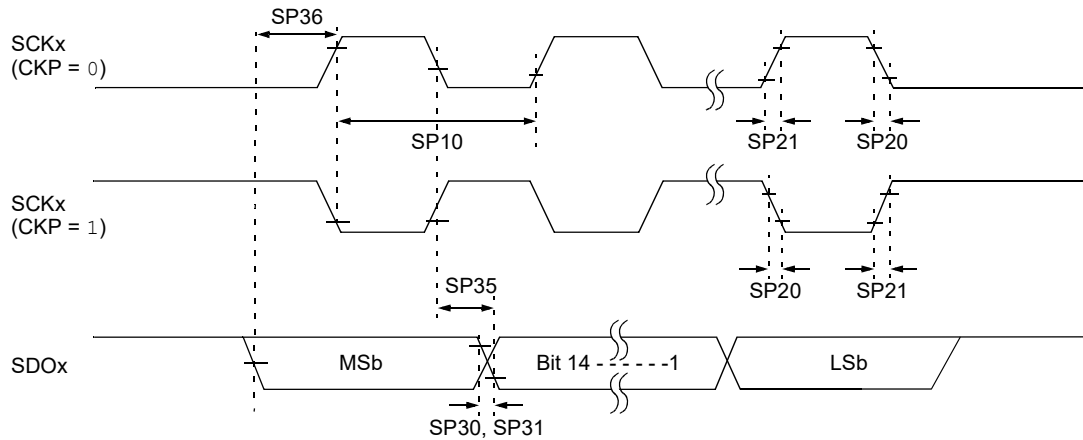


Table 33-32. SPIx Host Mode (Half-Duplex, Transmit Only) Timing Requirements

Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated)

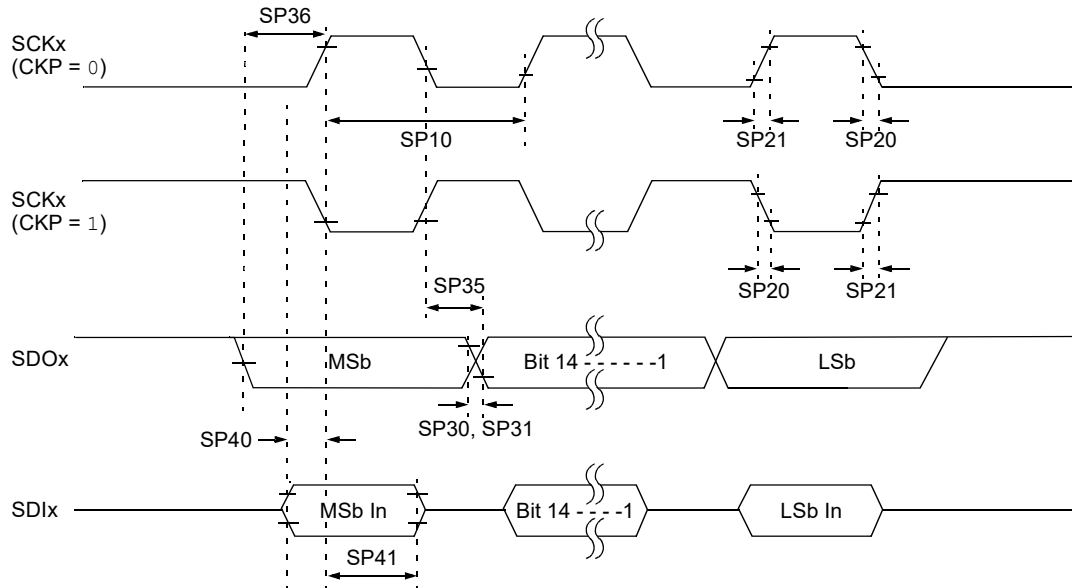
Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial

$-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended

Param No.	Symbol	Characteristic ⁽¹⁾	Min.	Typ. ⁽²⁾	Max.	Units	Conditions
SP10	F _{ScP}	Maximum SCKx Frequency	—	—	15	MHz	Using PPS pins
			—	—	40	MHz	SPI2 dedicated pins
SP20	T _{ScF}	SCKx Output Fall Time	—	—	—	ns	See Parameter DO32 (Note 3)
SP21	T _{ScR}	SCKx Output Rise Time	—	—	—	ns	See Parameter DO31 (Note 3)
SP30	T _{DOF}	SDOx Data Output Fall Time	—	—	—	ns	See Parameter DO32 (Note 3)
SP31	T _{DOR}	SDOx Data Output Rise Time	—	—	—	ns	See Parameter DO31 (Note 3)
SP35	T _{ScH2DOV} , T _{ScL2DOV}	SDOx Data Output Valid After SCKx Edge	—	6	20	ns	
SP36	T _{DI} V2ScH, T _{DI} V2ScL	SDOx Data Output Setup to First SCKx Edge	30	—	—	ns	Using PPS pins
			3	—	—	ns	SPI2 dedicated pins

Notes:

- These parameters are characterized but not tested in manufacturing.
- Data in "Typ." column are at 3.3V, +25°C unless otherwise stated.
- Assumes 50 pF load on all SPIx pins.

Figure 33-9. SPIx Host Mode (Full-Duplex, CKE = 1, CKP = x, SMP = 1)**Timing Characteristics****Table 33-33. SPIx Host Mode (Full-Duplex, CKE = 1, CKP = x, SMP = 1)****Timing Requirements**

Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated)

Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial $-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended

Param No.	Symbol	Characteristic ⁽¹⁾	Min.	Typ. ⁽²⁾	Max.	Units	Conditions
SP10	F_{SCP}	Maximum SCKx Frequency	—	—	15	MHz	Using PPS pins
			—	—	40	MHz	SPI2 dedicated pins
SP20	T_{SCF}	SCKx Output Fall Time	—	—	—	ns	See Parameter DO32 (Note 3)
SP21	T_{SCR}	SCKx Output Rise Time	—	—	—	ns	See Parameter DO31 (Note 3)
SP30	T_{DOF}	SDOx Data Output Fall Time	—	—	—	ns	See Parameter DO32 (Note 3)
SP31	T_{DOR}	SDOx Data Output Rise Time	—	—	—	ns	See Parameter DO31 (Note 3)

Notes:

- These parameters are characterized but not tested in manufacturing.
- Data in "Typ." column are at 3.3V, +25°C unless otherwise stated.
- Assumes 50 pF load on all SPIx pins.

.....continued

Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated)

Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial $-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended

Param No.	Symbol	Characteristic ⁽¹⁾	Min.	Typ. ⁽²⁾	Max.	Units	Conditions
SP35	$T_{SC}H2_{DOV}$, $T_{SC}L2_{DOV}$	SDOx Data Output Valid After SCKx Edge	—	6	20	ns	
SP36	$T_{DO}V2_{SC}$, $T_{DO}V2_{SCL}$	SDOx Data Output Setup to First SCKx Edge	30	—	—	ns	Using PPS pins
			3	—	—	ns	SPI2 dedicated pins
SP40	$T_{DI}V2_{SCH}$, $T_{DI}V2_{SCL}$	Setup Time of SDIx Data Input to SCKx Edge	30	—	—	ns	Using PPS pins
			20	—	—	ns	SPI2 dedicated pins
SP41	$T_{SC}H2_{DIL}$, $T_{SC}L2_{DIL}$	Hold Time of SDIx Data Input to SCKx Edge	30	—	—	ns	Using PPS pins
			15	—	—	ns	SPI2 dedicated pins

Notes:

- These parameters are characterized but not tested in manufacturing.
- Data in "Typ." column are at 3.3V, +25°C unless otherwise stated.
- Assumes 50 pF load on all SPIx pins.

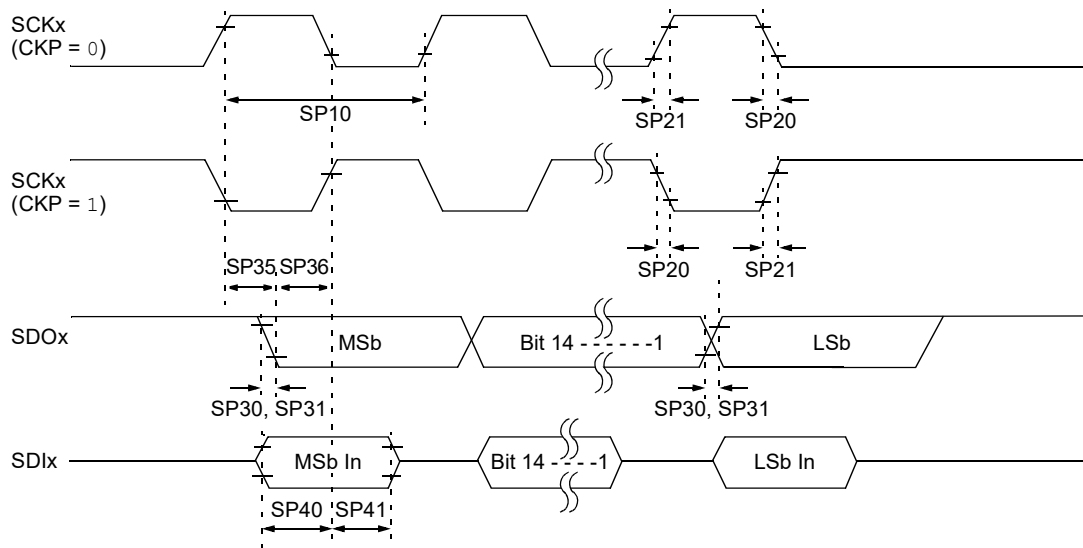
Figure 33-10. SPIx Host Mode (Full-Duplex, CKE = 0, CKP = x, SMP = 1)**Timing Characteristics**

Table 33-34. SPIx Host Mode (Full-Duplex, CKE = 0, CKP = x, SMP = 1)

Timing Requirements

Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated)

Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial $-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended

Param No.	Symbol	Characteristic ⁽¹⁾	Min.	Typ. ⁽²⁾	Max.	Units	Conditions
SP10	F _{SCP}	Maximum SCKx Frequency	—	—	15	MHz	Using PPS pins
			—	—	40	MHz	SPI2 dedicated pins
SP20	T _{SCF}	SCKx Output Fall Time	—	—	—	ns	See Parameter DO32 (Note 3)
SP21	T _{SCR}	SCKx Output Rise Time	—	—	—	ns	See Parameter DO31 (Note 3)
SP30	T _{DOF}	SDOx Data Output Fall Time	—	—	—	ns	See Parameter DO32 (Note 3)
SP31	T _{DOR}	SDOx Data Output Rise Time	—	—	—	ns	See Parameter DO31 (Note 3)
SP35	T _{SCH2DOV} , T _{SC L2DOV}	SDOx Data Output Valid After SCKx Edge	—	6	20	ns	
SP36	T _{DOV2SCH} , T _{DOV2SCL}	SDOx Data Output Setup to First SCKx Edge	30	—	—	ns	Using PPS pins
			20	—	—	ns	SPI2 dedicated pins
SP40	T _{diV2SCH} , T _{diV2SCL}	Setup Time of SDIx Data Input to SCKx Edge	30	—	—	ns	Using PPS pins
			10	—	—	ns	SPI2 dedicated pins
SP41	T _{SCH2DIL} , T _{SC L2DIL}	Hold Time of SDIx Data Input to SCKx Edge	30	—	—	ns	Using PPS pins
			15	—	—	ns	SPI2 dedicated pins

Notes:

1. These parameters are characterized but not tested in manufacturing.
2. Data in "Typ." column are at 3.3V, +25°C unless otherwise stated.
3. Assumes 50 pF load on all SPIx pins.

Figure 33-11. SPIx Client Mode (Full-Duplex, CKE = 0, CKP = x, SMP = 0)
Timing Characteristics

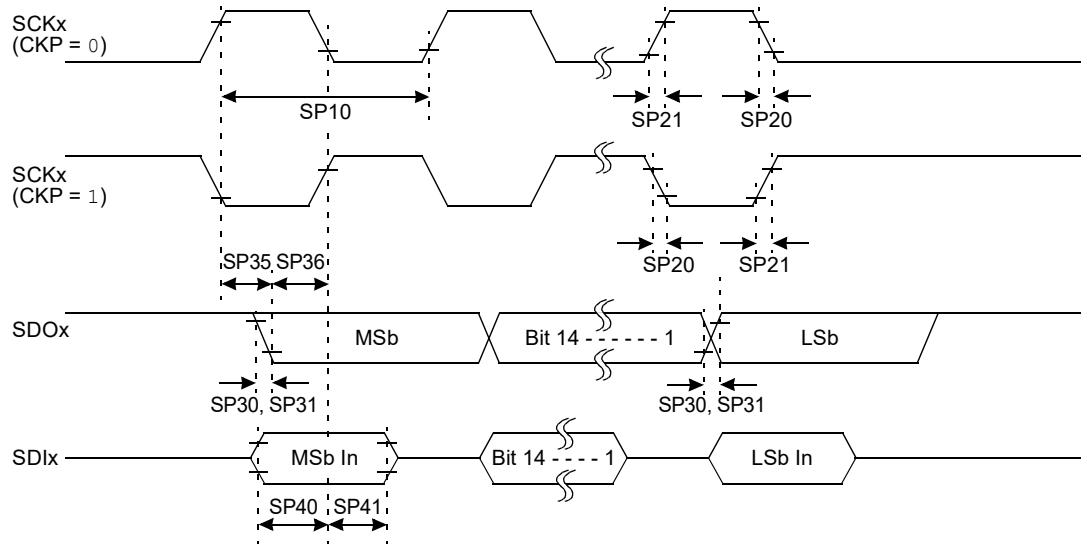


Table 33-35. SPIx Client Mode (Full-Duplex, CKE = 0, CKP = x, SMP = 0)
Timing Requirements

Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated)

Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial

$-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended

Param No.	Symbol	Char. ⁽¹⁾	Min.	Typ. ⁽²⁾	Max.	Units	Conditions
SP10	F _{SCP}	Maximum SCKx Input Frequency	—	—	15	MHz	Using PPS pins
			—	—	40	MHz	SPI2 dedicated pins
SP72	T _{SCF}	SCKx Input Fall Time	—	—	—	ns	See Parameter DO32 (Note 3)
SP73	T _{SCR}	SCKx Input Rise Time	—	—	—	ns	See Parameter DO31 (Note 3)
SP30	T _{DOF}	SDOx Data Output Fall Time	—	—	—	ns	See Parameter DO32 (Note 3)
SP31	T _{DOF}	SDOx Data Output Rise Time	—	—	—	ns	See Parameter DO31 (Note 3)
SP35	T _{SCH2DOV} , T _{SC L2DOV}	SDOx Data Output Valid After SCKx Edge	—	6	20	ns	

Notes:

- These parameters are characterized but not tested in manufacturing.
- Data in "Typ." column are at 3.3V, +25°C unless otherwise stated.
- Assumes 50 pF load on all SPIx pins.

.....continued

Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated)

Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial $-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended

Param No.	Symbol	Char. ⁽¹⁾	Min.	Typ. ⁽²⁾	Max.	Units	Conditions
SP36	$T_{DOV2_{SC}H}$, $T_{DOV2_{SC}L}$	SDOx Data Output Setup to First SCKx Edge	30	—	—	ns	Using PPS pins
			20	—	—	ns	SPI2 dedicated pins
SP40	$T_{DlV2_{SC}H}$, $T_{DlV2_{SC}L}$	Setup Time of SDIx Data Input to SCKx Edge	30	—	—	ns	Using PPS pins
			10	—	—	ns	SPI2 dedicated pins
SP41	$T_{SCH2_{Dl}L}$, $T_{SCH2_{Dl}L}$	Hold Time of SDIx Data Input to SCKx Edge	30	—	—	ns	Using PPS pins
			15	—	—	ns	SPI2 dedicated pins
SP50	$T_{SSL2_{SC}H}$, $T_{SSL2_{SC}L}$	$\overline{SSx} \downarrow$ to SCKx $\uparrow$ or SCKx $\downarrow$ Input	120	—	—	ns	
SP51	$T_{SSL2_{DOZ}}$	$\overline{SSx} \uparrow$ to SDOx Output High-Impedance	8	—	50	ns	Note 3
SP52	$T_{SCH2_{SS}H}$, $T_{SCH2_{SS}H}$	$\overline{SSx} \uparrow$ After SCKx Edge	$1.5 T_{CY} + 40$	—	—	ns	Note 3

Notes:

1. These parameters are characterized but not tested in manufacturing.
2. Data in "Typ." column are at 3.3V, +25°C unless otherwise stated.
3. Assumes 50 pF load on all SPIx pins.

Figure 33-12. SPIx Client Mode (Full-Duplex, CKE = 1, CKP = x, SMP = 0)
Timing Characteristics

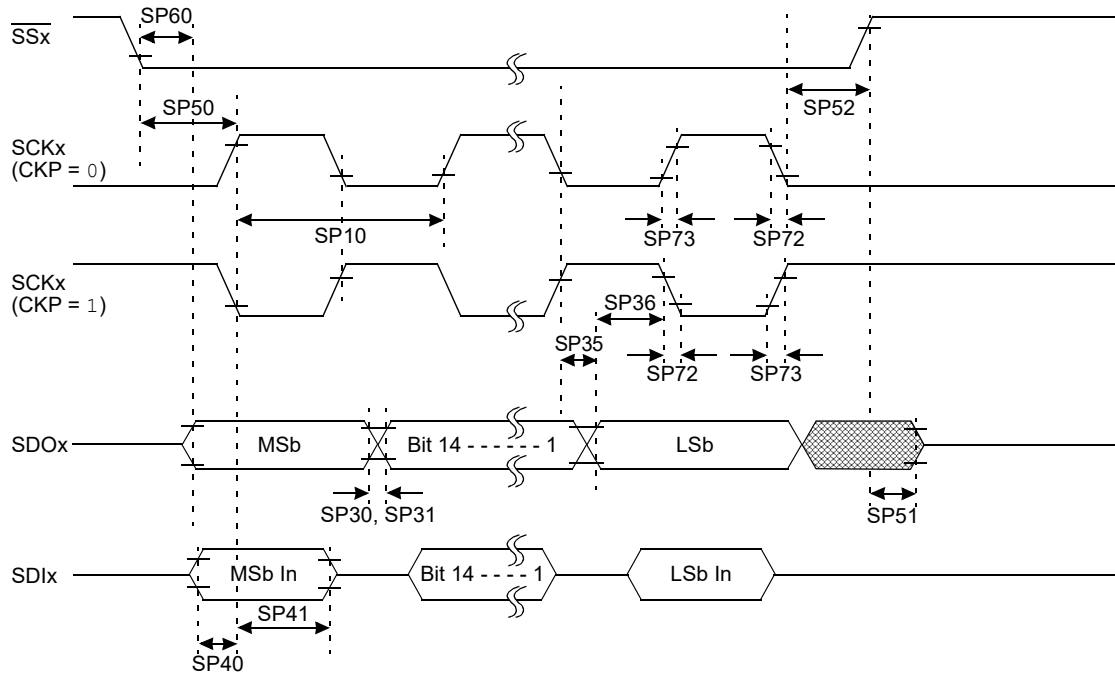


Table 33-36. SPIx Client Mode (Full-Duplex, CKE = 1, CKP = x, SMP = 0)
Timing Requirements

Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated)							
Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial							
$-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended							
Param No.	Symbol	Characteristic ⁽¹⁾	Min.	Typ. ⁽²⁾	Max.	Units	Conditions
SP10	F _{SCP}	Maximum SCKx Input Frequency	—	—	15	MHz	Using PPS pins
			—	—	40	MHz	SPI2 dedicated pins
SP72	T _{SCF}	SCKx Input Fall Time	—	—	—	ns	See Parameter DO32 (Note 3)
SP73	T _{SCR}	SCKx Input Rise Time	—	—	—	ns	See Parameter DO31 (Note 3)
SP30	T _{DOF}	SDOx Data Output Fall Time	—	—	—	ns	See Parameter DO32 (Note 3)
SP31	T _{DOR}	SDOx Data Output Rise Time	—	—	—	ns	See Parameter DO31 (Note 3)
Notes:							
1. These parameters are characterized but not tested in manufacturing.							
2. Data in "Typ." column are at 3.3V, +25°C unless otherwise stated.							
3. Assumes 50 pF load on all SPIx pins.							

.....continued

Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated)

Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial $-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended

Param No.	Symbol	Characteristic ⁽¹⁾	Min.	Typ. ⁽²⁾	Max.	Units	Conditions
SP35	T _{SCH2DOV} , T _{SCL2DOV}	SDOx Data Output Valid After SCKx Edge	—	6	20	ns	
SP36	T _{DOV2scH} , T _{DOV2sCL}	SDOx Data Output Setup to First SCKx Edge	30	—	—	ns	Using PPS pins
			20	—	—	ns	SPI2 dedicated pins
SP40	T _{DIV2SCH} , T _{DIV2sCL}	Setup Time of SDIx Data Input to SCKx Edge	30	—	—	ns	Using PPS pins
			10	—	—	ns	SPI2 dedicated pins
SP41	T _{SCH2diL} , T _{SCL2diL}	Hold Time of SDIx Data Input to SCKx Edge	30	—	—	ns	Using PPS pins
			15	—	—	ns	SPI2 dedicated pins
SP50	T _{SSL2SCH} , T _{SSL2sCL}	$\overline{\text{SSx}}$ ↓ to SCKx ↑ or SCKx ↓ Input	120	—	—	ns	
SP51	T _{SSH2doZ}	$\overline{\text{SSx}}$ ↑ to SDOx Output High-Impedance	8	—	50	ns	Note 3
SP52	T _{SCH2SSH} , T _{SCL2SSH}	$\overline{\text{SSx}}$ ↑ After SCKx Edge	1.5 T _{CY} + 40	—	—	ns	Note 3
SP60	T _{SSL2DOV}	SDOx Data Output Valid After $\overline{\text{SSx}}$ Edge	—	—	50	ns	

Notes:

1. These parameters are characterized but not tested in manufacturing.
2. Data in "Typ." column are at 3.3V, +25°C unless otherwise stated.
3. Assumes 50 pF load on all SPIx pins.

Figure 33-13. I2Cx Bus Start/Stop Bits Timing Characteristics (Host Mode)

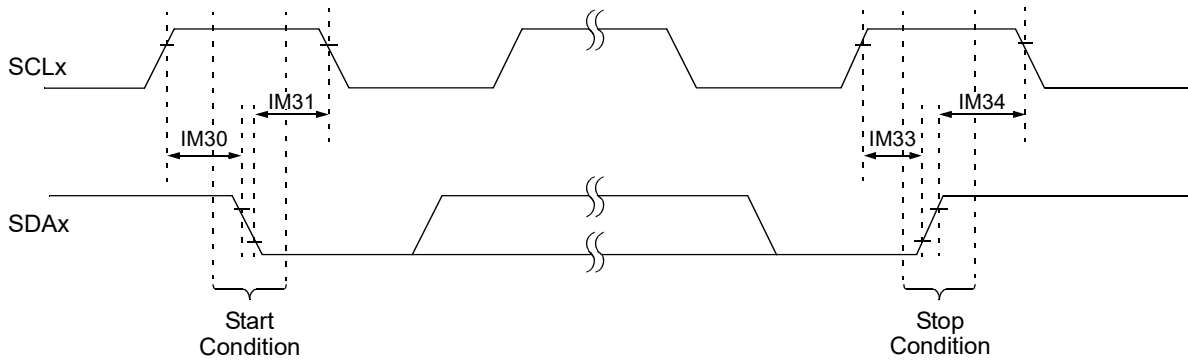


Figure 33-14. I2Cx Bus Data Timing Characteristics (Host Mode)

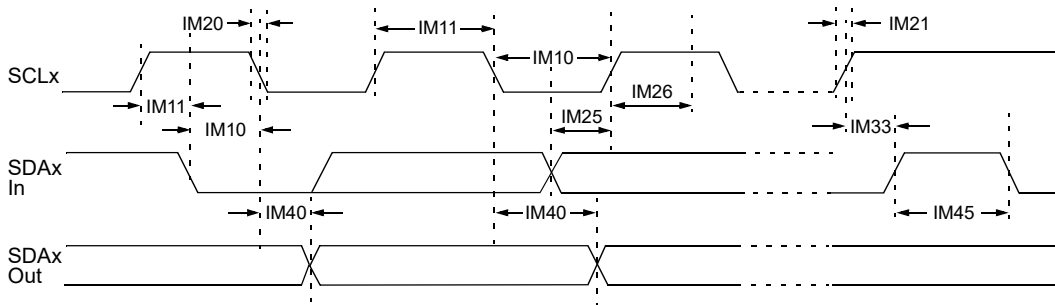


Table 33-37. I2Cx Bus Data Timing Requirements (Host Mode)

Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated)

Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial

$-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended

Param No.	Symbol	Characteristic ⁽⁴⁾		Min. ⁽¹⁾	Max.	Units	Conditions
IM10	Tlo:scl	Clock Low Time	100 kHz mode	T _{CY} (BRG + 1)	—	μs	
			400 kHz mode	T _{CY} (BRG + 1)	—	μs	
			1 MHz mode ⁽²⁾	T _{CY} (BRG + 1)	—	μs	

Notes:

1. BRG is the value of the I²C Baud Rate Generator.

2. Maximum Pin Capacitance = 10 pF for all I2Cx pins (for 1 MHz mode only).

3. Typical value for this parameter is 130 ns.

4. These parameters are characterized but not tested in manufacturing.

.....continued

Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated)

Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial $-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended

Param No.	Symbol	Characteristic ⁽⁴⁾	Min. ⁽¹⁾	Max.	Units	Conditions
IM11	Thi:scl	Clock High Time	100 kHz mode	$T_{CY}(\text{BRG} + 1)$	—	μs
			400 kHz mode	$T_{CY}(\text{BRG} + 1)$	—	μs
			1 MHz mode ⁽²⁾	$T_{CY}(\text{BRG} + 1)$	—	μs
IM20	Tf:scl	SDAx and SCLx Fall Time	100 kHz mode	—	300	ns
			400 kHz mode	$20 \times (V_{DD}/5.5\text{V})$	300	ns
			1 MHz mode ⁽²⁾	—	120	ns
IM21	Tr:scl	SDAx and SCLx Rise Time	100 kHz mode	—	1000	ns
			400 kHz mode	$20 + 0.1 C_b$	300	ns
			1 MHz mode ⁽²⁾	—	120	ns
IM25	Tsu:dat	Data Input Setup Time	100 kHz mode	250	—	ns
			400 kHz mode	100	—	ns
			1 MHz mode ⁽²⁾	50	—	ns
IM26	Thd:dat	Data Input Hold Time	100 kHz mode	0	—	μs
			400 kHz mode	0	0.9	μs
			1 MHz mode ⁽²⁾	0	0.3	μs

Notes:

1. BRG is the value of the I²C Baud Rate Generator.
2. Maximum Pin Capacitance = 10 pF for all I²Cx pins (for 1 MHz mode only).
3. Typical value for this parameter is 130 ns.
4. These parameters are characterized but not tested in manufacturing.

.....continued

Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated)

Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial $-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended

Param No.	Symbol	Characteristic ⁽⁴⁾		Min. ⁽¹⁾	Max.	Units	Conditions
IM30	Tsu:sta	Start Condition Setup Time	100 kHz mode	$T_{CY} (BRG + 1)$	—	μs	Only relevant for Repeated Start condition
			400 kHz mode	$T_{CY} (BRG + 1)$	—	μs	
			1 MHz mode ⁽²⁾	$T_{CY} (BRG + 1)$	—	μs	
IM31	Thd:sta	Start Condition Hold Time	100 kHz mode	$T_{CY} (BRG + 1)$	—	μs	After this period, the first clock pulse is generated
			400 kHz mode	$T_{CY} (BRG + 1)$	—	μs	
			1 MHz mode ⁽²⁾	$T_{CY} (BRG + 1)$	—	μs	
IM33	Tsu:sto	Stop Condition Setup Time	100 kHz mode	$T_{CY} (BRG + 1)$	—	μs	
			400 kHz mode	$T_{CY} (BRG + 1)$	—	μs	
			1 MHz mode ⁽²⁾	$T_{CY} (BRG + 1)$	—	μs	
IM34	Thd:sto	Stop Condition Hold Time	100 kHz mode	$T_{CY} (BRG + 1)$	—	μs	
			400 kHz mode	$T_{CY} (BRG + 1)$	—	μs	
			1 MHz mode ⁽²⁾	$T_{CY} (BRG + 1)$	—	μs	
IM40	Taa:scl	Output Valid from Clock	100 kHz mode	—	3450	ns	
			400 kHz mode	—	900	ns	
			1 MHz mode ⁽²⁾	—	450	ns	

Notes:

1. BRG is the value of the I²C Baud Rate Generator.
2. Maximum Pin Capacitance = 10 pF for all I2Cx pins (for 1 MHz mode only).
3. Typical value for this parameter is 130 ns.
4. These parameters are characterized but not tested in manufacturing.

.....continued

Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated)

Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial $-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended

Param No.	Symbol	Characteristic ⁽⁴⁾		Min. ⁽¹⁾	Max.	Units	Conditions
IM45	Tbf:sda	Bus Free Time	100 kHz mode	4.7	—	μs	Time the bus must be free before a new transmission can start
			400 kHz mode	1.3	—	μs	
			1 MHz mode ⁽²⁾	0.5	—	μs	
IM50	Cb	Bus Capacitive Loading		—	400	pF	
IM51	Tpgd	Pulse Gobbler Delay		65	390	ns	Note 3

Notes:

1. BRG is the value of the I²C Baud Rate Generator.
2. Maximum Pin Capacitance = 10 pF for all I2Cx pins (for 1 MHz mode only).
3. Typical value for this parameter is 130 ns.
4. These parameters are characterized but not tested in manufacturing.

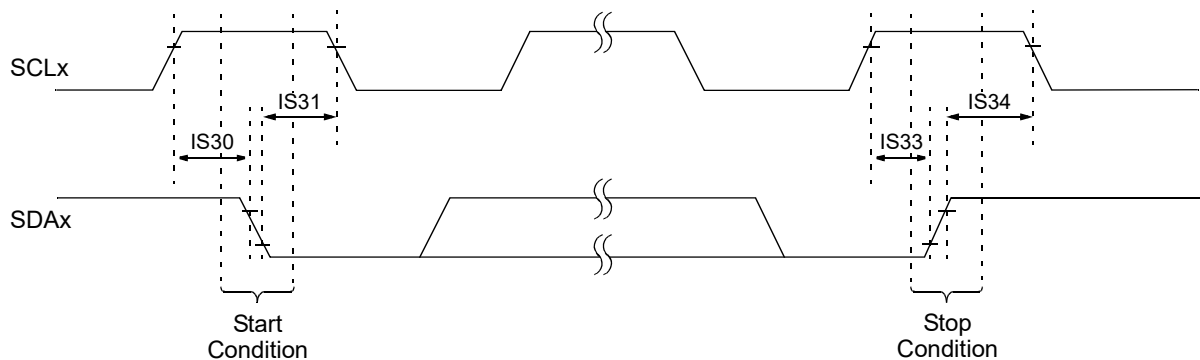
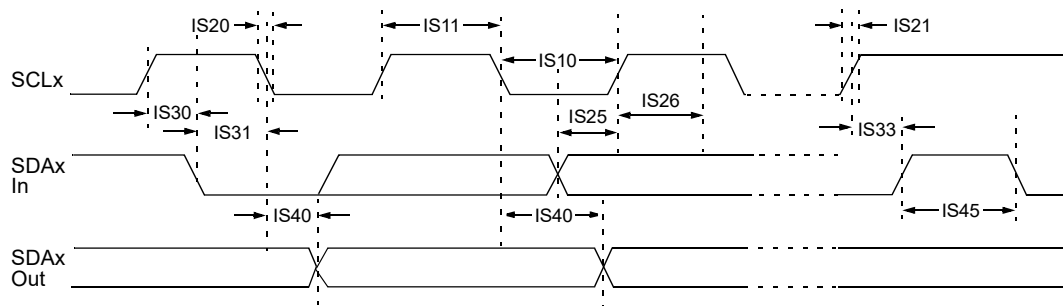
Figure 33-15. I2Cx Bus Start/Stop Bits Timing Characteristics (Client Mode)**Figure 33-16. I2Cx Bus Data Timing Characteristics (Client Mode)**

Table 33-38. I2Cx Bus Data Timing Requirements (Client Mode)

Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated)							
Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial							
$-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended							
Param No.	Symbol	Characteristic ⁽³⁾		Min.	Max.	Units	Conditions
IS10	Tlo:scl	Clock Low Time	100 kHz mode	4.7	—	μs	
			400 kHz mode	1.3	—	μs	
			1 MHz mode ⁽¹⁾	0.5	—	μs	
IS11	Thi:scl	Clock High Time	100 kHz mode	4.0	—	μs	Device must operate at a minimum of 1.5 MHz
			400 kHz mode	0.6	—	μs	Device must operate at a minimum of 10 MHz
			1 MHz mode ⁽¹⁾	0.28	—	μs	
IS20	Tf:scl	SDAx and SCLx Fall Time	100 kHz mode	—	300	ns	Cb is specified to be from 10 to 400 pF
			400 kHz mode	$20 \times (V_{DD}/5.5\text{V})$	300	ns	
			1 MHz mode ⁽¹⁾	$20 \times (V_{DD}/5.5\text{V})$	120	ns	
IS21	Tr:scl	SDAx and SCLx Rise Time	100 kHz mode	$20 + 0.1 \text{ Cb}$	1000	ns	Cb is specified to be from 10 to 400 pF
			400 kHz mode	—	300	ns	
			1 MHz mode ⁽¹⁾	—	120	ns	
IS25	Tsu:dat	Data Input Setup Time	100 kHz mode	250	—	ns	
			400 kHz mode	100	—	ns	
			1 MHz mode ⁽¹⁾	50	—	ns	

Notes:

1. Maximum Pin Capacitance = 10 pF for all I2Cx pins (for 1 MHz mode only).
2. Typical value for this parameter is 130 ns.
3. These parameters are characterized but not tested in manufacturing.

.....continued

Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated)

Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial $-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended

Param No.	Symbol	Characteristic ⁽³⁾		Min.	Max.	Units	Conditions
IS26	Thd:dat	Data Input Hold Time	100 kHz mode	0	—	μs	
			400 kHz mode	0	0.9	μs	
			1 MHz mode ⁽¹⁾	0	0.3	μs	
IS30	Tsu:sta	Start Condition Setup Time	100 kHz mode	4.7	—	μs	Only relevant for Repeated Start condition
			400 kHz mode	0.6	—	μs	
			1 MHz mode ⁽¹⁾	0.26	—	μs	
IS31	Thd:sta	Start Condition Hold Time	100 kHz mode	4.0	—	μs	After this period, the first clock pulse is generated
			400 kHz mode	0.6	—	μs	
			1 MHz mode ⁽¹⁾	0.26	—	μs	
IS33	Tsu:sto	Stop Condition Setup Time	100 kHz mode	4	—	μs	
			400 kHz mode	0.6	—	μs	
			1 MHz mode ⁽¹⁾	0.26	—	μs	
IS34	Thd:sto	Stop Condition Hold Time	100 kHz mode	> 0	—	μs	
			400 kHz mode	> 0	—	μs	
			1 MHz mode ⁽¹⁾	> 0	—	μs	

Notes:

1. Maximum Pin Capacitance = 10 pF for all I2Cx pins (for 1 MHz mode only).
2. Typical value for this parameter is 130 ns.
3. These parameters are characterized but not tested in manufacturing.

.....continued

Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated)

Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial $-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended

Param No.	Symbol	Characteristic ⁽³⁾	Min.	Max.	Units	Conditions
IS40	Taa:scl	Output Valid from Clock	100 kHz mode	0	3540	ns
			400 kHz mode	0	900	ns
			1 MHz mode ⁽¹⁾	0	400	ns
IS45	Tbf:sda	Bus Free Time	100 kHz mode	4.7	—	μs
			400 kHz mode	1.3	—	μs
			1 MHz mode ⁽¹⁾	0.5	—	μs
IS50	C _B	Bus Capacitive Loading	—	400	pF	
IS51	T _{PGD}	Pulse Gobbler Delay	65	390	ns	Note 2

Notes:

- Maximum Pin Capacitance = 10 pF for all I2Cx pins (for 1 MHz mode only).
- Typical value for this parameter is 130 ns.
- These parameters are characterized but not tested in manufacturing.

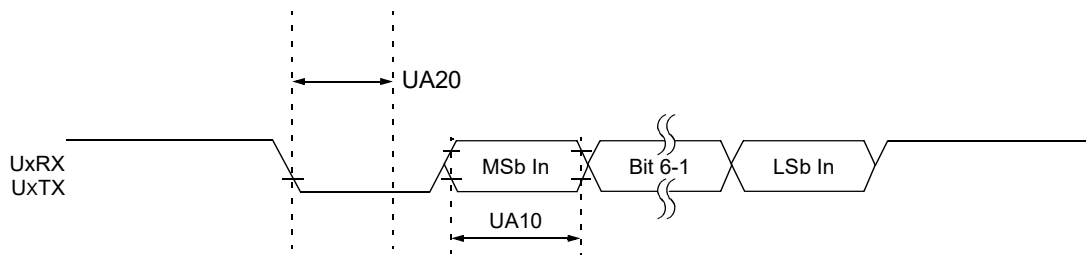
Figure 33-17. UARTx Module I/O Timing Characteristics

Table 33-39. UARTx Module I/O Timing Requirements

Standard Operating Conditions: 3.0V to 3.6V (unless otherwise stated) Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$							
Param No.	Symbol	Characteristic ⁽¹⁾	Min.	Typ. ⁽²⁾	Max.	Units	Conditions
UA10	T_{UABAUD}	UARTx Baud Time	40	—	—	ns	
UA11	F_{BAUD}	UARTx Baud Frequency	—	—	15	Mbps	
UA20	T_{CWF}	Start Bit Pulse Width to Trigger UARTx Wake-up	50	—	—	ns	
Notes: - These parameters are characterized but not tested in manufacturing. - Data in "Typ." column are at 3.3V, +25°C unless otherwise stated. Parameters are for design guidance only and are not tested.							

Table 33-40. ADC Module Specifications

Operating Conditions: 3.0V to 3.6V (unless otherwise stated) ⁽⁴⁾ Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial $-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended							
Param No.	Symbol	Characteristics	Min.	Typical ⁽⁷⁾	Max.	Units	Conditions
Clock Requirements							
AD9	F_{SRC}	ADC Module Input Frequency	—	—	500	MHz	Clock frequency selected by the CLKSELx bits
AD10	$F_{CORESRC}$	ADC Control Clock Frequency	—	—	250	MHz	Clock frequency after the first divider controlled by the CLKDIVx bits
AD11	F_{ADCORE}	ADC SAR Core Clock Frequency	—	—	70	MHz	SAR core frequency after the second divider controlled by the ADCSx or SHRADCs bits
Analog Input							
AD12	$V_{INH} - V_{INL}$	Full-Scale Input Span	AV_{SS}	—	AV_{DD}	V	
AD14	V_{IN}	Absolute Input Voltage	$AV_{SS} - 0.3$	—	$AV_{DD} + 0.3$	V	
AD17	R_{IN}	Recommended Impedance of Analog Voltage Source	—	100	—	Ω	For minimum sampling time (Note 1)
AD60	C_{HOLD}	Capacitance	—	5	—	pF	Dedicated cores (Note 1)
Notes: - These parameters are not characterized or tested in manufacturing. - These parameters are characterized but not tested in manufacturing. - Characterized with a 1 kHz sine wave. - The ADC module is functional at $V_{BORMIN} < V_{DD} < V_{DDMIN}$, but with degraded performance. Unless otherwise stated, module functionality is ensured, but not characterized. - For the dedicated core, the throughput includes 4 T_{ADCORE} sampling time and 13 T_{ADCORE} conversion time. - For the shared core, the throughput includes 10 T_{ADCORE} sampling time and 13 T_{ADCORE} conversion time. - Data in the "Typical" column are at 3.3V, +25°C. Parameters are for design guidance only and are not tested.							

dsPIC33CK512MPT608 Family

Electrical Characteristics

.....continued

Operating Conditions: 3.0V to 3.6V (unless otherwise stated)⁽⁴⁾

Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial

$-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended

Param No.	Symbol	Characteristics	Min.	Typical ⁽⁷⁾	Max.	Units	Conditions
AD61	CHOLD	Capacitance	—	18	—	pF	Shared core (Note 1)
AD62	R _{IC}	Input resistance	—	500	1000	Ω	Includes R _{SS} (Note 1)
AD66	V _{BG}	Internal Voltage Reference Source	1.14	1.2	1.26	V	
AD67	FSRC	ADC Module Input Frequency	—	—	500	MHz	Clock frequency selected by the CLKSELx bits
	FCORESRC	ADC Control Clock Frequency	—	—	250	MHz	Clock frequency after the first divider controlled by the CLKDIVx bits
	FADCORE	ADC SAR Core Clock Frequency	—	—	70	MHz	SAR core frequency after the second divider controlled by the ADCSx or SHRADCSx bits
ADC Accuracy							
AD20c	N _R	Resolution	12 data bits			bits	
AD21a	INL_1D	Dedicated Core Integral Nonlinearity (1 Active Core)	-3.5	-1.5/+1.5	+3.5	LSb	3.5 Msps ⁽⁵⁾ , T _{ADC} = 4 nS (250 MHz), T _{CORESRC} = 8 nS (125 MHz), T _{ADCORE} = 16 nS (62.5 MHz), Sampling Time = 4 T _{ADCORE} , V _{DD} = 3.3V, AV _{DD} = 3.3V
AD22a	DNL_1D	Dedicated Core Differential Nonlinearity (1 Active Core)	-1.0	-1.5/+1.5	+3.5	LSb	
AD23a	GERR_1D	Dedicated Core Gain Error (1 Active Core)	—	+4	—	LSb	
AD24a	OERR_1D	Dedicated Core Offset Error (1 Active Core)	—	-4	—	LSb	
AD21b	INL_1S	Shared Core Integral Nonlinearity (1 Active Core)	-3.5	-1.5/+1.5	+3.5	LSb	
AD22b	DNL_1S	Shared Core Differential Nonlinearity (1 Active Core)	-1.0	-1.5/+1.5	+3.5	LSb	2.7 Msps ⁽⁶⁾ , T _{ADC} = 4 nS (250 MHz), T _{CORESRC} = 8 nS (125 MHz), T _{ADCORE} = 16 nS (62.5 MHz), Sampling Time = 10 T _{ADCORE} , V _{DD} = 3.3V, AV _{DD} = 3.3V
AD23b	GERR_1S	Shared Core Gain Error (1 Active Core)	—	+4	—	LSb	
AD24b	OERR_1S	Shared Core Offset Error (1 Active Core)	—	-4	—	LSb	

Notes:

1. These parameters are not characterized or tested in manufacturing.
2. These parameters are characterized but not tested in manufacturing.
3. Characterized with a 1 kHz sine wave.
4. The ADC module is functional at $V_{BORMIN} < V_{DD} < V_{DDMIN}$, but with degraded performance. Unless otherwise stated, module functionality is ensured, but not characterized.
5. For the dedicated core, the throughput includes 4 T_{ADCORE} sampling time and 13 T_{ADCORE} conversion time.
6. For the shared core, the throughput includes 10 T_{ADCORE} sampling time and 13 T_{ADCORE} conversion time.
7. Data in the "Typical" column are at 3.3V, +25°C. Parameters are for design guidance only and are not tested.

.....continued

Operating Conditions: 3.0V to 3.6V (unless otherwise stated)⁽⁴⁾Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial $-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended

Param No.	Symbol	Characteristics	Min.	Typical ⁽⁷⁾	Max.	Units	Conditions
AD21c	INL_5D	Dedicated Core Integral Nonlinearity (5 Active Cores)	—	-8/+8	—	LSb	3.5 Msps ⁽⁵⁾ , $T_{ADC} = 4 \text{ nS}$ (250 MHz), $T_{CORESRC} = 8 \text{ nS}$ (125 MHz), $T_{ADCORE} = 16 \text{ nS}$ (62.5 MHz), Sampling Time = 4 T_{ADCORE} , $V_{DD} = 3.3\text{V}$, $AV_{DD} = 3.3\text{V}$, all core conversions are started simultaneously
AD22c	DNL_5D	Dedicated Core Differential Nonlinearity (5 Active Cores)	—	-1.5/+3	—	LSb	
AD23c	GERR_5D	Dedicated Core Gain Error (5 Active Cores)	—	+9.5	—	LSb	
AD24c	OERR_5D	Dedicated Core Offset Error (5 Active Cores)	—	-9.5	—	LSb	
AD21d	INL_5S	Shared Core Integral Nonlinearity (5 Active Cores)	—	-8/+8	—	LSb	2.7 Msps ⁽⁶⁾ , $T_{ADC} = 4 \text{ nS}$ (250 MHz), $T_{CORESRC} = 8 \text{ nS}$ (125 MHz), $T_{ADCORE} = 16 \text{ nS}$ (62.5 MHz), Sampling Time = 10 T_{ADCORE} , $V_{DD} = 3.3\text{V}$, $AV_{DD} = 3.3\text{V}$, all core conversions are started simultaneously
AD22d	DNL_5S	Shared Core Differential Nonlinearity (5 Active Cores)	—	-1.5/+3	—	LSb	
AD23d	GERR_5S	Shared Core Gain Error (5 Active Cores)	—	+9.5	—	LSb	
AD24d	OERR_5S	Shared Core Offset Error (5 Active Cores)	—	-9.5	—	LSb	
AD25c	—	Monotonicity	—	—	—	LSb	Guaranteed
Dynamic Performance							
AD31b	SINAD	Signal-to-Noise and Distortion	56	—	70	dB	Notes 2, 3
AD34b	ENOB	Effective Number of Bits	9.8	10.2	11.4	bits	Notes 2, 3
AD50	T_{AD}	ADC Clock Period	14.3	—	—	ns	
AD51	F_{TP}	Throughput Rate	—	—	3.5	Msps	Dedicated cores
			—	—	2.7	Msps	Shared core

Notes:

- These parameters are not characterized or tested in manufacturing.
- These parameters are characterized but not tested in manufacturing.
- Characterized with a 1 kHz sine wave.
- The ADC module is functional at $V_{BORMIN} < V_{DD} < V_{DDMIN}$, but with degraded performance. Unless otherwise stated, module functionality is ensured, but not characterized.
- For the dedicated core, the throughput includes 4 T_{ADCORE} sampling time and 13 T_{ADCORE} conversion time.
- For the shared core, the throughput includes 10 T_{ADCORE} sampling time and 13 T_{ADCORE} conversion time.
- Data in the "Typical" column are at 3.3V, +25°C. Parameters are for design guidance only and are not tested.

Table 33-41. High-Speed Analog Comparator Module Specifications

Operating Conditions: 3.0V to 3.6V (unless otherwise stated) ⁽²⁾							
Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial							
$-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended							
Param No.	Symbol	Characteristic	Min.	Typ.	Max.	Units	Comments
CM09	F_{IN}	Input Frequency	400	500	550	MHz	
CM10	V_{IOFF}	Input Offset Voltage	-20	—	+20	mV	
CM11	V_{ICM}	Input Common-Mode Voltage Range ⁽¹⁾	AV_{SS}	—	AV_{DD}	V	Note 1
CM13	CMRR	Common-Mode Rejection Ratio	60	—	—	dB	Note 1
CM14	T_{RESP}	Large Signal Response	—	15	—	ns	V+ input step of 100 mV while V- input is held at $AV_{\text{DD}}/2$
CM15	V_{HYST}	Input Hysteresis	15	30	45	mV	Depends on HYSSEL[1:0] (Note 1)
Notes:							
1. These parameters are for design guidance only and are not tested in manufacturing.							
2. The comparator module is functional at $V_{\text{BORMIN}} < V_{\text{DD}} < V_{\text{DDMIN}}$, but with degraded performance. Unless otherwise stated, module functionality is tested, but not characterized.							

Table 33-42. Die Temperature Diode Specifications

Operating Conditions: 3.0V to 3.6V (unless otherwise stated)							
Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial							
$-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended							
Param No.	Symbol	Characteristic	Min.	Typ.	Max.	Units	Comments
TD01	T_{COEFF}		—	1.5	—	mV/C	Note 1
Note:							
1. These parameters are not characterized or tested in manufacturing.							

Table 33-43. DACx Module Specifications

Operating Conditions: 3.0V to 3.6V (unless otherwise stated)							
Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial							
$-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended							
Param No.	Symbol	Characteristic	Min.	Typ. ⁽¹⁾	Max.	Units	Comments
DA02	CV _{RES}	Resolution	12			bits	
DA03	INL	Integral Nonlinearity Error	-43	—	0	LSB	
DA04	DNL	Differential Nonlinearity Error	-5	—	5	LSB	
DA05	EOFF	Offset Error	-3.5	—	25	LSB	Internal node at comparator input
DA06	EG	Gain Error	0	—	41	%	Internal node at comparator input
DA07	T _{SET}	Settling Time	600	750	2000	ns	Output with 1% of desired output voltage with a 5-95% or 95-5% step (Note 1)
DA08	V _{OUT}	Voltage Output Range	0.165	—	3.135	V	V _{DD} = 3.3V
DA09	T _{TR}	Transition Time	340	—	—	ns	Note 1
DA10	T _{SS}	Steady-State Time	550	—	—	ns	Note 1
Note:							
1. Parameters are for design guidance only and are not tested in manufacturing.							

Table 33-44. DACx Output (DACOUT1 and DACOUT2 Pins) Specifications

Operating Conditions: 3.0V to 3.6V (unless otherwise stated) ^(1,2,3)							
Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial							
$-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended							
Param No.	Symbol	Characteristic	Min.	Typ.	Max.	Units	Comments
DA11	R_{LOAD}	Resistive Output Load Impedance	10K	—	—	Ohm	
DA11a	C_{LOAD}	Output Load Capacitance	—	—	30	pF	Including output pin capacitance
DA12	I_{OUT}	Output Current Drive Strength	—	3	—	mA	Sink and source
Notes:							
1. Parameters are for design guidance only and are not tested in manufacturing.							
2. The DACx module is functional at $V_{\text{BORMIN}} < V_{\text{DD}} < V_{\text{DDMIN}}$, but with degraded performance. Unless otherwise stated, module functionality is tested, but not characterized.							
3. Using other pin functions may degrade DAC performance.							

Table 33-45. Constant-Current Source Specifications

Operating Conditions: 3.0V to 3.6V (unless otherwise stated) ⁽¹⁾							
Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial							
$-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended							
Param No.	Symbol	Characteristic	Min.	Typ.	Max.	Units	Conditions
CC02	I_{REG}	Current Regulation	—	± 3	—	%	IBIASx pin
CC03	$I_{10\text{SRC}}$	10 μA Source Current	8.8	—	11.2	μA	ISRCx pin
CC04	$I_{50\text{SRC}}$	50 μA Source Current	44	—	56	μA	IBIASx pin
CC05	$I_{50\text{SNK}}$	50 μA Sink Current	-44	—	-56	μA	IBIASx pin
Note:							
1. The constant-current source module is functional at $V_{\text{BORMIN}} < V_{\text{DD}} < V_{\text{DDMIN}}$, but with degraded performance. Unless otherwise stated, module functionality is tested, but not characterized.							

Table 33-46. Operational Amplifier Specifications

Operating Conditions: 3.0V to 3.6V (unless otherwise stated)							
Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial							
$-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended							
Param No.	Sym	Characteristic	Min	Typ ⁽¹⁾	Max	Units	Comments
OA01	GBWP	Gain Bandwidth Product	—	20	—	MHz	
OA02	SR	Slew Rate	—	40	—	V/ μs	

.....continued

Operating Conditions: 3.0V to 3.6V (unless otherwise stated)

Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial $-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended

Param No.	Sym	Characteristic	Min	Typ ⁽¹⁾	Max	Units	Comments
OA03	V _{IOFF}	Input Offset Voltage	-3 ⁽³⁾	-1/+1	+3 ⁽³⁾	mV	Unity gain configuration
			-8	-3/+3	+8	mV	Open-loop configuration
OA04	V _{IBC}	Input Bias Current	—	—	—	nA	Note 2
OA05	V _{ICM}	Common-Mode Input Voltage Range	V _{SS}	—	V _{DD}	V	NCHDISx = 0
			V _{SS}	—	V _{DD} - 1.4	V	NCHDISx = 1
OA07	CMRR	Common-Mode Rejection Ratio	—	68	—	dB	
OA08	PSRR	Power Supply Rejection Ratio	—	74	—	dB	
OA09	V _{OR}	Output Voltage Range	V _{SS}	—	V _{DD}	mV	0.5V input overdrive, no output loading (Note 1)
OA11	C _{LOAD}	Output Load Capacitance	—	—	30	pF	Including output pin capacitance (Note 1)
OA12	I _{OUT}	Output Current Drive Strength	—	3	—	mA	Sink and source (Note 1)
OA13	P _{MARGIN}	Phase Margin	44	—	—	degree	Unity gain (Note 1)
OA14	G _{MARGIN}	Gain Margin	7	—	—	dB	Unity gain (Note 1)
OA15	OLG	Open-Loop Gain	68	75	—	dB	Note 1

Notes:

1. These parameters are for design guidance only and are not tested in manufacturing.
2. The op amps use CMOS input circuitry with negligible input bias current. The maximum “effective bias current” is the I/O pin leakage specified by electrical Parameter [DI50](#).
3. Parameters are characterized but not tested in manufacturing.

Figure 33-18. QEA/QEB Input Characteristics

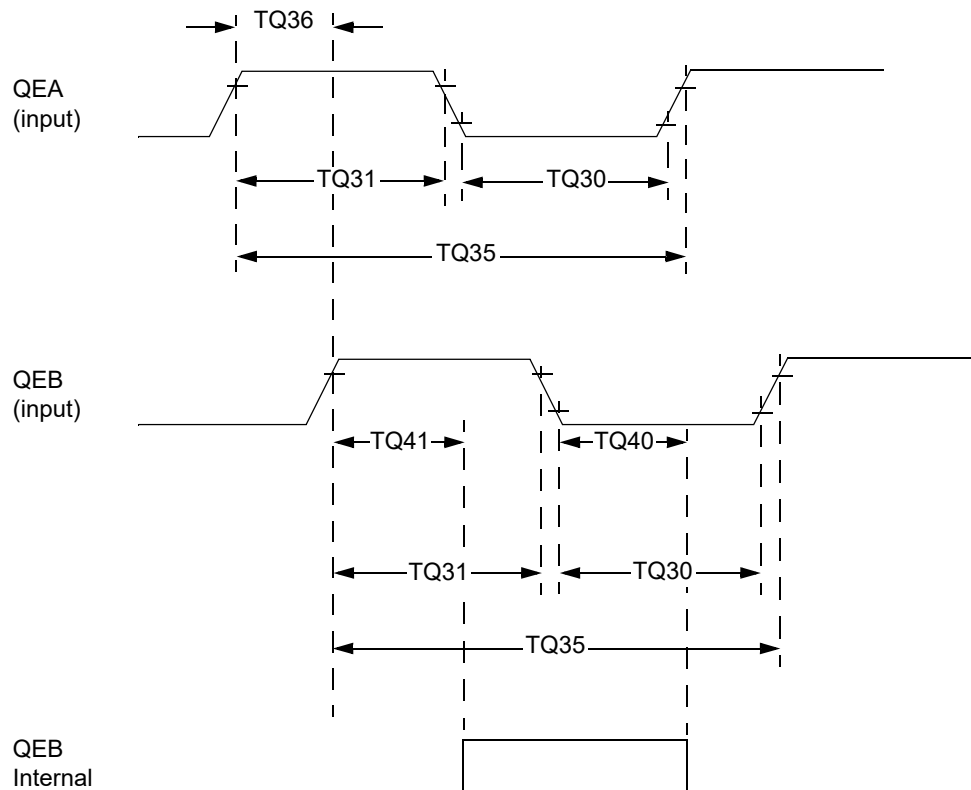


Table 33-47. Quadrature Decoding Timing Requirements

Operating Conditions: 3.0V to 3.6V (unless otherwise stated)						
Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial						
$-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended						
Param.	Symbol	Characteristic ⁽¹⁾	Typ. ⁽²⁾	Max.	Units	Conditions
TQ30	T _{QU} L	Quadrature Input Low Time	6 T _{CY}	—	ns	
TQ31	T _{QU} H	Quadrature Input High Time	6 T _{CY}	—	ns	
TQ35	T _{QU} IN	Quadrature Input Period	12 T _{CY}	—	ns	
TQ36	T _{QU} P	Quadrature Phase Period	3 T _{CY}	—	ns	

Notes:

- These parameters are characterized but not tested in manufacturing.
- Data in "Typ" column are at 3.3V, +25°C unless otherwise stated. Parameters are for design guidance only and are not tested.
- N = Index Channel Digital Filter Clock Divide Select bits. Refer to "**Quadrature Encoder Interface (QEI)**" (**DS70000601**) in the "*dsPIC33E/PIC24E Family Reference Manual*." Please see the Microchip web site for the latest family reference manual sections.

.....continued

Operating Conditions: 3.0V to 3.6V (unless otherwise stated)

Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial $-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended

Param.	Symbol	Characteristic ⁽¹⁾	Typ. ⁽²⁾	Max.	Units	Conditions
TQ40	T_{QUFL}	Filter Time to Recognize Low with Digital Filter	$3 * N * T_{CY}$	—	ns	N = 1, 2, 4, 16, 32, 64, 128 and 256 (Note 3)
TQ41	T_{QUFH}	Filter Time to Recognize High with Digital Filter	$3 * N * T_{CY}$	—	ns	N = 1, 2, 4, 16, 32, 64, 128 and 256 (Note 3)

Notes:

1. These parameters are characterized but not tested in manufacturing.
2. Data in “Typ” column are at 3.3V, +25°C unless otherwise stated. Parameters are for design guidance only and are not tested.
3. N = Index Channel Digital Filter Clock Divide Select bits. Refer to “**Quadrature Encoder Interface (QEI)**” (**DS70000601**) in the “*dsPIC33E/PIC24E Family Reference Manual*.” Please see the Microchip web site for the latest family reference manual sections.

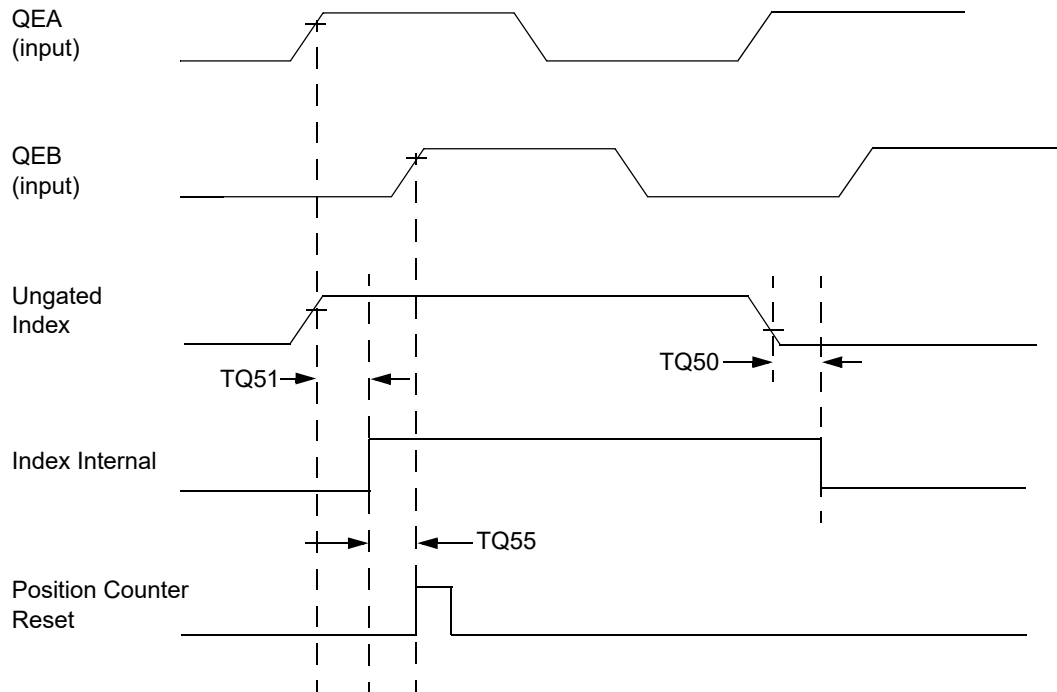
Figure 33-19. QEI Module Index Pulse Timing Characteristics

Table 33-48. QEI Index Pulse Timing Requirements

Operating Conditions: 3.0V to 3.6V (unless otherwise stated)						
Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial						
$-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended						
Param.	Symbol	Characteristic ⁽¹⁾	Typ. ⁽²⁾	Max.	Units	Conditions
TQ50	T_{QIL}	Filter Time to Recognize Low with Digital Filter	$3 * N * T_{CY}$	—	ns	$N = 1, 2, 4, 16, 32, 64, 128$ and 256 (Note 2)
TQ51	T_{QIH}	Filter Time to Recognize High with Digital Filter	$3 * N * T_{CY}$	—	ns	$N = 1, 2, 4, 16, 32, 64, 128$ and 256 (Note 2)
TQ55	T_{QIDXR}	Index Pulse Recognized to Position Counter Reset (ungated index)	$3 T_{CY}$	—	ns	

Notes:

- These parameters are characterized but not tested in manufacturing.
- Alignment of index pulses to QEA and QEB is shown for position counter Reset timing only. Shown for forward direction only (QEA leads QEB). The same timing applies for reverse direction (QEA lags QEB), but index pulse recognition occurs on the falling edge.

Figure 33-20. QEI External Clock Timing Characteristics

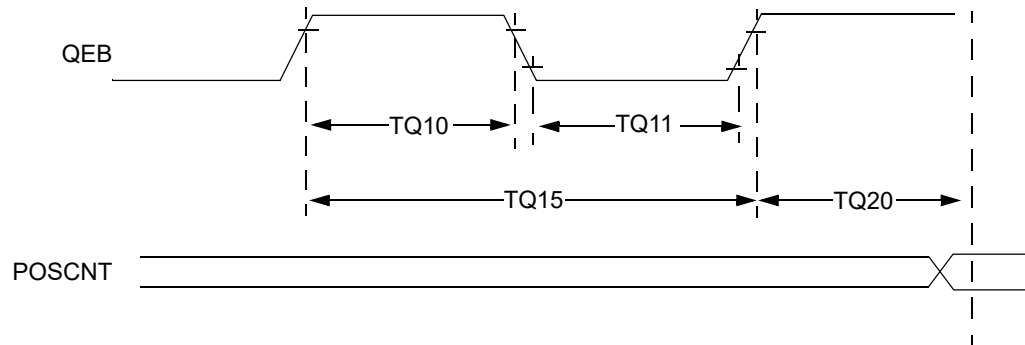


Table 33-49. QEI Module External Clock Timing Requirements

Operating Conditions: 3.0V to 3.6V (unless otherwise stated)

Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial

$-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended

Param.	Symbol	Characteristic ⁽¹⁾		Typ.	Max.	Units	Conditions
TQ10	T _{TQH}	TQCK High Time	Synchronous, with Prescaler	—	—	ns	Must also meet Parameter TQ15

Note:

- These parameters are characterized but not tested in manufacturing.

.....continued

Operating Conditions: 3.0V to 3.6V (unless otherwise stated)

Operating temperature $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$ for Industrial $-40^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ for Extended

Param.	Symbol	Characteristic ⁽¹⁾		Typ.	Max.	Units	Conditions
TQ11	T _{TQL}	TQCK Low Time	Synchronous, with Prescaler	—	—	ns	Must also meet Parameter TQ15
TQ15	T _{TQP}	TQCP Input Period	Synchronous, with Prescaler	—	—	ns	
TQ20	T _{CKEXTMRL}	Delay from External TxCK Clock Edge to Timer Increment		1	T _{CY}	—	

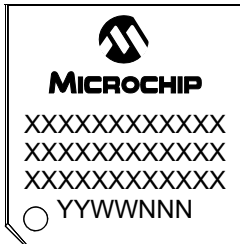
Note:

- These parameters are characterized but not tested in manufacturing.

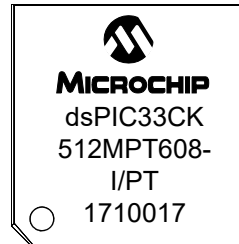
34. Packaging Information

34.1 Package Marking Information

100-Lead TQFP (12x12x1)



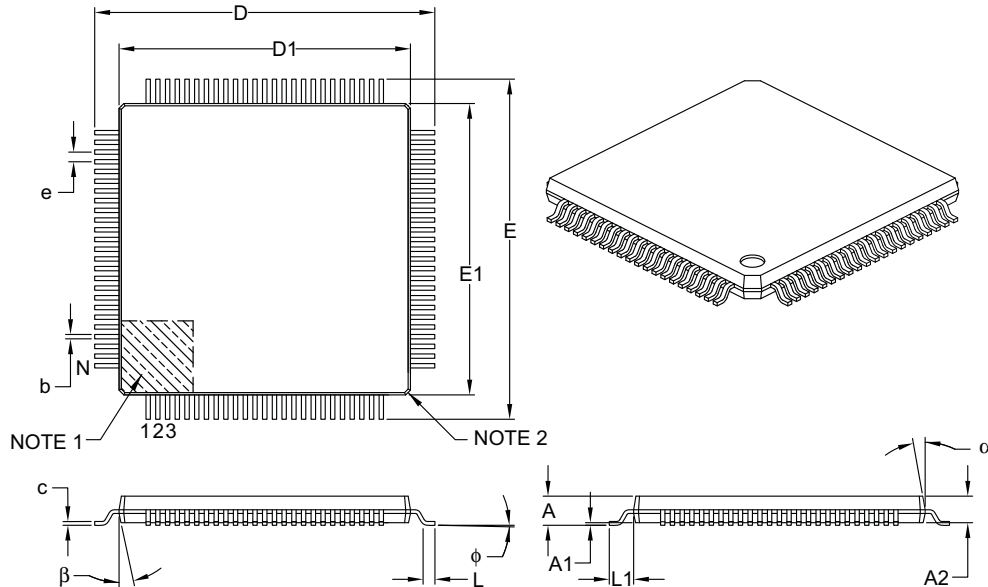
Example



34.2 Package Details

100-Lead Plastic Thin Quad Flatpack (PT) – 12x12x1 mm Body, 2.00 mm [TQFP]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Dimension Limits	Units	MILLIMETERS		
		MIN	NOM	MAX
Number of Leads	N	100		
Lead Pitch	e	0.40 BSC		
Overall Height	A	–	–	1.20
Molded Package Thickness	A2	0.95	1.00	1.05
Standoff	A1	0.05	–	0.15
Foot Length	L	0.45	0.60	0.75
Footprint	L1	1.00 REF		
Foot Angle	φ	0°	3.5°	7°
Overall Width	E	14.00 BSC		
Overall Length	D	14.00 BSC		
Molded Package Width	E1	12.00 BSC		
Molded Package Length	D1	12.00 BSC		
Lead Thickness	c	0.09	–	0.20
Lead Width	b	0.13	0.18	0.23
Mold Draft Angle Top	α	11°	12°	13°
Mold Draft Angle Bottom	β	11°	12°	13°

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Chamfers at corners are optional; size may vary.
- Dimensions D1 and E1 do not include mold flash or protrusions. Mold flash or protrusions shall not exceed 0.25 mm per side.
- Dimensioning and tolerancing per ASME Y14.5M.

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

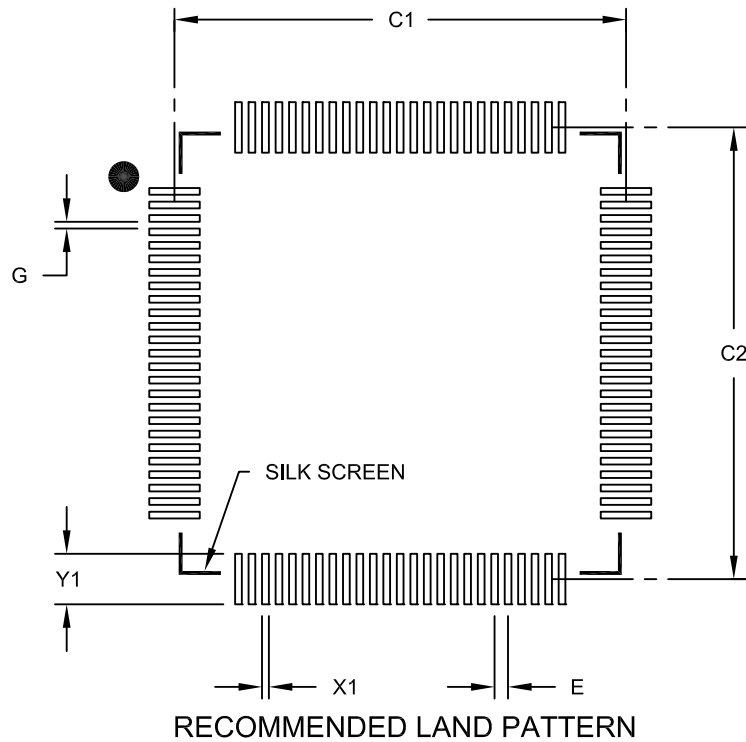
REF: Reference Dimension, usually without tolerance, for information purposes only.

Microchip Technology Drawing C04-100B

dsPIC33CK512MPT608 Family

Packaging Information

100-Lead Plastic Thin Quad Flatpack (PT)-12x12x1mm Body, 2.00 mm Footprint [TQFP]



Units	Dimension Limits	MILLIMETERS		
		MIN	NOM	MAX
Contact Pitch	E		0.40 BSC	
Contact Pad Spacing	C1		13.40	
Contact Pad Spacing	C2		13.40	
Contact Pad Width (X100)	X1			0.20
Contact Pad Length (X100)	Y1			1.50
Distance Between Pads	G	0.20		

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing No. C04-2100B

35. Revision History

Revision A (May 2022)

This is the initial version of the document.

Revision B (August 2022)

This revision incorporates the following updates:

- Sections:
 - **6.2 Features, 6.3 SPI Clock Configuration and 30.6 Brown-out Reset (BOR).**
- Tables
 - **Table 33-22. Secure Subsystem Δ Current.**
- Figures:
 - **Figure 6-1. Secure Subsystem Block Diagram.**

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PART NO. [X]⁽¹⁾ -X /XX
 Device Tape and Reel Temperature Range Package

Device:		
Tape & Reel Option:	Blank	= Tube
	T	= Tape & Reel
Temperature Range:	I	= -40°C to +85°C (Industrial)
	E	= -40°C to +125°C (Extended)
Package:	PT	= 100-Lead TQFP

Examples:

- dsPIC33CK512MPT608-I/PT: Industrial Temperature, 100-Lead TQFP Package

Notes:

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